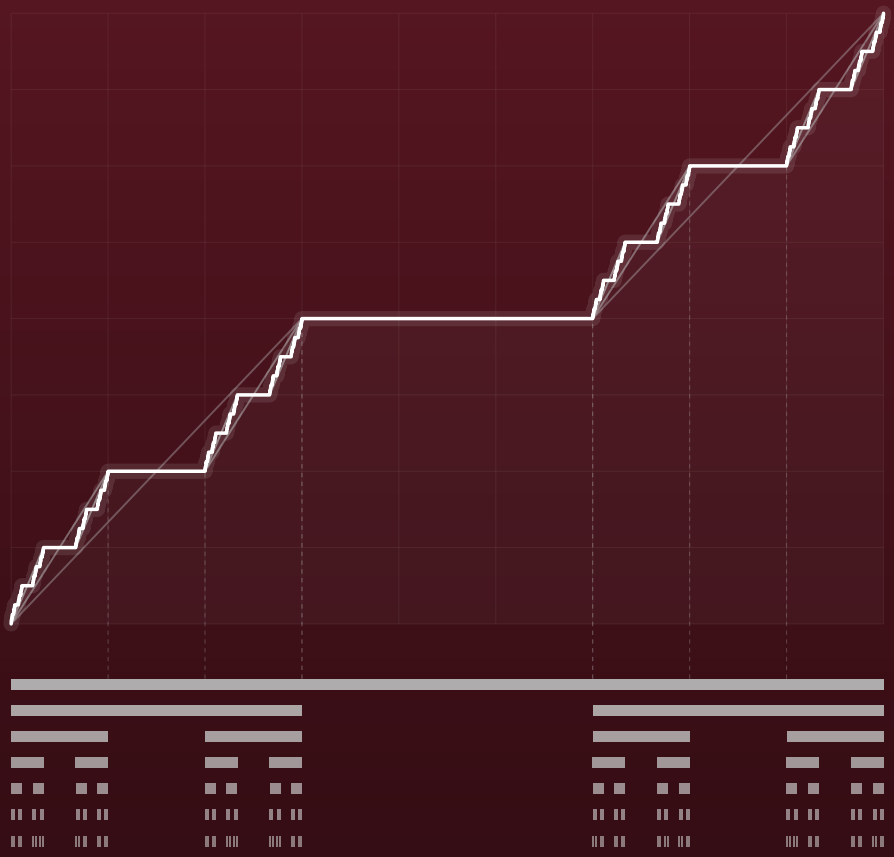
An Open Textbook

Real Analysis

A Rigorous Treatment from First Principles to the Lebesgue Integral

Written for anyone who has finished calculus. No proofs course required.



Jeffrey T. Neugebauer

Eastern Kentucky University

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Version 0.1, October 2026

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This is a draft. Please send corrections to the author.

This book is a living document. It will change over time, and the changes will be listed on its website, https://jneugemath.github.io/real-analysis/. Corrections and suggestions can be sent with the form at https://jneugemath.github.io/real-analysis/feedback.html.

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If you use this book in a course, please let me know with the form at https://jneugemath.github.io/real-analysis/feedback.html, and I will add your name here.

Preface

Who this book is for. This book is for a reader wanting to learn real analysis from the very beginning. It covers introductory real analysis through measure theory and Lebesgue integrals. It does not assume a course in proofs. A reader who has finished calculus and is willing to work should be able to read it.

Most students encounter two difficulties when taking real analysis. The ideas are new, and so is writing proofs. Many textbooks assume the second problem was solved in an earlier course. This book does not. Chapter 1 starts with sets and the integers. Each proof technique is explained the first time it is used. By the time the ε-N definition of convergence arrives in Chapter 3, the reader has already seen direct proofs, proofs by contradiction, proofs by cases and induction, all on simpler material.

The book is also meant to be complete. Results are proven, not quoted. When a fact is assumed, the text says so. The goal is to give a student a book that can be read without a lecture to fill in the gaps.

How the book is organized. The book has three parts. Part I, Numbers, Limits and Derivatives, starts from the axioms for the real numbers and goes through sequences, the topology of the real line, continuity and derivatives. Part II, Riemann Integrals and Series, covers functions of bounded variation, the Riemann and Riemann–Stieltjes integrals, series, and sequences and series of functions. Parts I and II are written for a two-semester undergraduate sequence. At Eastern Kentucky University that sequence is MAT 315, which covers Part I (Chapters 1–7), and MAT 415, which covers Part II (Chapters 8–12).

Part III, Measure and Lebesgue Integrals, covers measure theory and the Lebesgue integral. It is written for a first graduate course in real analysis, MAT 815 at Eastern Kentucky University, which sadly no longer exists. It assumes Parts I and II.

Appendices A and B construct the real numbers from the rational numbers in two different ways. Each one proves Theorem 1.8.2, which Chapter 1 uses as a starting point. They can be read any time after Chapter 1. Appendix B also uses the idea of a Cauchy sequence from Section 3.5.

Chapters and sections marked with a star ($*$) can be skipped. They are Section 1.3, Section 2.4, Chapter 5, Sections 6.6, 7.3 and 7.4, Chapter 8, and Sections 9.4, 9.5, 9.8, 9.9, 10.4, 10.5, 11.3 to 11.5, 12.4 and 12.5 in Parts I and II, and Sections 14.4, 15.4, 17.1, 17.6 and 17.7 and Chapter 18 in Part III, and Appendix B.5 in Appendix B. Nothing else in the book depends on these sections, with a few small exceptions. Example 9.2.8 uses the Cantor set. Section 10.5 and Exercise 1.4.4 use Section 1.3. Chapter 17 uses the statement of the Vitali Covering Lemma from Section 17.1, but not its proof. Section 9.9 uses functions of bounded variation from Chapter 8, and

Sections 17.3 and 17.5 recall its definitions and use a few of its results. Partitions are defined again in Section 9.1, so Chapter 9 can be read without Chapter 8. Appendices A and B construct $\mathbb{R}$ from $\mathbb{Q}$ in two ways. They can also be skipped. Sections 9.4, 9.5, 12.4 and 12.5 build $\ln x$, e^x, $\sin x$ and $\cos x$, two ways each. The rest of the book uses these functions freely, so a reader who skips those sections should assume their familiar properties, as Remark 3.1.1 does.

How to use this book. Read the Proof Technique boxes. They are listed after the Contents, so they can be referenced again later. Every section ends with exercises. Trying these exercises is the only way to learn to write proofs. Exercises marked [Hint] or [Solution] have hints or solutions in Appendix C. Always try problems before reading the hints and solutions. The Notation list gives the page where each symbol is introduced.

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Sources. This book grew out of lecture notes for MAT 315, MAT 415 and MAT 815 at Eastern Kentucky University. The notes for Parts I and II were adapted from Rudin's *Principles of Mathematical Analysis* [**62**] and Stoll's *Introduction to Real Analysis* [**71**]. The notes for Part III were adapted from Folland's *Real Analysis* [**31**], Royden's *Real Analysis* [**61**] and Rudin's *Real and Complex Analysis* [**63**]. Many definitions and proofs here follow their treatments.

Use of AI. The definitions, theorems, proofs, examples and exercises in this book began as my lecture notes. They were written and typed before any AI tool was used. After that, I used Claude Opus 5.5, an AI model made by Anthropic, to help turn the notes into a book. It standardized the notation and cross-references, found and fixed errors, and filled in missing steps in proofs. It suggested where to insert Proof Technique boxes. It also drafted new material. This includes some exposition, the Notes at the end of each chapter, the figures, and some theorems, examples and exercises. I reviewed and revised all of this material. I am responsible for every word, and for any errors that remain.

Notation

The page number points to where the notation is introduced.

Symbol	Meaning	Page
$x \in A$, $x \notin A$	x is (is not) an element of A	3
$\emptyset$	the empty set	3
$\{x \in A : P(x)\}$	the set of $x \in A$ for which $P(x)$ is true	3
$A \subset B$	every element of A is an element of B ($A = B$ is allowed)	3
$A \subsetneq B$	$A \subset B$ and $A \neq B$	3
$A \cup B$, $A \cap B$	union, intersection	4
$A \setminus B$	the elements of A that are not in B	4
A^c	the complement of A	4
$\{A_\alpha\}_{\alpha \in \Lambda}$	a family of sets with index set Λ	6
$\bigcup_{\alpha \in \Lambda} A_\alpha$, $\bigcap_{\alpha \in \Lambda} A_\alpha$	union and intersection of a family	6
(a, b)	an ordered pair (also an open interval)	7
$A \times B$	the Cartesian product of A and B	7
$\mathcal{P}(A)$	the power set of A	7
$\mathbb{N}$	the natural numbers $\{1, 2, 3, \dots\}$	8
$\mathbb{N}_0$	the whole numbers $\{0, 1, 2, \dots\}$	8
$\mathbb{Z}$	the integers	8
a^n	the nth power of a	10
$\sum_{i=1}^{n} a_i$	the sum $a_1 + a_2 + \cdots + a_n$	10
$n!$	n factorial	10
$\mathbb{Q}$	the rational numbers	15
$x \leq y$	$x < y$ or $x = y$	16
$\sup E$, $\operatorname{lub} E$	the supremum (least upper bound) of E	17
$\inf E$, $\operatorname{glb} E$	the infimum (greatest lower bound) of E	17
$\mathbb{F}$	a field	19
$-x$, x^{-1}	the additive and multiplicative inverses of x	19
$x - y$, $\frac{x}{y}$	$x + (-y)$ and $x \cdot y^{-1}$	19
$\lvert x \rvert$	the absolute value of x	22
$\mathbb{R}$	the real numbers	23
$\mathbb{R} \setminus \mathbb{Q}$	the irrational numbers	23
$[a, b]$, (a, b), $[a, \infty)$, $\dots$	intervals	23
$\lfloor x \rfloor$	the greatest integer less than or equal to x	25
$\sqrt[n]{x}$, $\sqrt{x}$	the positive nth root and square root of $x > 0$	25
$f : A \to B$	a function from A to B	29

Symbol	Meaning	Page
$f[E]$, Range f	the image of E, the range of f	29
$f^{-1}[G]$	the preimage of G	29
$g \circ f$	the composition of g and f	29
$A \sim B$, $\lvert A\rvert = \lvert B\rvert$	A and B have the same cardinality	30
$\mathbb{N}_n$	the set $\{1, 2, \ldots, n\}$	30
$\mathbb{E}$, $\mathbb{O}$	the even integers, the odd integers	31
$\{x_n\}$, $\{x_n\}_{n=1}^{\infty}$	a sequence	32
$2^{\mathbb{N}}$, $\{0,1\}^{\mathbb{N}}$	the set of sequences of 0's and 1's	34
$A \lesssim B$, $B \gtrsim A$	there is a one-to-one function from A into B	35
$\lim_{n\to\infty} x_n = a$, $x_n \to a$	the sequence $\{x_n\}$ converges to a	39
$x_n \to \pm\infty$	$\{x_n\}$ diverges to $\pm\infty$	52
$\{x_{n_k}\}$	a subsequence of $\{x_n\}$	48
$\limsup x_n$, $\liminf x_n$	the limit superior and limit inferior of $\{x_n\}$	52
E°	the interior of E	55
E'	the set of limit points of E	60
$\overline{E}$	the closure of E	60
$\{U_\alpha\}_{\alpha\in\Lambda}$, $\mathcal{U}$	an open cover	61
C, C_n, $J_{n,i}$	the Cantor set and the sets used to build it	67
$.n_1n_2n_3\ldots$	a ternary expansion	70
$\lim_{x\to c} f(x) = L$	the limit of f at c is L	73
$\lim_{x\to c^+} f(x)$, $\lim_{x\to c^-} f(x)$	the right-hand and left-hand limits of f at c	78
$\lim_{x\to c} f(x) = \pm\infty$	f has an infinite limit at c	80
$\lim_{x\to\pm\infty} f(x)$	the limit of f at $\pm\infty$	80
$f(c^-)$, $f(c^+)$	one-sided limits of a monotone function	88
$\lim_{x\to c^\pm} f(x) = \pm\infty$	one-sided infinite limits	80
$C(E)$, $C[a,b]$	the continuous functions on E, on $[a,b]$	83
f^{-1}	the inverse of a one-to-one function f	89
$f'(c)$, $\frac{d}{dx}f(x)$	the derivative of f at c	93
$f''(c)$, $f^{(n)}(c)$	the second and nth derivatives of f at c	93
$C^{(1)}(I)$, $C^{(n)}(I)$	continuously differentiable functions, and their analogues of order n	94
$\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$	a partition of $[a,b]$	109
Δx_i, $\lVert\mathcal{P}\rVert$	$x_i - x_{i-1}$, and the mesh of $\mathcal{P}$	109
$\mathcal{P}^*$	a refinement of $\mathcal{P}$	109
$V(\mathcal{P})$, $V_f[a,b]$	a variation sum, and the total variation of f on $[a,b]$	109
$BV[a,b]$	the functions of bounded variation on $[a,b]$	109
$V_f(x)$	the variation function $V_f[a,x]$	112
m_i, M_i	the infimum and supremum of f on $[x_{i-1}, x_i]$	115
$\mathcal{U}(\mathcal{P}, f)$, $\mathcal{L}(\mathcal{P}, f)$	the upper and lower sums of f for $\mathcal{P}$	115
$\overline{\int_a^b} f$, $\underline{\int_a^b} f$	the upper and lower integrals of f	116
$\int_a^b f$, $\int_a^b f(x)\,dx$	the Riemann integral of f over $[a,b]$	117

Symbol	Meaning	Page
$\mathcal{R}[a,b]$	the Riemann integrable functions on $[a,b]$	117
$m(E)=0$, $\ell(I)$	E has measure zero; the length of an interval I	123
$\mathcal{I}(\mathcal{P},f)$	a Riemann sum	127
$\lim_{\|\mathcal{P}\|\to 0}\mathcal{I}(\mathcal{P},f)$	the limit of Riemann sums	127
$\binom{n}{k}$	a binomial coefficient	131
e	Euler's number	131
$\ln x$	the natural logarithm	131
$\exp(x)$, e^x	the exponential function	133
a^x	a power with a real exponent	134
$\log_a x$	the logarithm to the base a	134
$\arctan x$, $\tan x$	the inverse tangent and the tangent function	137
π	the number π	137
$\sin x$, $\cos x$	the sine and cosine functions	138
$\arcsin x$, $\arccos x$	the inverse sine and inverse cosine	140
$\mathcal{R}_{\text{loc}}(I)$	the locally integrable functions on I	143
$\text{P.V.}\int$	a principal value	145
$\mathcal{U}(\mathcal{P},f,\alpha)$, $\mathcal{L}(\mathcal{P},f,\alpha)$	upper and lower Riemann–Stieltjes sums	151
$\int_a^b f\,d\alpha$, $\mathcal{R}(\alpha)$	the Riemann–Stieltjes integral, and the functions that have one	152
$I(x-c)$, I_c	the unit jump function at c	152
$\sum_{k=1}^{\infty} a_k$, $\sum a_k$	an infinite series, and its sum when it converges	163
s_n	the nth partial sum of a series	163
γ	Euler's constant	168
$\sum f_k$	a series of functions	183
$B_n f$	the Bernstein polynomial of f of degree n	198
R	the radius of convergence of a power series	204
$C^\infty(I)$	the infinitely differentiable functions on I	209
$T_n(f,c)$, $T(f,c)$	the Taylor polynomial of order n and the Taylor series of f at c	212
$R_n(f,c)$	the remainder $f-T_n(f,c)$	213
$\mathcal{A}$, $\mathcal{M}$	an algebra and a σ-algebra of subsets of X	231
$\sigma(\mathcal{E})$	the σ-algebra generated by $\mathcal{E}$	233
$\mathcal{B}_{\mathbb{R}}$	the Borel sets	235
G_δ, F_σ	countable intersections of open sets, countable unions of closed sets	235
$(X,\mathcal{M},\mu)$	a measure space	236
$m^*(A)$	the outer measure of A	241
$\mathcal{M}$, $m(E)$	the Lebesgue measurable sets, and Lebesgue measure	244
$E+y$	the translate $\{x+y : x\in E\}$	241
$x \overset{\circ}{+} y$, $E \overset{\circ}{+} y$	sum and translation modulo 1	249
$[-\infty,\infty]$	the extended real numbers	251
$\{f>\alpha\}$	the set $\{x : f(x)>\alpha\}$	251
χ_A	the characteristic function of A	253

Symbol	Meaning	Page
a.e.	almost everywhere	253
$\int \varphi$, $\int_E \varphi$	the integral of a simple function	261
$\int_E f$, $\int_E f\,dm$	the Lebesgue integral of f over E	265
$\omega_f(x)$	the oscillation of f at x	269
f^+, f^-	the positive and negative parts of f	277
$L^1(E)$, $L^1[a,b]$	the functions integrable over E, and over $[a,b]$	277
D^+f, D^-f, D_+f, D_-f	the Dini derivatives of f	285
$P_f[a,b]$, $N_f[a,b]$	the positive and negative variations of f on $[a,b]$	288
$AC[a,b]$	the absolutely continuous functions on $[a,b]$	294
ϕ	the Cantor function	298
$L^p(E)$, $\|f\|_p$	the pth power integrable functions on E, and the L^p norm	303
$L^\infty(E)$, $\|f\|_\infty$	the essentially bounded functions on E, and their norm	303
q^*	the rational cut $\{p \in \mathbb{Q} : p < q\}$	309
α^+	the positive elements of a cut $\alpha > 0^*$	313
$\mathcal{C}$	the Cauchy sequences of rational numbers	319
$x \sim y$, $[x]$	equivalent Cauchy sequences, the class of x	320
$\iota_{\mathbb{F}}$	the copy of $\mathbb{Q}$ in an ordered field $\mathbb{F}$	325

Part I

Numbers, Limits and Derivatives

Part I builds the real numbers from axioms and studies limits. Limits of sequences come first, then the topology of the real line, then limits of functions, continuity, and derivatives. It assumes the reader has taken courses in the calculus sequence, but it does not assume the reader has taken a course in proofs.

CHAPTER 1

The Real Number System

Calculus uses the real numbers without ever defining them. Limits exist. Continuous functions on closed intervals have maximum values. Equations like $x^2 = 2$ have solutions. Behind each of these facts is a property of the real numbers that the rational numbers do not have. This chapter finds that property.

The chapter starts from the beginning. Section 1.1 introduces sets and their properties. Section 1.2 covers the integers and the two main tools for proving facts about integers, the Well-Ordering Principle and the Principle of Mathematical Induction. The starred Section 1.3 uses them to prove the Fundamental Theorem of Arithmetic. Section 1.4 shows the rational numbers have a hole where $\sqrt{2}$ should be. Sections 1.5 to 1.7 build the language needed to describe that hole precisely. Section 1.8 introduces the real numbers, which have no holes.

This chapter is also about how to read and write proofs. Each new proof technique is explained the first time it is used.

1.1. Sets

Almost every statement in this book is about sets. Sequences converge to points. Functions are continuous on intervals. Sets have suprema. This section fixes the notation for sets and the basic ways to combine them. It also contains the first proofs in the book. They are short, which makes them a good place to see how a proof is built.

Definition 1.1.1. A *set* is a collection of objects. The objects in a set are called its *elements*. If x is an element of the set A, write $x \in A$. If x is not an element of A, write $x \notin A$. The set with no elements is called the *empty set* and is denoted $\emptyset$.

A set can be described by listing its elements, such as $A = \{1, 2, 3\}$. A set can also be described by a property its elements satisfy. If $P(x)$ is a statement about x, then

$$\{x \in A : P(x)\}$$

is the set of all elements x of A for which $P(x)$ is true. It is read "the set of all x in A such that $P(x)$."

Example 1.1.2. Let $A = \{1, 2, 3, 4, 5, 6\}$. Then $\{x \in A : x > 3\} = \{4, 5, 6\}$. Also, $\{1, 2, 3\} = \{3, 1, 2\}$. The order the elements are listed in does not matter.

Definition 1.1.3. Let A and B be sets. A is a *subset* of B, written $A \subset B$, if every element of A is an element of B. A and B are *equal*, written $A = B$, if $A \subset B$ and $B \subset A$. If $A \subset B$ and $A \neq B$, A is a *proper subset* of B, written $A \subsetneq B$.

Remark 1.1.4. Some books write $A \subseteq B$ for a subset and $A \subset B$ for a proper subset. In this book, $A \subset B$ allows $A = B$.

Example 1.1.5. $\{1,2\} \subsetneq \{1,2,3\}$. However, $\{1,4\} \not\subset \{1,2,3\}$, since $4 \in \{1,4\}$ but $4 \notin \{1,2,3\}$.

Proof Technique (Proving One Set Is a Subset of Another). To prove $A \subset B$, let x be an arbitrary element of A. Then show $x \in B$. The word arbitrary is important. Nothing can be assumed about x except that $x \in A$. Since x was any element of A, every element of A is in B.

Theorem 1.1.6. *Let A be a set. Then*

(1) $A \subset A$, and
(2) $\emptyset \subset A$.

PROOF. (1) Let $x \in A$. Then $x \in A$. So $A \subset A$.
(2) Let $x \in \emptyset$. The goal is to show $x \in A$. But $\emptyset$ has no elements, so there is no such x and nothing to show. So $\emptyset \subset A$. □

Proof Technique (Vacuous Truth). The proof of Theorem 1.1.6 (2) never checks anything. The statement "for all $x \in \emptyset$, $x \in A$" cannot fail, because there is no x for which it could fail. A statement of the form "for all $x \in S$, $P(x)$" with $S = \emptyset$ is called *vacuously true*. It is true no matter what $P(x)$ says. For example, every element of $\emptyset$ is a negative integer. Every element of $\emptyset$ is also a positive integer. Both statements are true. The same idea applies to a statement "if P, then Q" when P is false. Vacuous truth comes up again in Example 4.1.16, which shows that $\emptyset$ is open.

Proof Technique (Proving Two Sets Are Equal). To prove $A = B$, prove two things. First show $A \subset B$. Then show $B \subset A$. This is called a double inclusion argument. The proof of Theorem 1.1.10 (3) below is an example.

Definition 1.1.7. Let A and B be sets.

(1) The *union* of A and B is $A \cup B = \{x : x \in A \text{ or } x \in B\}$.
(2) The *intersection* of A and B is $A \cap B = \{x : x \in A \text{ and } x \in B\}$.
(3) The *difference* of A and B is $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$.

If $A \cap B = \emptyset$, A and B are *disjoint*. If every set under discussion is a subset of a fixed set X, then X is called the *universal set*. In this case, the *complement* of A is $A^c = X \setminus A$.

Remark 1.1.8. In mathematics, "or" always allows both. The statement "$x \in A$ or $x \in B$" is true when x is in both sets.

Example 1.1.9. Let $X = \{1,2,3,4,5,6\}$, $A = \{1,2,3,4\}$, and $B = \{3,4,5\}$. Then

$$A \cup B = \{1,2,3,4,5\}, \quad A \cap B = \{3,4\}, \quad A \setminus B = \{1,2\}, \quad B \setminus A = \{5\}.$$

Also, $A^c = \{5, 6\}$ and $B^c = \{1, 2, 6\}$. Notice $A^c \cap B^c = \{6\} = (A \cup B)^c$. This is a special case of De Morgan's Laws (Theorem 1.1.11).

Proof Technique (Direct Proof). Most theorems have the form "If P, then Q." The statement P is the *hypothesis*. The statement Q is the *conclusion*. In a direct proof, assume P is true. Then use definitions and results that are already proved to show Q is true. Each step must follow from the steps before it. A good first step is to write out what the hypothesis means using definitions. Then write out what the conclusion means. The proof connects the two. The proofs of Theorem 1.1.10 are direct proofs.

Theorem 1.1.10. *Let A, B and C be subsets of a universal set X.*

(1) $A \cap B \subset A$ *and* $A \subset A \cup B$.
(2) $A \cap B = B \cap A$ *and* $A \cup B = B \cup A$.
(3) $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.
(4) $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.
(5) $A \setminus B = A \cap B^c$.

Proof. (1) Let $x \in A \cap B$. Then $x \in A$ and $x \in B$. So $x \in A$. Thus $A \cap B \subset A$.

Next, let $x \in A$. Then $x \in A$ or $x \in B$. So $x \in A \cup B$. Thus $A \subset A \cup B$.

(2) This part is an exercise.

(3) Let $x \in A \cap (B \cup C)$. Then $x \in A$ and $x \in B \cup C$. So $x \in B$ or $x \in C$. If $x \in B$, then $x \in A \cap B$. If $x \in C$, then $x \in A \cap C$. In either case, $x \in (A \cap B) \cup (A \cap C)$. So $A \cap (B \cup C) \subset (A \cap B) \cup (A \cap C)$.

Now let $x \in (A \cap B) \cup (A \cap C)$. Then $x \in A \cap B$ or $x \in A \cap C$. If $x \in A \cap B$, then $x \in A$ and $x \in B$. If $x \in A \cap C$, then $x \in A$ and $x \in C$. In either case, $x \in A$ and $x \in B \cup C$. So $x \in A \cap (B \cup C)$. Thus $(A \cap B) \cup (A \cap C) \subset A \cap (B \cup C)$.

Therefore $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

(4) This part is an exercise.

(5) This part is an exercise. □

Proof Technique (Proof by Cases). Sometimes a proof splits into cases. Each case must be handled, and together the cases must cover every possibility. In the proof of Theorem 1.1.10 (3), $x \in B \cup C$ means $x \in B$ or $x \in C$. So there are two cases. The proof handles each one. Then it notices both cases give the same conclusion.

Theorem 1.1.11 (De Morgan's Laws)**.** *Let A and B be subsets of a universal set X. Then*

(1) $(A \cup B)^c = A^c \cap B^c$*; and*
(2) $(A \cap B)^c = A^c \cup B^c$.

PROOF. (1) Let $x \in (A \cup B)^c$. Then $x \in X$ and $x \notin A \cup B$. If $x \in A$, then $x \in A \cup B$. So $x \notin A$. Similarly, $x \notin B$. Thus $x \in A^c$ and $x \in B^c$. So $x \in A^c \cap B^c$. Therefore $(A \cup B)^c \subset A^c \cap B^c$.

Now let $x \in A^c \cap B^c$. Then $x \notin A$ and $x \notin B$. So x is not in $A \cup B$. Thus $x \in (A \cup B)^c$. Therefore $A^c \cap B^c \subset (A \cup B)^c$. So $(A \cup B)^c = A^c \cap B^c$.

(2) This part is an exercise. □

Definition 1.1.12. Let Λ be a nonempty set. Suppose for each $\alpha \in \Lambda$, A_α is a set. Then $\{A_\alpha\}_{\alpha\in\Lambda}$ is called a *family of sets* and Λ is called an *index set*. Define the union and intersection of the family by

$$\bigcup_{\alpha\in\Lambda} A_\alpha = \{x : x \in A_\alpha \text{ for some } \alpha \in \Lambda\}$$

and

$$\bigcap_{\alpha\in\Lambda} A_\alpha = \{x : x \in A_\alpha \text{ for all } \alpha \in \Lambda\}.$$

If $\Lambda = \{1, 2, \dots, n\}$, write $\bigcup_{i=1}^n A_i$ and $\bigcap_{i=1}^n A_i$. If $\Lambda = \{1, 2, 3, \dots\}$, write $\bigcup_{n=1}^\infty A_n$ and $\bigcap_{n=1}^\infty A_n$.

Example 1.1.13. For each $n \in \{1, 2, 3, \dots\}$, let $A_n = \{1, 2, \dots, n\}$. Then $\bigcup_{n=1}^\infty A_n = \{1, 2, 3, \dots\}$ and $\bigcap_{n=1}^\infty A_n = \{1\}$.

For each $n \in \{1, 2, 3, \dots\}$, let $B_n = \{n, n+1, n+2, \dots\}$. Then $\bigcap_{n=1}^\infty B_n = \emptyset$. To see this, let k be any positive integer. Then $k \notin B_{k+1}$. So k is not in every B_n.

Proof Technique (Negating Statements). The *negation* of a statement P is the statement "P is false," or "not P." Many proofs need the negation of a complicated statement. These rules give it.

- The negation of "P and Q" is "not P or not Q."
- The negation of "P or Q" is "not P and not Q."
- The negation of "for all x, $P(x)$" is "there exists an x such that not $P(x)$."
- The negation of "there exists an x such that $P(x)$" is "for all x, not $P(x)$."
- The negation of "if P, then Q" is "P and not Q."

For example, $A \subset B$ means every element of A is in B. So $A \not\subset B$ means there exists an element of A that is not in B. Notice De Morgan's Laws match the first two rules.

Statements in analysis often have several "for all" and "there exists" phrases in a row. To negate one, work from left to right. Change each "for all" to "there exists." Change each "there exists" to "for all." Then negate the last statement. Exercise 1.1.10 gives practice.

Theorem 1.1.14 (De Morgan's Laws)**.** *Let $\{A_\alpha\}_{\alpha\in\Lambda}$ be a family of subsets of a universal set X. Then*

(1) $\left(\bigcup_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcap_{\alpha\in\Lambda} A_\alpha^c$*; and*
(2) $\left(\bigcap_{\alpha\in\Lambda} A_\alpha\right)^c = \bigcup_{\alpha\in\Lambda} A_\alpha^c$.

PROOF. (1) Let $x \in \left(\bigcup_{\alpha\in\Lambda} A_\alpha\right)^c$. Then $x \notin \bigcup_{\alpha\in\Lambda} A_\alpha$. So it is false that $x \in A_\alpha$ for some $\alpha \in \Lambda$. Negating this statement, $x \notin A_\alpha$ for all $\alpha \in \Lambda$. So $x \in A_\alpha^c$ for all $\alpha \in \Lambda$. Thus $x \in \bigcap_{\alpha\in\Lambda} A_\alpha^c$.

Now let $x \in \bigcap_{\alpha\in\Lambda} A_\alpha^c$. Then $x \notin A_\alpha$ for all $\alpha \in \Lambda$. So there is no $\alpha \in \Lambda$ with $x \in A_\alpha$. Thus $x \notin \bigcup_{\alpha\in\Lambda} A_\alpha$. So $x \in \left(\bigcup_{\alpha\in\Lambda} A_\alpha\right)^c$.

(2) This part is an exercise. □

Definition 1.1.15. Let A and B be sets. An *ordered pair* (a, b) is a pair of objects where the order matters. Two ordered pairs (a, b) and (c, d) are equal if $a = c$ and $b = d$. The *Cartesian product* of A and B is

$$A \times B = \{(a, b) : a \in A,\ b \in B\}.$$

Example 1.1.16. $\{1, 2\} \times \{3, 4, 5\} = \{(1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 5)\}$.

Definition 1.1.17. A *relation* on a set A is a subset R of $A \times A$. If $(x, y) \in R$, write $x\,R\,y$. A relation $\sim$ on A is an *equivalence relation* if

(1) $x \sim x$ for all $x \in A$ (reflexive);
(2) if $x \sim y$, then $y \sim x$ (symmetric); and
(3) if $x \sim y$ and $y \sim z$, then $x \sim z$ (transitive).

Proof Technique (Counterexample). A statement of the form "for all x, $P(x)$" is false if $P(x)$ fails for just one x. An x for which $P(x)$ fails is called a *counterexample*. One counterexample is enough to show a "for all" statement is false. However, examples where $P(x)$ is true never prove a "for all" statement. Example 1.1.18 uses a counterexample.

Example 1.1.18. Equality is an equivalence relation on any set. Now consider the relation $\leq$ on $\{1, 2, 3\}$. It is reflexive and transitive. However, it is not symmetric. To see this, notice $1 \leq 2$ but $2 \not\leq 1$. So $x = 1$, $y = 2$ is a counterexample. Thus $\leq$ is not an equivalence relation.

Definition 1.1.19. The *power set* of a set A, denoted $\mathcal{P}(A)$, is the set of all subsets of A.

Example 1.1.20. $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

Exercises

1.1.1. Let $X = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $A = \{1, 2, 3\}$, $B = \{2, 4, 6, 8\}$, and $C = \{3, 6\}$. Find $A \cup B$, $A \cap C$, $B \setminus C$, $(A \cup B)^c$, and $A^c \cap C$. [Solution, p. 329]

1.1.2. Prove Theorem 1.1.10 (2), (4) and (5).

1.1.3. Prove Theorem 1.1.11 (2). [Solution, p. 329]

1.1.4. Prove Theorem 1.1.14 (2).

1.1.5. Prove that if $A \subset B$ and $B \subset C$, then $A \subset C$. [Solution, p. 329]

1.1.6. Prove that if $A \subset B$, then $A \cap B = A$ and $A \cup B = B$.

1.1.7. Find a counterexample to the statement "If $A \cup B = A \cup C$, then $B = C$." [Solution, p. 329]

1.1.8. Find $\mathcal{P}(\{1,2,3\})$. How many elements does it have?

1.1.9. For each $n \in \{1,2,3,\dots\}$, let $A_n = \{-n,\dots,-1,0,1,\dots,n\}$. Find $\bigcup_{n=1}^{\infty} A_n$ and $\bigcap_{n=1}^{\infty} A_n$.

1.1.10. Negate each statement.

(a) Every element of A is less than 5.
(b) There exists an $x \in A$ such that $x > 3$.
(c) For all $x \in A$, there exists a $y \in B$ such that $x < y$.
(d) For all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - a| < \varepsilon$.

1.2. The Integers

The integers are the starting point for every number system in this chapter. This section does not prove the rules of integer arithmetic. It takes them as known. Instead it proves facts about even numbers, odd numbers and division. These facts are needed later, for example to show $\sqrt{2}$ is irrational. Along the way, the section introduces the Well-Ordering Principle and induction. They are the main tools for proving a statement about every natural number at once.

Definition 1.2.1. Define the *natural numbers* $\mathbb{N} = \{1,2,3,\dots\}$, the *whole numbers* $\mathbb{N}_0 = \{0,1,2,\dots\}$ and the *integers* $\mathbb{Z} = \{\dots,-3,-2,-1,0,1,2,3,\dots\}$.

Remark 1.2.2. This book assumes the usual arithmetic and order of the integers. Sums, differences and products of integers are integers. The rules of algebra, such as $m(n+k) = mn + mk$, hold. Also, if $n \in \mathbb{Z}$ and $n > 0$, then $n \geq 1$. These facts can be proved from a short list of axioms, but that is not done here.

Definition 1.2.3. An integer n is *even* if $n = 2k$ for some $k \in \mathbb{Z}$. An integer n is *odd* if $n = 2k+1$ for some $k \in \mathbb{Z}$.

Example 1.2.4. $0 = 2 \cdot 0$ is even. $-7 = 2(-4) + 1$ is odd.

Theorem 1.2.5. *Let $m, n \in \mathbb{Z}$.*

(1) If m and n are even, then $m+n$ is even.
(2) If m and n are odd, then $m+n$ is even.
(3) If m is even, then mn is even.
(4) If m and n are odd, then mn is odd.

PROOF. (1) Since m and n are even, $m = 2j$ and $n = 2k$ for some $j, k \in \mathbb{Z}$. Then

$$m + n = 2j + 2k = 2(j+k).$$

Since $j + k \in \mathbb{Z}$, $m + n$ is even.

(2) Since m and n are odd, $m = 2j+1$ and $n = 2k+1$ for some $j, k \in \mathbb{Z}$. Then

$$m + n = 2j + 2k + 2 = 2(j+k+1).$$

Since $j + k + 1 \in \mathbb{Z}$, $m + n$ is even.

(3) This part is an exercise.

(4) Since m and n are odd, $m = 2j+1$ and $n = 2k+1$ for some $j, k \in \mathbb{Z}$. Then

$$mn = 4jk + 2j + 2k + 1 = 2(2jk + j + k) + 1.$$

Since $2jk + j + k \in \mathbb{Z}$, mn is odd. □

Remark 1.2.6. The proof of (1) writes $m = 2j$ and $n = 2k$. Writing $m = 2k$ and $n = 2k$ would assume $m = n$. Each new object in a proof gets its own letter.

Proof Technique (Converse). The *converse* of "If P, then Q" is "If Q, then P." A statement and its converse are different statements. One can be true while the other is false. For example, the converse of Theorem 1.2.5 (1) is "If $m + n$ is even, then m and n are even." This is false. A counterexample is $m = n = 1$. Then $m + n = 2$ is even, but 1 is not even, since $1 = 2k$ has no solution $k \in \mathbb{Z}$.

The next axiom looks obvious. It is still the key property of $\mathbb{N}$. It fails for $\mathbb{Z}$, which has no smallest element. It also fails for the positive rational numbers, since $\frac{q}{2}$ is a smaller positive rational than q.

Axiom 1.2.7 (The Well-Ordering Principle)**.** *Every nonempty subset of* $\mathbb{N}$ *has a smallest element.*

Corollary 1.2.8. *Every nonempty subset of* $\mathbb{N}_0$ *has a smallest element.*

Proof. Let $S \subset \mathbb{N}_0$ with $S \neq \emptyset$. If $0 \in S$, then 0 is the smallest element of S, since every element of $\mathbb{N}_0$ is at least 0. If $0 \notin S$, then $S \subset \mathbb{N}$. So by the Well-Ordering Principle, S has a smallest element. □

Proof Technique (Proof by Contradiction). To prove a statement P by *contradiction*, assume P is false. Then use this assumption to reach a statement that is impossible. Examples are $1 < 0$, or $x \in A$ and $x \notin A$. This impossible statement is called a contradiction. A false statement leads to a contradiction, but a true statement never does. So the assumption that P is false must be wrong, and P is true. Proofs by contradiction in this book start with "Assume by way of contradiction." The proof of Theorem 1.2.9 is by contradiction.

Theorem 1.2.9 (Principle of Mathematical Induction)**.** *For each* $n \in \mathbb{N}$*, let* $P(n)$ *be a statement about the positive integer* n*. If*

(1) $P(1)$ *is true, and*

(2) $P(k+1)$ *is true whenever* $P(k)$ *is true,*

then $P(n)$ *is true for all* $n \in \mathbb{N}$*.*

Proof. Assume by way of contradiction that there exists a positive integer n such that the statement $P(n)$ is false. Let

$$A = \{k \in \mathbb{N} : P(k) \text{ is false}\}.$$

Now $n \in A$, so $A \neq \emptyset$, and A has a smallest element m. Since $P(1)$ is true, $1 \notin A$, and $m > 1$. So $m - 1 \in \mathbb{N}$. Also, since m is the smallest element of A, $m - 1 \notin A$.

So $P(m-1)$ is true. But then, by (2), $P(m)$ is also true. So $m \notin A$, a contradiction. Consequently, $P(n)$ must be true for all $n \in \mathbb{N}$. □

Induction works like a row of dominoes. Condition (1) knocks over the first domino. Condition (2) says each domino knocks over the next one. Then every domino falls.

Proof Technique (Proof by Induction). To prove $P(n)$ is true for all $n \in \mathbb{N}$ by induction, do two steps. First, prove $P(1)$ is true. This is the *base case*. Second, let $k \in \mathbb{N}$ and assume $P(k)$ is true. Then prove $P(k+1)$ is true. This is the *inductive step*. The assumption that $P(k)$ is true is called the *inductive hypothesis*. The inductive step does not assume what is being proved. It only shows that each statement implies the next one.

Induction can start at any integer n_0. To prove $P(n)$ for all integers $n \geq n_0$, prove $P(n_0)$. Then prove $P(k)$ implies $P(k+1)$ for all $k \geq n_0$. This follows from Theorem 1.2.9 applied to the statement $Q(n) = P(n + n_0 - 1)$.

Definition 1.2.10. Many objects are defined one step at a time. This is called a *recursive definition*. Let $a, a_1, a_2, \ldots$ be numbers.

(1) The *powers* of a are $a^1 = a$ and $a^{n+1} = a^n \cdot a$ for $n \in \mathbb{N}$. If $a \neq 0$, $a^0 = 1$.
(2) The *sum* of $a_1, \ldots, a_n$ is $\sum_{i=1}^{1} a_i = a_1$ and $\sum_{i=1}^{n+1} a_i = \sum_{i=1}^{n} a_i + a_{n+1}$ for $n \in \mathbb{N}$. So $\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n$. Sums starting at an index other than 1 are defined the same way.
(3) The *factorial* of $n \in \mathbb{N}_0$ is $0! = 1$ and $(n+1)! = (n+1) \cdot n!$. So $n! = 1 \cdot 2 \cdots n$ for $n \in \mathbb{N}$.

Example 1.2.11. For all $n \in \mathbb{N}$,

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}.$$

To prove this by induction, first notice that when $n = 1$, the left side is 1 and the right side is $\frac{1 \cdot 2}{2} = 1$. So the statement is true for $n = 1$. Now let $k \in \mathbb{N}$ and assume $\sum_{i=1}^{k} i = \frac{k(k+1)}{2}$. Then

$$\sum_{i=1}^{k+1} i = \sum_{i=1}^{k} i + (k+1) = \frac{k(k+1)}{2} + (k+1) = \frac{(k+1)(k+2)}{2}.$$

So the statement is true for $k+1$. By induction, it is true for all $n \in \mathbb{N}$.

Proof Technique (Proving Uniqueness). To prove there is exactly one object with some property, prove two things. *Existence* shows there is at least one such object. *Uniqueness* shows there is at most one. To prove uniqueness, assume two objects have the property. Then show they are equal. The proof of Theorem 1.2.12 does both.

Theorem 1.2.12 (Division Algorithm)**.** *Let $n \in \mathbb{Z}$ and $d \in \mathbb{N}$. Then there exist unique $q, r \in \mathbb{Z}$ such that $n = qd + r$ and $0 \leq r < d$.*

PROOF. There are two things to prove. The integers q and r exist, and they are unique. To show that they exist, let

$$S = \{n - kd : k \in \mathbb{Z},\ n - kd \geq 0\}.$$

Let $k = -|n|$, where $|n| = n$ if $n \geq 0$ and $|n| = -n$ if $n < 0$. Then $n - kd = n + |n|d \geq n + |n| \geq 0$. So S is a nonempty subset of $\mathbb{N}_0$. By Corollary 1.2.8, S has a smallest element r. So $r = n - qd$ for some $q \in \mathbb{Z}$, and $r \geq 0$. Assume by way of contradiction that $r \geq d$. Then

$$r - d = n - (q + 1)d \geq 0.$$

So $r - d \in S$. But $r - d < r$, a contradiction to the fact that r is the smallest element of S. So $0 \leq r < d$ and $n = qd + r$.

To show that q and r are unique, suppose $n = qd + r = q'd + r'$ with $0 \leq r < d$ and $0 \leq r' < d$. Without loss of generality, assume $r \leq r'$. Then

$$r' - r = (q - q')d.$$

Also, $0 \leq r' - r < d$. So $0 \leq (q - q')d < d$. Dividing by d, $0 \leq q - q' < 1$. Since $q - q'$ is an integer, $q - q' = 0$. So $q = q'$, and then $r = r'$. □

Proof Technique (Without Loss of Generality). Sometimes two cases are the same after renaming. In the proof of Theorem 1.2.12, the cases $r \leq r'$ and $r' \leq r$ are the same with the roles of (q, r) and (q', r') switched. So only one case needs a proof. Writing "without loss of generality" tells the reader this. Only use it when the other cases really are the same.

Corollary 1.2.13. *Every integer is either even or odd, but not both.*

PROOF. Let $n \in \mathbb{Z}$. By the Division Algorithm with $d = 2$, $n = 2q + r$ where $q \in \mathbb{Z}$ and $r \in \{0, 1\}$. If $r = 0$, n is even. If $r = 1$, n is odd. If n were both even and odd, then $n = 2q + 0 = 2q' + 1$ for some $q, q' \in \mathbb{Z}$. This contradicts the uniqueness part of the Division Algorithm. □

Example 1.2.14. For every $n \in \mathbb{Z}$, $n^2 + n$ is even. By Corollary 1.2.13, there are two cases. If n is even, then $n = 2k$ for some $k \in \mathbb{Z}$. So $n^2 + n = 4k^2 + 2k = 2(2k^2 + k)$ is even. If n is odd, then $n = 2k + 1$ for some $k \in \mathbb{Z}$. So

$$n^2 + n = 4k^2 + 4k + 1 + 2k + 1 = 2(2k^2 + 3k + 1)$$

is even.

Proof Technique (If and Only If). The statement "P if and only if Q" means both "If P, then Q" and "If Q, then P" are true. So a proof of an *if and only if* statement has two parts. The ($\Rightarrow$) part assumes P and proves Q. The ($\Leftarrow$) part assumes Q and proves P.

Proof Technique (Contrapositive). The *contrapositive* of "If P, then Q" is "If not Q, then not P." A statement and its contrapositive are either both true or both false. So to prove a statement, it is enough to prove its contrapositive.

In the ($\Leftarrow$) part of the proof of Theorem 1.2.15, it is hard to start from "n^2 is even." It is easier to start from "n is not even."

Theorem 1.2.15. *Let $n \in \mathbb{Z}$. Then n is even if and only if n^2 is even.*

PROOF. ($\Rightarrow$) Suppose n is even. Then by Theorem 1.2.5 (3), $n^2 = n \cdot n$ is even.

($\Leftarrow$) This part is proved by contrapositive. Suppose n is not even. By Corollary 1.2.13, n is odd. So by Theorem 1.2.5 (4), $n^2 = n \cdot n$ is odd. By Corollary 1.2.13, n^2 is not even. So if n^2 is even, then n is even. □

Proof Technique (Smallest Counterexample). Here is another way to use the Well-Ordering Principle. To prove a statement for all $n \in \mathbb{N}$, assume by way of contradiction that it is false for some n. Then the set of such n is a nonempty subset of $\mathbb{N}$. So it has a smallest element. Then use the fact that the statement is true for every smaller number to reach a contradiction. The existence part of the proof of Theorem 1.2.16 works this way.

Theorem 1.2.16. *For each $k \in \mathbb{N}$, there exist unique $m, n \in \mathbb{N}$ such that $k = 2^{m-1}(2n-1)$.*

PROOF. There are two things to prove. Every k has this form, and m and n are unique. To show that every k has this form, assume by way of contradiction that some $k \in \mathbb{N}$ cannot be written this way. Let S be the set of all such k. Then S is a nonempty subset of $\mathbb{N}$. So by the Well-Ordering Principle, S has a smallest element k.

If k is odd, then $k = 2j + 1$ for some $j \in \mathbb{N}_0$. So $k = 2^{1-1}\big(2(j+1)-1\big)$ with $1, j+1 \in \mathbb{N}$. So $k \notin S$, a contradiction.

If k is even, then $k = 2j$ for some $j \in \mathbb{N}$, and $j < k$. Since k is the smallest element of S, $j \notin S$. So $j = 2^{m-1}(2n-1)$ for some $m, n \in \mathbb{N}$. Then $k = 2j = 2^{(m+1)-1}(2n-1)$ with $m+1, n \in \mathbb{N}$. So $k \notin S$, a contradiction.

By Corollary 1.2.13, these are the only two cases. So $S = \emptyset$.

To show that m and n are unique, suppose $2^{m-1}(2n-1) = 2^{p-1}(2q-1)$ with $m, n, p, q \in \mathbb{N}$. Without loss of generality, assume $m \leq p$. Dividing by 2^{m-1},

$$2n - 1 = 2^{p-m}(2q-1).$$

If $p > m$, then the right side is even by Theorem 1.2.5 (3), and the left side is odd. This contradicts Corollary 1.2.13. So $p = m$. Then $2n - 1 = 2q - 1$, and $n = q$. □

Exercises

1.2.1. Prove Theorem 1.2.5 (3).

1.2.2. Prove that the sum of an even integer and an odd integer is odd. [Solution, p. 329]

1.2.3. Prove that $n \in \mathbb{Z}$ is odd if and only if n^2 is odd. [Solution, p. 329]

1.2.4. Let $n \in \mathbb{Z}$. Use the Division Algorithm with $d = 3$ to prove that if $n^2 = 3k$ for some $k \in \mathbb{Z}$, then $n = 3j$ for some $j \in \mathbb{Z}$. [Hint, p. 329]

1.2.5. Let $a_0, a_1, \ldots$ be real numbers. Use induction to prove that for all $n \in \mathbb{N}$,

$$\sum_{i=1}^{n} (a_i - a_{i-1}) = a_n - a_0.$$

1.2.6. Use induction to prove that for all $n \in \mathbb{N}$,

$$\sum_{i=1}^{n} i^2 = \frac{n(n+1)(2n+1)}{6}.$$

[Solution, p. 330]

1.2.7. Let $r \neq 1$. Use induction to prove that for all $n \in \mathbb{N}_0$,

$$\sum_{i=0}^{n} r^i = \frac{1 - r^{n+1}}{1 - r}.$$

1.2.8. Let a and b be real numbers. Use induction to prove that for all $n \in \mathbb{N}$,

$$b^n - a^n = (b-a)\left(b^{n-1} + b^{n-2}a + \cdots + ba^{n-2} + a^{n-1}\right) = (b-a)\sum_{i=0}^{n-1} b^{n-1-i}a^i.$$

1.2.9. Use induction to prove that $2^n > n$ for all $n \in \mathbb{N}$. [Solution, p. 330]

1.2.10. Use induction to prove that $n! \geq 2^{n-1}$ for all $n \in \mathbb{N}$. [Hint, p. 330]

1.3. Prime Numbers*

Theorem 1.2.16 pulls every factor of 2 out of a natural number, and it does so in only one way. Doing the same with every prime gives the Fundamental Theorem of Arithmetic. Its proof uses both main tools of Section 1.2, the Well-Ordering Principle and the Division Algorithm. This section can be skipped. Later, only Exercise 1.4.4 and the starred Section 10.5 use it.

Definition 1.3.1. Let $d \in \mathbb{N}$ and $n \in \mathbb{Z}$. Then d *divides* n if $n = dm$ for some $m \in \mathbb{Z}$. A natural number $p \geq 2$ is *prime* if the only natural numbers that divide p are 1 and p.

Lemma 1.3.2. *Every natural number $n \geq 2$ is a product of primes. That is, $n = q_1 q_2 \cdots q_r$ for some $r \in \mathbb{N}$ and some primes $q_1, \ldots, q_r$, which need not be different.*

Proof. Assume by way of contradiction that some $n \geq 2$ is not a product of primes. By Axiom 1.2.7, there is a smallest such n. A prime is a product of one prime, so n is not prime. So $n = dm$ for some $d \in \mathbb{N}$ with $d \neq 1$ and $d \neq n$, and some $m \in \mathbb{Z}$. Since n and d are positive, $m \in \mathbb{N}$. Then $d \geq 2$. Also $m \geq 2$, since $m = 1$ would give $d = n$. So $d = \frac{n}{m} \leq \frac{n}{2} < n$, and in the same way $m < n$. Since n is the smallest natural number that is at least 2 and is not a product of primes, d and m are products of primes. Then so is $n = dm$, a contradiction. □

Lemma 1.3.3 (Euclid's Lemma)**.** *Let p be prime, and let $a, b \in \mathbb{N}$. If p divides ab, then p divides a or p divides b.*

PROOF. Suppose p divides ab but does not divide a. The goal is to show that p divides b. Let

$$S = \{x \in \mathbb{N} : p \text{ divides } xb\}.$$

Then $p \in S$ and $a \in S$. By Axiom 1.2.7, S has a smallest element d.

The first step shows that d divides every element of S. Let $x \in S$. By Theorem 1.2.12, $x = qd + r$ for some $q, r \in \mathbb{Z}$ with $0 \le r < d$. Write $xb = pu$ and $db = pv$ with $u, v \in \mathbb{Z}$. Then

$$rb = xb - qdb = p(u - qv),$$

so p divides rb. If $r > 0$, then $r \in S$ and $r < d$, which contradicts the choice of d. So $r = 0$, and $x = qd$.

In particular, d divides p and d divides a. Since p is prime, $d = 1$ or $d = p$. If $d = p$, then p divides a, which is false. So $d = 1$. Since $1 \in S$, p divides $1 \cdot b = b$. □

Theorem 1.3.4 (Fundamental Theorem of Arithmetic)**.** *Every natural number $n \ge 2$ is a product of primes, and the product is unique apart from the order of the factors. That is, if*

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$$

with all p_i and q_j prime, then $r = s$, and the q_j can be renumbered so that $q_i = p_i$ for each i.

PROOF. Existence is Lemma 1.3.2. Uniqueness needs one consequence of Lemma 1.3.3. If a prime p divides a product $a_1 a_2 \cdots a_k$ of natural numbers, then p divides some a_i. This follows by induction on k, since $a_1 \cdots a_k = (a_1 \cdots a_{k-1})a_k$.

To show uniqueness, assume by way of contradiction that some $n \ge 2$ has two factorizations that are not rearrangements of each other. By Axiom 1.2.7, there is a smallest such n, with

$$n = p_1 p_2 \cdots p_r = q_1 q_2 \cdots q_s$$

as above. The prime p_1 divides $q_1 q_2 \cdots q_s$, so p_1 divides some q_j. Renumber the q_j so that p_1 divides q_1. Since q_1 is prime and $p_1 \ge 2$, $p_1 = q_1$. If $r = 1$, then $q_2 \cdots q_s = 1$. Since each $q_j \ge 2$, this forces $s = 1$, and then the two factorizations are the same, which is false. In the same way, $s \ne 1$. So $r, s \ge 2$. Dividing by p_1,

$$m = p_2 \cdots p_r = q_2 \cdots q_s,$$

and $2 \le m < n$. By the choice of n, $r - 1 = s - 1$, and $q_2, \ldots, q_s$ can be renumbered so that $q_i = p_i$ for $i \ge 2$. Together with $q_1 = p_1$, this shows that the two factorizations of n are rearrangements of each other, a contradiction. □

Exercises

1.3.1. Let p be prime and $n \in \mathbb{N}$. Prove that if p divides n^2, then p divides n. This generalizes Exercise 1.2.4. [Solution, p. 330]

1.3.2. Let $n \ge 2$ be a natural number that is not prime. Prove that n has a prime factor p with $p^2 \le n$. Use this to show that 97 is prime. [Hint, p. 330]

1.4. The Rational Numbers

The rational numbers are the fractions. They can be added, subtracted, multiplied and divided, and the answers are still rational. So for arithmetic, $\mathbb{Q}$ is enough. The trouble is that $\mathbb{Q}$ has holes. The side of a square with area 2 should have a length p with $p^2 = 2$. Theorem 1.4.4 shows no rational number does this. Theorem 1.4.5 shows something stronger. The rational numbers with $p^2 < 2$ and the ones with $p^2 > 2$ crowd right up against each other, and no rational number sits between them.

Definition 1.4.1. Define the *rational numbers* $\mathbb{Q} = \left\{\frac{m}{n} : m, n \in \mathbb{Z},\ n \neq 0\right\}$.

Remark 1.4.2. Two fractions $\frac{m}{n}$ and $\frac{p}{q}$ are equal if $mq = np$. So $\frac{1}{2} = \frac{2}{4}$. Every integer m is rational, since $m = \frac{m}{1}$. So $\mathbb{N} \subset \mathbb{N}_0 \subset \mathbb{Z} \subset \mathbb{Q}$. The usual arithmetic and order of $\mathbb{Q}$ are assumed for now. Section 1.6 lists the rules they satisfy.

Lemma 1.4.3. *Let $p \in \mathbb{Q}$ with $p > 0$. Then there exist $m, n \in \mathbb{N}$ such that $p = \frac{m}{n}$ and m and n are not both even.*

Proof. Since $p \in \mathbb{Q}$, $p = \frac{a}{b}$ for some $a, b \in \mathbb{Z}$ with $b \neq 0$. Since $p > 0$, a and b are both positive or both negative. If both are negative, then $p = \frac{-a}{-b}$. So $p = \frac{a}{b}$ with $a, b \in \mathbb{N}$. Let

$$S = \left\{n \in \mathbb{N} : p = \frac{m}{n} \text{ for some } m \in \mathbb{N}\right\}.$$

Then $b \in S$, so $S \neq \emptyset$. By the Well-Ordering Principle, S has a smallest element n. So $p = \frac{m}{n}$ for some $m \in \mathbb{N}$. Assume by way of contradiction that m and n are both even. Then $m = 2j$ and $n = 2k$ for some $j, k \in \mathbb{N}$. So $p = \frac{2j}{2k} = \frac{j}{k}$, and $k \in S$. But $k < n$, a contradiction to the fact that n is the smallest element of S. So m and n are not both even. □

Theorem 1.4.4. *There exists no $p \in \mathbb{Q}$ with $p^2 = 2$.*

Proof. Assume by way of contradiction that there exists a $p \in \mathbb{Q}$ such that $p^2 = 2$. Since $(-p)^2 = p^2$ and $0^2 \neq 2$, assume $p > 0$. By Lemma 1.4.3, $p = \frac{m}{n}$ with $m, n \in \mathbb{N}$, where m and n are not both even. So $p^2 = \frac{m^2}{n^2}$. Since $p^2 = 2$, $2n^2 = m^2$. Therefore, since $n^2 \in \mathbb{N}$, m^2 is even. So by Theorem 1.2.15, m must be even. Thus, there exists a $k \in \mathbb{N}$ such that $m = 2k$, implying $4k^2 = m^2$, or $2n^2 = 4k^2$. Thus $n^2 = 2k^2$ and n^2 is even. So by Theorem 1.2.15, n is also even, a contradiction to the fact that m and n are not both even. So there exists no such p. □

> **Proof Technique (Proving Existence by Construction).** The statement "A has no largest element" means for every $p \in A$, there exists a $q \in A$ with $q > p$. To prove a "there exists" statement, it is often best to build the object. The proof of Theorem 1.4.5 builds q_p from p with a formula. Finding the formula is scratch work. It is not part of the proof. The proof only has to check that q_p works.

Theorem 1.4.5. *The set $A = \{p \in \mathbb{Q} : p > 0,\ p^2 < 2\}$ has no largest element and the set $B = \{p \in \mathbb{Q} : p > 0,\ p^2 > 2\}$ has no smallest element.*

PROOF. For each $p \in \mathbb{Q}$, $p > 0$, define

$$q_p = p - \frac{p^2 - 2}{p + 2} = \frac{2p + 2}{p + 2}.$$

Notice $q_p \in \mathbb{Q}$ and $q_p > 0$. Then

$$q_p^2 - 2 = \frac{2(p^2 - 2)}{(p + 2)^2}.$$

Let $p \in A$. Then $p^2 - 2 < 0$. So $q_p > p$ and $q_p^2 < 2$. So $q_p \in A$.
Next, let $p \in B$. Then $p^2 - 2 > 0$. So $q_p < p$ and $q_p^2 > 2$. Thus $q_p \in B$. □

Where does the formula for q_p come from? If $p \in A$, the goal is a number a little bigger than p whose square is still less than 2. The step up should shrink as p^2 gets close to 2. So it makes sense to step by a multiple of $2 - p^2$. Dividing by $p + 2$ keeps the step small enough that q_p^2 stays below 2. The identity for $q_p^2 - 2$ in the proof checks this. The same formula moves each element of B down.

So A and B split the positive rational numbers into two pieces, as in Figure 1.1. There is no largest number on the left and no smallest number on the right. Section 1.5 gives the language to say exactly what is missing.

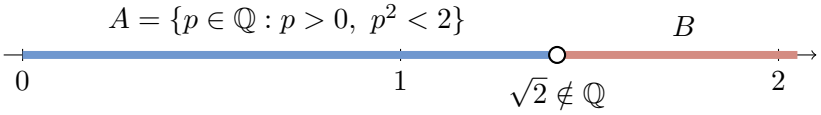


FIGURE 1.1. The sets A and B of Theorem 1.4.5. They meet at a point that is not a rational number.

Exercises

1.4.1. Prove there is no $q \in \mathbb{Q}$ with $q^2 = 3$. [Hint, p. 330]

1.4.2. Prove there is no $q \in \mathbb{Q}$ with $q^2 = 6$. [Hint, p. 330]

1.4.3. Let $p, q \in \mathbb{Q}$ with $p < q$. Prove that $\frac{p+q}{2} \in \mathbb{Q}$ and $p < \frac{p+q}{2} < q$. [Solution, p. 330]

1.4.4. If p is prime, prove that there is no $q \in \mathbb{Q}$ with $q^2 = p$. This exercise uses Theorem 1.3.4 from the starred Section 1.3. [Solution, p. 330]

1.5. Ordered Sets

To describe the hole in $\mathbb{Q}$ precisely, the language of order is needed. This section works with any set that has an order, not just $\mathbb{Q}$. Its main definition is the supremum, or least upper bound. It is the most important definition in this chapter.

Definition 1.5.1. Let S be a set. A relation $<$ on S is an *order* on S if it has the following two properties.

(1) (Trichotomy) For all $x, y \in S$, exactly one of $x < y$, $x = y$, $y < x$ is true.
(2) (Transitivity) For all $x, y, z \in S$, if $x < y$ and $y < z$, then $x < z$.

Write $y > x$ to mean $x < y$. Write $x \leq y$ to mean $x < y$ or $x = y$, and $y \geq x$ to mean $x \leq y$. A set together with an order is called an *ordered set*.

Example 1.5.2. $\mathbb{Q}$ is an ordered set if for $r, s \in \mathbb{Q}$, $r < s$ is defined as $s - r$ is a positive rational number.

Definition 1.5.3. Suppose S is an ordered set and $E \subset S$. If there exists a $\beta \in S$ such that $x \leq \beta$ for all $x \in E$, E is said to be *bounded above*, and β is called an *upper bound* of E.

If there exists an $\alpha \in S$ such that $x \geq \alpha$ for all $x \in E$, E is said to be *bounded below* and α is called a *lower bound* of E.

Definition 1.5.4. Let S be an ordered set and $E \subset S$. An element $\beta \in S$ is the *supremum*, or *least upper bound*, of E if

(1) β is an upper bound of E, and
(2) no element of S smaller than β is an upper bound of E. That is, if $\gamma < \beta$, then there exists an $x \in E$ with $x > \gamma$.

In this case, write

$$\beta = \sup E = \operatorname{lub} E.$$

Similarly, an element $\alpha \in S$ is the *infimum*, or *greatest lower bound*, of E if

(1) α is a lower bound of E, and
(2) no element of S larger than α is a lower bound of E. That is, if $\alpha < \gamma$, then there exists an $x \in E$ with $x < \gamma$.

In this case, write

$$\alpha = \inf E = \operatorname{glb} E.$$

The supremum is the smallest upper bound. It does not have to be an element of E. If E has a largest element, then that element is $\sup E$. So the supremum takes the place of a largest element when there is none. Example 1.5.6 gives a set whose infimum is not in the set.

Theorem 1.5.5. *Suppose S is an ordered set. If $E \subset S$, $E \neq \emptyset$, has a supremum, it is unique. Similarly, if E has an infimum, it is unique.*

Proof. Let both $\beta_1, \beta_2 \in S$ be a supremum of the set E. Since S is an ordered set, $\beta_1 < \beta_2$, $\beta_1 > \beta_2$, or $\beta_1 = \beta_2$. If $\beta_1 < \beta_2$, then by the definition of β_2, β_1 is not an upper bound of E, contradicting the fact that β_1 is a supremum of E. So $\beta_1 \geq \beta_2$. If $\beta_2 < \beta_1$, then by the definition of β_1, β_2 is not an upper bound of E, contradicting the fact that β_2 is a supremum of E. So $\beta_1 = \beta_2$.

Let both $\alpha_1, \alpha_2 \in S$ be an infimum of the set E. Since S is an ordered set, $\alpha_1 < \alpha_2$, $\alpha_1 > \alpha_2$, or $\alpha_1 = \alpha_2$. If $\alpha_1 < \alpha_2$, then by the definition of α_1, α_2 is not a lower bound of E, contradicting the fact that α_2 is an infimum of E. So $\alpha_1 \geq \alpha_2$. If $\alpha_2 < \alpha_1$, then by the definition of α_2, α_1 is not a lower bound of E, contradicting the fact that α_1 is an infimum of E. So $\alpha_1 = \alpha_2$. □

Example 1.5.6. Let $E = \{1/n : n \in \mathbb{N}\} \subset \mathbb{Q}$. Then $\sup E = 1 \in E$ and $\inf E = 0 \notin E$. To see this, first notice $\frac{1}{n} \leq 1$ for all $n \in \mathbb{N}$. So 1 is an upper bound of E. If $\gamma < 1$, then γ is not an upper bound of E, since $1 \in E$. So $\sup E = 1$.

Next, $\frac{1}{n} > 0$ for all $n \in \mathbb{N}$. So 0 is a lower bound of E. Let $\gamma \in \mathbb{Q}$ with $\gamma > 0$. Then $\gamma = \frac{a}{b}$ for some $a, b \in \mathbb{N}$. So $\frac{1}{b+1} < \frac{1}{b} \leq \frac{a}{b} = \gamma$, and γ is not a lower bound of E. So $\inf E = 0$.

Example 1.5.7. Let A and B be the sets in Theorem 1.4.5. Every element of B is an upper bound of A. To see this, let $q \in B$ and $p \in A$. If $q \leq p$, then $q^2 \leq p^2 < 2 < q^2$, which is impossible. So $p < q$. In particular, 2, 5, and 3/2 are upper bounds of A.

However, $\sup A$ does not exist in $\mathbb{Q}$. Assume by way of contradiction that $s = \sup A \in \mathbb{Q}$. Since $1 \in A$, $s \geq 1 > 0$. By Theorem 1.4.4, $s^2 \neq 2$. If $s^2 < 2$, then $s \in A$. By Theorem 1.4.5, there exists a $q \in A$ with $q > s$. This contradicts the fact that s is an upper bound of A. If $s^2 > 2$, then $s \in B$. By Theorem 1.4.5, there exists a $q \in B$ with $q < s$. But q is an upper bound of A. This contradicts the fact that s is the least upper bound of A. So $\sup A$ does not exist in $\mathbb{Q}$.

Definition 1.5.8. An ordered set S is said to have the *least upper bound property* if for all $E \subset S$ with $E \neq \emptyset$, E bounded above, $\sup E \in S$.

Example 1.5.9. By Example 1.5.7, $\mathbb{Q}$ does not have the least upper bound property.

This is the precise form of the hole in $\mathbb{Q}$. The set A is nonempty and bounded above in $\mathbb{Q}$, but nothing in $\mathbb{Q}$ is its least upper bound. The real numbers are built to fix exactly this problem.

Theorem 1.5.10. *Suppose S is an ordered set with the least upper bound property, $E \subset S$, $E \neq \emptyset$ and E is bounded below. Then* $\inf E$ *exists in S.*

Proof. Let L be the set of all lower bounds of E. That is,

$$L = \{y \in S : y \leq x \text{ for all } x \in E\}.$$

Since E is bounded below, $L \neq \emptyset$. By the definition of L, every element of E is an upper bound of L. Since $E \neq \emptyset$, L is bounded above. By the least upper bound property, $\alpha = \sup L$ exists in S. The claim is that $\alpha = \inf E$.

First, α is a lower bound of E. Let $x \in E$. Then x is an upper bound of L. If $x < \alpha$, then x would not be an upper bound of L, by the definition of α. So $\alpha \leq x$.

Second, let $\gamma \in S$ with $\alpha < \gamma$. Since α is an upper bound of L, $\gamma \notin L$. So γ is not a lower bound of E. Therefore $\alpha = \inf E$. □

Exercises

1.5.1. Suppose S is an ordered set, $E \subset S$ and $E \neq \emptyset$. Prove if β is an upper bound of E and if, for all upper bounds κ of E, $\beta \leq \kappa$, then $\sup E = \beta$. A similar proof shows if α is a lower bound of E and if, for all lower bounds η of E, $\alpha \geq \eta$, then $\inf E = \alpha$.

1.5.2. Suppose S is an ordered set and $E \subset S$ is nonempty. Prove that if $\inf E$ and $\sup E$ both exist, then $\inf E \leq \sup E$. [Solution, p. 330]

1.5.3. Let $E = \{1 - \frac{1}{n} : n \in \mathbb{N}\} \subset \mathbb{Q}$. Find $\sup E$ and $\inf E$. Prove your answers. [Hint, p. 330]

1.6. Fields and Ordered Fields

The rational numbers and the real numbers follow the same rules of arithmetic. This section lists those rules as axioms. A set with an addition and a multiplication that satisfy them is called a field. Anything proved from the axioms is then true in $\mathbb{Q}$, in $\mathbb{R}$, and in every other field. That is the advantage of working from axioms. One proof covers many number systems.

Definition 1.6.1. A *field* is a set $\mathbb{F}$ with two operations $+$ and $\cdot$ which satisfy the following axioms.

(1) If $x \in \mathbb{F}$ and $y \in \mathbb{F}$, then $x + y \in \mathbb{F}$.
(2) $x + y = y + x$ for all $x, y \in \mathbb{F}$.
(3) $(x + y) + z = x + (y + z)$ for all $x, y, z \in \mathbb{F}$.
(4) There exists a $0 \in \mathbb{F}$ such that $0 + x = x$ for all $x \in \mathbb{F}$.
(5) For all $x \in \mathbb{F}$, there exists a $-x \in \mathbb{F}$ such that $x + (-x) = 0$.
(6) If $x \in \mathbb{F}$ and $y \in \mathbb{F}$, then $x \cdot y \in \mathbb{F}$.
(7) $x \cdot y = y \cdot x$ for all $x, y \in \mathbb{F}$.
(8) $(x \cdot y) \cdot z = x \cdot (y \cdot z)$ for all $x, y, z \in \mathbb{F}$.
(9) There exists a $1 \in \mathbb{F}$ with $1 \neq 0$ such that $1 \cdot x = x$ for all $x \in \mathbb{F}$.
(10) For all $x \in \mathbb{F} \setminus \{0\}$, there exists an $x^{-1} \in \mathbb{F}$ such that $x \cdot x^{-1} = 1$.
(11) $x \cdot (y + z) = x \cdot y + x \cdot z$ for all $x, y, z \in \mathbb{F}$.

Remark 1.6.2. In a field, $x - y$ means $x + (-y)$. If $y \neq 0$, $\frac{x}{y}$ means $x \cdot y^{-1}$. Also xy means $x \cdot y$, and powers x^n are defined as in Definition 1.2.10. The elements $1 + 1$, $1 + 1 + 1$, and so on are written 2, 3, and so on.

Example 1.6.3. $\mathbb{Q}$ with $+$ and $\cdot$ defined as addition and multiplication is a field.

Example 1.6.4. $\mathbb{Z}$ is not a field. The integer 2 has no multiplicative inverse in $\mathbb{Z}$. If $2n = 1$ for some $n \in \mathbb{Z}$, then 1 is even, which contradicts Corollary 1.2.13.

Theorem 1.6.5. *If $\mathbb{F}$ is a field and $x, y, z \in \mathbb{F}$, each of the following hold.*

(1) If $x + y = x + z$, then $y = z$.
(2) If $x + y = x$, then $y = 0$.
(3) If $x + y = 0$, then $y = -x$.
(4) $-(-x) = x$.
(5) If $x \neq 0$ and $x \cdot y = x \cdot z$, then $y = z$.
(6) If $x \neq 0$ and $x \cdot y = x$, then $y = 1$.
(7) If $x \neq 0$ and $x \cdot y = 1$, then $y = x^{-1}$.
(8) If $x \neq 0$, $(x^{-1})^{-1} = x$.
(9) $0 \cdot x = 0$.
(10) If $x \neq 0$ and $y \neq 0$, then $x \cdot y \neq 0$.
(11) $(-x) \cdot y = -(x \cdot y) = x \cdot (-y)$.
(12) $(-x) \cdot (-y) = x \cdot y$.

Proof. (1) Let $x + y = x + z$. Add $-x$ to both sides and use the axioms of Definition 1.6.1. Then

$$y = 0 + y \qquad \text{by (4)}$$

$$\begin{aligned} &= (-x + x) + y && \text{by (2) and (5)} \\ &= -x + (x + y) && \text{by (3)} \\ &= -x + (x + z) && \text{since } x + y = x + z \\ &= (-x + x) + z && \text{by (3)} \\ &= 0 + z = z && \text{by (2), (5) and (4).} \end{aligned}$$

(2) This part is an exercise.
(3) This part is an exercise.
(4) This part is an exercise.
(5) This proof is similar to the proof of (1).
(6) This proof is similar to the proof of (2).
(7) This proof is similar to the proof of (3).
(8) This proof is similar to the proof of (4).
(9) Now

$$0 \cdot x + 0 \cdot x = (0 + 0) \cdot x = 0 \cdot x.$$

Hence by (2), $0 \cdot x = 0$.
(10) Assume by way of contradiction that $x \neq 0$, $y \neq 0$, but $x \cdot y = 0$. Then by (9),

$$1 = (x \cdot x^{-1}) \cdot (y \cdot y^{-1}) = x^{-1} \cdot y^{-1} \cdot x \cdot y = x^{-1} \cdot y^{-1} \cdot 0 = 0,$$

a contradiction to the fact that $1 \neq 0$. So $x \cdot y \neq 0$.
(11) Now

$$(-x) \cdot y + (x \cdot y) = (-x + x) \cdot y = 0 \cdot y = 0.$$

So by (3), $(-x) \cdot y = -(x \cdot y)$. The argument that $x \cdot (-y) = -(x \cdot y)$ is similar.
(12) By (11),

$$(-x) \cdot (-y) = -(x \cdot (-y)) = -(-(x \cdot y)).$$

By (4), $-(-(x \cdot y)) = x \cdot y$. □

Definition 1.6.6. An *ordered field* is a field $\mathbb{F}$ which is also an ordered set with the following properties.

(1) If $x, y \in \mathbb{F}$ with $x \leq y$, then $x + z \leq y + z$ for all $z \in \mathbb{F}$.
(2) If $x, y \in \mathbb{F}$ with $0 \leq x$ and $0 \leq y$, then $0 \leq x \cdot y$.

If $x > 0$, x is *positive*. If $x < 0$, x is *negative*.

Example 1.6.7. $\mathbb{Q}$, with the usual operations and order, is an ordered field.

Theorem 1.6.8. *If $\mathbb{F}$ is an ordered field and $x, y, z \in \mathbb{F}$, the following hold.*

(1) If $x > 0$, then $-x < 0$ and vice versa.
(2) $x < y$ if and only if $-y < -x$.
(3) If $x > 0$ and $y < z$, then $x \cdot y < x \cdot z$.
(4) If $x < 0$ and $y < z$, $x \cdot y > x \cdot z$.
(5) If $x \neq 0$, then $x \cdot x = x^2 > 0$. In particular, $1 > 0$.
(6) If $0 < x < y$, $0 < y^{-1} < x^{-1}$.

PROOF. (1) Let $x > 0$. By Definition 1.6.6 (1) with $z = -x$, $0 + (-x) \leq x + (-x)$. So $-x \leq 0$. If $-x = 0$, then $x = -(-x) = 0$ by Theorem 1.6.5 (4), which is false. So $-x < 0$. The proof that $x < 0$ implies $-x > 0$ is similar.

(2) This part is an exercise.

(3) This part is an exercise.

(4) This part is an exercise.

(5) Suppose $x > 0$. By (3) with $y = 0$ and $z = x$, $x \cdot 0 < x \cdot x$. Since $x \cdot 0 = 0$, $x^2 > 0$. Now suppose $x < 0$. By (1), $-x > 0$, so $(-x) \cdot (-x) > 0$ in the same way. By Theorem 1.6.5 (12), $(-x) \cdot (-x) = x^2$. So $x^2 > 0$ in both cases.

Since $1 \neq 0$ and $1 = 1^2$, $1 > 0$.

(6) Now $y^{-1} \neq 0$. If $y^{-1} < 0$, then by (3), $1 = y \cdot y^{-1} < y \cdot 0 = 0$, contradicting (5). So $y^{-1} > 0$. In the same way, $x^{-1} > 0$. So $x^{-1}y^{-1} \geq 0$ by Definition 1.6.6 (2), and $x^{-1}y^{-1} \neq 0$ by Theorem 1.6.5 (10). Thus $x^{-1}y^{-1} > 0$. Multiplying $x < y$ by $x^{-1}y^{-1}$, (3) gives $x^{-1}y^{-1}x < x^{-1}y^{-1}y$. That is, $y^{-1} < x^{-1}$. □

Exercises

1.6.1. Prove Theorem 1.6.5 (2)–(4). [Solution, p. 330]

1.6.2. Prove Theorem 1.6.8 (2)–(4). [Solution, p. 331]

1.6.3. Let $\mathbb{F}$ be an ordered field and $x, y, z, w \in \mathbb{F}$. Prove each of the following.

(a) If $x < y$ and $z < w$, then $x + z < y + w$.

(b) If $x \leq y$ and $z \leq w$, then $x + z \leq y + w$.

(c) If $0 < x < y$, then $x^2 < y^2$.

(d) If $x < y$, then $x < \frac{x+y}{2} < y$.

1.6.4. Let $\mathbb{F}$ be an ordered field and $a, b \in \mathbb{F}$ with $0 < a < b$. Use induction to prove that $a^n < b^n$ for all $n \in \mathbb{N}$.

1.6.5. Let $\mathbb{F}$ be an ordered field and $a \in \mathbb{F}$. Use induction to prove that if $a \geq 1$, then for all $n \in \mathbb{N}$, $a \leq a^n$. Also prove that if $0 < a < 1$, then for all $n \in \mathbb{N}$, $0 < a^n \leq a$.

1.6.6. (Bernoulli's Inequality) Let $\mathbb{F}$ be an ordered field and $x \in \mathbb{F}$ with $x \geq -1$. Use induction to prove that $(1 + x)^n \geq 1 + nx$ for all $n \in \mathbb{N}$. [Hint, p. 331]

1.6.7. Let $\mathbb{F}$ be an ordered field with operations $+$ and $\cdot$ and order $<$ which has the least upper bound property.

(a) Prove that if $E \subset \mathbb{F}$, $E \neq \emptyset$, and E is bounded below, then the set $-E := \{-x \in \mathbb{F} : x \in E\}$ is bounded above.

(b) Prove $\inf E = -\sup(-E)$.

[Solution, p. 331]

1.7. The Absolute Value Function

Distance on the number line is measured with absolute values. The distance between x and y is $|x - y|$. Most estimates in analysis are statements about distances, so the facts in this section are used on almost every page of the rest of the book.

Definition 1.7.1. If $\mathbb{F}$ is an ordered field and $x \in \mathbb{F}$, the *absolute value* of x, denoted $|x|$, is defined by

$$|x| = \begin{cases} -x, & x < 0, \\ x, & x \geq 0. \end{cases}$$

Theorem 1.7.2. *Let $\mathbb{F}$ be an ordered field and $x, b \in \mathbb{F}$.*

(1) $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$.
(2) $|-x| = |x|$.
(3) $-|x| \leq x \leq |x|$.
(4) If $b \geq 0$, then $|x| \leq b$ if and only if $-b \leq x \leq b$.
(5) If $b > 0$, then $|x| < b$ if and only if $-b < x < b$.

PROOF. (1) If $x > 0$, then $|x| = x > 0$. If $x < 0$, then $|x| = -x > 0$ by Theorem 1.6.8 (1). If $x = 0$, then $|x| = 0$. So $|x| \geq 0$, and $|x| = 0$ if and only if $x = 0$.
(2) This part is an exercise.
(3) If $x \geq 0$, then $-|x| = -x \leq 0 \leq x = |x|$. If $x < 0$, then $-|x| = x < 0 < -x = |x|$.
(4) ($\Rightarrow$) Suppose $|x| \leq b$. By (3), $x \leq |x| \leq b$. Also, by (2) and (3), $-x \leq |-x| = |x| \leq b$. So by Theorem 1.6.8 (2), $-b \leq x$. Thus $-b \leq x \leq b$.
($\Leftarrow$) Suppose $-b \leq x \leq b$. If $x \geq 0$, then $|x| = x \leq b$. If $x < 0$, then $|x| = -x \leq b$, since $-b \leq x$.
(5) This part is an exercise. □

Proof Technique (Adding Zero). A useful trick is to add and subtract the same quantity. This does not change the value of an expression, but it can make a known result apply. In the proof of Theorem 1.7.3 (3) below, $x = (x - y) + y$. Then the Triangle Inequality applies to $(x - y) + y$. This trick appears often in later chapters.

Theorem 1.7.3. *If $\mathbb{F}$ is an ordered field with $x, y \in \mathbb{F}$, the following hold.*

(1) $|xy| = |x||y|$.
(2) (Triangle Inequality) *$|x + y| \leq |x| + |y|$.*
(3) (Reverse Triangle Inequality) *$\big||x| - |y|\big| \leq |x - y|$.*

PROOF. (1) This part is an exercise.
(2) By Theorem 1.7.2 (3), $-|x| \leq x \leq |x|$ and $-|y| \leq y \leq |y|$. Adding these inequalities (Exercise 1.6.3),

$$-(|x| + |y|) \leq x + y \leq |x| + |y|.$$

Since $|x| + |y| \geq 0$, Theorem 1.7.2 (4) gives $|x + y| \leq |x| + |y|$.
(3) By (2),

$$|x| = |(x - y) + y| \leq |x - y| + |y|.$$

So $|x| - |y| \leq |x - y|$. Similarly, $|y| - |x| \leq |y - x| = |x - y|$. So $-|x - y| \leq |x| - |y| \leq |x - y|$. By Theorem 1.7.2 (4), $\big||x| - |y|\big| \leq |x - y|$. □

The name comes from distances. By Exercise 1.7.2, $|x-y| \leq |x-z| + |z-y|$. Going straight from x to y is never longer than stopping at z on the way. This is the form of the Triangle Inequality used most often in later chapters.

Exercises

1.7.1. Prove Theorem 1.7.2 (2) and (5), and Theorem 1.7.3 (1).

1.7.2. Let $\mathbb{F}$ be an ordered field and $x, y, z \in \mathbb{F}$. Prove $|x-y| \leq |x-z| + |z-y|$.
[Solution, p. 331]

1.7.3. Let $\mathbb{F}$ be an ordered field and $x_1, \ldots, x_n \in \mathbb{F}$. Use induction to prove $|\sum_{i=1}^{n} x_i| \leq \sum_{i=1}^{n} |x_i|$.

1.8. The Completeness of the Real Numbers

The rational numbers are an ordered field without the least upper bound property. The real numbers are an ordered field with it. That one property is what makes analysis work. This section introduces $\mathbb{R}$ and proves the first consequences of the least upper bound property. There are no infinitely large real numbers, there is a rational number between any two real numbers, and positive real numbers have nth roots.

Definition 1.8.1. If $\mathbb{F}$ is an ordered field with operations $+$ and $\cdot$ and order $<$, $\mathbb{H}$ is a *subfield* of $\mathbb{F}$ if $\mathbb{H} \subset \mathbb{F}$ and $\mathbb{H}$ is an ordered field with operations $+$ and $\cdot$ and order $<$.

Theorem 1.8.2. *There is an ordered field $\mathbb{R}$ with the least upper bound property such that $\mathbb{Q}$, with its usual operations and order, is a subfield of $\mathbb{R}$.*

Proof. See Appendix A. A second proof is in Appendix B. □

Remark 1.8.3. The proof of Theorem 1.8.2 builds $\mathbb{R}$ out of $\mathbb{Q}$. Appendix A gives a construction using cuts, and Appendix B gives a second one using Cauchy sequences. Everything in Sections 1.5 to 1.7 applies to $\mathbb{R}$, since $\mathbb{R}$ is an ordered field. The least upper bound property of $\mathbb{R}$ is also called *completeness*. It is the one property that separates $\mathbb{R}$ from $\mathbb{Q}$. Every theorem in this book that says something exists, such as a limit, a maximum value or a root, depends on it.

Definition 1.8.4. The elements of the field $\mathbb{R}$ are called *real numbers*. The elements of $\mathbb{R} \setminus \mathbb{Q}$ are called *irrational numbers*.

Definition 1.8.5. Let $J \subset \mathbb{R}$, $J \neq \emptyset$. If for all $x, y \in J$ with $x \leq y$ and for all $z \in \mathbb{R}$ such that $x \leq z \leq y$, $z \in J$, J is called an *interval*.

Definition 1.8.6. Let $a, b \in \mathbb{R}$ with $a \leq b$. Define

$$
\begin{aligned}
[a,b] &= \{x \in \mathbb{R} : a \leq x \leq b\}, & (a,b) &= \{x \in \mathbb{R} : a < x < b\},\\
[a,b) &= \{x \in \mathbb{R} : a \leq x < b\}, & (a,b] &= \{x \in \mathbb{R} : a < x \leq b\},\\
[a,\infty) &= \{x \in \mathbb{R} : x \geq a\}, & (a,\infty) &= \{x \in \mathbb{R} : x > a\},\\
(-\infty,b] &= \{x \in \mathbb{R} : x \leq b\}, & (-\infty,b) &= \{x \in \mathbb{R} : x < b\},
\end{aligned}
$$

and $(-\infty, \infty) = \mathbb{R}$. The interval $[a, b]$ is called a *closed interval*. If $a < b$, (a, b) is called an *open interval*. The symbols ∞ and $-\infty$ are not real numbers.

Proof Technique (Working with Suprema). Two facts about $\beta = \sup E$ are used again and again. First, β is an upper bound of E. So $x \leq \beta$ for all $x \in E$. Second, nothing smaller than β is an upper bound of E. So for every $\varepsilon > 0$, some element of E is bigger than $\beta - \varepsilon$. This second fact is Theorem 1.8.7. To prove a number is $\sup E$, prove both facts.

Theorem 1.8.7. *Let $E \subset \mathbb{R}$ be nonempty and bounded above, and let $\beta = \sup E$. Then for all $\varepsilon > 0$, there exists an $x \in E$ such that $\beta - \varepsilon < x \leq \beta$. Similarly, if E is nonempty and bounded below and $\alpha = \inf E$, then for all $\varepsilon > 0$, there exists an $x \in E$ such that $\alpha \leq x < \alpha + \varepsilon$.*

PROOF. Let $\varepsilon > 0$. Then $\beta - \varepsilon < \beta$. So $\beta - \varepsilon$ is not an upper bound of E. Thus there exists an $x \in E$ with $x > \beta - \varepsilon$. Since β is an upper bound of E, $x \leq \beta$. The proof for $\inf E$ is an exercise. □

The next theorem says $\mathbb{R}$ has no infinitely large numbers. However small $x > 0$ is and however large y is, adding x to itself enough times gets past y. Taking $x = 1$, the natural numbers are not bounded above in $\mathbb{R}$. Taking $y = 1$, for every $\varepsilon > 0$ there is an $N \in \mathbb{N}$ with $\frac{1}{N} < \varepsilon$. That last form is used constantly in later chapters.

Theorem 1.8.8 (Archimedean Principle)**.** *If $x, y \in \mathbb{R}$ and $x > 0$, then there exists an $n \in \mathbb{N}$ such that $nx > y$. Hence, for all $x \in \mathbb{R}$, there exists an $n \in \mathbb{N}$ such that $x < n$.*

PROOF. Let $x, y \in \mathbb{R}$ with $x > 0$. Assume by way of contradiction that $nx \leq y$ for every $n \in \mathbb{N}$. Then the set $A = \{nx : n \in \mathbb{N}\}$ is nonempty and bounded above by y. By the least upper bound property, $\alpha = \sup A$ exists. Since $x > 0$, $\alpha - x < \alpha$. By Theorem 1.8.7 with $\varepsilon = x$, there is an element of A larger than $\alpha - x$. That is, $mx > \alpha - x$ for some $m \in \mathbb{N}$. Then $(m + 1)x > \alpha$. But $(m + 1)x \in A$, so this contradicts the fact that α is an upper bound of A. So $nx > y$ for some $n \in \mathbb{N}$.

Taking $x = 1$ and y any real number gives the second statement. □

Theorem 1.8.9. *For all $x \in \mathbb{R}$, there exists a unique $n \in \mathbb{Z}$ such that $n - 1 \leq x < n$.*

PROOF. There are two things to prove. Such an integer n exists, and it is unique. The proof of existence finds n using the smallest element of a set of natural numbers. By the Archimedean Principle, there exist $m_1, m_2 \in \mathbb{N}$ such that $m_1 > x$ and $m_2 > -x$. Let

$$S = \{k \in \mathbb{N} : k - m_2 > x\}.$$

Since $m_1 + m_2 - m_2 = m_1 > x$, $m_1 + m_2 \in S$. So $S \neq \emptyset$. By the Well-Ordering Principle, S has a smallest element j. Let $n = j - m_2$. Then $n \in \mathbb{Z}$ and $n > x$. If $j = 1$, then $n - 1 = -m_2 < x$. If $j > 1$, then $j - 1 \in \mathbb{N}$ and $j - 1 \notin S$, since j is the smallest element of S. So $n - 1 = (j - 1) - m_2 \leq x$. In either case, $n - 1 \leq x < n$.

To show that n is unique, suppose $n, n' \in \mathbb{Z}$ with $n-1 \leq x < n$ and $n'-1 \leq x < n'$. Then $n' > x \geq n-1$. So $n' > n-1$, and since n' and n are integers, $n' \geq n$. Similarly, $n \geq n'$. So $n = n'$. □

Definition 1.8.10. For $x \in \mathbb{R}$, the *greatest integer* less than or equal to x is denoted $\lfloor x \rfloor$. By Theorem 1.8.9, $\lfloor x \rfloor$ is the unique integer with $\lfloor x \rfloor \leq x < \lfloor x \rfloor + 1$.

Theorem 1.8.11 (Density of the Rational Numbers)**.** *For all $x, y \in \mathbb{R}$ with $x < y$, there exists a $q \in \mathbb{Q}$ such that $x < q < y$.*

PROOF. Since $x < y$, $y - x > 0$, and so by the Archimedean Principle (Theorem 1.8.8), there exists an $n \in \mathbb{N}$ such that $n(y - x) > 1$. By Theorem 1.8.9, there exists an $m \in \mathbb{Z}$ such that

$$m - 1 \leq nx < m.$$

Combining inequalities gives

$$nx < m \leq 1 + nx < ny.$$

Since $n > 0$,

$$x < \frac{m}{n} < y.$$

Since $m \in \mathbb{Z}$, $n \in \mathbb{N}$, $\frac{m}{n} \in \mathbb{Q}$. □

So every open interval, however short, contains a rational number. The rational numbers are spread throughout $\mathbb{R}$. Even so, they have holes and $\mathbb{R}$ does not. By Exercise 1.8.5, the irrational numbers are spread throughout $\mathbb{R}$ as well.

Lemma 1.8.12. *Let $n \in \mathbb{N}$ and $0 < a < b$. Then $b^n - a^n \leq nb^{n-1}(b - a)$.*

PROOF. By Exercise 1.2.8,

$$b^n - a^n = (b - a) \sum_{i=0}^{n-1} b^{n-1-i} a^i.$$

By Exercise 1.6.4, $a^i \leq b^i$ for each i. So each term of the sum satisfies $b^{n-1-i}a^i \leq b^{n-1-i}b^i = b^{n-1}$. There are n terms. So the sum is at most nb^{n-1}. Since $b - a > 0$, $b^n - a^n \leq nb^{n-1}(b - a)$. □

The next theorem shows $\mathbb{R}$ has no hole at $\sqrt{2}$, or at any other nth root. The idea is the one Section 1.4 suggests. Let E be the set of positive t with $t^n < x$, and let $y = \sup E$. If $y^n < x$, a number slightly bigger than y is still in E. If $y^n > x$, a number slightly smaller than y is still an upper bound of E. Both are impossible. Lemma 1.8.12 controls how much t^n changes when t changes a little.

Theorem 1.8.13. *For every real $x > 0$ and every integer $n > 0$, there is a unique positive real number y with $y^n = x$. This number is denoted $\sqrt[n]{x}$. If $n = 2$, it is denoted $\sqrt{x}$.*

PROOF. There are two things to prove. Such a y exists, and it is unique. The candidate for y is the supremum of the positive numbers whose nth power is less than x. So let $E = \{t \in \mathbb{R} : t > 0,\ t^n < x\}$. Let $t_0 = \frac{1}{2}\min\{1, x\}$. Then $0 < t_0 < 1$, so $t_0^n \leq t_0 < x$ by Exercise 1.6.5. So $t_0 \in E$ and $E \neq \emptyset$. Let $M = \max\{1, x\}$. If $t > M$, then $t > 1$ and $t > x$. So $t^n \geq t > x$ by Exercise 1.6.5, and $t \notin E$. So M is an upper bound of E. By the least upper bound property, $y = \sup E$ exists. Since $t_0 \in E$, $y \geq t_0 > 0$.

It remains to show that $y^n = x$. The proof rules out $y^n < x$ and then $y^n > x$. Assume by way of contradiction that $y^n < x$. Choose h with $0 < h < 1$ and $hn(y+1)^{n-1} < x - y^n$. For example, $h = \min\left\{\frac{1}{2}, \frac{x-y^n}{2n(y+1)^{n-1}}\right\}$ works. By Lemma 1.8.12 with $a = y$ and $b = y + h$,

$$(y+h)^n - y^n \le n(y+h)^{n-1}h \le n(y+1)^{n-1}h < x - y^n.$$

So $(y+h)^n < x$, and $y + h \in E$. Since $y + h > y$, this contradicts the fact that y is an upper bound of E. So $y^n \ge x$.

Assume by way of contradiction that $y^n > x$. Let

$$k = \frac{y^n - x}{ny^{n-1}}.$$

Then $k > 0$. Also $k < y$, since $y^n - x < y^n \le ny^n$. By Lemma 1.8.12 with $a = y - k$ and $b = y$,

$$y^n - (y-k)^n \le ny^{n-1}k = y^n - x.$$

So $(y-k)^n \ge x$. Now let $t \in E$. If $t \ge y - k$, then $t^n \ge (y-k)^n \ge x$ by Exercise 1.6.4, which is false since $t \in E$. So $t < y - k$. Thus $y - k$ is an upper bound of E. But $y - k < y$, which contradicts $y = \sup E$. So $y^n \le x$. Therefore $y^n = x$.

It remains to show that y is unique. If $0 < y_1 < y_2$, then $y_1^n < y_2^n$ by Exercise 1.6.4. So two different positive numbers cannot both have nth power equal to x. □

Exercises

1.8.1. Prove that if $|x| \le \varepsilon$ for all $\varepsilon > 0$, then $x = 0$. [Hint, p. 331]

1.8.2. Prove the statement about $\inf E$ in Theorem 1.8.7.

1.8.3. Define $E = \left\{(-1)^n \frac{n}{n+1} : n \in \mathbb{N}_0\right\} \subset \mathbb{R}$. Prove $\inf E = -1$. [Hint, p. 331]

1.8.4. Prove that if $r \in \mathbb{Q} \setminus \{0\}$ and $x \in \mathbb{R} \setminus \mathbb{Q}$, then $r + x \in \mathbb{R} \setminus \mathbb{Q}$ and $rx \in \mathbb{R} \setminus \mathbb{Q}$. [Hint, p. 331]

1.8.5. (a) Prove that between any two rational numbers there exists an irrational number.

(b) Prove that between any two real numbers there exists an irrational number.

[Hint, p. 331]

1.8.6. Let $A = \{p \in \mathbb{Q} : p > 0,\ p^2 < 2\}$. Prove that $\sup A = \sqrt{2}$ in $\mathbb{R}$. [Hint, p. 331]

1.8.7. Let L and U be nonempty subsets of $\mathbb{R}$ with $\mathbb{R} = L \cup U$ such that for each $l \in L$ and $u \in U$, $l < u$. Prove either $\sup L \in L$ or $\inf U \in U$. [Solution, p. 331]

Notes

Augustus De Morgan stated his laws in his 1847 book on logic [**22**]. Induction became one of the axioms for the natural numbers in the work of Richard Dedekind in 1888 [**21**] and Giuseppe Peano in 1889 [**57**]. The Division Algorithm is the first step of the Euclidean algorithm, in Book VII of Euclid's *Elements* [**28**]. The Pythagoreans

knew that $\sqrt{2}$ is irrational. Aristotle mentions a proof by odd and even numbers, like the one in Theorem 1.4.4. The Archimedean Principle appears in Book V of the *Elements*, and as an assumption in Archimedes' *On the Sphere and Cylinder* [**2**]. Bernoulli's Inequality (Exercise 1.6.6) is named for Jacob Bernoulli, who published it in 1689 [**4**]. Bernard Bolzano proved a form of the least upper bound property in 1817 [**5**]. In 1872, Dedekind [**20**] and Georg Cantor [**8**] each published a construction of $\mathbb{R}$ from $\mathbb{Q}$. Dedekind used cuts, and Cantor used Cauchy sequences. These are the constructions in Appendices A and B.

CHAPTER 2

Cardinality

When do two sets have the same cardinality? For finite sets, count the elements. For infinite sets, counting never ends, so a different idea is needed. Two sets are said to have the same cardinality if their elements can be paired off, one to one, with nothing left over on either side. This chapter makes that idea precise.

The results are surprising. The natural numbers, the integers and the rational numbers all have the same cardinality, even though each set is a proper subset of the next. The real numbers are different. No list $x_1, x_2, x_3, \ldots$ of real numbers contains every real number. So two infinite sets do not always have the same cardinality. The starred Section 2.4 shows how to compare the sizes of infinite sets, and that there is no largest one.

2.1. Countable and Uncountable Sets

Functions are the tool for pairing off the elements of two sets. This section sets up the language of functions, then uses one-to-one and onto functions to define what it means for two sets to have the same cardinality.

Definition 2.1.1. Let A and B be two sets. A *function* $f : A \to B$ is a rule that assigns each element in A exactly one element in B. The set A is the *domain* of f and B is the *codomain* of f. The element of B assigned to $a \in A$ is denoted $f(a)$.

Definition 2.1.2. Let $f : A \to B$, $E \subset A$ and $G \subset B$. The *image* of E under f is

$$f[E] = \{f(x) : x \in E\}.$$

The image $f[A]$ is called the *range* of f and is also written $\operatorname{Range} f$. The *preimage* of G under f is

$$f^{-1}[G] = \{x \in A : f(x) \in G\}.$$

Definition 2.1.3. Let $f : A \to B$ and $g : B \to C$. The *composition* of g and f is the function $g \circ f : A \to C$ defined by $(g \circ f)(a) = g(f(a))$.

Definition 2.1.4. A function $f : A \to B$ is *one-to-one* (1-1), or *injective*, if for $a_1, a_2 \in A$ with $a_1 \neq a_2$, $f(a_1) \neq f(a_2)$. f is *onto*, or *surjective*, if for each $b \in B$, there exists an $a \in A$ with $f(a) = b$. f is *bijective* if f is both injective and surjective. A bijective function is called a *bijection*.

Proof Technique (Proving a Function Is One-to-One). The definition says different inputs give different outputs. Its contrapositive is easier to use. If $f(a_1) = f(a_2)$, then $a_1 = a_2$. So to prove f is one-to-one, let $a_1, a_2 \in A$ with

$f(a_1) = f(a_2)$. Then show $a_1 = a_2$. To prove f is not one-to-one, find $a_1 \neq a_2$ with $f(a_1) = f(a_2)$.

Proof Technique (Proving a Function Is Onto). To prove $f : A \to B$ is onto, let b be an arbitrary element of B. Then find an $a \in A$ with $f(a) = b$. Finding a is usually scratch work. Solve the equation $f(a) = b$ for a. Then the proof states a and checks two things. First, $a \in A$. Second, $f(a) = b$. To prove f is not onto, find a $b \in B$ that is not $f(a)$ for any $a \in A$.

Example 2.1.5. Define $f : \mathbb{Z} \to \mathbb{Z}$ by $f(n) = n+3$. Let $m, n \in \mathbb{Z}$ with $f(m) = f(n)$. Then $m + 3 = n + 3$, or $m = n$. So f is one-to-one. Now let $k \in \mathbb{Z}$. Then $k - 3 \in \mathbb{Z}$ and $f(k - 3) = k$. So f is onto. Thus f is a bijection.

Example 2.1.6. Define $f : \mathbb{Z} \to \mathbb{Z}$ by $f(n) = n^2$. Then $f(-1) = 1 = f(1)$, so f is not one-to-one. Also, $n^2 \neq 2$ for all $n \in \mathbb{Z}$, so f is not onto. Notice $f[\{-1, 0, 1, 2\}] = \{0, 1, 4\}$, $f^{-1}[\{1, 4\}] = \{-2, -1, 1, 2\}$, and $f^{-1}[\{2\}] = \emptyset$.

Definition 2.1.7. Two sets A and B are said to have the *same cardinality*, written $A \sim B$ or $|A| = |B|$, if there exists a bijection from A to B.

Theorem 2.1.8. *Cardinal equivalence is an equivalence relation; i.e.,*

(1) $A \sim A$;
(2) if $A \sim B$, then $B \sim A$; and
(3) if $A \sim B$ and $B \sim C$, then $A \sim C$.

PROOF. (1) Define $f : A \to A$ by $f(a) = a$. Let $a_1, a_2 \in A$ with $f(a_1) = f(a_2)$. Since $f(a_1) = a_1$ and $f(a_2) = a_2$, f is one-to-one. Now let $a \in A$. Then $f(a) = a$ and so f is onto. Thus f is a bijection. So $A \sim A$.

(2) Since $A \sim B$, there exists a bijection $f : A \to B$. Define $g : B \to A$ by $g(b) = a$ if and only if $f(a) = b$. Since f is onto and 1-1, g is a well defined function. Let $b_1, b_2 \in B$ with $g(b_1) = g(b_2) = a$. Then $f(a) = b_1$ and $f(a) = b_2$. Since f is a function, $b_1 = b_2$. So g is one-to-one. Now let $a \in A$. So $f(a) = b$ for some $b \in B$, implying $g(b) = a$. So g is onto. Thus g is a bijection. So $B \sim A$.

(3) Since $A \sim B$, there exists a bijection $f : A \to B$. Since $B \sim C$, there exists a bijection $g : B \to C$. Define $h : A \to C$ by $h(a) = g(f(a))$. Let $a_1, a_2 \in A$ with $h(a_1) = h(a_2)$. Then $g(f(a_1)) = g(f(a_2))$. Since g is 1-1, $f(a_1) = f(a_2)$. Since f is one-to-one, $a_1 = a_2$. So h is one-to-one. Now let $c \in C$. Since g is onto, there exists a $b \in B$ such that $g(b) = c$. Since f is onto, there exists an $a \in A$ with $f(a) = b$. So $h(a) = g(f(a)) = g(b) = c$, implying h is onto. So h is a bijection and $A \sim C$. □

Definition 2.1.9. For each $n \in \mathbb{N}$, let $\mathbb{N}_n = \{1, 2, \ldots, n\}$. A set A is *finite* if $A = \emptyset$ or $A \sim \mathbb{N}_n$ for some $n \in \mathbb{N}$. A set that is not finite is *infinite*. A set A is *countable* if $A \sim \mathbb{N}$. An infinite set that is not countable is *uncountable*. A set is *at most countable* if it is finite or countable.

A countable set is one whose elements can be listed in a sequence $a_1, a_2, a_3, \ldots$ that includes every element exactly once. The bijection $f : \mathbb{N} \to A$ is the list, with $a_n = f(n)$.

Example 2.1.10. Let $S = \{1^2, 2^2, 3^2, \ldots\}$. Define $f : \mathbb{N} \to S$ by $f(n) = n^2$. Let $m, n \in \mathbb{N}$ with $f(m) = f(n)$. Then $m^2 = n^2$, or $m = n$. So f is one-to-one. Now let $s \in S$. So $\sqrt{s} \in \mathbb{N}$ with $f(\sqrt{s}) = s$. So f is onto. Thus f is a bijection and $S \sim \mathbb{N}$, implying S is countable.

Example 2.1.11. $\mathbb{Z}$ is countable. The proof is an exercise.

Example 2.1.12. Denote $\mathbb{E}$ as the set of even integers. Define $f : \mathbb{Z} \to \mathbb{E}$ by $f(n) = 2n$. Let $m, n \in \mathbb{Z}$ with $f(m) = f(n)$. Then $2m = 2n$, or $m = n$. So f is one-to-one. Now let $n \in \mathbb{E}$. Then $\frac{n}{2} \in \mathbb{Z}$ and $f\left(\frac{n}{2}\right) = 2 \cdot \frac{n}{2} = n$. So f is onto. Thus f is a bijection and $\mathbb{E} \sim \mathbb{Z}$, implying $\mathbb{E}$ is countable.

Exercises

2.1.1. Prove that $\mathbb{Z}$ is countable by proving $f : \mathbb{N} \to \mathbb{Z}$ defined by

$$f(n) = \begin{cases} \dfrac{n}{2}, & n \text{ even}, \\ \dfrac{-(n-1)}{2}, & n \text{ odd}, \end{cases}$$

is a bijection. [Solution, p. 332]

2.1.2. Let $\mathbb{O}$ denote the set of odd integers. Prove $\mathbb{O}$ is countable. [Solution, p. 332]

2.1.3. Prove that $f : \mathbb{R} \to \mathbb{R}$ defined by $f(x) = 5x - 7$ is a bijection.

2.1.4. Let $f : A \to B$, $E_1, E_2 \subset A$ and $G_1, G_2 \subset B$. Prove each of the following.

(a) $f[E_1 \cup E_2] = f[E_1] \cup f[E_2]$.
(b) $f^{-1}[G_1 \cup G_2] = f^{-1}[G_1] \cup f^{-1}[G_2]$.
(c) $f^{-1}[G_1 \cap G_2] = f^{-1}[G_1] \cap f^{-1}[G_2]$.
(d) $f^{-1}[B \setminus G_1] = A \setminus f^{-1}[G_1]$.

2.1.5. Find a function f and sets E_1 and E_2 with $f[E_1 \cap E_2] \neq f[E_1] \cap f[E_2]$. [Hint, p. 332]

2.2. Countability of the Rationals

At first, it seems that $\mathbb{N} \times \mathbb{N}$ could not have the same cardinality as $\mathbb{N}$. It contains a whole copy of $\mathbb{N}$ in each row. The next theorem shows that it does. The idea is to give each pair (m, n) its own natural number, using every natural number exactly once. The table shows the assignment $(m, n) \mapsto 2^{m-1}(2n - 1)$. Row $m = 1$ lists the odd numbers. Row $m = 2$ lists twice the odd numbers, and so on. By Theorem 1.2.16, every natural number appears in the table exactly once.

	$n = 1$	$n = 2$	$n = 3$	$n = 4$	$n = 5$
$m = 1$	1	3	5	7	9
$m = 2$	2	6	10	14	18
$m = 3$	4	12	20	28	36
$m = 4$	8	24	40	56	72

Theorem 2.2.1. $\mathbb{N} \times \mathbb{N}$ *is countable.*

PROOF. By Theorem 1.2.16, for each $k \in \mathbb{N}$, there exists unique $m, n \in \mathbb{N}$ such that $k = 2^{m-1}(2n-1)$. Define $f : \mathbb{N} \times \mathbb{N} \to \mathbb{N}$ by $f(m,n) = 2^{m-1}(2n-1)$. The existence gives that f is onto and the uniqueness gives that f is one-to-one. Thus f is a bijection. So $\mathbb{N} \times \mathbb{N}$ is countable. □

Definition 2.2.2. Let A be a set. A *sequence* is a function $f : \mathbb{N} \to A$. For each $n \in \mathbb{N}$, denote $x_n = f(n)$. Then x_n is called the *nth term* of the sequence. Sequences are denoted by $\{x_n\}_{n=1}^{\infty}$ or $\{x_n\}$.

Definition 2.2.3. Let A be a countable set. A bijection $f : \mathbb{N} \to A$ is called an *enumeration* of A and is written $A = \{x_n : n = 1, 2, \dots\} = \{x_1, x_2, \dots\}$.

Theorem 2.2.4. *Every infinite subset of a countable set is countable.*

PROOF. Let A be a countable set and let $\{x_n : n = 1, 2, \dots\}$ be an enumeration of A. Let $E \subset A$ be infinite. For each $x \in E$, $x \in A$. So x must have the form x_k for some $k \in \mathbb{N}$. Let

$$n_1 = \min\{k : x_k \in E\},$$

which must exist by the Well-Ordering Principle (Axiom 1.2.7). Relabel this x_k as x_{n_1}. Let

$$n_2 = \min\big\{k : x_k \in E \setminus \{x_{n_1}\}\big\}.$$

Relabel this x_k as x_{n_2}. In general, let

$$n_m = \min\big\{k : x_k \in E \setminus \{x_{n_1}, \dots, x_{n_{m-1}}\}\big\}.$$

Relabel this x_k as x_{n_m}. Define $f(k) = x_{n_k}$. Since E is infinite, f is defined for all $k \in \mathbb{N}$.

Let $m, k \in \mathbb{N}$ with $m \neq k$. Assume without loss of generality that $m > k$. Then $n_m > n_k$ and so $x_{n_m} \neq x_{n_k}$. So $f(m) \neq f(k)$ and f is one-to-one. If $x \in E$, $x \in A$ and so $x = x_j$ for some $j \in \mathbb{N}$. By construction, $n_k = j$ for some $k \in \mathbb{N}$, and so $f(k) = x$. Thus f is onto and is a bijection. So E is countable. □

Theorem 2.2.5. *If f maps $\mathbb{N}$ onto A, then A is at most countable.*

PROOF. If A is finite, the conclusion follows immediately. Suppose A is infinite. Since f maps $\mathbb{N}$ onto A, for each $a \in A$, there exists an $n \in \mathbb{N}$ such that $f(n) = a$. Also, by the Well-Ordering Principle, for each $a \in A$,

$$f^{-1}[\{a\}] = \{n \in \mathbb{N} : f(n) = a\}$$

has a smallest integer, call it n_a. Then the mapping $a \mapsto n_a$ is one-to-one, since f is a function. So $A \sim \{n_a : a \in A\}$. This set is an infinite subset of $\mathbb{N}$. So by Theorem 2.2.4, A is countable. □

Theorem 2.2.6. *Let $E_1, E_2, \dots$ be countably many countable sets. Then*

$$S = \bigcup_{n=1}^{\infty} E_n$$

is countable.

PROOF. Since E_n is countable for each $n \in \mathbb{N}$, E_n can be enumerated as $E_n = \{x_{n,k} : k = 1, 2, \dots\}$. Since $E_1 \subset S$ and E_1 is infinite, S is infinite. Define $h : \mathbb{N} \times \mathbb{N} \to S$ by

$$h(n, k) = x_{n,k}.$$

The function h is onto. Since $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$, there exists a mapping from $\mathbb{N}$ onto S. Since S is infinite, by Theorem 2.2.5, S is countable. □

Corollary 2.2.7. *$\mathbb{Q}$ is countable.*

PROOF. For each $m \in \mathbb{N}$, define

$$E_m = \left\{ \frac{n}{m} : n \in \mathbb{Z} \right\}.$$

Then E_m is countable and since

$$\mathbb{Q} = \bigcup_{m=1}^{\infty} E_m,$$

by Theorem 2.2.6, $\mathbb{Q}$ is countable. □

This is surprising at first. Between any two rational numbers there is another one, by Exercise 1.4.3. So there are infinitely many rational numbers in every interval. Even so, all of them can be listed in a single sequence $q_1, q_2, q_3, \dots$.

Proof Technique (Proving a Set Is Countable). There are two common ways to prove a set S is countable. One is to build a bijection from $\mathbb{N}$ to S, or from S to $\mathbb{N}$. The other is to use the theorems in this section. If S is an infinite subset of a countable set, Theorem 2.2.4 applies. If S is a countable union of countable sets, Theorem 2.2.6 applies. The proof of Corollary 2.2.7 uses the second way.

Exercises

2.2.1. Let $n \in \mathbb{N}$. Prove that

$$\mathbb{N}^n = \underbrace{\mathbb{N} \times \mathbb{N} \times \cdots \times \mathbb{N}}_{n \text{ times}} = \{(x_1, x_2, \dots, x_n) : x_i \in \mathbb{N} \text{ for all } i = 1, \dots, n\}$$

is countable. [Hint, p. 332]

2.2.2. A real number a is *algebraic* if there exists a nonzero polynomial p with integer coefficients such that $p(a) = 0$.

(a) Denote the set of polynomials of degree n with integer coefficients by P_n. Prove P_n is countable.

(b) Prove that the set of all polynomials with integer coefficients $P = \bigcup_{n \in \mathbb{N}} P_n$ is countable.

(c) Let $p \in P$. Let R_p be the set of roots of p. Prove that the set of algebraic numbers $A = \bigcup_{p \in P} R_p$ is countable.

2.3. Uncountability of the Reals

Every infinite set so far has turned out to be countable. This section shows $[0,1]$, and so $\mathbb{R}$, is not. Since $\mathbb{Q}$ is countable, it follows that the set of irrational numbers is uncountable. See Exercise 2.3.2.

Remark 2.3.1. Every $x \in [0,1]$ has a *decimal expansion* $x = 0.d_1d_2d_3\ldots$, where each $d_k \in \{0,1,\ldots,9\}$. This means $x = \sum_{k=1}^{\infty} \frac{d_k}{10^k}$. Some numbers have two decimal expansions. For example, $0.5000\cdots = 0.4999\ldots$. This happens exactly when one expansion ends in all 0's and the other ends in all 9's. These facts are assumed in this section. Infinite series are studied in Chapter 10.

Proof Technique (Diagonal Argument). The proof of Theorem 2.3.2 shows no list $x_1, x_2, x_3, \ldots$ of numbers in $[0,1]$ can contain all of $[0,1]$. It builds a number y whose nth decimal digit is different from the nth decimal digit of x_n. So $y \neq x_n$ for every n. This is called a *diagonal argument*, since it uses the digits $x_{1,1}, x_{2,2}, x_{3,3}, \ldots$ on the diagonal of the list. Theorem 2.3.3 uses the same idea.

Theorem 2.3.2. *The closed interval* $[0,1]$ *is uncountable.*

Proof. Let $E = \{x_n : n = 1,2,\ldots\}$ be a countable subset of $[0,1]$. Then each x_n has the decimal expansion

$$x_n = 0.x_{n,1}x_{n,2}x_{n,3}\ldots,$$

where for each $k \in \mathbb{N}$, $x_{n,k} \in \{0,1,\ldots,9\}$. Define a number

$$y = 0.y_1y_2y_3\ldots$$

in the following way. If $x_{n,n} \leq 5$, define $y_n = 6$. If $x_{n,n} \geq 6$, define $y_n = 3$. Then $y \in [0,1]$ and since $y_n \neq 0, 9$, y is not one of the real numbers with two decimal expansions. Also, since for each $n \in \mathbb{N}$, $y_n \neq x_{n,n}$, $y \neq x_n$ for any n. So $y \notin E$. Thus no countable subset of $[0,1]$ is all of $[0,1]$. Since $[0,1]$ is infinite, $[0,1]$ is uncountable. □

Theorem 2.3.3. *The set of all sequences whose elements are* 0 *or* 1*, denoted as* $2^{\mathbb{N}}$ *or* $\{0,1\}^{\mathbb{N}}$*, is uncountable.*

Proof. Let $E = \{s_n : n = 1,2,\ldots\}$ be an enumeration of E, a countable subset of $2^{\mathbb{N}}$. For each $n \in \mathbb{N}$, s_n is a sequence of 0's and 1's.

Now, for each $k \in \mathbb{N}$, let

$$s(k) = 1 - s_k(k).$$

Thus, if $s_k(k) = 0$, $s(k) = 1$, and if $s_k(k) = 1$, $s(k) = 0$. Thus $s \in 2^{\mathbb{N}}$. Since, for each $k \in \mathbb{N}$, $s(k) \neq s_k(k)$, $s \neq s_n$ for all $n \in \mathbb{N}$. So $s \notin E$. Thus no countable subset of $2^{\mathbb{N}}$ is all of $2^{\mathbb{N}}$. Since $2^{\mathbb{N}}$ is infinite, $2^{\mathbb{N}}$ is uncountable. □

Exercises

2.3.1. Prove that $\mathbb{R}$ is uncountable.

2.3.2. Prove that the set $\mathbb{R} \setminus \mathbb{Q}$ of irrational numbers is uncountable. [Hint, p. 332]

2.4. Comparing Infinite Sets*

Section 2.3 showed that $\mathbb{N}$ and $\mathbb{R}$ do not have the same cardinality. It is natural to say $\mathbb{R}$ is bigger. This section makes "bigger" precise. It proves the Schröder–Bernstein Theorem, which makes it easy to show two sets have the same cardinality. It also proves Cantor's Theorem, which says there is no largest infinite set. This section can be skipped. Nothing later depends on it.

Definition 2.4.1. Let A and B be sets. Write $A \lesssim B$, or $B \gtrsim A$, if there exists a one-to-one function $f : A \to B$.

Think of $A \lesssim B$ as saying B is at least as big as A. If $A \subset B$, then $A \lesssim B$, using the function $f(a) = a$. If $A \sim B$, then $A \lesssim B$ and $B \lesssim A$. If $A \lesssim B$ and $B \lesssim C$, then $A \lesssim C$, since a composition of one-to-one functions is one-to-one. See Exercise 2.4.1.

The next theorem is the converse of the second fact. It is the main tool for comparing infinite sets.

Theorem 2.4.2 (Schröder–Bernstein Theorem)**.** *If $A \lesssim B$ and $B \lesssim A$, then $A \sim B$.*

Proof. Let $f : A \to B$ and $g : B \to A$ be one-to-one. The plan is to use f on part of A and the inverse of g on the rest.

Let $C_0 = A \setminus g[B]$, the set of points of A that are not values of g. For $n \in \mathbb{N}_0$, let $C_{n+1} = g\big[f[C_n]\big]$. Let $C = \bigcup_{n=0}^{\infty} C_n$. Every point of $A \setminus C$ is in $g[B]$, since it is not in C_0. Since g is one-to-one, each such point is $g(b)$ for exactly one $b \in B$. Define $h : A \to B$ by

$$h(a) = \begin{cases} f(a), & a \in C, \\ \text{the } b \in B \text{ with } g(b) = a, & a \notin C. \end{cases}$$

It remains to show that h is one-to-one and onto. First, h is one-to-one. To see this, let $a, a' \in A$ with $h(a) = h(a')$. If $a, a' \in C$, then $f(a) = f(a')$, so $a = a'$. If $a, a' \notin C$, then $a = g(h(a)) = g(h(a')) = a'$. Suppose $a \in C$ and $a' \notin C$. Then $a' = g(h(a')) = g(h(a)) = g(f(a))$. Since $a \in C_n$ for some n, $a' \in C_{n+1} \subset C$. This contradicts $a' \notin C$. So this case cannot happen.

Next, h is onto. To see this, let $b \in B$, and let $a = g(b)$. If $a \notin C$, then $h(a) = b$. If $a \in C$, then $a \in C_n$ for some n. Since $a \in g[B]$, $n \neq 0$. So $a \in C_n = g\big[f[C_{n-1}]\big]$, and $a = g(f(a'))$ for some $a' \in C_{n-1}$. Since g is one-to-one, $b = f(a') = h(a')$.

So h is a bijection, and $A \sim B$. □

Proof Technique (Proving Two Sets Have the Same Cardinality). To prove $A \sim B$, a bijection is needed. Writing one down can be hard. By Theorem 2.4.2, it is enough to find two one-to-one functions, one from A into B and one from B into A. Neither has to be onto. That is usually much easier.

Example 2.4.3. $(0,1) \sim [0,1]$. Since $(0,1) \subset [0,1]$, $(0,1) \lesssim [0,1]$. The function $f(x) = \frac{x+1}{3}$ is one-to-one and maps $[0,1]$ into $\left[\frac{1}{3}, \frac{2}{3}\right] \subset (0,1)$. So $[0,1] \lesssim (0,1)$. By Theorem 2.4.2, $(0,1) \sim [0,1]$. A bijection exists, but it is not easy to write down.

Theorem 2.4.4. $[0,1] \sim 2^{\mathbb{N}}$.

PROOF. By Theorem 2.4.2, it is enough to show $2^{\mathbb{N}} \lesssim [0,1]$ and $[0,1] \lesssim 2^{\mathbb{N}}$. To show the first, let $s \in 2^{\mathbb{N}}$, and let $F(s) = \sum_{k=1}^{\infty} \frac{2s(k)}{3^k}$. The series converges by comparison with $\sum \frac{2}{3^k} = 1$, and $0 \le F(s) \le 1$. (Series are studied in Chapter 10. As in Remark 2.3.1, these facts are assumed here.) (These are the points of the Cantor set of Chapter 5.) Let $s \ne t$, and let k be the first index with $s(k) \ne t(k)$. Say $s(k) = 1$ and $t(k) = 0$. Then

$$F(s) - F(t) \ge \frac{2}{3^k} - \sum_{j=k+1}^{\infty} \frac{2}{3^j} = \frac{2}{3^k} - \frac{1}{3^k} > 0.$$

So $F(s) \ne F(t)$, and F is one-to-one.

To show $[0,1] \lesssim 2^{\mathbb{N}}$, use binary expansions. As with decimal expansions in Remark 2.3.1, every $x \in [0,1)$ has a binary expansion $x = \sum_{k=1}^{\infty} \frac{d_k}{2^k}$ with each $d_k \in \{0,1\}$, and it can be chosen so that it does not end in all 1's. Then it is unique. Let $G(x)$ be the sequence $\{d_k\}$ of that expansion, and let $G(1)$ be the sequence of all 1's. Different numbers in $[0,1)$ have different expansions, and no expansion used for $x < 1$ ends in all 1's. So G is one-to-one.

By Theorem 2.4.2, $[0,1] \sim 2^{\mathbb{N}}$. □

So $\mathbb{R}$, $[0,1]$ and $2^{\mathbb{N}}$ all have the same cardinality (Exercise 2.4.3). The set $2^{\mathbb{N}}$ is really the set of all subsets of $\mathbb{N}$. A sequence of 0's and 1's marks which natural numbers belong to a subset (Exercise 2.4.4). The last theorem shows that the step from $\mathbb{N}$ to its set of subsets always gives a bigger set, and that the same step works for every set.

Theorem 2.4.5 (Cantor's Theorem)**.** *Let A be a set and let $\mathcal{P}(A)$ be its power set. Then $A \lesssim \mathcal{P}(A)$, but $A \not\sim \mathcal{P}(A)$.*

PROOF. The function $a \mapsto \{a\}$ is one-to-one from A to $\mathcal{P}(A)$. So $A \lesssim \mathcal{P}(A)$. Let $f : A \to \mathcal{P}(A)$ be any function. Let

$$D = \{a \in A : a \notin f(a)\}.$$

Then $D \in \mathcal{P}(A)$. Suppose $D = f(d)$ for some $d \in A$. If $d \in D$, then $d \notin f(d) = D$. If $d \notin D$, then $d \in f(d) = D$. Both are impossible. So D is not a value of f, and f is not onto. So there is no bijection from A to $\mathcal{P}(A)$. □

This is the diagonal argument again. The set D differs from $f(a)$ at the element a, just as the number y in Theorem 2.3.2 differs from x_n in the nth digit.

Applying Theorem 2.4.5 over and over gives

$$\mathbb{N} \lesssim \mathcal{P}(\mathbb{N}) \lesssim \mathcal{P}(\mathcal{P}(\mathbb{N})) \lesssim \cdots,$$

and no two of these sets have the same cardinality (Exercise 2.4.5). So there are infinitely many different sizes of infinite sets, and none is the largest.

Remark 2.4.6. Is there a set $S \subset \mathbb{R}$ that is bigger than $\mathbb{N}$ but smaller than $\mathbb{R}$? That is, is there an uncountable $S \subset \mathbb{R}$ with $S \not\sim \mathbb{R}$? Cantor believed not. This is the *continuum hypothesis*. It cannot be settled from the usual axioms of set theory. Kurt Gödel showed in 1940 that it cannot be disproved from them, and Paul Cohen showed in 1963 that it cannot be proved from them.

Exercises

2.4.1. Let A, B and C be sets. Prove that $A \lesssim A$, that $A \subset B$ implies $A \lesssim B$, and that $A \lesssim B$ and $B \lesssim C$ imply $A \lesssim C$. [Hint, p. 332]

2.4.2. Let $a < b$. Use Theorem 2.4.2 to prove that (a, b), $[a, b)$, $(a, b]$ and $[a, b]$ all have the same cardinality as $[0, 1]$. [Hint, p. 333]

2.4.3. Prove that $f(x) = \frac{x}{1+|x|}$ is a bijection from $\mathbb{R}$ to $(-1, 1)$. Conclude that $\mathbb{R} \sim [0, 1] \sim 2^{\mathbb{N}}$. [Solution, p. 333]

2.4.4. For $E \subset \mathbb{N}$, let χ_E be the sequence with $\chi_E(k) = 1$ if $k \in E$ and $\chi_E(k) = 0$ if $k \notin E$. Prove that $E \mapsto \chi_E$ is a bijection from $\mathcal{P}(\mathbb{N})$ to $2^{\mathbb{N}}$. Conclude that $\mathcal{P}(\mathbb{N}) \sim \mathbb{R}$. [Solution, p. 333]

2.4.5. Let $A_0 = \mathbb{N}$ and $A_{n+1} = \mathcal{P}(A_n)$. Prove that if $m < n$, then $A_m \lesssim A_n$ and $A_m \not\sim A_n$. [Hint, p. 333]

2.4.6. For $s, t \in 2^{\mathbb{N}}$, define the sequence u by $u(2k-1) = s(k)$ and $u(2k) = t(k)$. Prove that $(s, t) \mapsto u$ is a bijection from $2^{\mathbb{N}} \times 2^{\mathbb{N}}$ to $2^{\mathbb{N}}$. Conclude that $\mathbb{R} \times \mathbb{R} \sim \mathbb{R}$. So the plane has no more points than a line. [Hint, p. 333]

2.4.7. Let $\mathcal{F}$ be the set of all functions from $\mathbb{R}$ to $\mathbb{R}$. Prove that $\mathbb{R} \lesssim \mathcal{F}$ and $\mathbb{R} \not\sim \mathcal{F}$. [Hint, p. 333]

Notes

Galileo noticed in 1638 that the perfect squares can be paired off with all of the natural numbers [**34**]. He concluded that "equal," "greater" and "less" do not apply to infinite collections. In 1888, Dedekind turned this around. He defined a set to be infinite if it has the same cardinality as one of its proper subsets [**21**]. Georg Cantor proved in 1874 that the algebraic numbers are countable and that $\mathbb{R}$ is uncountable [**9**]. Together these show that numbers that are not algebraic exist. Joseph Liouville had already built explicit examples, in work announced in 1844 and published in full in 1851 [**49**]. In 1878, Cantor defined two sets to have the same cardinality when there is a bijection between them [**10**]. His diagonal argument appeared in 1891 [**13**]. Cantor proved Theorem 2.4.5 in the same 1891 paper [**13**]. He also proved in 1878 that $\mathbb{R} \times \mathbb{R} \sim \mathbb{R}$ (Exercise 2.4.6), and wrote to Dedekind, "I see it, but I do not believe it." Cantor stated Theorem 2.4.2 without proof. Dedekind proved it in 1887 but did not publish his proof. Ernst Schröder published a proof in 1896 that turned out to have a gap, and Felix Bernstein gave a correct proof in 1897, published in 1898.

CHAPTER 3

Sequences of Real Numbers

Limits are the central idea of analysis. Derivatives, integrals and infinite series are all defined as limits. This chapter studies the simplest kind of limit, the limit of a sequence.

The definition of convergence is the first one in the book with the pattern "for all $\varepsilon > 0$, there exists an N." Learning to read and use that pattern is the main skill of this chapter. The rest of the chapter answers one question in several ways. How can a sequence be shown to converge? Bounded monotone sequences converge. Bounded sequences have convergent subsequences. Cauchy sequences converge. Each of these answers uses the least upper bound property of $\mathbb{R}$. None of them is true in $\mathbb{Q}$.

3.1. Convergence of a Sequence

Remark 3.1.1. Examples in this chapter use familiar functions such as $\sqrt{x}$, $\ln x$, $\sin x$ and $\cos x$. Their basic properties are assumed for now. The functions $\ln x$ and e^x are defined carefully in Chapter 9. The functions $\sin x$ and $\cos x$ are defined in Sections 9.5 and 12.4.

Definition 3.1.2. A sequence $\{x_n\}_{n=1}^{\infty}$ in $\mathbb{R}$ is said to *converge* to the real number a, denoted by $\lim_{n\to\infty} x_n = a$, or $x_n \to a$, if for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - a| < \varepsilon$.

In words, $x_n \to a$ means the terms x_n are eventually as close to a as anyone could ask. The number ε says how close. Once ε is given, there must be a place N in the sequence after which every term is within ε of a. Smaller values of ε usually need larger values of N. So N depends on ε. Figure 3.1 shows this for the sequence in Example 3.1.3 with $\varepsilon = \frac{1}{4}$. Every term from the Nth on lies in the band between $a - \varepsilon$ and $a + \varepsilon$.

Proof Technique (Proving a Limit with the Definition). To prove $x_n \to a$ from the definition, start with an arbitrary $\varepsilon > 0$. Then find an N that works. Finding N is scratch work done before writing the proof. Start with $|x_n - a| < \varepsilon$ and simplify until it reads $n >$ (something). Then choose N larger than that something. The Archimedean Principle (Theorem 1.8.8) says such an $N \in \mathbb{N}$ exists. The proof then runs forward. It states the choice of N and checks that $|x_n - a| < \varepsilon$ for all $n \geq N$. In Example 3.1.3, the scratch work is $\left|x_n - \frac{3}{2}\right| = \frac{2}{n} < \varepsilon$, which holds when $n > \frac{2}{\varepsilon}$.

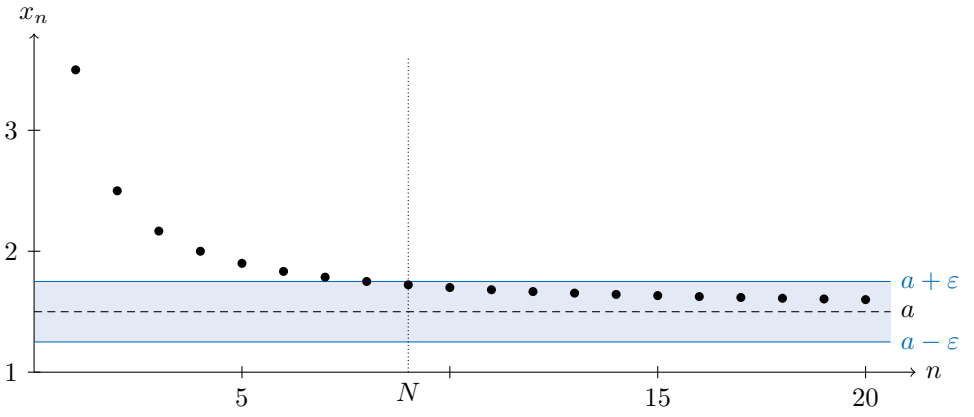


FIGURE 3.1. The sequence $x_n = \frac{3n+4}{2n}$ converges to $a = \frac{3}{2}$. With $\varepsilon = \frac{1}{4}$, every term from $N = 9$ on lies inside the shaded band.

Example 3.1.3. Consider the sequence $\{x_n\}$ defined by $x_n = \frac{3n+4}{2n}$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{2}{\varepsilon}$. So for all $n \geq N$,

$$\begin{aligned} \left| x_n - \frac{3}{2} \right| &= \left| \frac{3n+4}{2n} - \frac{3}{2} \right| \\ &= \left| \frac{3n+4-3n}{2n} \right| \\ &= \frac{2}{n} \\ &< \frac{2}{2/\varepsilon} = \varepsilon. \end{aligned}$$

So $x_n \to \frac{3}{2}$.

Example 3.1.4. Consider the sequence $\{x_n\}$ defined by $x_n = \frac{1}{n}$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon}$. Thus for all $n \geq N$,

$$\left| \frac{1}{n} - 0 \right| = \frac{1}{n} < \frac{1}{1/\varepsilon} = \varepsilon.$$

So $x_n \to 0$.

Example 3.1.5. Consider the sequence $\{x_n\}$ defined by $x_n = \frac{3n-7\sqrt{n}}{n}$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{49}{\varepsilon^2}$. Now for all $n \geq N$,

$$\begin{aligned} |x_n - 3| &= \left| \frac{3n-7\sqrt{n}}{n} - 3 \right| \\ &= \left| \frac{3n-7\sqrt{n}-3n}{n} \right| \\ &= \frac{7}{\sqrt{n}} \end{aligned}$$

$$< \frac{7}{\sqrt{49/\varepsilon^2}} = \varepsilon.$$

So $x_n \to 3$.

Proof Technique (Simplifying with Inequalities). Sometimes $|x_n - a| < \varepsilon$ is hard to solve for n. It is enough to find something larger and simpler that is still less than ε. In Example 3.1.6, $\frac{1}{6+8n} < \frac{1}{8n}$. Making the denominator smaller makes the fraction larger. Then $\frac{1}{8n} < \varepsilon$ is easy to solve. The bound does not need to be the best one. Example 3.1.9 uses this idea twice.

Example 3.1.6. Consider the sequence $\{x_n\}$ defined by $x_n = \frac{1+2n}{3+4n}$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{8}{\varepsilon}$. Then for all $n \geq N$,

$$\begin{aligned}\left|x_n - \frac{1}{2}\right| &= \left|\frac{1+2n}{3+4n} - \frac{1}{2}\right| \\ &= \left|\frac{2(1+2n) - (3+4n)}{2(3+4n)}\right| \\ &= \frac{1}{6+8n} \\ &< \frac{1}{8n} \\ &< \frac{1}{8(8/\varepsilon)} = \frac{\varepsilon}{64} < \varepsilon.\end{aligned}$$

So $x_n \to \frac{1}{2}$.

Example 3.1.7. Suppose $0 < |c| < 1$. Consider the sequence $\{c^n\}$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ with $N > \frac{\ln \varepsilon}{\ln |c|}$. Then for all $n \geq N$,

$$\ln |c|^n = n \ln |c| < \ln \varepsilon.$$

So for all $n \geq N$,

$$|c^n - 0| = |c|^n < \varepsilon.$$

Proof Technique (Showing a Sequence Diverges). A sequence $\{x_n\}$ *diverges* if it does not converge to any $a \in \mathbb{R}$. Negating the definition, $\{x_n\}$ does not converge to a means there exists an $\varepsilon > 0$ such that for all $N \in \mathbb{N}$, there exists an $n \geq N$ with $|x_n - a| \geq \varepsilon$. To show $\{x_n\}$ diverges, this must hold for every $a \in \mathbb{R}$. A proof by contradiction is often easier. Assume $x_n \to a$, pick a small ε, and find a contradiction. Example 3.1.8 picks $\varepsilon = 1$.

Example 3.1.8. Consider the sequence $\{(-1)^n\}$. Assume by way of contradiction that $(-1)^n \to a \in \mathbb{R}$. Let $\varepsilon = 1$. So there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - a| < 1$. Thus, for any odd integer $n \geq N$, $|x_n - a| < 1$, and for any even integer $n \geq N$, $|x_n - a| < 1$. Choose $n_0 > N$. Then

$$2 = |x_{n_0} - x_{n_0+1}| \leq |x_{n_0} - a| + |x_{n_0+1} - a| < 2,$$

a contradiction.

Example 3.1.9. Consider the sequence $\{x_n\}$ defined by $x_n = \frac{\cos n + 3\sin n}{n^2+n-60}$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \max\left\{\frac{2}{\sqrt{\varepsilon}}, 60\right\}$. Then for all $n \geq N$,

$$\begin{aligned}\left|\frac{\cos n + 3\sin n}{n^2+n-60} - 0\right| &\leq \frac{|\cos n| + 3|\sin n|}{|n^2+n-60|} \\ &\leq \frac{4}{|n^2+n-60|} \\ &= \frac{4}{n^2+n-60} \\ &\leq \frac{4}{n^2} \\ &< \frac{4}{(2/\sqrt{\varepsilon})^2} = \varepsilon.\end{aligned}$$

So $x_n \to 0$.

Exercises

3.1.1. Prove, using the ε-N definition, the following sequences $\{x_n\}$ converge to the given number a.

(a) $x_n = \dfrac{5n}{3n+2}$, $a = \dfrac{5}{3}$

(b) $x_n = \dfrac{12n^2 + 5\sqrt{n}}{2n^2}$, $a = 6$

(c) $x_n = 1 - \dfrac{(-1)^n}{n}$, $a = 1$

(d) $x_n = \dfrac{(1+4n)^2}{6n^2+n-75}$, $a = \dfrac{8}{3}$

[Solution, p. 333]

3.1.2. Define the sequence $\{x_n\}$ by $x_n = \frac{\sin(n\pi/2)}{8}$. Prove, using the ε-N definition, that this sequence diverges.

3.2. Properties of Convergent Sequences

Proving every limit from the definition would be slow. This section proves general rules instead. Limits add, multiply and divide as expected, and limits respect inequalities. With these rules, many limits can be found without going back to ε and N.

Proof Technique (Using the Definition with a Specific ε). The definition of $x_n \to a$ holds for every $\varepsilon > 0$. So a proof can use any specific ε it wants. The proof of Theorem 3.2.1 uses $\varepsilon = \frac{|a_1 - a_2|}{2}$. Also, if one statement holds for all $n \geq N_1$ and another holds for all $n \geq N_2$, then both hold for all $n \geq \max\{N_1, N_2\}$.

Theorem 3.2.1. *The limit of a convergent sequence is unique.*

PROOF. Assume by way of contradiction that there exists a sequence $\{x_n\}$ and $a_1, a_2 \in \mathbb{R}$ such that $x_n \to a_1$ and $x_n \to a_2$ with $a_1 \neq a_2$. Since $a_1 \neq a_2$, $|a_1 - a_2| > 0$. Since $x_n \to a_1$, there exists an $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$,

$$|x_n - a_1| < \frac{|a_1 - a_2|}{2}.$$

Since $x_n \to a_2$, there exists an $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$,

$$|x_n - a_2| < \frac{|a_1 - a_2|}{2}.$$

Set $N = \max\{N_1, N_2\}$. Then

$$\begin{aligned} |a_1 - a_2| &= |a_1 - x_N + x_N - a_2| \\ &\leq |a_1 - x_N| + |a_2 - x_N| \\ &< \frac{|a_1 - a_2|}{2} + \frac{|a_1 - a_2|}{2} \\ &= |a_1 - a_2|, \end{aligned}$$

a contradiction. □

Definition 3.2.2. A sequence $\{x_n\}$ is said to be *bounded* if there exists an $M > 0$ such that $|x_n| \leq M$ for all $n \in \mathbb{N}$.

Theorem 3.2.3. *A convergent sequence is bounded.*

PROOF. Let $\{x_n\}$ be a sequence in $\mathbb{R}$ with $x_n \to a \in \mathbb{R}$. Let $\varepsilon = 1$. So there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - a| < 1$. Then, by the reverse triangle inequality, for all $n \geq N$

$$1 > |x_n - a| \geq \big||x_n| - |a|\big| \geq |x_n| - |a|.$$

So for all $n \geq N$, $|x_n| < 1 + |a|$. Define

$$M = \max\{|x_1|, |x_2|, \ldots, |x_{N-1}|, |a| + 1\}.$$

Then $|x_n| \leq M$ for all $n \in \mathbb{N}$. So $\{x_n\}$ is bounded. □

Proof Technique (Splitting ε). In the proof of Theorem 3.2.4 (1), the goal is $|(x_n + y_n) - (a + b)| < \varepsilon$. The Triangle Inequality splits this into $|x_n - a| + |y_n - b|$. If each piece is less than $\frac{\varepsilon}{2}$, the sum is less than ε. This is allowed because the definition of $x_n \to a$ works for every positive number, including $\frac{\varepsilon}{2}$. The proof of (2) uses the same idea with $\frac{\varepsilon}{2M}$ and $\frac{\varepsilon}{2(|b|+1)}$. The $+1$ in $|b| + 1$ avoids dividing by 0 when $b = 0$. The proof of (2) also adds and subtracts $x_n b$, as in Section 1.7.

Theorem 3.2.4. *Suppose that $x_n \to a$ and $y_n \to b$. Then*

(1) $x_n + y_n \to a + b$;
(2) $x_n y_n \to ab$; and
(3) if $b \neq 0$ and $y_n \neq 0$ for all $n \in \mathbb{N}$, $\frac{x_n}{y_n} \to \frac{a}{b}$.

PROOF. (1) Let $\varepsilon > 0$. Since $x_n \to a$, there exists an $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|x_n - a| < \frac{\varepsilon}{2}$. Since $y_n \to b$, there exists an $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|y_n - b| < \frac{\varepsilon}{2}$.

Let $N = \max\{N_1, N_2\}$. So for all $n \geq N$,

$$\begin{aligned} |(x_n + y_n) - (a + b)| &= |(x_n - a) + (y_n - b)| \\ &\leq |x_n - a| + |y_n - b| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

So $x_n + y_n \to a + b$.

(2) Since $\{x_n\}$ converges, it is bounded, and so there exists an $M > 0$ such that $|x_n| \leq M$ for all $n \in \mathbb{N}$. Let $\varepsilon > 0$. Since $x_n \to a$, there exists an $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|x_n - a| < \frac{\varepsilon}{2(|b|+1)}$. Since $y_n \to b$, there exists an $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|y_n - b| < \frac{\varepsilon}{2M}$.

Let $N = \max\{N_1, N_2\}$. So for all $n \geq N$,

$$\begin{aligned} |x_n y_n - ab| &= |x_n y_n - x_n b + x_n b - ab| \\ &\leq |x_n y_n - x_n b| + |x_n b - ab| \\ &= |x_n||y_n - b| + |b||x_n - a| \\ &< M\left(\frac{\varepsilon}{2M}\right) + |b|\left(\frac{\varepsilon}{2(|b|+1)}\right) \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

(3) Since $y_n \to b$, there exists an $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|y_n - b| < \frac{|b|}{2}$. Notice the reverse triangle inequality implies that for $n \geq N_1$,

$$|b| - |y_n| \leq |b - y_n| = |y_n - b| < \frac{|b|}{2}.$$

Thus for all $n \geq N_1$, $\frac{|b|}{2} \leq |y_n|$ and $\frac{1}{|y_n|} \leq \frac{2}{|b|}$.

Let $\varepsilon > 0$. Then, again, since $y_n \to b$, there exists an $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|y_n - b| < \frac{\varepsilon|b|^2}{2}$.

Let $N = \max\{N_1, N_2\}$. Then for all $n \geq N$,

$$\begin{aligned} \left|\frac{1}{y_n} - \frac{1}{b}\right| &= \frac{|y_n - b|}{|y_n||b|} \\ &\leq \frac{2}{|b|^2}|y_n - b| \\ &< \frac{2}{|b|^2} \cdot \frac{\varepsilon|b|^2}{2} = \varepsilon. \end{aligned}$$

So $\frac{1}{y_n} \to \frac{1}{b}$. By (2), $\frac{x_n}{y_n} \to \frac{a}{b}$. □

Corollary 3.2.5. *If $\{x_n\}$ is a convergent sequence in $\mathbb{R}$ with $x_n \to a$, then for all $c \in \mathbb{R}$,*

(1) $x_n + c \to a + c$; and
(2) $cx_n \to ca$.

PROOF. Let $\{c_n\}$ be defined by $c_n = c$ for all $n \in \mathbb{N}$. The result then follows directly from Theorem 3.2.4. □

Corollary 3.2.6. *Let $\{x_n\}$ and $\{y_n\}$ be sequences of real numbers with $\{y_n\}$ bounded and $x_n \to 0$. Then $x_n y_n \to 0$.*

PROOF. See Exercise 3.2.2. □

Theorem 3.2.7. *Suppose that $\{x_n\}$ and $\{y_n\}$ are convergent sequences in $\mathbb{R}$ with $x_n \to a$ and $y_n \to b$. If $x_n \leq y_n$ for all $n \in \mathbb{N}$, $a \leq b$.*

PROOF. Assume by way of contradiction that $a > b$. Thus $\frac{a-b}{2} > 0$. Since $x_n \to a$, there exists an $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|x_n - a| < \frac{a-b}{2}$. So $x_n - a > \frac{-a+b}{2}$ for all $n \geq N_1$. Also, since $y_n \to b$, there exists an $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|y_n - b| < \frac{a-b}{2}$. So for all $n \geq N_2$, $y_n - b < \frac{a-b}{2}$.

Set $N = \max\{N_1, N_2\}$. Then

$$y_N < \frac{a-b}{2} + b = \frac{a+b}{2} = a + \frac{-a+b}{2} < x_N,$$

a contradiction. So $a \leq b$. □

Corollary 3.2.8. *Suppose that $\{x_n\}$ and $\{y_n\}$ are convergent sequences in $\mathbb{R}$ with $x_n \to a$ and $y_n \to b$. If there exists an $N \in \mathbb{N}$ such that $x_n \leq y_n$ for all $n \geq N$, $a \leq b$.*

Corollary 3.2.9. *If $x_n \to c$ and if there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $a \leq x_n \leq b$, $a \leq c \leq b$.*

Theorem 3.2.10 (Squeeze Theorem)**.** *Suppose $\{x_n\}$, $\{y_n\}$, and $\{z_n\}$ are sequences in $\mathbb{R}$ with $x_n \to a$, $y_n \to a$. Then if $x_n \leq z_n \leq y_n$ for all $n \in \mathbb{N}$, $z_n \to a$.*

PROOF. Let $\varepsilon > 0$. Since $x_n \to a$, there exists an $N_1 \in \mathbb{N}$ such that for all $n \geq N_1$, $|x_n - a| < \varepsilon$. Thus for all $n \geq N_1$,

$$-\varepsilon < x_n - a < \varepsilon,$$

or

$$a - \varepsilon < x_n < a + \varepsilon.$$

Likewise, since $y_n \to a$, there exists an $N_2 \in \mathbb{N}$ such that for all $n \geq N_2$, $|y_n - a| < \varepsilon$. Thus for all $n \geq N_2$,

$$a - \varepsilon < y_n < a + \varepsilon.$$

Set $N = \max\{N_1, N_2\}$. Then for all $n \geq N$,

$$a - \varepsilon < x_n \leq z_n \leq y_n < a + \varepsilon.$$

So for all $n \geq N$, $|z_n - a| < \varepsilon$, implying $z_n \to a$. □

Corollary 3.2.11 (Squeeze Theorem)**.** *Suppose $\{x_n\}$, $\{y_n\}$, and $\{z_n\}$ are sequences in $\mathbb{R}$ with $x_n \to a$, $y_n \to a$. Then if there exists an $N \in \mathbb{N}$ such that $x_n \leq z_n \leq y_n$ for all $n \geq N$, $z_n \to a$.*

Definition 3.2.12. A sequence $\{x_n\}$ is

(1) *increasing* if $x_{n+1} \geq x_n$ for all $n \in \mathbb{N}$;

(2) *strictly increasing* if $x_{n+1} > x_n$ for all $n \in \mathbb{N}$;
(3) *decreasing* if $x_{n+1} \leq x_n$ for all $n \in \mathbb{N}$;
(4) *strictly decreasing* if $x_{n+1} < x_n$ for all $n \in \mathbb{N}$; and
(5) *monotonic* if $\{x_n\}$ is either increasing or decreasing.

Theorem 3.2.13. *Suppose $\{x_n\}$ is monotonic. Then $\{x_n\}$ converges if and only if $\{x_n\}$ is bounded.*

PROOF. ($\Rightarrow$) This result is an immediate consequence of Theorem 3.2.3.

($\Leftarrow$) Assume $\{x_n\}$ is bounded, and without loss of generality, assume $\{x_n\}$ is increasing. Let $A = \{x_n : n \in \mathbb{N}\}$. Now since $\{x_n\}$ is bounded, A has an upper bound. So A has a least upper bound in $\mathbb{R}$, say $\alpha = \sup A$.

Let $\varepsilon > 0$. Since $\alpha = \sup A$, by Theorem 1.8.7, there exists an $a \in A$ such that $a > \alpha - \varepsilon$; i.e., there exists an $N \in \mathbb{N}$ such that $x_N = a > \alpha - \varepsilon$. But $\{x_n\}$ is increasing. So for all $n \geq N$, $\alpha \geq x_n > \alpha - \varepsilon$. So for all $n \geq N$, $|x_n - \alpha| < \varepsilon$. So $x_n \to \alpha$. □

This theorem depends on the least upper bound property. It is false in $\mathbb{Q}$. For example, let x_n be the decimal expansion of $\sqrt{2}$ cut off after n digits, so $x_1 = 1.4$, $x_2 = 1.41$, $x_3 = 1.414$, and so on. Each x_n is rational, the sequence is increasing, and it is bounded above by 2. Its limit is $\sqrt{2}$, which is not in $\mathbb{Q}$.

Corollary 3.2.14. *If $\{x_n\}$ is bounded and if there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $x_{n+1} \geq x_n$, or for all $n \geq N$, $x_{n+1} \leq x_n$ (x_n is eventually monotonic), $\{x_n\}$ converges.*

Exercises

3.2.1. Suppose that $x_n \to a$. Prove by induction that, for $k \in \mathbb{N}$, $(x_n)^k \to a^k$.

3.2.2. Suppose $\{x_n\}$ and $\{y_n\}$ are sequences in $\mathbb{R}$ with $\{y_n\}$ bounded and $x_n \to 0$. Prove $x_n y_n \to 0$. (Note: $\{y_n\}$ does not necessarily converge.) Find an example to show the condition that $\{y_n\}$ is bounded is necessary.

[Solution, p. 334]

3.2.3. Prove that a bounded decreasing sequence $\{x_n\}$ converges by appealing to the sequence $\{-x_n\}$.

3.3. Recursively Defined Sequences

Many sequences are defined by a rule that gives each term from the one before it. Such sequences come up in numerical methods for solving equations. Example 3.3.4 below is Newton's method for finding $\sqrt{3}$. Often there is no formula for x_n at all. So the theorems of Section 3.2 are the main tools.

Example 3.3.1. The sequence $\{x_n\}$ where x_n is the finite decimal expansion consisting of n 1s after the decimal converges to $\frac{1}{9}$. To see this, first, notice that $x_1 = 0.1$, $x_2 = 0.11$, $x_3 = 0.111$, and so on. Since $\{x_n\}$ is bounded and increasing, there exists an $a \in \mathbb{R}$ such that $x_n \to a$. So $x_{n+1} \to a$ as well. But

$$10x_{n+1} - x_n = 1.$$

So

$$\lim_{n\to\infty}(10x_{n+1}-x_n)=1.$$

Therefore $10a-a=1$. So $a=\frac{1}{9}$.

Proof Technique (Recursive Sequences). Many recursive sequences are handled in three steps. First, use induction to show the sequence is bounded. Second, use induction to show the sequence is monotone. Then by Theorem 3.2.13, the sequence converges, say $x_n \to a$. Third, take the limit of both sides of the recursive formula and solve for a. The order matters. The third step only makes sense once the limit is known to exist. For example, let $x_1 = 1$ and $x_{n+1} = 2x_n$. Taking limits of both sides gives $a = 2a$, so $a = 0$. But $x_n = 2^{n-1}$, which diverges.

Example 3.3.2. The sequence defined recursively by $x_1 = 1$ and $x_{n+1} = \frac{2x_n+3}{4}$ converges to $\frac{3}{2}$. First, notice $x_1 = 1 < 2$. Using induction, assume there is a $k \in \mathbb{N}$ with $x_k < 2$. So

$$x_{k+1}=\frac{2x_k+3}{4}\leq\frac{7}{4}<2.$$

By induction, $\{x_n\}$ is bounded above by 2. Next, notice $x_1 = 1$ and $x_2 = \frac{5}{4} > 1 = x_1$. Again, using induction, assume there is a $k \in \mathbb{N}$, $k > 1$ with $x_k > x_{k-1}$. Then

$$x_{k+1}=\frac{2x_k+3}{4}>\frac{2x_{k-1}+3}{4}=x_k.$$

So by induction, $\{x_n\}$ is increasing. Thus $\{x_n\}$ is bounded and monotonic, so it converges, say $x_n \to a$.

Now, since $x_n \to a$, $x_{n+1} \to a$. But $x_{n+1} = \frac{2x_n+3}{4}$, implying

$$\lim_{n\to\infty}x_{n+1}=\lim_{n\to\infty}\frac{2x_n+3}{4},$$

or $a = \frac{2a+3}{4}$. So $a = \frac{3}{2}$.

Example 3.3.3. The sequence defined recursively by $x_1 = 5$ and $x_{n+1} = \sqrt{4+x_n}$, $n \geq 1$ converges to $\frac{1+\sqrt{17}}{2}$. Here, notice $x_1 = 5 > 0$. Now assume by way of induction that there is a $k \in \mathbb{N}$ with $x_k > 0$. Then

$$x_{k+1}=\sqrt{4+x_k}>\sqrt{4}>0.$$

So by induction, $\{x_n\}$ is bounded below by zero. Also, $x_1 = 5$ and $x_2 = 3 < x_1$. Again, by way of induction, assume there is a $k \in \mathbb{N}$, $k > 1$, with $x_k < x_{k-1}$. Then

$$x_{k+1}=\sqrt{4+x_k}<\sqrt{4+x_{k-1}}=x_k.$$

So by induction, $\{x_k\}$ is decreasing. Thus $\{x_k\}$ is bounded and decreasing, so it converges, say $x_n \to a$.

Now, since $x_n \to a$, $x_{n+1} \to a$. But since $x_{n+1} = \sqrt{4+x_n}$,

$$\lim_{n\to\infty}x_{n+1}=\lim_{n\to\infty}\sqrt{4+x_n}.$$

So $a = \sqrt{4+a}$. Since $x_n > 0$ for all $n \in \mathbb{N}$, $a \geq 0$ by Theorem 3.2.7. Thus $a = \frac{1+\sqrt{17}}{2}$.

Example 3.3.4. The sequence defined recursively by $x_1 = 1$ and $x_{n+1} = \frac{x_n^2+3}{2x_n}$ converges to $\sqrt{3}$. Notice $x_2 = 2 \geq \sqrt{3}$. Assume by way of induction that for $k \in \mathbb{N}$, $k > 1$, $x_k > \sqrt{3}$. Notice that the quadratic $x^2 - 2\sqrt{3}x + 3 > 0$ for all $x \in \mathbb{R}$ with $x \neq \sqrt{3}$. Then

$$x_{k+1} = \frac{x_k^2 + 3}{2x_k} > \sqrt{3}.$$

So by induction, for $n \geq 2$, $\{x_n\}$ is bounded below by $\sqrt{3}$. Now, notice $x_2 = 2$ and $x_3 = \frac{7}{4} < x_2$. Again, by way of induction, assume that for $k \in \mathbb{N}$, $k > 2$, $x_k < x_{k-1}$. Then

$$x_{k+1} = \frac{x_k^2 + 3}{2x_k} < \frac{x_k^2 + x_k^2}{2x_k} = x_k.$$

Thus $\{x_k\}$ is bounded and eventually decreasing. So x_n converges, say $x_n \to a$.

Now, since $x_n \to a$, $x_{n+1} \to a$. But since $x_{n+1} = \frac{x_n^2+3}{2x_n}$,

$$\lim_{n\to\infty} x_{n+1} = \lim_{n\to\infty} \frac{x_n^2 + 3}{2x_n}.$$

So $a = \frac{a^2+3}{2a}$. Since $x_n > \sqrt{3}$ for $n \geq 2$, $a \geq \sqrt{3} > 0$. Thus $a = \sqrt{3}$.

Exercises

3.3.1. Prove the sequence defined recursively by $x_1 = 2$, $x_{n+1} = \frac{4x_n+7}{10}$ converges, and determine the limit. [Solution, p. 334]

3.3.2. Prove the sequence defined recursively by $x_1 = 5$, $x_{n+1} = \frac{2x_n^2+5}{4x_n}$ converges, and determine the limit.

3.4. Subsequences

A sequence that does not converge can still have parts that do. For example, $\{(-1)^n\}$ diverges, but every even-numbered term is 1. Subsequences make this idea precise. The main result of the section is the Bolzano–Weierstrass Theorem. Every bounded sequence has a convergent subsequence.

Definition 3.4.1. If $\{n_k\}_{k=1}^{\infty}$ is a strictly increasing sequence of positive integers, then the sequence $\{x_{n_k}\}_{k=1}^{\infty}$ is called a *subsequence* of $\{x_n\}$.

Example 3.4.2. The sequence $\left\{\sin\left(\frac{n\pi}{2}\right)\right\}$ has subsequences

$$\{x_{2k}\} = \{0, 0, 0, 0, \dots\},$$

$$\{x_{2k-1}\} = \{1, -1, 1, -1, \dots\},$$

and

$$\{x_{4k+1}\} = \{1, 1, 1, 1, \dots\}.$$

Theorem 3.4.3. *Suppose $\{x_n\}$ is a sequence with $x_n \to a$. Then every subsequence of $\{x_n\}$ converges to a.*

PROOF. Let $\{x_{n_k}\}$ be a subsequence of $\{x_n\}$. Let $\varepsilon > 0$. Since $x_n \to a$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - a| < \varepsilon$. Now there exists a $K \in \mathbb{N}$ such that $n_k \geq N$ for all $k \geq K$. Thus for all $k \geq K$, $|x_{n_k} - a| < \varepsilon$. □

Theorem 3.4.4 (Nested Interval Theorem)**.** *Suppose $\{I_n\}$, $I_n = [a_n, b_n]$, is a sequence of closed intervals such that for all $n \in \mathbb{N}$, $I_{n+1} \subset I_n$. The sequences $\{a_n\}$ and $\{b_n\}$ converge, and $\bigcap_{n=1}^{\infty} I_n = [\alpha, \beta]$, where $a_n \to \alpha$ and $b_n \to \beta$. Moreover, if $b_n - a_n \to 0$, then $\alpha = \beta$ is the only common point to every interval I_n.*

PROOF. Since $I_{n+1} \subset I_n$, $a_n \leq a_{n+1}$ and $b_{n+1} \leq b_n$. So $\{a_n\}$ is an increasing sequence and $\{b_n\}$ is a decreasing sequence. Furthermore, since for all $n \in \mathbb{N}$, $a_1 \leq a_n \leq b_n \leq b_1$, $\{a_n\}$ and $\{b_n\}$ are bounded sequences. Thus $\{a_n\}$ and $\{b_n\}$ converge, say $a_n \to \alpha$ and $b_n \to \beta$.

By Theorem 3.2.7, $\alpha \leq \beta$, and since $\{a_n\}$ is increasing and $\{b_n\}$ is decreasing, for all $n \in \mathbb{N}$, $a_n \leq \alpha \leq \beta \leq b_n$. So $[\alpha, \beta] \subset I_n$ for all $n \in \mathbb{N}$. Thus $[\alpha, \beta] \subset \bigcap_{n=1}^{\infty} I_n$. If $x \in \bigcap_{n=1}^{\infty} I_n$, then for all $n \in \mathbb{N}$, $a_n \leq x$ and $x \leq b_n$. So by Corollary 3.2.9, $\alpha \leq x \leq \beta$. So $x \in [\alpha, \beta]$ and $\bigcap_{n=1}^{\infty} I_n \subset [\alpha, \beta]$. Therefore $[\alpha, \beta] = \bigcap_{n=1}^{\infty} I_n$.

If

$$\beta - \alpha = \lim_{n \to \infty} (b_n - a_n) = 0,$$

then $\alpha = \beta$. So $\bigcap_{n=1}^{\infty} I_n = [\alpha, \beta] = \{\alpha\}$. □

Both conditions on the intervals matter. If the intervals are not closed, the intersection can be empty. For example, $\bigcap_{n=1}^{\infty} \left(0, \frac{1}{n}\right) = \emptyset$ by the Archimedean Principle. If the intervals are not bounded, the intersection can also be empty. For example, $\bigcap_{n=1}^{\infty} [n, \infty) = \emptyset$.

Proof Technique (Bisection). The proof of the Bolzano–Weierstrass Theorem cuts an interval in half again and again. At each step it keeps a half that still contains infinitely many terms of the sequence. The lengths of the intervals go to 0. The Nested Interval Theorem then gives one point common to all the intervals. This technique is called *bisection*. It appears again in later chapters. Figure 3.2 shows the first few steps.

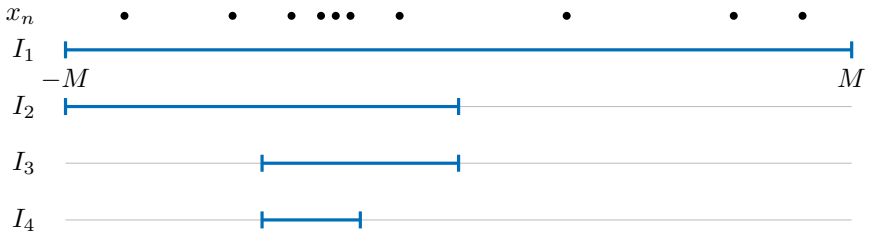


FIGURE 3.2. Bisection. Each interval is a half of the one above it that still contains infinitely many terms of the sequence.

Theorem 3.4.5 (Bolzano–Weierstrass Theorem)**.** *Every bounded sequence in $\mathbb{R}$ has a convergent subsequence.*

PROOF. Let $\{x_n\}$ be a bounded sequence in $\mathbb{R}$. So there exists an $M > 0$ such that $|x_n| \leq M$ for all $n \in \mathbb{N}$. Let $I_1 = [-M, M] = [a_1, b_1]$ and define $c_1 = \frac{a_1+b_1}{2}$.

Since the sequence has infinitely many terms, it follows that there are infinitely many terms contained in either $[a_1, c_1]$ or $[c_1, b_1]$ (or both). Choose one that contains infinitely many terms and denote it by $I_2 = [a_2, b_2]$. In general, for $k \in \mathbb{N}$, $I_k = [a_k, b_k]$, define $c_k = \frac{a_k+b_k}{2}$ and define I_{k+1} to be $[a_k, c_k]$ if it contains infinitely many terms of $\{x_n\}$ or $[c_k, b_k]$ if it contains infinitely many terms of $\{x_n\}$ and $[a_k, c_k]$ does not. Notice $I_{k+1} \subset I_k$.

This construction gives a nested sequence of intervals $\{I_n\}$. By the Nested Interval Theorem (Theorem 3.4.4), the sequences $\{a_n\}$ and $\{b_n\}$ converge, say $a_n \to \alpha$ and $b_n \to \beta$.

Also,

$$b_n - a_n = \frac{b_{n-1} - a_{n-1}}{2} = \frac{b_1 - a_1}{2^{n-1}} = \frac{2M}{2^{n-1}} \to 0.$$

So $\bigcap_{n=1}^{\infty} I_n = \{\alpha\}$.

Now, choose an $x \in I_1 \cap \{x_n : n \in \mathbb{N}\}$ and label this x as x_{n_1}. Next, choose an $x \in I_2 \cap \{x_n : n \in \mathbb{N},\ n > n_1\}$. In general, for $k \in \mathbb{N}$, choose an $x \in I_k \cap \{x_n : n \in \mathbb{N},\ n > n_{k-1}\}$ and label this x as x_{n_k}. So $\{x_{n_k}\}$ is a subsequence of $\{x_n\}$. For each $k \in \mathbb{N}$, $a_k \leq x_{n_k} \leq b_k$. So by the Squeeze Theorem (Theorem 3.2.10), $x_{n_k} \to \alpha$. □

Boundedness is needed. The sequence $\{n\}$ has no convergent subsequence. The theorem is also false in $\mathbb{Q}$. A sequence of rational numbers converging to $\sqrt{2}$ is bounded. By Theorem 3.4.3, each of its subsequences converges to $\sqrt{2}$, so none converges to a rational number.

Corollary 3.4.6. *Suppose $\{x_n\}$ is a sequence contained in $[a, b]$ where $a, b \in \mathbb{R}$, $a \leq b$. Then $\{x_n\}$ has a subsequence converging to a point in $[a, b]$.*

Exercises

3.4.1. Suppose that $\{x_n\}$ is a sequence that is not bounded above. Show that there is a subsequence $\{x_{n_k}\}$ such that $x_{n_k} > k$ for all $k \in \mathbb{N}$. [Hint, p. 334]

3.4.2. Let $\varepsilon > 0$ and $a \in \mathbb{R}$. Suppose that a sequence $\{x_n\}$ satisfies $|x_n - a| \geq \varepsilon$ for all $n \geq 1$. Prove that no subsequence of $\{x_n\}$ can converge to a.

3.5. Cauchy Sequences

To show a sequence converges from the definition, the limit has to be known first. Cauchy's idea was to check instead that the terms get close to each other. In $\mathbb{R}$ this is enough. In $\mathbb{Q}$ it is not, which is one more sign of the hole at $\sqrt{2}$.

Definition 3.5.1. A sequence $\{x_n\}$ is a *Cauchy sequence* if for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $m, n \geq N$, $|x_n - x_m| < \varepsilon$.

Theorem 3.5.2 (Cauchy Criterion)**.** *A sequence $\{x_n\}$ of real numbers is Cauchy if and only if it converges.*

PROOF. ($\Leftarrow$) Let $\{x_n\}$ be a convergent sequence with $x_n \to a$. Let $\varepsilon > 0$. So there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - a| < \frac{\varepsilon}{2}$. So for all $m, n \geq N$,

$$\begin{aligned} |x_n - x_m| &= |x_n - a + a - x_m| \\ &\leq |x_n - a| + |x_m - a| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

So $\{x_n\}$ is Cauchy.

($\Rightarrow$) Suppose $\{x_n\}$ is Cauchy. So there exists an $N \in \mathbb{N}$ such that for all $m, n \geq N$, $|x_n - x_m| < 1$. Therefore, $|x_n| < 1 + |x_N|$ for all $n \geq N$. Thus, if

$$M = \max\{|x_1|, |x_2|, \ldots, |x_{N-1}|, 1 + |x_N|\},$$

$|x_n| \leq M$ for all $n \in \mathbb{N}$. So $\{x_n\}$ is bounded. Therefore, by the Bolzano–Weierstrass Theorem (Theorem 3.4.5), there exists a convergent subsequence $\{x_{n_k}\}$ with $x_{n_k} \to a$.

Let $\varepsilon > 0$. So there exists an $N_1 \in \mathbb{N}$ such that for all $m, n \geq N_1$, $|x_n - x_m| < \frac{\varepsilon}{2}$. Since $x_{n_k} \to a$, there exists a $K \in \mathbb{N}$ with $n_K > N_1$ such that for all $k \geq K$, $|x_{n_k} - a| < \frac{\varepsilon}{2}$. Then for all $n \geq N_1$,

$$\begin{aligned} |x_n - a| &= |x_n - x_{n_K} + x_{n_K} - a| \\ &\leq |x_n - x_{n_K}| + |x_{n_K} - a| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

So $x_n \to a$. □

Appendix B uses Cauchy sequences of rational numbers to build $\mathbb{R}$ out of $\mathbb{Q}$.

Proof Technique (Proving Convergence Without Knowing the Limit). To use the definition of convergence, the limit a must be known. The Cauchy Criterion avoids this. It only compares terms of the sequence with each other. So it can prove a sequence converges even when the limit is unknown. Exercise 3.5.1 uses this idea.

Exercises

3.5.1. Show the sequence $\{x_n\}$ defined by

$$x_n = \sum_{k=1}^{n} \frac{1}{k!} = 1 + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \cdots + \frac{1}{n!}$$

converges by showing it is a Cauchy sequence. [Hint, p. 335]

3.5.2. Let $x_n = \sum_{k=1}^{n} \frac{1}{k}$. Prove that $x_{2n} - x_n \geq \frac{1}{2}$ for all $n \in \mathbb{N}$. Conclude that $\{x_n\}$ is not a Cauchy sequence, so it diverges. [Solution, p. 335]

3.6. Limit Superior and Limit Inferior of a Sequence

A bounded sequence does not have to converge. It still has a largest and a smallest cluster point. These are the limit superior and the limit inferior. They exist for every sequence, if $\pm\infty$ are allowed, and they describe what the sequence does in the long run. The sequence converges exactly when the two are equal and finite.

Definition 3.6.1. For a real sequence $\{x_n\}$, $a \in \mathbb{R}$ is a *cluster point* of $\{x_n\}$ if there is a subsequence of $\{x_n\}$ converging to a.

Example 3.6.2. $\{(-1)^n\}$ has cluster points 1 and -1.

Example 3.6.3. $\left\{\sin\left(\frac{n\pi}{4}\right)\right\}$ has cluster points 0, 1, -1, $\frac{\sqrt{2}}{2}$, and $-\frac{\sqrt{2}}{2}$.

Example 3.6.4. $\left\{\frac{1}{n}\sin\left(\frac{n\pi}{2}\right)\right\}$ has limit 0 and hence its only cluster point is 0.

Definition 3.6.5. A sequence $\{x_n\}$ *diverges to* ∞ if for all $M > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n > M$. $\{x_n\}$ *diverges to* $-\infty$ if for all $M < 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n < M$. In these cases, write $\lim_{n\to\infty} x_n = \infty$ or $x_n \to \infty$, and $\lim_{n\to\infty} x_n = -\infty$ or $x_n \to -\infty$.

Definition 3.6.6. ∞ is a *cluster point* of $\{x_n\}$ if for all $M > 0$ and for all $N \in \mathbb{N}$, there exists an $n \geq N$ such that $x_n > M$. $-\infty$ is a cluster point of $\{x_n\}$ if for all $M < 0$ and for all $N \in \mathbb{N}$, there exists an $n \geq N$ such that $x_n < M$.

Remark 3.6.7. If a set $E \subset \mathbb{R}$ is not bounded above, write $\sup E = \infty$. If E is not bounded below, write $\inf E = -\infty$. With this convention, $\sup\{x_k : k \geq n\}$ always makes sense. Let $b_n = \sup\{x_k : k \geq n\}$. Then $\{b_n\}$ is decreasing, so $\lim_{n\to\infty} b_n$ exists. It is a real number if $\{b_n\}$ is bounded below, by Theorem 3.2.13. It is $-\infty$ if $\{b_n\}$ is not bounded below. It is ∞ if every b_n is ∞. The same holds for $\inf\{x_k : k \geq n\}$.

Definition 3.6.8. If $\{x_n\}$ is a sequence of real numbers, then the *limit superior* of $\{x_n\}$ is

$$\limsup x_n = \lim_{n\to\infty} (\sup\{x_k : k \geq n\}) = \inf_{n\in\mathbb{N}} (\sup\{x_k : k \geq n\}).$$

The *limit inferior* of $\{x_n\}$ is

$$\liminf x_n = \lim_{n\to\infty} (\inf\{x_k : k \geq n\}) = \sup_{n\in\mathbb{N}} (\inf\{x_k : k \geq n\}).$$

Example 3.6.9. Let $x_n = (-1)^n\left(1 + \frac{1}{n}\right)$. The terms with k even are positive and decrease toward 1. The terms with k odd are negative and increase toward -1. So for each n, $\sup\{x_k : k \geq n\} = 1 + \frac{1}{k}$, where k is the first even number with $k \geq n$. These suprema decrease to 1. So $\limsup x_n = 1$. In the same way, $\liminf x_n = -1$. The sequence diverges, and its cluster points are 1 and -1.

Theorem 3.6.10. *Let $\{x_n\}$ be a sequence of real numbers. Then*

(1) $\limsup(-x_n) = -\liminf x_n$*; and*
(2) $\liminf x_n \leq \limsup x_n$.

Proof. (1) Since for all $n \in \mathbb{N}$, $\sup\{-x_k : k \geq n\} = -\inf\{x_k : k \geq n\}$, $\limsup(-x_n) = -\liminf x_n$.

(2) For all $n \in \mathbb{N}$, $\inf\{x_k : k \geq n\} \leq \sup\{x_k : k \geq n\}$, implying $\liminf x_n \leq \limsup x_n$. □

Theorem 3.6.11. *Let $\{x_n\}$ be a sequence of real numbers.* $\limsup x_n = a \in \mathbb{R}$ *if and only if for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n < a + \varepsilon$ and for all $\varepsilon > 0$ and for all $N \in \mathbb{N}$, there exists an $n \geq N$ such that $x_n > a - \varepsilon$. Similarly,* $\liminf x_n = a \in \mathbb{R}$ *if and only if for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n > a - \varepsilon$ and for all $\varepsilon > 0$ and for all $N \in \mathbb{N}$, there exists an $n \geq N$ such that $x_n < a + \varepsilon$.*

PROOF. ($\Rightarrow$) Suppose $a \in \mathbb{R}$ with $a = \limsup x_n = \lim_{n\to\infty} b_n$, where $b_n = \sup\{x_k : k \geq n\}$. Let $\varepsilon > 0$. Since $\lim_{n\to\infty} b_n = a$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $b_n < a + \varepsilon$. Since for each $n \geq N$, $x_n \leq b_n$, $x_n < a + \varepsilon$ for all $n \geq N$.

Let $\varepsilon > 0$, $N \in \mathbb{N}$. Since $\lim_{n\to\infty} b_n = a$ and $\{b_n\}$ is decreasing, $b_n \geq a$ for all $n \in \mathbb{N}$. In particular, $b_N \geq a$. But since $b_N = \sup\{x_k : k \geq N\}$, there exists an $n \geq N$ such that $x_n > b_N - \varepsilon \geq a - \varepsilon$.

($\Leftarrow$) Let $\varepsilon > 0$. Again, define $b_n = \sup\{x_k : k \geq n\}$. Since there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n < a + \varepsilon$, $b_N \leq a + \varepsilon$. Since $\{b_n\}$ is decreasing, $b_n \leq a + \varepsilon$ for all $n \geq N$. So $\limsup x_n \leq a + \varepsilon$ for arbitrary ε, implying $\limsup x_n \leq a$. Assume by way of contradiction that $\limsup x_n = a' < a$. Choose $\varepsilon > 0$ such that $a' < a - 2\varepsilon$. By the proof of ($\Rightarrow$), there exists an $N \in \mathbb{N}$ such that $x_n < a' + \varepsilon < a - \varepsilon$ for all $n \geq N$, contradicting the hypothesis that for all $\varepsilon > 0$ and for all $N \in \mathbb{N}$, there exists an $n \geq N$ such that $x_n > a - \varepsilon$. If $\limsup x_n = -\infty$, then $b_n < a - \varepsilon$ for all large n, and the same contradiction follows. So $\limsup x_n = a$. □

Theorem 3.6.12. *Let $\{x_n\}$ be a sequence of real numbers.*

(1) $\limsup x_n = \infty$ *if and only if ∞ is a cluster point of $\{x_n\}$.*
(2) $\limsup x_n = -\infty$ *if and only if* $\lim_{n\to\infty} x_n = -\infty$.
(3) $\liminf x_n = -\infty$ *if and only if $-\infty$ is a cluster point of $\{x_n\}$.*
(4) $\liminf x_n = \infty$ *if and only if* $\lim_{n\to\infty} x_n = \infty$.

PROOF. (1) ($\Rightarrow$) Suppose $\limsup x_n = \infty$. Then, by definition, $\lim_{n\to\infty} b_n = \infty$, where $b_n = \sup\{x_k : k \geq n\}$. Let $M > 0$, $N \in \mathbb{N}$. Then there exists a $K \in \mathbb{N}$, $K \geq N$, such that for all $n \geq K$, $b_n > M + 1$. In particular, $b_K > M + 1$. Therefore $\sup\{x_k : k \geq K\} > M + 1$. So there exists an $n_0 \geq K \geq N$ with $x_{n_0} > M$. So ∞ is a cluster point of $\{x_n\}$.

($\Leftarrow$) Suppose ∞ is a cluster point of $\{x_n\}$. Let $M > 0$, $N \in \mathbb{N}$. Then there exists a $n_0 \geq N$ such that $x_{n_0} > M$. Therefore $\sup\{x_k : k \geq N\} > M$. So $b_N > M$. Since $N \in \mathbb{N}$ was arbitrary, $b_n > M$ for all $n \in \mathbb{N}$. So $\lim_{n\to\infty} b_n = \infty$, or $\limsup x_n = \infty$.

(2) ($\Rightarrow$) Suppose $\limsup x_n = -\infty$. Then, by definition, $\lim_{n\to\infty} b_n = -\infty$, where $b_n = \sup\{x_k : k \geq n\}$. Let $M < 0$. There exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $b_n < M$. In particular, $b_N < M$. So $\sup\{x_k : k \geq N\} < M$. So M is an upper bound of $\{x_k : k \geq N\}$. Therefore, $x_n < M$ for all $n \geq N$, implying $\lim_{n\to\infty} x_n = -\infty$.

($\Leftarrow$) Suppose $\lim_{n\to\infty} x_n = -\infty$. Let $M < 0$. So there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n < M$. Thus $\sup\{x_k : k \geq N\} \leq M$. Therefore

for all $n \geq N$, $\sup\{x_k : k \geq n\} \leq M$, implying for all $n \geq N$, $b_n \leq M$. So $\lim_{n\to\infty} b_n = -\infty$. Therefore $\limsup x_n = -\infty$.

(3) This proof is similar to the proof of (1).

(4) This proof is similar to the proof of (2). □

Corollary 3.6.13. *Let $\{x_n\}$ be a sequence of real numbers.* $\limsup x_n$ *and* $\liminf x_n$ *are the largest and smallest cluster points, respectively, of $\{x_n\}$.*

PROOF. See Exercise 3.6.1. □

Corollary 3.6.14. *Let $\{x_n\}$ be a sequence of real numbers. $\{x_n\}$ converges to $a \in \mathbb{R}$ or diverges to $\pm\infty$ if and only if* $\limsup x_n = \liminf x_n = a$ *or* $\limsup x_n = \liminf x_n = \pm\infty$.

PROOF. If $a \in \mathbb{R}$, this follows directly from Theorem 3.6.11. For the $\pm\infty$ case, this follows directly from Theorem 3.6.12. □

Exercises

3.6.1. Let $\{x_n\}$ be a sequence of real numbers. Prove $\limsup x_n$ and $\liminf x_n$ are the largest and smallest cluster points, respectively, of $\{x_n\}$. Make sure to prove all the cases. [Solution, p. 335]

3.6.2. True or false? If $\{x_n\}$ is a sequence in $\mathbb{R}$ with $x_n \geq 0$ for all $n \in \mathbb{N}$ and $\limsup x_n = 0$, then $x_n \to 0$. If true, prove it. If false, provide a counterexample.

Notes

The Cauchy Criterion is named for Augustin-Louis Cauchy, whose *Cours d'analyse* of 1821 made limits the foundation of calculus [**15**]. Bernard Bolzano stated the same criterion in 1817 [**5**]. The Bolzano–Weierstrass Theorem is named for Bolzano and for Karl Weierstrass. Bolzano proved it in 1817, as a step toward the Intermediate Value Theorem. Weierstrass proved it again about fifty years later, in his lectures in Berlin, and made it a basic tool of analysis.

The recursion $x_{n+1} = \frac{1}{2}\left(x_n + \frac{3}{x_n}\right)$ in Example 3.3.4 is very old. For square roots it is often called Heron's method, after Heron of Alexandria, who used it in the first century. It is a special case of Newton's method. Isaac Newton described his method in *De analysi*, written in 1669 and published in 1711 [**55**]. Joseph Raphson published a simpler version in 1690 [**58**].

CHAPTER 4

Topological Properties of the Real Line

Chapter 3 asked which points a sequence gets close to. This chapter asks the same question about sets. An open set is a set in which every point has some room around it. A closed set is a set that contains every point it gets close to. These ideas are called the *topology* of $\mathbb{R}$.

The most important idea in the chapter is compactness. A set is compact if every open cover of it can be cut down to a finite one. That sounds abstract. What it does is turn a fact that holds near each point into a fact that holds on the whole set. The main result, Theorem 4.2.15, says a subset of $\mathbb{R}$ is compact exactly when it is closed and bounded. Chapter 6 uses compactness to prove that a continuous function on $[a, b]$ is uniformly continuous.

4.1. Open and Closed Sets

Every point of an open interval (a, b) has a little room on both sides that stays inside the interval. An open set is any set with this property. A closed set is the complement of an open set.

Definition 4.1.1. A set $U \subset \mathbb{R}$ is said to be *open* if for all $x \in U$, there exists an $\varepsilon(x) > 0$ such that $(x - \varepsilon, x + \varepsilon) \subset U$.

The notation $\varepsilon(x)$ is a reminder that ε may depend on x. Points near the edge of U usually need a smaller ε.

Definition 4.1.2. Let $E \subset \mathbb{R}$. A point $x \in E$ is said to be an *interior point* of E if there exists an $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \subset E$. The set of interior points of E is denoted E°.

Theorem 4.1.3. *A set $U \subset \mathbb{R}$ is open if and only if $U = U^\circ$.*

Proof. See Exercise 4.1.1. □

Proof Technique (Proving a Set Is Open). To prove a set U is open, start with an arbitrary $x \in U$. Then find an $\varepsilon > 0$ that works for that x. The choice of ε is allowed to depend on x. A good first guess is the distance from x to the nearest point outside U, or a fraction of it. Then check that every y with $|y - x| < \varepsilon$ is in U. Example 4.1.4 does this for $(0, 1)$. There, ε is half the distance from x to the nearer endpoint.

Example 4.1.4. The set $(0,1)$ is open. By letting $x \in (0,1)$ and defining $\varepsilon = \min\left\{\frac{x}{2}, \frac{1-x}{2}\right\}$,

$$x - \varepsilon \geq x - \frac{x}{2} = \frac{x}{2} > 0,$$

and

$$x + \varepsilon \leq x + \frac{1-x}{2} = \frac{1+x}{2} < 1.$$

So $(x-\varepsilon, x+\varepsilon) \subset (0,1)$ and $(0,1)$ is open. See Figure 4.1.

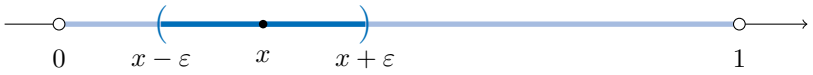


FIGURE 4.1. The interval $(x-\varepsilon, x+\varepsilon)$ in Example 4.1.4, with $x = 0.3$ and $\varepsilon = \min\{0.15, 0.35\} = 0.15$. It stays inside $(0,1)$.

Example 4.1.5. If $E = (0,42]$, $E^\circ = (0,42)$. The point 42 is not an interior point, since every interval $(42-\varepsilon, 42+\varepsilon)$ contains numbers greater than 42.

Example 4.1.6. If $E = \left\{\frac{1}{n} : n \in \mathbb{N}\right\}$, $E^\circ = \emptyset$. Every interval $(x-\varepsilon, x+\varepsilon)$ contains an irrational number by Exercise 1.8.5, and E has no irrational numbers. So no point of E is an interior point.

Theorem 4.1.7. *Every open interval in $\mathbb{R}$ is an open subset of $\mathbb{R}$.*

PROOF. See Exercise 4.1.2. □

By Definition 1.8.6, (a,b) is called an open interval only when $a < b$ are real numbers. The sets (a,∞), $(-\infty,b)$ and $(-\infty,\infty) = \mathbb{R}$ are called *unbounded open intervals*. They are also open sets. See Exercise 4.1.4 and Example 4.1.16.

Families of sets $\{A_\alpha\}_{\alpha\in\Lambda}$ and index sets were defined in Definition 1.1.12. The index set Λ can be finite, countable or uncountable.

Example 4.1.8. An enumeration of a set S gives an index set $J \subset \mathbb{N}$.

Example 4.1.9. Any countable set can be indexed by $\mathbb{N}$.

Example 4.1.10. Uncountable sets can also be used to index. For example, with $U_x = (x-1, x+1)$, $\mathbb{R}$ is the index set of the collection $\{U_x\}_{x\in\mathbb{R}}$.

Theorem 4.1.11. *(1) If $\{U_\alpha\}_{\alpha\in\Lambda}$ is a collection of open subsets of $\mathbb{R}$, then $\bigcup_{\alpha\in\Lambda} U_\alpha$ is open.*

(2) If $\{U_1, \ldots, U_n\}$ is a finite collection of open subsets of $\mathbb{R}$, then $\bigcap_{i=1}^n U_i$ is open.

PROOF. (1) Let $x \in \bigcup_{\alpha\in\Lambda} U_\alpha$. So there exists an $\alpha_0 \in \Lambda$ such that $x \in U_{\alpha_0}$. Since U_{α_0} is open, there exists an $\varepsilon > 0$ such that

$$(x-\varepsilon, x+\varepsilon) \subset U_{\alpha_0} \subset \bigcup_{\alpha\in\Lambda} U_\alpha.$$

So $\bigcup_{\alpha\in\Lambda} U_\alpha$ is open.

(2) Let $x \in \bigcap_{i=1}^{n} U_i$. So $x \in U_i$ for $i = 1, \ldots, n$. Since each U_i is open, there exists an $\varepsilon_i > 0$ such that $(x - \varepsilon_i, x + \varepsilon_i) \subset U_i$ for $i = 1, \ldots, n$. Set $\varepsilon = \min\{\varepsilon_1, \ldots, \varepsilon_n\} > 0$. Then $(x - \varepsilon, x + \varepsilon) \subset U_i$ for $i = 1, \ldots, n$, implying $(x - \varepsilon, x + \varepsilon) \subset \bigcap_{i=1}^{n} U_i$. So $\bigcap_{i=1}^{n} U_i$ is open. □

Part (2) needs the collection to be finite. The proof takes the minimum of $\varepsilon_1, \ldots, \varepsilon_n$. The minimum of finitely many positive numbers is positive, but infinitely many positive numbers can have infimum 0. Example 4.1.18 gives infinitely many open sets whose intersection is not open.

Theorem 4.1.12. *Every nonempty open set of real numbers is the union of an at most countable collection of disjoint open intervals, bounded or unbounded.*

Proof. Let $U \subset \mathbb{R}$ be a nonempty open set, and let $x \in U$. Since U is open, there exists an $\varepsilon > 0$ with $(x - \varepsilon, x + \varepsilon) \subset U$. Let

$$a_x = \inf\{z < x : (z, x) \subset U\} \quad \text{and} \quad b_x = \sup\{y > x : (x, y) \subset U\}.$$

These sets are nonempty, since $x - \varepsilon$ is in the first one and $x + \varepsilon$ is in the second one. If the first set is not bounded below, $a_x = -\infty$. If the second set is not bounded above, $b_x = \infty$. (See Remark 3.6.7.) So $a_x \leq x - \varepsilon < x < x + \varepsilon \leq b_x$. Let $I_x = (a_x, b_x)$. It is an open interval, possibly unbounded, and $x \in I_x$.

First, $I_x \subset U$. Let $w \in I_x$. If $w = x$, then $w \in U$. Suppose $x < w < b_x$. Since w is less than the supremum b_x, w is not an upper bound of $\{y > x : (x, y) \subset U\}$. So there exists a $y > w$ with $(x, y) \subset U$. Thus $w \in (x, y) \subset U$. The case $a_x < w < x$ is similar.

Second, if b_x is a real number, then $b_x \notin U$. Suppose $b_x \in U$. Since U is open, there exists a $\delta > 0$ with $(b_x - \delta, b_x + \delta) \subset U$. Since $(x, b_x) \subset I_x \subset U$, this gives $(x, b_x + \delta) \subset U$. So $b_x + \delta$ is in the set whose supremum is b_x, a contradiction. In the same way, if a_x is a real number, then $a_x \notin U$.

Third, $U = \bigcup_{x \in U} I_x$. Each $x \in U$ lies in I_x, and each I_x is a subset of U.

Fourth, any two of the intervals I_x are either equal or disjoint. Suppose $I_x = (a, b)$ and $I_y = (c, d)$ have a point w in common. Suppose $a < c$. Then c is a real number and $a < c < w < b$. So $c \in I_x \subset U$. But $c \notin U$ by the second step, a contradiction. So $c \leq a$. Switching the roles of I_x and I_y gives $a \leq c$. So $a = c$. The same argument gives $b = d$. Thus $I_x = I_y$.

Last, the collection of distinct intervals I_x is at most countable. Each of them contains a rational number by Theorem 1.8.11. Choose one rational number in each of them. Since distinct intervals are disjoint, distinct intervals get distinct rational numbers. So the collection is in one-to-one correspondence with a subset of $\mathbb{Q}$. Since $\mathbb{Q}$ is countable by Corollary 2.2.7, the collection is at most countable by Theorem 2.2.4. □

Corollary 4.1.13. *Every nonempty open set of real numbers is the union of an at most countable collection of open intervals.*

Proof. From the proof of Theorem 4.1.12, each open set is the disjoint union of open intervals. But an open interval $(a, \infty) = \bigcup_{n \in \mathbb{Z}, n > a} (a, n)$ and an open interval $(-\infty, b) = \bigcup_{n \in \mathbb{Z}, n < b} (n, b)$. Also, $(-\infty, \infty) = \bigcup_{n \in \mathbb{N}} (-n, n)$. So each open set is the

union of a countable collection of open intervals. The collection is at most countable by Theorem 2.2.6. □

Theorem 4.1.14 (Lindelöf)**.** *Let $\{U_\alpha\}_{\alpha\in\Lambda}$ be a collection of open sets of real numbers. Then there is a countable subcollection $\{U_n\}_{n\in\mathbb{N}}$ of $\{U_\alpha\}_{\alpha\in\Lambda}$ such that $\bigcup_{\alpha\in\Lambda} U_\alpha = \bigcup_{n\in\mathbb{N}} U_n$.*

PROOF. Let $U = \bigcup_{\alpha\in\Lambda} U_\alpha$. If $U = \emptyset$, then every U_α is empty, and any one of them, repeated, works. So assume $U \neq \emptyset$.

Let $\mathcal{J}$ be the collection of open intervals (p, q) with $p, q \in \mathbb{Q}$ and $p < q$. Each interval in $\mathcal{J}$ is determined by its endpoints. So sending the interval (p, q) to the ordered pair (p, q) gives a one-to-one map from $\mathcal{J}$ into $\mathbb{Q} \times \mathbb{Q}$. Since $\mathbb{Q} \sim \mathbb{N}$ by Corollary 2.2.7, $\mathbb{Q} \times \mathbb{Q} \sim \mathbb{N} \times \mathbb{N}$, which is countable by Theorem 2.2.1. So $\mathcal{J}$ is at most countable by Theorem 2.2.4.

Let $\mathcal{J}_0$ be the collection of $J \in \mathcal{J}$ such that $J \subset U_\alpha$ for some $\alpha \in \Lambda$. For each $J \in \mathcal{J}_0$, choose one $\alpha_J \in \Lambda$ with $J \subset U_{\alpha_J}$. Then

$$\bigcup_{J\in\mathcal{J}_0} U_{\alpha_J} \subset U.$$

Now let $x \in U$. Then $x \in U_\alpha$ for some $\alpha \in \Lambda$. Since U_α is open, there exists an $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \subset U_\alpha$. By Theorem 1.8.11, there exist $p, q \in \mathbb{Q}$ with $x - \varepsilon < p < x < q < x + \varepsilon$. Then $J = (p, q) \subset U_\alpha$, so $J \in \mathcal{J}_0$. Thus $x \in J \subset U_{\alpha_J}$. So

$$U = \bigcup_{J\in\mathcal{J}_0} U_{\alpha_J}.$$

The collection $\mathcal{J}_0$ is nonempty and at most countable. So the sets U_{α_J}, $J \in \mathcal{J}_0$, can be listed as $U_1, U_2, U_3, \ldots$, repeating sets if there are only finitely many. Then $U = \bigcup_{n\in\mathbb{N}} U_n$. □

Definition 4.1.15. A set $F \subset \mathbb{R}$ is said to be *closed* if $F^c = \mathbb{R} \setminus F$ is open.

The empty set satisfies the definition of an open set because the condition is vacuously true. See the Vacuous Truth box in Section 1.1.

Example 4.1.16. $\mathbb{R}$ and $\emptyset$ are both open and closed. Let $x \in \mathbb{R}$. Then $(x-1, x+1) \subset \mathbb{R}$. So $\mathbb{R}$ is open. So $\emptyset = \mathbb{R}^c$ is closed.

Now, let $x \in \emptyset$. So for $\varepsilon = 1$, $(x - 1, x + 1) \subset \emptyset$. This is vacuously true, since there is no $x \in \emptyset$. So $\emptyset$ is open. Thus $\mathbb{R} = \emptyset^c$ is closed.

"Closed" does not mean "not open." The sets $\mathbb{R}$ and $\emptyset$ are both. The interval $[0, 1)$ is neither. It is not open, since every interval $(-\varepsilon, \varepsilon)$ around 0 contains negative numbers. It is not closed, since its complement contains 1 and every interval $(1 - \varepsilon, 1 + \varepsilon)$ contains points of $[0, 1)$.

Theorem 4.1.17. *(1) If $\{F_\alpha\}_{\alpha\in\Lambda}$ is a collection of closed subsets of $\mathbb{R}$, then $\bigcap_{\alpha\in\Lambda} F_\alpha$ is closed.*

(2) If $\{F_1, \ldots, F_n\}$ is a finite collection of closed subsets of $\mathbb{R}$, then $\bigcup_{i=1}^n F_i$ is closed.

PROOF. This follows directly from Theorem 4.1.11 and De Morgan's Laws (Theorem 1.1.14) and is an exercise. □

Proof Technique (Using Complements). Every statement about closed sets is a statement about open sets in disguise. To prove something about a closed set F, look at the open set F^c. De Morgan's Laws turn the complement of a union into an intersection of complements. So facts about unions of open sets become facts about intersections of closed sets. Exercise 4.1.5 uses this idea to prove Theorem 4.1.17.

Example 4.1.18. Let $U_n = \left(-\frac{1}{n}, \frac{1}{n}\right)$. Then $\bigcap_{n=1}^{\infty} U_n = \{0\}$ is closed. To see the intersection is $\{0\}$, notice $0 \in U_n$ for all n. If $x \neq 0$, then by the Archimedean Principle (Theorem 1.8.8) there is an $n \in \mathbb{N}$ with $\frac{1}{n} < |x|$, and $x \notin U_n$. So an intersection of infinitely many open sets need not be open.

Example 4.1.19. Let $G_n = \left(0, 1 + \frac{1}{n}\right)$. Then $\bigcap_{n=1}^{\infty} G_n = (0, 1]$. This set is neither open nor closed.

Definition 4.1.20. Let $E \subset \mathbb{R}$.

(1) A point $a \in \mathbb{R}$ is called a *limit point* of E if for all $\varepsilon > 0$, there exists an $x \in E$, $x \neq a$, with $|x - a| < \varepsilon$.
(2) A point $a \in E$ is called an *isolated point* of E if it is not a limit point.

Example 4.1.21. $(0, 1)$ has limit points $[0, 1]$.

Example 4.1.22. $\left\{\frac{1}{n} : n \in \mathbb{N}\right\}$ has limit point 0. Every point of this set is an isolated point.

A limit point of E does not have to be in E, as both examples show. A limit point of a set is also not the same as a cluster point of a sequence (Definition 3.6.1). The constant sequence $x_n = 1$ has cluster point 1. The set $\{x_n : n \in \mathbb{N}\} = \{1\}$ has no limit points.

Theorem 4.1.23. *Let $E \subset \mathbb{R}$. The point $a \in \mathbb{R}$ is a limit point of E if and only if there exists a sequence $\{x_n\}$ in E with $x_n \neq a$ for all $n \in \mathbb{N}$ such that $\lim_{n\to\infty} x_n = a$.*

Proof. ($\Rightarrow$) Let a be a limit point of E. Then for all $n \in \mathbb{N}$, there exists an $x_n \in E$ such that $x_n \neq a$ and $|x_n - a| < \frac{1}{n}$. Then by the Squeeze Theorem (Theorem 3.2.10), $x_n \to a$.

($\Leftarrow$) Suppose there exists a sequence $\{x_n\}$ in E with $x_n \neq a$ for all $n \in \mathbb{N}$ such that $\lim_{n\to\infty} x_n = a$. Let $\varepsilon > 0$. Then there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - a| < \varepsilon$. In particular, $x_N \in E$, $x_N \neq a$, and $|x_N - a| < \varepsilon$. So a is a limit point of E. □

Theorem 4.1.24. *The set $F \subset \mathbb{R}$ is closed if and only if F contains all of its limit points.*

Proof. ($\Rightarrow$) Let F be closed. Then F^c is open. So if $x \in F^c$, there exists an $\varepsilon > 0$ such that $(x - \varepsilon, x + \varepsilon) \subset F^c$. So $(x - \varepsilon, x + \varepsilon) \cap F = \emptyset$, and therefore x is not a limit point of F. So F contains all of its limit points.

($\Leftarrow$) Let $F \subset \mathbb{R}$ be a set that contains all of its limit points. Let $x \in F^c$. Since F contains all of its limit points, x is not a limit point of F. So there exists an $\varepsilon > 0$

such that $(x-\varepsilon, x+\varepsilon) \cap F = \emptyset$. So $(x-\varepsilon, x+\varepsilon) \subset F^c$ and F^c is open. So F is closed. □

Definition 4.1.25. For $E \subset \mathbb{R}$, E' denotes the set of limit points of E. The *closure* of E, $\overline{E}$, is

$$\overline{E} = E \cup E'.$$

Theorem 4.1.26. *If $E \subset \mathbb{R}$, then*

(1) $\overline{E}$ is closed;
(2) $E = \overline{E}$ if and only if E is closed; and
(3) $\overline{E} \subset F$ for every closed set $F \subset \mathbb{R}$ such that $E \subset F$.

PROOF. (1) Let $x \in \overline{E}^c$. Then $x \notin E$ and x is also not a limit point of E. Thus there exists an $\varepsilon > 0$ such that $(x-\varepsilon, x+\varepsilon) \cap E = \emptyset$.

Assume by way of contradiction that $(x-\varepsilon, x+\varepsilon) \cap E' \neq \emptyset$. So there exists a $y \in (x-\varepsilon, x+\varepsilon) \cap E'$. So y is a limit point of E. Since $(x-\varepsilon, x+\varepsilon)$ is open, there exists a $\delta > 0$ such that $(y-\delta, y+\delta) \subset (x-\varepsilon, x+\varepsilon)$. But since $y \in E'$, $(y-\delta, y+\delta) \cap E \neq \emptyset$, a contradiction. So $(x-\varepsilon, x+\varepsilon) \cap \overline{E} = \emptyset$ and $(x-\varepsilon, x+\varepsilon) \subset \overline{E}^c$. So $\overline{E}^c$ is open and $\overline{E}$ is closed.

(2) If $E = \overline{E}$, then by (1), E is closed. If E is closed, $E' \subset E$ by Theorem 4.1.24 and $\overline{E} = E$.

(3) If $E \subset F$ and F is closed, then $E' \subset F$. To see this, notice every limit point of E is a limit point of F, since $E \subset F$. So $E' \subset F' \subset F$ by Theorem 4.1.24. Thus $\overline{E} \subset F$. □

So $\overline{E}$ is the smallest closed set containing E.

Definition 4.1.27. A set $D \subset \mathbb{R}$ is *dense* in $\mathbb{R}$ if $\overline{D} = \mathbb{R}$.

Example 4.1.28. $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are dense in $\mathbb{R}$. Let $x \in \mathbb{R}$ and $\varepsilon > 0$. By Theorem 1.8.11, there is a rational number in $(x, x+\varepsilon)$. It is not x. So x is a limit point of $\mathbb{Q}$, and $\overline{\mathbb{Q}} = \mathbb{R}$. Exercise 1.8.5 gives the same for $\mathbb{R} \setminus \mathbb{Q}$.

Definition 4.1.29. A set $A \subset \mathbb{R}$ is *connected* if there do not exist two disjoint open sets U and V such that $A \cap U \neq \emptyset$, $A \cap V \neq \emptyset$, and $(A \cap U) \cup (A \cap V) = A$.

Theorem 4.1.30. *A nonempty subset A of $\mathbb{R}$ is connected if and only if it is an interval.*

PROOF. ($\Rightarrow$) Assume A is not an interval. Then there exist $a, b \in A$ with $a < b$ and $t \in (a, b)$ with $t \notin A$. Define $U = (-\infty, t)$ and $V = (t, \infty)$. Then $U \cap V = \emptyset$. Also $a \in A \cap U$ and $b \in A \cap V$, so $A \cap U \neq \emptyset$, $A \cap V \neq \emptyset$. Last, since $t \notin A$, $(A \cap U) \cup (A \cap V) = A$, so A is not connected.

($\Leftarrow$) This direction is an exercise. □

Exercises

4.1.1. Let $U \subset \mathbb{R}$. Prove U is open if and only if $U = U^\circ$. [Solution, p. 336]

4.1.2. Prove that an open interval in $\mathbb{R}$ is an open set. [Solution, p. 336]

4.1.3. Find an example of an infinite collection of closed sets whose union is not closed.

4.1.4. Let $a, b \in \mathbb{R}$. Prove that (a, ∞) and $(-\infty, b)$ are open sets.

4.1.5. Prove Theorem 4.1.17.

4.1.6. Prove that $\mathbb{Q}$ is neither open nor closed. [Hint, p. 336]

4.1.7. Prove that $x \in \overline{E}$ if and only if there is a sequence $\{x_n\}$ with $x_n \in E$ for all $n \in \mathbb{N}$ such that $\lim_{n\to\infty} x_n = x$. [Solution, p. 336]

4.1.8. Let $E = \left\{\frac{1}{n} : n \in \mathbb{N}\right\} \cup (2, 3)$. Find E°, E' and $\overline{E}$.

4.1.9. Let $E \subset \mathbb{R}$ and let a be a limit point of E. Prove that for every $\varepsilon > 0$, the interval $(a - \varepsilon, a + \varepsilon)$ contains infinitely many points of E. Conclude that a finite set has no limit points. [Hint, p. 336]

4.1.10. Prove the ($\Leftarrow$) direction of Theorem 4.1.30. [Hint, p. 336]

4.2. Compact Sets

Many facts are easy for finite sets and false for infinite ones. A finite set of numbers has a largest element. A finite set of positive numbers has a positive minimum. Compact sets are infinite sets that behave like finite sets in one precise way. This section defines compactness and proves that the compact subsets of $\mathbb{R}$ are exactly the closed and bounded sets.

Definition 4.2.1. Let $E \subset \mathbb{R}$. A collection $\mathcal{U} = \{U_\alpha\}_{\alpha\in\Lambda}$ of open subsets of $\mathbb{R}$ is called an *open cover* of E if $E \subset \bigcup_{\alpha\in\Lambda} U_\alpha$.

Example 4.2.2. Let $E = (0, 1)$ and let $U_n = \left(0, 1 - \frac{1}{n}\right)$. Then $\{U_n\}_{n\in\mathbb{N}}$ is an open cover of E. Each $x \in (0, 1)$ is in U_n once $\frac{1}{n} < 1 - x$.

Example 4.2.3. Let $E = \mathbb{R}$ and let $U_x = (x - 1, x + 1)$. Then $\{U_x\}_{x\in\mathbb{R}}$ is an open cover of $\mathbb{R}$.

Definition 4.2.4. Let $E \subset \mathbb{R}$ and $\{U_\alpha\}_{\alpha\in\Lambda}$ be an open cover of E. If there exists a $\Lambda_0 \subset \Lambda$ such that $\{U_\beta\}_{\beta\in\Lambda_0}$ is an open cover of E, $\{U_\beta\}_{\beta\in\Lambda_0}$ is called a *subcover* of $\{U_\alpha\}_{\alpha\in\Lambda}$. If Λ_0 is finite, it is called a *finite subcover*.

Definition 4.2.5. A subset $K \subset \mathbb{R}$ is *compact* if every open cover of K has a finite subcover.

Proof Technique (Proving a Set Is Not Compact). To prove a set K is compact, every open cover of K has to be handled. To prove K is not compact, one open cover is enough. Find an open cover of K with no finite subcover. A good choice is a sequence of open sets $U_1 \subset U_2 \subset U_3 \subset \cdots$ whose union contains K but none of which contains K. The union of finitely many of them is the one with the largest index, and it misses part of K. Example 4.2.6 does this.

Example 4.2.6. $(0,1)$ is not compact. To show this, assume by way of contradiction that $(0,1)$ is compact. Then, since $\{U_n\}_{n\in\mathbb{N}} = \left\{\left(0, 1-\frac{1}{n}\right)\right\}_{n\in\mathbb{N}}$ covers $(0,1)$, there exists $n_1 < n_2 < \cdots < n_k$ such that

$$(0,1) \subset \bigcup_{i=1}^{k} \left(0, 1-\frac{1}{n_i}\right) = \left(0, 1-\frac{1}{n_k}\right),$$

a contradiction, since $1 - \frac{1}{2n_k}$ is in $(0,1)$ but not in $\left(0, 1-\frac{1}{n_k}\right)$. So $(0,1)$ is not compact.

Example 4.2.7. $[0,\infty)$ is not compact. See Exercise 4.2.1.

Example 4.2.8. A finite set is compact. To see this, let $P = \{p_1, \ldots, p_n\}$ be a finite set. Let $\{U_\alpha\}_{\alpha\in\Lambda}$ be an open cover of P. So for each $p_k \in P$, $k = 1, \ldots, n$, there exists an $\alpha_k \in \Lambda$ such that $p_k \in U_{\alpha_k}$. So $\{U_{\alpha_k}\}_{k=1}^{n}$ is a finite subcover of P and hence, P is compact.

> **Proof Technique (Finite Subcover Arguments).** Many proofs about a compact set K follow one pattern. For each point $y \in K$, choose an open set around y with some useful property. These sets form an open cover of K. Compactness gives finitely many of them that still cover K. Then take a minimum or a maximum over the finitely many choices. The minimum of finitely many positive numbers is positive. The minimum of infinitely many might not exist, and their infimum might be 0. The proof of Theorem 4.2.9 uses this pattern twice.

Theorem 4.2.9. *A compact subset of* $\mathbb{R}$ *is closed and bounded.*

Proof. Let $K \subset \mathbb{R}$ be compact. If $K = \emptyset$, then K is closed and bounded. So assume $K \neq \emptyset$.

First, K is bounded. Every real number is in $(-n, n)$ for some $n \in \mathbb{N}$, by the Archimedean Principle. So the intervals $(-n, n)$, $n \in \mathbb{N}$, form an open cover of K. Since K is compact, finitely many of them cover K, say $(-n_1, n_1), \ldots, (-n_m, n_m)$. Let $N = \max\{n_1, \ldots, n_m\}$. Each of these intervals is a subset of $(-N, N)$. So $K \subset (-N, N)$, and K is bounded.

Next, K is closed. Let $x \in K^c$. For each $y \in K$, $y \neq x$. Let $\varepsilon_y = \frac{1}{2}|x - y| > 0$. The intervals $(x - \varepsilon_y, x + \varepsilon_y)$ and $(y - \varepsilon_y, y + \varepsilon_y)$ are disjoint. Indeed, if z were in both, then

$$|x - y| \leq |x - z| + |z - y| < 2\varepsilon_y = |x - y|.$$

The intervals $(y - \varepsilon_y, y + \varepsilon_y)$, $y \in K$, form an open cover of K. Since K is compact, there exist $y_1, \ldots, y_n \in K$ such that

$$K \subset \bigcup_{i=1}^{n} (y_i - \varepsilon_{y_i}, y_i + \varepsilon_{y_i}).$$

Let $\varepsilon = \min\{\varepsilon_{y_1}, \ldots, \varepsilon_{y_n}\} > 0$. For each i, $(x - \varepsilon, x + \varepsilon) \subset (x - \varepsilon_{y_i}, x + \varepsilon_{y_i})$, which is disjoint from $(y_i - \varepsilon_{y_i}, y_i + \varepsilon_{y_i})$. So $(x - \varepsilon, x + \varepsilon) \cap K = \emptyset$, and $(x - \varepsilon, x + \varepsilon) \subset K^c$. Thus K^c is open and K is closed. □

Lemma 4.2.10. *Every closed subset of a compact set is compact.*

PROOF. Let K be compact and let $F \subset K$ be closed. Let $\{U_\alpha\}_{\alpha\in\Lambda}$ be an open cover of F. Since F is closed, F^c is open. Each point of K is either in F, and so in some U_α, or in F^c. So the sets U_α, $\alpha \in \Lambda$, together with F^c form an open cover of K. Since K is compact, finitely many of these sets cover K. Adding F^c to the list if needed,

$$K \subset U_{\alpha_1} \cup \cdots \cup U_{\alpha_n} \cup F^c$$

for some $\alpha_1, \ldots, \alpha_n \in \Lambda$. No point of F is in F^c. So $F \subset U_{\alpha_1} \cup \cdots \cup U_{\alpha_n}$, and F is compact. □

Theorem 4.2.11 (Bolzano–Weierstrass Property)**.** *If S is an infinite subset of a compact set K, then S has a limit point in K.*

PROOF. Assume by way of contradiction that no point of K is a limit point of S. Then for each $x \in K$, there exists an $\varepsilon_x > 0$ such that $(x - \varepsilon_x, x + \varepsilon_x)$ contains no point of S other than x. So each interval $(x - \varepsilon_x, x + \varepsilon_x)$ contains at most one point of S. These intervals, for $x \in K$, form an open cover of K. Since K is compact, finitely many of them cover K, and so they cover S. Since each of them contains at most one point of S, S is finite. This is a contradiction. So S has a limit point in K. □

Theorem 4.2.12 (Cantor's Intersection Theorem)**.** *If $\{K_n\}_{n=1}^\infty$ is a sequence of nonempty compact subsets of $\mathbb{R}$ with $K_n \supset K_{n+1}$ for all $n \in \mathbb{N}$, then $K = \bigcap_{n=1}^\infty K_n$ is nonempty and compact.*

PROOF. Let $U_n = K_n^c$. By Theorem 4.2.9, K_n is closed and therefore U_n is open. Assume by way of contradiction that $K = \emptyset$. Then $\bigcup_{n=1}^\infty U_n = \mathbb{R}$. So $\{U_n\}$ is an open cover of $\mathbb{R}$ and hence K_1. So there exists $n_1 < n_2 < \cdots < n_k$ such that $K_1 \subset \bigcup_{j=1}^k U_{n_j}$. This implies $K_1^c \supset \bigcap_{j=1}^k K_{n_j}$, or

$$K_1 \cap K_{n_1} \cap K_{n_2} \cap \cdots \cap K_{n_k} = \emptyset,$$

a contradiction, since the sets are nested and so this intersection is K_{n_k}, which is nonempty. Thus $K \neq \emptyset$. By Theorem 4.1.17, K is closed and since $K \subset K_1$, by Lemma 4.2.10, K is compact. □

A collection of sets has the *finite intersection property* if every finite subcollection has a nonempty intersection. The nested compact sets in Cantor's Intersection Theorem have it. The next corollary drops the nesting.

Corollary 4.2.13. *Let $\{F_\alpha\}_{\alpha\in\Lambda}$ be a collection of closed sets with the property that every finite subcollection of $\{F_\alpha\}_{\alpha\in\Lambda}$ has a nonempty intersection. Suppose one of the sets F_α is bounded. Then $\bigcap_{\alpha\in\Lambda} F_\alpha \neq \emptyset$.*

PROOF. See Exercise 4.2.7. □

This extends the Nested Interval Theorem (Theorem 3.4.4) from closed intervals to compact sets. Compactness cannot be dropped. For example, $\bigcap_{n=1}^\infty [n, \infty) = \emptyset$.

Theorem 4.2.14 (Heine–Borel)**.** *Every closed and bounded interval $[a, b]$ is compact.*

PROOF. Let $\mathcal{U} = \{U_\alpha\}_{\alpha\in\Lambda}$ be an open cover of $[a,b]$. Call a point $r \in [a,b]$ *good* if $[a,r]$ is covered by finitely many of the sets in $\mathcal{U}$. Let E be the set of good points. Some U_α contains a, so $[a,a] = \{a\}$ is covered by one set in $\mathcal{U}$. So $a \in E$, and $E \neq \emptyset$. Also, E is bounded above by b. By the least upper bound property, $c = \sup E$ exists, and $a \leq c \leq b$.

Since $c \in [a,b]$, there exists an $\alpha_c \in \Lambda$ with $c \in U_{\alpha_c}$. Since U_{α_c} is open, there exists an $\varepsilon > 0$ such that $(c-\varepsilon, c+\varepsilon) \subset U_{\alpha_c}$. By Theorem 1.8.7, there exists a good point r with $c - \varepsilon < r \leq c$. So

$$[a,r] \subset U_{\alpha_1} \cup \cdots \cup U_{\alpha_n}$$

for some $\alpha_1, \ldots, \alpha_n \in \Lambda$. Let $t = \min\left\{b, c + \frac{\varepsilon}{2}\right\}$. Then $[r,t] \subset (c-\varepsilon, c+\varepsilon) \subset U_{\alpha_c}$. So

$$[a,t] \subset U_{\alpha_1} \cup \cdots \cup U_{\alpha_n} \cup U_{\alpha_c},$$

and t is good. Since $c = \sup E$, $t \leq c$. If $c < b$, then $t > c$, a contradiction. So $c = b$, and then $t = b$. Thus b is good. That is, $[a,b]$ is covered by finitely many of the sets in $\mathcal{U}$. □

Proof Technique (Proving Several Statements Are Equivalent). To prove statements (1), (2) and (3) are equivalent, it is enough to prove (1)⇒(2), (2)⇒(3) and (3)⇒(1). This is a cycle of implications. Any one statement implies any other by following the cycle. For example, (2)⇒(1) follows from (2)⇒(3) and (3)⇒(1). Three proofs replace six. Theorem 4.2.15 is proved this way.

Theorem 4.2.15 (Heine–Borel–Bolzano–Weierstrass)**.** *Let $K \subset \mathbb{R}$. The following are equivalent.*

(1) K is closed and bounded.
(2) K is compact.
(3) Every infinite subset of K has a limit point in K.

PROOF. (1)⇒(2) Since K is bounded, there exists an $M > 0$ such that $K \subset [-M, M]$. Now, by the Heine–Borel Theorem (Theorem 4.2.14), $[-M, M]$ is compact. So by Lemma 4.2.10, K is compact.

(2)⇒(3) Theorem 4.2.11.

(3)⇒(1) Suppose every infinite subset of K has a limit point in K.

First, K is bounded. Assume by way of contradiction that K is not bounded. Choose $x_1 \in K$ with $|x_1| > 1$. Having chosen $x_1, \ldots, x_n$, choose $x_{n+1} \in K$ with $|x_{n+1}| > \max\{n+1, |x_n|\}$. This is possible since K is not bounded. Then $|x_1| < |x_2| < |x_3| < \cdots$, so the points x_n are distinct, and $|x_n| > n$ for all n. Let $S = \{x_n : n \in \mathbb{N}\}$. Then S is an infinite subset of K. Let $y \in \mathbb{R}$, and choose $N \in \mathbb{N}$ with $N > |y| + 1$. By the Reverse Triangle Inequality (Theorem 1.7.3), for all $n \geq N$,

$$|x_n - y| \geq |x_n| - |y| > n - |y| \geq N - |y| > 1.$$

So $(y-1, y+1)$ contains at most the finitely many points $x_1, \ldots, x_{N-1}$ of S. By Exercise 4.1.9, y is not a limit point of S. So S has no limit point in $\mathbb{R}$, and so none in K. This contradicts (3). So K is bounded.

Next, K is closed. Let x be a limit point of K. By Theorem 4.1.24, it is enough to show $x \in K$. For each $n \in \mathbb{N}$, there exists an $x_n \in K$ with $x_n \neq x$ and $|x_n - x| < \frac{1}{n}$.

Let $S = \{x_n : n \in \mathbb{N}\}$. If S were finite, then $\delta = \min\{|s - x| : s \in S\}$ would be positive, and $|x_n - x| < \frac{1}{n} < \delta$ for large n. This is impossible, since each x_n is in S. So S is infinite. By (3), S has a limit point $z \in K$. Suppose $z \neq x$. Let $\varepsilon = \frac{1}{2}|z - x|$, and choose $N \in \mathbb{N}$ with $\frac{1}{N} < \varepsilon$. For all $n \geq N$,

$$|x_n - z| \geq |z - x| - |x_n - x| > 2\varepsilon - \varepsilon = \varepsilon.$$

So $(z - \varepsilon, z + \varepsilon)$ contains at most the finitely many points $x_1, \ldots, x_{N-1}$ of S. By Exercise 4.1.9, z is not a limit point of S, a contradiction. So $z = x$. Thus $x \in K$, and K is closed. □

Closed and bounded is the easiest of the three conditions to check. Compactness is the most useful in proofs. The third condition connects compactness to sequences, as Corollary 4.2.16 shows.

Corollary 4.2.16. *Let K be a nonempty compact subset of $\mathbb{R}$. Then every sequence in K has a convergent subsequence that converges to a point in K.*

Proof. See Exercise 4.2.3. □

Exercises

4.2.1. Use the definition of a compact set to prove the interval $[0, \infty)$ is not compact.

4.2.2. Let $\{x_n\}$ be a sequence in $\mathbb{R}$ with $x_n \to a$. Use the definition of a compact set to prove the set $\{x_n : n \in \mathbb{N}\} \cup \{a\}$ is compact. [Solution, p. 337]

4.2.3. Prove Corollary 4.2.16. [Hint, p. 337]

4.2.4. Give an example of a closed set that is not compact, and an example of a bounded set that is not compact.

4.2.5. Use the definition of a compact set to prove that the union of finitely many compact sets is compact. [Hint, p. 337]

4.2.6. Prove that the intersection of any collection of compact sets is compact. [Hint, p. 337]

4.2.7. Prove Corollary 4.2.13. Give an example to show that the conclusion may be false if all of the sets are unbounded. [Solution, p. 337]

Notes

Peter Gustav Lejeune Dirichlet used the idea behind the Heine–Borel Theorem in his 1852 lectures, to prove that a continuous function on a closed interval is uniformly continuous. Eduard Heine published that result in 1872 [38]. Émile Borel proved the Heine–Borel Theorem for countable open covers in 1895 [6]. Pierre Cousin proved a version for arbitrary covers the same year [17], and Henri Lebesgue gave a proof for arbitrary covers in his 1904 lectures [43]. The name Heine–Borel Theorem is due to Arthur Schoenflies [65]. Ernst Lindelöf proved his theorem in 1903 [48].

CHAPTER 5

The Cantor Set*

This short chapter builds one example. It can be skipped. Later chapters use it only in Example 9.2.8, a few exercises, and the starred Section 17.6, which builds the Cantor function. Start with $[0,1]$. Remove the open middle third. Then remove the open middle third of each of the two pieces that are left. Keep going forever. What is left at the end is the Cantor set.

The Cantor set is a test case for intuition. It is compact. It has no isolated points. The intervals removed to build it have total length 1, so it seems that almost nothing is left. It contains no open intervals at all. Yet it is uncountable. It returns in Chapter 9 as an example of a set of measure zero.

5.1. Construction of the Cantor Set

Definition 5.1.1 (Construction of the Cantor Set)**.** Define

$$\begin{aligned} C_0 &= [0,1] \\ &= J_{0,1}, \end{aligned}$$

$$\begin{aligned} C_1 &= [0,1/3] \cup [2/3,1] \\ &= J_{1,1} \cup J_{1,2}, \end{aligned}$$

$$\begin{aligned} C_2 &= [0,1/9] \cup [2/9,1/3] \cup [2/3,7/9] \cup [8/9,1] \\ &= J_{2,1} \cup J_{2,2} \cup J_{2,3} \cup J_{2,4}, \end{aligned}$$

and in general, C_{n+1} is obtained from C_n by removing the open middle third of each closed interval in C_n. That is, each interval $[a,b]$ in C_n is replaced by the two intervals $\left[a, a+\frac{b-a}{3}\right]$ and $\left[b-\frac{b-a}{3}, b\right]$. This gives

$$C_n = \bigcup_{i=1}^{2^n} J_{n,i}, \quad \text{where } J_{n,i} = \left[\frac{k_{n,i}}{3^n}, \frac{k_{n,i}+1}{3^n}\right]$$

for some $k_{n,i} \in \mathbb{N}_0$. The intervals are numbered from left to right. Define the *Cantor set* to be

$$C = \bigcap_{n=0}^{\infty} C_n.$$

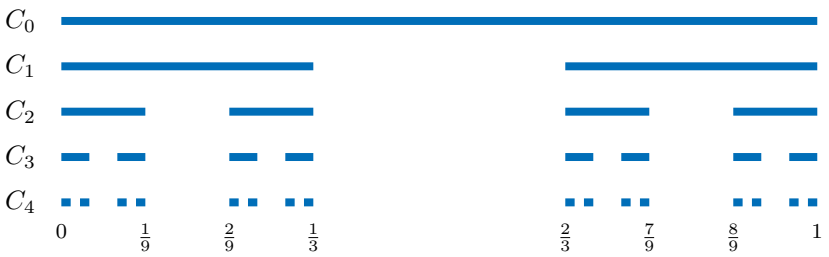


FIGURE 5.1. The sets $C_0, C_1, \ldots, C_4$. Each one is built from the one above it by removing the open middle third of each interval.

Lemma 5.1.2. *For each $n \in \mathbb{N}_0$, the following hold.*

(1) The intervals $J_{n,1}, \ldots, J_{n,2^n}$ are pairwise disjoint, and each has length $\frac{1}{3^n}$.
(2) The left endpoints of $J_{n,1}, \ldots, J_{n,2^n}$ are exactly the numbers $\sum_{k=1}^{n} \frac{d_k}{3^k}$ with each $d_k \in \{0, 2\}$. (For $n = 0$, this sum is 0.)
(3) The endpoints of each $J_{n,i}$ are in C.

PROOF. Parts (1) and (2) are proved together by induction on n. For $n = 0$, $C_0 = J_{0,1} = [0, 1]$ has length $1 = \frac{1}{3^0}$ and left endpoint 0. Suppose (1) and (2) hold for n. Let $J_{n,i} = \left[s, s + \frac{1}{3^n}\right]$ be one of the intervals of C_n. Removing its open middle third leaves

$$\left[s, s + \frac{1}{3^{n+1}}\right] \quad \text{and} \quad \left[s + \frac{2}{3^{n+1}}, s + \frac{1}{3^n}\right].$$

Each has length $\frac{1}{3^{n+1}}$. They are disjoint, and they lie inside $J_{n,i}$. Since the intervals of C_n are pairwise disjoint, all $2 \cdot 2^n = 2^{n+1}$ intervals of C_{n+1} are pairwise disjoint. Their left endpoints are $s + \frac{0}{3^{n+1}}$ and $s + \frac{2}{3^{n+1}}$, where $s = \sum_{k=1}^{n} \frac{d_k}{3^k}$ with each $d_k \in \{0, 2\}$. These are exactly the sums $\sum_{k=1}^{n+1} \frac{d_k}{3^k}$ with each $d_k \in \{0, 2\}$. So (1) and (2) hold for $n + 1$.

For (3), let p be an endpoint of $J_{n,i}$. The display above shows p is also an endpoint of one of the two intervals of C_{n+1} inside $J_{n,i}$. By induction, p is an endpoint of an interval of C_m, and so $p \in C_m$, for every $m \geq n$. Also $p \in C_n \subset C_m$ for every $m \leq n$. So $p \in C$. □

Theorem 5.1.3. *C is compact and nonempty.*

PROOF. The endpoints of $J_{n,i}$, $n = 1, 2, \ldots$ are in C by Lemma 5.1.2, so $C \neq \emptyset$. Also, each $J_{n,i}$ is compact by the Heine–Borel Theorem (Theorem 4.2.14), so C_n is compact for $n = 1, 2, \ldots$ by Exercise 4.2.5. Also, $C_{n+1} \subset C_n$ for all n. Therefore, by Theorem 4.2.12, C is compact. □

Theorem 5.1.4. *Every point of C is a limit point of C.*

PROOF. Let $\varepsilon > 0$. Let $x \in C$. Choose $m \in \mathbb{N}$ such that $(1/3)^m < \varepsilon$. Since $x \in C_m$, $x \in J_{m,i}$ for some i, $i = 1, \ldots, 2^m$. But

$$J_{m,i} = \left[\frac{k_{m,i}}{3^m}, \frac{k_{m,i} + 1}{3^m}\right],$$

so the length of $J_{m,i}$ is $(1/3)^m < \varepsilon$. Thus $J_{m,i} \subset (x-\varepsilon, x+\varepsilon)$. Therefore, both end points of $J_{m,i}$ are in $C \cap (x-\varepsilon, x+\varepsilon)$ by Lemma 5.1.2. At least one of them is not x, and thus $\big(C \cap (x-\varepsilon, x+\varepsilon)\big) \setminus \{x\} \neq \emptyset$. □

So C has no isolated points. A closed set with no isolated points is called *perfect*. The Cantor set is perfect.

Theorem 5.1.5. *The sum of the lengths of the intervals removed in constructing C_1 from C_0, C_2 from C_1, and so on, is 1.*

Proof. When constructing C_1 from C_0, an interval of length $1/3$ was removed. At step n, 2^{n-1} intervals of length $(1/3)^n$ were removed. Thus the sum of the length of the intervals removed is

$$2^0\frac{1}{3} + 2\frac{1}{3^2} + \cdots + 2^{n-1}\frac{1}{3^n} + \cdots = \sum_{n=1}^{\infty} \frac{2^{n-1}}{3^n} = \frac{1}{3}\sum_{n=0}^{\infty}\left(\frac{2}{3}\right)^n = \frac{1}{3}\cdot\frac{1}{1-2/3} = 1.$$

Here the infinite sum means the limit of the sums of the first n terms. By Exercise 1.2.7,

$$\sum_{k=1}^{n} \frac{2^{k-1}}{3^k} = \frac{1}{3}\sum_{i=0}^{n-1}\left(\frac{2}{3}\right)^i = \frac{1}{3}\cdot\frac{1-(2/3)^n}{1-2/3} = 1 - \left(\frac{2}{3}\right)^n,$$

which converges to 1 by Example 3.1.7. The same argument shows $\sum_{i=0}^{\infty} r^i = \frac{1}{1-r}$ whenever $|r| < 1$. This formula is used again below. Infinite series are studied in Chapter 10. □

So the intervals removed have total length 1, the length of $[0,1]$. In this sense C has length 0. Example 9.2.8 makes this precise. It shows C has measure zero.

Exercises

5.1.1. Prove that the sum of the lengths of the intervals in C_n is $\left(\frac{2}{3}\right)^n$. Use this to give another proof of Theorem 5.1.5. [Solution, p. 337]

5.1.2. Let P be the set of all endpoints of the intervals $J_{n,i}$, for $n \in \mathbb{N}_0$ and $i = 1, \ldots, 2^n$. Prove that P is countable. [Hint, p. 337]

5.2. Ternary Expansion of Elements of the Cantor Set

A decimal expansion writes a number using powers of 10. A ternary expansion uses powers of 3. The Cantor set is built by cutting intervals into thirds, so ternary expansions describe it well. A number is in C exactly when it has a ternary expansion that uses only the digits 0 and 2. This description shows C is uncountable.

Theorem 5.2.1. *C contains no open intervals.*

Proof. Assume by way of contradiction that $(a,b) \subset C$ for some $a < b$. By the Archimedean Principle (Theorem 1.8.8), choose $n \in \mathbb{N}$ with $\frac{1}{n} < \frac{b-a}{2}$. Since $3^n > n$, $\frac{1}{3^n} < \frac{b-a}{2}$. Let $m = \frac{a+b}{2}$. Since $m \in C \subset C_n$, $m \in J_{n,i}$ for some i. Every point of $J_{n,i}$ is within $\frac{1}{3^n} < \frac{b-a}{2}$ of m. So $J_{n,i} \subset (a,b) \subset C$.

Let M be the open middle third of $J_{n,i}$. It is removed in constructing C_{n+1}. So the points of M are not in the two intervals of C_{n+1} inside $J_{n,i}$. They are not in the other intervals of C_{n+1} either. Those lie inside the intervals $J_{n,j}$ with $j \neq i$, which are disjoint from $J_{n,i}$ by Lemma 5.1.2. So $M \cap C_{n+1} = \emptyset$. But M is nonempty and $M \subset J_{n,i} \subset C \subset C_{n+1}$. This is a contradiction. □

Definition 5.2.2. Let $x \in [0,1]$. Let $n_1 \in \{0,1,2\}$ be the largest integer such that

$$\frac{n_1}{3} < x.$$

(If $x = 0$, no such n_1 exists. In that case, use $.000\ldots$.) Having chosen $n_1, \ldots, n_k$, let $n_{k+1} \in \{0,1,2\}$ be the largest integer such that

$$\frac{n_1}{3} + \frac{n_2}{3^2} + \cdots + \frac{n_k}{3^k} + \frac{n_{k+1}}{3^{k+1}} < x.$$

The expression $.n_1n_2n_3\ldots$ is called a *ternary expansion* of x. In general, if $x = \sum_{k=1}^{\infty} \frac{n_k}{3^k}$, where $n_k \in \{0,1,2\}$ for all $k \in \mathbb{N}$, $.n_1n_2n_3\ldots$ is called a ternary expansion of x.

Remark 5.2.3. The sum $\sum_{k=1}^{\infty} \frac{n_k}{3^k}$ means the limit of the partial sums $s_m = \sum_{k=1}^{m} \frac{n_k}{3^k}$. This limit always exists. The partial sums increase, and by Exercise 1.2.7,

$$s_m \leq \sum_{k=1}^{m} \frac{2}{3^k} = 1 - \frac{1}{3^m} < 1.$$

So $\{s_m\}$ converges by the Monotone Convergence Theorem (Theorem 3.2.13). The digits chosen in the first part of Definition 5.2.2 satisfy $s_m < x \leq s_m + \frac{1}{3^m}$ for every m, by Exercise 5.2.1. So $s_m \to x$, and $.n_1n_2n_3\ldots$ really is a ternary expansion of x.

Example 5.2.4. Let $x = \frac{1}{3}$. For

$$\frac{n_1}{3} < \frac{1}{3}, \quad n_1 = 0.$$

For

$$\frac{0}{3} + \frac{n_2}{3^2} < \frac{1}{3}, \quad n_2 = 2.$$

For

$$\frac{0}{3} + \frac{n_2}{3^2} + \frac{n_3}{3^3} < \frac{1}{3}, \quad n_3 = 2.$$

Now

$$\begin{aligned} .0222\ldots &= \sum_{k=2}^{\infty} \frac{2}{3^k} \\ &= \frac{2}{9} \sum_{k=0}^{\infty} \left(\frac{1}{3}\right)^k \\ &= \frac{2}{9} \cdot \frac{1}{1 - 1/3} = \frac{1}{3}. \end{aligned}$$

So $.0222\ldots$ is the ternary expansion of $\frac{1}{3}$. Notice $\frac{1}{3} = .100\ldots$ (base 3). So $\frac{1}{3}$ has two ternary expansions.

Definition 5.2.5. The ternary expansion $x = .n_1n_2\ldots$ is *finite* or *terminating* if there exists an $m \in \mathbb{N}$ such that $n_k = 0$ for all $k > m$.

Example 5.2.6. Notice if x has a terminating ternary expansion, its ternary expansion is not unique. This happens when $x = \frac{a}{3^m}$, $a \in \mathbb{N}$, $0 < a < 3^m$, and a is not a multiple of 3. In this case,

$$x = \frac{a_1}{3} + \cdots + \frac{a_m}{3^m}, \quad a_m \in \{1, 2\}.$$

If $a_m = 1$,

$$x = \frac{a_1}{3} + \cdots + \frac{0}{3^m} + \sum_{k=m+1}^{\infty} \frac{2}{3^k},$$

and if $a_m = 2$,

$$x = \frac{a_1}{3} + \cdots + \frac{1}{3^m} + \sum_{k=m+1}^{\infty} \frac{2}{3^k}.$$

If $a_m = 1$, consider only the infinite expansion. If $a_m = 2$, consider only the finite expansion.

Theorem 5.2.7. *Let $x \in [0, 1]$. Then $x \in C$ if and only if x has a ternary expansion $.n_1n_2n_3\ldots$ with $n_k \in \{0, 2\}$ for all $k \in \mathbb{N}$.*

Proof. ($\Leftarrow$) Suppose $x = \sum_{k=1}^{\infty} \frac{n_k}{3^k}$ with each $n_k \in \{0, 2\}$. Let $m \in \mathbb{N}_0$ and let $s_m = \sum_{k=1}^{m} \frac{n_k}{3^k}$. For $p > m$, by Exercise 1.2.7,

$$0 \leq s_p - s_m = \sum_{k=m+1}^{p} \frac{n_k}{3^k} \leq \sum_{k=m+1}^{p} \frac{2}{3^k} = \frac{1}{3^m} - \frac{1}{3^p} < \frac{1}{3^m}.$$

Letting $p \to \infty$, Theorem 3.2.7 gives $s_m \leq x \leq s_m + \frac{1}{3^m}$. By Lemma 5.1.2, $\left[s_m, s_m + \frac{1}{3^m}\right]$ is one of the intervals of C_m. So $x \in C_m$. Since m was arbitrary, $x \in C$.

($\Rightarrow$) Suppose $x \in C$. For each $m \in \mathbb{N}_0$, $x \in C_m$. Since the intervals of C_m are disjoint, x is in exactly one of them. Call it $\left[s_m, s_m + \frac{1}{3^m}\right]$. Then $s_0 = 0$. The interval of C_{m+1} containing x lies inside an interval of C_m that contains x, so it lies inside $\left[s_m, s_m + \frac{1}{3^m}\right]$. By the proof of Lemma 5.1.2, its left endpoint s_{m+1} is s_m or $s_m + \frac{2}{3^{m+1}}$. So $n_{m+1} = 3^{m+1}(s_{m+1} - s_m)$ is 0 or 2. By induction, $s_m = \sum_{k=1}^{m} \frac{n_k}{3^k}$ for all m. Since $s_m \leq x \leq s_m + \frac{1}{3^m}$, $|x - s_m| \leq \frac{1}{3^m}$, which converges to 0 by Example 3.1.7. So $s_m \to x$, and $x = \sum_{k=1}^{\infty} \frac{n_k}{3^k}$ with each $n_k \in \{0, 2\}$. □

For example, $\frac{1}{3} = .1000\cdots = .0222\ldots$ is in C, since its second expansion uses only 0's and 2's.

Theorem 5.2.8. *C is uncountable.*

Proof. By Theorem 5.2.7, $C \sim 2^{\mathbb{N}}$. So C is uncountable.

In more detail, define $f : 2^{\mathbb{N}} \to C$ by $f(\{b_k\}) = \sum_{k=1}^{\infty} \frac{2b_k}{3^k}$. By Theorem 5.2.7, f maps $2^{\mathbb{N}}$ onto C. To see f is one-to-one, let $\{b_k\} \neq \{c_k\}$. Let j be the first index

with $b_j \neq c_j$, and say $b_j = 1$ and $c_j = 0$. The estimate in the proof of Theorem 5.2.7 gives

$$f(\{b_k\}) - f(\{c_k\}) \geq \frac{2}{3^j} - \sum_{k=j+1}^{\infty} \frac{2}{3^k} = \frac{2}{3^j} - \frac{1}{3^j} > 0.$$

So f is a bijection, and $C \sim 2^{\mathbb{N}}$. Since $2^{\mathbb{N}}$ is uncountable by Theorem 2.3.3, so is C. □

Example 5.2.9. Since $\frac{1}{4} = .020202\ldots$, $\frac{1}{4} \in C$. To check the expansion, notice

$$.020202\cdots = \sum_{j=1}^{\infty} \frac{2}{3^{2j}} = \frac{2}{9} \sum_{j=0}^{\infty} \left(\frac{1}{9}\right)^j = \frac{2}{9} \cdot \frac{1}{1 - 1/9} = \frac{1}{4}.$$

Every endpoint of every $J_{n,i}$ has the form $\frac{k}{3^n}$. The number $\frac{1}{4}$ does not, since $3^n = 4k$ is impossible when 3^n is odd and $4k$ is even. So C contains points that are not endpoints.

Exercises

5.2.1. Let $x \in (0, 1]$ and let $n_1, n_2, n_3, \ldots$ be the digits chosen in Definition 5.2.2. Let $s_0 = 0$ and $s_m = \sum_{k=1}^{m} \frac{n_k}{3^k}$. Prove by induction that $s_m < x \leq s_m + \frac{1}{3^m}$ for all $m \in \mathbb{N}_0$.

5.2.2. Find the ternary expansions of $\frac{1}{9}$ and $\frac{2}{3}$ given by Definition 5.2.2. Then find a second ternary expansion of each.

5.2.3. Prove that $\frac{3}{4} \in C$. [Hint, p. 337]

5.2.4. Find 4 points in the Cantor set, 2 of which are not endpoints of the intervals $J_{n,i}$. [Solution, p. 338]

5.2.5. Prove that $C^\circ = \emptyset$ and $C' = C$. [Hint, p. 338]

Notes

Henry Smith described a set like this one in 1875 [67]. Georg Cantor described it in 1883 [11].

CHAPTER 6

Functions and Continuity

Chapter 3 defined the limit of a sequence. This chapter defines the limit of a function at a point and uses it to define continuity. The definitions follow the same pattern as before. The number ε still says how close the outputs must be. A new number δ says how close the inputs must be.

The main results are about continuous functions on a closed and bounded interval $[a, b]$. Such a function takes every value between any two of its values. This is the Intermediate Value Theorem. It has a largest and a smallest value. This is the Extreme Value Theorem. It is also uniformly continuous. Each of these facts uses the completeness of $\mathbb{R}$, and each fails over $\mathbb{Q}$. For example, $f(x) = x^2 - 2$ is negative at 0 and positive at 2, but it has no rational zero. The starred Section 6.6 asks which sets can be the set of points where a function is continuous.

6.1. Limits

The limit of f at c describes what $f(x)$ does as x gets close to c. The value $f(c)$ plays no role. The function does not even have to be defined at c. This is why the definition uses $0 < |x - c| < \delta$.

Definition 6.1.1. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. Let c be a limit point of E. Then f has a *limit* of L at c, denoted $\lim_{x\to c} f(x) = L$ or $f(x) \to L$ as $x \to c$, if for all $\varepsilon > 0$, there exists a $\delta(c, \varepsilon) > 0$ such that for all $x \in E$ with $0 < |x - c| < \delta$, $|f(x) - L| < \varepsilon$.

The point c must be a limit point of E. Then for every $\delta > 0$ there are points $x \in E$ with $0 < |x - c| < \delta$. Without this, every number L would satisfy the definition. The notation $\delta(c, \varepsilon)$ is a reminder that δ may depend on both c and ε.

Example 6.1.2. The limit $\lim_{x\to 3}(5x + 7) = 22$. To show this, let $\varepsilon > 0$. Define $\delta = \frac{\varepsilon}{5}$. Then for all $x \in \mathbb{R}$ with $0 < |x - 3| < \delta$,

$$\begin{aligned} |(5x + 7) - 22| &= |5x - 15| \\ &= 5|x - 3| \\ &< 5\delta = \frac{5\varepsilon}{5} = \varepsilon. \end{aligned}$$

So $5x + 7 \to 22$ as $x \to 3$.

Example 6.1.3. The limit $\lim_{x\to 6}(x^2 + 2x - 7) = 41$. To see this, let $\varepsilon > 0$. Define $\delta = \min\left\{1, \frac{\varepsilon}{15}\right\}$. Then for all $x \in \mathbb{R}$ with $0 < |x - 6| < \delta$,

$$\begin{aligned} |(x^2 + 2x - 7) - 41| &= |x^2 + 2x - 48| \\ &= |x - 6||x + 8| \end{aligned}$$

$$\begin{aligned} &\leq 15|x-6| \\ &< 15\delta \leq \frac{15\varepsilon}{15} = \varepsilon. \end{aligned}$$

So $x^2 + 2x - 7 \to 41$ as $x \to 6$.

Proof Technique (Finding δ). The scratch work for a limit of a function is like the scratch work for a sequence. Start with $|f(x) - L|$ and pull out a factor of $|x - c|$. In Example 6.1.2, $|f(x) - L| = 5|x - 3|$, so $\delta = \frac{\varepsilon}{5}$ works. In Example 6.1.3, the other factor $|x + 8|$ depends on x. The trick is to first make sure $\delta \leq 1$. Then $|x - 6| < 1$, so $5 < x < 7$, and $|x + 8| < 15$. Now $|f(x) - L| \leq 15|x - 6|$, and $\delta \leq \frac{\varepsilon}{15}$ finishes the job. Taking $\delta = \min\left\{1, \frac{\varepsilon}{15}\right\}$ makes both conditions hold at once. Example 6.1.7 shows that $\delta \leq 1$ is not always the right first restriction.

Example 6.1.4. The limit $\lim_{x\to 2}(x^2 - 7x) = -10$. Let $\varepsilon > 0$. Define $\delta = \min\left\{1, \frac{\varepsilon}{4}\right\}$. Then for all $x \in \mathbb{R}$ with $0 < |x - 2| < \delta$,

$$\begin{aligned} |(x^2 - 7x) + 10| &= |x - 2||x - 5| \\ &\leq 4|x - 2| \\ &< 4\delta \leq \frac{4\varepsilon}{4} = \varepsilon. \end{aligned}$$

So $x^2 - 7x \to -10$ as $x \to 2$.

Example 6.1.5. $\lim_{x\to 7} \frac{1}{x-4} = \frac{1}{3}$. Let $\varepsilon > 0$. Define $\delta = \min\{1, \varepsilon\}$. Then for all $x \in \mathbb{R} \setminus \{4\}$ with $0 < |x - 7| < \delta$,

$$\begin{aligned} \left|\frac{1}{x-4} - \frac{1}{3}\right| &= \left|\frac{3 - (x-4)}{3(x-4)}\right| \\ &= \left|\frac{-x+7}{3(x-4)}\right| \\ &= \frac{|x-7|}{3|x-4|} \\ &\leq \frac{|x-7|}{6} \\ &< |x-7| < \delta \leq \varepsilon. \end{aligned}$$

The fourth line holds since $|x - 7| < 1$ gives $x > 6$, so $3|x - 4| > 6$. So $\frac{1}{x-4} \to \frac{1}{3}$ as $x \to 7$.

Example 6.1.6. $\lim_{x\to 2} \frac{x^2-4}{x-2} = 4$. Let $\varepsilon > 0$. Define $\delta = \varepsilon$. Then for all $x \in \mathbb{R} \setminus \{2\}$ with $0 < |x - 2| < \delta$,

$$\begin{aligned} \left|\frac{x^2-4}{x-2} - 4\right| &= |x + 2 - 4| \\ &= |x - 2| < \delta = \varepsilon. \end{aligned}$$

So $\frac{x^2-4}{x-2} \to 4$ as $x \to 2$.

Proof Technique (Keeping δ Away from Trouble). The next example uses $\delta = \min\left\{\frac{1}{2}, \frac{\varepsilon}{10}\right\}$ instead of $\min\{1, \dots\}$. The first restriction on δ has to keep x away from every point where the other factor is unbounded. Usually that is a point where a denominator is 0. In Example 6.1.7, the other factor is $\frac{|x+1|}{x^2}$, and the trouble is at $x = 0$. The restriction $\delta \leq 1$ only gives $0 < x < 2$. Then x^2 can be arbitrarily close to 0, and $\frac{|x+1|}{x^2}$ has no upper bound. The restriction $\delta \leq \frac{1}{2}$ gives $\frac{1}{2} < x < \frac{3}{2}$. Then $x^2 > \frac{1}{4}$ and $|x+1| < \frac{5}{2}$, so the factor is less than 10.

A safe rule is to make the first restriction at most half the distance from c to the nearest trouble point. Here $c = 1$ and the trouble point is 0, so $\frac{1}{2}$ works. In Example 6.1.5, $c = 7$ and the trouble point is 4, so $\delta \leq 1$ is already safe. It gives $6 < x < 8$ and $x - 4 > 2$. Exercise 6.1.1 (g) needs the same care, since its denominator is 0 at $x = 3$ and $c = 4$.

Example 6.1.7. $\lim_{x\to 1} \frac{1}{x^2} = 1$. Let $\varepsilon > 0$. Define $\delta = \min\left\{\frac{1}{2}, \frac{\varepsilon}{10}\right\}$. Then for all $x \in \mathbb{R} \setminus \{0\}$ with $0 < |x-1| < \delta$,

$$\begin{aligned}\left|\frac{1}{x^2} - 1\right| &= \left|\frac{1-x^2}{x^2}\right| \\ &= \frac{|x+1||x-1|}{x^2} \\ &< \frac{\frac{5}{2}|x-1|}{\frac{1}{4}} \\ &= 10|x-1| \\ &< 10\delta \leq 10 \cdot \frac{\varepsilon}{10} = \varepsilon.\end{aligned}$$

The third line holds since $|x-1| < \frac{1}{2}$ gives $\frac{1}{2} < x < \frac{3}{2}$, so $|x+1| < \frac{5}{2}$ and $x^2 > \frac{1}{4}$. So $\frac{1}{x^2} \to 1$ as $x \to 1$.

Example 6.1.8. $\lim_{x\to c} x^2 = c^2$. Let $\varepsilon > 0$. Define $\delta = \min\left\{1, \frac{\varepsilon}{2|c|+1}\right\}$. Then for all x with $0 < |x-c| < \delta$,

$$\begin{aligned}|x^2 - c^2| &= |x+c||x-c| \\ &\leq (|x| + |c|)|x-c| \\ &< (2|c|+1)|x-c| \\ &< (2|c|+1)\delta \leq (2|c|+1)\frac{\varepsilon}{2|c|+1} = \varepsilon.\end{aligned}$$

The third line holds since $|x| \leq |x-c| + |c| < 1 + |c|$. So $x^2 \to c^2$ as $x \to c$.

Example 6.1.9. The function $f : \mathbb{R} \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}, \end{cases}$$

does not have a limit at any point. To show this, start by letting $c \in \mathbb{R}$. Assume by way of contradiction that $f(x) \to L \in \mathbb{R}$ as $x \to c$. Let $\varepsilon = \frac{1}{2}$. So there exists a

$\delta > 0$ such that for all $x \in \mathbb{R}$ with $0 < |x - c| < \delta$, $|f(x) - L| < \frac{1}{2}$. Now there exists an $x_1 \in \mathbb{Q}$ and there exists an $x_2 \in \mathbb{R} \setminus \mathbb{Q}$ such that $x_1, x_2 \in (c - \delta, c + \delta) \setminus \{c\}$ by Theorem 1.8.11 and Exercise 1.8.5. Then

$$1 = |f(x_1) - f(x_2)| = |f(x_1) - L + L - f(x_2)| \leq |f(x_1) - L| + |f(x_2) - L| < \frac{1}{2} + \frac{1}{2} = 1,$$

a contradiction. So $\lim_{x \to c} f(x)$ does not exist. Since c was arbitrary, $\lim_{x \to c} f(x)$ does not exist for all $c \in \mathbb{R}$. This function is called the *Dirichlet function*.

Theorem 6.1.10. *Let $E \subset \mathbb{R}$, $f : E \to \mathbb{R}$, and c a limit point of E. If $\lim_{x \to c} f(x)$ exists, it is unique.*

PROOF. Suppose there exist $L_1, L_2 \in \mathbb{R}$ with $\lim_{x \to c} f(x) = L_1$ and $\lim_{x \to c} f(x) = L_2$. Assume by way of contradiction that $L_1 \neq L_2$. Let $\varepsilon = \frac{|L_1 - L_2|}{2} > 0$. So there exists a $\delta_1 > 0$ such that for all $x \in E$ with $0 < |x - c| < \delta_1$, $|f(x) - L_1| < \frac{|L_1 - L_2|}{2}$ and there exists a $\delta_2 > 0$ such that for all $x \in E$ with $0 < |x - c| < \delta_2$, $|f(x) - L_2| < \frac{|L_1 - L_2|}{2}$. Since c is a limit point of E, it is possible to choose $x_0 \in E$ such that $0 < |x_0 - c| < \min\{\delta_1, \delta_2\}$. Then

$$\begin{aligned} |L_1 - L_2| &= |L_1 - f(x_0) + f(x_0) - L_2| \\ &\leq |f(x_0) - L_1| + |f(x_0) - L_2| \\ &< \frac{|L_1 - L_2|}{2} + \frac{|L_1 - L_2|}{2} = |L_1 - L_2|, \end{aligned}$$

a contradiction. So $L_1 = L_2$. □

Lemma 6.1.11. *Let $E \subset \mathbb{R}$, $f : E \to \mathbb{R}$, and c a limit point of E. Suppose $\lim_{x \to c} f(x) = L \neq 0$. If $L > 0$, then there exists a $\delta > 0$ such that $f(x) > \frac{L}{2} > 0$ for all $x \in E$ with $0 < |x - c| < \delta$. If $L < 0$, then there exists a $\delta > 0$ such that $f(x) < \frac{L}{2} < 0$ for all $x \in E$ with $0 < |x - c| < \delta$.*

PROOF. See Exercise 6.1.3. □

Lemma 6.1.12. *Let $E \subset \mathbb{R}$, $f : E \to \mathbb{R}$, and c a limit point of E. Suppose $\lim_{x \to c} f(x)$ exists. Then there exist $M > 0$ and $\delta > 0$ such that $|f(x)| \leq M$ for all $x \in E$ with $0 < |x - c| < \delta$.*

PROOF. See Exercise 6.1.4. □

Theorem 6.1.13. *Let $E \subset \mathbb{R}$, $f, g : E \to \mathbb{R}$, and c a limit point of E. Suppose $f(x) \to L_1$ and $g(x) \to L_2$ as $x \to c$. Then*

(1) $f(x) + g(x) \to L_1 + L_2$ as $x \to c$;
(2) $f(x)g(x) \to L_1 L_2$ as $x \to c$; and
(3) if $L_2 \neq 0$, $\frac{f(x)}{g(x)} \to \frac{L_1}{L_2}$ as $x \to c$.

PROOF. (1) This part is an exercise.
(2) This part is an exercise.
(3) By Lemma 6.1.11, there exists a $\delta_1 > 0$ such that for all $x \in E$ with $0 < |x - c| < \delta_1$, $|g(x)| > \frac{|L_2|}{2}$, and hence, $\frac{1}{|g(x)|} < \frac{2}{|L_2|}$. In particular, $g(x) \neq 0$, so $\frac{f(x)}{g(x)}$ is defined for these x. Let $\varepsilon > 0$. So there exists a $\delta_2 > 0$

such that for all $x \in E$ with $0 < |x - c| < \delta_2$, $|g(x) - L_2| < \frac{|L_2|^2 \varepsilon}{2}$. Set $\delta = \min\{\delta_1, \delta_2\}$. So for all $x \in E$ with $0 < |x - c| < \delta$,

$$\left|\frac{1}{g(x)} - \frac{1}{L_2}\right| = \frac{|g(x) - L_2|}{|g(x)||L_2|} < \frac{2}{|L_2|} \cdot \frac{1}{|L_2|} \cdot \frac{|L_2|^2}{2}\varepsilon = \varepsilon.$$

So $\frac{1}{g(x)} \to \frac{1}{L_2}$ as $x \to c$. Then by (2), $\frac{f(x)}{g(x)} \to \frac{L_1}{L_2}$ as $x \to c$. □

Corollary 6.1.14. *Let $p(x)$ and $q(x)$ be polynomials with $q(c) \neq 0$. Then*

(1) $\lim_{x\to c} p(x) = p(c)$; and
(2) $\lim_{x\to c} \frac{p(x)}{q(x)} = \frac{p(c)}{q(c)}$.

Proof. By Exercise 6.1.1, $\lim_{x\to c} k = k$ for every constant k and $\lim_{x\to c} x = c$. Applying Theorem 6.1.13 (2) repeatedly gives $\lim_{x\to c} x^j = c^j$ for every $j \in \mathbb{N}$, and then $\lim_{x\to c} a_j x^j = a_j c^j$ for every $a_j \in \mathbb{R}$. A polynomial $p(x) = a_n x^n + \cdots + a_1 x + a_0$ is a finite sum of such terms, so (1) follows from Theorem 6.1.13 (1). Part (2) follows from (1) and Theorem 6.1.13 (3). □

Theorem 6.1.15. *Let $E \subset \mathbb{R}$, $f, g : E \to \mathbb{R}$, and c a limit point of E. Suppose there exists a $\delta_0 > 0$ such that $f(x) \leq g(x)$ for all $x \in E$ with $0 < |x - c| < \delta_0$. Then if $f(x) \to L_1$ and $g(x) \to L_2$ as $x \to c$, $L_1 \leq L_2$.*

Proof. Assume by way of contradiction that $L_1 > L_2$. Then $\lim_{x\to c}(f(x) - g(x)) = L_1 - L_2 > 0$. So by Lemma 6.1.11, there exists a $\delta_1 > 0$ such that $f(x) - g(x) > \frac{L_1 - L_2}{2} > 0$ for all $x \in E$ with $0 < |x - c| < \delta_1$. Since c is a limit point of E, there exists an $x \in E$ with $0 < |x - c| < \min\{\delta_0, \delta_1\}$. For this x, $f(x) \leq g(x)$ and $f(x) > g(x)$, a contradiction. □

Theorem 6.1.16 (Squeeze/Sandwich Theorem)**.** *Let $E \subset \mathbb{R}$, $f, g, h : E \to \mathbb{R}$, and c a limit point of E. Suppose there exists a $\delta_0 > 0$ such that $f(x) \leq g(x) \leq h(x)$ for all $x \in E$ with $0 < |x - c| < \delta_0$. If $f(x) \to L$ and $h(x) \to L$ as $x \to c$, then $g(x) \to L$ as $x \to c$.*

Proof. Let $\varepsilon > 0$. Since $f(x) \to L$ as $x \to c$, there exists a $\delta_1 > 0$ such that for all $x \in E$ with $0 < |x - c| < \delta_1$, $|f(x) - L| < \varepsilon$. So for all $x \in E$ with $0 < |x - c| < \min\{\delta_0, \delta_1\}$, $-\varepsilon < f(x) - L \leq g(x) - L$. Also, since $h(x) \to L$ as $x \to c$, there exists a $\delta_2 > 0$ such that for all $x \in E$ with $0 < |x - c| < \delta_2$, $|h(x) - L| < \varepsilon$. So for all $x \in E$ with $0 < |x - c| < \min\{\delta_0, \delta_2\}$, $g(x) - L \leq h(x) - L < \varepsilon$. Set $\delta = \min\{\delta_0, \delta_1, \delta_2\}$. So for all $x \in E$ with $0 < |x - c| < \delta$,

$$-\varepsilon < f(x) - L \leq g(x) - L \leq h(x) - L < \varepsilon;$$

i.e. for all $x \in E$ with $0 < |x - c| < \delta$, $|g(x) - L| < \varepsilon$. So $g(x) \to L$ as $x \to c$. □

Example 6.1.17. The limit $\lim_{x\to 0} x \sin\frac{1}{x} = 0$. To use the Squeeze Theorem, notice for $x \neq 0$,

$$-1 \leq \sin\frac{1}{x} \leq 1,$$

and so

$$-|x| \leq x \sin \frac{1}{x} \leq |x|.$$

Now $-|x| \to 0$ and $|x| \to 0$ as $x \to 0$ by Exercise 6.1.1, so by the Squeeze Theorem (Theorem 6.1.16), $\lim_{x\to 0} x \sin \frac{1}{x} = 0$.

Theorem 6.1.18. *Let $E \subset \mathbb{R}$, $f : E \to \mathbb{R}$, and c a limit point of E. Then $\lim_{x\to c} f(x) = L$ if and only if $\lim_{n\to\infty} f(x_n) = L$ for every sequence $\{x_n\}$ in E with $x_n \neq c$ for all $n \in \mathbb{N}$ and $x_n \to c$.*

Proof. ($\Rightarrow$) Suppose $\lim_{x\to c} f(x) = L$. Let $\{x_n\}$ be any sequence in E with $x_n \neq c$ for all $n \in \mathbb{N}$ and $x_n \to c$. Let $\varepsilon > 0$. Since $\lim_{x\to c} f(x) = L$, there exists a $\delta > 0$ such that for all $x \in E$ with $0 < |x - c| < \delta$, $|f(x) - L| < \varepsilon$. Since $x_n \to c$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $0 < |x_n - c| < \delta$. Thus for all $n \geq N$, $|f(x_n) - L| < \varepsilon$. Therefore $f(x_n) \to L$.

($\Leftarrow$) Conversely, suppose $f(x_n) \to L$ for every sequence $\{x_n\}$ in E with $x_n \neq c$ for all $n \in \mathbb{N}$ and $x_n \to c$. Assume by way of contradiction $\lim_{x\to c} f(x) \neq L$. Then there exists an $\varepsilon > 0$ such that for all $\delta > 0$, there exists an $x \in E$ with $0 < |x - c| < \delta$ and $|f(x) - L| \geq \varepsilon$. For each $n \in \mathbb{N}$, define $\delta_n = \frac{1}{n}$. So for all $n \in \mathbb{N}$, there exists an $x_n \in E$ with $0 < |x_n - c| < \frac{1}{n}$ and $|f(x_n) - L| \geq \varepsilon$. Thus $x_n \to c$ but $\{f(x_n)\}$ does not converge to L, a contradiction. So $\lim_{x\to c} f(x) = L$. □

Example 6.1.19. Let $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ be defined by $f(x) = \sin \frac{1}{x}$. Define $x_n = \frac{2}{(2n-1)\pi}$. Then $f(x_n) = \sin \frac{(2n-1)\pi}{2} = (-1)^{n+1}$. So $\{f(x_n)\}$ does not converge. But $x_n \to 0$. So $\lim_{x\to 0} \sin \frac{1}{x}$ does not exist.

Proof Technique (Showing a Limit Does Not Exist). By Theorem 6.1.18, $\lim_{x\to c} f(x) = L$ means $f(x_n) \to L$ for every sequence $\{x_n\}$ in E with $x_n \neq c$ and $x_n \to c$. So to show the limit does not exist, find one such sequence for which $\{f(x_n)\}$ diverges. Or find two such sequences for which $\{f(x_n)\}$ has two different limits. Example 6.1.19 uses one sequence. The Dirichlet function in Example 6.1.9 could be handled with two, one of rational numbers and one of irrational numbers.

Definition 6.1.20. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. Suppose $(c, b) \subset E$ for some $b > c$. Then L is the *right-hand limit* of f at c, denoted $\lim_{x\to c^+} f(x) = L$ or $f(x) \to L$ as $x \to c^+$, if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in E$ with $0 < x - c < \delta$, $|f(x) - L| < \varepsilon$.

Suppose $(a, c) \subset E$ for some $a < c$. Then L is the *left-hand limit* of f at c, denoted $\lim_{x\to c^-} f(x) = L$ or $f(x) \to L$ as $x \to c^-$, if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in E$ with $-\delta < x - c < 0$, $|f(x) - L| < \varepsilon$.

Example 6.1.21. $\lim_{x\to 0^+} \sqrt{x} = 0$ and $\lim_{x\to 0} \sqrt{x} = 0$. The second limit makes sense since the domain of $\sqrt{x}$ is $E = [0, \infty)$, and 0 is a limit point of E. Only $x \in E$ count in Definition 6.1.1.

Example 6.1.22. $\lim_{x\to 5^-} \lfloor x \rfloor = 4$ and $\lim_{x\to 5^+} \lfloor x \rfloor = 5$.

Theorem 6.1.23. *Let $E \subset \mathbb{R}$, $f : E \to \mathbb{R}$, and c a limit point of E. Suppose $(c-r, c) \cup (c, c+r) \subset E$ for some $r > 0$. Then $\lim_{x\to c} f(x)$ exists and equals L if and only if $\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = L$.*

PROOF. See Exercise 6.1.8. □

Exercises

6.1.1. Suppose c and k are real numbers. Prove each of the following limits using the ε-δ definition of limits.

(a) $\lim_{x\to c} k = k$;
(b) $\lim_{x\to c} x = c$;
(c) $\lim_{x\to c} |x| = |c|$;
(d) $\lim_{x\to c} \sqrt{x} = \sqrt{c}$ for $c > 0$.
(e) $\lim_{x\to 3} (6x + 7) = 25$
(f) $\lim_{x\to 2} (x^2 + 5x) = 14$
(g) $\lim_{x\to 4} \dfrac{1}{x^2 + 5x - 24} = \dfrac{1}{12}$
(h) $\lim_{x\to 3} \dfrac{1}{x - 2} = 1$

[Hint, p. 338]

6.1.2. Prove the function $f : \mathbb{R} \setminus \{0\} \to \mathbb{R}$ defined by $f(x) = \frac{|x|}{x}$ does not have a limit at $c = 0$.

6.1.3. Prove Lemma 6.1.11. [Solution, p. 339]

6.1.4. Prove Lemma 6.1.12.

6.1.5. Prove Theorem 6.1.13 (1)–(2). [Solution, p. 339]

6.1.6. Let $E \subset \mathbb{R}$, $f : E \to \mathbb{R}$, and c a limit point of E. Let $k \in \mathbb{R}$. Prove that if there exists a $\delta > 0$ such that $f(x) > k$ for all $x \in E$ with $0 < |x - c| < \delta$, and if $\lim_{x\to c} f(x)$ exists, then $\lim_{x\to c} f(x) \geq k$.

6.1.7. Prove, using the Squeeze Theorem, that $\lim_{x\to 0} \frac{\sin x}{x} = 1$. Use this result to prove $\lim_{x\to 0} \frac{\cos x - 1}{x} = 0$. [Hint, p. 339]

6.1.8. Prove Theorem 6.1.23. [Solution, p. 339]

6.1.9. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = x$ if $x \in \mathbb{Q}$ and $f(x) = 0$ if $x \notin \mathbb{Q}$. Prove that $\lim_{x\to 0} f(x) = 0$ and that $\lim_{x\to c} f(x)$ does not exist for every $c \neq 0$.

[Hint, p. 340]

6.1.10. Prove that $\lim_{x\to 0} \cos \frac{1}{x}$ does not exist. [Hint, p. 340]

6.2. Infinite Limits and Limits at Infinity

Section 6.1 gives meaning to $\lim_{x\to c} f(x) = L$ when c and L are real numbers. This section allows c or L to be ∞ or $-\infty$. Each definition follows the same pattern. When $L = \infty$, "within ε of L" becomes "greater than M." When $c = \infty$, "within δ of c" becomes "greater than d." Remember that ∞ and $-\infty$ are not real numbers

(Definition 1.8.6). A statement such as $\lim_{x\to c} f(x) = \infty$ describes how f behaves. It does not say the limit exists in the sense of Definition 6.1.1.

Definition 6.2.1. Let $E \subset \mathbb{R}$, $f : E \to \mathbb{R}$, and c a limit point of E. Then $\lim_{x\to c} f(x) = \infty$ if for all $M > 0$, there exists a $\delta > 0$ such that for all $x \in E$ with $0 < |x - c| < \delta$, $f(x) > M$.

$\lim_{x\to c} f(x) = -\infty$ if for all $M < 0$, there exists a $\delta > 0$ such that for all $x \in E$ with $0 < |x - c| < \delta$, $f(x) < M$.

Definition 6.2.2. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. Suppose $(c, b) \subset E$ for some $b > c$. Then $\lim_{x\to c^+} f(x) = \infty$ if for all $M > 0$, there exists a $\delta > 0$ such that for all $x \in E$ with $0 < x - c < \delta$, $f(x) > M$. Also, $\lim_{x\to c^+} f(x) = -\infty$ if for all $M < 0$, there exists a $\delta > 0$ such that for all $x \in E$ with $0 < x - c < \delta$, $f(x) < M$. If $(a, c) \subset E$ for some $a < c$, then $\lim_{x\to c^-} f(x) = \infty$ and $\lim_{x\to c^-} f(x) = -\infty$ are defined in the same way, using $-\delta < x - c < 0$.

Example 6.2.3. $\lim_{x\to 0} \frac{1}{x^2} = \infty$. Let $M > 0$. Choose $\delta = \frac{1}{\sqrt{M}}$. Then for all $x \in \mathbb{R} \setminus \{0\}$ with $0 < |x| < \delta$,

$$\frac{1}{x^2} > \frac{1}{\delta^2} = \frac{1}{(1/\sqrt{M})^2} = M.$$

So $\lim_{x\to 0} \frac{1}{x^2} = \infty$.

Example 6.2.4. $\lim_{x\to 0^+} \frac{1}{x} = \infty$ and $\lim_{x\to 0^-} \frac{1}{x} = -\infty$. For the first, let $M > 0$ and choose $\delta = \frac{1}{M}$. Then for all x with $0 < x < \delta$, $\frac{1}{x} > \frac{1}{\delta} = M$. The second is similar. So $\frac{1}{x}$ has neither a limit nor an infinite limit at 0.

Definition 6.2.5. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. Suppose $(a, \infty) \subset E$ for some $a \in \mathbb{R}$. Then $\lim_{x\to\infty} f(x) = L$ if for all $\varepsilon > 0$, there exists a $d > 0$ such that for all $x \in E$ with $x > d$, $|f(x) - L| < \varepsilon$.

Suppose $(-\infty, b) \subset E$ for some $b \in \mathbb{R}$. Then $\lim_{x\to-\infty} f(x) = L$ if for all $\varepsilon > 0$, there exists a $d < 0$ such that for all $x \in E$ with $x < d$, $|f(x) - L| < \varepsilon$.

Definition 6.2.6. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. Suppose $(a, \infty) \subset E$ for some $a \in \mathbb{R}$. Then $\lim_{x\to\infty} f(x) = \infty$ if for all $M > 0$, there exists a $d > 0$ such that for all $x \in E$ with $x > d$, $f(x) > M$. The statements $\lim_{x\to\infty} f(x) = -\infty$, $\lim_{x\to-\infty} f(x) = \infty$ and $\lim_{x\to-\infty} f(x) = -\infty$ are defined in the same way.

Remark 6.2.7. The limit laws (Theorem 6.1.13), Theorem 6.1.15 and the Squeeze Theorem (Theorem 6.1.16) also hold for one-sided limits and for limits as $x \to \infty$ or $x \to -\infty$, when the limits L_1, L_2 and L are real numbers. The proofs are the same, with $0 < |x - c| < \delta$ replaced by $0 < x - c < \delta$, by $-\delta < x - c < 0$, or by $x > d$ or $x < d$.

Theorem 6.2.8. *Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. Theorem 6.1.18 remains true in each of these cases.*

(1) For a right-hand limit, using sequences in E with $x_n > c$ and $x_n \to c$. For a left-hand limit, using sequences with $x_n < c$.

(2) *When $L = \infty$ or $L = -\infty$, where $f(x_n) \to \pm\infty$ is meant in the sense of Definition 3.6.5.*
(3) *For limits as $x \to \infty$ or $x \to -\infty$, using sequences in E with $x_n \to \infty$ or $x_n \to -\infty$.*

PROOF. The proofs follow the proof of Theorem 6.1.18. Here is one case. The others are an exercise. Suppose $(c, b) \subset E$. The claim is that $\lim_{x\to c^+} f(x) = \infty$ if and only if $f(x_n) \to \infty$ for every sequence $\{x_n\}$ in E with $x_n > c$ and $x_n \to c$.

($\Rightarrow$) Let $\{x_n\}$ be such a sequence and let $M > 0$. Choose $\delta > 0$ such that $f(x) > M$ for all $x \in E$ with $0 < x - c < \delta$. Since $x_n \to c$ and $x_n > c$, there exists an $N \in \mathbb{N}$ such that $0 < x_n - c < \delta$ for all $n \geq N$. So $f(x_n) > M$ for all $n \geq N$. Thus $f(x_n) \to \infty$.

($\Leftarrow$) Suppose $\lim_{x\to c^+} f(x) \neq \infty$. Negating the definition, there exists an $M > 0$ such that for every $\delta > 0$, there exists an $x \in E$ with $0 < x - c < \delta$ and $f(x) \leq M$. Taking $\delta = \frac{1}{n}$ gives, for each $n \in \mathbb{N}$, an $x_n \in E$ with $c < x_n < c + \frac{1}{n}$ and $f(x_n) \leq M$. Then $x_n > c$ and $x_n \to c$, but $\{f(x_n)\}$ does not diverge to ∞. □

Exercises

6.2.1. Prove $\lim_{x\to\infty} \frac{1}{x} = 0$. [Solution, p. 340]

6.2.2. Prove that $\lim_{x\to\infty} x^2 = \infty$ and $\lim_{x\to-\infty} x^3 = -\infty$.

6.2.3. Prove $\lim_{x\to\infty} \frac{2x+1}{x+3} = 2$. [Hint, p. 340]

6.2.4. Prove the remaining cases of Theorem 6.2.8.

6.2.5. Prove that Theorem 6.1.23 holds when $L = \infty$ or $L = -\infty$.

6.3. Continuity

In calculus, a function is continuous if its graph can be drawn without lifting the pencil. The definition here says $f(x)$ is close to $f(c)$ whenever x is close to c. The precise definition can behave in ways the pencil picture does not predict. Example 6.3.2 is a function that is continuous at every irrational number and at no rational number.

Definition 6.3.1. Let $E \subset \mathbb{R}$, $f : E \to \mathbb{R}$, and $c \in E$. Suppose $(c - r, c + r) \subset E$ for some $r > 0$. Then f is *continuous* at c if $\lim_{x\to c} f(x) = f(c)$. If f is continuous at every point in an interval $I \subset E$, f is continuous on I.

Example 6.3.2. The function $f : (0, 1) \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0, & x \in (0,1) \setminus \mathbb{Q}, \\ \frac{1}{n}, & x \in \mathbb{Q} \cap (0,1) \text{ with } x = \frac{m}{n} \text{ in lowest terms,} \end{cases}$$

is continuous at every irrational number in $(0, 1)$ and is not continuous at any rational number in $(0, 1)$. Here $x = \frac{m}{n}$ in lowest terms means n is the smallest positive integer such that $x = \frac{m}{n}$ for some integer m. Such an n exists by the Well-Ordering Principle.

Let $c \in (0, 1)$. Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$. There exist only finitely many $x \in (0, 1) \cap \mathbb{Q}$ with $x = \frac{m}{n}$ in lowest terms and $n \leq N$. For each $n \leq N$, there

are at most $n-1$ of them, namely some of $\frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}$. Denote these rationals by $r_1, \ldots, r_k$. Set

$$\delta = \min\{|r_i - c| : i = 1, \ldots, k,\ r_i \neq c\}.$$

(If no r_i differs from c, let $\delta = 1$.) Then for all $x \in (0,1)$ with $0 < |x-c| < \delta$, $f(x) = 0$ or $f(x) = \frac{1}{n}$ with $n \geq N$. So for $c \notin \mathbb{Q}$, $|f(x) - f(c)| = 0$ or $|f(x) - f(c)| = \frac{1}{n} < \varepsilon$. So $\lim_{x\to c} f(x) = 0 = f(c)$. So f is continuous at c. The same argument shows $\lim_{x\to c} f(x) = 0$ for every $c \in (0,1)$, rational or not. Since $\lim_{x\to c} f(x) = 0$, if $c \in \mathbb{Q}$, $\lim_{x\to c} f(x) \neq f(c)$. So f is not continuous at c.

This function is called *Thomae's function*. See Figure 6.1.

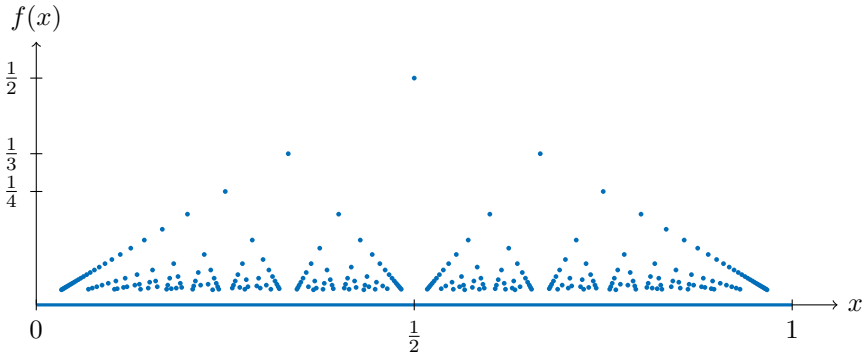


FIGURE 6.1. Thomae's function, showing the points $\left(\frac{m}{n}, \frac{1}{n}\right)$ with $n \leq 30$. At irrational numbers the value is 0, shown by the line along the bottom.

Theorem 6.3.3. *Let $E \subset \mathbb{R}$, $f, g : E \to \mathbb{R}$, and $c \in E$. Suppose f and g are continuous at c. Then the functions $f+g$, fg, and $\frac{f}{g}$, if $g(c) \neq 0$, are continuous at c.*

PROOF. This follows from Theorem 6.1.13. For example, $\lim_{x\to c}(f(x) + g(x)) = f(c) + g(c)$, so $f + g$ is continuous at c. □

Theorem 6.3.4. *Let $D, E \subset \mathbb{R}$, $g : E \to \mathbb{R}$ with $g[E] \subset D$, $f : D \to \mathbb{R}$, and $c \in E$. Suppose g is continuous at c and f is continuous at $g(c)$. Then $f \circ g$ is continuous at c.*

PROOF. Let $\varepsilon > 0$. Because $\lim_{y\to g(c)} f(y) = f(g(c))$, there exists a $\delta_1 > 0$ such that $|f(y) - f(g(c))| < \varepsilon$ for all $y \in D$ with $|y - g(c)| < \delta_1$. Since $\lim_{x\to c} g(x) = g(c)$, there exists a $\delta > 0$ such that for all $x \in E$ with $|x - c| < \delta$, $|g(x) - g(c)| < \delta_1$. So for all $x \in E$ with $|x - c| < \delta$, $|f(g(x)) - f(g(c))| < \varepsilon$. □

Definition 6.3.5. Let $f : [a,b] \to \mathbb{R}$. Then f is continuous at a if $\lim_{x\to a^+} f(x) = f(a)$. Similarly, f is continuous at b if $\lim_{x\to b^-} f(x) = f(b)$.

Definitions 6.3.1 and 6.3.5 cover functions defined on intervals. Later results, such as Theorem 6.4.7, need continuity on other sets. The next definition covers every case at once.

Definition 6.3.6. Let $E \subset \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $c \in E$. Then f is *continuous at c relative to E* if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for all $x \in E$ with $|x - c| < \delta$, $|f(x) - f(c)| < \varepsilon$. If f is continuous relative to E at every point of E, then f is *continuous on E*.

Remark 6.3.7. If $(c - r, c + r) \subset E$ for some $r > 0$, this agrees with Definition 6.3.1. If $E = [a, b]$ and c is a or b, it agrees with Definition 6.3.5. If c is a limit point of E, then f is continuous at c relative to E if and only if $\lim_{x \to c} f(x) = f(c)$. If c is an isolated point of E, every function on E is continuous at c. There is a $\delta > 0$ such that the only $x \in E$ with $|x - c| < \delta$ is $x = c$. Theorems 6.3.3 and 6.3.4 hold for this definition too, with the same proofs.

Definition 6.3.8. Let $E \subset \mathbb{R}$. The set of all functions $f : E \to \mathbb{R}$ that are continuous on E is denoted $C(E)$. For intervals, write $C[a, b]$ for $C([a, b])$, $C(a, b)$ for $C((a, b))$, and so on. By Theorem 6.3.3, if $f, g \in C(E)$, then $f + g \in C(E)$ and $fg \in C(E)$.

Theorem 6.3.9. *Let $E \subset \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $c \in E$. Then f is continuous at c relative to E if and only if $f(x_n) \to f(c)$ for every sequence $\{x_n\}$ in E with $x_n \to c$.*

Proof. ($\Rightarrow$) Let $\{x_n\}$ be a sequence in E with $x_n \to c$, and let $\varepsilon > 0$. Choose $\delta > 0$ such that $|f(x) - f(c)| < \varepsilon$ for all $x \in E$ with $|x - c| < \delta$. Since $x_n \to c$, there exists an $N \in \mathbb{N}$ such that $|x_n - c| < \delta$ for all $n \geq N$. So $|f(x_n) - f(c)| < \varepsilon$ for all $n \geq N$. Thus $f(x_n) \to f(c)$.

($\Leftarrow$) Suppose f is not continuous at c relative to E. Then there exists an $\varepsilon > 0$ such that for all $\delta > 0$, there exists an $x \in E$ with $|x - c| < \delta$ and $|f(x) - f(c)| \geq \varepsilon$. Taking $\delta = \frac{1}{n}$, for each $n \in \mathbb{N}$ there exists an $x_n \in E$ with $|x_n - c| < \frac{1}{n}$ and $|f(x_n) - f(c)| \geq \varepsilon$. Then $x_n \to c$, but $\{f(x_n)\}$ does not converge to $f(c)$. □

Unlike Theorem 6.1.18, this theorem allows $x_n = c$. That is because continuity at c is about $f(c)$ itself.

Proof Technique (Locating a Point with a Supremum). To find a point with some property, it often helps to define a set S of points that are "on one side" and let $c = \sup S$. Then use the two facts about suprema from Section 1.8. Points of S come arbitrarily close to c from the left. Points to the right of c are not in S. Each fact gives an inequality, and together they pin down c. The proof of the Intermediate Value Theorem below works this way. So did the proof of the Heine–Borel Theorem (Theorem 4.2.14).

Theorem 6.3.10 (Intermediate Value Theorem)**.** *Suppose that $f : [a, b] \to \mathbb{R}$ is continuous, $f(a) \neq f(b)$, and k is any number between $f(a)$ and $f(b)$. Then there exists a $c \in (a, b)$ such that $f(c) = k$.*

Proof. Suppose $f(a) < k < f(b)$. Define the set $S = \{x \in [a, b] : f(x) \leq k\}$. Then $a \in S$, so $S \neq \emptyset$, and since $S \subset [a, b]$, S is bounded above. Thus, $\sup S \in \mathbb{R}$, say $c = \sup S$.

If $c = b$, then $f(c) = f(b) > k$. If $c < b$, then since c is an upper bound of S, for all x with $c < x \leq b$, $x \notin S$. So for all x with $c < x \leq b$, $f(x) > k$. Therefore,

$$f(c) = \lim_{x \to c} f(x) = \lim_{x \to c^+} f(x) \geq k$$

by Exercise 6.1.6.

Now, since c is the supremum of S, for all $n \in \mathbb{N}$, there exists an $x_n \in S$ with $c - \frac{1}{n} < x_n \leq c$. By the Squeeze Theorem, $x_n \to c$. Also, $f(x_n) \leq k$ for all $n \in \mathbb{N}$, since for all $n \in \mathbb{N}$, $x_n \in S$. Therefore,

$$f(c) = \lim_{n \to \infty} f(x_n) \leq k$$

by Theorems 3.2.7 and 6.3.9. So $f(c) = k$. Since $f(a) < k < f(b)$, $c \neq a$ and $c \neq b$. So $c \in (a, b)$.

Now suppose $f(b) < k < f(a)$. Define $g(x) = -f(x)$. Then g is continuous on $[a, b]$ and $-f(a) = g(a) < -k < g(b) = -f(b)$. So there exists a $c \in (a, b)$ such that $g(c) = -k$. So $f(c) = k$. □

Corollary 6.3.11. *If $I \subset \mathbb{R}$ is an interval and $f : I \to \mathbb{R}$ is continuous on I, then $f[I]$ is an interval.*

Proof. If I consists of a single point, $f[I]$ is a single point and the proof is complete. Assume now there exists $s, t \in f[I]$ with $s < t$. Then there exist $x, y \in I$ with $f(x) = s$, $f(y) = t$. Choose $k \in (s, t)$. Then there exists a c between x and y with $f(c) = k$ by the Intermediate Value Theorem, applied on the closed interval with endpoints x and y. This interval is a subset of I, since I is an interval. So $k \in f[I]$ and $f[I]$ is an interval. □

Corollary 6.3.12. *If $f : [0, 1] \to [0, 1]$ is continuous, there exists a $c \in [0, 1]$ such that $f(c) = c$.*

Proof. Let $g(x) = f(x) - x$. Then $g(0) = f(0) \geq 0$ and $g(1) = f(1) - 1 \leq 0$. Thus there exists a $c \in [0, 1]$ such that $g(c) = 0$; i.e. where $f(c) = c$. If $g(0) = 0$ or $g(1) = 0$, take $c = 0$ or $c = 1$. Otherwise $g(1) < 0 < g(0)$, and the Intermediate Value Theorem gives c. □

Definition 6.3.13. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. Then f is *bounded above* on E if there exists an $M \in \mathbb{R}$ such that $f(x) \leq M$ for all $x \in E$. f is *bounded below* on E if there exists an $M \in \mathbb{R}$ such that $f(x) \geq M$ for all $x \in E$. f is *bounded* if it is both bounded above and below.

Theorem 6.3.14. *Let $f : [a, b] \to \mathbb{R}$ be continuous. Then f is bounded on $[a, b]$.*

Proof. Assume by way of contradiction that f is not bounded above on $[a, b]$. Then for all $n \in \mathbb{N}$, there exists an $x_n \in [a, b]$ such that $f(x_n) \geq n$. But $\{x_n\}$ is contained in $[a, b]$, and so by the Bolzano–Weierstrass Theorem (Corollary 3.4.6), $\{x_n\}$ has a convergent subsequence, say $x_{n_k} \to c \in [a, b]$. Then $\lim_{k \to \infty} f(x_{n_k}) = f(c)$ by Theorem 6.3.9, a contradiction. The contradiction is that $f(x_{n_k}) \geq n_k \geq k$ (since $\{n_k\}$ is a strictly increasing sequence of positive integers), so $\{f(x_{n_k})\}$ is not bounded, but convergent sequences are bounded by Theorem 3.2.3. So f is bounded above on $[a, b]$. Similarly, f is bounded below on $[a, b]$. □

Theorem 6.3.15 (Extreme Value Theorem)**.** *Suppose that $f : [a, b] \to \mathbb{R}$ is continuous. Then f attains its maximum and minimum values on $[a, b]$; i.e. there exist $c_1, c_2 \in [a, b]$ such that $f(c_1) \leq f(x) \leq f(c_2)$ for all $x \in [a, b]$.*

PROOF. Let $W = \{f(x) : a \leq x \leq b\}$. From Theorem 6.3.14, W is bounded. Also, $W \neq \emptyset$. So $\inf W, \sup W \in \mathbb{R}$, say $\alpha = \inf W$, $\beta = \sup W$.

Since $\alpha = \inf W$, for all $n \in \mathbb{N}$, there exists a $y_n \in W$ such that $\alpha \leq y_n < \alpha + \frac{1}{n}$. By the way W is defined, there exists an $x_n \in [a, b]$ such that $y_n = f(x_n)$. The sequence $\{x_n\}$ is contained in $[a, b]$, so it has a convergent subsequence, say $x_{n_k} \to c_1 \in [a, b]$. Since f is continuous at c_1, $f(c_1) = \lim_{k\to\infty} f(x_{n_k})$ by Theorem 6.3.9. But for all $k \in \mathbb{N}$,

$$\alpha \leq f(x_{n_k}) < \alpha + \frac{1}{n_k} \leq \alpha + \frac{1}{k}.$$

Here $n_k \geq k$, since $\{n_k\}$ is a strictly increasing sequence of positive integers. So $f(c_1) = \alpha$ by the Squeeze Theorem (Theorem 3.2.10). Similarly, there exists a $c_2 \in [a, b]$ with $f(c_2) = \beta$. □

Both hypotheses matter. The function $f(x) = x$ is continuous on $(0, 1)$ but has no largest or smallest value there. The function $f(x) = \frac{1}{x}$ is continuous on $(0, 1]$ but is not bounded. The function equal to x on $[0, 1)$ and to 0 at 1 is bounded on $[0, 1]$ but has no largest value.

Continuity can be described using open sets alone, with no ε or δ. This description is the one used in topology, and Part III uses it to define measurable functions.

Theorem 6.3.16. *Let $f : \mathbb{R} \to \mathbb{R}$. Then f is continuous on $\mathbb{R}$ if and only if for each open set $U \subset \mathbb{R}$, $f^{-1}[U]$ is an open set.*

PROOF. ($\Leftarrow$) Let $c \in \mathbb{R}$. Let $\varepsilon > 0$. The interval $I = (f(c) - \varepsilon, f(c) + \varepsilon)$ is open and so $f^{-1}[I]$ is open. Since $c \in f^{-1}[I]$ and $f^{-1}[I]$ is open, there exists a $\delta > 0$ such that $(c - \delta, c + \delta) \subset f^{-1}[I]$. So if $x \in (c - \delta, c + \delta)$, $f(x) \in (f(c) - \varepsilon, f(c) + \varepsilon)$; i.e. if $|x - c| < \delta$, $|f(x) - f(c)| < \varepsilon$. So f is continuous at c.

($\Rightarrow$) Suppose f is continuous and let $U \subset \mathbb{R}$ be open. Let $c \in f^{-1}[U]$. Then $f(c) \in U$ and since U is open, there exists an $\varepsilon > 0$ such that $(f(c) - \varepsilon, f(c) + \varepsilon) \subset U$. Since f is continuous at c, there exists a $\delta > 0$ such that for all x with $|x - c| < \delta$, $|f(x) - f(c)| < \varepsilon$. Thus for every $x \in (c - \delta, c + \delta)$, $f(x) \in (f(c) - \varepsilon, f(c) + \varepsilon) \subset U$. Hence $(c - \delta, c + \delta) \subset f^{-1}[U]$. So $f^{-1}[U]$ is open. □

Exercises

6.3.1. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = x \sin \frac{1}{x}$ for $x \neq 0$ and $f(0) = 0$. Prove f is continuous at 0. [Hint, p. 340]

6.3.2. Let f be the function in Exercise 6.1.9. Prove that f is continuous at 0 and at no other point.

6.3.3. Prove that the equation $x^5 - 3x + 1 = 0$ has a solution in $(0, 1)$.

6.3.4. Let f and g be continuous on $[a, b]$ with $f(a) < g(a)$ and $f(b) > g(b)$. Prove there exists a $c \in (a, b)$ with $f(c) = g(c)$. [Hint, p. 340]

6.3.5. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous with $f(q) = 0$ for every $q \in \mathbb{Q}$. Prove $f(x) = 0$ for every $x \in \mathbb{R}$. [Hint, p. 340]

6.3.6. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. Prove that f is continuous on E if and only if for each open set O, there is an open set U such that $f^{-1}[O] = E \cap U$. (This extends Theorem 6.3.16 to functions on any set E.) [Solution, p. 340]

6.4. Uniform Continuity

If f is continuous on E, then for each $x \in E$ and each $\varepsilon > 0$ there is a δ. That δ may depend on x. Uniform continuity asks for one δ that works at every point of E at once. In symbols, the only change is the order of "for all x" and "there exists a δ."

Definition 6.4.1. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. f is *uniformly continuous* on E if for all $\varepsilon > 0$, there exists a $\delta(\varepsilon) > 0$ such that for all $x, y \in E$ with $|x - y| < \delta$, $|f(x) - f(y)| < \varepsilon$.

Example 6.4.2. Let $E \subset \mathbb{R}$ be bounded. Then $f(x) = x^2$ is uniformly continuous on E. Since E is bounded, there exists an $M > 0$ such that $|x| < M$ for all $x \in E$. Let $\varepsilon > 0$. Define $\delta = \frac{\varepsilon}{2M}$. Then for all $x, y \in E$ with $|x - y| < \delta$,

$$|f(x) - f(y)| = |x^2 - y^2| = |x + y||x - y| < 2M|x - y| < 2M\delta = \varepsilon.$$

Example 6.4.3. $f(x) = \sin x$ is uniformly continuous on $\mathbb{R}$. Let $\varepsilon > 0$. Define $\delta = \varepsilon$. Then for all $x, y \in \mathbb{R}$ with $|x - y| < \delta$,

$$\begin{aligned} |f(x) - f(y)| &= |\sin x - \sin y| \\ &= 2\left|\cos\left[\tfrac{1}{2}(x + y)\right] \sin\left[\tfrac{1}{2}(x - y)\right]\right| \\ &\leq 2\left|\sin\left[\tfrac{1}{2}(x - y)\right]\right| \\ &\leq 2\left|\tfrac{1}{2}(x - y)\right| \\ &= |x - y| < \delta = \varepsilon. \end{aligned}$$

This uses the identity $\sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$ and the inequality $|\sin t| \leq |t|$. See Remark 3.1.1.

Example 6.4.4. The function $f(x) = \frac{1}{x}$ is not uniformly continuous on $(0, 1)$. Assume by way of contradiction that f is uniformly continuous on $(0, 1)$. Let $\varepsilon = 1$. Then there exists a $\delta > 0$ such that for all $x, y \in (0, 1)$ with $|x - y| < \delta$, $|f(x) - f(y)| < \varepsilon$. Choose $N \in \mathbb{N}$ with $N > \frac{1}{\delta}$ and $N \geq 2$. Define $x = \frac{1}{N}$, $y = \frac{1}{N+1}$. Then

$$|x - y| = \left|\frac{1}{N} - \frac{1}{N + 1}\right| = \frac{1}{N(N + 1)} < \frac{1}{N} < \delta,$$

but $|f(x) - f(y)| = |N - (N + 1)| = 1 = \varepsilon$, a contradiction. So f is not uniformly continuous on $(0, 1)$.

Proof Technique (Proving a Function Is Not Uniformly Continuous). Negate the definition. A function f is not uniformly continuous on E if there exists an $\varepsilon > 0$ such that for every $\delta > 0$, there exist $x, y \in E$ with $|x - y| < \delta$ and $|f(x) - f(y)| \geq \varepsilon$. So fix one ε. Then, for an arbitrary δ, find two points

closer than δ whose values are at least ε apart. The points usually go where the graph is steepest. In Example 6.4.4, they go near 0.

Definition 6.4.5. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. f is *Lipschitz continuous* on E (or satisfies a *Lipschitz condition* on E) if there exists an $M > 0$ such that for all $x, y \in E$, $|f(x) - f(y)| \leq M|x - y|$.

Theorem 6.4.6. *Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. If f is Lipschitz continuous on E, f is uniformly continuous on E.*

Proof. See Exercise 6.4.3. □

Theorem 6.4.7. *Let $K \subset \mathbb{R}$ be compact and $f : K \to \mathbb{R}$. If f is continuous on K, f is uniformly continuous on K.*

Proof. Let $\varepsilon > 0$. Since f is continuous on K, for all $x \in K$, there exists a $\delta_x > 0$ such that for all $y \in K$ with $0 < |y - x| < 2\delta_x$, $|f(y) - f(x)| < \frac{\varepsilon}{2}$. The collection $\{(x-\delta_x, x+\delta_x)\}_{x\in K}$ is an open cover of K. Since K is compact, there exists $x_1, \ldots, x_n$ such that $K \subset \bigcup_{i=1}^{n}(x_i - \delta_{x_i}, x_i + \delta_{x_i})$. Define $\delta = \min\{\delta_{x_1}, \ldots, \delta_{x_n}\} > 0$. Suppose $x, y \in K$ with $|x - y| < \delta$. Since $x \in K$, there exists an x_i such that $x \in (x_i - \delta_{x_i}, x_i + \delta_{x_i})$. Since $|x - y| < \delta$, $y \in (x_i - 2\delta_{x_i}, x_i + 2\delta_{x_i})$. Indeed, $|y - x_i| \leq |y - x| + |x - x_i| < \delta + \delta_{x_i} \leq 2\delta_{x_i}$. Then

$$|f(x) - f(y)| \leq |f(x) - f(x_i)| + |f(y) - f(x_i)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So f is uniformly continuous on K. □

Corollary 6.4.8. *If f is continuous on $[a, b]$, f is uniformly continuous on $[a, b]$.*

Proof. This is an immediate consequence of Theorem 6.4.7 and the Heine–Borel Theorem (Theorem 4.2.14). □

Exercises

6.4.1. Let $a > 0$. Prove that $f(x) = \frac{1}{x}$ is uniformly continuous on $[a, \infty)$. [Solution, p. 341]

6.4.2. Prove $f(x) = x^2$ is not uniformly continuous on $\mathbb{R}$. [Solution, p. 341]

6.4.3. Prove Theorem 6.4.6.

6.4.4. Prove that $f(x) = \sqrt{x}$ is uniformly continuous on $[0, \infty)$. Show that f is not Lipschitz continuous on $[0, 1]$. [Hint, p. 341]

6.4.5. Let f be uniformly continuous on E and let $\{x_n\}$ be a Cauchy sequence in E. Prove that $\{f(x_n)\}$ is a Cauchy sequence. Use this to give another proof that $\frac{1}{x}$ is not uniformly continuous on $(0, 1)$. [Hint, p. 341]

6.4.6. Prove that $f(x) = \sin\frac{1}{x}$ is not uniformly continuous on $(0, 1)$. [Hint, p. 341]

6.5. Monotonicity

Monotone functions are simpler than general functions. Their one-sided limits always exist, so the only way they can be discontinuous is to jump. This section shows they can jump at only countably many points. It also shows the inverse of a continuous strictly monotone function is continuous. That gives the continuity of $\sqrt[n]{x}$ with no extra work.

Definition 6.5.1. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. Then

(1) f is *increasing* on E if $f(x) \leq f(y)$ for all $x, y \in E$ with $x < y$;
(2) f is *strictly increasing* on E if $f(x) < f(y)$ for all $x, y \in E$ with $x < y$;
(3) f is *decreasing* on E if $f(x) \geq f(y)$ for all $x, y \in E$ with $x < y$;
(4) f is *strictly decreasing* on E if $f(x) > f(y)$ for all $x, y \in E$ with $x < y$;
(5) f is *monotone* on E if f is increasing or decreasing on E; and
(6) f is *strictly monotone* on E if f is strictly increasing or strictly decreasing on E.

Theorem 6.5.2. *Let $I \subset \mathbb{R}$ be an open interval and $f : I \to \mathbb{R}$ increasing on I. Then for all $c \in I$, $\lim_{x\to c^+} f(x)$ and $\lim_{x\to c^-} f(x)$ exist and*

$$\begin{aligned}\sup\{f(x) : x < c,\ x \in I\} &= \lim_{x\to c^-} f(x) \leq f(c)\\ &\leq \lim_{x\to c^+} f(x) = \inf\{f(x) : x > c,\ x \in I\}.\end{aligned}$$

Furthermore, if $c, d \in I$ with $c < d$, $\lim_{x\to c^+} f(x) \leq \lim_{x\to d^-} f(x)$.

Proof. Let $c \in I$ and let $A = \{f(x) : x \in I,\ x < c\}$. Since I is open, $A \neq \emptyset$. Since f is increasing, $f(c)$ is an upper bound of A. So $L = \sup A$ exists and $L \leq f(c)$. Let $\varepsilon > 0$. By Theorem 1.8.7, there exists an $x_0 \in I$ with $x_0 < c$ and $L - \varepsilon < f(x_0)$. Let $\delta = c - x_0$. If $c - \delta < x < c$, then $x \in I$, since I is an interval. So $f(x_0) \leq f(x) \leq L$, and $L - \varepsilon < f(x) \leq L$. Thus $|f(x) - L| < \varepsilon$. So $\lim_{x\to c^-} f(x) = L$. The same argument, with infima in place of suprema, shows $\lim_{x\to c^+} f(x) = \inf\{f(x) : x \in I,\ x > c\}$ and that this infimum is at least $f(c)$.

Now let $c, d \in I$ with $c < d$. Every x with $c < x < d$ is in I. For such an x,

$$\inf\{f(t) : t \in I,\ t > c\} \leq f(x) \leq \sup\{f(t) : t \in I,\ t < d\}.$$

There is at least one such x. So $\lim_{x\to c^+} f(x) \leq \lim_{x\to d^-} f(x)$. □

Corollary 6.5.3. *If f is monotone on an open interval I, then the set of discontinuities of f is at most countable.*

Proof. If f is decreasing, then $-f$ is increasing and has the same points of discontinuity. So assume f is increasing. For $c \in I$, write $f(c^-) = \lim_{x\to c^-} f(x)$ and $f(c^+) = \lim_{x\to c^+} f(x)$. These exist and $f(c^-) \leq f(c) \leq f(c^+)$ by Theorem 6.5.2. If $f(c^-) = f(c^+)$, then both equal $f(c)$, and $\lim_{x\to c} f(x) = f(c)$ by Theorem 6.1.23. So f is continuous at c.

Let E be the set of points of I where f is discontinuous. For each $c \in E$, $f(c^-) < f(c^+)$. By Theorem 1.8.11, choose $r_c \in \mathbb{Q}$ with $f(c^-) < r_c < f(c^+)$. If $c, d \in E$ with $c < d$, then

$$r_c < f(c^+) \leq f(d^-) < r_d$$

by Theorem 6.5.2. So $c \mapsto r_c$ is a one-to-one map from E into $\mathbb{Q}$. Since $\mathbb{Q}$ is countable, E is at most countable by Theorem 2.2.4. □

Definition 6.5.4. Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. If f is one-to-one, define f^{-1} on $f[E]$ by $f^{-1}(y) = x$ if $f(x) = y$.

Since f is one-to-one, each $y \in f[E]$ equals $f(x)$ for exactly one x. So f^{-1} is a function. The proof of Theorem 2.1.8 (2) used the same construction. Do not confuse $f^{-1}(y)$ with the preimage $f^{-1}[G]$ of Definition 2.1.2, which is defined for every function.

Lemma 6.5.5. *If f is one-to-one, then f^{-1} is one-to-one.*

Proof. See Exercise 6.5.1. □

Lemma 6.5.6. *Let $E \subset \mathbb{R}$ and $f : E \to \mathbb{R}$. If f is strictly monotone, f is one-to-one.*

Proof. See Exercise 6.5.2. □

Theorem 6.5.7. *Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ strictly monotone and continuous on I. Then f^{-1} is strictly monotone and continuous on $f[I] = J$.*

Proof. Without loss of generality, assume f is strictly increasing. Since f is continuous, by Corollary 6.3.11, $f[I] = J$ is an interval. Furthermore, by Lemma 6.5.6, since f is strictly increasing, f is one-to-one. Hence by Lemma 6.5.5, f^{-1} is one-to-one.

Now suppose $y_1, y_2 \in J$ with $y_1 < y_2$. Then there exists $x_1, x_2 \in I$, $x_1 \neq x_2$ such that $f(x_1) = y_1$ and $f(x_2) = y_2$. Since f is strictly increasing, $x_1 < x_2$. Otherwise $x_1 > x_2$ would give $y_1 > y_2$. Thus

$$f^{-1}(y_1) = x_1 < x_2 = f^{-1}(y_2).$$

So f^{-1} is strictly increasing on J.

Choose $d \in J \setminus \{\inf J\}$. So there exists an $c \in I$ with $f(c) = d$. Also, note $(-\infty, c) \cap I \neq \emptyset$. Otherwise c would be the smallest element of I, and $d = f(c)$ would be the smallest element of J. Let $\varepsilon > 0$. Choose $a \in I$ with $a < c$. Then $[a, c] \subset I$, since I is an interval. A δ that works for $\min\{\varepsilon, c - a\}$ also works for ε. So it is enough to take $\varepsilon \leq c - a$. Then $c - \varepsilon \in I$. Define $\delta_1 = f(c) - f(c - \varepsilon)$. Since f is continuous and strictly increasing,

$$f[(c - \varepsilon, c]] = (f(c - \varepsilon), f(c)] = (d - \delta_1, d]$$

by Corollary 6.3.11 applied on $[c - \varepsilon, c]$. Thus, since f^{-1} is strictly increasing, for all $y \in (d - \delta_1, d)$,

$$c - \varepsilon = f^{-1}(d - \delta_1) < f^{-1}(y) < f^{-1}(d) = c.$$

Hence for all $y \in J$ with $d - \delta_1 < y \leq d$, $|f^{-1}(d) - f^{-1}(y)| < \varepsilon$. So $\lim_{y \to d^-} f^{-1}(y) = f^{-1}(d)$. Notice if $\sup J \in J$, this implies f^{-1} is continuous at $\sup J$. Using a similar argument, for $d \in J \setminus \{\sup J\}$, there exists a $\delta_2 > 0$ such that for all $y \in J$ with $d \leq y < d + \delta_2$, $|f^{-1}(d) - f^{-1}(y)| < \varepsilon$. If $\inf J \in J$, this implies f^{-1} is continuous at $\inf J$. Let $d \in J^\circ$. Set $\delta = \min\{\delta_1, \delta_2\}$. Then for all $y \in J$ with $|y - d| < \delta$, $|f^{-1}(d) - f^{-1}(y)| < \varepsilon$. So f^{-1} is continuous at d. □

Exercises

6.5.1. Prove Lemma 6.5.5. [Solution, p. 341]

6.5.2. Prove Lemma 6.5.6. [Solution, p. 341]

6.5.3. State and prove a version of Theorem 6.5.2 for decreasing functions.

6.5.4. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = x + \lfloor x \rfloor$. Prove that f is strictly increasing on $\mathbb{R}$ and find every point where f is discontinuous.

6.5.5. Let $n \in \mathbb{N}$. Use Theorem 6.5.7 to prove that $f(x) = \sqrt[n]{x}$ is continuous on $[0, \infty)$.

6.6. Where Can a Function Be Continuous?*

Thomae's function (Example 6.3.2) is continuous at every irrational number and at no rational number. Is there a function that does the reverse, continuous at every rational number and at no irrational number? This section shows the answer is no. The proof uses the Baire Category Theorem, which says $\mathbb{R}$ cannot be built from countably many "thin" sets. This section can be skipped. Nothing later depends on it.

Definition 6.6.1. A set $G \subset \mathbb{R}$ is a *G_δ set* if it is the intersection of countably many open sets.

Every open set is a G_δ set. So is every closed interval, since $[a, b] = \bigcap_{n=1}^{\infty} \left(a - \frac{1}{n}, b + \frac{1}{n}\right)$. The irrational numbers form a G_δ set (Exercise 6.6.2). The first theorem says the points where a function is continuous always form a G_δ set.

Theorem 6.6.2. *Let $f : \mathbb{R} \to \mathbb{R}$. For $n \in \mathbb{N}$, let U_n be the set of $c \in \mathbb{R}$ for which there is a $\delta > 0$ with $|f(x) - f(y)| < \frac{1}{n}$ for all $x, y \in (c - \delta, c + \delta)$. Then each U_n is open, and f is continuous at c if and only if $c \in \bigcap_{n=1}^{\infty} U_n$. So the set of points where f is continuous is a G_δ set.*

Proof. The first step shows that each U_n is open. Let $c \in U_n$, with δ as in the definition. Let $c' \in (c - \delta, c + \delta)$, and let $\delta' = \delta - |c' - c| > 0$. Then $(c' - \delta', c' + \delta') \subset (c - \delta, c + \delta)$. So $|f(x) - f(y)| < \frac{1}{n}$ for all x, y in $(c' - \delta', c' + \delta')$, and $c' \in U_n$. Thus $(c - \delta, c + \delta) \subset U_n$.

It remains to show that f is continuous at c exactly when c is in every U_n. Suppose f is continuous at c, and let $n \in \mathbb{N}$. Choose $\delta > 0$ with $|f(x) - f(c)| < \frac{1}{2n}$ whenever $|x - c| < \delta$. For $x, y \in (c - \delta, c + \delta)$, $|f(x) - f(y)| \leq |f(x) - f(c)| + |f(c) - f(y)| < \frac{1}{n}$. So $c \in U_n$. Conversely, suppose $c \in U_n$ for every n, and let $\varepsilon > 0$. Choose n with $\frac{1}{n} < \varepsilon$, and δ for this n. If $|x - c| < \delta$, take $y = c$. Then $|f(x) - f(c)| < \frac{1}{n} < \varepsilon$. So f is continuous at c. □

Theorem 6.6.3 (Baire Category Theorem)**.** *Let $U_1, U_2, U_3, \ldots$ be open sets that are dense in $\mathbb{R}$. Then $\bigcap_{n=1}^{\infty} U_n$ is dense in $\mathbb{R}$. In particular, it is not empty.*

Proof. First, if U is dense, then every open interval (a, b) meets U. Otherwise no point of (a, b) is in U or is a limit point of U, so $(a, b) \cap \overline{U} = \emptyset$, and $\overline{U} \neq \mathbb{R}$.

Let (a, b) be an open interval. It is enough to find a point of $\bigcap U_n$ in (a, b). The set $(a, b) \cap U_1$ is open and not empty. So it contains an open interval, and inside

that a closed interval $[a_1, b_1]$ with $a_1 < b_1$. Suppose $[a_n, b_n] \subset U_n$ has been chosen, with $a_n < b_n$. Then $(a_n, b_n) \cap U_{n+1}$ is open and not empty, so it contains a closed interval $[a_{n+1}, b_{n+1}]$ with $a_{n+1} < b_{n+1}$. This gives nested closed intervals

$$(a, b) \supset [a_1, b_1] \supset [a_2, b_2] \supset \cdots$$

with $[a_n, b_n] \subset U_n$. By Theorem 3.4.4, some x lies in every $[a_n, b_n]$. Then $x \in (a, b)$ and $x \in U_n$ for every n. □

The name comes from Baire's language. A set is *nowhere dense* if its closure contains no open interval. Sets that are countable unions of nowhere dense sets are said to be of the *first category*. Taking complements, Theorem 6.6.3 says $\mathbb{R}$ is not of the first category (Exercise 6.6.5). Countable sets are of the first category, and so is the Cantor set.

Theorem 6.6.4. *The set $\mathbb{Q}$ is not a G_δ set.*

Proof. Suppose $\mathbb{Q} = \bigcap_{n=1}^{\infty} U_n$ with each U_n open. Each U_n contains $\mathbb{Q}$, so it is dense. Let $q_1, q_2, q_3, \ldots$ be an enumeration of $\mathbb{Q}$ (Corollary 2.2.7), and let $V_k = \mathbb{R} \setminus \{q_k\}$. Each V_k is open, and it is dense since it misses only one point. The sets $U_1, V_1, U_2, V_2, \ldots$ are countably many dense open sets. By Theorem 6.6.3, they have a point x in common. Since $x \in \bigcap U_n$, $x \in \mathbb{Q}$. Since $x \in V_k$ for every k, $x \neq q_k$ for every k, so $x \notin \mathbb{Q}$. This is a contradiction. □

Corollary 6.6.5. *There is no function $f : \mathbb{R} \to \mathbb{R}$ that is continuous at every rational number and discontinuous at every irrational number.*

Proof. For such an f, the set of points where f is continuous would be $\mathbb{Q}$. By Theorem 6.6.2, it is a G_δ set. This contradicts Theorem 6.6.4. □

Exercises

6.6.1. Find the sets U_n of Theorem 6.6.2 for the Dirichlet function of Example 6.1.9 and for the function that is 0 for $x < 0$ and 1 for $x \geq 0$. [Hint, p. 341]

6.6.2. Prove that $\mathbb{R} \setminus \mathbb{Q}$ is a G_δ set. Then extend Thomae's function to $\mathbb{R}$ by $f(x) = \frac{1}{n}$ when $x = \frac{m}{n}$ in lowest terms with $n \in \mathbb{N}$, and $f(x) = 0$ when x is irrational. Explain why f is continuous exactly at the irrational numbers. [Solution, p. 341]

6.6.3. Use Theorem 6.6.3 to give another proof that $\mathbb{R}$ is uncountable. [Solution, p. 341]

6.6.4. Let G and H be G_δ sets that are dense in $\mathbb{R}$. Prove that $G \cap H$ is dense in $\mathbb{R}$. Conclude that a dense G_δ set is uncountable. [Hint, p. 341]

6.6.5. Prove that $\mathbb{R}$ is not a countable union of nowhere dense sets. Then prove that the Cantor set of Chapter 5 is nowhere dense. [Hint, p. 342]

Notes

Bernard Bolzano proved the Intermediate Value Theorem in 1817 [**5**]. He also proved the Extreme Value Theorem in the 1830s, but that work was not published until 1930. Karl Weierstrass found it independently around 1860. Augustin-Louis Cauchy defined continuity in his *Cours d'analyse* of 1821 [**15**]. The ε-δ form used here is due to Weierstrass, in his lectures in Berlin. Peter Gustav Lejeune Dirichlet described the function in Example 6.1.9 in 1829 [**24**]. Carl Johannes Thomae described the function in Example 6.3.2 in 1875 [**73**]. Eduard Heine defined uniform continuity and proved Theorem 6.4.7 in 1872 [**38**]. Rudolf Lipschitz used his condition in 1876 to prove existence theorems for differential equations [**50**]. The fact that a monotone function has only countably many discontinuities (Corollary 6.5.3) is sometimes called Froda's Theorem, after Alexandru Froda's 1929 thesis [**33**]. Froda himself noted that it was already well known. Vito Volterra showed in 1881 that no function can be continuous exactly at the rational numbers. René Baire proved Theorem 6.6.3 in his 1899 thesis. William Osgood had found the case of $\mathbb{R}$ in 1897.

CHAPTER 7

The Derivative

The derivative is the first new idea built from the limits of Chapter 6. It is a limit of difference quotients. Section 7.1 proves the rules from calculus, including the Chain Rule. Section 7.2 proves the Mean Value Theorem. It is the most important result in the chapter, because it turns information about f' into information about f. Almost everything after it, here and in later chapters, depends on it.

Sections 7.3 and 7.4 are marked with a star. They can be skipped. Nothing later in the book depends on them.

Examples and exercises use the derivatives of $\sin x$, $\cos x$, $\ln x$ and e^x from calculus. As in Remark 3.1.1, these are assumed for now. They are proved in the starred Sections 9.4, 9.5, 12.4 and 12.5.

7.1. Properties of the Derivative

Definition 7.1.1. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. For $c \in I$, the *derivative* of f at c, denoted $f'(c)$, is

$$f'(c) = \lim_{x \to c} \frac{f(x) - f(c)}{x - c},$$

provided the limit exists. If $f'(c)$ exists, f is *differentiable* at c. If f is differentiable at every point of I, then f is differentiable on I, and f' is the function $c \mapsto f'(c)$ on I. If f' is differentiable at c, its derivative is the *second derivative* $f''(c)$. In general, $f^{(0)} = f$, and $f^{(n)} = \left(f^{(n-1)}\right)'$ is the *nth derivative* when it exists. The derivative is also written $\frac{d}{dx} f(x)$.

The difference quotient $\frac{f(x)-f(c)}{x-c}$ is a function of x on $I \setminus \{c\}$. The limit makes sense because c is a limit point of $I \setminus \{c\}$. This uses that I is an interval with more than one point.

Remark 7.1.2. If $c \in I^\circ$, there exists an $r > 0$ such that $(c - r, c + r) \subset I$. Set $x = c + h$ with $0 < |h| < r$. Then

$$f'(c) = \lim_{h \to 0} \frac{f(c + h) - f(c)}{h},$$

provided the limit exists.

If $c \in I$ is a left endpoint, then

$$f'(c) = \lim_{h \to 0^+} \frac{f(c + h) - f(c)}{h},$$

provided the limit exists.

If $c \in I$ is a right endpoint, then

$$f'(c) = \lim_{h \to 0^-} \frac{f(c+h) - f(c)}{h},$$

provided the limit exists.

Theorem 7.1.3. *Let $I \subset \mathbb{R}$ be an interval, $f : I \to \mathbb{R}$, and $c \in I$. If f is differentiable at c, then f is continuous at c.*

PROOF. For $x \neq c$, $x \in I$,

$$f(x) - f(c) = \left(\frac{f(x) - f(c)}{x - c} \right) (x - c).$$

Since $f'(c)$ exists, by Theorem 6.1.13 (2),

$$\lim_{x \to c} (f(x) - f(c)) = \lim_{x \to c} \left(\frac{f(x) - f(c)}{x - c} \right) (x - c) = f'(c) \cdot 0 = 0.$$

So $\lim_{x \to c} f(x) = f(c)$ and f is continuous at c (relative to I, as in Remark 6.3.7). □

The converse is false. The function $f(x) = |x|$ is continuous at 0 but not differentiable there. See Exercise 7.1.4.

Definition 7.1.4. Let $I \subset \mathbb{R}$ be an interval. A function $f : I \to \mathbb{R}$ is *continuously differentiable* on I if f is differentiable on I and $f' \in C(I)$. The set of such functions is denoted $C^{(1)}(I)$. More generally, for $n \in \mathbb{N}$, $C^{(n)}(I)$ is the set of $f : I \to \mathbb{R}$ such that $f^{(n)}$ exists on I and $f^{(n)} \in C(I)$. Write $C^{(n)}[a, b]$ for $C^{(n)}([a, b])$.

By Theorem 7.1.3, $C^{(1)}(I) \subset C(I)$. A differentiable function need not be in $C^{(1)}(I)$. See Exercise 7.1.5.

Theorem 7.1.5. *Let $I \subset \mathbb{R}$ be an interval and $f, g : I \to \mathbb{R}$. If f and g are differentiable at $c \in I$, then $f + g$, fg, and f/g (if $g(c) \neq 0$) are differentiable at c and*

(1) $(f + g)'(c) = f'(c) + g'(c)$;
(2) $(fg)'(c) = f'(c)g(c) + f(c)g'(c)$; and
(3) $(f/g)'(c) = \frac{f'(c)g(c) - f(c)g'(c)}{[g(c)]^2}$.

PROOF. (1) This part is an exercise.

(2) By adding and subtracting $f(c + h)g(c)$, $h \neq 0$,

$$\frac{(fg)(c+h) - (fg)(c)}{h} = f(c+h) \left(\frac{g(c+h) - g(c)}{h} \right) + \left(\frac{f(c+h) - f(c)}{h} \right) g(c).$$

So

$$\begin{aligned}
(fg)'(c) &= \lim_{h \to 0} \frac{(fg)(c+h) - (fg)(c)}{h} \\
&= \lim_{h \to 0} \left[f(c+h) \left(\frac{g(c+h) - g(c)}{h} \right) + \left(\frac{f(c+h) - f(c)}{h} \right) g(c) \right] \\
&= f(c)g'(c) + f'(c)g(c).
\end{aligned}$$

The last step uses $f(c+h) \to f(c)$ as $h \to 0$, since f is continuous at c by Theorem 7.1.3.

(3) First, consider $(1/g)'(c)$. Since $g(c) \neq 0$ and g is continuous at c, there exists a $\delta > 0$ such that for all h with $|h| < \delta$ and $c+h \in I$, $g(c+h) \neq 0$ by Lemma 6.1.11. Thus, for $0 < |h| < \delta$ with $c+h \in I$,

$$\frac{\frac{1}{g(c+h)} - \frac{1}{g(c)}}{h} = \frac{-(g(c+h) - g(c))}{h} \left(\frac{1}{g(c)g(c+h)} \right).$$

So

$$\begin{aligned} \left(\frac{1}{g}\right)'(c) &= \lim_{h \to 0} \frac{\frac{1}{g(c+h)} - \frac{1}{g(c)}}{h} \\ &= \lim_{h \to 0} \frac{-(g(c+h) - g(c))}{h} \left(\frac{1}{g(c)g(c+h)} \right) \\ &= -g'(c) \cdot \frac{1}{[g(c)]^2}. \end{aligned}$$

Now apply the product rule to obtain

$$\begin{aligned} \left(\frac{f}{g}\right)'(c) &= \left(f \cdot \frac{1}{g}\right)'(c) \\ &= f'(c) \cdot \frac{1}{g(c)} + f(c) \cdot \frac{-g'(c)}{[g(c)]^2} \\ &= \frac{f'(c)g(c) - f(c)g'(c)}{[g(c)]^2}. \end{aligned}$$

□

Proof Technique (Avoiding Division by Zero). A tempting proof of the Chain Rule writes

$$\frac{g(f(t)) - g(f(c))}{t-c} = \frac{g(f(t)) - g(f(c))}{f(t) - f(c)} \cdot \frac{f(t) - f(c)}{t-c}$$

and takes limits. This fails if $f(t) = f(c)$ for t arbitrarily close to c, since then the first fraction is $0/0$. For example, with $c = 0$, $f(t) = t^2 \sin \frac{1}{t}$ (and $f(0) = 0$) has $f(t) = 0$ at $t = \frac{1}{n\pi}$ for every $n \in \mathbb{N}$. The proof below avoids the division. It writes each difference as a product, $f(t) - f(c) = (t-c)\big(f'(c) + u(t)\big)$, where $u(t) \to 0$. Products can be substituted into each other without dividing by anything. Whenever a proof divides, check that the denominator cannot be 0.

Theorem 7.1.6 (Chain Rule)**.** *Let $I, J \subset \mathbb{R}$ be intervals, $f : I \to \mathbb{R}$ with $f[I] \subset J$, and $g : J \to \mathbb{R}$. If f is differentiable at $c \in I$ and g is differentiable at $f(c) \in J$, then $g \circ f$ is differentiable at c and*

$$(g \circ f)'(c) = g'(f(c))f'(c).$$

PROOF. Let $d = f(c)$. Define $u : I \to \mathbb{R}$ and $v : J \to \mathbb{R}$ by

$$u(t) = \begin{cases} \frac{f(t)-f(c)}{t-c} - f'(c), & t \neq c, \\ 0, & t = c, \end{cases} \qquad v(s) = \begin{cases} \frac{g(s)-g(d)}{s-d} - g'(d), & s \neq d, \\ 0, & s = d. \end{cases}$$

Since $f'(c)$ and $g'(d)$ exist, $\lim_{t\to c} u(t) = 0 = u(c)$ and $\lim_{s\to d} v(s) = 0 = v(d)$. So u is continuous at c and v is continuous at d. For every $t \in I$ and $s \in J$,

$$f(t) - f(c) = (t-c)\big(f'(c) + u(t)\big) \quad \text{and} \quad g(s) - g(d) = (s-d)\big(g'(d) + v(s)\big).$$

Both sides of the first equation are 0 when $t = c$, and both sides of the second are 0 when $s = d$. Put $s = f(t)$ in the second equation, and then use the first. This gives

$$\begin{aligned}(g \circ f)(t) - (g \circ f)(c) &= \big(f(t) - f(c)\big)\big(g'(d) + v(f(t))\big) \\ &= (t-c)\big(f'(c) + u(t)\big)\big(g'(d) + v(f(t))\big).\end{aligned}$$

So for $t \in I$ with $t \neq c$,

$$\frac{(g \circ f)(t) - (g \circ f)(c)}{t-c} = \big(f'(c) + u(t)\big)\big(g'(d) + v(f(t))\big).$$

By Theorem 7.1.3, f is continuous at c. Since v is continuous at $d = f(c)$, $v \circ f$ is continuous at c by Theorem 6.3.4. So $v(f(t)) \to v(d) = 0$ as $t \to c$. Also $u(t) \to 0$. By Theorem 6.1.13, the right side converges to $f'(c)g'(d)$ as $t \to c$. Thus

$$(g \circ f)'(c) = g'(d)f'(c) = g'(f(c))f'(c).$$

□

Definition 7.1.7. Let $S \subset \mathbb{R}$ and $f : S \to \mathbb{R}$.

(1) f has a *relative maximum* at a point $c \in S$ if there exists a $\delta > 0$ such that $f(x) \leq f(c)$ for all $x \in S$ with $|x - c| < \delta$.
(2) f has a *relative minimum* at a point $c \in S$ if there exists a $\delta > 0$ such that $f(x) \geq f(c)$ for all $x \in S$ with $|x - c| < \delta$.
(3) f has an *absolute maximum* at $c \in S$ if $f(x) \leq f(c)$ for all $x \in S$.
(4) f has an *absolute minimum* at $c \in S$ if $f(x) \geq f(c)$ for all $x \in S$.
(5) f has a *relative extremum* at $c \in S$ if f has a relative maximum or minimum at c.
(6) f has an *absolute extremum* at $c \in S$ if f has an absolute maximum or minimum at c.

A relative maximum or minimum is also called a *local* maximum or minimum.

Theorem 7.1.8. *Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. Suppose f has a relative extremum at $c \in I^\circ$. If f is differentiable at c, then $f'(c) = 0$.*

PROOF. Assume without loss of generality that f has a local maximum at c. (Otherwise, replace f by $-f$.) Then there exists a $\delta > 0$ such that $f(x) \leq f(c)$ for all $x \in I$ with $|x - c| < \delta$. In particular, if $c < x < c + \delta$, $x \in I$, then

$$\frac{f(x) - f(c)}{x - c} \leq 0.$$

Similarly if $c - \delta < x < c$, $x \in I$, then

$$\frac{f(x) - f(c)}{x - c} \geq 0.$$

Since $f'(c)$ exists, $\lim_{x \to c} \frac{f(x)-f(c)}{x-c}$ exists and

$$0 \geq \lim_{x \to c^+} \frac{f(x) - f(c)}{x - c} = \lim_{x \to c^-} \frac{f(x) - f(c)}{x - c} \geq 0,$$

by Theorems 6.1.15 and 6.1.23. Here $c \in I^\circ$ is used, so that both one-sided limits make sense. So $\lim_{x \to c} \frac{f(x)-f(c)}{x-c} = 0$. So $f'(c) = 0$. □

Corollary 7.1.9. *Let $f : [a, b] \to \mathbb{R}$ be continuous. If f has a relative extremum at $c \in (a, b)$, then either $f'(c)$ does not exist or $f'(c) = 0$.*

Proof. If $f'(c)$ exists, then $f'(c) = 0$ by Theorem 7.1.8, since $c \in (a, b) = [a, b]^\circ$. □

Exercises

7.1.1. Prove Theorem 7.1.5 (1).

7.1.2. Prove $\frac{d}{dx} x^n = nx^{n-1}$ for all $n \in \mathbb{N}_0$. [Hint, p. 342]

7.1.3. Prove $\frac{d}{dx} x^n = nx^{n-1}$ for all $n \in \mathbb{Z}$ and $x \neq 0$. [Hint, p. 342]

7.1.4. Prove that $f : \mathbb{R} \to \mathbb{R}$, $f(x) = |x|$, is not differentiable at 0. [Solution, p. 342]

7.1.5. Define $f : \mathbb{R} \to \mathbb{R}$ by $f(x) = x^2 \sin \frac{1}{x}$ for $x \neq 0$ and $f(0) = 0$. Prove that f is differentiable on $\mathbb{R}$ with $f'(0) = 0$, and that f' is not continuous at 0. [Hint, p. 342]

7.2. The Mean Value Theorem

The Mean Value Theorem says that somewhere between a and b, the tangent line to the graph of f is parallel to the secant line through $(a, f(a))$ and $(b, f(b))$. See Figure 7.1. Rolle's Theorem is the special case where the secant line is horizontal.

Theorem 7.2.1 (Rolle's Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b), with $f(a) = f(b)$. Then there exists a $c \in (a, b)$ such that $f'(c) = 0$.*

Proof. If f is constant on $[a, b]$, then $f'(x) = 0$ for all $x \in (a, b)$.

Now assume f is not constant. By the Extreme Value Theorem (Theorem 6.3.15), f has a maximum and a minimum on $[a, b]$. Choose some $t \in (a, b)$ such that $f(t) \neq f(a)$. If $f(t) > f(a)$, then f has a maximum at some $c \in (a, b)$. The maximum value is at least $f(t) > f(a) = f(b)$, so it is not attained at a or b. An absolute maximum at $c \in (a, b)$ is also a relative maximum. Thus, by Theorem 7.1.8, $f'(c) = 0$. If $f(t) < f(a)$, then f has a local minimum at some $c \in (a, b)$. Thus $f'(c) = 0$. □

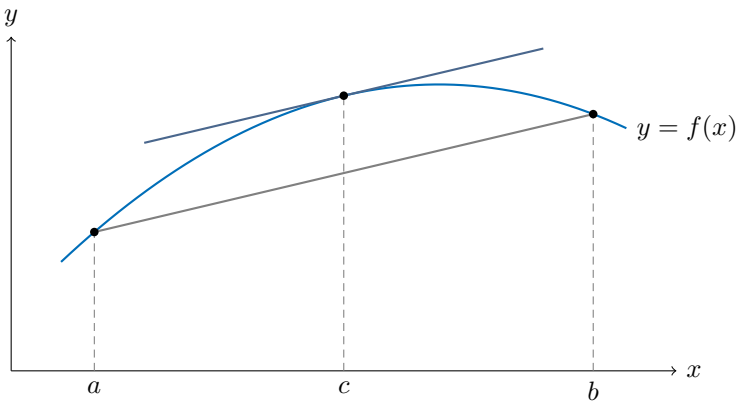


FIGURE 7.1. The Mean Value Theorem. The tangent line at c is parallel to the secant line through $(a, f(a))$ and $(b, f(b))$.

Example 7.2.2. Continuity in Rolle's Theorem is required. Consider $f : [0, 1] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} x, & 0 \leq x < 1, \\ 0, & x = 1. \end{cases}$$

f is differentiable on $(0, 1)$ and $f(0) = f(1)$, but $f'(x) \neq 0$ for all $x \in (0, 1)$.

Example 7.2.3. Differentiability at the endpoints in Rolle's Theorem is not required. Consider $f : [-2, 2] \to \mathbb{R}$, $f(x) = \sqrt{4 - x^2}$. Then $f(-2) = 0 = f(2)$, and $f'(x) = \frac{-x}{\sqrt{4-x^2}}$ does not exist at -2 and 2. But $f'(0) = 0$.

Theorem 7.2.4 (Mean Value Theorem)**.** *Let $f : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists a $c \in (a, b)$ such that*

$$f(b) - f(a) = f'(c)(b - a).$$

PROOF. Define the function $g : [a, b] \to \mathbb{R}$ by

$$g(x) = f(x) - f(a) - \left[\frac{f(b) - f(a)}{b - a}\right](x - a).$$

Then g is continuous on $[a, b]$, differentiable on (a, b), and $g(a) = g(b)$. Thus, by Rolle's Theorem, there exists a $c \in (a, b)$ such that $g'(c) = 0$. But

$$g'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}$$

for all $x \in (a, b)$. So

$$g'(c) = f'(c) - \frac{f(b) - f(a)}{b - a} = 0.$$

Thus $f'(c) = \frac{f(b)-f(a)}{b-a}$. So $f(b) - f(a) = f'(c)(b - a)$. □

Proof Technique (Auxiliary Functions). The proof of the Mean Value Theorem does not work with f directly. It builds a new function g that satisfies

the hypotheses of Rolle's Theorem, applies Rolle's Theorem to g, and translates the answer back to f. The function g is f minus the secant line, chosen so that $g(a) = g(b)$. A function built for one proof like this is called an *auxiliary function*. Finding it is the creative step. The proofs of Theorems 7.2.6 and 7.2.8 use the same idea.

Example 7.2.5. The inequality $\frac{x}{1+x} \leq \ln(1+x) \leq x$ holds for all $x > -1$. To show this, assume $\frac{d}{dx} \ln x = \frac{1}{x}$. Let $f : (-1, \infty) \to \mathbb{R}$, $f(x) = \ln(1+x)$. Then $f(0) = 0$. If $x > 0$, then by the Mean Value Theorem, there exists a $c \in (0, x)$ such that

$$\ln(1+x) = f(x) - f(0) = f'(c)x.$$

But $f'(c) = \frac{1}{1+c}$ and $\frac{1}{1+x} < \frac{1}{1+c} < 1$ for all $c \in (0, x)$. Therefore, $\frac{x}{1+x} < f'(c)x < x$. So $\frac{x}{1+x} \leq \ln(1+x) \leq x$ for all $x \geq 0$.

Now suppose $-1 < x < 0$. Then again, by the Mean Value Theorem, there exists a $c \in (x, 0)$ such that

$$\ln(1+x) = f(x) - f(0) = \frac{x}{1+c}.$$

But since $x < c < 0$, $1 < \frac{1}{1+c} < \frac{1}{1+x}$, and since $x < 0$, $\frac{x}{1+x} < \frac{x}{1+c} < x$. So

$$\frac{x}{1+x} < \ln(1+x) < x$$

for $-1 < x < 0$.

Theorem 7.2.6 (Cauchy Mean Value Theorem)**.** *Let $f, g : [a, b] \to \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists a $c \in (a, b)$ such that*

$$[f(b) - f(a)]g'(c) = [g(b) - g(a)]f'(c).$$

Proof. Let $h(x) = [f(b) - f(a)]g(x) - [g(b) - g(a)]f(x)$. Then h is continuous on $[a, b]$, differentiable on (a, b), and

$$h(a) = f(b)g(a) - f(a)g(b) = h(b).$$

Thus by Rolle's Theorem, there exists a $c \in (a, b)$ such that $h'(c) = 0$. Therefore,

$$[f(b) - f(a)]g'(c) - [g(b) - g(a)]f'(c) = 0,$$

which gives the result. □

Theorem 7.2.7. *Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be differentiable on I.*

(1) If $f'(x) \geq 0$ for all $x \in I$, then f is increasing on I.
(2) If $f'(x) > 0$ for all $x \in I$, then f is strictly increasing on I.
(3) If $f'(x) \leq 0$ for all $x \in I$, then f is decreasing on I.
(4) If $f'(x) < 0$ for all $x \in I$, then f is strictly decreasing on I.
(5) If $f'(x) = 0$ for all $x \in I$, then f is constant on I.

Proof. Suppose $x_1, x_2 \in I$ with $x_1 < x_2$. By the Mean Value Theorem, there exists a $c \in (x_1, x_2)$ such that

$$f(x_2) - f(x_1) = f'(c)(x_2 - x_1).$$

The results for (1)–(5) follow directly. For example, in (1), $f'(c) \geq 0$ and $x_2 - x_1 > 0$, so $f(x_2) \geq f(x_1)$. The Mean Value Theorem applies on $[x_1, x_2]$ since f is continuous there by Theorem 7.1.3. □

Theorem 7.2.8 (Intermediate Value Theorem for Derivatives)**.** *Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be differentiable on I. Then, given $a, b \in I$ with $a < b$ and a real number k between $f'(a)$ and $f'(b)$, there exists a $c \in (a, b)$ such that $f'(c) = k$.*

Proof. Define $g : I \to \mathbb{R}$ by $g(x) = f(x) - kx$. Then g is differentiable on I with $g'(x) = f'(x) - k$.

Assume without loss of generality that $f'(a) < k < f'(b)$. Then $g'(a) < 0$ and $g'(b) > 0$. Since $g'(a) < 0$, there exists an $x_1 \in (a, b)$ such that $g(x_1) < g(a)$. Indeed, by Lemma 6.1.11 there exists a $\delta > 0$ such that $\frac{g(x)-g(a)}{x-a} < 0$ for all $x \in I$ with $0 < |x - a| < \delta$. Any $x_1 \in (a, b)$ with $x_1 < a + \delta$ works. Also, since $g'(b) > 0$, there exists an $x_2 \in (a, b)$ such that $g(x_2) < g(b)$. So g has an absolute minimum at some point $c \in (a, b)$. By the Extreme Value Theorem, g has an absolute minimum on $[a, b]$ at some c. It is not at a, since $g(x_1) < g(a)$, and not at b, since $g(x_2) < g(b)$. By Theorem 7.1.8, $g'(c) = f'(c) - k = 0$. So $f'(c) = k$. □

This result is also called *Darboux's Theorem*. A derivative does not have to be continuous (Exercise 7.1.5). Even so, it has the intermediate value property, just like a continuous function. One consequence is that a derivative cannot have a jump. See Exercise 7.2.8.

Theorem 7.2.9 (Inverse Function Theorem)**.** *Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be differentiable on I with $f'(x) \neq 0$ for all $x \in I$. Then f is one-to-one on I and the inverse function f^{-1} is continuous and differentiable on $J = f[I]$ with*

$$(f^{-1})'(f(c)) = \frac{1}{f'(c)}$$

for all $c \in I$.

Proof. Since $f'(x) \neq 0$ for all $x \in I$, by Theorem 7.2.8, f' is either always positive or always negative on all of I. Assume without loss of generality that $f'(x) > 0$ for all $x \in I$. Then by Theorem 7.2.7, f is strictly increasing on I, and by Theorem 6.5.7, f^{-1} is continuous on $J = f[I]$.

Let $d \in J$ and let $\{y_n\}$ be a sequence in J with $y_n \to d$, $y_n \neq d$ for all $n \in \mathbb{N}$. For each $n \in \mathbb{N}$, there exists an $x_n \in I$ with $f(x_n) = y_n$. Since f^{-1} is continuous, $x_n \to c = f^{-1}(d)$. Since f^{-1} is one-to-one, $x_n \neq c$ for all n. So $\frac{f(x_n)-f(c)}{x_n-c} \to f'(c) \neq 0$ by Theorem 6.1.18. Hence

$$\lim_{n\to\infty} \frac{f^{-1}(y_n) - f^{-1}(d)}{y_n - d} = \lim_{n\to\infty} \frac{x_n - c}{f(x_n) - f(c)} = \frac{1}{f'(c)}.$$

Since this holds for any sequence $\{y_n\}$ in J such that $y_n \to d$, $y_n \neq d$ for all $n \in \mathbb{N}$, by Theorem 6.1.18,

$$(f^{-1})'(d) = \lim_{y\to d} \frac{f^{-1}(y) - f^{-1}(d)}{y - d} = \frac{1}{f'(c)}.$$

□

Example 7.2.10. Let $f : [0, \pi] \to \mathbb{R}$, $f(x) = \cos x$. Since $f'(x) = -\sin x < 0$ for all $x \in (0, \pi)$, $f(x) = \cos x$ is strictly decreasing on $[0, \pi]$. Also, $f[[0, \pi]] = [-1, 1]$ and so $f^{-1}(y) = \arccos y$ exists on $[-1, 1]$. Since $f'(0) = f'(\pi) = 0$, apply Theorem 7.2.9 on $I = (0, \pi)$, where $J = (-1, 1)$. For $x \in (0, \pi)$, $\sin x > 0$, so $\sin x = \sqrt{1 - \cos^2 x}$. Now

$$(f^{-1})'(f(x)) = \frac{1}{f'(x)} = \frac{1}{-\sin x} = \frac{-1}{\sqrt{1 - \cos^2 x}} = \frac{-1}{\sqrt{1 - [f(x)]^2}}.$$

So

$$(f^{-1})'(y) = \frac{-1}{\sqrt{1 - y^2}}$$

for $y \in (-1, 1)$.

Exercises

7.2.1. Use the Mean Value Theorem to establish the inequality $e^x \geq 1 + x$. You may assume $\frac{d}{dx} e^x = e^x$. [Hint, p. 342]

7.2.2. Suppose $f : (a, b) \to \mathbb{R}$ satisfies $|f(x) - f(y)| \leq M|x - y|^\alpha$ for some $\alpha > 1$ and for all $x, y \in (a, b)$. Prove that f is a constant function on (a, b). [Hint, p. 342]

7.2.3. (The First Derivative Test) Let $f : [a, b] \to \mathbb{R}$ be differentiable on (a, b). Suppose $c \in (a, b)$ is such that $f'(c) = 0$.

(a) If there exists a $\delta > 0$ such that $f'(x) > 0$ for all $x \in (c - \delta, c)$ and $f'(x) < 0$ for all $x \in (c, c + \delta)$, prove f has a local maximum at $x = c$.

(b) If there exists a $\delta > 0$ such that $f'(x) < 0$ for all $x \in (c - \delta, c)$ and $f'(x) > 0$ for all $x \in (c, c + \delta)$, prove f has a local minimum at $x = c$.

7.2.4. (The Second Derivative Test) Let $f : [a, b] \to \mathbb{R}$ be differentiable on (a, b). Suppose $c \in (a, b)$ is such that $f'(c) = 0$, and $f''(c)$ exists.

(a) If $f''(c) > 0$, prove that f has a local minimum at c.

(b) If $f''(c) < 0$, prove that f has a local maximum at c.

(c) Show, using two specific examples, that no conclusion can be made if $f''(c) = 0$.

7.2.5. Prove $\frac{d}{dx} x^{1/n} = \frac{1}{n} x^{1/n-1}$ for all $x \in (0, \infty)$, $n \in \mathbb{N}$. [Hint, p. 343]

7.2.6. Prove $\frac{d}{dx} x^{m/n} = \frac{d}{dx} (x^{1/n})^m = \frac{m}{n} x^{m/n-1}$ for all $x \in (0, \infty)$, $m \in \mathbb{Z}$, $n \in \mathbb{N}$.

7.2.7. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be differentiable on I with $|f'(x)| \leq M$ for all $x \in I$. Prove that f is Lipschitz continuous on I. Use this to show $|\sin x - \sin y| \leq |x - y|$ for all $x, y \in \mathbb{R}$. [Solution, p. 343]

7.2.8. Define $g : \mathbb{R} \to \mathbb{R}$ by $g(x) = 0$ for $x < 0$ and $g(x) = 1$ for $x \geq 0$. Prove that there is no differentiable function $f : \mathbb{R} \to \mathbb{R}$ with $f' = g$. [Hint, p. 343]

7.3. L'Hospital's Rule*

This section can be skipped. Nothing later in the book depends on it.

L'Hospital's Rule computes limits of the forms $\frac{0}{0}$ and $\frac{\infty}{\infty}$. The statement allows $a = -\infty$ and $L = \pm\infty$. The limits involved are defined in Section 6.2.

Theorem 7.3.1 (L'Hospital's Rule)**.** *Let $-\infty \le a < b \le \infty$, and let $f, g : (a, b) \to \mathbb{R}$ be differentiable on (a, b) with $g'(x) \neq 0$ for all $x \in (a, b)$. Suppose*

$$\lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L,$$

where $L \in \mathbb{R} \cup \{-\infty, \infty\}$. If

(1) $\lim_{x\to a^+} f(x) = \lim_{x\to a^+} g(x) = 0$, or
(2) $\lim_{x\to a^+} g(x) = \pm\infty$,

then

$$\lim_{x\to a^+} \frac{f(x)}{g(x)} = L.$$

The analogous result is true for $x \to b^-$. When $a = -\infty$, $x \to a^+$ means $x \to -\infty$.

PROOF. (1) Suppose a is finite. Let $\{x_n\}$ be a sequence in (a, b) with $x_n \to a$ and $x_n \neq a$ for all $n \in \mathbb{N}$. Define $f(a) = g(a) = 0$. Then f and g are continuous at a as functions on $[a, b)$. Thus by Cauchy's Mean Value Theorem on $[a, x_n]$, for each $n \in \mathbb{N}$, there exists a $c_n \in (a, x_n)$ such that

$$[f(x_n) - f(a)]g'(c_n) = [g(x_n) - g(a)]f'(c_n),$$

implying

$$\frac{f(x_n)}{g(x_n)} = \frac{f'(c_n)}{g'(c_n)}.$$

Since $g'(x) \neq 0$ for all $x \in (a, b)$, $g(x_n) \neq g(a)$ for all $n \in \mathbb{N}$ by Rolle's Theorem. As $n \to \infty$, $c_n \to a$ by the Squeeze Theorem. Thus by Theorem 6.1.18 and Theorem 6.2.8,

$$\lim_{n\to\infty} \frac{f(x_n)}{g(x_n)} = \lim_{n\to\infty} \frac{f'(c_n)}{g'(c_n)} = \lim_{x\to a^+} \frac{f'(x)}{g'(x)} = L.$$

Since the above holds for every sequence $\{x_n\}$ in (a, b) with $x_n \to a$ and $x_n \neq a$ for all $n \in \mathbb{N}$, by Theorem 6.1.18, the result follows. Here the domain is (a, b), so the limit as $x \to a$ over (a, b) is the right-hand limit.

Suppose $a = -\infty$. Make the substitution $x = -\frac{1}{t}$. Then as $t \to 0^+$, $x \to -\infty$. For some $c > 0$, define the functions $\varphi(t)$ and $\psi(t)$ on $(0, c)$ by

$$\varphi(t) = f\left(-\frac{1}{t}\right) \text{ and } \psi(t) = g\left(-\frac{1}{t}\right).$$

See Exercise 7.3.1 for the choice of c. By the Chain Rule, $\varphi'(t) = \frac{1}{t^2} f'\left(-\frac{1}{t}\right)$ and $\psi'(t) = \frac{1}{t^2} g'\left(-\frac{1}{t}\right) \neq 0$. Then

$$\lim_{t\to 0^+} \frac{\varphi'(t)}{\psi'(t)} = \lim_{x\to-\infty} \frac{f'(x)}{g'(x)} = L,$$

and

$$\lim_{t\to 0^+} \varphi(t) = \lim_{t\to 0^+} \psi(t) = 0.$$

Since 0 is finite,

$$\lim_{x\to-\infty} \frac{f(x)}{g(x)} = \lim_{t\to 0^+} \frac{\varphi(t)}{\psi(t)} = \lim_{t\to 0^+} \frac{\varphi'(t)}{\psi'(t)} = L.$$

(2) Suppose $\lim_{x\to a^+} g(x) = \infty$. If the limit is $-\infty$, apply this case to $-f$ and $-g$, since $\frac{-f}{-g} = \frac{f}{g}$ and $\frac{(-f)'}{(-g)'} = \frac{f'}{g'}$. The proof works for $a = -\infty$ as written, reading (a, c) as $(-\infty, c)$.

First suppose $L < \infty$. The claim is that for every $\beta > L$, there exists a $c \in (a, b)$ such that $\frac{f(x)}{g(x)} < \beta$ for all $x \in (a, c)$. Let $\beta > L$ and choose r with $L < r < \beta$. Since $\frac{f'(t)}{g'(t)} \to L < r$ as $t \to a^+$, there exists a $c_1 \in (a, b)$ such that $\frac{f'(t)}{g'(t)} < r$ for all $t \in (a, c_1)$. Fix $y \in (a, c_1)$. Since $g(x) \to \infty$ as $x \to a^+$, there exists a $c_2 \in (a, y)$ such that $g(x) > \max\{g(y), 0\}$ for all $x \in (a, c_2)$. Let $x \in (a, c_2)$. By the Cauchy Mean Value Theorem on $[x, y]$, there exists a $t \in (x, y)$ with $[f(y) - f(x)]g'(t) = [g(y) - g(x)]f'(t)$. Since $g'(t) \neq 0$ and $g(x) \neq g(y)$,

$$\frac{f(x) - f(y)}{g(x) - g(y)} = \frac{f'(t)}{g'(t)} < r.$$

The number $\frac{g(x)-g(y)}{g(x)}$ is positive. Multiplying by it gives

$$\frac{f(x) - f(y)}{g(x)} < r\left(1 - \frac{g(y)}{g(x)}\right), \quad \text{so} \quad \frac{f(x)}{g(x)} < r + \frac{f(y) - rg(y)}{g(x)}$$

for all $x \in (a, c_2)$. Since y is fixed and $g(x) \to \infty$, the last fraction converges to 0 as $x \to a^+$. So the right side converges to $r < \beta$. Hence there exists a $c \in (a, c_2)$ such that the right side is less than β, and so $\frac{f(x)}{g(x)} < \beta$, for all $x \in (a, c)$.

In the same way, if $L > -\infty$, then for every $\alpha < L$, there exists a $c' \in (a, b)$ such that $\frac{f(x)}{g(x)} > \alpha$ for all $x \in (a, c')$.

If L is finite, let $\varepsilon > 0$ and apply both statements with $\alpha = L - \varepsilon$ and $\beta = L + \varepsilon$. Then $\left|\frac{f(x)}{g(x)} - L\right| < \varepsilon$ for all $x \in (a, \min\{c, c'\})$. If $L = -\infty$, the first statement holds for every $\beta \in \mathbb{R}$, which says $\frac{f(x)}{g(x)} \to -\infty$. If $L = \infty$, the second statement holds for every $\alpha \in \mathbb{R}$, which says $\frac{f(x)}{g(x)} \to \infty$. □

Proof Technique (Reducing to a Case Already Proved). The proof of L'Hospital's Rule handles $a = -\infty$ by substituting $x = -\frac{1}{t}$. That turns a limit as $x \to -\infty$ into a limit as $t \to 0^+$, which is a finite case already proved. It handles $g \to -\infty$ by replacing f and g with $-f$ and $-g$. When a new case looks hard, look for a change of variables or a simple transformation that turns it into a case already done.

Corollary 7.3.2 (L'Hospital's Rule)**.** *Let $-\infty \le a < c < b \le \infty$, and let $f, g : (a, c) \cup (c, b) \to \mathbb{R}$ be differentiable with $g'(x) \neq 0$ for all $x \in (a, c) \cup (c, b)$. Suppose*

$$\lim_{x\to c} \frac{f'(x)}{g'(x)} = L,$$

where $L \in \mathbb{R} \cup \{-\infty, \infty\}$. If

(1) $\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = 0$, or

(2) $\lim_{x\to c} g(x) = \pm\infty$,

then

$$\lim_{x\to c} \frac{f(x)}{g(x)} = L.$$

PROOF. Apply Theorem 7.3.1 on (c, b) to get the right-hand limit, and its analogue for $x \to b^-$ on (a, c) to get the left-hand limit. Both equal L. So the limit is L by Theorem 6.1.23 (or by Exercise 6.2.5, if $L = \pm\infty$). □

Exercises

7.3.1. Let $h : (-\infty, b) \to \mathbb{R}$. Show that there exists a $c > 0$ such that $\varphi(t) = h\left(-\frac{1}{t}\right)$ is defined on $(0, c)$ and that $\lim_{x\to-\infty} h(x) = \lim_{t\to 0^+} \varphi(t)$.

7.3.2. Use L'Hospital's Rule to prove the following limits.

(a) $\lim_{x\to 0^+} \dfrac{\ln(1+x)}{x} = 1$

(b) $\lim_{x\to 0^+} \dfrac{1-\cos x}{x^2} = \dfrac{1}{2}$

(c) $\lim_{x\to 0^+} \dfrac{e^{-1/x}}{x} = 0$

[Solution, p. 343]

7.4. Concavity and Convexity*

This section can be skipped. Nothing later in the book depends on it.

Definition 7.4.1. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. Then f is a *convex* function on I if for every $a, b \in I$ and $0 \le t \le 1$,

$$f(ta + (1-t)b) \le tf(a) + (1-t)f(b).$$

Also, f is a *concave* function on I if for every $a, b \in I$ and $0 \le t \le 1$,

$$f(ta + (1-t)b) \ge tf(a) + (1-t)f(b).$$

As t runs from 0 to 1, the point $ta + (1-t)b$ runs over the interval between a and b, and $tf(a) + (1-t)f(b)$ runs over the chord joining $(a, f(a))$ and $(b, f(b))$. So f is convex when the graph lies on or below every chord.

Theorem 7.4.2. *Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. Then f is convex on I if and only if for all $a, x, b \in I$ with $a < x < b$,*

$$\frac{f(x) - f(a)}{x - a} \le \frac{f(b) - f(x)}{b - x}.$$

Similarly, f is concave on I if and only if for all $a, x, b \in I$ with $a < x < b$,

$$\frac{f(x) - f(a)}{x - a} \ge \frac{f(b) - f(x)}{b - x}.$$

PROOF. See Exercise 7.4.1. □

Theorem 7.4.3. *Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be differentiable on I. Then f is convex on I if and only if f' is increasing on I. Similarly, f is concave on I if and only if f' is decreasing on I.*

PROOF. ($\Rightarrow$) Let $a, b \in I$ with $a < b$. Then

$$\begin{aligned} f'(a) &= \lim_{x \to a^+} \frac{f(x) - f(a)}{x - a} \\ &\leq \lim_{x \to a^+} \frac{f(b) - f(x)}{b - x} \\ &= \frac{f(b) - f(a)}{b - a} \\ &= \lim_{x \to b^-} \frac{f(x) - f(a)}{x - a} \\ &\leq \lim_{x \to b^-} \frac{f(b) - f(x)}{b - x} \\ &= f'(b). \end{aligned}$$

The inequalities come from Theorem 7.4.2 with $a < x < b$. The middle equalities use the continuity of f at a and b (Theorem 7.1.3).

($\Leftarrow$) Let $a, x, b \in I$ with $a < x < b$. Then by the Mean Value Theorem, there exists $r \in (a, x)$ and $s \in (x, b)$ such that

$$\frac{f(x) - f(a)}{x - a} = f'(r) \text{ and } \frac{f(b) - f(x)}{b - x} = f'(s).$$

Since f' is increasing, $f'(r) \leq f'(s)$. So

$$\frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(x)}{b - x},$$

and by Theorem 7.4.2, f is convex. □

Theorem 7.4.4. *Let $f : (a, b) \to \mathbb{R}$. If f is convex on (a, b), then f is continuous on (a, b). Similarly, if f is concave on (a, b), then f is continuous on (a, b).*

PROOF. Let $c \in (a, b)$. Choose $u, w \in (a, b)$ with $u < c < w$, and let $c < v < w$. Then

$$\frac{f(c) - f(u)}{c - u} \leq \frac{f(v) - f(c)}{v - c} \leq \frac{f(w) - f(c)}{w - c}.$$

The first inequality is Theorem 7.4.2. For the second, write $v = tc + (1 - t)w$ with $t = \frac{w - v}{w - c}$. By convexity, $f(v) \leq tf(c) + (1 - t)f(w)$, so $f(v) - f(c) \leq (1 - t)\big(f(w) - f(c)\big) = \frac{v - c}{w - c}\big(f(w) - f(c)\big)$. Solving for $f(v)$ gives

$$\frac{f(c) - f(u)}{c - u}(v - c) + f(c) \leq f(v) \leq \frac{f(w) - f(c)}{w - c}(v - c) + f(c).$$

Then by the Squeeze Theorem, since u and w are fixed,

$$\lim_{v \to c^+} f(v) = f(c).$$

A similar argument shows

$$\lim_{v \to c^-} f(v) = f(c).$$

So f is continuous at c. □

Example 7.4.5. The function $f : [0,1] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 0, & 0 \le x < 1, \\ 1, & x = 1, \end{cases}$$

is convex on $[0,1]$ but is not continuous at 1. So Theorem 7.4.4 needs an open interval.

Exercises

7.4.1. Prove Theorem 7.4.2. [Solution, p. 343]

7.4.2. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$ be twice differentiable on I. Prove that f is convex on I if and only if $f''(x) \ge 0$ for all $x \in I$. [Hint, p. 344]

7.4.3. Let $f, g : I \to \mathbb{R}$ be convex on an interval I. Prove that $f + g$ and $h(x) = \max\{f(x), g(x)\}$ are convex on I.

Notes

Michel Rolle proved his theorem for polynomials in 1691 [**60**]. The Mean Value Theorem is often called Lagrange's theorem, after Joseph-Louis Lagrange's 1797 book [**41**]. Augustin-Louis Cauchy stated and proved it in its modern form in 1823 [**16**]. L'Hospital's Rule appeared in the first calculus textbook, written by the Marquis de l'Hôpital in 1696 [**47**]. The result is due to his teacher, Johann Bernoulli. Gaston Darboux proved Theorem 7.2.8 in 1875 [**19**].

Part II

Riemann Integrals and Series

Part II is about integrals and infinite sums. It defines the Riemann and Riemann–Stieltjes integrals, studies series of numbers and series of functions, and ends with power series and Taylor's Theorem. It assumes Part I. Chapter 8 is marked with a star and can be skipped, since Chapter 9 repeats the definitions it needs.

CHAPTER 8

Functions of Bounded Variation*

This chapter can be skipped. Only Section 9.9 uses it. Partitions, defined in Section 8.1, are defined again in Section 9.1.

Imagine walking along the graph of f from $x = a$ to $x = b$ and recording only the vertical distance traveled, up or down. That total is the variation of f. A function of bounded variation is one for which it is finite. Monotone functions have bounded variation, and so do functions with bounded derivatives. The main result, the Jordan Decomposition Theorem, says every function of bounded variation is a difference of two increasing functions. These functions are the natural integrators for the Riemann–Stieltjes integral in Section 9.9, and they return in Part III.

8.1. Partitions and Functions of Bounded Variation

Definition 8.1.1. A *partition* $\mathcal{P}$ of $[a, b]$ is a finite collection $\{x_0, x_1, \ldots, x_n\}$ with $a = x_0 < x_1 < \cdots < x_n = b$.

Definition 8.1.2. The *mesh* of a partition $\mathcal{P}$ is defined as $\|\mathcal{P}\| = \max_i \Delta x_i$, where $\Delta x_i = x_i - x_{i-1}$.

Definition 8.1.3. A partition $\mathcal{P}$ of $[a, b]$ is *regular* if $x_k - x_{k-1} = \frac{b-a}{n}$ for all $k = 1, 2, \ldots, n$.

Definition 8.1.4. A partition $\mathcal{P}^*$ of $[a, b]$ is a *refinement* of the partition $\mathcal{P}$ if $\mathcal{P} \subset \mathcal{P}^*$.

Definition 8.1.5. Suppose $f : [a, b] \to \mathbb{R}$. If $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ is a partition of $[a, b]$, define

$$V(\mathcal{P}) = \sum_{k=1}^{n} |f(x_k) - f(x_{k-1})|.$$

Define the *total variation* of f on $[a, b]$ by

$$V_f[a, b] = \sup_{\mathcal{P}} V(\mathcal{P}),$$

where the supremum is taken over all possible partitions of $[a, b]$. If $V_f[a, b] < \infty$, f is of *bounded variation* on $[a, b]$. The set of functions of bounded variation on $[a, b]$ is denoted $BV[a, b]$.

Here $V_f[a, b] = \infty$ means the set of sums $V(\mathcal{P})$ is not bounded above, as in Remark 3.6.7. When more than one function or interval is involved, the context says which one $V(\mathcal{P})$ refers to.

Remark 8.1.6. Notice that by picking $\mathcal{P} = \{a, b\}$, $V(\mathcal{P}) = |f(b) - f(a)|$. So $V_f[a, b] \geq |f(b) - f(a)|$.

Lemma 8.1.7. *Let $f : [a, b] \to \mathbb{R}$. If $\mathcal{P}^*$ is a refinement of the partition $\mathcal{P}$ of $[a, b]$, then $V(\mathcal{P}) \leq V(\mathcal{P}^*)$.*

PROOF. First suppose $\mathcal{P}^* = \mathcal{P} \cup \{x^*\}$ has one more point than $\mathcal{P} = \{x_0, \ldots, x_n\}$. Then $x_{k-1} < x^* < x_k$ for some k. By the Triangle Inequality,

$$|f(x_k) - f(x_{k-1})| \leq |f(x_k) - f(x^*)| + |f(x^*) - f(x_{k-1})|.$$

The left side is the term of $V(\mathcal{P})$ for $[x_{k-1}, x_k]$. The right side is the sum of the two terms of $V(\mathcal{P}^*)$ that replace it. All other terms are the same. So $V(\mathcal{P}) \leq V(\mathcal{P}^*)$. In general, $\mathcal{P}^*$ has finitely many more points than $\mathcal{P}$. Adding them one at a time gives the result. □

Proof Technique (Telescoping Sums). A sum of the form $\sum_{k=1}^{n}(a_k - a_{k-1})$ collapses. Writing it out, $(a_1 - a_0) + (a_2 - a_1) + \cdots + (a_n - a_{n-1})$, every term cancels except $a_n - a_0$. Such a sum is called *telescoping*. The proof of Theorem 8.1.8 uses it with $a_k = f(x_k)$. The proof of Theorem 8.1.9 uses it with $a_k = x_k$, which gives $\sum_{k=1}^{n} \Delta x_k = b - a$ for every partition of $[a, b]$.

Theorem 8.1.8. *Let $f : [a, b] \to \mathbb{R}$ be monotone on $[a, b]$. Then f is of bounded variation on $[a, b]$ with $V_f[a, b] = |f(b) - f(a)|$.*

PROOF. Assume without loss of generality that f is increasing on $[a, b]$. (Otherwise, replace f by $-f$, which has the same variation.) Let $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. Then

$$\begin{aligned}\sum_{k=1}^{n} |f(x_k) - f(x_{k-1})| &= \sum_{k=1}^{n} \big(f(x_k) - f(x_{k-1})\big) \\ &= f(x_1) - f(x_0) + f(x_2) - f(x_1) + \cdots + f(x_n) - f(x_{n-1}) \\ &= f(x_n) - f(x_0) \\ &= f(b) - f(a).\end{aligned}$$

The first equality holds since $f(x_k) \geq f(x_{k-1})$ for each k. So every sum $V(\mathcal{P})$ equals $f(b) - f(a)$. So $V_f[a, b] = f(b) - f(a)$. □

Theorem 8.1.9. *Let $f : [a, b] \to \mathbb{R}$ be differentiable on $[a, b]$ with f' bounded on $[a, b]$. Then f is of bounded variation on $[a, b]$.*

PROOF. Since f' is bounded on $[a, b]$, there exists an $M > 0$ such that $|f'(x)| \leq M$ for all $x \in [a, b]$. Let $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. By the Mean Value Theorem, for each $k = 1, 2, \ldots, n$, there exists a $c_k \in (x_{k-1}, x_k)$ such that

$$f(x_k) - f(x_{k-1}) = f'(c_k)(x_k - x_{k-1}).$$

So

$$|f(x_k) - f(x_{k-1})| = |f'(c_k)||x_k - x_{k-1}| \leq M|x_k - x_{k-1}|.$$

This implies

$$\sum_{k=1}^{n} |f(x_k) - f(x_{k-1})| \le \sum_{k=1}^{n} M|x_k - x_{k-1}| = M(b-a).$$

Hence $V_f[a,b] \le M(b-a)$ and the function f is of bounded variation on $[a,b]$. □

Corollary 8.1.10. *Let $f \in C^{(1)}[a,b]$. Then f is of bounded variation on $[a,b]$.*

PROOF. By Theorem 6.3.14, f' is bounded on $[a,b]$. So Theorem 8.1.9 applies. □

Example 8.1.11. Notice that $f : [-1,1] \to \mathbb{R}$, $f(x) = \sqrt[3]{x}$, is of bounded variation on $[-1,1]$ since it is strictly increasing on $[-1,1]$. However, its derivative is unbounded on $[-1,1] \setminus \{0\}$, and $f'(0)$ does not exist. So the converse of Theorem 8.1.9 is false. (For $x < 0$, $\sqrt[3]{x} = -\sqrt[3]{-x}$.)

Theorem 8.1.12. *Let $f : [a,b] \to \mathbb{R}$ satisfy a Lipschitz condition on $[a,b]$. Then f is of bounded variation on $[a,b]$.*

PROOF. See Exercise 8.1.2. □

Exercises

8.1.1. Prove if $\alpha \ge \beta + 1$ and $\beta > 0$, then $f : [0,1] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} x^\alpha \sin \frac{1}{x^\beta}, & 0 < x \le 1, \\ 0, & x = 0, \end{cases}$$

is of bounded variation on $[0,1]$. [Hint, p. 344]

8.1.2. Prove Theorem 8.1.12. [Hint, p. 344]

8.1.3. Let $f : [a,b] \to \mathbb{R}$ be of bounded variation on $[a,b]$. Prove that $|f(x)| \le |f(a)| + V_f[a,b]$ for all $x \in [a,b]$. So f is bounded. [Hint, p. 344]

8.1.4. Define $f : [0,1] \to \mathbb{R}$ by $f(x) = x \sin \frac{1}{x}$ for $x > 0$ and $f(0) = 0$. Prove that f is continuous on $[0,1]$ but is not of bounded variation on $[0,1]$. [Hint, p. 344]

8.1.5. Prove that the Dirichlet function of Example 6.1.9, restricted to $[0,1]$, is not of bounded variation on $[0,1]$. [Hint, p. 344]

8.2. Jordan Decomposition Theorem

This section shows that a function of bounded variation is the difference of two increasing functions. The two functions are built from the variation of f on $[a,x]$, viewed as a function of x.

Theorem 8.2.1. *Let $f : [a,b] \to \mathbb{R}$. If f is of bounded variation on $[a,b]$ and $c \in (a,b)$, then f is of bounded variation on $[a,c]$ and $[c,b]$, and $V_f[a,b] = V_f[a,c] + V_f[c,b]$.*

PROOF. Let $\mathcal{P}_a = \{x_0, x_1, \ldots, x_j\}$ be a partition of $[a, c]$ and let $\mathcal{P}_b = \{y_0, y_1, \ldots, y_n\}$ be a partition of $[c, b]$. Then $\mathcal{P} = \mathcal{P}_a \cup \mathcal{P}_b$ is a partition of $[a, b]$ with

$$V(\mathcal{P}) = \sum_{k=1}^{j} |f(x_k) - f(x_{k-1})| + \sum_{k=1}^{n} |f(y_k) - f(y_{k-1})| \leq \sup_{\mathcal{P}} V(\mathcal{P}) = V_f[a, b].$$

Take the supremum over all partitions of $[a, c]$. Then

$$V_f[a, c] + \sum_{k=1}^{n} |f(y_k) - f(y_{k-1})| \leq \sup_{\mathcal{P}} V(\mathcal{P}) = V_f[a, b].$$

So f is of bounded variation on $[a, c]$. Then taking the supremum over all partitions of $[c, b]$,

$$V_f[a, c] + V_f[c, b] \leq \sup_{\mathcal{P}} V(\mathcal{P}) = V_f[a, b].$$

So f is of bounded variation on $[c, b]$.

Now let $\mathcal{P} = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. Without loss of generality assume $c \in \mathcal{P}$. If not, add it. By Lemma 8.1.7, this does not decrease $V(\mathcal{P})$. So there exists a $k \in \{1, \ldots, n\}$ with $c = x_k$. Then $\mathcal{P}_a = \{x_0, \ldots, x_k\}$ is a partition of $[a, c]$ and $\mathcal{P}_b = \{x_k, \ldots, x_n\}$ is a partition of $[c, b]$. Now

$$V(\mathcal{P}) = V(\mathcal{P}_a) + V(\mathcal{P}_b) \leq V_f[a, c] + V_f[c, b].$$

By taking the supremum of all partitions of $[a, b]$,

$$V_f[a, b] \leq V_f[a, c] + V_f[c, b].$$

So

$$V_f[a, b] = V_f[a, c] + V_f[c, b].$$

□

Proof Technique (Taking Suprema One at a Time). The first half of the proof of Theorem 8.2.1 has two independent choices, a partition $\mathcal{P}_a$ of $[a, c]$ and a partition $\mathcal{P}_b$ of $[c, b]$. It knows $V(\mathcal{P}_a) + V(\mathcal{P}_b) \leq V_f[a, b]$ for every pair. To get $V_f[a, c] + V_f[c, b] \leq V_f[a, b]$, it fixes $\mathcal{P}_b$ and takes the supremum over $\mathcal{P}_a$. The result still holds for every $\mathcal{P}_b$, so it then takes the supremum over $\mathcal{P}_b$. Each step uses only that a supremum is the least upper bound. If every element of a set is at most K, then so is the supremum.

Theorem 8.2.2. *Suppose that $f : [a, b] \to \mathbb{R}$ is of bounded variation on $[a, b]$. Define $V_f(x) = V_f[a, x]$ for $a < x \leq b$, and $V_f(a) = 0$. Then the function V_f is increasing on $[a, b]$.*

PROOF. By Theorem 8.2.1, V_f is a well-defined function. Now let $x_1, x_2 \in [a, b]$ with $x_1 < x_2$. If $x_1 = a$, then $V_f(x_1) = 0 \leq V_f(x_2)$. So assume $x_1 > a$. By Theorem 8.2.1 on $[a, x_2]$,

$$V_f[a, x_2] = V_f[a, x_1] + V_f[x_1, x_2].$$

So

$$V_f(x_2) - V_f(x_1) = V_f[x_1, x_2] \geq 0.$$

Thus, $V_f(x_1) \leq V_f(x_2)$ and V_f is increasing on $[a, b]$. □

Theorem 8.2.3. *Suppose that $f : [a,b] \to \mathbb{R}$ is of bounded variation on $[a,b]$. Then f is continuous at $c \in [a,b]$ if and only if V_f is continuous at c.*

PROOF. ($\Leftarrow$) Let $x \in [a,b]$. If $c < x$, then by Remark 8.1.6 and Theorem 8.2.1,

$$|f(x) - f(c)| \leq V_f[c,x] = V_f(x) - V_f(c).$$

If $x < c$, the same argument gives $|f(c) - f(x)| \leq V_f(c) - V_f(x)$. So $|f(x) - f(c)| \leq |V_f(x) - V_f(c)|$ for all $x \in [a,b]$. Let $\varepsilon > 0$. Since V_f is continuous at c, there exists a $\delta > 0$ such that $|V_f(x) - V_f(c)| < \varepsilon$ for all $x \in [a,b]$ with $|x - c| < \delta$. For these x, $|f(x) - f(c)| < \varepsilon$. So f is continuous at c.

($\Rightarrow$) Suppose f is continuous at c. First suppose $c < b$. To show $\lim_{x \to c^+} V_f(x) = V_f(c)$, let $\varepsilon > 0$. Since f is continuous at c, there exists a $\delta > 0$ such that $|f(x) - f(c)| < \frac{\varepsilon}{2}$ for all $x \in [a,b]$ with $|x - c| < \delta$. Since $V_f[c,b] - \frac{\varepsilon}{2}$ is less than the supremum $V_f[c,b]$, there exists a partition $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ of $[c,b]$ with $V(\mathcal{P}) > V_f[c,b] - \frac{\varepsilon}{2}$. By Lemma 8.1.7, adding points to $\mathcal{P}$ does not decrease $V(\mathcal{P})$. So assume $x_1 - c < \delta$ and $x_1 < b$. Then

$$V_f[c,b] - \frac{\varepsilon}{2} < V(\mathcal{P}) = |f(x_1) - f(c)| + \sum_{k=2}^{n} |f(x_k) - f(x_{k-1})| < \frac{\varepsilon}{2} + V_f[x_1,b].$$

By Theorem 8.2.1, $V_f[c,b] - V_f[x_1,b] = V_f[c,x_1]$. So $V_f[c,x_1] < \varepsilon$. Now let $c < x < x_1$. Since V_f is increasing,

$$0 \leq V_f(x) - V_f(c) = V_f[c,x] \leq V_f[c,x_1] < \varepsilon,$$

where the last step uses Theorem 8.2.2 on $[c,b]$. So $|V_f(x) - V_f(c)| < \varepsilon$ for all x with $c < x < x_1$. Thus $\lim_{x \to c^+} V_f(x) = V_f(c)$.

If $c > a$, a similar argument with a partition of $[a,c]$ shows $\lim_{x \to c^-} V_f(x) = V_f(c)$. Together, these show V_f is continuous at c. (At $c = a$ only the right-hand limit is needed, and at $c = b$ only the left-hand limit.) □

Theorem 8.2.4. *Suppose that $f : [a,b] \to \mathbb{R}$ is of bounded variation on $[a,b]$. Then the function $V_f - f$ is increasing on $[a,b]$.*

PROOF. Let $x_1, x_2 \in [a,b]$ with $x_1 < x_2$. Then

$$\begin{aligned}(V_f - f)(x_2) - (V_f - f)(x_1) &= V_f(x_2) - V_f(x_1) - (f(x_2) - f(x_1)) \\ &= V_f[x_1,x_2] - (f(x_2) - f(x_1)) \\ &\geq |f(x_2) - f(x_1)| - (f(x_2) - f(x_1)) \\ &\geq 0.\end{aligned}$$

The first inequality is Remark 8.1.6 on $[x_1,x_2]$. Thus $(V_f - f)(x_2) \geq (V_f - f)(x_1)$ and $V_f - f$ is increasing on $[a,b]$. □

Theorem 8.2.5 (Jordan Decomposition Theorem)**.** *Suppose that $f : [a,b] \to \mathbb{R}$. Then f is of bounded variation on $[a,b]$ if and only if f can be written as the difference of two increasing functions.*

PROOF. ($\Leftarrow$) Suppose that $g, h : [a,b] \to \mathbb{R}$ are increasing functions and f is defined by $f = g - h$. By Theorem 8.1.8, g and h are of bounded variation with

$V_g[a,b] = g(b) - g(a)$ and $V_h[a,b] = h(b) - h(a)$. Let $\mathcal{P} = \{x_0, \ldots, x_n\}$ be a partition of $[a,b]$. Then

$$\begin{aligned} V(\mathcal{P}) &= \sum_{k=1}^{n} |f(x_k) - f(x_{k-1})| \\ &= \sum_{k=1}^{n} |g(x_k) - g(x_{k-1}) - (h(x_k) - h(x_{k-1}))| \\ &\leq \sum_{k=1}^{n} |g(x_k) - g(x_{k-1})| + \sum_{k=1}^{n} |h(x_k) - h(x_{k-1})| \\ &= g(b) - g(a) + h(b) - h(a). \end{aligned}$$

So $V_f[a,b] \leq g(b) - g(a) + h(b) - h(a) < \infty$ and f is of bounded variation on $[a,b]$.

($\Rightarrow$) If f is of bounded variation on $[a,b]$, then by Theorems 8.2.2 and 8.2.4, V_f and $V_f - f$ are increasing functions. But $f = V_f - (V_f - f)$. So f is the difference of two increasing functions. □

Exercises

8.2.1. Write $f(x) = x^2$ on $[-1,1]$ as the difference of two monotone functions by finding $V_f(x)$. [Hint, p. 344]

8.2.2. Prove if f, g are of bounded variation on $[a,b]$, then so are $f+g$ and $f-g$. [Solution, p. 345]

8.2.3. Prove if f, g are of bounded variation on $[a,b]$, then so is fg. [Hint, p. 345]

8.2.4. Let $f : [a,b] \to \mathbb{R}$ be continuous and of bounded variation on $[a,b]$. Prove that f is the difference of two continuous increasing functions. [Hint, p. 345]

Notes

Camille Jordan introduced functions of bounded variation in 1881, in his work on Fourier series [**39**]. Section 17.3 in Part III treats bounded variation again, using positive, negative and total variation.

CHAPTER 9

Integrals

This chapter defines the Riemann integral. The approach, due to Darboux, traps the area under a graph between upper sums and lower sums. A function is integrable when the two can be made as close as anyone likes. Continuous functions and monotone functions are integrable. The integral is linear, and the Fundamental Theorem of Calculus connects it to the derivative. The starred Sections 9.4 and 9.5 use the integral to define the logarithm, the exponential function and the trigonometric functions. The chapter then treats improper integrals. The starred Section 9.8 uses them to prove Wallis's product, Stirling's formula, Gauss's integral and $\sum \frac{1}{k^2} = \frac{\pi^2}{6}$. Section 9.9, which is marked with a star, generalizes the integral by measuring length with an increasing function α in place of x.

The Riemann integral has limits. Lebesgue's Theorem (Theorem 9.2.9) says exactly which bounded functions it can integrate. Part III builds the Lebesgue integral, which integrates many more.

9.1. Darboux's Definition of the Integral

Partitions were defined in Section 8.1. For readers who skipped Chapter 8, here are the definitions this chapter needs.

Definition 9.1.1. A *partition* $\mathcal{P}$ of $[a, b]$ is a finite set $\{x_0, x_1, \ldots, x_n\}$ with $a = x_0 < x_1 < \cdots < x_n = b$. Write $\Delta x_i = x_i - x_{i-1}$. The *mesh* of $\mathcal{P}$ is $\|\mathcal{P}\| = \max_i \Delta x_i$. A partition $\mathcal{P}^*$ of $[a, b]$ is a *refinement* of $\mathcal{P}$ if $\mathcal{P} \subset \mathcal{P}^*$. (This repeats Definitions 8.1.1, 8.1.2 and 8.1.4.)

Definition 9.1.2. Let $f : [a, b] \to \mathbb{R}$ be bounded. Given a partition $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ of $[a, b]$, for each $i = 1, 2, \ldots, n$, define

$$m_i = \inf\{f(x) : x_{i-1} \le x \le x_i\}$$

and

$$M_i = \sup\{f(x) : x_{i-1} \le x \le x_i\},$$

which are well-defined since f is bounded. Then the *upper sum* $\mathcal{U}(\mathcal{P}, f)$ for the partition $\mathcal{P}$ and the function f is

$$\mathcal{U}(\mathcal{P}, f) = \sum_{i=1}^{n} M_i \Delta x_i$$

and the *lower sum* $\mathcal{L}(\mathcal{P}, f)$ is

$$\mathcal{L}(\mathcal{P}, f) = \sum_{i=1}^{n} m_i \Delta x_i.$$

Theorem 9.1.3. *Let $f : [a, b] \to \mathbb{R}$. If $m \le f(x) \le M$ for all $x \in [a, b]$, then*

$$m(b-a) \le \mathcal{L}(\mathcal{P}, f) \le \mathcal{U}(\mathcal{P}, f) \le M(b-a)$$

for any partition $\mathcal{P}$ of $[a, b]$.

PROOF. See Exercise 9.1.2. □

Definition 9.1.4 (Darboux)**.** Let $f : [a, b] \to \mathbb{R}$ be bounded. The *upper and lower integrals* of f, denoted $\overline{\int_a^b} f$ and $\underline{\int_a^b} f$, respectively, are defined by

$$\overline{\int_a^b} f = \inf\{\mathcal{U}(\mathcal{P}, f) : \mathcal{P} \text{ is a partition of } [a, b]\}$$

and

$$\underline{\int_a^b} f = \sup\{\mathcal{L}(\mathcal{P}, f) : \mathcal{P} \text{ is a partition of } [a, b]\}.$$

Lemma 9.1.5. *Let $f : [a, b] \to \mathbb{R}$ be bounded. If $\mathcal{P}^*$ is a refinement of $\mathcal{P}$, then*

$$\mathcal{L}(\mathcal{P}, f) \le \mathcal{L}(\mathcal{P}^*, f) \le \mathcal{U}(\mathcal{P}^*, f) \le \mathcal{U}(\mathcal{P}, f).$$

PROOF. Suppose $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ and that $\mathcal{P}^*$ has one more point than $\mathcal{P}$, $\mathcal{P}^* = \mathcal{P} \cup \{x^*\}$ where $x^* \ne x_j$ for all $j = 0, 1, \ldots, n$. Then there exists a $k \in \{1, \ldots, n\}$ with $x_{k-1} < x^* < x_k$. Define

$$M_k^1 = \sup\{f(x) : x_{k-1} \le x \le x^*\}$$

and

$$M_k^2 = \sup\{f(x) : x^* \le x \le x_k\}.$$

Then $M_k^1 \le M_k$ and $M_k^2 \le M_k$. So

$$\begin{aligned}
\mathcal{U}(\mathcal{P}^*, f) &= \sum_{i=1}^{k-1} M_i \Delta x_i + M_k^1 (x^* - x_{k-1}) + M_k^2 (x_k - x^*) + \sum_{i=k+1}^{n} M_i \Delta x_i \\
&\le \sum_{i=1}^{k-1} M_i \Delta x_i + M_k (x^* - x_{k-1}) + M_k (x_k - x^*) + \sum_{i=k+1}^{n} M_i \Delta x_i \\
&= \sum_{i=1}^{k-1} M_i \Delta x_i + M_k (x_k - x_{k-1}) + \sum_{i=k+1}^{n} M_i \Delta x_i \\
&= \mathcal{U}(\mathcal{P}, f).
\end{aligned}$$

The proof for $\mathcal{L}(\mathcal{P}^*, f)$ is similar. The middle inequality $\mathcal{L}(\mathcal{P}^*, f) \le \mathcal{U}(\mathcal{P}^*, f)$ holds since $m_i \le M_i$ for every subinterval.

Repeat the argument for any additional points in $\mathcal{P}^*$. □

> **Proof Technique (Common Refinements).** Two partitions $\mathcal{P}$ and $\mathcal{Q}$ of $[a, b]$ can be compared through their *common refinement* $\mathcal{P} \cup \mathcal{Q}$. It refines both. By Lemma 9.1.5, its lower sum is at least each of the two lower sums, and its upper sum is at most each of the two upper sums. So if $\mathcal{P}$ has one useful property and

$\mathcal{Q}$ has another, $\mathcal{P} \cup \mathcal{Q}$ has both. The proofs of Theorems 9.1.6, 9.2.1 and 9.2.11 all use this idea.

Theorem 9.1.6. *Let $f : [a,b] \to \mathbb{R}$ be bounded. Then $\underline{\int_a^b} f \le \overline{\int_a^b} f$.*

PROOF. Given any two partitions $\mathcal{P}$, $\mathcal{Q}$ of $[a,b]$,

$$\mathcal{L}(\mathcal{P}, f) \le \mathcal{L}(\mathcal{P} \cup \mathcal{Q}, f) \le \mathcal{U}(\mathcal{P} \cup \mathcal{Q}, f) \le \mathcal{U}(\mathcal{Q}, f).$$

Thus $\mathcal{L}(\mathcal{P}, f) \le \mathcal{U}(\mathcal{Q}, f)$ for any partitions $\mathcal{P}, \mathcal{Q}$. Hence

$$\underline{\int_a^b} f = \sup_{\mathcal{P}} \mathcal{L}(\mathcal{P}, f) \le \mathcal{U}(\mathcal{Q}, f)$$

for any partition $\mathcal{Q}$. Since $\underline{\int_a^b} f$ is independent of the partition $\mathcal{Q}$,

$$\underline{\int_a^b} f \le \inf_{\mathcal{Q}} \mathcal{U}(\mathcal{Q}, f) = \overline{\int_a^b} f.$$

□

Definition 9.1.7. Let $f : [a,b] \to \mathbb{R}$ be bounded. If $\underline{\int_a^b} f = \overline{\int_a^b} f$, then f is *(Riemann) integrable* on $[a,b]$. The common value is denoted $\int_a^b f$ or $\int_a^b f(x)\,dx$ and is called the *(Riemann) integral* of f over $[a,b]$. Denote the set of Riemann integrable functions on $[a,b]$ by $\mathcal{R}[a,b]$. In addition, if $f \in \mathcal{R}[a,b]$, define $\int_b^a f = -\int_a^b f$. Also define $\int_a^a f = 0$.

Remark 9.1.8. Part III defines the Lebesgue integral and the set $L^1[a,b]$ of Lebesgue integrable functions. These are different from $\int_a^b f$ and $\mathcal{R}[a,b]$. Every $f \in \mathcal{R}[a,b]$ is Lebesgue integrable, with the same integral. But $L^1[a,b]$ is much larger. For example, the Dirichlet function in Example 9.1.9 is in $L^1[0,1]$ but not in $\mathcal{R}[0,1]$.

Example 9.1.9. Consider the Dirichlet function (Example 6.1.9) $f : [0,1] \to \mathbb{R}$ defined by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Let $\mathcal{P} = \{x_0, \ldots, x_n\}$ be any partition of $[0,1]$. Then $m_i = 0$ and $M_i = 1$ for $i = 1, \ldots, n$. Then $\mathcal{L}(\mathcal{P}, f) = 0$ and $\mathcal{U}(\mathcal{P}, f) = 1$. Therefore,

$$\underline{\int_0^1} f = 0 \text{ and } \overline{\int_0^1} f = 1.$$

So $f \notin \mathcal{R}[0,1]$.

Proof Technique (Computing an Integral from the Definition). To show $f \in \mathcal{R}[a,b]$ with $\int_a^b f = I$, it is enough to do two things. First, show $\mathcal{L}(\mathcal{P}, f) \le I \le \mathcal{U}(\mathcal{P}, f)$ for every partition $\mathcal{P}$. Then $\underline{\int_a^b} f \le I \le \overline{\int_a^b} f$. Second, show $\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f)$ can be made smaller than any $\varepsilon > 0$ by choosing $\mathcal{P}$.

Then the upper and lower integrals are within every ε of each other, so they are equal, and both equal I. Examples 9.1.10 and 9.1.11 work this way. Often $\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f)$ is bounded by a multiple of the mesh $\|\mathcal{P}\|$.

Example 9.1.10. Let $f : [0, 1] \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 0, & x \in [0, 1/2), \\ 1, & x \in [1/2, 1]. \end{cases}$$

Let $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[0, 1]$. Then there is a $k \in \{1, \ldots, n\}$ such that $x_{k-1} < 1/2 \le x_k$. So

$$m_i = \begin{cases} 0, & i = 1, \ldots, k, \\ 1, & i = k+1, \ldots, n, \end{cases} \quad \text{and} \quad M_i = \begin{cases} 0, & i = 1, \ldots, k-1, \\ 1, & i = k, \ldots, n. \end{cases}$$

Thus

$$\mathcal{L}(\mathcal{P}, f) = \sum_{i=k+1}^{n} \Delta x_i = 1 - x_k$$

and

$$\mathcal{U}(\mathcal{P}, f) = \sum_{i=k}^{n} \Delta x_i = 1 - x_{k-1}.$$

Since

$$1 - x_k \le \frac{1}{2} < 1 - x_{k-1},$$

$\mathcal{L}(\mathcal{P}, f) \le \frac{1}{2} \le \mathcal{U}(\mathcal{P}, f)$ for all partitions $\mathcal{P}$ of $[0, 1]$. Thus

$$\underline{\int_0^1} f \le \frac{1}{2} \le \overline{\int_0^1} f.$$

Since

$$1 - x_{k-1} = (1 - x_k) + (x_k - x_{k-1}),$$

$\mathcal{U}(\mathcal{P}, f) = \mathcal{L}(\mathcal{P}, f) + (x_k - x_{k-1})$.

Let $\varepsilon > 0$. Let $\mathcal{P}$ be a partition with $\|\mathcal{P}\| < \varepsilon$. Then $\mathcal{U}(\mathcal{P}, f) \le \mathcal{L}(\mathcal{P}, f) + \varepsilon$. Thus

$$\begin{aligned} 0 &\le \overline{\int_0^1} f - \underline{\int_0^1} f \\ &\le \mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) \\ &\le \varepsilon. \end{aligned}$$

Thus, $\underline{\int_0^1} f = \overline{\int_0^1} f$. Since

$$\underline{\int_0^1} f \le \frac{1}{2} \le \overline{\int_0^1} f,$$

$\int_0^1 f = \frac{1}{2}$.

Example 9.1.11. Consider $f : [a, b] \to \mathbb{R}$ defined by $f(x) = x$. Let $\mathcal{P} = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. Then $m_i = f(x_{i-1}) = x_{i-1}$ and $M_i = f(x_i) = x_i$. Thus

$$\mathcal{L}(\mathcal{P}, f) = \sum_{i=1}^{n} x_{i-1}\Delta x_i \text{ and } \mathcal{U}(\mathcal{P}, f) = \sum_{i=1}^{n} x_i \Delta x_i.$$

For each $i = 1, \ldots, n$,

$$x_{i-1} \leq \frac{1}{2}(x_{i-1} + x_i) \leq x_i.$$

Multiplying by $\Delta x_i = x_i - x_{i-1}$ gives

$$\mathcal{L}(\mathcal{P}, f) \leq \sum_{i=1}^{n} \frac{1}{2}(x_i^2 - x_{i-1}^2) \leq \mathcal{U}(\mathcal{P}, f).$$

But $\sum_{i=1}^{n} \frac{1}{2}(x_i^2 - x_{i-1}^2)$ is a telescoping sum, so

$$\sum_{i=1}^{n} \frac{1}{2}(x_i^2 - x_{i-1}^2) = \frac{1}{2}(x_n^2 - x_0^2) = \frac{1}{2}(b^2 - a^2).$$

Thus, by taking the supremum and infimum over all partitions $\mathcal{P}$ of $[a, b]$,

$$\underline{\int_a^b} f \leq \frac{1}{2}(b^2 - a^2) \leq \overline{\int_a^b} f.$$

Now, for any partition $\mathcal{P}$ of $[a, b]$,

$$\begin{aligned} 0 &\leq \overline{\int_a^b} f - \underline{\int_a^b} f \\ &\leq \mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) \\ &= \sum_{i=1}^{n} (x_i - x_{i-1})\Delta x_i \\ &\leq \|\mathcal{P}\|(b - a). \end{aligned}$$

Let $\varepsilon > 0$. Choose a partition $\mathcal{P}$ such that $\|\mathcal{P}\| < \frac{\varepsilon}{b-a+1}$. Then

$$\overline{\int_a^b} f - \underline{\int_a^b} f < \varepsilon.$$

So $\overline{\int_a^b} f = \underline{\int_a^b} f$. Since

$$\underline{\int_a^b} f \leq \frac{1}{2}(b^2 - a^2) \leq \overline{\int_a^b} f,$$

$$\int_a^b f = \frac{1}{2}(b^2 - a^2).$$

Example 9.1.12. Consider $f : [0, 1] \to \mathbb{R}$ defined by $f(x) = x^2$. For $n \in \mathbb{N}$, let $\mathcal{P}_n$ be the partition $\left\{0, \frac{1}{n}, \frac{2}{n}, \ldots, 1\right\}$. Then

$$m_i = \frac{(i-1)^2}{n^2} \text{ and } M_i = \frac{i^2}{n^2}.$$

Since $\Delta x_i = \frac{1}{n}$ for all i,

$$\mathcal{L}(\mathcal{P}_n, f) = \frac{1}{n^3}[1^2 + 2^2 + \cdots + (n-1)^2]$$

and

$$\mathcal{U}(\mathcal{P}_n, f) = \frac{1}{n^3}[1^2 + 2^2 + \cdots + n^2].$$

Since

$$1^2 + 2^2 + \cdots + m^2 = \frac{1}{6}m(m+1)(2m+1)$$

by Exercise 1.2.6,

$$\mathcal{L}(\mathcal{P}_n, f) = \frac{1}{6}\left(1 - \frac{1}{n}\right)\left(2 - \frac{1}{n}\right)$$

and

$$\mathcal{U}(\mathcal{P}_n, f) = \frac{1}{6}\left(1 + \frac{1}{n}\right)\left(2 + \frac{1}{n}\right).$$

Then $\sup_n \mathcal{L}(\mathcal{P}_n, f) = \frac{1}{3}$ and $\inf_n \mathcal{U}(\mathcal{P}_n, f) = \frac{1}{3}$. But

$$\{\mathcal{P}_n : n \in \mathbb{N}\} \subset \{\mathcal{P} : \mathcal{P} \text{ is a partition of } [0,1]\}.$$

So

$$\frac{1}{3} = \sup_n \mathcal{L}(\mathcal{P}_n, f) \leq \sup_{\mathcal{P}} \mathcal{L}(\mathcal{P}, f) = \underline{\int_0^1} f,$$

and

$$\frac{1}{3} = \inf_n \mathcal{U}(\mathcal{P}_n, f) \geq \inf_{\mathcal{P}} \mathcal{U}(\mathcal{P}, f) = \overline{\int_0^1} f.$$

So $\int_0^1 f = \frac{1}{3}$ by Theorem 9.1.6.

Exercises

9.1.1. Let $f : [-1, 2] \to \mathbb{R}$, $f(x) = 1 - x^2$. For $\mathcal{P} = \{-1, 0, 1, 2\}$, calculate $\mathcal{L}(\mathcal{P}, f)$ and $\mathcal{U}(\mathcal{P}, f)$.

9.1.2. Prove Theorem 9.1.3. [Hint, p. 345]

9.1.3. Let $f : [0, 1] \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} 0, & x \in \mathbb{Q}, \\ x, & x \notin \mathbb{Q}. \end{cases}$$

Compute $\underline{\int_0^1} f$ and $\overline{\int_0^1} f$. Is $f \in \mathcal{R}[0, 1]$? [Hint, p. 345]

9.1.4. Use the definition of the integral to calculate $\int_0^1 \lfloor 3x \rfloor \, dx$. [Hint, p. 345]

9.1.5. Use the definition of the integral to calculate $\int_0^1 (2x + 1) \, dx$. [Hint, p. 345]

9.1.6. Use the definition of the integral to calculate, for $0 \leq a < b$, $\int_a^b x^2 \, dx$. [Hint, p. 346]

9.2. Properties of the Integral

Computing upper and lower integrals directly is hard. This section gives a test for integrability that needs only one good partition, and uses it to show that continuous functions and monotone functions are integrable. It then proves the algebraic properties of the integral.

Theorem 9.2.1 (Riemann's Criterion for Integrability)**.** *Let $f : [a,b] \to \mathbb{R}$ be bounded. The function $f \in \mathcal{R}[a,b]$ if and only if for all $\varepsilon > 0$, there exists a partition $\mathcal{P}$ of $[a,b]$ such that*

$$\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) < \varepsilon.$$

Furthermore, if $\mathcal{P}$ is a partition of $[a,b]$ for which

$$\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) < \varepsilon,$$

then the inequality holds for all refinements of $\mathcal{P}$.

Proof. ($\Leftarrow$) Let $\varepsilon > 0$. Suppose there exists a partition $\mathcal{P}$ of $[a,b]$ with

$$\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) < \varepsilon.$$

Since

$$\underline{\int_a^b} f \geq \mathcal{L}(\mathcal{P}, f)$$

and

$$\mathcal{U}(\mathcal{P}, f) \geq \overline{\int_a^b} f,$$

$$0 \leq \overline{\int_a^b} f - \underline{\int_a^b} f \leq \mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $f \in \mathcal{R}[a,b]$.

($\Rightarrow$) Let $f \in \mathcal{R}[a,b]$ and let $\varepsilon > 0$. Write $I = \int_a^b f$. Since I is the infimum of the upper sums, Theorem 1.8.7 gives a partition $\mathcal{P}_1$ with $\mathcal{U}(\mathcal{P}_1, f) < I + \frac{\varepsilon}{2}$. Since I is also the supremum of the lower sums, it gives a partition $\mathcal{P}_2$ with $\mathcal{L}(\mathcal{P}_2, f) > I - \frac{\varepsilon}{2}$. The common refinement $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$ keeps both properties, by Lemma 9.1.5. So

$$\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) \leq \mathcal{U}(\mathcal{P}_1, f) - \mathcal{L}(\mathcal{P}_2, f) < \left(I + \frac{\varepsilon}{2}\right) - \left(I - \frac{\varepsilon}{2}\right) = \varepsilon.$$

Finally, if $\mathcal{P}^*$ refines $\mathcal{P}$, then $\mathcal{U}(\mathcal{P}^*, f) - \mathcal{L}(\mathcal{P}^*, f) \leq \mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f)$ by Lemma 9.1.5. So the inequality holds for all refinements of $\mathcal{P}$. □

Theorem 9.2.2. *Let $f : [a,b] \to \mathbb{R}$.*

(1) If f is continuous on $[a,b]$, $f \in \mathcal{R}[a,b]$.
(2) If f is monotone on $[a,b]$, $f \in \mathcal{R}[a,b]$.

Proof. (1) Let $\varepsilon > 0$. By Corollary 6.4.8, f is uniformly continuous on $[a,b]$. So there exists a $\delta > 0$ such that $|f(x) - f(y)| < \frac{\varepsilon}{b-a}$ for all $x, y \in [a,b]$ with $|x - y| < \delta$. Choose a partition $\mathcal{P}$ of $[a,b]$ with $\|\mathcal{P}\| < \delta$.

Each subinterval $[x_{i-1}, x_i]$ has length less than δ, so any two values of f on it differ by less than $\frac{\varepsilon}{b-a}$. So $M_i - m_i \leq \frac{\varepsilon}{b-a}$ for each i, and

$$\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) = \sum_{i=1}^{n} (M_i - m_i)\Delta x_i \leq \frac{\varepsilon}{b-a} \sum_{i=1}^{n} \Delta x_i = \varepsilon.$$

This holds for every $\varepsilon > 0$, so $f \in \mathcal{R}[a, b]$ by Riemann's Criterion.

(2) Suppose f is increasing. (If f is decreasing, apply this to $-f$ and use Theorem 9.2.11 (2).) Let $\varepsilon > 0$, and choose $n \in \mathbb{N}$ with $\frac{b-a}{n}\big(f(b) - f(a)\big) < \varepsilon$. Let $\mathcal{P}$ be the partition of $[a, b]$ into n subintervals of length $\frac{b-a}{n}$. Since f is increasing, the smallest and largest values of f on $[x_{i-1}, x_i]$ are at the endpoints. So $m_i = f(x_{i-1})$ and $M_i = f(x_i)$, and the sum telescopes.

$$\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) = \frac{b-a}{n} \sum_{i=1}^{n} \big(f(x_i) - f(x_{i-1})\big) = \frac{b-a}{n}\big(f(b) - f(a)\big) < \varepsilon.$$

So $f \in \mathcal{R}[a, b]$ by Riemann's Criterion. □

Proof Technique (Splitting into Good and Bad Pieces). Some estimates work on most of a partition but not all of it. Then split the subintervals into two groups. On the good subintervals, $M_i - m_i$ is small. On the bad ones, $M_i - m_i$ is only bounded, but the bad subintervals have small total length. Each group then contributes a small amount to $\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f)$. The proof of Theorem 9.2.3 works this way.

Theorem 9.2.3. *Let $f \in \mathcal{R}[a, b]$, with $f[[a, b]] \subset [c, d]$. If $\varphi : [c, d] \to \mathbb{R}$ is continuous on $[c, d]$, then $\varphi \circ f \in \mathcal{R}[a, b]$.*

Proof. By Theorem 6.3.14, there exists a $K > 0$ with $|\varphi(y)| \leq K$ for all $y \in [c, d]$. Let $\varepsilon > 0$, and let $\eta = \frac{\varepsilon}{b-a+2K}$. By Corollary 6.4.8, φ is uniformly continuous on $[c, d]$. So there exists a δ with $0 < \delta < \eta$ such that $|\varphi(y) - \varphi(z)| < \eta$ for all $y, z \in [c, d]$ with $|y - z| < \delta$. By Riemann's Criterion, there exists a partition $\mathcal{P} = \{x_0, \ldots, x_n\}$ of $[a, b]$ with $\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) < \delta^2$.

Write m_i, M_i for the infimum and supremum of f on $[x_{i-1}, x_i]$, and m_i^*, M_i^* for those of $\varphi \circ f$. Call i good if $M_i - m_i < \delta$ and bad otherwise. If i is good, then any two values $f(x)$ and $f(y)$ with $x, y \in [x_{i-1}, x_i]$ differ by less than δ. So $|\varphi(f(x)) - \varphi(f(y))| < \eta$, and $M_i^* - m_i^* \leq \eta$. If i is bad, then $M_i^* - m_i^* \leq 2K$. The bad subintervals have small total length, since

$$\delta \sum_{i \text{ bad}} \Delta x_i \leq \sum_{i \text{ bad}} (M_i - m_i)\Delta x_i \leq \mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) < \delta^2,$$

so $\sum_{i \text{ bad}} \Delta x_i < \delta$. Therefore

$$\begin{aligned} \mathcal{U}(\mathcal{P}, \varphi \circ f) - \mathcal{L}(\mathcal{P}, \varphi \circ f) &= \sum_{i \text{ good}} (M_i^* - m_i^*)\Delta x_i + \sum_{i \text{ bad}} (M_i^* - m_i^*)\Delta x_i \\ &\leq \eta(b-a) + 2K\delta \\ &< \eta(b - a + 2K) = \varepsilon. \end{aligned}$$

So $\varphi \circ f \in \mathcal{R}[a,b]$ by Riemann's Criterion. □

Corollary 9.2.4. *If $f \in \mathcal{R}[a,b]$, then $|f|$ and $f^2 \in \mathcal{R}[a,b]$.*

PROOF. Since f is bounded, $f[[a,b]] \subset [-M,M]$ for some $M > 0$. Apply Theorem 9.2.3 with $\varphi(y) = |y|$ and with $\varphi(y) = y^2$ on $[-M,M]$. □

Definition 9.2.5. The set $E \subset \mathbb{R}$ has *measure zero*, written $m(E) = 0$, if for all $\varepsilon > 0$, there exists a finite or countable collection $\{I_n\}_n$ of open intervals such that

$$E \subset \bigcup_n I_n \text{ and } \sum_n \ell(I_n) < \varepsilon,$$

where $\ell(I_n)$ denotes the length of I_n. For a countable collection, the condition $\sum_n \ell(I_n) < \varepsilon$ means $\sum_{n=1}^N \ell(I_n) < \varepsilon$ for every $N \in \mathbb{N}$. Infinite series are studied in Chapter 10.

Example 9.2.6. Every finite set $E = \{x_1, \ldots, x_n\}$ has measure zero. To see this, let $\varepsilon > 0$ and set $I_k = \left(x_k - \frac{\varepsilon}{4n}, x_k + \frac{\varepsilon}{4n}\right)$. Then $E \subset \bigcup_{k=1}^n I_k$ and $\sum_{k=1}^n \ell(I_k) = \varepsilon/2 < \varepsilon$.

Example 9.2.7. Every countable set $E = \{x_n\}_{n=1}^\infty$ has measure zero. To see this, let $\varepsilon > 0$ and set $I_n = \left(x_n - \frac{\varepsilon}{2^{n+2}}, x_n + \frac{\varepsilon}{2^{n+2}}\right)$. Then $E \subset \bigcup_n I_n$ and $\ell(I_n) = \frac{\varepsilon}{2^{n+1}}$. By Exercise 1.2.7, for every $N \in \mathbb{N}$,

$$\sum_{n=1}^N \ell(I_n) = \frac{\varepsilon}{2}\left(1 - \frac{1}{2^N}\right) < \varepsilon.$$

Example 9.2.8. The Cantor set has measure zero. This example uses Chapter 5, which is optional. Let $\varepsilon > 0$. For each n, $C \subset C_n$, and C_n is the union of 2^n closed intervals of length $\frac{1}{3^n}$ by Lemma 5.1.2. Enlarge each to an open interval of length $\frac{2}{3^n}$ with the same midpoint. These 2^n open intervals cover C, and their total length is $2\left(\frac{2}{3}\right)^n$. By Example 3.1.7, this is less than ε for large n.

Theorem 9.2.9 (Lebesgue's Theorem)**.** *Let $f : [a,b] \to \mathbb{R}$ be bounded. Then $f \in \mathcal{R}[a,b]$ if and only if the set of points of $[a,b]$ at which f is discontinuous has measure zero.*

The proof of this theorem needs Lebesgue measure. It is given in Part III, as Theorem 16.3.11.

Remark 9.2.10. Notice that Theorem 9.2.2 is an immediate consequence of Lebesgue's Theorem. If f is continuous, it has no discontinuities. If f is monotone, by Corollary 6.5.3, f has at most countably many discontinuities in (a,b), and so at most countably many in $[a,b]$. These form a set of measure zero by Examples 9.2.6 and 9.2.7.

Theorem 9.2.11. *Let $f, g \in \mathcal{R}[a,b]$. Then*

(1) $f + g \in \mathcal{R}[a,b]$ with $\int_a^b (f+g) = \int_a^b f + \int_a^b g$;
(2) $cf \in \mathcal{R}[a,b]$ for all $c \in \mathbb{R}$ with $\int_a^b cf = c\int_a^b f$; and
(3) $fg \in \mathcal{R}[a,b]$.

PROOF. (1) Let $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a, b]$. For each $i = 1, \ldots, n$, let

$$M_i(f) = \sup\{f(x) : x \in [x_{i-1}, x_i]\},$$

and

$$M_i(g) = \sup\{g(x) : x \in [x_{i-1}, x_i]\}.$$

Then

$$f(x) + g(x) \leq M_i(f) + M_i(g)$$

for all $x \in [x_{i-1}, x_i]$. Therefore,

$$\sup\{f(x) + g(x) : x \in [x_{i-1}, x_i]\} \leq M_i(f) + M_i(g),$$

which implies

$$\mathcal{U}(\mathcal{P}, f + g) \leq \mathcal{U}(\mathcal{P}, f) + \mathcal{U}(\mathcal{P}, g).$$

Let $\varepsilon > 0$. Since $f, g \in \mathcal{R}[a, b]$, there exist partitions $\mathcal{P}_f$ and $\mathcal{P}_g$ of $[a, b]$ such that

$$\mathcal{U}(\mathcal{P}_f, f) < \int_a^b f + \frac{\varepsilon}{2} \text{ and } \mathcal{U}(\mathcal{P}_g, g) < \int_a^b g + \frac{\varepsilon}{2}.$$

Let $\mathcal{Q} = \mathcal{P}_f \cup \mathcal{P}_g$. Since $\mathcal{Q}$ is a refinement of both $\mathcal{P}_f$ and $\mathcal{P}_g$,

$$\mathcal{U}(\mathcal{Q}, f + g) \leq \mathcal{U}(\mathcal{P}_f, f) + \mathcal{U}(\mathcal{P}_g, g) < \int_a^b f + \int_a^b g + \varepsilon.$$

Therefore,

$$\overline{\int_a^b} (f + g) < \int_a^b f + \int_a^b g + \varepsilon,$$

implying

$$\overline{\int_a^b} (f + g) \leq \int_a^b f + \int_a^b g.$$

Similarly,

$$\underline{\int_a^b} (f + g) \geq \int_a^b f + \int_a^b g,$$

which implies $f + g \in \mathcal{R}[a, b]$ with

$$\int_a^b (f + g) = \int_a^b f + \int_a^b g.$$

(The last step uses Theorem 9.1.6.)

(2) This part is an exercise. See Exercise 9.2.1.

(3) Notice

$$fg = \frac{1}{4}\left[(f + g)^2 - (f - g)^2\right].$$

By (2), $-g \in \mathcal{R}[a, b]$. So by (1), $f + g, f - g \in \mathcal{R}[a, b]$. By Corollary 9.2.4, $(f + g)^2, (f - g)^2 \in \mathcal{R}[a, b]$. By (2), $-(f - g)^2 \in \mathcal{R}[a, b]$. By (1), $(f + g)^2 - (f - g)^2 \in \mathcal{R}[a, b]$. Finally, by (2), $\frac{1}{4}[(f + g)^2 - (f - g)^2] \in \mathcal{R}[a, b]$. Thus, $fg \in \mathcal{R}[a, b]$. □

Theorem 9.2.12. *If $f \in \mathcal{R}[a, b]$, then $|f| \in \mathcal{R}[a, b]$ with $\left|\int_a^b f\right| \leq \int_a^b |f|$.*

PROOF. Since $f \in \mathcal{R}[a,b]$, by Corollary 9.2.4, $|f| \in \mathcal{R}[a,b]$. Let $c \in \{\pm 1\}$ be such that

$$\left|\int_a^b f\right| = c\int_a^b f.$$

Since $cf(x) \le |f(x)|$ for all $x \in [a,b]$, $\mathcal{U}(\mathcal{P}, cf) \le \mathcal{U}(\mathcal{P}, |f|)$ for any partition $\mathcal{P}$ of $[a,b]$. Therefore, since both $cf, |f| \in \mathcal{R}[a,b]$,

$$\int_a^b cf \le \int_a^b |f|$$

(see also Exercise 9.2.3). Combining the above gives

$$\left|\int_a^b f\right| = c\int_a^b f = \int_a^b cf \le \int_a^b |f|.$$

□

Theorem 9.2.13. *Let $f : [a,b] \to \mathbb{R}$ be bounded and suppose $c \in (a,b)$. Then $f \in \mathcal{R}[a,b]$ if and only if $f \in \mathcal{R}[a,c]$ and $f \in \mathcal{R}[c,b]$. If this is the case,*

$$\int_a^b f = \int_a^c f + \int_c^b f.$$

PROOF. Let $c \in (a,b)$. Suppose $\mathcal{P}_1$ and $\mathcal{P}_2$ are partitions of $[a,c]$ and $[c,b]$, respectively. Then $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$ is a partition of $[a,b]$ with $c \in \mathcal{P}$. Conversely, if $\mathcal{P}$ is a partition of $[a,b]$ with $c \in \mathcal{P}$, then $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$, where $\mathcal{P}_1$ and $\mathcal{P}_2$ are partitions of $[a,c]$ and $[c,b]$, respectively.

For any such partition $\mathcal{P}$,

$$\mathcal{U}(\mathcal{P}, f) = \mathcal{U}(\mathcal{P}_1, f) + \mathcal{U}(\mathcal{P}_2, f) \ge \overline{\int_a^c} f + \overline{\int_c^b} f.$$

If $\mathcal{Q}$ is any partition of $[a,b]$, then $\mathcal{P} = \mathcal{Q} \cup \{c\}$ is a refinement of $\mathcal{Q}$ containing c. Therefore,

$$\mathcal{U}(\mathcal{Q}, f) \ge \mathcal{U}(\mathcal{P}, f) \ge \overline{\int_a^c} f + \overline{\int_c^b} f.$$

Taking the infimum over all partitions $\mathcal{Q}$ of $[a,b]$ gives

$$\overline{\int_a^b} f \ge \overline{\int_a^c} f + \overline{\int_c^b} f.$$

Next, let $\varepsilon > 0$. Then there exist partitions $\mathcal{P}_1$ and $\mathcal{P}_2$ of $[a,c]$ and $[c,b]$, respectively, such that

$$\mathcal{U}(\mathcal{P}_1, f) < \overline{\int_a^c} f + \frac{\varepsilon}{2} \text{ and } \mathcal{U}(\mathcal{P}_2, f) < \overline{\int_c^b} f + \frac{\varepsilon}{2}.$$

Let $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$. Then

$$\begin{aligned}\overline{\int_a^b} f &\le \mathcal{U}(\mathcal{P}, f) \\ &= \mathcal{U}(\mathcal{P}_1, f) + \mathcal{U}(\mathcal{P}_2, f)\end{aligned}$$

$$< \overline{\int_a^c} f + \overline{\int_c^b} f + \varepsilon.$$

So

$$\overline{\int_a^b} f \leq \overline{\int_a^c} f + \overline{\int_c^b} f.$$

Therefore,

$$\overline{\int_a^b} f = \overline{\int_a^c} f + \overline{\int_c^b} f.$$

A similar argument gives

$$\underline{\int_a^b} f = \underline{\int_a^c} f + \underline{\int_c^b} f.$$

($\Leftarrow$) If $f \in \mathcal{R}[a,c]$ and $f \in \mathcal{R}[c,b]$, then

$$\int_a^c f + \int_c^b f = \underline{\int_a^b} f \leq \overline{\int_a^b} f = \int_a^c f + \int_c^b f.$$

So $f \in \mathcal{R}[a,b]$ with

$$\int_a^c f + \int_c^b f = \int_a^b f.$$

($\Rightarrow$) If $f \in \mathcal{R}[a,b]$, then

$$\underline{\int_a^c} f + \underline{\int_c^b} f = \overline{\int_a^c} f + \overline{\int_c^b} f.$$

But

$$\underline{\int_a^c} f \leq \overline{\int_a^c} f \text{ and } \underline{\int_c^b} f \leq \overline{\int_c^b} f.$$

So

$$\underline{\int_a^c} f = \overline{\int_a^c} f \text{ and } \underline{\int_c^b} f = \overline{\int_c^b} f.$$

Thus $f \in \mathcal{R}[a,c]$ and $f \in \mathcal{R}[c,b]$ with

$$\int_a^c f + \int_c^b f = \int_a^b f.$$

□

Exercises

9.2.1. Prove Theorem 9.2.11 (2). [Hint, p. 346]

9.2.2. Assume $f(x) = 0$ for all $x \neq c_1, \ldots, c_n \in [a,b]$. Prove $f \in \mathcal{R}[a,b]$ and $\int_a^b f = 0$. [Hint, p. 346]

9.2.3. Let $f, g \in \mathcal{R}[a,b]$ with $f(x) \leq g(x)$ for all $x \in [a,b]$. Prove that $\int_a^b f \leq \int_a^b g$. [Hint, p. 346]

9.2.4. Let $f \in \mathcal{R}[a,b]$, and let $g : [a,b] \to \mathbb{R}$ be equal to f except at finitely many points. Prove that $g \in \mathcal{R}[a,b]$ and $\int_a^b g = \int_a^b f$. [Hint, p. 347]

9.2.5. Let $f \in C[a,b]$ with $f(x) \geq 0$ for all $x \in [a,b]$ and $\int_a^b f = 0$. Prove that $f(x) = 0$ for all $x \in [a,b]$. [Hint, p. 347]

9.3. The Fundamental Theorem of Calculus

Calculus courses define the integral as a limit of Riemann sums. Theorem 9.3.3 shows that this gives the same integral as Darboux's definition. The section then proves the two parts of the Fundamental Theorem of Calculus, and uses the second to define the natural logarithm.

Definition 9.3.1. Let $f : [a,b] \to \mathbb{R}$ be bounded and let $\mathcal{P} = \{x_0, \ldots, x_n\}$ be a partition of $[a,b]$. For each $i = 1, 2, \ldots, n$, choose $t_i \in [x_{i-1}, x_i]$. The sum

$$\mathcal{I}(\mathcal{P}, f) = \sum_{i=1}^{n} f(t_i)\Delta x_i$$

is called the *Riemann sum* of f with respect to the partition $\mathcal{P}$ and the points $\{t_i\}$.

Definition 9.3.2. Let $f : [a,b] \to \mathbb{R}$ be bounded. Then

$$\lim_{\|\mathcal{P}\| \to 0} \mathcal{I}(\mathcal{P}, f) = I$$

if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\left| \sum_{i=1}^{n} f(t_i)\Delta x_i - I \right| < \varepsilon$$

for all partitions $\mathcal{P}$ of $[a,b]$ with $\|\mathcal{P}\| < \delta$ and for all choices of $t_i \in [x_{i-1}, x_i]$.

Theorem 9.3.3. *Let $f : [a,b] \to \mathbb{R}$ be bounded. If $\lim_{\|\mathcal{P}\| \to 0} \mathcal{I}(\mathcal{P}, f) = I$, then $f \in \mathcal{R}[a,b]$ with $\int_a^b f = I$. Conversely, if $f \in \mathcal{R}[a,b]$, then $\lim_{\|\mathcal{P}\| \to 0} \mathcal{I}(\mathcal{P}, f)$ exists with*

$$\lim_{\|\mathcal{P}\| \to 0} \mathcal{I}(\mathcal{P}, f) = \int_a^b f.$$

Proof. ($\Rightarrow$) Suppose $\lim_{\|\mathcal{P}\| \to 0} \mathcal{I}(\mathcal{P}, f) = I$. Let $\varepsilon > 0$. Then there exists a $\delta > 0$ such that $|\mathcal{I}(\mathcal{P}, f) - I| < \varepsilon/4$ for all partitions $\mathcal{P} = \{x_0, \ldots, x_n\}$ of $[a,b]$ with $\|\mathcal{P}\| < \delta$, and all $t_i \in [x_{i-1}, x_i]$. Choose a partition $\mathcal{P}$ with $\|\mathcal{P}\| < \delta$. By the definition of M_i, for each $i = 1, \ldots, n$, there exists $\xi_i \in [x_{i-1}, x_i]$ such that $f(\xi_i) > M_i - \frac{\varepsilon}{4(b-a+1)}$. So

$$\begin{aligned}
\mathcal{U}(\mathcal{P}, f) &= \sum_{i=1}^{n} M_i \Delta x_i \\
&< \sum_{i=1}^{n} f(\xi_i)\Delta x_i + \sum_{i=1}^{n} \frac{\varepsilon}{4(b-a+1)} \Delta x_i \\
&< I + \frac{\varepsilon}{4} + \frac{\varepsilon}{4} \\
&= I + \frac{\varepsilon}{2}.
\end{aligned}$$

Similarly, $\mathcal{L}(\mathcal{P}, f) > I - \frac{\varepsilon}{2}$. So

$$\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) < \varepsilon,$$

and so, by Riemann's Criterion for Integrability, $f \in \mathcal{R}[a, b]$ with $\int_a^b f = I$. The value is I since $\mathcal{L}(\mathcal{P}, f) \le \int_a^b f \le \mathcal{U}(\mathcal{P}, f)$, so $\left| \int_a^b f - I \right| < \frac{\varepsilon}{2}$ for every $\varepsilon > 0$.

($\Leftarrow$) Now suppose $f \in \mathcal{R}[a, b]$. Let $M > 0$ be such that $|f(x)| \le M$ for all $x \in [a, b]$. Let $\varepsilon > 0$. Since $f \in \mathcal{R}[a, b]$, there exists a partition $\mathcal{Q}$ of $[a, b]$ such that

$$\int_a^b f - \frac{\varepsilon}{2} < \mathcal{L}(\mathcal{Q}, f) \le \mathcal{U}(\mathcal{Q}, f) < \int_a^b f + \frac{\varepsilon}{2}.$$

Suppose $\mathcal{Q} = \{x_0, \ldots, x_N\}$. Let $\delta = \frac{\varepsilon}{4MN}$ and let $\mathcal{P} = \{y_0, \ldots, y_n\}$ be a partition of $[a, b]$ with $\|\mathcal{P}\| < \delta$. Let

$$M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}, \quad i = 1, \ldots, N.$$

Consider an interval $[y_{k-1}, y_k]$, $k \in \{1, \ldots, n\}$. This interval may or may not contain any points $x_i \in \mathcal{Q}$. Since $\mathcal{Q}$ contains $N + 1$ points, at most $N - 1$ intervals $[y_{k-1}, y_k]$ contain an $x_i \in \mathcal{Q}$, $i \ne 0, N$. Suppose $x_j, x_{j+1}, \ldots, x_{j+m} \in [y_{k-1}, y_k]$. If $x_j \ne y_{k-1}$, set

$$M_k^1 = \sup\{f(x) : x \in [y_{k-1}, x_j]\}.$$

If $x_{j+m} \ne y_k$, set

$$M_k^2 = \sup\{f(x) : x \in [x_{j+m}, y_k]\}.$$

Let $t_k \in [y_{k-1}, y_k]$ be arbitrary. Since $|f(t) - f(s)| \le 2M$ for all $t, s \in [y_{k-1}, y_k]$,

$$f(t_k) \le 2M + M_{j+p}, \quad p = 1, \ldots, m,$$

and

$$f(t_k) \le 2M + M_k^i, \quad i = 1, 2.$$

Therefore,

$$\begin{aligned} f(t_k)\Delta y_k &= f(t_k)(x_j - y_{k-1}) + \sum_{p=1}^{m} f(t_k)\Delta x_{j+p} + f(t_k)(y_k - x_{j+m}) \\ &\le 2M\Delta y_k + M_k^1(x_j - y_{k-1}) + \sum_{p=1}^{m} M_{j+p}\Delta x_{j+p} + M_k^2(y_k - x_{j+m}) \\ &< 2M\delta + \mathcal{U}(\mathcal{P}_k, f), \end{aligned}$$

where $\mathcal{P}_k = \{y_{k-1}, x_j, \ldots, x_{j+m}, y_k\}$ is a partition of $[y_{k-1}, y_k]$. If $[y_{k-1}, y_k] \cap \mathcal{Q} = \emptyset$, then set $\mathcal{P}_k = \{y_{k-1}, y_k\}$. In that case $f(t_k)\Delta y_k \le \mathcal{U}(\mathcal{P}_k, f)$, with no extra term. Let $\mathcal{P}' = \bigcup_k \mathcal{P}_k$. Then $\mathcal{P}'$ is a refinement of $\mathcal{Q}$. Since at most $N - 1$ intervals $[y_{k-1}, y_k]$ contain a point in $\mathcal{Q}$ other than x_0 and x_N,

$$\begin{aligned} \mathcal{I}(\mathcal{P}, f) &= \sum_{k=1}^{n} f(t_k)\Delta y_k \\ &< 2M(N-1)\delta + \sum_{k=1}^{n} \mathcal{U}(\mathcal{P}_k, f) \\ &< \frac{\varepsilon}{2} + \mathcal{U}(\mathcal{P}', f) \end{aligned}$$

$$\leq \frac{\varepsilon}{2} + \mathcal{U}(\mathcal{Q}, f)$$
$$< \varepsilon + \int_a^b f.$$

A similar argument shows

$$\mathcal{I}(\mathcal{P}, f) > \int_a^b f - \varepsilon.$$

Thus

$$\left| \mathcal{I}(\mathcal{P}, f) - \int_a^b f \right| < \varepsilon$$

for any partition $\mathcal{P}$ of $[a, b]$ with $\|\mathcal{P}\| < \delta$ and any choice $t_k \in [y_{k-1}, y_k]$, $k = 1, \ldots, n$. Thus $\lim_{\|\mathcal{P}\| \to 0} \mathcal{I}(\mathcal{P}, f) = \int_a^b f$. □

Example 9.3.4. Consider $\int_a^b x\, dx$. Let $\mathcal{P} = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. Choose $t_i = \frac{x_{i-1}+x_i}{2}$. Then

$$\mathcal{I}(\mathcal{P}, f) = \frac{1}{2} \sum_{i=1}^{n} (x_{i-1} + x_i)(x_i - x_{i-1}) = \frac{1}{2}(b^2 - a^2).$$

Since $f(x) = x$ is continuous, $f \in \mathcal{R}[a, b]$, and by Theorem 9.3.3 the Riemann sums converge to $\int_a^b f$ as $\|\mathcal{P}\| \to 0$. Every one of these midpoint sums equals $\frac{1}{2}(b^2 - a^2)$. So

$$\int_a^b x\, dx = \frac{1}{2}(b^2 - a^2).$$

Definition 9.3.5. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. A function $F : I \to \mathbb{R}$ is an *antiderivative* of f on I if $F'(x) = f(x)$ for all $x \in I$.

Theorem 9.3.6 (Fundamental Theorem of Calculus)**.** *If $f \in \mathcal{R}[a, b]$ and if F is an antiderivative of f on $[a, b]$, then*

$$\int_a^b f = F(b) - F(a).$$

PROOF. Let $\mathcal{P} = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. If F is an antiderivative of f, then by the Mean Value Theorem, for each $i = 1, \ldots, n$, there exists a $t_i \in (x_{i-1}, x_i)$ such that

$$F(x_i) - F(x_{i-1}) = f(t_i)\Delta x_i.$$

Therefore,

$$\sum_{i=1}^{n} f(t_i)\Delta x_i = \sum_{i=1}^{n} (F(x_i) - F(x_{i-1})) = F(b) - F(a).$$

Since

$$\mathcal{L}(\mathcal{P}, f) \leq \sum_{i=1}^{n} f(t_i)\Delta x_i \leq \mathcal{U}(\mathcal{P}, f),$$

$\mathcal{L}(\mathcal{P}, f) \le F(b) - F(a) \le \mathcal{U}(\mathcal{P}, f)$ for every partition $\mathcal{P}$. So

$$\underline{\int_a^b} f \le F(b) - F(a) \le \overline{\int_a^b} f.$$

Since $f \in \mathcal{R}[a, b]$,

$$\int_a^b f = F(b) - F(a).$$

□

Theorem 9.3.7 (Fundamental Theorem of Calculus)**.** *Let $f \in \mathcal{R}[a, b]$. Define $F : [a, b] \to \mathbb{R}$ by $F(x) = \int_a^x f(t)\,dt$. Then F is continuous on $[a, b]$. Furthermore, if f is continuous at a point $c \in [a, b]$, then F is differentiable at c and $F'(c) = f(c)$.*

Proof. The proof that F is continuous is an exercise. See Exercise 9.3.2. Now suppose f is continuous at $c \in [a, b)$. Let $h > 0$ with $c + h < b$. Then by Theorem 9.2.13,

$$\begin{aligned} F(c+h) - F(c) &= \int_a^{c+h} f(t)\,dt - \int_a^c f(t)\,dt \\ &= \int_c^{c+h} f(t)\,dt. \end{aligned}$$

Therefore,

$$\begin{aligned} \frac{F(c+h) - F(c)}{h} - f(c) &= \frac{1}{h}\int_c^{c+h} f(t)\,dt - f(c) \\ &= \frac{1}{h}\int_c^{c+h} [f(t) - f(c)]\,dt. \end{aligned}$$

Let $\varepsilon > 0$. Since f is continuous at c, there exists a $\delta > 0$ such that for all $t \in [a, b]$ with $|t - c| < \delta$, $|f(t) - f(c)| < \varepsilon$. Therefore, for $0 < h < \delta$,

$$\begin{aligned} \left|\frac{F(c+h) - F(c)}{h} - f(c)\right| &\le \frac{1}{h}\int_c^{c+h} |f(t) - f(c)|\,dt \\ &< \frac{1}{h}\int_c^{c+h} \varepsilon\,dt = \varepsilon. \end{aligned}$$

(The first step uses Theorem 9.2.12, and the second uses Exercise 9.2.3.) So $\lim_{h\to 0^+} \frac{F(c+h)-F(c)}{h} = f(c)$. Similarly, for $c \in (a, b]$, $\lim_{h\to 0^-} \frac{F(c+h)-F(c)}{h} = f(c)$. So $F'(c) = f(c)$. □

Exercises

9.3.1. Define $f : [0, 2] \to \mathbb{R}$ by $f(t) = \begin{cases} t, & 0 \le t < 1, \\ b - t^2, & 1 \le t \le 2. \end{cases}$ Define $F(x) = \int_0^x f(t)\,dt$. For what values of b is F differentiable for all $x \in [0, 2]$?
[Hint, p. 347]

9.3.2. Let $f \in \mathcal{R}[a, b]$. Define $F : [a, b] \to \mathbb{R}$ by $F(x) = \int_a^x f(t)\,dt$. Prove F is continuous on $[a, b]$.
[Hint, p. 347]

9.3.3. For $n \in \mathbb{N}_0$ and $k = 0, 1, \ldots, n$, let $\binom{n}{k} = \frac{n!}{k!(n-k)!}$. Prove that $\binom{n}{k-1} + \binom{n}{k} = \binom{n+1}{k}$ for $1 \leq k \leq n$. Then use induction to prove the *Binomial Theorem*, that $(a+b)^n = \sum_{k=0}^{n} \binom{n}{k} a^{n-k} b^k$ for all $a, b \in \mathbb{R}$ and $n \in \mathbb{N}$. [Hint, p. 347]

9.4. Logarithms and Exponential Functions*

Until now, $\ln x$ and e^x have been used in examples with their properties assumed. See Remark 3.1.1. This section defines them and proves those properties. The natural logarithm comes first, as an integral. The exponential function is its inverse. Then a^x is defined for every $a > 0$ and every real x, and so is x^α for every real α. Before this section, powers with irrational exponents had no meaning. Section 12.5 builds the same functions a second way, starting from the power series of e^x. A reader willing to assume the familiar properties of $\ln x$ and e^x can skip both sections.

Example 9.4.1 (The Number e)**.** Define a sequence $t_n = \left(1 + \frac{1}{n}\right)^n$. By the Binomial Theorem (Exercise 9.3.3),

$$t_n = 1 + n \cdot \frac{1}{n} + \frac{n(n-1)}{1 \cdot 2} \cdot \frac{1}{n^2} + \cdots + \frac{n(n-1)\cdots 1}{1 \cdot 2 \cdots n} \cdot \frac{1}{n^n}.$$

Notice there are $n+1$ terms. For $k = 1, \ldots, n$, the $(k+1)$st term is

$$\frac{n(n-1)\cdots(n-k+1)}{1 \cdot 2 \cdots k} \cdot \frac{1}{n^k} = \frac{1}{1 \cdot 2 \cdots k}\left(1 - \frac{1}{n}\right)\left(1 - \frac{2}{n}\right)\cdots\left(1 - \frac{k-1}{n}\right).$$

Notice if t_{n+1} is expanded in the same way, there are $n+2$ terms, and for $k = 1, 2, \ldots, n$, the $(k+1)$st term is

$$\frac{1}{1 \cdot 2 \cdots k}\left(1 - \frac{1}{n+1}\right)\left(1 - \frac{2}{n+1}\right)\cdots\left(1 - \frac{k-1}{n+1}\right),$$

which is greater than or equal to the $(k+1)$st term in t_n. Thus $t_n < t_{n+1}$ for all $n \in \mathbb{N}$. So $\{t_n\}$ is an increasing sequence. Also,

$$\begin{aligned} t_n &\leq 1 + 1 + \frac{1}{1 \cdot 2} + \frac{1}{1 \cdot 2 \cdot 3} + \cdots + \frac{1}{1 \cdot 2 \cdots n} \\ &\leq 1 + 1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-1}} \\ &= 1 + 2\left(1 - \frac{1}{2^n}\right) \\ &< 3. \end{aligned}$$

The second line uses $k! \geq 2^{k-1}$ (Exercise 1.2.10), and the third uses Exercise 1.2.7. So $\{t_n\}$ is bounded. Therefore, $\{t_n\}$ converges by the Monotone Convergence Theorem. Define

$$e = \lim_{n \to \infty} \left(1 + \frac{1}{n}\right)^n.$$

Definition 9.4.2. The *natural logarithm* is the function $\ln : (0, \infty) \to \mathbb{R}$ defined by

$$\ln x = \int_1^x \frac{1}{t}\, dt.$$

Remark 9.4.3. Here ln is the name of a function, just as f is, and $\ln x$ is its value at x, just as $f(x)$ is. By tradition the parentheses are usually left off, so $\ln x$ means $\ln(x)$. Writing ln alone, as in $\ln : (0,\infty) \to \mathbb{R}$ or "ln is increasing," refers to the whole function. The same goes for exp below, and for sin, cos and arctan in Section 9.5. Parentheses are used when the input is more than a single symbol, as in $\ln(1+x)$. Products are an exception by custom, so $\ln ab$ means $\ln(ab)$.

Theorem 9.4.4. *The natural logarithm* $\ln : (0,\infty) \to \mathbb{R}$ *has the following properties.*

(1) ln *is differentiable,* $\frac{d}{dx}\ln x = \frac{1}{x}$*, and* ln *is strictly increasing. Also* $\ln 1 = 0$ *and* $\ln e = 1$*.*
(2) $\ln ab = \ln a + \ln b$ *and* $\ln \frac{a}{b} = \ln a - \ln b$ *for all* $a, b > 0$*.*
(3) $\ln(b^r) = r \ln b$ *for all* $b > 0$ *and* $r \in \mathbb{Q}$*.*
(4) ln *is concave on* $(0,\infty)$*.*
(5) $\lim_{x\to\infty} \ln x = \infty$ *and* $\lim_{x\to 0^+} \ln x = -\infty$*.*
(6) ln *maps* $(0,\infty)$ *onto* $\mathbb{R}$*.*

Proof. (1) Since $f(t) = \frac{1}{t}$ is continuous on $(0,\infty)$, $\frac{d}{dx}\ln x = \frac{1}{x}$ for all $x > 0$ by Theorem 9.3.7, applied on any closed interval in $(0,\infty)$ that contains 1 and x. So $\ln x$ is a strictly increasing function by Theorem 7.2.7. Also $\ln 1 = \int_1^1 \frac{1}{t}\,dt = 0$. That $\ln e = 1$ is shown at the end of the proof.

(2) Notice

$$\frac{d}{dx}\ln ax = \frac{1}{ax}a = \frac{1}{x} = \frac{d}{dx}\ln x.$$

So by Theorem 7.2.7 (5), $\ln ax = \ln x + C$. Now $\ln 1 = 0$, so $C = \ln a$ and therefore,

$$\ln ab = \ln a + \ln b$$

for $a, b > 0$.

Next, notice

$$\frac{d}{dx}\ln\frac{1}{x} = \frac{1}{1/x}\cdot\frac{-1}{x^2} = \frac{-1}{x} = -\frac{d}{dx}\ln x.$$

So $\ln \frac{1}{x} = -\ln x + C$, and since $\ln 1 = 0$, $C = 0$. Thus

$$\ln\frac{1}{x} = -\ln x.$$

For the second half of (2), $\ln \frac{a}{b} = \ln a + \ln \frac{1}{b} = \ln a - \ln b$.

(3) Now, note if $n \in \mathbb{N}$, $\ln(b^n) = n \ln b$ by induction. Also,

$$\ln(b^{-n}) = \ln\left(\left(\frac{1}{b}\right)^n\right) = n\ln\frac{1}{b} = -n\ln b.$$

So $\ln(b^n) = n \ln b$ for all $n \in \mathbb{Z}$. Also, since

$$n\ln(\sqrt[n]{b}) = \ln b,$$

$$\ln(\sqrt[n]{b}) = \frac{1}{n}\ln b.$$

Thus $\ln(b^r) = r \ln b$ for all $r \in \mathbb{Q}$.

It remains to prove $\ln e = 1$, the last claim in (1). Last,

$$\begin{aligned} 1 = \frac{d}{dx} \ln x \Big|_{x=1} &= \lim_{n\to\infty} \frac{\ln\left(1+\frac{1}{n}\right) - \ln 1}{1/n} \\ &= \lim_{n\to\infty} n \ln\left(1+\frac{1}{n}\right) \\ &= \lim_{n\to\infty} \ln\left(\left(1+\frac{1}{n}\right)^n\right) \\ &= \ln e. \end{aligned}$$

The first line uses Theorem 6.1.18 with $x_n = 1 + \frac{1}{n}$. The last line uses the continuity of ln and Theorem 6.3.9.

(4) The derivative $\frac{1}{x}$ is decreasing on $(0,\infty)$. So ln is concave by Theorem 7.4.3.

(5) Since ln is strictly increasing and $\ln 1 = 0$, $\ln 2 > 0$. Let $M \in \mathbb{R}$. By Theorem 1.8.8, there is an $n \in \mathbb{N}$ with $n \ln 2 > M$. If $x > 2^n$, then $\ln x > \ln(2^n) = n \ln 2 > M$. So $\ln x \to \infty$ as $x \to \infty$. Next, let $M \in \mathbb{R}$ and choose $n \in \mathbb{N}$ with $n \ln 2 > -M$. If $0 < x < 2^{-n}$, then $\frac{1}{x} > 2^n$, so $\ln x = -\ln \frac{1}{x} < -n \ln 2 < M$. So $\ln x \to -\infty$ as $x \to 0^+$.

(6) Let $y \in \mathbb{R}$. By (5), there are $a, b > 0$ with $\ln a < y < \ln b$. Then $a < b$, and ln is continuous on $[a,b]$ by Theorem 7.1.3. By Theorem 6.3.10, $\ln c = y$ for some $c \in (a,b)$. □

Definition 9.4.5. The *exponential function* $\exp : \mathbb{R} \to (0,\infty)$ is the inverse of ln. That is, $\exp(y) = x$ exactly when $\ln x = y$.

By Theorem 9.4.4, ln is one-to-one and onto $\mathbb{R}$. So exp is defined on all of $\mathbb{R}$.

Theorem 9.4.6. *The exponential function has the following properties.*

(1) exp *is strictly increasing and* $\exp(x) > 0$ *for all* $x \in \mathbb{R}$. *Also* $\exp(0) = 1$ *and* $\exp(1) = e$.
(2) exp *is differentiable and* $\exp'(x) = \exp(x)$ *for all* $x \in \mathbb{R}$.
(3) $\exp(x+y) = \exp(x)\exp(y)$ *and* $\exp(-x) = \frac{1}{\exp(x)}$ *for all* $x, y \in \mathbb{R}$.
(4) $\exp(r) = e^r$ *for all* $r \in \mathbb{Q}$.
(5) $\lim_{x\to\infty} \exp(x) = \infty$ *and* $\lim_{x\to-\infty} \exp(x) = 0$.
(6) exp *is convex on* $\mathbb{R}$.

Proof. (1) By Theorem 6.5.7, exp is strictly increasing and continuous. Its values lie in $(0,\infty)$, the domain of ln. Since $\ln 1 = 0$ and $\ln e = 1$, $\exp(0) = 1$ and $\exp(1) = e$.

(2) Apply Theorem 7.2.9 to $f = \ln$ on $I = (0,\infty)$. Here $f'(c) = \frac{1}{c} \neq 0$ and $f[I] = \mathbb{R}$. So exp is differentiable on $\mathbb{R}$, and for $c > 0$,

$$\exp'(\ln c) = \frac{1}{1/c} = c = \exp(\ln c).$$

Every $x \in \mathbb{R}$ is $\ln c$ for some $c > 0$. So $\exp'(x) = \exp(x)$.

(3) Let $a = \exp(x)$ and $b = \exp(y)$. Then $\ln ab = \ln a + \ln b = x + y$, so $\exp(x+y) = ab$. Taking $y = -x$ gives $\exp(x)\exp(-x) = \exp(0) = 1$.

(4) Let $r \in \mathbb{Q}$. Then $\ln(e^r) = r \ln e = r$ by Theorem 9.4.4 (3). So $\exp(r) = e^r$.

(5) Let $M > 0$. If $x > \ln M$, then $\exp(x) > \exp(\ln M) = M$. So $\exp(x) \to \infty$ as $x \to \infty$. Let $\varepsilon > 0$. If $x < \ln \varepsilon$, then $0 < \exp(x) < \varepsilon$. So $\exp(x) \to 0$ as $x \to -\infty$.

(6) By (1) and (2), $\exp'$ is increasing. So exp is convex by Theorem 7.4.3. □

By Theorem 9.4.6 (4), $\exp(x)$ agrees with e^x whenever x is rational. So from now on, e^x means $\exp(x)$ for every $x \in \mathbb{R}$. With this notation, part (3) reads $e^{x+y} = e^x e^y$, the usual law of exponents. The same idea defines every power.

Definition 9.4.7. Let $a > 0$ and $x \in \mathbb{R}$. Define

$$a^x = e^{x \ln a}.$$

This does not conflict with the powers already defined. If $r \in \mathbb{Q}$, then $\ln(a^r) = r \ln a$ by Theorem 9.4.4 (3), where a^r has its old meaning. So the old a^r is $e^{r \ln a}$. Also e^x has the same meaning as before, since $\ln e = 1$.

Theorem 9.4.8 (Laws of Exponents)**.** *Let $a, b > 0$ and $x, y \in \mathbb{R}$.*

(1) $\ln(a^x) = x \ln a$.
(2) $a^{x+y} = a^x a^y$ *and* $a^{-x} = \frac{1}{a^x}$.
(3) $(a^x)^y = a^{xy}$.
(4) $(ab)^x = a^x b^x$.
(5) $\frac{d}{dx} a^x = a^x \ln a$.
(6) If $\alpha \in \mathbb{R}$, then $\frac{d}{dx} x^\alpha = \alpha x^{\alpha-1}$ *for all $x > 0$.*

Proof. (1) $\ln(a^x) = \ln(e^{x \ln a}) = x \ln a$, since ln and exp are inverses.

(2) By Theorem 9.4.6 (3),

$$a^{x+y} = e^{(x+y)\ln a} = e^{x \ln a + y \ln a} = e^{x \ln a} e^{y \ln a} = a^x a^y.$$

Taking $y = -x$ gives $a^x a^{-x} = a^0 = 1$.

(3) By (1), $(a^x)^y = e^{y \ln(a^x)} = e^{xy \ln a} = a^{xy}$.

(4) $(ab)^x = e^{x \ln ab} = e^{x \ln a + x \ln b} = a^x b^x$.

(5) By Theorem 7.1.6 and Theorem 9.4.6 (2), $\frac{d}{dx} e^{x \ln a} = e^{x \ln a} \ln a = a^x \ln a$.

(6) For $x > 0$, $x^\alpha = e^{\alpha \ln x}$. By the Chain Rule and (2),

$$\frac{d}{dx} x^\alpha = e^{\alpha \ln x} \cdot \frac{\alpha}{x} = \alpha x^\alpha x^{-1} = \alpha x^{\alpha-1}.$$ □

Part (6) extends the power rules of Exercises 7.1.3 and 7.2.6 to every real exponent.

Definition 9.4.9. Let $a > 0$ with $a \neq 1$. The *logarithm to the base a*, written $\log_a$, is the inverse of the function $x \mapsto a^x$.

This makes sense. By Theorem 9.4.8 (5), the derivative of a^x is $a^x \ln a$, which is never 0 since $\ln a \neq 0$. So a^x is strictly monotone. It maps $\mathbb{R}$ onto $(0, \infty)$, since $x \mapsto x \ln a$ maps $\mathbb{R}$ onto $\mathbb{R}$ and exp maps $\mathbb{R}$ onto $(0, \infty)$. So $\log_a : (0, \infty) \to \mathbb{R}$. Notice $\log_e = \ln$.

Theorem 9.4.10. *Let $a > 0$ with $a \neq 1$. Then* $\log_a x = \frac{\ln x}{\ln a}$ *for all $x > 0$.*

Proof. Let $y = \log_a x$. Then $a^y = x$. By Theorem 9.4.8 (1), $y \ln a = \ln x$. So $y = \frac{\ln x}{\ln a}$. □

Two inequalities are used often. The first says the graph of e^x lies above its tangent line at 0. The second says the graph of $\ln x$ lies below its tangent line at 1.

Theorem 9.4.11. *(1) $e^x \geq 1 + x$ for all $x \in \mathbb{R}$, with equality only when $x = 0$.*
(2) $\ln x \leq x - 1$ for all $x > 0$, with equality only when $x = 1$.

PROOF. (1) Let $g(x) = e^x - 1 - x$. Then $g'(x) = e^x - 1$. This is negative for $x < 0$ and positive for $x > 0$, since exp is strictly increasing and $e^0 = 1$. By Theorem 7.2.7, g is strictly decreasing on $(-\infty, 0]$ and strictly increasing on $[0, \infty)$. So $g(x) > g(0) = 0$ for every $x \neq 0$.

(2) Let $x > 0$ and apply (1) to $\ln x$. Then $x = e^{\ln x} \geq 1 + \ln x$, with equality only when $\ln x = 0$, that is, when $x = 1$. □

Compare Exercise 7.2.1, which proves (1) with the Mean Value Theorem.

The number e was defined as $\lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$. The same kind of limit gives every value of e^x.

Theorem 9.4.12. *For every $x \in \mathbb{R}$,*

$$\lim_{n\to\infty} \left(1 + \frac{x}{n}\right)^n = e^x.$$

PROOF. If $x = 0$, both sides are 1. Let $x \neq 0$. For $n > |x|$, $1 + \frac{x}{n} > 0$. So by Theorem 9.4.4 (3),

$$\left(1 + \frac{x}{n}\right)^n = e^{n \ln(1+x/n)}.$$

Let $h_n = \frac{x}{n}$. Then $h_n \neq 0$ and $h_n \to 0$. Since the derivative of ln at 1 is 1, Theorem 6.1.18 gives

$$n \ln\left(1 + \frac{x}{n}\right) = x \cdot \frac{\ln(1 + h_n) - \ln 1}{h_n} \to x \cdot 1 = x.$$

Since exp is continuous, $e^{n \ln(1+x/n)} \to e^x$ by Theorem 6.3.9. □

Last, e^x grows faster than every power of x, and $\ln x$ grows slower than every positive power of x.

Theorem 9.4.13. *Let $\alpha > 0$. Then*

(1) $\lim_{x\to\infty} \frac{x^\alpha}{e^x} = 0$,
(2) $\lim_{x\to\infty} \frac{\ln x}{x^\alpha} = 0$, *and*
(3) $\lim_{x\to 0^+} x^\alpha \ln x = 0$.

PROOF. (1) Choose $m \in \mathbb{N}$ with $m \geq \alpha$, and let $k = m + 1$. Let $x > 1$. Then $x^\alpha \leq x^m$ by Theorem 9.4.8 (2), since $x^{m-\alpha} \geq 1$. By Theorem 9.4.11 (1), $e^{x/k} > \frac{x}{k}$. By Theorem 9.4.8 (3), $e^x = \left(e^{x/k}\right)^k > \left(\frac{x}{k}\right)^k$. So

$$0 < \frac{x^\alpha}{e^x} < x^m \cdot \frac{k^k}{x^k} = \frac{k^k}{x}.$$

Since $\frac{k^k}{x} \to 0$ as $x \to \infty$, the Squeeze Theorem gives (1).

(2) Let $\{x_n\}$ be a sequence with $x_n > 1$ and $x_n \to \infty$. Let $t_n = \alpha \ln x_n$. Then $t_n > 0$, and $t_n \to \infty$ by Theorem 9.4.4 (5) and Theorem 6.2.8. Also $x_n^\alpha = e^{t_n}$. So

$$\frac{\ln x_n}{x_n^\alpha} = \frac{1}{\alpha} \cdot \frac{t_n}{e^{t_n}} \to 0$$

by (1) with exponent 1 and Theorem 6.2.8. Since this holds for every such sequence, (2) follows from Theorem 6.2.8.

(3) Let $\{x_n\}$ be a sequence with $0 < x_n < 1$ and $x_n \to 0$. Let $y_n = \frac{1}{x_n}$. Then $y_n \to \infty$, and

$$x_n^\alpha \ln x_n = -\frac{\ln y_n}{y_n^\alpha} \to 0$$

by (2). So (3) follows from Theorem 6.2.8. □

Exercises

9.4.1. Let $a, b > 0$ with $a \neq 1$ and $b \neq 1$. Prove that $\log_a(xy) = \log_a x + \log_a y$ and $\log_a(x^r) = r \log_a x$ for all $x, y > 0$ and $r \in \mathbb{R}$. Then prove that $\log_b x = \frac{\log_a x}{\log_a b}$ for all $x > 0$. [Hint, p. 347]

9.4.2. Use the definition $\ln x = \int_1^x \frac{1}{t}\, dt$ to show that $\ln 2 < 1$. Then use a lower sum with 8 subintervals of equal length to show that $\ln 3 > 1$. Conclude that $2 < e < 3$. [Hint, p. 347]

9.4.3. Define $f : (0, \infty) \to \mathbb{R}$ by $f(x) = x^x$.
(a) Find $f'(x)$.
(b) Find the smallest value of f on $(0, \infty)$.
(c) Prove that $\lim_{x \to 0^+} x^x = 1$.
[Solution, p. 347]

9.4.4. Let $f : \mathbb{R} \to \mathbb{R}$ be differentiable with $f' = f$ and $f(0) = 1$. Prove that $f(x) = e^x$ for all $x \in \mathbb{R}$. [Solution, p. 348]

9.4.5. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous, not identically 0, with $f(x + y) = f(x)f(y)$ for all $x, y \in \mathbb{R}$. Prove that $a = f(1) > 0$ and that $f(x) = a^x$ for all $x \in \mathbb{R}$. [Hint, p. 348]

9.4.6. Define $\cosh x = \frac{e^x + e^{-x}}{2}$ and $\sinh x = \frac{e^x - e^{-x}}{2}$ for $x \in \mathbb{R}$.
(a) Prove that $\frac{d}{dx} \cosh x = \sinh x$, $\frac{d}{dx} \sinh x = \cosh x$, and $\cosh^2 x - \sinh^2 x = 1$.
(b) Prove that sinh is one-to-one and maps $\mathbb{R}$ onto $\mathbb{R}$.
(c) Prove that the inverse of sinh is $x \mapsto \ln\bigl(x + \sqrt{x^2 + 1}\bigr)$.
[Hint, p. 348]

9.4.7. Prove that $\lim_{x \to 0}(1 + x)^{1/x} = e$ and $\lim_{x \to \infty} \left(1 + \frac{1}{x}\right)^x = e$. [Solution, p. 348]

9.4.8. Let $a, b > 0$ and $0 \leq t \leq 1$. Use the concavity of ln to prove that $a^t b^{1-t} \leq ta + (1 - t)b$. Conclude that $\sqrt{ab} \leq \frac{a+b}{2}$. [Hint, p. 348]

9.5. Trigonometric Functions from Integrals*

This section builds sin and cos from an integral, the way Section 9.4 built ln from $\int_1^x \frac{1}{t}\,dt$. Section 12.4 builds the same functions a second way, from power series. Either section can be read without the other. A reader willing to assume the familiar properties of sin and cos can skip both.

Calculus says $\arctan x = \int_0^x \frac{dt}{1+t^2}$. Here that formula becomes the definition. The integrand is continuous on all of $\mathbb{R}$, so no improper integrals are needed. Then tan is the inverse of arctan, and sin and cos are built from tan. In this section, when $x < 0$, $\int_0^x f$ means $-\int_x^0 f$.

Definition 9.5.1. For $x \in \mathbb{R}$, define

$$\arctan x = \int_0^x \frac{dt}{1+t^2}.$$

Theorem 9.5.2. *(1)* arctan *is differentiable on* $\mathbb{R}$, $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$, *and* arctan *is strictly increasing.*
(2) $\arctan(-x) = -\arctan x$ *for all* $x \in \mathbb{R}$.
(3) $|\arctan x| < 2$ *for all* $x \in \mathbb{R}$.

Proof. (1) Let $x \in \mathbb{R}$, and choose $a < b$ with $a < \min\{0, x\}$ and $b > \max\{0, x\}$. On $[a, b]$, $\arctan x = \int_a^x \frac{dt}{1+t^2} - \int_a^0 \frac{dt}{1+t^2}$ by Theorem 9.2.13. The integrand is continuous, so Theorem 9.3.7 gives $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$. This is positive, so arctan is strictly increasing by Theorem 7.2.7.

(2) Let $g(x) = \arctan x + \arctan(-x)$. By the Chain Rule, $g'(x) = \frac{1}{1+x^2} - \frac{1}{1+x^2} = 0$. So g is constant, and $g(x) = g(0) = 0$.

(3) By (2), it is enough to take $x \geq 0$. If $0 \leq x \leq 1$, then $\frac{1}{1+t^2} \leq 1$, so $\arctan x \leq x \leq 1$ by Exercise 9.2.3. If $x > 1$, then $\frac{1}{1+t^2} < \frac{1}{t^2}$ for $t \geq 1$, and $-\frac{1}{t}$ is an antiderivative of $\frac{1}{t^2}$. So by Theorem 9.3.6,

$$\arctan x = \arctan 1 + \int_1^x \frac{dt}{1+t^2} \leq 1 + \int_1^x \frac{dt}{t^2} = 1 + 1 - \frac{1}{x} < 2. \qquad \square$$

Definition 9.5.3. Define π by

$$\frac{\pi}{2} = \sup\{\arctan x : x \in \mathbb{R}\}.$$

The supremum exists by Theorem 9.5.2 (3). Since $\arctan 1 > \arctan 0 = 0$, $0 < \pi \leq 4$.

Theorem 9.5.4. $\lim_{x\to\infty} \arctan x = \frac{\pi}{2}$ *and* $\lim_{x\to-\infty} \arctan x = -\frac{\pi}{2}$. *Also* arctan *maps* $\mathbb{R}$ *onto* $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

Proof. Let $\varepsilon > 0$. Since $\frac{\pi}{2} - \varepsilon$ is not an upper bound, $\arctan x_0 > \frac{\pi}{2} - \varepsilon$ for some x_0. If $x > x_0$, then $\frac{\pi}{2} - \varepsilon < \arctan x_0 < \arctan x \leq \frac{\pi}{2}$. So $\arctan x \to \frac{\pi}{2}$ as $x \to \infty$. By Theorem 9.5.2 (2), $\arctan x \to -\frac{\pi}{2}$ as $x \to -\infty$. Also $\arctan x < \arctan(x+1) \leq \frac{\pi}{2}$, so every value is less than $\frac{\pi}{2}$, and similarly greater than $-\frac{\pi}{2}$. Let $y \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. By the two limits, there are $a < b$ with $\arctan a < y < \arctan b$. By Theorem 6.3.10, $\arctan c = y$ for some $c \in (a, b)$. $\square$

Definition 9.5.5. The *tangent function* $\tan : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to \mathbb{R}$ is the inverse of arctan.

Theorem 9.5.6. *The function* tan *is differentiable and strictly increasing on* $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, *with* $\frac{d}{dx} \tan x = 1 + \tan^2 x$. *Also* $\tan 0 = 0$, $\tan(-x) = -\tan x$, $\tan x \to \infty$ *as* $x \to \frac{\pi}{2}^-$, *and* $\tan x \to -\infty$ *as* $x \to -\frac{\pi}{2}^+$.

Proof. By Theorem 7.2.9 and Theorem 9.5.4, tan is differentiable on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. If $x = \arctan y$, then $\tan'(x) = \frac{1}{\arctan'(y)} = 1 + y^2 = 1 + \tan^2 x$. Since $\tan' > 0$, tan is strictly increasing. Since $\arctan 0 = 0$, $\tan 0 = 0$, and $\tan(-x) = -\tan x$ follows from Theorem 9.5.2 (2). Let $M \in \mathbb{R}$. If $\arctan M < x < \frac{\pi}{2}$, then $\tan x > M$. So $\tan x \to \infty$ as $x \to \frac{\pi}{2}^-$. The other limit follows since tan is odd. □

In calculus, $\cos x = \frac{1}{\sqrt{1+\tan^2 x}}$ and $\sin x = \tan x \cos x$ when $|x| < \frac{\pi}{2}$. These formulas define sin and cos on that interval. Then $\sin(x + \pi) = -\sin x$ and $\cos(x + \pi) = -\cos x$ extend them to all of $\mathbb{R}$.

Definition 9.5.7. For $|x| < \frac{\pi}{2}$, define

$$\cos x = \frac{1}{\sqrt{1 + \tan^2 x}} \quad \text{and} \quad \sin x = \tan x \cos x.$$

Define $\cos\left(\pm\frac{\pi}{2}\right) = 0$ and $\sin\left(\pm\frac{\pi}{2}\right) = \pm 1$. For any other $x \in \mathbb{R}$, write $x = y + k\pi$, where $k \in \mathbb{Z}$ and $-\frac{\pi}{2} < y \leq \frac{\pi}{2}$, and define

$$\sin x = (-1)^k \sin y \quad \text{and} \quad \cos x = (-1)^k \cos y.$$

Remark 9.5.8. Here sin and cos are names of functions, just as f is, and $\sin x$ and $\cos x$ are their values at x, just as $f(x)$ is. By tradition the parentheses are usually left off, so $\sin x$ means $\sin(x)$. Writing sin alone, as in "$\sin' = \cos$," refers to the whole function. Parentheses are used when the input is more than a single symbol, as in $\sin(x + y)$. Also by tradition, $\sin^2 x$ means $(\sin x)^2$, not $\sin(\sin x)$. The same conventions hold for tan and arctan.

There is exactly one such k, because the intervals $\left(-\frac{\pi}{2} + k\pi, \frac{\pi}{2} + k\pi\right]$, $k \in \mathbb{Z}$, cover $\mathbb{R}$ and do not overlap. Notice $y + (k+1)\pi$ uses $k + 1$. So

$$\sin(x + \pi) = -\sin x \quad \text{and} \quad \cos(x + \pi) = -\cos x$$

for every $x \in \mathbb{R}$. The definition at $-\frac{\pi}{2}$ agrees with this rule, since $-\frac{\pi}{2} = \frac{\pi}{2} - \pi$.

The derivatives at the points $\frac{\pi}{2} + k\pi$ need a lemma. It says a derivative has no removable discontinuities.

Lemma 9.5.9. *Let* $f : (a, b) \to \mathbb{R}$ *be continuous, let* $c \in (a, b)$, *and suppose* f *is differentiable at every point of* (a, b) *except perhaps* c. *If* $\lim_{x \to c} f'(x) = L$, *then* f *is differentiable at* c *and* $f'(c) = L$.

Proof. Let $\varepsilon > 0$. Choose $\delta > 0$ so that $|f'(t) - L| < \varepsilon$ whenever $0 < |t - c| < \delta$ and $t \in (a, b)$. Let $x \in (a, b)$ with $0 < |x - c| < \delta$. By Theorem 7.2.4 on the interval between c and x, $\frac{f(x)-f(c)}{x-c} = f'(\xi)$ for some ξ strictly between c and x. Then $0 < |\xi - c| < \delta$, so $\left|\frac{f(x)-f(c)}{x-c} - L\right| < \varepsilon$. □

Theorem 9.5.10. *The functions* sin *and* cos *are differentiable on* $\mathbb{R}$*, and*

$$\frac{d}{dx}\sin x = \cos x \quad \textit{and} \quad \frac{d}{dx}\cos x = -\sin x.$$

Also $\sin 0 = 0$*,* $\cos 0 = 1$*,* $\sin \frac{\pi}{2} = 1$*,* $\cos \frac{\pi}{2} = 0$ *and* $\sin^2 x + \cos^2 x = 1$ *for all* $x \in \mathbb{R}$*.*

PROOF. The proof starts with the values. Since $\tan 0 = 0$, the definition gives $\cos 0 = 1$ and $\sin 0 = 0$. The values at $\frac{\pi}{2}$ are part of the definition. For $|x| < \frac{\pi}{2}$,

$$\sin^2 x + \cos^2 x = \frac{\tan^2 x + 1}{1 + \tan^2 x} = 1.$$

This also holds at $\pm\frac{\pi}{2}$, and multiplying by $(-1)^{2k} = 1$ does not change it. So it holds on $\mathbb{R}$.

Next come the derivatives on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. There cos and sin are given directly in terms of tan. By Theorem 7.2.9 applied to x^2 on $(0, \infty)$, $\sqrt{u}$ is differentiable for $u > 0$ with derivative $\frac{1}{2\sqrt{u}}$. Let $u = 1 + \tan^2 x$, so $u' = 2\tan x(1 + \tan^2 x)$. By the Chain Rule and the Quotient Rule,

$$\frac{d}{dx}\cos x = -\frac{1}{u} \cdot \frac{u'}{2\sqrt{u}} = -\frac{\tan x}{\sqrt{1 + \tan^2 x}} = -\sin x.$$

By the Product Rule,

$$\frac{d}{dx}\sin x = (1 + \tan^2 x)\cos x - \tan x \sin x = (1 + \tan^2 x)\cos x - \tan^2 x \cos x = \cos x.$$

The other open intervals follow by shifting. On $\left(-\frac{\pi}{2} + k\pi, \frac{\pi}{2} + k\pi\right)$, $\sin x = (-1)^k \sin(x - k\pi)$ with $x - k\pi$ in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. So $\sin'(x) = (-1)^k \cos(x - k\pi) = \cos x$. Similarly $\cos' = -\sin$.

It remains to check the points $\frac{\pi}{2} + k\pi$, where tan is not defined. First take $c = \frac{\pi}{2}$. As $x \to \frac{\pi}{2}^-$, $\tan x \to \infty$, so

$$\cos x = \frac{1}{\sqrt{1 + \tan^2 x}} \to 0 \quad \text{and} \quad \sin x = \frac{1}{\sqrt{1 + 1/\tan^2 x}} \to 1.$$

As $x \to \frac{\pi}{2}^+$, $x - \pi \to -\frac{\pi}{2}^+$. The same computation, with $\tan(x - \pi) \to -\infty$, gives $\cos(x - \pi) \to 0$ and $\sin(x - \pi) \to -1$. So $\cos x = -\cos(x - \pi) \to 0$ and $\sin x = -\sin(x - \pi) \to 1$. Thus sin and cos are continuous at $\frac{\pi}{2}$, and so on $(0, \pi)$. Also $\sin'(x) = \cos x \to 0 = \cos \frac{\pi}{2}$ and $\cos'(x) = -\sin x \to -1 = -\sin \frac{\pi}{2}$ as $x \to \frac{\pi}{2}$. By Lemma 9.5.9, $\sin'\left(\frac{\pi}{2}\right) = \cos \frac{\pi}{2}$ and $\cos'\left(\frac{\pi}{2}\right) = -\sin \frac{\pi}{2}$. For $c = \frac{\pi}{2} + k\pi$, use $\sin x = (-1)^k \sin(x - k\pi)$ and $\cos x = (-1)^k \cos(x - k\pi)$ near c. □

So $\sin'' = -\sin$ and $\cos'' = -\cos$. The identities now follow from one fact about the equation $y'' + y = 0$. The same argument appears in Section 12.4, so that each section can be read alone.

Theorem 9.5.11. *(1) If* $y : \mathbb{R} \to \mathbb{R}$ *is twice differentiable with* $y'' + y = 0$ *on* $\mathbb{R}$*, then* $y(x) = y(0)\cos x + y'(0)\sin x$ *for all* x*.*

(2) $\sin(x + y) = \sin x \cos y + \cos x \sin y$ *and* $\cos(x + y) = \cos x \cos y - \sin x \sin y$ *for all* $x, y \in \mathbb{R}$*.*

(3) $\sin(-x) = -\sin x$*,* $\cos(-x) = \cos x$*,* $\sin\left(x + \frac{\pi}{2}\right) = \cos x$ *and* $\sin(x + 2\pi) = \sin x$*,* $\cos(x + 2\pi) = \cos x$ *for all* x*.*

(4) $\sin x > 0$ *for* $0 < x < \pi$. *Also* $\sin x = 0$ *exactly when* $x = k\pi$, *and* $\cos x = 0$ *exactly when* $x = \frac{\pi}{2} + k\pi$, *for some* $k \in \mathbb{Z}$.

PROOF. (1) Let $z(x) = y(x) - y(0)\cos x - y'(0)\sin x$. Then $z'' + z = 0$, $z(0) = 0$ and $z'(0) = 0$. Let $E = z^2 + (z')^2$. Then $E' = 2z'(z + z'') = 0$, so E is constant by Theorem 7.2.7, and $E = E(0) = 0$. So $z = 0$.

(2) Fix y. The function $f(x) = \sin(x + y)$ has $f'' = -f$, $f(0) = \sin y$ and $f'(0) = \cos y$. By (1), $f(x) = \sin y \cos x + \cos y \sin x$. For $\cos(x + y)$, use $g(x) = \cos(x + y)$, with $g(0) = \cos y$ and $g'(0) = -\sin y$.

(3) The functions $-\sin(-x)$ and $\cos(-x)$ solve $y'' + y = 0$ with the same values and derivatives at 0 as sin and cos. So they equal sin and cos by (1). Similarly, $\sin\left(x + \frac{\pi}{2}\right)$ solves the equation with value 1 and derivative $\cos\frac{\pi}{2} = 0$ at 0, so it equals $\cos x$. Applying $\sin(x + \pi) = -\sin x$ and $\cos(x + \pi) = -\cos x$ twice gives period 2π.

(4) If $0 < x < \frac{\pi}{2}$, then $\tan x > 0$ and $\cos x > 0$, so $\sin x > 0$. Also $\sin\frac{\pi}{2} = 1$. If $\frac{\pi}{2} < x < \pi$, then $y = x - \pi \in \left(-\frac{\pi}{2}, 0\right)$ and $\sin x = -\sin y = -\tan y \cos y > 0$. On $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right]$, $\sin y = 0$ only when $\tan y = 0$, that is, $y = 0$, and $\cos y = 0$ only when $y = \frac{\pi}{2}$, since $\cos y > 0$ for $|y| < \frac{\pi}{2}$. The rest follows from the definition, since $\sin(y + k\pi) = (-1)^k \sin y$ and $\cos(y + k\pi) = (-1)^k \cos y$. □

Last come the inverse sine and inverse cosine. They have integral formulas too. The integral for arcsin is improper at ± 1, which is one reason this section started from arctan.

Theorem 9.5.12. *(1)* sin *is strictly increasing on* $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ *and maps it onto* $[-1, 1]$. *Its inverse* $\arcsin : [-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ *satisfies*

$$\arcsin x = \int_0^x \frac{dt}{\sqrt{1 - t^2}} \quad \text{for } -1 < x < 1.$$

(2) cos *is strictly decreasing on* $[0, \pi]$ *and maps it onto* $[-1, 1]$. *Its inverse* $\arccos : [-1, 1] \to [0, \pi]$ *satisfies* $\arccos x = \frac{\pi}{2} - \arcsin x$ *for* $-1 \le x \le 1$.

PROOF. (1) On $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, $\sin' = \cos > 0$. If $-\frac{\pi}{2} \le a < b \le \frac{\pi}{2}$, then by Theorem 7.2.4, $\sin b - \sin a = \cos\xi\,(b - a) > 0$ for some $\xi \in (a, b)$. So sin is strictly increasing on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Since $\sin\left(\pm\frac{\pi}{2}\right) = \pm 1$, it maps onto $[-1, 1]$ by Theorem 6.3.10. By Theorem 7.2.9, arcsin is differentiable on $(-1, 1)$. If $y = \sin x$ with $|x| < \frac{\pi}{2}$, then $\cos x > 0$, so $\cos x = \sqrt{1 - y^2}$ and $\arcsin'(y) = \frac{1}{\cos x} = \frac{1}{\sqrt{1-y^2}}$. The integral $G(x) = \int_0^x \frac{dt}{\sqrt{1-t^2}}$ has the same derivative on $(-1, 1)$, by Theorem 9.3.7. Both are 0 at 0. So $G = \arcsin$ on $(-1, 1)$ by Theorem 7.2.7 (5).

(2) On $(0, \pi)$, $\cos' = -\sin < 0$ by Theorem 9.5.11 (4). As in (1), cos is strictly decreasing on $[0, \pi]$ and maps it onto $[\cos\pi, \cos 0] = [-1, 1]$. Let $y \in [-1, 1]$ and $x = \arcsin y$. Then $\frac{\pi}{2} - x \in [0, \pi]$. By Theorem 9.5.11 (2) and (3), $\cos\left(\frac{\pi}{2} - x\right) = \cos\frac{\pi}{2}\cos x + \sin\frac{\pi}{2}\sin x = \sin x = y$. So $\arccos y = \frac{\pi}{2} - x$. □

The inverse cosine here is the one in Example 7.2.10. For a reader who knows arc length, $\int_0^x \frac{dt}{\sqrt{1-t^2}}$ is the length of the arc of the unit circle above $[0, x]$. So $\pi = 2\arcsin 1$ is half the length of the unit circle, which is how π is usually introduced.

Exercises

9.5.1. Prove that $\arctan x + \arctan \frac{1}{x} = \frac{\pi}{2}$ for all $x > 0$. Conclude that $\arctan 1 = \frac{\pi}{4}$. [Solution, p. 348]

9.5.2. Let $x, y \in \mathbb{R}$ with $xy < 1$. Prove that $\arctan x + \arctan y = \arctan \frac{x+y}{1-xy}$. [Hint, p. 348]

9.5.3. Prove that $1 - t^2 \le \frac{1}{1+t^2} \le 1 - t^2 + t^4$ for all $t \in \mathbb{R}$. Use Exercise 9.5.1 to conclude that $\frac{8}{3} \le \pi \le \frac{52}{15}$. [Hint, p. 348]

9.5.4. Prove that $\sin x < x$ for all $x > 0$. Then prove that $\lim_{x\to 0} \frac{\sin x}{x} = 1$. [Hint, p. 349]

9.5.5. Define $\sec x = \frac{1}{\cos x}$ where $\cos x \neq 0$. Prove that $\frac{d}{dx} \sec x = \sec x \tan x$ and $\sec^2 x = 1 + \tan^2 x$ for $|x| < \frac{\pi}{2}$. [Hint, p. 349]

9.5.6. Let $F(x) = \frac{1}{2}\left(x\sqrt{1-x^2} + \arcsin x\right)$ for $-1 \le x \le 1$. Prove that $F'(x) = \sqrt{1-x^2}$ for $-1 < x < 1$. Then prove that $\int_0^1 \sqrt{1-t^2}\, dt = \frac{\pi}{4}$. So the area of the unit disk is π. [Solution, p. 349]

9.6. More Properties of the Integral

Theorem 9.6.1 (Mean Value Theorem for Integrals)**.** *Let $f \in C[a,b]$. Then there exists a $c \in [a,b]$ such that*

$$\int_a^b f = f(c)(b-a).$$

PROOF. Define $F(x) = \int_a^x f$. Since f is continuous on $[a,b]$, $F'(x) = f(x)$ for all $x \in [a,b]$ by Theorem 9.3.7. Thus by the Mean Value Theorem, there exists a $c \in (a,b)$ such that

$$\int_a^b f = F(b) - F(a) = F'(c)(b-a) = f(c)(b-a).$$

□

Theorem 9.6.2 (Integration by Parts)**.** *Let $f, g : [a,b] \to \mathbb{R}$ be differentiable on $[a,b]$ with $f', g' \in \mathcal{R}[a,b]$. Then*

$$\int_a^b fg' = f(b)g(b) - f(a)g(a) - \int_a^b gf'.$$

PROOF. Since f, g are continuous, $f, g \in \mathcal{R}[a,b]$. So $fg', gf' \in \mathcal{R}[a,b]$ by Theorem 9.2.11 (3). Since

$$(fg)' = f'g + fg',$$

$(fg)' \in \mathcal{R}[a,b]$. Thus by Theorems 9.2.11 and 9.3.6,

$$f(b)g(b) - f(a)g(a) = \int_a^b (fg)' = \int_a^b gf' + \int_a^b fg'.$$

□

Theorem 9.6.3 (u-Substitution)**.** *Let $\varphi : [a,b] \to \mathbb{R}$ be differentiable on $[a,b]$ with $\varphi' \in \mathcal{R}[a,b]$. Let $I = \varphi[[a,b]]$. If $f : I \to \mathbb{R}$ is continuous on I, then*

$$\int_a^b f(\varphi(t))\varphi'(t)\,dt = \int_{\varphi(a)}^{\varphi(b)} f(x)\,dx.$$

PROOF. Since φ is continuous, by Corollary 6.3.11, I is an interval. It contains its endpoints, since φ attains a maximum and a minimum on $[a,b]$ by the Extreme Value Theorem. So I is a closed and bounded interval. Also, since $f \circ \varphi$ is continuous and $\varphi' \in \mathcal{R}[a,b]$, $(f \circ \varphi)\varphi' \in \mathcal{R}[a,b]$. If I is a single point, then φ is constant on $[a,b]$. Thus $\varphi'(t) = 0$ for all $t \in [a,b]$ and the equation holds since both integrals are zero. Otherwise, for $x \in I$, define

$$F(x) = \int_{\varphi(a)}^{x} f(s)\,ds.$$

Since f is continuous, $F'(x) = f(x)$ for all $x \in I$ by Theorem 9.3.7 and Theorem 9.2.13. By the Chain Rule,

$$\frac{d}{dt}F(\varphi(t)) = F'(\varphi(t))\varphi'(t) = f(\varphi(t))\varphi'(t),$$

for all $t \in [a,b]$. Thus by the Fundamental Theorem of Calculus,

$$\int_a^b f(\varphi(t))\varphi'(t)\,dt = F(\varphi(b)) - F(\varphi(a)) = \int_{\varphi(a)}^{\varphi(b)} f(s)\,ds.$$

□

Exercises

9.6.1. Let $f \in C[a,b]$ and $g \in \mathcal{R}[a,b]$ with $g(x) \geq 0$ for all $x \in [a,b]$. Prove there exists a $c \in [a,b]$ such that

$$\int_a^b f(x)g(x)\,dx = f(c)\int_a^b g(x)\,dx.$$

[Hint, p. 349]

9.6.2. Let $f \in \mathcal{R}[-a,a]$, $a > 0$. Prove each of the following. Here f is even if $f(-x) = f(x)$ for all x, and odd if $f(-x) = -f(x)$ for all x.
(a) If f is even, then $\int_{-a}^{a} f = 2\int_0^a f$.
(b) If f is odd, then $\int_{-a}^{a} f = 0$.

[Hint, p. 349]

9.7. Improper Integrals

The Riemann integral needs a bounded function on a closed and bounded interval. An improper integral removes one of these restrictions by taking a limit. Throughout this section, x^p for real p and $x > 0$ means $e^{p \ln x}$, as in Definition 9.4.7. Its derivative is px^{p-1}.

Definition 9.7.1. Let $I \subset \mathbb{R}$ be an interval and $f : I \to \mathbb{R}$. If f is integrable on every closed and bounded subinterval of I, f is *locally integrable* on I, denoted $f \in \mathcal{R}_{\text{loc}}(I)$.

Definition 9.7.2. Let $f \in \mathcal{R}_{\text{loc}}[a, b)$, where $a \in \mathbb{R}$ and $b \in \mathbb{R} \cup \{\infty\}$. Then the *improper Riemann integral* of f on $[a, b)$ is defined as

$$\int_a^b f = \lim_{s \to b^-} \int_a^s f,$$

provided the limit exists. If $f \in \mathcal{R}_{\text{loc}}(a, b]$, where $a \in \mathbb{R} \cup \{-\infty\}$ and $b \in \mathbb{R}$, the improper integral of f on $(a, b]$ is defined as

$$\int_a^b f = \lim_{t \to a^+} \int_t^b f,$$

provided the limit exists. If the integral exists, f is integrable on $[a, b)$ or $(a, b]$ and the integral *converges*. Otherwise, the integral *diverges*. When $b = \infty$, $s \to b^-$ means $s \to \infty$, and when $a = -\infty$, $t \to a^+$ means $t \to -\infty$. These limits are defined in Section 6.2.

Example 9.7.3. Let $f : (0, 1] \to \mathbb{R}$, $f(x) = \frac{1}{\sqrt{x}}$. Then

$$\int_0^1 f = \lim_{t \to 0^+} \int_t^1 \frac{1}{\sqrt{x}}\, dx = \lim_{t \to 0^+} \left(2 - 2\sqrt{t}\right) = 2.$$

However, notice

$$\int_0^1 f^2 = \lim_{t \to 0^+} \ln x|_{x=t}^{x=1} = \lim_{t \to 0^+} (-\ln t),$$

which does not exist. So the improper integral of f converging does not guarantee the improper integral of f^2 converging.

Definition 9.7.4. Let $f \in \mathcal{R}_{\text{loc}}(a, b)$ with $a \in \mathbb{R} \cup \{-\infty\}$ and $b \in \mathbb{R} \cup \{\infty\}$, and let $c \in (a, b)$. Then the improper integral of f on (a, b) is defined as

$$\int_a^b f = \lim_{t \to a^+} \int_t^c f + \lim_{s \to b^-} \int_c^s f,$$

provided both limits exist. If the integral exists, f is integrable on (a, b) and the integral converges. Otherwise, the integral diverges. The value does not depend on c. See Exercise 9.7.5.

Example 9.7.5. Even though $\lim_{t \to \infty} \int_{-t}^t x^3\, dx = 0$, $\int_{-\infty}^{\infty} x^3\, dx$ does not exist, since

$$\int_0^\infty x^3\, dx = \lim_{s \to \infty} \int_0^s x^3\, dx$$

does not exist.

Definition 9.7.6. Let $f \in \mathcal{R}_{\text{loc}}[a, c)$ and $f \in \mathcal{R}_{\text{loc}}(c, b]$, where $a, b, c \in \mathbb{R}$ with $a < c < b$. Then the improper integral of f on $[a, b]$ is defined as

$$\int_a^b f = \lim_{s \to c^-} \int_a^s f + \lim_{t \to c^+} \int_t^b f,$$

provided both limits exist. If the integral exists, f is integrable on $[a,b]$ and the integral converges. Otherwise, the integral diverges.

Theorem 9.7.7 (p-Test)**.** *Let $a>0$. Then $\int_a^\infty \frac{1}{x^p}\,dx$ converges if and only if $p>1$. Also, $\int_0^a \frac{1}{x^p}\,dx$ converges if and only if $p<1$.*

PROOF. See Exercise 9.7.1. □

Theorem 9.7.8 (Comparison Test)**.** *Let $I\subset\mathbb{R}$ be an interval and let $f,g\in\mathcal{R}_{\text{loc}}(I)$ with $0\le f(x)\le g(x)$ for all $x\in I$. If g is integrable on I, then f is integrable on I. If f is not integrable on I, then g is not integrable on I.*

PROOF. Let $I=[a,b)$. If $0\le f\le g$ on I and g is integrable on I, then $\int_a^s f$ is an increasing function of s and is bounded above by $\int_a^b g$. Indeed, for $a\le s<b$, $\int_a^s f\le\int_a^s g\le\int_a^b g$, since $\int_a^s g$ increases with s to its limit $\int_a^b g$. So by the argument of Theorem 6.5.2,

$$\int_a^b f=\lim_{s\to b^-}\int_a^s f$$

exists. That argument shows an increasing function that is bounded above on $[a,b)$ has a left-hand limit at b equal to its supremum, and it works for $b=\infty$ too. The cases where $I=(a,b]$ and $I=(a,b)$ are similar. The second statement is the contrapositive of the first. □

Theorem 9.7.9 (Limit Comparison Test)**.** *Suppose $f,g\in\mathcal{R}_{\text{loc}}(a,b]$ with $f(x),g(x)>0$ for all $x\in(a,b]$.*

(1) If $\lim_{x\to a^+}\frac{g(x)}{f(x)}=L>0$, then either both $\int_a^b f$ and $\int_a^b g$ converge or both diverge.

(2) If $\lim_{x\to a^+}\frac{g(x)}{f(x)}=0$, then if $\int_a^b f$ converges, $\int_a^b g$ converges.

(3) If $\lim_{x\to a^+}\frac{g(x)}{f(x)}=\infty$ then if $\int_a^b f$ diverges, $\int_a^b g$ diverges.

If $I=[a,b)$, consider $\lim_{x\to b^-}\frac{g(x)}{f(x)}$ and the results are similar.

PROOF. (1) Suppose $\lim_{x\to a^+}\frac{g(x)}{f(x)}=L>0$. Set $\varepsilon=\frac{L}{2}$. Then there exists a $\delta>0$ with $a+\delta<b$ such that for every $x\in(a,a+\delta)$,

$$\frac{L}{2}=L-\varepsilon<\frac{g(x)}{f(x)}<L+\varepsilon=\frac{3L}{2}.$$

Thus $\frac{L}{2}f(x)\le g(x)\le\frac{3L}{2}f(x)$ for all $x\in(a,a+\delta)$. By the Comparison Test on $(a,a+\delta]$, $\int_a^{a+\delta} f$ and $\int_a^{a+\delta} g$ both converge or both diverge. Since $f,g\in\mathcal{R}[a+\delta,b]$, the same is true of $\int_a^b f$ and $\int_a^b g$.

(2) Suppose $\lim_{x\to a^+}\frac{g(x)}{f(x)}=0$. Then there exists a $\delta>0$ such that for every $x\in(a,a+\delta)$, $0<\frac{g(x)}{f(x)}<1$, which implies $0<g(x)<f(x)$ for all $x\in(a,a+\delta)$. The result then follows from the Comparison Test.

(3) Suppose $\lim_{x\to a^+}\frac{g(x)}{f(x)}=\infty$. Then there exists a $\delta>0$ such that for every $x\in(a,a+\delta)$, $\frac{g(x)}{f(x)}>1$, which implies $g(x)>f(x)$ for all $x\in(a,a+\delta)$. The result then follows from the Comparison Test. □

Theorem 9.7.10 (Absolute Convergence Test)**.** *If $f \in \mathcal{R}_{\text{loc}}(I)$ and if $|f|$ is integrable on I, then f is integrable on I.*

PROOF. Since $-|f(x)| \le f(x) \le |f(x)|$ for all $x \in I$, $0 \le f(x) + |f(x)| \le 2|f(x)|$ for all $x \in I$. By the Comparison Test, $f + |f|$ is integrable on I. Therefore, $f = (f + |f|) - |f|$ is integrable on I. The last step uses the limit laws for one-sided limits (Remark 6.2.7). □

Example 9.7.11. Using Integration by Parts, notice

$$\int_\pi^c \frac{\sin x}{x}\,dx = -\frac{\cos c}{c} - \frac{1}{\pi} - \int_\pi^c \frac{\cos x}{x^2}\,dx.$$

So

$$\lim_{c\to\infty} \int_\pi^c \frac{\sin x}{x}\,dx = -\frac{1}{\pi} - \int_\pi^\infty \frac{\cos x}{x^2}\,dx.$$

But $\left|\frac{\cos x}{x^2}\right| \le \frac{1}{x^2}$, so by the Comparison Test and the Absolute Convergence Test, $\int_\pi^\infty \frac{\cos x}{x^2}\,dx$ converges. Thus $\int_\pi^\infty \frac{\sin x}{x}\,dx$ converges.

However, since

$$\int_{(k+1/4)\pi}^{(k+3/4)\pi} \frac{|\sin x|}{x}\,dx \ge \frac{\sqrt{2}}{2}\cdot\frac{1}{(k+1)\pi}\left[\left(k+\frac{3}{4}\right)\pi - \left(k+\frac{1}{4}\right)\pi\right]$$

$$= \frac{\sqrt{2}}{4}\cdot\frac{1}{k+1},$$

$$\int_\pi^{(n+1)\pi} \frac{|\sin x|}{x}\,dx \ge \frac{\sqrt{2}}{4}\sum_{k=1}^{n}\frac{1}{k+1},$$

which implies $\int_\pi^\infty \frac{|\sin x|}{x}\,dx$ does not exist. The last step uses that the sums $\sum_{k=1}^n \frac{1}{k+1}$ are not bounded. See the hint in Exercise 8.1.4, or Example 10.2.12 and Exercise 10.1.1.

Definition 9.7.12. Let $f \in \mathcal{R}_{\text{loc}}(-\infty,\infty)$. The *principal value* of $\int_{-\infty}^\infty f$ is

$$\text{P. V.} \int_{-\infty}^\infty f = \lim_{t\to\infty}\int_{-t}^t f.$$

Example 9.7.13. Let $n \in \mathbb{N}$. Then

$$\text{P. V.} \int_{-\infty}^\infty x^{2n-1}\,dx = 0.$$

However, $\int_{-\infty}^\infty x^{2n-1}\,dx$ does not exist.

Example 9.7.14. Notice

$$\text{P. V.} \int_{-\infty}^\infty \frac{1}{1+x^2}\,dx = \lim_{t\to\infty}(2\arctan t) = \pi.$$

Also,

$$\int_{-\infty}^\infty \frac{1}{1+x^2}\,dx = \lim_{t\to-\infty}\int_t^0 \frac{1}{1+x^2}\,dx + \lim_{s\to\infty}\int_0^s \frac{1}{1+x^2}\,dx$$

$$= \lim_{t\to-\infty}(-\arctan t) + \lim_{s\to\infty}\arctan s$$
$$= \pi.$$

So

$$\text{P. V.}\int_{-\infty}^{\infty}\frac{1}{1+x^2}\,dx = \int_{-\infty}^{\infty}\frac{1}{1+x^2}\,dx.$$

Theorem 9.7.15. *Let $f \in \mathcal{R}_{\text{loc}}(-\infty,\infty)$. If $\int_{-\infty}^{\infty} f$ converges, then* P. V. $\int_{-\infty}^{\infty} f$ *converges and*

$$\int_{-\infty}^{\infty} f = \text{P. V.}\int_{-\infty}^{\infty} f.$$

PROOF. Since $\int_{-\infty}^{\infty} f$ exists, both $\lim_{t\to-\infty}\int_t^0 f$ and $\lim_{s\to\infty}\int_0^s f$ exist. So

$$\begin{aligned}\text{P. V.}\int_{-\infty}^{\infty} f &= \lim_{t\to\infty}\int_{-t}^{t} f\\ &= \lim_{t\to\infty}\left(\int_{-t}^{0} f + \int_0^t f\right)\\ &= \lim_{t\to-\infty}\int_t^0 f + \lim_{s\to\infty}\int_0^s f\\ &= \int_{-\infty}^{\infty} f.\end{aligned}$$

□

Definition 9.7.16. Let $f \in \mathcal{R}_{\text{loc}}[a,c)$ and $f \in \mathcal{R}_{\text{loc}}(c,b]$ where $a,b,c \in \mathbb{R}$ with $a < c < b$. Then the principal value of $\int_a^b f$ is defined as

$$\text{P. V.}\int_a^b f = \lim_{\varepsilon\to0^+}\left(\int_a^{c-\varepsilon} f + \int_{c+\varepsilon}^b f\right).$$

Example 9.7.17. Define $f : [-1,0)\cup(0,1] \to \mathbb{R}$ by $f(x) = \frac{1}{x}$. Notice that since f is continuous on $[-1,0)\cup(0,1]$, $f \in \mathcal{R}_{\text{loc}}[-1,0)$ and $f \in \mathcal{R}_{\text{loc}}(0,1]$. By the p-Test, $\int_0^1 f$ diverges, and so $\int_{-1}^1 f$ diverges. However,

$$\text{P. V.}\int_{-1}^{1}\frac{1}{x}\,dx = \lim_{\varepsilon\to0^+}\left(\int_{-1}^{-\varepsilon}\frac{1}{x}\,dx + \int_{\varepsilon}^{1}\frac{1}{x}\,dx\right) = \lim_{\varepsilon\to0^+}(\ln|-\varepsilon| - \ln|\varepsilon|) = 0.$$

Theorem 9.7.18. *Let $f \in \mathcal{R}_{\text{loc}}[a,c)$ and $f \in \mathcal{R}_{\text{loc}}(c,b]$. If $\int_a^b f$ converges, then* P. V. $\int_a^b f$ *converges with*

$$\int_a^b f = \text{P. V.}\int_a^b f.$$

PROOF. See Exercise 9.7.2. □

Exercises

9.7.1. Prove Theorem 9.7.7. [Hint, p. 349]

9.7.2. Prove Theorem 9.7.18. [Hint, p. 350]

9.7.3. For what values of p does $\int_2^\infty \frac{1}{x(\ln x)^p}\,dx$ converge? [Hint, p. 350]

9.7.4. For $x > 0$, define $\Gamma(x) = \int_0^\infty e^{-t}t^{x-1}\,dt$.

(a) Show the improper integral converges for all $x > 0$.

(b) Use integration by parts to show that $\Gamma(x+1) = x\Gamma(x)$, $x > 0$.

(c) Show that $\Gamma(1) = 1$.

(d) For $n \in \mathbb{N}$, prove that $\Gamma(n+1) = n!$.

[Hint, p. 350]

9.7.5. Show that the value of the improper integral in Definition 9.7.4 does not depend on the choice of $c \in (a, b)$. [Hint, p. 350]

9.8. Four Famous Formulas*

This section uses integrals to prove four classical formulas. Wallis's product writes $\frac{\pi}{2}$ as an infinite product. Stirling's formula says how fast $n!$ grows. Gauss's integral is $\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}$. Euler's formula is $\sum_{k=1}^\infty \frac{1}{k^2} = \frac{\pi^2}{6}$. All four come from the integrals of powers of $\sin x$ and $\cos x$. The section uses sin, cos, ln and e^x with their familiar properties (see Sections 9.4 and 9.5). It can be skipped. Nothing later depends on it.

Lemma 9.8.1. *For $n \in \mathbb{N}_0$, let $I_n = \int_0^{\pi/2} \sin^n x\,dx$. Then*

(1) $I_0 = \frac{\pi}{2}$, $I_1 = 1$, and $I_n = \frac{n-1}{n} I_{n-2}$ for $n \geq 2$,

(2) $I_{2m} = \frac{(2m)!}{4^m(m!)^2} \cdot \frac{\pi}{2}$ and $I_{2m+1} = \frac{4^m(m!)^2}{(2m+1)!}$ for $m \in \mathbb{N}_0$,

(3) $0 < I_{n+1} \leq I_n$ for all n,

(4) $nI_nI_{n-1} = \frac{\pi}{2}$ for $n \geq 1$, and

(5) $\lim_{n\to\infty} \sqrt{n}\, I_n = \sqrt{\frac{\pi}{2}}$.

Proof. (1) The values of I_0 and I_1 follow from Theorem 9.3.6. For $n \geq 2$, integrate by parts (Theorem 9.6.2) with $u = \sin^{n-1} x$ and $v = -\cos x$. The boundary term $-\sin^{n-1} x \cos x$ is 0 at both ends. So

$$\begin{aligned} I_n &= (n-1)\int_0^{\pi/2} \sin^{n-2} x \cos^2 x\,dx \\ &= (n-1)\int_0^{\pi/2} \sin^{n-2} x\,(1 - \sin^2 x)\,dx = (n-1)(I_{n-2} - I_n). \end{aligned}$$

Solving for I_n gives $I_n = \frac{n-1}{n} I_{n-2}$.

(2) This follows from (1) by induction on m. For example, $I_{2m} = \frac{2m-1}{2m} I_{2m-2}$, and $\frac{(2m)!}{4^m(m!)^2} = \frac{(2m)(2m-1)}{4m^2} \cdot \frac{(2m-2)!}{4^{m-1}((m-1)!)^2}$.

(3) On $\left(0, \frac{\pi}{2}\right)$, $0 < \sin x < 1$, so $0 < \sin^{n+1} x < \sin^n x$. Integrate, using Exercise 9.2.3.

(4) By (1), $nI_nI_{n-1} = (n-1)I_{n-1}I_{n-2}$ for $n \geq 2$. So $nI_nI_{n-1} = 1 \cdot I_1I_0 = \frac{\pi}{2}$.

(5) By (3) and (1), $1 \le \frac{I_{n-1}}{I_n} \le \frac{I_{n-2}}{I_n} = \frac{n}{n-1}$ for $n \ge 2$. So $\frac{I_{n-1}}{I_n} \to 1$ by the Squeeze Theorem. By (4), $nI_n^2 = \frac{\pi}{2} \cdot \frac{I_n}{I_{n-1}} \to \frac{\pi}{2}$. Take square roots. □

Theorem 9.8.2 (Wallis's Product)**.**

$$\frac{\pi}{2} = \lim_{m\to\infty} \prod_{k=1}^{m} \frac{(2k)^2}{(2k-1)(2k+1)} = \lim_{m\to\infty} \frac{2}{1}\cdot\frac{2}{3}\cdot\frac{4}{3}\cdot\frac{4}{5}\cdot\frac{6}{5}\cdot\frac{6}{7}\cdots\frac{2m}{2m-1}\cdot\frac{2m}{2m+1}.$$

PROOF. The numerators multiply to $(2^m m!)^2 = 4^m (m!)^2$. The odd numbers $1\cdot 3\cdots(2m-1)$ multiply to $\frac{(2m)!}{2^m m!}$. So the denominators multiply to $\left(\frac{(2m)!}{2^m m!}\right)^2 (2m+1)$, and the product is

$$P_m = \frac{16^m (m!)^4}{((2m)!)^2(2m+1)} = \frac{\pi}{2}\cdot\frac{I_{2m+1}}{I_{2m}}$$

by Lemma 9.8.1 (2). In the proof of Lemma 9.8.1 (5), $\frac{I_{n-1}}{I_n} \to 1$. So $P_m \to \frac{\pi}{2}$. □

Next comes Stirling's formula. It compares $n!$ with $\sqrt{2\pi n}\left(\frac{n}{e}\right)^n$. The proof has two steps. First, a monotone sequence argument shows the ratio has a positive limit. Then Wallis's integrals identify the limit.

Theorem 9.8.3 (Stirling's Formula)**.**

$$\lim_{n\to\infty} \frac{n!}{\sqrt{2\pi n}\left(\frac{n}{e}\right)^n} = 1.$$

PROOF. Let $a_n = \frac{n!\, e^n}{n^{n+1/2}}$. Then

$$\ln\frac{a_n}{a_{n+1}} = \left(n+\tfrac{1}{2}\right)\ln\left(1+\tfrac{1}{n}\right) - 1 = \left(n+\tfrac{1}{2}\right)\int_n^{n+1}\frac{dt}{t} - 1.$$

Let $m = n + \frac{1}{2}$. For $t > 0$,

$$\frac{1}{t} - \left(\frac{1}{m} - \frac{t-m}{m^2}\right) = \frac{(t-m)^2}{m^2 t} \ge 0,$$

with equality only at $t = m$. The function in parentheses is linear, and its integral over $[n, n+1]$ is $\frac{1}{m}$, since m is the midpoint. So $\int_n^{n+1}\frac{dt}{t} > \frac{1}{m}$, and $\ln\frac{a_n}{a_{n+1}} > 0$. Also, on $[n, n+1]$,

$$\frac{2n+1-t}{n(n+1)} - \frac{1}{t} = \frac{(t-n)(n+1-t)}{n(n+1)t} \ge 0.$$

The integral of the linear function on the left over $[n, n+1]$ is $\frac{1}{2}\left(\frac{1}{n} + \frac{1}{n+1}\right)$. So

$$\ln\frac{a_n}{a_{n+1}} < \frac{2n+1}{2}\cdot\frac{2n+1}{2n(n+1)} - 1 = \frac{1}{4n(n+1)} = \frac{1}{4}\left(\frac{1}{n} - \frac{1}{n+1}\right).$$

Thus $\{a_n\}$ is decreasing, and $b_n = a_n e^{-1/(4n)}$ is increasing, since $\ln b_n - \ln b_{n+1} = \ln\frac{a_n}{a_{n+1}} - \frac{1}{4}\left(\frac{1}{n} - \frac{1}{n+1}\right) < 0$. So $b_1 \le b_n \le a_n$, and $\{a_n\}$ is decreasing and bounded below by $b_1 > 0$. By the Monotone Convergence Theorem, $a_n \to L$ for some $L \ge b_1 > 0$.

To find L, use Lemma 9.8.1 (2) and (5). Writing $n! = a_n n^{n+1/2} e^{-n}$,

$$\frac{(2m)!}{4^m (m!)^2} = \frac{a_{2m}(2m)^{2m+1/2} e^{-2m}}{4^m a_m^2 m^{2m+1} e^{-2m}} = \frac{a_{2m}}{a_m^2} \cdot \frac{\sqrt{2}}{\sqrt{m}}.$$

So

$$\sqrt{2m}\, I_{2m} = \sqrt{2m} \cdot \frac{a_{2m}}{a_m^2} \cdot \frac{\sqrt{2}}{\sqrt{m}} \cdot \frac{\pi}{2} = \frac{a_{2m}}{a_m^2} \cdot \pi \to \frac{L}{L^2} \cdot \pi = \frac{\pi}{L}.$$

By Lemma 9.8.1 (5), the left side tends to $\sqrt{\frac{\pi}{2}}$. So $\frac{\pi}{L} = \sqrt{\frac{\pi}{2}}$, and $L = \sqrt{2\pi}$. That is, $\frac{n!}{\sqrt{2\pi n}(n/e)^n} = \frac{a_n}{\sqrt{2\pi}} \to 1$. □

The proof gives more than the limit. Since a_n decreases to $\sqrt{2\pi}$ and b_n increases to it, $\sqrt{2\pi n}\left(\frac{n}{e}\right)^n < n! < \sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{1/(4n)}$. See Exercise 9.8.3.

Theorem 9.8.4 (Gauss's Integral)**.** *The improper integral* $\int_0^\infty e^{-x^2}\,dx$ *converges, and*

$$\int_0^\infty e^{-x^2}\,dx = \frac{\sqrt{\pi}}{2}.$$

PROOF. For $x \geq 1$, $0 < e^{-x^2} \leq e^{-x}$, and $\int_1^\infty e^{-x}\,dx$ converges. So the integral converges by Theorem 9.7.8. Call its value G. By the substitution $u = \sqrt{n}\,x$ (Theorem 9.6.3, on $[0, b]$, then $b \to \infty$),

$$\int_0^\infty e^{-nx^2}\,dx = \frac{G}{\sqrt{n}}.$$

Since $e^u \geq 1 + u$ for all u (Theorem 9.4.11), $1 - x^2 \leq e^{-x^2}$ and $1 + x^2 \leq e^{x^2}$. Raising to the power n,

$$(1 - x^2)^n \leq e^{-nx^2} \quad (0 \leq x \leq 1) \qquad \text{and} \qquad e^{-nx^2} \leq \frac{1}{(1 + x^2)^n} \quad (x \geq 0).$$

Integrate. The substitution $x = \sin\theta$ gives $\int_0^1 (1 - x^2)^n\,dx = \int_0^{\pi/2} \cos^{2n+1}\theta\,d\theta$. The substitution $x = \tan\theta$ on $[0, b]$, with $b \to \infty$, gives $\int_0^\infty \frac{dx}{(1+x^2)^n} = \int_0^{\pi/2} \cos^{2n-2}\theta\,d\theta$. Integrals of powers of cos over $\left[0, \frac{\pi}{2}\right]$ equal those of sin (Exercise 9.8.1). So

$$I_{2n+1} \leq \frac{G}{\sqrt{n}} \leq I_{2n-2}.$$

Multiply by $\sqrt{n}$. Then $\sqrt{n} I_{2n+1} = \sqrt{\frac{n}{2n+1}}\sqrt{2n+1}\, I_{2n+1} \to \frac{1}{\sqrt{2}}\sqrt{\frac{\pi}{2}} = \frac{\sqrt{\pi}}{2}$ by Lemma 9.8.1 (5). In the same way, $\sqrt{n} I_{2n-2} \to \frac{\sqrt{\pi}}{2}$. By the Squeeze Theorem, $G = \frac{\sqrt{\pi}}{2}$. □

Theorem 9.8.5 (Euler's Formula for $\sum \frac{1}{k^2}$)**.**

$$\sum_{k=1}^{\infty} \frac{1}{k^2} = \frac{\pi^2}{6}.$$

PROOF. For $n \in \mathbb{N}_0$, let $A_n = \int_0^{\pi/2} \cos^{2n} x\,dx$ and $B_n = \int_0^{\pi/2} x^2 \cos^{2n} x\,dx$. By Exercise 9.8.1 and Lemma 9.8.1 (1), $A_n = I_{2n}$ and $A_n = \frac{2n-1}{2n} A_{n-1}$ for $n \geq 1$.

Let $n \geq 1$. Integrate A_n by parts twice. First, with $u = \cos^{2n} x$ and $v = x$,

$$A_n = 2n \int_0^{\pi/2} x \sin x \cos^{2n-1} x\,dx.$$

Next, with $u = \sin x \cos^{2n-1} x$ and $v = \frac{x^2}{2}$. Here

$$u' = \cos^{2n} x - (2n-1)\sin^2 x \cos^{2n-2} x = 2n\cos^{2n} x - (2n-1)\cos^{2n-2} x.$$

The boundary terms are 0 both times. So

$$A_n = -n \int_0^{\pi/2} x^2 \big(2n\cos^{2n} x - (2n-1)\cos^{2n-2} x\big)\,dx = n(2n-1)B_{n-1} - 2n^2 B_n.$$

Divide by $n^2 A_n$, and use $n^2 A_n = \frac{n(2n-1)}{2} A_{n-1}$. This gives

$$\frac{1}{n^2} = \frac{2B_{n-1}}{A_{n-1}} - \frac{2B_n}{A_n}.$$

The sum telescopes. Since $A_0 = \frac{\pi}{2}$ and $B_0 = \frac{\pi^3}{24}$,

$$\sum_{k=1}^{N} \frac{1}{k^2} = \frac{2B_0}{A_0} - \frac{2B_N}{A_N} = \frac{\pi^2}{6} - \frac{2B_N}{A_N}.$$

It remains to show $\frac{B_N}{A_N} \to 0$. Let $g(x) = \sin x - \frac{2x}{\pi}$. Then $g(0) = g\left(\frac{\pi}{2}\right) = 0$, and $g'(x) = \cos x - \frac{2}{\pi}$ is decreasing on $\left[0, \frac{\pi}{2}\right]$, positive at 0 and negative at $\frac{\pi}{2}$. So g increases and then decreases, and $g \geq 0$ on $\left[0, \frac{\pi}{2}\right]$. That is, $x \leq \frac{\pi}{2}\sin x$ there. So

$$B_N \leq \frac{\pi^2}{4} \int_0^{\pi/2} \sin^2 x \cos^{2N} x\,dx = \frac{\pi^2}{4}(A_N - A_{N+1}) = \frac{\pi^2}{4} \cdot \frac{A_N}{2N+2}.$$

Thus $0 \leq \frac{B_N}{A_N} \leq \frac{\pi^2}{8(N+1)} \to 0$. □

Euler found this sum in 1734, solving the Basel problem, which had been open for almost a century. His argument treated $\frac{\sin x}{x}$ as a polynomial with infinitely many roots. Making that argument rigorous takes more machinery than the proof above.

Exercises

9.8.1. Use the substitution $x = \frac{\pi}{2} - t$ to prove that $\int_0^{\pi/2} \cos^n x\,dx = \int_0^{\pi/2} \sin^n x\,dx$ for all $n \in \mathbb{N}_0$. [Hint, p. 350]

9.8.2. Use Theorem 9.8.3 to prove that $\lim_{n\to\infty} \frac{\sqrt[n]{n!}}{n} = \frac{1}{e}$. [Solution, p. 350]

9.8.3. Prove that $\sqrt{2\pi n}\left(\frac{n}{e}\right)^n < n! < \sqrt{2\pi n}\left(\frac{n}{e}\right)^n e^{1/(4n)}$ for all $n \in \mathbb{N}$. How close is the left side to $10! = 3628800$? [Hint, p. 351]

9.8.4. Let Γ be the Gamma function of Exercise 9.7.4. Use the substitution $t = u^2$ to prove that $\Gamma\left(\frac{1}{2}\right) = \sqrt{\pi}$. Then find $\Gamma\left(\frac{5}{2}\right)$. [Solution, p. 351]

9.8.5. Prove that $\int_{-\infty}^{\infty} e^{-x^2}\,dx = \sqrt{\pi}$ and $\int_0^{\infty} x^2 e^{-x^2}\,dx = \frac{\sqrt{\pi}}{4}$. [Solution, p. 351]

9.8.6. Use Theorem 9.8.5 to prove that $\sum_{k=1}^{\infty} \frac{1}{(2k-1)^2} = \frac{\pi^2}{8}$ and $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k^2} = \frac{\pi^2}{12}$.
[Hint, p. 351]

9.8.7. This exercise proves that π is irrational. Suppose $\pi = \frac{a}{b}$ with $a, b \in \mathbb{N}$. For $n \in \mathbb{N}$, let $f(x) = \frac{x^n(a-bx)^n}{n!}$.

(a) Prove that $f(\pi - x) = f(x)$, and that $f^{(j)}(0)$ and $f^{(j)}(\pi)$ are integers for all $j \in \mathbb{N}_0$.

(b) Let $F = f - f'' + f^{(4)} - \cdots + (-1)^n f^{(2n)}$. Prove that $F'' + F = f$, and that $\frac{d}{dx}\big(F'(x)\sin x - F(x)\cos x\big) = f(x)\sin x$. Conclude that $\int_0^\pi f(x)\sin x\,dx$ is an integer.

(c) Prove that $0 < \int_0^\pi f(x)\sin x\,dx < \frac{\pi^{n+1}a^n}{n!}$, and get a contradiction for large n.

[Hint, p. 351]

9.9. Riemann–Stieltjes Integrals*

This section can be skipped. Nothing later in the book depends on it. The end of the section uses functions of bounded variation from Chapter 8.

In a Riemann sum, each subinterval $[x_{i-1}, x_i]$ is weighted by its length Δx_i. In a Riemann–Stieltjes sum, it is weighted by $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$, where α is an increasing function. When $\alpha(x) = x$, this is the Riemann integral. When α jumps, the integral picks out values of f at the jumps, so sums and integrals become special cases of one idea.

Definition 9.9.1. Let $\alpha : [a,b] \to \mathbb{R}$ be increasing and let $f : [a,b] \to \mathbb{R}$ be bounded. For each partition $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ of $[a,b]$, set $\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1})$, $i = 1, \ldots, n$. Set

$$m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}$$

and

$$M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}.$$

Define the *upper Riemann–Stieltjes sum* of f with respect to α by

$$\mathcal{U}(\mathcal{P}, f, \alpha) = \sum_{i=1}^{n} M_i \Delta\alpha_i,$$

and define the *lower Riemann–Stieltjes sum* of f with respect to α by

$$\mathcal{L}(\mathcal{P}, f, \alpha) = \sum_{i=1}^{n} m_i \Delta\alpha_i.$$

Lemma 9.9.2. *Let $\alpha : [a,b] \to \mathbb{R}$ be increasing, let $\mathcal{P}$ be a partition of $[a,b]$, and let $f : [a,b] \to \mathbb{R}$ with $m \le f(x) \le M$ for all $x \in [a,b]$. Then*

$$m[\alpha(b) - \alpha(a)] \le \mathcal{L}(\mathcal{P}, f, \alpha) \le \mathcal{U}(\mathcal{P}, f, \alpha) \le M[\alpha(b) - \alpha(a)].$$

Proof. The result follows directly from the fact that $m \le m_i \le M_i \le M$ and $\Delta\alpha_i \ge 0$. The sum $\sum_{i=1}^{n} \Delta\alpha_i$ telescopes to $\alpha(b) - \alpha(a)$. □

Definition 9.9.3. Define the upper and lower Riemann–Stieltjes integrals of f with respect to α over $[a,b]$, denoted $\overline{\int_a^b} f\,d\alpha$ and $\underline{\int_a^b} f\,d\alpha$ respectively, by

$$\overline{\int_a^b} f\,d\alpha = \inf\{\mathcal{U}(\mathcal{P},f,\alpha) : \mathcal{P} \text{ is a partition of } [a,b]\}$$

and

$$\underline{\int_a^b} f\,d\alpha = \sup\{\mathcal{L}(\mathcal{P},f,\alpha) : \mathcal{P} \text{ is a partition of } [a,b]\}.$$

Theorem 9.9.4. *Let $f : [a,b] \to \mathbb{R}$ be bounded and $\alpha : [a,b] \to \mathbb{R}$ increasing. Then*

$$\underline{\int_a^b} f\,d\alpha \le \overline{\int_a^b} f\,d\alpha.$$

Proof. If $\mathcal{P}^*$ is a refinement of $\mathcal{P}$, then

$$\mathcal{L}(\mathcal{P},f,\alpha) \le \mathcal{L}(\mathcal{P}^*,f,\alpha) \le \mathcal{U}(\mathcal{P}^*,f,\alpha) \le \mathcal{U}(\mathcal{P},f,\alpha)$$

by the proof of Lemma 9.1.5, with Δx_i replaced by $\Delta\alpha_i$. Thus, if $\mathcal{P}$ and $\mathcal{Q}$ are any two partitions of $[a,b]$,

$$\mathcal{L}(\mathcal{P},f,\alpha) \le \mathcal{L}(\mathcal{P}\cup\mathcal{Q},f,\alpha) \le \mathcal{U}(\mathcal{P}\cup\mathcal{Q},f,\alpha) \le \mathcal{U}(\mathcal{Q},f,\alpha).$$

Thus

$$\mathcal{L}(\mathcal{P},f,\alpha) \le \mathcal{U}(\mathcal{Q},f,\alpha)$$

for any partitions $\mathcal{P},\mathcal{Q}$ of $[a,b]$. Hence

$$\underline{\int_a^b} f\,d\alpha = \sup_{\mathcal{P}} \mathcal{L}(\mathcal{P},f,\alpha) \le \mathcal{U}(\mathcal{Q},f,\alpha)$$

for any partition $\mathcal{Q}$ of $[a,b]$. Thus

$$\underline{\int_a^b} f\,d\alpha \le \inf_{\mathcal{Q}} \mathcal{U}(\mathcal{Q},f,\alpha) = \overline{\int_a^b} f\,d\alpha.$$

□

Definition 9.9.5. Let $f : [a,b] \to \mathbb{R}$ be bounded, and $\alpha : [a,b] \to \mathbb{R}$ increasing. If $\underline{\int_a^b} f\,d\alpha = \overline{\int_a^b} f\,d\alpha$, then f is said to be *Riemann integrable with respect to* α on $[a,b]$, written $f \in \mathcal{R}(\alpha)$. The common value is written as $\int_a^b f\,d\alpha$ and is called the *Riemann–Stieltjes integral* of f with respect to α.

Example 9.9.6. Fix $a < c \le b$. Let $I_c(x) = I(x-c)$ be the *unit jump function* defined by

$$I_c(x) = \begin{cases} 0, & x < c, \\ 1, & x \ge c. \end{cases}$$

Let $f \in C[a,b]$. Set $\alpha(x) = I_c(x)$, which is clearly increasing on $[a,b]$. Let $\mathcal{P} = \{x_0,\dots,x_n\}$ be a partition of $[a,b]$. Since $a < c \le b$, there exists a $k \in \{1,\dots,n\}$ such that $x_{k-1} < c \le x_k$. Then,

$$\Delta\alpha_k = \alpha(x_k) - \alpha(x_{k-1}) = 1 - 0 = 1 \text{ and } \Delta\alpha_i = 0,\ i = 1,\dots,n,\ i \ne k.$$

Therefore,

$$\mathcal{U}(\mathcal{P}, f, \alpha) = M_k \Delta\alpha_k = M_k = \sup\{f(x) : x_{k-1} \le x \le x_k\}$$

and

$$\mathcal{L}(\mathcal{P}, f, \alpha) = m_k \Delta\alpha_k = m_k = \inf\{f(x) : x_{k-1} \le x \le x_k\}.$$

Let $\varepsilon > 0$. Since f is continuous at c, there exists a $\delta > 0$ such that

$$f(c) - \varepsilon < f(x) < f(c) + \varepsilon$$

for all $x \in [a, b]$ with $|x - c| < \delta$. Thus, for $\|\mathcal{P}\| < \delta$,

$$f(c) - \varepsilon \le m_k \le M_k \le f(c) + \varepsilon.$$

Therefore

$$f(c) - \varepsilon \le \mathcal{L}(\mathcal{P}, f, \alpha) \le \mathcal{U}(\mathcal{P}, f, \alpha) \le f(c) + \varepsilon.$$

So

$$f(c) - \varepsilon \le \underline{\int_a^b} f \, d\alpha \le \overline{\int_a^b} f \, d\alpha \le f(c) + \varepsilon.$$

This implies $f \in \mathcal{R}(I_c)$ and

$$\int_a^b f \, d\alpha = f(c).$$

Example 9.9.7. Define $f : [a, b] \to \mathbb{R}$ by

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q}, \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Let $\alpha : [a, b] \to \mathbb{R}$ be a nonconstant increasing function, $a < b$. Let $\mathcal{P} = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. Then $m_i = 0$ and $M_i = 1$ for all $i = 1, \ldots, n$. Thus $\mathcal{L}(\mathcal{P}, f, \alpha) = 0$ and $\mathcal{U}(\mathcal{P}, f, \alpha) = \alpha(b) - \alpha(a) \ne 0$. Thus $f \notin \mathcal{R}(\alpha)$ for any nonconstant α.

Theorem 9.9.8 (Riemann's Criterion for Integrability)**.** *Let $\alpha : [a, b] \to \mathbb{R}$ be increasing and $f : [a, b] \to \mathbb{R}$ be bounded. Then $f \in \mathcal{R}(\alpha)$ if and only if for all $\varepsilon > 0$, there exists a partition $\mathcal{P}$ of $[a, b]$ such that*

$$\mathcal{U}(\mathcal{P}, f, \alpha) - \mathcal{L}(\mathcal{P}, f, \alpha) < \varepsilon.$$

Proof. The proof is similar to the proof of Theorem 9.2.1 and is therefore omitted. □

Theorem 9.9.9. *Let $f : [a, b] \to \mathbb{R}$ and $\alpha : [a, b] \to \mathbb{R}$ be increasing.*

(1) If f is continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$.

(2) If f is monotone on $[a, b]$ and α is continuous on $[a, b]$, then $f \in \mathcal{R}(\alpha)$.

Proof. (1) Let $\varepsilon > 0$. As in the proof of Theorem 9.2.2 (1), uniform continuity gives a partition $\mathcal{P}$ with $M_i - m_i \le \varepsilon$ for every i. Then

$$\mathcal{U}(\mathcal{P}, f, \alpha) - \mathcal{L}(\mathcal{P}, f, \alpha) = \sum_{i=1}^{n} (M_i - m_i) \Delta\alpha_i \le \varepsilon \sum_{i=1}^{n} \Delta\alpha_i = \varepsilon[\alpha(b) - \alpha(a)].$$

Since $\varepsilon > 0$ was arbitrary, Riemann's Criterion gives $f \in \mathcal{R}(\alpha)$.

(2) Suppose f is increasing. (If f is decreasing, apply this to $-f$.) Let $\varepsilon > 0$, and choose $n \in \mathbb{N}$ with $\frac{\alpha(b)-\alpha(a)}{n}[f(b) - f(a)] < \varepsilon$. Since α is continuous, Exercise 9.9.2 gives a partition $\mathcal{P} = \{x_0, \ldots, x_n\}$ of $[a, b]$ with $\Delta\alpha_i = \frac{\alpha(b)-\alpha(a)}{n}$ for every i. Since f is increasing, $M_i = f(x_i)$ and $m_i = f(x_{i-1})$. The sum telescopes.

$$\begin{aligned}\mathcal{U}(\mathcal{P}, f, \alpha) - \mathcal{L}(\mathcal{P}, f, \alpha) &= \frac{\alpha(b) - \alpha(a)}{n} \sum_{i=1}^{n} [f(x_i) - f(x_{i-1})] \\ &= \frac{\alpha(b) - \alpha(a)}{n}[f(b) - f(a)] < \varepsilon.\end{aligned}$$

So $f \in \mathcal{R}(\alpha)$ by Riemann's Criterion. □

Theorem 9.9.10. *Let $\alpha, \alpha_1, \alpha_2 : [a, b] \to \mathbb{R}$ be increasing.*

(1) If $f, g \in \mathcal{R}(\alpha)$, then $f + g \in \mathcal{R}(\alpha)$ with

$$\int_a^b (f + g)\, d\alpha = \int_a^b f\, d\alpha + \int_a^b g\, d\alpha.$$

(2) If $c \in \mathbb{R}$ and $f \in \mathcal{R}(\alpha)$, then $cf \in \mathcal{R}(\alpha)$ with

$$\int_a^b cf\, d\alpha = c \int_a^b f\, d\alpha.$$

(3) If $f \in \mathcal{R}(\alpha_i)$, $i = 1, 2$, then $f \in \mathcal{R}(\alpha_1 + \alpha_2)$ with

$$\int_a^b f\, d(\alpha_1 + \alpha_2) = \int_a^b f\, d\alpha_1 + \int_a^b f\, d\alpha_2.$$

(4) If $f \in \mathcal{R}(\alpha)$ on $[a, b]$ and $a < c < b$, then $f \in \mathcal{R}(\alpha)$ on $[a, c]$ and on $[c, b]$ with

$$\int_a^b f\, d\alpha = \int_a^c f\, d\alpha + \int_c^b f\, d\alpha.$$

(5) If $f, g \in \mathcal{R}(\alpha)$ with $f(x) \le g(x)$ for all $x \in [a, b]$, then

$$\int_a^b f\, d\alpha \le \int_a^b g\, d\alpha.$$

(6) If $|f(x)| \le M$ for all $x \in [a, b]$ and $f \in \mathcal{R}(\alpha)$, then $|f| \in \mathcal{R}(\alpha)$ and

$$\left| \int_a^b f\, d\alpha \right| \le \int_a^b |f|\, d\alpha \le M[\alpha(b) - \alpha(a)].$$

Proof. Parts (1), (2), (4) and (5) are proved as for the Riemann integral (Theorems 9.2.11 and 9.2.13 and Exercise 9.2.3). See Exercise 9.9.8.

(3) Let $\varepsilon > 0$. Since $f \in \mathcal{R}(\alpha_i)$, $i = 1, 2$, there exist partitions $\mathcal{P}_i$, $i = 1, 2$ such that

$$\mathcal{U}(\mathcal{P}_i, f, \alpha_i) - \mathcal{L}(\mathcal{P}_i, f, \alpha_i) < \frac{\varepsilon}{2}.$$

Let $\mathcal{P} = \mathcal{P}_1 \cup \mathcal{P}_2$. Since $\mathcal{P}$ is a refinement of $\mathcal{P}_1$ and $\mathcal{P}_2$,

$$\mathcal{U}(\mathcal{P}, f, \alpha_i) - \mathcal{L}(\mathcal{P}, f, \alpha_i) < \frac{\varepsilon}{2}.$$

Since

$$\Delta(\alpha_1+\alpha_2)_i = \Delta(\alpha_1)_i + \Delta(\alpha_2)_i$$

for all $i = 1, \dots, n$,

$$\begin{aligned}\mathcal{U}(\mathcal{P}, f, \alpha_1+\alpha_2) - \mathcal{L}(\mathcal{P}, f, \alpha_1+\alpha_2) &= \mathcal{U}(\mathcal{P}, f, \alpha_1) - \mathcal{L}(\mathcal{P}, f, \alpha_1) \\ &\quad + \mathcal{U}(\mathcal{P}, f, \alpha_2) - \mathcal{L}(\mathcal{P}, f, \alpha_2) \\ &< \varepsilon/2 + \varepsilon/2 = \varepsilon.\end{aligned}$$

Therefore, by Riemann's Criterion for Integrability, $f \in \mathcal{R}(\alpha_1+\alpha_2)$.

Furthermore, for any partition $\mathcal{P}$ of $[a, b]$,

$$\begin{aligned}\mathcal{L}(\mathcal{P}, f, \alpha_1+\alpha_2) &= \mathcal{L}(\mathcal{P}, f, \alpha_1) + \mathcal{L}(\mathcal{P}, f, \alpha_2) \\ &\le \int_a^b f\, d\alpha_1 + \int_a^b f\, d\alpha_2 \\ &\le \mathcal{U}(\mathcal{P}, f, \alpha_1) + \mathcal{U}(\mathcal{P}, f, \alpha_2) \\ &= \mathcal{U}(\mathcal{P}, f, \alpha_1+\alpha_2).\end{aligned}$$

So

$$\int_a^b f\, d(\alpha_1+\alpha_2) = \int_a^b f\, d\alpha_1 + \int_a^b f\, d\alpha_2.$$

(6) Suppose $f \in \mathcal{R}(\alpha)$ and $\mathcal{P} = \{x_0, \dots, x_n\}$ is a partition of $[a, b]$. For each $i = 1, \dots, n$, let

$$M_i = \sup\{f(x) : x \in [x_{i-1}, x_i]\}, \qquad M_i^* = \sup\{|f(x)| : x \in [x_{i-1}, x_i]\},$$
$$m_i = \inf\{f(x) : x \in [x_{i-1}, x_i]\}, \qquad m_i^* = \inf\{|f(x)| : x \in [x_{i-1}, x_i]\}.$$

If $x, y \in [x_{i-1}, x_i]$,

$$\big||f(x)| - |f(y)|\big| \le |f(x) - f(y)| \le M_i - m_i.$$

Thus

$$M_i^* - m_i^* \le M_i - m_i$$

for all $i = 1, \dots, n$. So

$$\mathcal{U}(\mathcal{P}, |f|, \alpha) - \mathcal{L}(\mathcal{P}, |f|, \alpha) \le \mathcal{U}(\mathcal{P}, f, \alpha) - \mathcal{L}(\mathcal{P}, f, \alpha).$$

Thus, by Riemann's Criterion for Integrability, $|f| \in \mathcal{R}(\alpha)$.

Choose $c \in \{\pm 1\}$ such that $c\int_a^b f\, d\alpha \ge 0$. Then

$$\begin{aligned}\left|\int_a^b f\, d\alpha\right| &= c\int_a^b f\, d\alpha \\ &= \int_a^b cf\, d\alpha \\ &\le \int_a^b |f|\, d\alpha \\ &\le M\int_a^b d\alpha \\ &= M[\alpha(b) - \alpha(a)].\end{aligned}$$

The last two steps use (5) and Lemma 9.9.2. □

Corollary 9.9.11. *Let $\alpha : [a, b] \to \mathbb{R}$ be continuous and increasing. If $f \in BV[a, b]$, then $f \in \mathcal{R}(\alpha)$.*

PROOF. By the Jordan Decomposition Theorem, $f = g - h$ where g and h are increasing functions. By Theorem 9.9.9 (2), $g, h \in \mathcal{R}(\alpha)$. So by Theorem 9.9.10 (1) and (2), $f \in \mathcal{R}(\alpha)$. □

Theorem 9.9.12 (Mean Value Theorem for Riemann–Stieltjes Integrals)**.** *Let $f \in C[a, b]$ and $\alpha : [a, b] \to \mathbb{R}$ be increasing. Then there exists a $c \in [a, b]$ such that*

$$\int_a^b f \, d\alpha = f(c)[\alpha(b) - \alpha(a)].$$

PROOF. Let m and M denote the minimum and maximum values, respectively, of f on $[a, b]$. Then by Lemma 9.9.2,

$$m[\alpha(b) - \alpha(a)] \leq \int_a^b f \, d\alpha \leq M[\alpha(b) - \alpha(a)].$$

(The integral lies between the lower and upper sums.) If $\alpha(b) - \alpha(a) = 0$, then any $c \in [a, b]$ satisfies the equation. If $\alpha(b) - \alpha(a) > 0$, then

$$m \leq \frac{1}{\alpha(b) - \alpha(a)} \int_a^b f \, d\alpha \leq M.$$

So by the Intermediate Value Theorem, there exists a $c \in [a, b]$ such that

$$f(c) = \frac{1}{\alpha(b) - \alpha(a)} \int_a^b f \, d\alpha.$$

(Apply it on the closed interval between a point where $f = m$ and a point where $f = M$. If $m = M$, f is constant.) □

Theorem 9.9.13 (Integration by Parts)**.** *Suppose $\alpha, \beta : [a, b] \to \mathbb{R}$ are increasing. Then $\alpha \in \mathcal{R}(\beta)$ if and only if $\beta \in \mathcal{R}(\alpha)$. If this is the case,*

$$\int_a^b \alpha \, d\beta = \alpha(b)\beta(b) - \alpha(a)\beta(a) - \int_a^b \beta \, d\alpha.$$

PROOF. An exercise (Exercise 9.9.3) will show that

$$\mathcal{U}(\mathcal{P}, \alpha, \beta) = \alpha(b)\beta(b) - \alpha(a)\beta(a) - \mathcal{L}(\mathcal{P}, \beta, \alpha)$$

and

$$\mathcal{L}(\mathcal{P}, \alpha, \beta) = \alpha(b)\beta(b) - \alpha(a)\beta(a) - \mathcal{U}(\mathcal{P}, \beta, \alpha).$$

Therefore,

$$\mathcal{U}(\mathcal{P}, \alpha, \beta) - \mathcal{L}(\mathcal{P}, \alpha, \beta) = \mathcal{U}(\mathcal{P}, \beta, \alpha) - \mathcal{L}(\mathcal{P}, \beta, \alpha).$$

It follows from Riemann's Criterion for Integrability that $\alpha \in \mathcal{R}(\beta)$ if and only if $\beta \in \mathcal{R}(\alpha)$.

Let $\varepsilon > 0$. If $\beta \in \mathcal{R}(\alpha)$, there exists a partition $\mathcal{P}$ of $[a, b]$ such that

$$\mathcal{L}(\mathcal{P}, \beta, \alpha) > \int_a^b \beta \, d\alpha - \varepsilon.$$

Hence

$$\int_a^b \alpha\, d\beta \le \mathcal{U}(\mathcal{P}, \alpha, \beta) < \alpha(b)\beta(b) - \alpha(a)\beta(a) - \int_a^b \beta\, d\alpha + \varepsilon.$$

Thus

$$\int_a^b \alpha\, d\beta \le \alpha(b)\beta(b) - \alpha(a)\beta(a) - \int_a^b \beta\, d\alpha.$$

Similarly, there exists a partition $\mathcal{Q}$ such that

$$\mathcal{U}(\mathcal{Q}, \beta, \alpha) < \int_a^b \beta\, d\alpha + \varepsilon.$$

So

$$\int_a^b \alpha\, d\beta \ge \mathcal{L}(\mathcal{Q}, \alpha, \beta) > \alpha(b)\beta(b) - \alpha(a)\beta(a) - \int_a^b \beta\, d\alpha - \varepsilon.$$

This implies

$$\int_a^b \alpha\, d\beta \ge \alpha(b)\beta(b) - \alpha(a)\beta(a) - \int_a^b \beta\, d\alpha.$$

Thus

$$\int_a^b \alpha\, d\beta = \alpha(b)\beta(b) - \alpha(a)\beta(a) - \int_a^b \beta\, d\alpha.$$

□

Theorem 9.9.14. *Let $f : [a, b] \to \mathbb{R}$ be bounded with only finitely many points of discontinuity on $[a, b]$, and let $\alpha : [a, b] \to \mathbb{R}$ be increasing and continuous at every point at which f is discontinuous. Then $f \in \mathcal{R}(\alpha)$.*

Proof. Let $\varepsilon > 0$ and let $M > 0$ with $|f(x)| \le M$ for all $x \in [a, b]$. Let E be the finite set of points where f is discontinuous. If $E = \emptyset$, then f is continuous and Theorem 9.9.9 (1) applies. So assume $E \neq \emptyset$. Since α is continuous at each point of E, each $p \in E$ is the center of a small closed interval on which α increases by less than $\frac{\varepsilon}{|E|}$, where $|E|$ is the number of points in E. Shrinking them if needed, these intervals are disjoint and lie in $[a, b]$, except that an interval around a or b is cut off at that endpoint. Call them $[u_1, v_1], \ldots, [u_k, v_k]$. Then $\sum_{j=1}^{k} [\alpha(v_j) - \alpha(u_j)] < \varepsilon$, and every point of E in (a, b) lies in some open interval (u_j, v_j).

Let K be what is left of $[a, b]$ after removing the open intervals (u_j, v_j). Then K is closed and bounded, so it is compact. The function f is continuous at every point of K, relative to K. By Theorem 6.4.7, f is uniformly continuous on K. So there exists a $\delta > 0$ such that $|f(x) - f(y)| < \varepsilon$ for all $x, y \in K$ with $|x - y| < \delta$.

Now build a partition $\mathcal{P}$ of $[a, b]$ that contains every u_j and every v_j, has no points inside any (u_j, v_j), and cuts the rest of K into subintervals of length less than δ. There are two kinds of subintervals. The subintervals $[u_j, v_j]$ have $M_i - m_i \le 2M$ and total $\Delta\alpha_i$ less than ε. Every other subinterval lies in K and has length less than δ, so $M_i - m_i \le \varepsilon$ on it. Therefore

$$\mathcal{U}(\mathcal{P}, f, \alpha) - \mathcal{L}(\mathcal{P}, f, \alpha) \le 2M\varepsilon + \varepsilon[\alpha(b) - \alpha(a)].$$

Since $\varepsilon > 0$ was arbitrary, $f \in \mathcal{R}(\alpha)$ by Riemann's Criterion. □

Theorem 9.9.15. *Let $\{s_1, \ldots, s_n\}$ be a finite subset of $(a, b]$ with $s_1 < \cdots < s_n$, $\{c_1, \ldots, c_n\}$ a finite collection of nonnegative real numbers, and $I(x - c)$ the unit jump function at $c \in \mathbb{R}$. Let $f \in C[a, b]$ and let α be the increasing function defined by $\alpha(x) = \sum_{i=1}^{n} c_i I(x - s_i)$. Then $f \in \mathcal{R}(\alpha)$ and*

$$\int_a^b f \, d\alpha = \sum_{i=1}^{n} c_i f(s_i).$$

PROOF. See Exercise 9.9.4. □

Theorem 9.9.16. *Let $\{s_n\}$ be a countable subset of $(a, b]$, $\{c_n\}$ a sequence of nonnegative real numbers for which $\sum_{n=1}^{\infty} c_n$ converges, and $I(x - c)$ the unit jump function at $c \in \mathbb{R}$. Let $f \in C[a, b]$ and let α be the increasing function defined by $\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n)$. Then $f \in \mathcal{R}(\alpha)$ and*

$$\int_a^b f \, d\alpha = \sum_{n=1}^{\infty} c_n f(s_n).$$

(This theorem uses convergent series from Chapter 10.)

PROOF. Since f is continuous, $f \in \mathcal{R}(\alpha)$. Let $\varepsilon > 0$. Choose an $N \in \mathbb{N}$ such that $\sum_{n=N+1}^{\infty} c_n < \varepsilon$. Define

$$\beta_1(x) = \sum_{n=1}^{N} c_n I(x - s_n)$$

and

$$\beta_2(x) = \sum_{n=N+1}^{\infty} c_n I(x - s_n).$$

Then

$$\int_a^b f \, d\beta_1 = \sum_{n=1}^{N} c_n f(s_n).$$

Also, $\alpha = \beta_1 + \beta_2$.

Let $M = \max_{x \in [a,b]} |f(x)|$. Then by Theorem 9.9.10 (3) and (6),

$$\begin{aligned} \left| \int_a^b f \, d\alpha - \sum_{n=1}^{N} c_n f(s_n) \right| &= \left| \int_a^b f \, d\beta_2 \right| \\ &\leq M[\beta_2(b) - \beta_2(a)] \\ &\leq M \sum_{n=N+1}^{\infty} c_n \\ &< M\varepsilon. \end{aligned}$$

Thus

$$\lim_{N \to \infty} \sum_{n=1}^{N} c_n f(s_n) = \int_a^b f \, d\alpha.$$

□

Theorem 9.9.17. *Suppose $f \in \mathcal{R}[a,b]$ and $\alpha : [a,b] \to \mathbb{R}$ is an increasing differentiable function with $\alpha' \in \mathcal{R}[a,b]$. Then $f \in \mathcal{R}(\alpha)$ and*

$$\int_a^b f\,d\alpha = \int_a^b f\alpha'.$$

Proof. Since $f, \alpha' \in \mathcal{R}[a,b]$, $f\alpha' \in \mathcal{R}[a,b]$. Let $M > 0$ with $|f(x)| \le M$ for all $x \in [a,b]$, and let $\varepsilon > 0$. By Riemann's Criterion, there is a partition $\mathcal{P}$ with $\mathcal{U}(\mathcal{P},\alpha') - \mathcal{L}(\mathcal{P},\alpha') < \varepsilon$, and this stays true for every refinement of $\mathcal{P}$.

Let $\mathcal{Q} = \{x_0, \ldots, x_n\}$ be any refinement of $\mathcal{P}$. By the Mean Value Theorem, $\Delta\alpha_i = \alpha'(t_i)\Delta x_i$ for some $t_i \in (x_{i-1}, x_i)$. Let $s_i \in [x_{i-1}, x_i]$ be arbitrary. Both $\alpha'(s_i)$ and $\alpha'(t_i)$ lie between the infimum and the supremum of α' on $[x_{i-1}, x_i]$. So

$$\begin{aligned}\left|\sum_{i=1}^n f(s_i)\Delta\alpha_i - \sum_{i=1}^n f(s_i)\alpha'(s_i)\Delta x_i\right| &\le M\sum_{i=1}^n |\alpha'(t_i) - \alpha'(s_i)|\Delta x_i \\ &\le M\big(\mathcal{U}(\mathcal{Q},\alpha') - \mathcal{L}(\mathcal{Q},\alpha')\big) < M\varepsilon.\end{aligned}$$

Since $\sum_{i=1}^n f(s_i)\alpha'(s_i)\Delta x_i \le \mathcal{U}(\mathcal{Q}, f\alpha')$, this gives $\sum_{i=1}^n f(s_i)\Delta\alpha_i < \mathcal{U}(\mathcal{Q}, f\alpha') + M\varepsilon$ for every choice of the s_i. The supremum of the left side over all choices of the s_i is $\mathcal{U}(\mathcal{Q}, f, \alpha)$, since each $\Delta\alpha_i \ge 0$. So $\mathcal{U}(\mathcal{Q}, f, \alpha) \le \mathcal{U}(\mathcal{Q}, f\alpha') + M\varepsilon$. In the same way, $\mathcal{U}(\mathcal{Q}, f\alpha') \le \mathcal{U}(\mathcal{Q}, f, \alpha) + M\varepsilon$.

Now choose the refinement $\mathcal{Q}$ so that also $\mathcal{U}(\mathcal{Q}, f\alpha') < \int_a^b f\alpha' + \varepsilon$. Then

$$\overline{\int_a^b} f\,d\alpha \le \mathcal{U}(\mathcal{Q}, f, \alpha) < \int_a^b f\alpha' + (M+1)\varepsilon.$$

Choosing $\mathcal{Q}$ instead so that $\mathcal{U}(\mathcal{Q}, f, \alpha) < \overline{\int_a^b} f\,d\alpha + \varepsilon$ gives

$$\int_a^b f\alpha' \le \mathcal{U}(\mathcal{Q}, f\alpha') < \overline{\int_a^b} f\,d\alpha + (M+1)\varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $\overline{\int_a^b} f\,d\alpha = \int_a^b f\alpha'$. The same argument with lower sums gives $\underline{\int_a^b} f\,d\alpha = \int_a^b f\alpha'$. So $f \in \mathcal{R}(\alpha)$ and $\int_a^b f\,d\alpha = \int_a^b f\alpha'$. □

Example 9.9.18. Consider $\int_0^2 e^x\,d\lfloor x\rfloor$. For $x \in [0,2]$, $\lfloor x\rfloor = I(x-1) + I(x-2)$. Thus, since e^x is continuous, $\int_0^2 e^x\,d\lfloor x\rfloor = e^1 + e^2$ by Theorem 9.9.15.

Example 9.9.19. Consider $\int_0^1 \sin\pi x\,d(x^2)$. Since x^2 is increasing on $[0,1]$ and its derivative is continuous, Theorem 9.9.17 gives

$$\begin{aligned}\int_0^1 \sin\pi x\,d(x^2) &= 2\int_0^1 x\sin\pi x\,dx \\ &= \left[\frac{-1}{\pi}x\cos\pi x + \frac{1}{\pi^2}\sin\pi x\right]\Bigg|_0^1 \\ &= \frac{1}{\pi}.\end{aligned}$$

Definition 9.9.20. Let $f : [a,b] \to \mathbb{R}$ be bounded and let $\alpha \in BV[a,b]$. Then f is Riemann–Stieltjes integrable with respect to α on $[a,b]$ if $f \in \mathcal{R}(V_\alpha)$ and $f \in \mathcal{R}(V_\alpha - \alpha)$. If this is the case, write $f \in \mathcal{R}(\alpha)$ and define

$$\int_a^b f\,d\alpha = \int_a^b f\,dV_\alpha - \int_a^b f\,d(V_\alpha - \alpha).$$

This works because $\alpha = V_\alpha - (V_\alpha - \alpha)$, and both V_α and $V_\alpha - \alpha$ are increasing by Theorems 8.2.2 and 8.2.4. If α is already increasing, then $V_\alpha = \alpha - \alpha(a)$ by Theorems 8.1.8 and 8.2.1, so $V_\alpha - \alpha$ is constant, and the definition agrees with Definition 9.9.5.

Theorem 9.9.21. *Let $\alpha \in C^{(1)}[a,b]$.*

(1) If $f \in C[a,b]$, then $f \in \mathcal{R}(\alpha)$.
(2) If $f \in BV[a,b]$, then $f \in \mathcal{R}(\alpha)$.

Proof. Since $\alpha \in C^{(1)}[a,b]$, by Corollary 8.1.10, $\alpha \in BV[a,b]$. Furthermore, since α is continuous on $[a,b]$, by Theorem 8.2.3, V_α is continuous on $[a,b]$. So V_α and $V_\alpha - \alpha$ are continuous, increasing functions on $[a,b]$.

(1) Since $f \in C[a,b]$, then by Theorem 9.9.9 (1), $f \in \mathcal{R}(V_\alpha)$ and $f \in \mathcal{R}(V_\alpha - \alpha)$. So $f \in \mathcal{R}(\alpha)$.
(2) Since $f \in BV[a,b]$, then by Corollary 9.9.11, $f \in \mathcal{R}(V_\alpha)$ and $f \in \mathcal{R}(V_\alpha - \alpha)$. So $f \in \mathcal{R}(\alpha)$. □

Theorem 9.9.22. *Let $\alpha \in C^{(1)}[a,b]$ and $f \in \mathcal{R}[a,b]$. Then $f \in \mathcal{R}(\alpha)$ and*

$$\int_a^b f\,d\alpha = \int_a^b f\alpha'.$$

Proof. See Exercise 9.9.7. □

Exercises

9.9.1. Let $f \in C[a,b]$. Fix $c \in (a,b]$. Show that if

$$\alpha(x) = \begin{cases} c_1, & a \le x < c, \\ c_2, & c \le x \le b, \end{cases}$$

where $c_1 < c_2$, then $\int_a^b f\,d\alpha = f(c)(c_2 - c_1)$. [Hint, p. 351]

9.9.2. Let $\alpha : [a,b] \to \mathbb{R}$ be increasing and continuous. Prove that for any $n \in \mathbb{N}$, there exists a partition $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ such that

$$\Delta\alpha_i = \alpha(x_i) - \alpha(x_{i-1}) = \frac{1}{n}[\alpha(b) - \alpha(a)].$$

[Hint, p. 351]

9.9.3. Suppose $\alpha, \beta : [a,b] \to \mathbb{R}$ are increasing and $\mathcal{P}$ is a partition of $[a,b]$. Prove that $\mathcal{U}(\mathcal{P}, \alpha, \beta) = \alpha(b)\beta(b) - \alpha(a)\beta(a) - \mathcal{L}(\mathcal{P}, \beta, \alpha)$. [Hint, p. 352]

9.9.4. Prove Theorem 9.9.15. [Hint, p. 352]

9.9.5. Compute $\int_1^5 x^2\,d\lfloor x^2 \rfloor$, where $\lfloor \cdot \rfloor$ is the greatest integer function. [Hint, p. 352]

9.9.6. Compute $\int_0^{\pi/2} \sin x \, d(\sin x)$. [Hint, p. 352]

9.9.7. Prove Theorem 9.9.22. [Hint, p. 352]

9.9.8. Prove Theorem 9.9.10 (1), (2), (4) and (5).

Notes

Bernhard Riemann defined his integral in 1854, in a paper published in 1868 [**59**]. Gaston Darboux gave the definition by upper and lower sums in 1875 [**19**]. Cauchy gave the first careful proof of the Fundamental Theorem of Calculus, for continuous functions, in 1823 [**16**]. Thomas Stieltjes introduced his integral in 1894, in a long paper on continued fractions [**69**]. Henri Lebesgue gave Theorem 9.2.9 in his 1904 lectures [**43**].

CHAPTER 10

Series of Real Numbers

An infinite series is a sum with infinitely many terms. Such a sum has no meaning until one is given, and the meaning is a limit. Add the first n terms, then let $n \to \infty$. So this chapter is really about a special kind of sequence, and the tools of Chapter 3 do most of the work.

Section 10.1 gives the definition and the two kinds of series whose sums are easy to find. It also shows that the harmonic series diverges. Section 10.2 proves the standard convergence tests. They decide whether a series converges without finding its sum. Section 10.3 separates series that converge because their terms are small from series that converge because their terms cancel. The difference matters. Reordering the terms of a series of the second kind can change its sum to any real number at all. The starred Section 10.4 shows how to multiply two series. The starred Section 10.5 uses the harmonic series to prove that there are infinitely many primes.

10.1. Types of Series

Definition 10.1.1. Let $\{a_n\}_{n=1}^{\infty}$ be a sequence of real numbers. Let $\{s_n\}_{n=1}^{\infty}$ be the sequence obtained from $\{a_n\}$, where for each $n \in \mathbb{N}$, $s_n = \sum_{k=1}^{n} a_k$. The sequence $\{s_n\}$ is called an *infinite series* and is denoted $\sum_{k=1}^{\infty} a_k$, $\sum a_k$, or $a_1 + a_2 + \cdots$. For each $n \in \mathbb{N}$, s_n is called the *nth partial sum* of the series and a_n is called the *nth term* of the series.

The series $\sum a_k$ *converges* if the sequence $\{s_n\}$ converges. If $s_n \to s$, then s is called the *sum* of the series, written $\sum a_k = s$. If the sequence $\{s_n\}$ diverges, the series $\sum a_k$ is said to *diverge*.

Remark 10.1.2. The symbol $\sum a_k$ stands for two things, the sequence of partial sums and, when it converges, its sum. The context always makes clear which one is meant. A series can also start at an index other than 1. For example, $\sum_{k=0}^{\infty} a_k$ has partial sums $s_n = \sum_{k=0}^{n} a_k$ for $n \geq 0$. Everything in this chapter holds for such series, with the obvious changes.

Example 10.1.3 (Geometric Series)**.** Consider, for $|r| < 1$, $\sum_{k=1}^{\infty} r^k$. Then for $n \in \mathbb{N}$,

$$s_n = r + r^2 + r^3 + \cdots + r^n.$$

Thus

$$(1 - r)s_n = s_n - rs_n = r - r^{n+1}.$$

Therefore,

$$s_n = \frac{r - r^{n+1}}{1 - r}.$$

But since $|r| < 1$, $r^n \to 0$ by Example 3.1.7. Thus

$$\lim_{n\to\infty} s_n = \frac{r}{1-r}.$$

Hence

$$\sum_{k=1}^{\infty} r^k = \frac{r}{1-r}.$$

Adding the term $r^0 = 1$ gives $\sum_{k=0}^{\infty} r^k = 1 + \frac{r}{1-r} = \frac{1}{1-r}$. If $|r| \geq 1$, the geometric series diverges, by Example 10.2.3.

Example 10.1.4 (Telescoping Series)**.** Consider $\sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1}\right)$. Then

$$s_n = \left(1 - \frac{1}{2}\right) + \left(\frac{1}{2} - \frac{1}{3}\right) + \cdots + \left(\frac{1}{n} - \frac{1}{n+1}\right) = 1 - \frac{1}{n+1}.$$

Thus $s_n \to 1$ and

$$\sum_{k=1}^{\infty} \left(\frac{1}{k} - \frac{1}{k+1}\right) = 1.$$

The partial sums have a simple formula because each term has the form $b_k - b_{k+1}$, here with $b_k = \frac{1}{k}$. A series of this form is called *telescoping*. Its partial sums are $s_n = b_1 - b_{n+1}$. So it converges exactly when $\{b_n\}$ converges, and then its sum is $b_1 - \lim_{n\to\infty} b_n$.

Example 10.1.5 (The Harmonic Series)**.** The *harmonic series* is $\sum_{k=1}^{\infty} \frac{1}{k}$. Its terms tend to 0, but it diverges. The idea is to group the terms in blocks whose lengths double. Each block adds at least $\frac{1}{2}$. For example,

$$\begin{aligned} s_8 &= 1 + \frac{1}{2} + \left(\frac{1}{3} + \frac{1}{4}\right) + \left(\frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8}\right) \\ &\geq 1 + \frac{1}{2} + \left(\frac{1}{4} + \frac{1}{4}\right) + \left(\frac{1}{8} + \frac{1}{8} + \frac{1}{8} + \frac{1}{8}\right) = 1 + \frac{3}{2}. \end{aligned}$$

In general, the claim is that

$$s_{2^n} \geq 1 + \frac{n}{2}$$

for all $n \in \mathbb{N}$. For $j \in \mathbb{N}$, the block of terms from $k = 2^{j-1} + 1$ to $k = 2^j$ has 2^{j-1} terms, and each is at least $\frac{1}{2^j}$. So

$$\sum_{k=2^{j-1}+1}^{2^j} \frac{1}{k} \geq 2^{j-1} \cdot \frac{1}{2^j} = \frac{1}{2}.$$

The partial sum s_{2^n} is the first term 1 plus the blocks for $j = 1, \ldots, n$. So $s_{2^n} \geq 1 + \frac{n}{2}$. For $n \geq 2$ the inequality is strict, since $\frac{1}{3} > \frac{1}{4}$.

So the partial sums are not bounded. Indeed, for each $M > 0$, Theorem 1.8.8 gives an $n \in \mathbb{N}$ with $n > 2M$, and then $s_{2^n} > M$. A convergent sequence is bounded, by Theorem 3.2.3. So $\{s_n\}$ diverges, and the harmonic series diverges.

Exercises

10.1.1. Let s_n be the nth partial sum of the harmonic series $\sum \frac{1}{k}$. Example 10.1.5 shows that $s_{2^n} \geq 1 + \frac{n}{2}$. Prove that $s_{2^n} \leq 1 + n$ for all $n \in \mathbb{N}$. So the partial sums grow without bound, but slowly.

10.1.2. Find the sum of each series.

(a) $\displaystyle\sum_{k=2}^{\infty} \left(\frac{2}{3}\right)^k$

(b) $\displaystyle\sum_{k=0}^{\infty} \frac{5(-1)^k}{3^k}$

(c) $\displaystyle\sum_{k=1}^{\infty} \frac{2^k + 3^k}{6^k}$

[Solution, p. 353]

10.1.3. The repeating decimal $0.272727\ldots$ means $\sum_{k=1}^{\infty} \frac{27}{100^k}$. Write it as a fraction. More generally, let $m \in \mathbb{N}$ and let D be an integer with $0 \leq D < 10^m$. Prove that $\sum_{k=1}^{\infty} \frac{D}{10^{mk}} = \frac{D}{10^m - 1}$. This is the repeating decimal whose repeating block is the m digits of D, so every such decimal is a rational number.

[Solution, p. 353]

10.1.4. A ball is dropped from a height of 10 feet. Each time it hits the ground, it bounces back up to $\frac{3}{4}$ of the height it just fell. Find the total distance the ball travels.

10.1.5. Prove that $\sum_{k=1}^{\infty} \frac{1}{k(k+2)} = \frac{3}{4}$. Start by writing $\frac{1}{k(k+2)} = \frac{1}{2}\left(\frac{1}{k} - \frac{1}{k+2}\right)$.

10.2. Convergence Tests

The geometric and telescoping series are rare. For most series there is no formula for s_n. A *convergence test* decides whether a series converges without finding its sum. This section proves the standard tests. All of them rest on the Cauchy Criterion or on the Monotone Convergence Theorem, applied to the partial sums.

Theorem 10.2.1 (Cauchy Criterion)**.** *The series $\sum a_k$ converges if and only if for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $m > n \geq N$,*

$$\left| \sum_{k=n+1}^{m} a_k \right| < \varepsilon.$$

Proof. ($\Rightarrow$) Since the series converges, the sequence of partial sums $\{s_n\}$ converges. Thus, by Theorem 3.5.2, the sequence of partial sums is Cauchy. Let $\varepsilon > 0$. So there exists an $N \in \mathbb{N}$ such that for all $m > n \geq N$, $|s_m - s_n| < \varepsilon$. But

$$\left| \sum_{k=n+1}^{m} a_k \right| = |s_m - s_n|.$$

So for all $m > n \geq N$,

$$\left| \sum_{k=n+1}^{m} a_k \right| < \varepsilon.$$

($\Leftarrow$) Let $\varepsilon > 0$. Then there exists an $N \in \mathbb{N}$ such that for all $m > n \geq N$,

$$\left| \sum_{k=n+1}^{m} a_k \right| = |s_m - s_n| < \varepsilon.$$

The same bound holds for $n > m \geq N$ by symmetry, and it is trivial for $m = n$. So $\{s_n\}$ is Cauchy and therefore, by Theorem 3.5.2, $\{s_n\}$ converges. □

Proof Technique (Working with Partial Sums). A series is a sequence in disguise. It is the sequence $\{s_n\}$ of its partial sums. So every question about a series is a question about a sequence, and every theorem of Chapter 3 applies. The Cauchy Criterion for series is the Cauchy Criterion for $\{s_n\}$, rewritten using $s_m - s_n = \sum_{k=n+1}^{m} a_k$. When the terms are nonnegative, $\{s_n\}$ is increasing, and the Monotone Convergence Theorem applies instead (Theorem 10.2.6). Before trying anything clever with a series, write down s_n and ask what is known about it.

Corollary 10.2.2 (nth Term Test)**.** *If $\sum a_k$ converges, then $a_k \to 0$. If $a_k \not\to 0$, $\sum a_k$ diverges.*

Proof. See Exercise 10.2.1. □

The converse is false. The terms of the harmonic series $\sum \frac{1}{k}$ tend to 0, but the series diverges, by Example 10.1.5. Example 10.2.12 gives a second proof.

Example 10.2.3. If $|r| \geq 1$, then $|r^k| = |r|^k \geq 1$ for all $k \in \mathbb{N}$. So $r^k \not\to 0$, and $\sum r^k$ diverges by the nth Term Test.

Theorem 10.2.4. *If $\sum a_k = \alpha$ and $\sum b_k = \beta$, then*

(1) $\sum ca_k = c\alpha$ for all $c \in \mathbb{R}$; and
(2) $\sum (a_k + b_k) = \alpha + \beta$.

Proof. (1) This part is an exercise. See Exercise 10.2.2.

(2) For each $n \in \mathbb{N}$, let $s_n = \sum_{k=1}^{n} a_k$ and $t_n = \sum_{k=1}^{n} b_k$. So $s_n \to \alpha$ and $t_n \to \beta$. So $s_n + t_n \to \alpha + \beta$ by Theorem 3.2.4, implying $\sum (a_k + b_k) = \alpha + \beta$. Here $s_n + t_n = \sum_{k=1}^{n} (a_k + b_k)$ is the nth partial sum of $\sum (a_k + b_k)$. □

Remark 10.2.5. Changing finitely many terms of a series does not change whether it converges. Suppose $a_k = b_k$ for all $k > K$. Then for $m > n \geq K$, $\sum_{k=n+1}^{m} a_k = \sum_{k=n+1}^{m} b_k$. So the condition in the Cauchy Criterion holds for $\sum a_k$ exactly when it holds for $\sum b_k$. The sums can differ, of course. In the same way, deleting the first K terms of a series does not change whether it converges. So a hypothesis such as "$a_k \geq 0$" in the tests below only needs to hold for all $k \geq K$, for some $K \in \mathbb{N}$.

Theorem 10.2.6. *Suppose $a_k \geq 0$ for all $k \in \mathbb{N}$. Then $\sum a_k$ converges if and only if its partial sums are bounded. If the series converges, then $s_n \leq \sum_{k=1}^{\infty} a_k$ for all $n \in \mathbb{N}$. If it diverges, then $s_n \to \infty$.*

Proof. Since $s_{n+1} - s_n = a_{n+1} \geq 0$, the sequence $\{s_n\}$ is increasing. By the Monotone Convergence Theorem (Theorem 3.2.13), it converges if and only

if it is bounded. If it converges to s, then $s = \sup\{s_n : n \in \mathbb{N}\}$, by the proof of Theorem 3.2.13. So $s_n \leq s$ for all n. If it diverges, it is not bounded above. So for each $M > 0$ there is an $N \in \mathbb{N}$ with $s_N > M$. Since $\{s_n\}$ is increasing, $s_n > M$ for all $n \geq N$. So $s_n \to \infty$. □

Theorem 10.2.7 (Comparison Test)**.** *Suppose $\sum a_k$ and $\sum b_k$ are two given series of nonnegative real numbers satisfying $0 \leq a_k \leq Mb_k$ for some positive constant M and all integers $k \geq K$ for some fixed $K \in \mathbb{N}$.*

(1) If $\sum b_k$ converges, then $\sum a_k$ converges.
(2) If $\sum a_k$ diverges, then $\sum b_k$ diverges.

PROOF. (1) For all $m > n \geq K$,

$$0 \leq \sum_{k=n+1}^{m} a_k \leq M \sum_{k=n+1}^{m} b_k.$$

Then by the Cauchy Criterion, if $\sum b_k$ converges, $\sum a_k$ converges. In detail, let $\varepsilon > 0$. Since $\sum b_k$ converges, there is an $N \geq K$ such that $\sum_{k=n+1}^{m} b_k < \frac{\varepsilon}{M}$ for all $m > n \geq N$. Then $\sum_{k=n+1}^{m} a_k < \varepsilon$ for all $m > n \geq N$.

(2) This statement is the contrapositive of (1). □

Corollary 10.2.8 (Limit Comparison Test)**.** *Suppose $\sum a_k$ and $\sum b_k$ are two given series of positive real numbers.*

(1) If $\lim_{k\to\infty} \frac{a_k}{b_k} = L$ with $0 < L < \infty$, then $\sum a_k$ converges if and only if $\sum b_k$ converges.
(2) If $\lim_{k\to\infty} \frac{a_k}{b_k} = 0$ and $\sum b_k$ converges, then $\sum a_k$ converges.

PROOF. See Exercise 10.2.3. □

Example 10.2.9. Consider $\sum_{k=1}^{\infty} \frac{k}{3^k}$. Notice $\lim_{k\to\infty} k\left(\frac{2}{3}\right)^k = 0$. To see this, use the Binomial Theorem (Exercise 9.3.3). For $k \geq 2$,

$$\left(\frac{3}{2}\right)^k = \left(1 + \frac{1}{2}\right)^k \geq \binom{k}{2}\frac{1}{4} = \frac{k(k-1)}{8}.$$

So $0 < k\left(\frac{2}{3}\right)^k \leq \frac{8}{k-1}$, and the Squeeze Theorem applies. So there exists a $K \in \mathbb{N}$ such that $k\left(\frac{2}{3}\right)^k < 1$ for all $k \geq K$. Thus

$$\frac{k}{3^k} = k\left(\frac{2}{3}\right)^k \frac{1}{2^k} \leq \frac{1}{2^k}$$

for all $k \geq K$. Thus, by the Comparison Test, since $\sum_{k=1}^{\infty} \frac{1}{2^k}$ converges, $\sum_{k=1}^{\infty} \frac{k}{3^k}$ converges.

Theorem 10.2.10 (Integral Test)**.** *Let $\{a_k\}$ be a decreasing sequence of nonnegative real numbers, and let $f : [1, \infty) \to \mathbb{R}$ be nonnegative and decreasing with $f(k) = a_k$ for all $k \in \mathbb{N}$. Then $\sum a_k$ converges if and only if $\int_1^{\infty} f$ converges.*

PROOF. Since f is monotone on $[1, \infty)$, then by Theorem 9.2.2, $f \in \mathcal{R}_{\text{loc}}[1, \infty)$. Let $n \in \mathbb{N}$, $n \geq 2$, and consider the partition $\mathcal{P} = \{1, 2, \ldots, n\}$ of $[1, n]$. Since f is

decreasing, for each $k = 2, 3, \ldots, n$, $M_k = a_{k-1}$ and $m_k = a_k$. Here M_k and m_k are the supremum and infimum of f on $[k-1, k]$, and $\Delta x_k = 1$. Thus

$$\begin{aligned}\sum_{k=2}^{n} a_k &= \mathcal{L}(\mathcal{P}, f) \\ &\leq \int_1^n f \\ &\leq \mathcal{U}(\mathcal{P}, f) \\ &= \sum_{k=1}^{n-1} a_k.\end{aligned}$$

See Figure 10.1.

Let $F(t) = \int_1^t f$ for $t \geq 1$. Since $f \geq 0$, F is increasing, by Theorem 9.2.13 and Exercise 9.2.3.

Suppose $\sum a_k$ converges, and let s be its sum. Let $t \geq 1$, and choose an integer $n \geq 2$ with $n \geq t$. Then $F(t) \leq F(n) \leq \sum_{k=1}^{n-1} a_k \leq s$, by Theorem 10.2.6. So F is increasing and bounded above on $[1, \infty)$. As in the proof of Theorem 9.7.8, $\lim_{t\to\infty} F(t)$ exists. So $\int_1^\infty f$ converges.

Suppose $\int_1^\infty f$ converges, and let I be its value. Since F is increasing, $F(n) \leq F(t)$ for all $t \geq n$, and letting $t \to \infty$ gives $F(n) \leq I$. So for all $n \geq 2$,

$$s_n = a_1 + \sum_{k=2}^{n} a_k \leq a_1 + F(n) \leq a_1 + I.$$

The partial sums are bounded, so $\sum a_k$ converges by Theorem 10.2.6. □

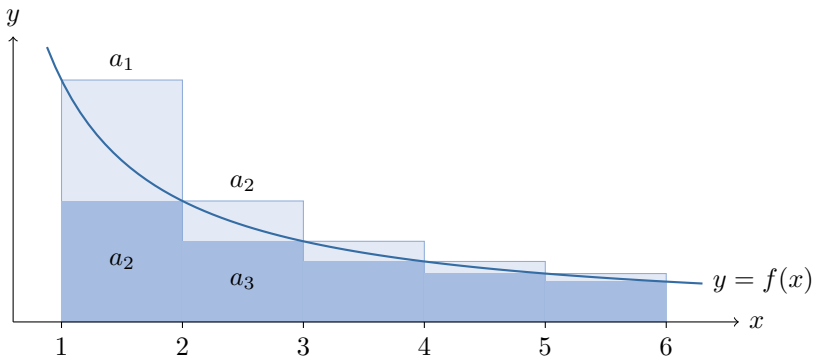


FIGURE 10.1. The Integral Test with $n = 6$. On each interval $[k-1, k]$, the light rectangle has height a_{k-1} and the dark rectangle has height a_k. So $\sum_{k=2}^{n} a_k \leq \int_1^n f \leq \sum_{k=1}^{n-1} a_k$.

Example 10.2.11 (Euler's Constant)**.** The harmonic series diverges, but its partial sums stay close to $\ln n$. Let $\gamma_n = \sum_{k=1}^{n} \frac{1}{k} - \ln n$. Then $\{\gamma_n\}$ converges. Its limit $\gamma \approx 0.5772$ is called *Euler's constant*.

The proof compares sums with integrals, as in Theorem 10.2.10. Since $\ln n = \int_1^n \frac{dt}{t}$,

$$\gamma_n = \sum_{k=1}^{n-1} \int_k^{k+1} \left(\frac{1}{k} - \frac{1}{t}\right) dt + \frac{1}{n}.$$

Each integrand is at least 0 on $[k, k+1]$, so $\gamma_n > 0$. Also

$$\gamma_n - \gamma_{n+1} = \ln(n+1) - \ln n - \frac{1}{n+1} = \int_n^{n+1} \left(\frac{1}{t} - \frac{1}{n+1}\right) dt \geq 0.$$

So $\{\gamma_n\}$ is decreasing and bounded below by 0. It converges by the Monotone Convergence Theorem. Nobody knows whether γ is rational.

Example 10.2.12 (p-series)**.** Consider the p-series $\sum_{k=1}^{\infty} \frac{1}{k^p}$. If $p \leq 0$, then $\frac{1}{k^p} \geq 1$ for all $k \in \mathbb{N}$, so the series diverges by the nth Term Test. If $p > 0$, then $f(x) = \frac{1}{x^p}$ is nonnegative and decreasing on $[1, \infty)$. By the p-Test (Theorem 9.7.7), $\int_1^{\infty} \frac{1}{x^p}\, dx$ converges if and only if $p > 1$. So by the Integral Test, $\sum_{k=1}^{\infty} \frac{1}{k^p}$ converges if and only if $p > 1$. The case $p = 1$ shows that the harmonic series $\sum \frac{1}{k}$ diverges.

Theorem 10.2.13 (Ratio Test)**.** *Let $\sum a_k$ be a series of positive terms and let*

$$R = \limsup \frac{a_{k+1}}{a_k} \quad \text{and} \quad r = \liminf \frac{a_{k+1}}{a_k}.$$

(1) If $R < 1$, then $\sum a_k$ converges.
(2) If $r > 1$, then $\sum a_k$ diverges.
(3) If $r \leq 1 \leq R$, the test is inconclusive.

Proof. (1) Choose $c \in (R, 1)$. Then there exists a $K \in \mathbb{N}$ such that for all $k \geq K$, $\frac{a_{k+1}}{a_k} \leq c$ by Theorem 3.6.11 with $\varepsilon = c - R$. In particular, $a_{K+1} \leq ca_K$. Also, $a_{K+2} \leq ca_{K+1} \leq c^2 a_K$. By induction, $a_{K+m} \leq c^m a_K$ for all $m \geq 0$. Therefore, $a_k \leq Mc^k$ for all $k \geq K$, where $k = K + m$ and $M = \frac{a_K}{c^K}$. Thus, since $0 < c < 1$, $\sum c^k$ converges, and hence, by the Comparison Test, $\sum a_k$ converges.

(2) Now choose $c \in (1, r)$. Then, using a similar argument to above, there exists a $K \in \mathbb{N}$ such that $a_k \geq Mc^k$ for some constant M and for all $k \geq K$. If $r < \infty$, the K comes from Theorem 3.6.11 with $\varepsilon = r - c$. If $r = \infty$, it comes from Theorem 3.6.12. The constant is $M = \frac{a_K}{c^K} > 0$. Since $c > 1$, $\sum c^k$ diverges, and thus by the Comparison Test, $\sum a_k$ diverges.

(3) Notice for $a_k = \frac{1}{k}$, $r = R = 1$, but $\sum \frac{1}{k}$ diverges. Also, for $a_k = \frac{1}{k^2}$, $r = R = 1$, but $\sum \frac{1}{k^2}$ converges. Thus the test is inconclusive if $r \leq 1 \leq R$. Both facts about these series come from Example 10.2.12. □

Often the ratios $\frac{a_{k+1}}{a_k}$ have a limit L. Then $R = r = L$ by Corollary 3.6.14. So the series converges if $L < 1$ and diverges if $L > 1$. The test says nothing if $L = 1$.

Theorem 10.2.14 (Root Test)**.** *Let $\sum a_k$ be a series of nonnegative terms, and let*

$$\alpha = \limsup \sqrt[k]{a_k}.$$

(Here $\sqrt[k]{0} = 0$.)

(1) If $\alpha < 1$, $\sum a_k$ converges.
(2) If $\alpha > 1$, $\sum a_k$ diverges.
(3) If $\alpha = 1$, the test is inconclusive.

PROOF. (1) Choose $c \in (\alpha, 1)$. Then there exists a $K \in \mathbb{N}$ such that $\sqrt[k]{a_k} < c$ for all $k \geq K$ by Theorem 3.6.11 with $\varepsilon = c - \alpha$. But then $a_k \leq c^k$ for all $k \geq K$. Thus $\sum a_k$ converges by the Comparison Test and Example 10.1.3.

(2) Notice $\sqrt[k]{a_k} > 1$ for infinitely many k. If $\alpha < \infty$, this follows from Theorem 3.6.11 with $\varepsilon = \alpha - 1$. If $\alpha = \infty$, it holds because ∞ is a cluster point of $\{\sqrt[k]{a_k}\}$, by Theorem 3.6.12. So $a_k > 1$ for infinitely many k. Thus $a_k \not\to 0$. So $\sum a_k$ diverges by the nth Term Test.

(3) This part is an exercise. See Exercise 10.2.4. □

Proof Technique (Choosing a Number in Between). The Ratio and Root Tests compare a series with a geometric series. The hypothesis $R < 1$ does not say $\frac{a_{k+1}}{a_k} \leq R$ for large k. It says the ratios are eventually below every number bigger than R. So the proof chooses a number c strictly between R and 1. Then $\frac{a_{k+1}}{a_k} \leq c$ for all large k, and $c < 1$ makes $\sum c^k$ converge. Choosing $c = R$ fails, because the ratios may stay above R forever. Choosing $c = 1$ fails too, because $\sum 1$ diverges. Whenever a strict inequality like $R < 1$ has to be used, try putting a number strictly between the two sides. Theorem 3.6.11 then turns $R < c$ into a statement about the terms.

Theorem 10.2.15 (Abel Partial Summation Formula)**.** *Let $\{a_k\}$ and $\{b_k\}$ be sequences of real numbers. Set $A_0 = 0$ and $A_n = \sum_{k=1}^{n} a_k$ for $n \geq 1$. Then if $1 \leq p \leq q$,*

$$\sum_{k=p}^{q} a_k b_k = \sum_{k=p}^{q-1} A_k(b_k - b_{k+1}) + A_q b_q - A_{p-1} b_p.$$

PROOF. Since $a_k = A_k - A_{k-1}$ for every $k \geq 1$,

$$\sum_{k=p}^{q} a_k b_k = \sum_{k=p}^{q} A_k b_k - \sum_{k=p}^{q} A_{k-1} b_k.$$

In the second sum on the right, shift the index down by one. It becomes $\sum_{k=p-1}^{q-1} A_k b_{k+1}$. Now take the term with $k = q$ out of the first sum, and the term with $k = p - 1$ out of the second. This gives

$$\sum_{k=p}^{q} a_k b_k = \left(\sum_{k=p}^{q-1} A_k b_k + A_q b_q \right) - \left(\sum_{k=p}^{q-1} A_k b_{k+1} + A_{p-1} b_p \right).$$

Combining the two sums from p to $q - 1$ gives the formula. □

Proof Technique (Summation by Parts). Abel's formula is a discrete version of integration by parts (Theorem 9.6.2). Think of A_k as an antiderivative of

a_k, since $A_k - A_{k-1} = a_k$. Then $\sum a_k b_k$ plays the role of $\int A'g$. The sum $\sum A_k(b_k - b_{k+1})$ plays the role of $-\int Ag'$, and $A_q b_q - A_{p-1}b_p$ is the boundary term. Use the formula when the terms a_k are hard to control but their partial sums A_k are easy, and the b_k change slowly. The Dirichlet Test is the main example. The terms $\sin k$ do not tend to 0, but their partial sums are bounded, as the proof of Theorem 10.2.21 shows.

Theorem 10.2.16 (Dirichlet Test)**.** *Suppose $\{a_k\}$ and $\{b_k\}$ are sequences of real numbers such that the partial sums $A_n = \sum_{k=1}^{n} a_k$ form a bounded sequence, and $\{b_k\}$ is a decreasing sequence of nonnegative numbers with $b_k \to 0$. Then $\sum a_k b_k$ converges.*

PROOF. Choose $M > 0$ with $|A_n| \le M$ for all $n \in \mathbb{N}$. This also holds for $A_0 = 0$. Let $\varepsilon > 0$. Since $b_k \to 0$, there is an $N \in \mathbb{N}$ with $b_N < \frac{\varepsilon}{2M}$. Let $m > n \ge N$. Apply Abel's formula with $p = n+1$ and $q = m$. Since $\{b_k\}$ is decreasing, each $b_k - b_{k+1}$ is nonnegative, and each b_k is nonnegative. So

$$\begin{aligned}\left|\sum_{k=n+1}^{m} a_k b_k\right| &\le \sum_{k=n+1}^{m-1} |A_k|(b_k - b_{k+1}) + |A_m|b_m + |A_n|b_{n+1} \\ &\le M\left(\sum_{k=n+1}^{m-1}(b_k - b_{k+1}) + b_m + b_{n+1}\right) \\ &= M\big((b_{n+1} - b_m) + b_m + b_{n+1}\big) \\ &= 2Mb_{n+1} \\ &\le 2Mb_N < \varepsilon.\end{aligned}$$

The sum in the second line telescopes. By the Cauchy Criterion (Theorem 10.2.1), $\sum a_k b_k$ converges. □

Theorem 10.2.17 (Alternating Series Test)**.** *If $\{b_k\}$ is a decreasing sequence of nonnegative real numbers with $b_k \to 0$, then $\sum(-1)^{k+1}b_k$ converges.*

PROOF. Let $a_k = (-1)^{k+1}$. Then $|A_n| \le 1$ for all $n \in \mathbb{N}$, and the Dirichlet Test applies. In fact $A_n = 1$ when n is odd and $A_n = 0$ when n is even. □

Example 10.2.18. The series $\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{2k-1}$ converges. Here $b_k = \frac{1}{2k-1}$ decreases to 0.

Remark 10.2.19. If $\{b_k\}$ is not decreasing, this test does not work. The conclusion can fail. See Exercise 10.2.6.

The proof of the next theorem shows more than the bound. The partial sums jump back and forth across the sum s, as in Figure 10.2. So stopping after n terms gives an error no bigger than the first term left out.

Theorem 10.2.20. *Consider the series $\sum(-1)^{k+1}b_k$, where $\{b_k\}$ is a decreasing sequence of nonnegative real numbers with $b_k \to 0$. Let*

$$s_n = \sum_{k=1}^{n}(-1)^{k+1}b_k \quad \text{and} \quad s = \sum_{k=1}^{\infty}(-1)^{k+1}b_k.$$

Then $|s - s_n| \le b_{n+1}$ for all $n \in \mathbb{N}$.

PROOF. Consider the sequence $\{s_{2n}\}$. Since

$$s_{2n} = \sum_{k=1}^{2n}(-1)^{k+1}b_k = (b_1 - b_2) + \cdots + (b_{2n-1} - b_{2n}),$$

and $b_{k-1} - b_k \ge 0$ for all $k \ge 2$, the sequence $\{s_{2n}\}$ is increasing. Similarly, $\{s_{2n+1}\}$ is decreasing. Indeed, $s_{2n+1} - s_{2n+3} = b_{2n+2} - b_{2n+3} \ge 0$. Since $\{s_n\}$ converges to s, so do the subsequences $\{s_{2n}\}$ and $\{s_{2n+1}\}$. Therefore, $s_{2n} \le s \le s_{2n+1}$ for all $n \in \mathbb{N}$. (An increasing sequence is at most its limit, and a decreasing sequence is at least its limit, by the proof of Theorem 3.2.13.) So s lies between s_k and s_{k+1} for every $k \in \mathbb{N}$. For $k = 2n$ this says $s_{2n} \le s \le s_{2n+1}$, and for $k = 2n + 1$ it says $s_{2n+2} \le s \le s_{2n+1}$. As a consequence,

$$|s - s_k| \le |s_{k+1} - s_k| = b_{k+1}$$

for all $k \in \mathbb{N}$. □

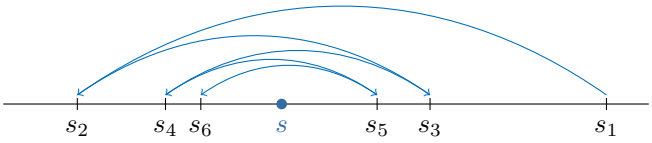


FIGURE 10.2. The partial sums of $\sum(-1)^{k+1}\frac{1}{k}$. The jumps alternate in direction, and each is shorter than the one before. So the sum s lies between any two consecutive partial sums.

Theorem 10.2.21 (Trigonometric Series Test)**.** *Suppose $\{b_k\}$ is a decreasing sequence of real numbers with $b_k \to 0$. Then*

(1) $\sum b_k \sin kt$ converges for all $t \in \mathbb{R}$, and
(2) $\sum b_k \cos kt$ converges for all $t \in \mathbb{R}$, except perhaps $t = 2p\pi$, $p \in \mathbb{Z}$.

PROOF. Since $\{b_k\}$ decreases to 0, $b_k \ge 0$ for all k.

(1) Set $A_n = \sum_{k=1}^{n} \sin kt$. Then, since

$$\sin A \sin B = \frac{1}{2}[\cos(A - B) - \cos(A + B)],$$

$$\sin(t/2)A_n = \sum_{k=1}^{n} \sin(t/2) \sin kt$$

$$= \frac{1}{2}\sum_{k=1}^{n}[\cos((k-1/2)t) - \cos((k+1/2)t)]$$
$$= \frac{1}{2}[\cos(t/2) - \cos((n+1/2)t)].$$

So for $t \neq 2p\pi$, $p \in \mathbb{Z}$,

$$A_n = \frac{\cos(t/2) - \cos((n+1/2)t)}{2\sin(t/2)}.$$

Therefore,

$$|A_n| \leq \frac{|\cos(t/2)| + |\cos((n+1/2)t)|}{2|\sin(t/2)|}$$
$$\leq \frac{1}{|\sin(t/2)|},$$

which is finite, provided $t \neq 2p\pi$, $p \in \mathbb{Z}$. Therefore, by the Dirichlet Test, the series $\sum b_k \sin kt$ converges for all $t \neq 2p\pi$, $p \in \mathbb{Z}$. If $t = 2p\pi$, $p \in \mathbb{Z}$, then $\sin kt = 0$ for all k. Thus the series converges.

(2) The proof is similar to the proof of (1) and is therefore omitted. See Exercise 10.2.9. Notice $\sum b_k \cos kt$ may or may not converge if $t = 2p\pi$, $p \in \mathbb{Z}$, since $\cos kt = 1$ for $t = 2p\pi$, $p \in \mathbb{Z}$. □

Example 10.2.22. The series $\sum_{k=1}^{\infty} \frac{1}{k}\cos kt$ converges for all $t \neq 2p\pi$, $p \in \mathbb{Z}$. However, when $t = 2p\pi$, $p \in \mathbb{Z}$, $\cos kt = 1$ and $\sum_{k=1}^{\infty} \frac{1}{k}$ diverges. However, the series $\sum_{k=1}^{\infty} \frac{1}{k^2}\cos kt$ converges for all $t \in \mathbb{R}$. For $t \neq 2p\pi$ this follows from the test. For $t = 2p\pi$ it is the p-series with $p = 2$.

Exercises

10.2.1. Prove Corollary 10.2.2. [Hint, p. 353]

10.2.2. Prove Theorem 10.2.4 (1).

10.2.3. Prove Corollary 10.2.8. [Hint, p. 354]

10.2.4. Prove Theorem 10.2.14 (3). [Hint, p. 354]

10.2.5. Prove the sequence $\left\{\frac{p^k}{k!}\right\}$, $0 < p < \infty$, converges to 0. Appeal to the series $\sum \frac{p^k}{k!}$ and the ratio test.

10.2.6. Show by example that the hypothesis of the Alternating Series Test cannot be replaced by $b_k \geq 0$ and $\lim_{k\to\infty} b_k = 0$; i.e., $\{b_k\}$ must be decreasing. [Hint, p. 354]

10.2.7. Prove that if $\sum a_k$ converges and $\{b_k\}$ is monotone and bounded, then $\sum a_k b_k$ converges. This is called *Abel's Test*. [Hint, p. 354]

10.2.8. Suppose $a_k \geq 0$ for all k and $\sum a_k$ converges. Prove that $\sum a_k^2$ converges. Show by example that $\sum \sqrt{a_k}$ can diverge. [Hint, p. 354]

10.2.9. Prove Theorem 10.2.21 (2). [Hint, p. 354]

10.2.10. Decide whether each series converges or diverges. Justify each answer.

(a) $\displaystyle\sum_{k=1}^{\infty} \frac{k^2}{2^k}$

(b) $\displaystyle\sum_{k=1}^{\infty} \frac{k!}{k^k}$.

(c) $\displaystyle\sum_{k=2}^{\infty} \frac{1}{k \ln k}$

(d) $\displaystyle\sum_{k=1}^{\infty} \left(\frac{k}{k+1}\right)^{k^2}$

(e) $\displaystyle\sum_{k=1}^{\infty} \frac{1}{\sqrt{k(k+1)}}$

(f) $\displaystyle\sum_{k=1}^{\infty} \frac{k+1}{k^3+1}$

(g) $\displaystyle\sum_{k=2}^{\infty} \frac{(-1)^k}{\ln k}$

[Hint, p. 355]

10.2.11. (Cauchy Condensation Test) Let $\{a_k\}$ be a decreasing sequence of nonnegative numbers. Prove that $\sum_{k=1}^{\infty} a_k$ converges if and only if $\sum_{j=0}^{\infty} 2^j a_{2^j}$ converges. Use this to prove again that $\sum \frac{1}{k^p}$ converges exactly when $p > 1$.

[Hint, p. 355]

10.3. Conditional and Absolute Convergence

A series can converge for two different reasons. Its terms can shrink so fast that even their absolute values have a finite sum. Or its positive and negative terms can cancel. This section shows that the first kind of convergence is much better behaved. A series that converges for the first reason can be added up in any order. A series that converges only for the second reason can be reordered to have any sum at all.

Definition 10.3.1. A series $\sum a_k$ is said to be *absolutely convergent* if $\sum |a_k|$ converges. The series is said to be *conditionally convergent* if it is convergent but not absolutely convergent.

Example 10.3.2. $\sum_{k=1}^{\infty} \frac{(-1)^k}{k}$ is conditionally convergent, while $\sum_{k=1}^{\infty} \frac{(-1)^k}{k^2}$ is absolutely convergent. The first series is -1 times the series $\sum \frac{(-1)^{k+1}}{k}$, which converges by the Alternating Series Test. But $\sum \frac{1}{k}$ diverges. The second converges absolutely because $\sum \frac{1}{k^2}$ converges (Example 10.2.12).

Theorem 10.3.3. *If $\sum a_k$ converges absolutely, $\sum a_k$ converges and*

$$\left|\sum a_k\right| \leq \sum |a_k|.$$

PROOF. Suppose $\sum |a_k|$ converges. By the Triangle Inequality, for $1 \leq p \leq q$,

$$\left| \sum_{k=p}^{q} a_k \right| \leq \sum_{k=p}^{q} |a_k|.$$

Thus an application of the Cauchy Criterion shows that $\sum a_k$ converges. Finally,

$$\begin{aligned} \left| \sum_{k=1}^{\infty} a_k \right| &= \lim_{q \to \infty} \left| \sum_{k=1}^{q} a_k \right| \\ &\leq \lim_{q \to \infty} \sum_{k=1}^{q} |a_k| \\ &= \sum_{k=1}^{\infty} |a_k|. \end{aligned}$$

The first equality holds because $\big||s_q| - |s|\big| \leq |s_q - s|$, where s_q is the qth partial sum and s is the sum. The inequality holds by Theorem 3.2.7. □

Theorem 10.3.4 (Ratio and Root Test)**.** *Let $\sum a_k$ be a series of real numbers, and let*

$$\alpha = \limsup \sqrt[k]{|a_k|}.$$

Also, if $a_k \neq 0$ for all $k \in \mathbb{N}$, let

$$R = \limsup \left| \frac{a_{k+1}}{a_k} \right| \quad \text{and} \quad r = \liminf \left| \frac{a_{k+1}}{a_k} \right|.$$

(1) If $\alpha < 1$ or $R < 1$, then $\sum a_k$ is absolutely convergent.
(2) If $\alpha > 1$ or $r > 1$, then $\sum a_k$ is divergent.
(3) If $\alpha = 1$ or $r \leq 1 \leq R$, then the test is inconclusive.

PROOF. (1) Apply the Root Test or the Ratio Test to the series $\sum |a_k|$.

(2) If $\alpha > 1$, the proof of Theorem 10.2.14 (2) shows that $|a_k| > 1$ for infinitely many k. If $r > 1$, choose $c \in (1, r)$. As in the proof of Theorem 10.2.13 (2), there is a $K \in \mathbb{N}$ with $|a_{k+1}| \geq c|a_k| > |a_k|$ for all $k \geq K$. So $|a_k| \geq |a_K| > 0$ for all $k \geq K$. In both cases $a_k \not\to 0$, so $\sum a_k$ diverges by the nth Term Test.

(3) The examples in the proof of Theorem 10.2.13 (3) and in Exercise 10.2.4 have positive terms, so they work here too. □

Example 10.3.5. Let p be a polynomial and $b \in \mathbb{R}$ with $|b| < 1$. Let $a_k = p(k)b^k$. If $b = 0$ or p is the zero polynomial, there is nothing to prove. Otherwise p has finitely many roots, so there is a $K \in \mathbb{N}$ with $p(k) \neq 0$ for all $k \geq K$. By Remark 10.2.5, it is enough to study $\sum_{k=K}^{\infty} a_k$. If p has degree d, dividing the numerator and denominator of $\frac{p(k+1)}{p(k)}$ by k^d shows that this ratio tends to 1. Then, since

$$\left| \frac{a_{k+1}}{a_k} \right| = |b| \left| \frac{p(k+1)}{p(k)} \right| \to |b| < 1,$$

the series $\sum p(k)b^k$ is absolutely convergent.

Definition 10.3.6. A series $\sum a'_k$ is a *rearrangement* of the series $\sum a_k$ if there exists a bijection $j : \mathbb{N} \to \mathbb{N}$ such that $a'_k = a_{j(k)}$ for all $k \in \mathbb{N}$.

Example 10.3.7. The alternating harmonic series

$$\sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{5} - \cdots$$

converges, but not absolutely. Let s be its sum. Now rearrange it so that two positive terms come before each negative term,

$$1 + \frac{1}{3} - \frac{1}{2} + \frac{1}{5} + \frac{1}{7} - \frac{1}{4} + \frac{1}{9} + \frac{1}{11} - \frac{1}{6} + \cdots .$$

Every term of the original series appears exactly once. This rearranged series also converges, but its sum is different.

First, $s < \frac{5}{6}$. Let s_n be the nth partial sum of the original series. The proof of Theorem 10.2.20 shows that $s \le s_{2n+1}$ for all n. Also $s_{2n+1} - s_{2n+3} = \frac{1}{2n+2} - \frac{1}{2n+3} > 0$. So

$$s \le s_5 < s_3 = 1 - \frac{1}{2} + \frac{1}{3} = \frac{5}{6}.$$

Next, let s'_n be the nth partial sum of the rearranged series. Its terms come in groups of three, and the kth group is

$$\frac{1}{4k-3} + \frac{1}{4k-1} - \frac{1}{2k} = \frac{8k-3}{2k(4k-1)(4k-3)}.$$

So s'_{3n} is the sum of the first n groups. Each group is positive, so $s'_3 < s'_6 < s'_9 < \cdots$. Since $4k-1 \ge 3k$ and $4k-3 \ge k$ for $k \ge 1$, each group is at most $\frac{8k}{2k \cdot 3k \cdot k} = \frac{4}{3k^2}$. So the series of groups converges, by the Comparison Test and Example 10.2.12. Let s' be its sum, so $s'_{3n} \to s'$. Since $\{s'_{3n}\}$ is strictly increasing, $s' \ge s'_6 > s'_3 = 1 + \frac{1}{3} - \frac{1}{2} = \frac{5}{6}$.

The other partial sums also tend to s', since

$$s'_{3n+1} = s'_{3n} + \frac{1}{4n+1} \quad \text{and} \quad s'_{3n+2} = s'_{3n} + \frac{1}{4n+1} + \frac{1}{4n+3}.$$

In detail, let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $|s'_{3n} - s'| < \frac{\varepsilon}{2}$ and $\frac{2}{4n+1} < \frac{\varepsilon}{2}$ for all $n \ge N$. Every $m \ge 3N$ has the form $3n$, $3n+1$ or $3n+2$ with $n \ge N$. So $|s'_m - s'| \le |s'_{3n} - s'| + \frac{2}{4n+1} < \varepsilon$. Thus the rearranged series converges to s'.

Since $s < \frac{5}{6} < s'$, rearranging the terms changed the sum.

Theorem 10.3.8. *If the series $\sum a_k$ converges absolutely, then every rearrangement of $\sum a_k$ converges to the same sum.*

PROOF. Let $\sum a'_k$ be a rearrangement of $\sum a_k$, with $a'_k = a_{j(k)}$ for a bijection $j : \mathbb{N} \to \mathbb{N}$. Let s_n and s'_n be the nth partial sums of the two series, and let $s = \sum_{k=1}^{\infty} a_k$. Let $\varepsilon > 0$. Since $s_n \to s$, and by the Cauchy Criterion for $\sum |a_k|$, there is an $N \in \mathbb{N}$ such that $|s_N - s| < \frac{\varepsilon}{2}$ and

$$\sum_{i=N+1}^{m} |a_i| < \frac{\varepsilon}{2} \quad \text{for all } m > N.$$

Since j is onto, each of $1, 2, \ldots, N$ is $j(k)$ for some k. Let P be the largest of these N values of k. Let $n \ge P$, and let $J_n = \{j(1), j(2), \ldots, j(n)\}$. Then $\{1, 2, \ldots, N\} \subset J_n$.

Since j is one-to-one, the indices in J_n are distinct. So the terms $a_1, a_2, \ldots, a_N$ cancel in the difference, and

$$s'_n - s_N = \sum_{k=1}^{n} a_{j(k)} - \sum_{i=1}^{N} a_i = \sum_{i \in J_n,\ i>N} a_i.$$

If no index in J_n is larger than N, this sum is 0. Otherwise let m be the largest index in J_n. Then every index in the sum lies between $N+1$ and m, so

$$|s'_n - s_N| \le \sum_{i=N+1}^{m} |a_i| < \frac{\varepsilon}{2}.$$

In both cases, $|s'_n - s| \le |s'_n - s_N| + |s_N - s| < \varepsilon$ for all $n \ge P$. So $s'_n \to s$. □

The next theorem is due to Riemann. It says that Theorem 10.3.8 fails as badly as possible for conditionally convergent series. The idea of the proof is simple. Add positive terms until the total passes α. Then add negative terms until it drops below α. Repeat forever. The overshoots shrink because the terms tend to 0.

Theorem 10.3.9 (Riemann Rearrangement Theorem)**.** *Let $\sum a_k$ be a conditionally convergent series of real numbers. Suppose $\alpha \in \mathbb{R}$. Then there exists a rearrangement $\sum a'_k$ of $\sum a_k$ which converges to α.*

Proof. The proof has three steps. The first step shows that the positive part and the negative part of the series both diverge. That is what lets the rearranged sums swing past α in both directions as often as needed. For each $n \in \mathbb{N}$, let P_n be the sum of the nonnegative terms among $a_1, \ldots, a_n$, and let Q_n be the sum of the absolute values of the negative terms among $a_1, \ldots, a_n$. Then

$$s_n = P_n - Q_n \quad \text{and} \quad \sum_{k=1}^{n} |a_k| = P_n + Q_n.$$

Suppose $\{P_n\}$ were bounded. Since $\{s_n\}$ converges, it is bounded, so $\{Q_n\} = \{P_n - s_n\}$ would be bounded too. Then the partial sums of $\sum |a_k|$ would be bounded, and $\sum |a_k|$ would converge by Theorem 10.2.6. That is false. So $\{P_n\}$ is not bounded. In the same way, $\{Q_n\}$ is not bounded. In particular, $\sum a_k$ has infinitely many nonnegative terms and infinitely many negative terms.

Let $p_1, p_2, p_3, \ldots$ be the nonnegative terms of $\sum a_k$, in the order they appear. Let $q_1, q_2, q_3, \ldots$ be the absolute values of the negative terms, in the order they appear. Each P_n is either 0 or a partial sum of $\sum p_i$. Since $\{P_n\}$ is not bounded, the partial sums of $\sum p_i$ are not bounded. So $\sum_{i=1}^{n} p_i \to \infty$ by Theorem 10.2.6. In the same way, $\sum_{i=1}^{n} q_i \to \infty$. Also $a_k \to 0$ by the nth Term Test. Since $\{p_i\}$ is a subsequence of $\{a_k\}$, and $\{q_i\}$ is a subsequence of $\{|a_k|\}$, both tend to 0.

The second step builds the rearrangement. It adds nonnegative terms until the total passes α, then negative terms until the total drops below α, and so on. Let m_1 be the smallest positive integer with

$$X_1 = p_1 + \cdots + p_{m_1} > \alpha.$$

It exists because $\sum_{i=1}^{n} p_i \to \infty$. Let n_1 be the smallest positive integer with

$$Y_1 = X_1 - q_1 - \cdots - q_{n_1} < \alpha.$$

It exists because $\sum_{i=1}^{n} q_i \to \infty$. Continue in the same way. Once m_k, n_k, X_k and Y_k are chosen, let m_{k+1} be the smallest integer greater than m_k with

$$X_{k+1} = Y_k + p_{m_k+1} + \cdots + p_{m_{k+1}} > \alpha,$$

and let n_{k+1} be the smallest integer greater than n_k with

$$Y_{k+1} = X_{k+1} - q_{n_k+1} - \cdots - q_{n_{k+1}} < \alpha.$$

The rearranged series is

$$p_1 + \cdots + p_{m_1} - q_1 - \cdots - q_{n_1} + p_{m_1+1} + \cdots + p_{m_2} - q_{n_1+1} - \cdots - q_{n_2} + \cdots .$$

Since $m_1 < m_2 < \cdots$ and $n_1 < n_2 < \cdots$, every p_i and every q_i is used, exactly once. So this is a rearrangement of $\sum a_k$.

The third step shows that the rearranged series converges to α. Let $k \geq 2$. Since m_k is the smallest choice, the total just before p_{m_k} was added was at most α. So $\alpha < X_k \leq \alpha + p_{m_k}$. In the same way, the total just before q_{n_k} was added was at least α, so $\alpha - q_{n_k} \leq Y_k < \alpha$. Since $m_k \geq k$ and $n_k \geq k$, the sequences $\{p_{m_k}\}$ and $\{q_{n_k}\}$ are subsequences of sequences that tend to 0. So $X_k \to \alpha$ and $Y_k \to \alpha$, by the Squeeze Theorem.

Let s'_n be the nth partial sum of the rearranged series. While positive terms are being added, the partial sums increase. While negative terms are being added, they decrease. So every partial sum in the block $-q_{n_{k-1}+1} - \cdots - q_{n_k}$ lies in $[Y_k, X_k]$, where $n_0 = 0$. Every partial sum in the block $p_{m_k+1} + \cdots + p_{m_{k+1}}$ lies in $[Y_k, X_{k+1}]$.

Let $\varepsilon > 0$. Choose an integer $K \geq 2$ such that $|X_k - \alpha| < \varepsilon$ and $|Y_k - \alpha| < \varepsilon$ for all $k \geq K$. Let $n_* = m_K + n_{K-1}$, which is the position of the last term of the Kth block of positive terms, where again $n_0 = 0$. Every partial sum s'_n with $n > n_*$ lies in $[Y_k, X_k]$ or in $[Y_k, X_{k+1}]$ for some $k \geq K$. So $|s'_n - \alpha| < \varepsilon$ for all $n > n_*$. Thus $s'_n \to \alpha$. □

Proof Technique (Greedy Constructions). The proof of Theorem 10.3.9 builds the rearrangement in stages, and at each stage it uses the fewest terms that do the job. A choice like this is called *greedy*. Taking the fewest terms is what gives control. The total passed α only at the last term added, so it overshot α by at most that term. A proof that took just some number of terms with total above α would have no such bound. The smallest such number exists by the Well-Ordering Principle (Axiom 1.2.7).

Exercises

10.3.1. Suppose that the sequence $\{b_n\}$ is decreasing with $\lim_{n\to\infty} b_n = 0$. If $\{a_n\}$ is a sequence in $\mathbb{R}$ satisfying $|a_n| \leq b_n - b_{n+1}$ for all $n \in \mathbb{N}$, prove $\sum a_k$ converges absolutely. [Hint, p. 355]

10.3.2. Prove that if $\lim_{k\to\infty} k^p a_k = A$ for some $p > 1$, then $\sum a_k$ converges absolutely. [Hint, p. 355]

10.3.3. Prove that if $\sum a_k$ converges absolutely, then $\sum a_k^2$ converges. Give an example of a convergent series $\sum a_k$ for which $\sum a_k^2$ diverges. [Hint, p. 355]

10.3.4. For each k, let $a_k^+ = \max\{a_k, 0\}$ and $a_k^- = \max\{-a_k, 0\}$. Prove that $\sum a_k$ converges absolutely if and only if both $\sum a_k^+$ and $\sum a_k^-$ converge. Prove that in this case $\sum a_k = \sum a_k^+ - \sum a_k^-$. [Hint, p. 355]

10.3.5. Find a rearrangement of $\sum \frac{(-1)^{k+1}}{k}$ whose partial sums tend to ∞. [Hint, p. 356]

10.4. The Cauchy Product*

Two convergent series can be added term by term. Multiplying them is harder, since every term of one series must be multiplied by every term of the other. The Cauchy product collects these products by the sum of their indices, the way polynomials are multiplied. In this section the series start at $k = 0$. This section can be skipped. Nothing later depends on it.

Definition 10.4.1. The *Cauchy product* of $\sum_{k=0}^{\infty} a_k$ and $\sum_{k=0}^{\infty} b_k$ is the series $\sum_{n=0}^{\infty} c_n$, where

$$c_n = \sum_{k=0}^{n} a_k b_{n-k} = a_0 b_n + a_1 b_{n-1} + \cdots + a_n b_0.$$

Theorem 10.4.2 (Mertens's Theorem)**.** *Suppose $\sum a_k$ converges absolutely to A and $\sum b_k$ converges to B. Then their Cauchy product converges to AB.*

Proof. Let $A_N = \sum_{k=0}^{N} a_k$, $B_N = \sum_{k=0}^{N} b_k$ and $C_N = \sum_{n=0}^{N} c_n$. Collecting the terms with the same a_k,

$$C_N = \sum_{k=0}^{N} a_k B_{N-k} = A_N B + \sum_{k=0}^{N} a_k \beta_{N-k},$$

where $\beta_j = B_j - B$. Since $A_N B \to AB$, it is enough to show $\sum_{k=0}^{N} a_k \beta_{N-k} \to 0$.

Since $\beta_j \to 0$, there is a K with $|\beta_j| \le K$ for all j. Let $\alpha = \sum_{k=0}^{\infty} |a_k|$, and let $\varepsilon > 0$. Choose J with $|\beta_j| < \varepsilon$ for $j \ge J$. By Theorem 10.2.1, there is an N_0 with $\sum_{k=m}^{\infty} |a_k| < \varepsilon$ for $m \ge N_0$. Let $N \ge N_0 + J$. In the terms with $k \le N - J$, the index $N - k$ is at least J. In the terms with $k > N - J$, $k > N_0$. So

$$\left| \sum_{k=0}^{N} a_k \beta_{N-k} \right| \le \sum_{k=0}^{N-J} |a_k| \varepsilon + \sum_{k=N-J+1}^{N} |a_k| K \le \alpha\varepsilon + K\varepsilon.$$

Since ε is arbitrary, the sum tends to 0. □

Example 10.4.3. For every x, the series $\sum_{k=0}^{\infty} \frac{x^k}{k!}$ converges absolutely, by the Ratio Test. The Cauchy product of the series for x and for y has terms

$$c_n = \sum_{k=0}^{n} \frac{x^k}{k!} \cdot \frac{y^{n-k}}{(n-k)!} = \frac{1}{n!} \sum_{k=0}^{n} \binom{n}{k} x^k y^{n-k} = \frac{(x+y)^n}{n!},$$

by the Binomial Theorem (Exercise 9.3.3). So by Theorem 10.4.2,

$$\left(\sum_{k=0}^{\infty}\frac{x^k}{k!}\right)\left(\sum_{k=0}^{\infty}\frac{y^k}{k!}\right)=\sum_{n=0}^{\infty}\frac{(x+y)^n}{n!}.$$

This is the law $e^x e^y = e^{x+y}$, read off from the series. Compare Theorem 12.5.2 (3).

Example 10.4.4. Absolute convergence of one series cannot be dropped. The series $\sum_{k=0}^{\infty}\frac{(-1)^k}{\sqrt{k+1}}$ converges by the Alternating Series Test, but not absolutely. Its Cauchy product with itself has

$$|c_n|=\sum_{k=0}^{n}\frac{1}{\sqrt{(k+1)(n-k+1)}}\geq\sum_{k=0}^{n}\frac{2}{n+2}=\frac{2(n+1)}{n+2}\geq 1,$$

since $\sqrt{uv}\leq\frac{u+v}{2}$ for $u,v\geq 0$, which is $(\sqrt{u}-\sqrt{v})^2\geq 0$. So $c_n\not\to 0$, and the Cauchy product diverges by the nth Term Test.

Exercises

10.4.1. Let $|x|<1$. Use Theorem 10.4.2 to prove that $\frac{1}{(1-x)^2}=\sum_{n=0}^{\infty}(n+1)x^n$.
[Solution, p. 356]

10.4.2. Suppose $\sum a_k$ and $\sum b_k$ both converge absolutely. Prove that their Cauchy product converges absolutely.
[Hint, p. 356]

10.5. Infinitely Many Primes*

Euclid proved that there are infinitely many primes around 300 BC. His proof is Exercise 10.5.1. Euler found a different proof in 1737, using series. Each prime p gives a geometric series

$$1+\frac{1}{p}+\frac{1}{p^2}+\frac{1}{p^3}+\cdots=\frac{1}{1-\frac{1}{p}}.$$

Multiply these series together, one for each prime. Every natural number is a product of primes, by Lemma 1.3.2 in the starred Section 1.3, so every $\frac{1}{n}$ appears when the product is multiplied out. Euler concluded that the product is the harmonic series. If there were only finitely many primes, the product would be a finite number. But the harmonic series diverges, by Example 10.1.5.

This section makes Euler's argument precise. It uses only finite sums and finite products. It can be skipped. Nothing later depends on it.

Theorem 10.5.1 (Euler)**.** *There are infinitely many primes.*

Proof. Assume by way of contradiction that there are only finitely many primes. Call them $p_1,\ldots,p_m$. The plan is to show that the partial sums $s_N=\sum_{n=1}^{N}\frac{1}{n}$ of the harmonic series are bounded.

Let $N \in \mathbb{N}$, and choose $K \in \mathbb{N}$ with $2^K > N$. Let $n \in \mathbb{N}$ with $n \leq N$. If $n \geq 2$, then n is a product of primes by Lemma 1.3.2, and each of these primes is one of $p_1, \ldots, p_m$. Collecting equal primes,

$$n = p_1^{a_1} p_2^{a_2} \cdots p_m^{a_m}$$

for some $a_1, \ldots, a_m \in \mathbb{N}_0$. This also holds for $n = 1$, with every $a_i = 0$. Since $p_i \geq 2$, $2^{a_i} \leq p_i^{a_i} \leq n < 2^K$. So $a_i < K$ for each i.

Now consider the product

$$P_K = \prod_{i=1}^{m} \left(1 + \frac{1}{p_i} + \frac{1}{p_i^2} + \cdots + \frac{1}{p_i^K}\right).$$

Multiplying it out with the distributive law, P_K is the sum of the numbers

$$\frac{1}{p_1^{a_1} p_2^{a_2} \cdots p_m^{a_m}}$$

over all choices of $a_1, \ldots, a_m \in \{0, 1, \ldots, K\}$. This follows by induction on m. Each $n \leq N$ gives one such choice. Different values of n give different choices, since the choice determines n. All the terms are positive. So

$$s_N = \sum_{n=1}^{N} \frac{1}{n} \leq P_K.$$

Each factor of P_K is a partial sum of a geometric series with positive terms. So by Example 10.1.3,

$$1 + \frac{1}{p_i} + \cdots + \frac{1}{p_i^K} < \sum_{k=0}^{\infty} \frac{1}{p_i^k} = \frac{1}{1 - \frac{1}{p_i}} = \frac{p_i}{p_i - 1}.$$

So $s_N < C$ for every $N \in \mathbb{N}$, where $C = \prod_{i=1}^{m} \frac{p_i}{p_i - 1}$. This contradicts Example 10.1.5, which shows that the partial sums of the harmonic series are not bounded. So there are infinitely many primes. □

Remark 10.5.2. Euler went further. Every $n \geq 2$ is a product of primes in only one way, apart from the order of the factors. This is the Fundamental Theorem of Arithmetic, Theorem 1.3.4. With it, each $\frac{1}{n}$ appears exactly once when the geometric series are multiplied out. Euler showed that for $\alpha > 1$,

$$\sum_{n=1}^{\infty} \frac{1}{n^\alpha} = \prod_{q \text{ prime}} \frac{1}{1 - q^{-\alpha}}.$$

Here the infinite product means the limit of the products over the primes $q \leq M$ as $M \to \infty$. This is called the *Euler product*. It ties the primes to analysis, and it is the starting point of analytic number theory.

Exercises

10.5.1. (Euclid) Let $p_1, \dots, p_m$ be primes, and let $n = p_1 p_2 \cdots p_m + 1$. Prove that no p_i divides n. Use Lemma 1.3.2 to conclude that there is a prime different from $p_1, \dots, p_m$. This gives another proof of Theorem 10.5.1. [Hint, p. 356]

10.5.2. For $K \in \mathbb{N}$, let T_K be the sum of $\frac{1}{2^a 3^b}$ over all $a, b \in \{0, 1, \dots, K\}$. Prove that $T_K \to 3$. The numbers $2^a 3^b$ are the natural numbers whose only prime factors are 2 and 3. [Solution, p. 356]

10.5.3. Let $q_1 < q_2 < q_3 < \cdots$ be the primes. Prove that $\sum \frac{1}{q_k}$ diverges. [Hint, p. 356]

Notes

Nicole Oresme proved that the harmonic series diverges, by the grouping in Example 10.1.5, around 1350 [**56**]. Colin Maclaurin gave the Integral Test in 1742 [**53**], and Cauchy gave it again later. The Ratio Test is due to Jean le Rond d'Alembert (1768) [**18**], and the Root Test and the Cauchy Criterion for series to Cauchy (1821) [**15**]. Niels Henrik Abel published his summation formula in 1826 [**1**]. Dirichlet observed in 1837, in his paper on primes in arithmetic progressions, that rearranging a conditionally convergent series can change its sum [**25**]. Riemann proved Theorem 10.3.9 in his 1854 work on trigonometric series, published in 1868 [**59**]. Euler introduced the constant γ of Example 10.2.11 in 1734. Cauchy defined the product of two series in his *Cours d'analyse* of 1821, and Franz Mertens proved Theorem 10.4.2 in 1875. Euclid's proof that there are infinitely many primes is Proposition 20 of Book IX of the *Elements* [**28**]. Euler's proof of Theorem 10.5.1 is in a paper he wrote in 1737 and published in 1744 [**29**]. The same paper has the Euler product of Remark 10.5.2 and the divergence of $\sum \frac{1}{q_k}$ in Exercise 10.5.3. Dirichlet's 1837 paper extended Euler's method to prove that every arithmetic progression $a, a+d, a+2d, \dots$ with $a, d \in \mathbb{N}$ and no common factor greater than 1 contains infinitely many primes [**25**].

CHAPTER 11

Sequences and Series of Functions

A sequence of functions can converge in two very different ways. Pointwise convergence asks only that $f_n(x) \to f(x)$ at each x separately. It is the obvious definition, and it is too weak. Section 11.1 shows that a pointwise limit of continuous functions need not be continuous, that the integrals of f_n need not converge to the integral of the limit, and that derivatives behave even worse. Section 11.2 introduces uniform convergence, which asks that f_n be close to f at every x at once. Uniform convergence fixes the first two problems. With an extra hypothesis, it fixes the third. Section 11.3, which is marked with a star, uses uniform convergence to build a continuous function that has a derivative at no point at all. The starred Section 11.4 shows that every continuous function on $[a, b]$ is a uniform limit of polynomials, and the starred Section 11.5 uses uniform convergence to solve differential equations.

11.1. The Problems with Pointwise Convergence

Definition 11.1.1. Let $E \subset \mathbb{R}$, and let $\{f_n\}_{n=1}^{\infty}$ be a sequence of functions $f_n : E \to \mathbb{R}$. The sequence $\{f_n\}$ *converges pointwise* on E if the sequence of real numbers $\{f_n(x)\}$ converges for every $x \in E$. If this is the case, then f defined by

$$f(x) = \lim_{n\to\infty} f_n(x), \quad x \in E,$$

defines a function on E. The function f is called the *limit* of the sequence $\{f_n\}$.

In other words, the sequence $\{f_n\}$ converges pointwise to f on E if for each $x \in E$ and for all $\varepsilon > 0$, there exists an $N(x, \varepsilon) \in \mathbb{N}$ such that for all $n \geq N$, $|f_n(x) - f(x)| < \varepsilon$.

Definition 11.1.2. Let $E \subset \mathbb{R}$, and let $\{f_n\}$ be a sequence of functions $f_n : E \to \mathbb{R}$. For each $n \in \mathbb{N}$, define a sequence of functions $\{S_n\}$ by

$$S_n(x) = \sum_{k=1}^{n} f_k(x).$$

Then $\{S_n\}$ is called a *series of functions* and is denoted by $\sum f_k$. The series of functions converges pointwise on E if for each $x \in E$, $\{S_n(x)\}$ converges. If the series $\{S_n\}$ converges pointwise to S, S is the *sum* of the series, written $S = \sum f_k$.

Remark 11.1.3. Let $\{f_n\}$ be a sequence of functions $f_n : [a, b] \to \mathbb{R}$ converging pointwise to f on $[a, b]$.

(1) If each f_n is continuous at $c \in [a, b]$, is f continuous at c; i.e., does

$$\lim_{x\to c} \left(\lim_{n\to\infty} f_n(x) \right) = \lim_{n\to\infty} \left(\lim_{x\to c} f_n(x) \right)?$$

(2) If each $f_n \in \mathcal{R}[a,b]$, is $f \in \mathcal{R}[a,b]$? If so, does

$$\int_a^b f = \lim_{n\to\infty} \int_a^b f_n,$$

i.e., does

$$\lim_{n\to\infty} \int_a^b f_n = \int_a^b \lim_{n\to\infty} f_n?$$

(3) If each f_n is differentiable at $c \in [a,b]$, is f differentiable at c? If so, does

$$f'(c) = \lim_{n\to\infty} f_n'(c)?$$

Each question asks whether two limit operations can be done in either order. The next five examples show that the answer to every question is no.

Example 11.1.4. Let $E = [0,1]$ and for each $x \in E$, $n \in \mathbb{N}$, define $f_n(x) = x^n$. Since $f_n(1) = 1$ for all $n \in \mathbb{N}$, $f_n(1) \to 1$. If $x \in [0,1)$, then $f_n(x) \to 0$ by Example 3.1.7. Therefore,

$$\lim_{n\to\infty} f_n(x) = f(x) = \begin{cases} 0, & x \in [0,1), \\ 1, & x = 1. \end{cases}$$

But each f_n is continuous and f is not. See Figure 11.1.

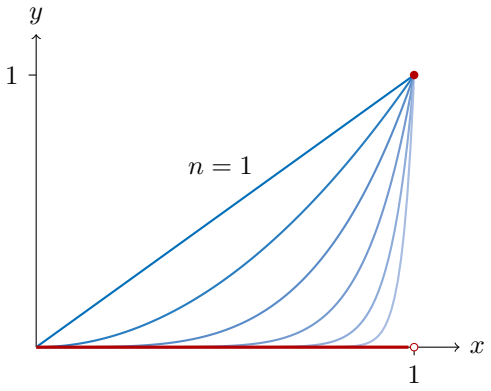


FIGURE 11.1. The functions $f_n(x) = x^n$ on $[0,1]$ for $n = 1, 2, 4, 8, 16, 32$. Each one is continuous. Their pointwise limit, shown in red, is 0 on $[0,1)$ and 1 at $x = 1$.

Example 11.1.5. Let $f_k(x) = \frac{x^2}{(1+x^2)^k}$ for $x \in \mathbb{R}$ and consider the series

$$S(x) = \sum_{k=0}^{\infty} f_k(x) = \sum_{k=0}^{\infty} x^2 \left(\frac{1}{1+x^2}\right)^k.$$

Notice each f_k is continuous on $\mathbb{R}$. If $x = 0$, then $f_k(0) = 0$ and thus $S(0) = 0$. If $x \neq 0$, $\frac{1}{1+x^2} < 1$ and hence by Example 10.1.3

$$x^2 \sum_{k=0}^{\infty} \left(\frac{1}{1+x^2}\right)^k = x^2 \left[\frac{1}{1 - \frac{1}{1+x^2}}\right] = 1 + x^2.$$

Thus $S(x) = 1 + x^2$ for $x \neq 0$. So

$$S(x) = \begin{cases} 0, & x = 0, \\ 1 + x^2, & x \neq 0, \end{cases}$$

which is not continuous on $\mathbb{R}$.

Example 11.1.6. Let $\{x_k\}$ be an enumeration of the rational numbers in $[0, 1]$. For each $n \in \mathbb{N}$, define $f_n : [0, 1] \to \mathbb{R}$ as

$$f_n(x) = \begin{cases} 0, & x = x_k, \quad 1 \leq k \leq n, \\ 1, & \text{otherwise.} \end{cases}$$

Since each f_n is continuous except at $x_1, \ldots, x_n$, $f_n \in \mathcal{R}[0, 1]$ with $\int_0^1 f_n = 1$. This follows from Exercise 9.2.4, since f_n equals the constant 1 except at n points. But

$$\lim_{n\to\infty} f_n(x) = f(x) = \begin{cases} 0, & x \in \mathbb{Q} \cap [0, 1], \\ 1, & x \in [0, 1] \setminus \mathbb{Q}, \end{cases}$$

and $f \notin \mathcal{R}[0, 1]$. As in Example 9.1.9, every lower sum of f is 0 and every upper sum is 1.

Example 11.1.7. For $x \in [0, 1]$, $n \in \mathbb{N}$, define $f_n(x) = nx(1 - x^2)^n$. Since each f_n is continuous, $f_n \in \mathcal{R}[0, 1]$. If $x = 0, 1$, then $f_n(x) = 0$. If $x \in (0, 1)$, then $0 < 1-x^2 < 1$, and so $f_n(x) \to 0$. Indeed, $\sum_n nx(1 - x^2)^n$ converges by Example 10.3.5, so its terms tend to 0. Thus for all $x \in [0, 1]$,

$$\lim_{n\to\infty} f_n(x) = f(x) = 0.$$

Thus $f \in \mathcal{R}[0, 1]$ as well. But the substitution $u = 1 - x^2$ (Theorem 9.6.3) gives

$$\int_0^1 f_n = \frac{1}{2} \cdot \frac{n}{n + 1},$$

implying

$$\lim_{n\to\infty} \int_0^1 f_n = \frac{1}{2} \neq 0 = \int_0^1 f.$$

Example 11.1.8. Consider $f_n(x) = \frac{\sin nx}{n}$ for $x \in \mathbb{R}$. Since $|\sin nx| \leq 1$ for all $x \in \mathbb{R}$ and $n \in \mathbb{N}$,

$$\lim_{n\to\infty} f_n(x) = f(x) = 0,$$

for all $x \in \mathbb{R}$. But $f_n'(x) = \cos nx$ which implies $f_n'(0) = 1$ for all $n \in \mathbb{N}$. Therefore

$$\lim_{n\to\infty} f_n'(0) = 1 \neq 0 = f'(0).$$

11.2. The Importance of Uniform Convergence

In each example of Section 11.1, the convergence is slower at some points than at others. For $f_n(x) = x^n$, the closer x is to 1, the larger n must be before x^n is small. No single n works for every x. Uniform convergence rules this out.

Definition 11.2.1. Let $E \subset \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $\{f_n\}$ be a sequence of functions $f_n : E \to \mathbb{R}$. The sequence $\{f_n\}$ *converges uniformly* to f on E if for all $\varepsilon > 0$, there exists an $N(\varepsilon) \in \mathbb{N}$ such that for all $n \geq N$, $|f_n(x) - f(x)| < \varepsilon$ for all $x \in E$. Similarly, $\sum f_k$ converges uniformly to S on E if the series $\{S_n\}$ converges uniformly to S on E.

Remark 11.2.2. The difference from pointwise convergence is the order of the quantifiers. In pointwise convergence, N may depend on x. In uniform convergence, one N works for every $x \in E$ at once. Geometrically, for $n \geq N$ the graph of f_n lies inside the band between the graphs of $f - \varepsilon$ and $f + \varepsilon$. See Figure 11.2. A sequence that converges uniformly to f also converges pointwise to f. So to decide whether a sequence converges uniformly, first find its pointwise limit. That is the only possible uniform limit.

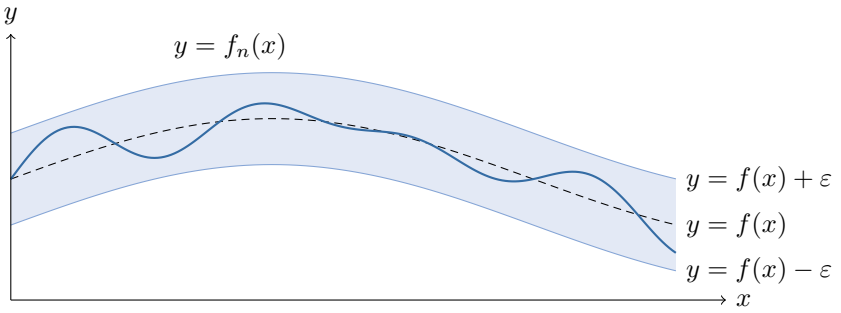


FIGURE 11.2. Uniform convergence. For every $n \geq N$, the whole graph of f_n lies inside the shaded band around the graph of f.

Example 11.2.3. For $x \in [0, 1]$, consider $f_n(x) = x^n$. Assume by way of contradiction that f_n converges uniformly to f, where

$$f(x) = \begin{cases} 0, & x \in [0, 1), \\ 1, & x = 1. \end{cases}$$

Let $\varepsilon > 0$ with $\varepsilon < 1$. So there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $x^n < \varepsilon$ for all $x \in [0, 1)$. In particular, $x^N < \varepsilon$ for all $x \in [0, 1)$. Since $0^N = 0$ and $1^N = 1$, by the Intermediate Value Theorem, there exists a $c \in (0, 1)$ such that $c^N = \varepsilon$, a contradiction.

Example 11.2.4. Consider the series

$$\sum_{k=1}^{\infty} \left[kxe^{-kx^2} - (k-1)xe^{-(k-1)x^2} \right]$$

for $x \in [0, \infty)$. Now $S_n(x) = nxe^{-nx^2}$, so $S(x) = 0$ for all $x \in [0, \infty)$. The partial sums telescope. For $x > 0$, $nxe^{-nx^2} = nx\, r^n$ with $r = e^{-x^2} \in (0, 1)$, which tends to 0 by Example 10.3.5 and the nth Term Test. Assume by way of contradiction this convergence is uniform. Let $\varepsilon = 1$. So there exists an $N \in \mathbb{N}$ such that for all $n \geq N$,

$$|S_n(x) - S(x)| = nxe^{-nx^2} < 1$$

for all $x \in [0, \infty)$. But

$$S_n'(x) = ne^{-nx^2} - 2n^2x^2e^{-nx^2} = 0$$

when $x = \sqrt{\frac{1}{2n}}$, implying S_n has a maximum value of

$$S_n\left(\sqrt{\frac{1}{2n}}\right) = \sqrt{\frac{n}{2e}} > 1$$

for $n \geq 6$, a contradiction. (The derivative is positive to the left of $\sqrt{1/(2n)}$ and negative to the right of it, so this is the maximum.)

Figure 11.3 shows S_n for $n = 1$, 4 and 16. Each graph has a bump. As n grows, the bump gets taller and moves toward 0. At each fixed $x > 0$ the bump eventually passes by, so $S_n(x) \to 0$. But the bump never shrinks.

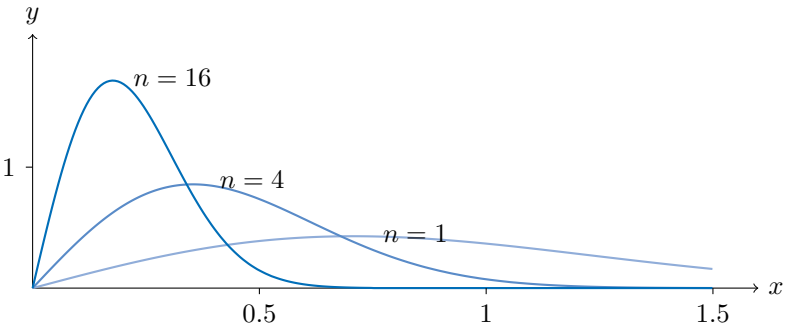


FIGURE 11.3. The partial sums $S_n(x) = nxe^{-nx^2}$ of Example 11.2.4 for $n = 1, 4, 16$. Each tends to 0 at every x, but the maximum $\sqrt{n/(2e)}$ grows without bound.

Theorem 11.2.5 (Cauchy Criterion)**.** *Let $E \subset \mathbb{R}$, and let $\{f_n\}$ be a sequence of functions $f_n : E \to \mathbb{R}$. Then $\{f_n\}$ converges uniformly on E to some function $f : E \to \mathbb{R}$ if and only if for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $m, n \geq N$, $|f_n(x) - f_m(x)| < \varepsilon$ for all $x \in E$.*

PROOF. ($\Rightarrow$) Suppose $\{f_n\}$ converges uniformly on E to f. Let $\varepsilon > 0$. So there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|f_n(x) - f(x)| < \frac{\varepsilon}{2}$ for all $x \in E$. So for all $m, n \geq N$ and all $x \in E$,

$$\begin{aligned} |f_n(x) - f_m(x)| &= |f_n(x) - f(x) + f(x) - f_m(x)| \\ &\leq |f_n(x) - f(x)| + |f(x) - f_m(x)| \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \end{aligned}$$

($\Leftarrow$) For each $x \in E$, the sequence $\{f_n(x)\}$ is Cauchy in $\mathbb{R}$ and therefore, by Theorem 3.5.2, converges. So

$$f(x) = \lim_{n \to \infty} f_n(x)$$

exists for every $x \in E$.

Let $\varepsilon > 0$. So there exists an $N \in \mathbb{N}$ such that for all $m, n \geq N$, $|f_n(x) - f_m(x)| < \frac{\varepsilon}{2}$ for all $x \in E$. Fix $M \geq N$. Then

$$|f(x) - f_M(x)| = \lim_{n\to\infty} |f_n(x) - f_M(x)| \leq \frac{\varepsilon}{2} < \varepsilon$$

for all $x \in E$. The equality holds because $\big||u| - |v|\big| \leq |u - v|$, and the inequality holds by Theorem 3.2.7. Since $M \geq N$ was arbitrary, $\{f_n\}$ converges uniformly to f. □

Like the Cauchy Criterion for sequences, this criterion proves uniform convergence without knowing the limit. The proof of the Weierstrass M-Test below uses it in this way.

Corollary 11.2.6. *Let $E \subset \mathbb{R}$, and let $\{f_k\}$ be a sequence of functions $f_k : E \to \mathbb{R}$. The series $\sum f_k$ converges uniformly on E if and only if for all $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that for all $m > n \geq N$ and for all $x \in E$,*

$$\left| \sum_{k=n+1}^{m} f_k(x) \right| < \varepsilon.$$

Proof. Apply Theorem 11.2.5 to the series $\{S_n\}$ where $S_n(x) = \sum_{k=1}^{n} f_k(x)$. □

Theorem 11.2.7. *Let $E \subset \mathbb{R}$, let $f : E \to \mathbb{R}$, and let $\{f_n\}$ be a sequence of functions $f_n : E \to \mathbb{R}$ that converges pointwise to f on E. For each $n \in \mathbb{N}$, set*

$$M_n = \sup_{x\in E} |f_n(x) - f(x)|.$$

Then $\{f_n\}$ converges uniformly to f on E if and only if $M_n \to 0$. (Some M_n may be ∞. Then $M_n \to 0$ means M_n is finite for all large n and tends to 0.)

Proof. See Exercise 11.2.1. □

Example 11.2.8. Consider $f_n(x) = x^n$, $x \in [0, 1]$. Then $\{f_n\}$ converges pointwise to

$$f(x) = \begin{cases} 0, & x \in [0, 1), \\ 1, & x = 1. \end{cases}$$

Notice

$$M_n = \sup_{x\in[0,1]} |f_n(x) - f(x)| = 1,$$

for all $n \in \mathbb{N}$. So $M_n \to 1$ and $\{f_n\}$ does not converge uniformly to f.

Example 11.2.9. Consider $S_n(x) = nxe^{-nx^2}$, which converges pointwise to $S(x) = 0$. Notice

$$M_n = \sup_{x\in[0,\infty)} |S_n(x) - S(x)| = \sqrt{\frac{n}{2e}}.$$

So $M_n \to \infty$ and $\{S_n\}$ does not converge uniformly to S.

Let $a > 0$ and consider $\{S_n\}$ on $[a, \infty)$. Then since S_n attains its maximum value at $x = \sqrt{\frac{1}{2n}} \to 0$ as $n \to \infty$, S_n attains its maximum value on $[a, \infty)$ at $x = a$ when $\sqrt{\frac{1}{2n}} < a$. This is because S_n is decreasing on $\left[\sqrt{1/(2n)}, \infty\right)$. So

$$M_n = \sup_{x \in [a,\infty)} |S_n(x) - S(x)| = nae^{-na^2} \to 0.$$

Thus $\{S_n\}$ converges uniformly to S on $[a, \infty)$.

Proof Technique (Showing Convergence Is Not Uniform). First find the pointwise limit f. If the convergence is uniform at all, it is uniform to f. Then there are three common ways to show it is not uniform. One is to compute or bound $M_n = \sup_{x \in E} |f_n(x) - f(x)|$ and show $M_n \not\to 0$, as in Examples 11.2.8 and 11.2.9. Another is to find one $\varepsilon > 0$ and, for infinitely many n, a point $x_n \in E$ with $|f_n(x_n) - f(x_n)| \geq \varepsilon$. In Example 11.2.4, the points $x_n = \sqrt{1/(2n)}$ work with $\varepsilon = 1$. The point x_n is allowed to move with n, and it usually has to. The third way applies when every f_n is continuous. If the limit f is not continuous, the convergence is not uniform, by Corollary 11.2.16 below.

Theorem 11.2.10 (Weierstrass M-Test)**.** *Let $E \subset \mathbb{R}$. Suppose $\{f_k\}$ is a sequence of functions $f_k : E \to \mathbb{R}$, and $\{M_k\}$ is a sequence of real numbers satisfying $|f_k(x)| \leq M_k$ for all $x \in E$, $k \in \mathbb{N}$. If $\sum M_k$ converges, then $\sum f_k$ converges uniformly and absolutely on E.*

Proof. The fact that $\sum f_k$ converges absolutely for each $x \in E$ follows from the Comparison Test. Let $S_n(x) = \sum_{k=1}^{n} f_k(x)$. Then, for $m > n$,

$$|S_n(x) - S_m(x)| = \left| \sum_{k=n+1}^{m} f_k(x) \right| \leq \sum_{k=n+1}^{m} |f_k(x)| \leq \sum_{k=n+1}^{m} M_k.$$

Uniform convergence now follows from the Cauchy Criterion. In detail, let $\varepsilon > 0$. By the Cauchy Criterion for series (Theorem 10.2.1), there is an $N \in \mathbb{N}$ with $\sum_{k=n+1}^{m} M_k < \varepsilon$ for all $m > n \geq N$. Then $|S_n(x) - S_m(x)| < \varepsilon$ for all $m > n \geq N$ and all $x \in E$. So $\sum f_k$ converges uniformly on E by Corollary 11.2.6. □

Example 11.2.11. Consider $\sum_{k=1}^{\infty} \frac{\cos kx}{k^p}$, $p > 1$, $x \in \mathbb{R}$. Then

$$\left| \frac{\cos kx}{k^p} \right| \leq \frac{1}{k^p} = M_k.$$

Since $\sum M_k$ is a p-series with $p > 1$, $\sum M_k$ converges. So $\sum_{k=1}^{\infty} \frac{\cos kx}{k^p}$ converges uniformly and absolutely on $\mathbb{R}$. Similarly, $\sum_{k=1}^{\infty} \frac{\sin kx}{k^p}$ converges uniformly and absolutely on $\mathbb{R}$.

Example 11.2.12. Consider the series $\sum_{k=1}^{\infty} \left(\frac{x}{2}\right)^k$. Let $a \in [0, 2)$. For all $x \in [-a, a]$,

$$\left| \left(\frac{x}{2}\right)^k \right| \leq \left(\frac{a}{2}\right)^k = M_k.$$

Since $\frac{a}{2} < 1$, $\sum M_k$ converges. So $\sum_{k=1}^{\infty} \left(\frac{x}{2}\right)^k$ converges uniformly on $[-a, a]$ for any $a \in [0, 2)$. However, an exercise (Exercise 11.2.2) will show this series does not converge uniformly on $(-2, 2)$.

Example 11.2.13. In general, uniform convergence does not imply absolute convergence. Consider the series $\sum_{k=1}^{\infty}(-1)^{k+1}\frac{x^k}{k}$, $x \in [0, 1]$. Set $b_k(x) = \frac{x^k}{k}$. Then, for $x \in [0, 1]$, $b_1(x) \geq b_2(x) \geq \cdots \geq 0$ and $b_k(x) \to 0$. Thus by the Alternating Series Test (Theorem 10.2.17), the series $\sum(-1)^{k+1}b_k(x)$ converges for all $x \in [0, 1]$. Now let $S(x) = \sum_{k=1}^{\infty}(-1)^{k+1}b_k(x)$, $S_n(x) = \sum_{k=1}^{n}(-1)^{k+1}b_k(x)$. Then by Theorem 10.2.20,

$$M_n = \sup_{x\in[0,1]} |S(x) - S_n(x)| \leq \sup_{x\in[0,1]} b_{n+1}(x) = \frac{1}{n+1}.$$

Since $M_n \to 0$, by Theorem 11.2.7, $\{S_n\}$ converges uniformly to S on $[0, 1]$. However, notice if $x = 1$, the series does not converge absolutely.

Example 11.2.14. In general, absolute convergence does not imply uniform convergence. Consider the series $\sum_{k=0}^{\infty} x^2(1 + x^2)^{-k}$. This series converges absolutely to

$$f(x) = \begin{cases} 0, & x = 0, \\ 1 + x^2, & x \neq 0, \end{cases}$$

on all of $\mathbb{R}$. However, Corollary 11.2.16 below will show the convergence is not uniform. (The terms are nonnegative, so convergence and absolute convergence are the same here. See Example 11.1.5.)

The rest of this section answers the three questions of Remark 11.1.3. Uniform convergence lets limits pass through limits, continuity and integrals. For derivatives, the derivatives themselves must converge uniformly.

Theorem 11.2.15. *Let $E \subset \mathbb{R}$, let $f : E \to \mathbb{R}$, and suppose $\{f_n\}$ is a sequence of functions $f_n : E \to \mathbb{R}$ that converges uniformly to f on E. Let c be a limit point of E, and suppose that for each $n \in \mathbb{N}$,*

$$\lim_{x\to c} f_n(x) = A_n.$$

Then the sequence $\{A_n\}$ converges and

$$\lim_{x\to c} f(x) = \lim_{n\to\infty} A_n;$$

i.e.,

$$\lim_{x\to c}\left(\lim_{n\to\infty} f_n(x)\right) = \lim_{n\to\infty}\left(\lim_{x\to c} f_n(x)\right).$$

Proof. The first step shows that the numbers A_n converge. Their limit is the candidate for the limit of f at c. Let $\varepsilon > 0$. By the Cauchy Criterion (Theorem 11.2.5), there is an $N \in \mathbb{N}$ such that $|f_n(x) - f_m(x)| < \varepsilon$ for all $x \in E$ and all $m, n \geq N$. Fix $m, n \geq N$. By the limit laws, $f_n(x) - f_m(x) \to A_n - A_m$ as $x \to c$, and so $|f_n(x) - f_m(x)| \to |A_n - A_m|$, since $\big||u| - |v|\big| \leq |u - v|$. So $|A_n - A_m| \leq \varepsilon$, by Theorem 6.1.15. Thus $\{A_n\}$ is a Cauchy sequence. By Theorem 3.5.2, it converges. Let $A = \lim_{n\to\infty} A_n$.

The second step shows that f has limit A at c. Let $\varepsilon > 0$. For all large n, both $|f(x) - f_n(x)| < \frac{\varepsilon}{3}$ for every $x \in E$, and $|A_n - A| < \frac{\varepsilon}{3}$. Fix one such n. Since $f_n(x) \to A_n$ as $x \to c$, there is a $\delta > 0$ such that $|f_n(x) - A_n| < \frac{\varepsilon}{3}$ for all $x \in E$ with $0 < |x - c| < \delta$. For those x,

$$\begin{aligned} |f(x) - A| &\leq |f(x) - f_n(x)| + |f_n(x) - A_n| + |A_n - A| \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon. \end{aligned}$$

So $\lim_{x \to c} f(x) = A$. □

Proof Technique (The $\varepsilon/3$ Argument). The second half of the proof of Theorem 11.2.15 is a pattern that appears again and again. To show f has a property, approximate f by a function f_n that is known to have it. Then go from f to f_n, use the property of f_n, and come back to the target. That is three steps, so give each step $\frac{\varepsilon}{3}$. The order of choices matters. First choose one n, using uniform convergence so that f_n is close to f at every point at once. Only then use the property of that one f_n. Pointwise convergence is not enough, because the δ for f_n would come first and n would then depend on x. The proofs of Corollary 11.2.16 and Theorem 11.2.18 use the same idea.

Corollary 11.2.16. *Let $E \subset \mathbb{R}$.*

(1) If $\{f_n\}$ is a sequence of continuous functions $f_n : E \to \mathbb{R}$, and if $\{f_n\}$ converges uniformly to f on E, then f is continuous on E.

(2) If $\{f_n\}$ is a sequence of continuous functions $f_n : E \to \mathbb{R}$, and if $\sum f_n$ converges uniformly on E, then $S(x) = \sum f_n(x)$ is continuous on E.

PROOF. (1) If $c \in E$ is an isolated point of E, then f is automatically continuous at c (Remark 6.3.7). Suppose $c \in E$ is a limit point of E. Then since f_n is continuous for each $n \in \mathbb{N}$,

$$\lim_{x \to c} f_n(x) = f_n(c),$$

for all $n \in \mathbb{N}$. Then by Theorem 11.2.15,

$$\lim_{x \to c} f(x) = \lim_{x \to c} \left(\lim_{n \to \infty} f_n(x) \right) = \lim_{n \to \infty} \left(\lim_{x \to c} f_n(x) \right) = \lim_{n \to \infty} f_n(c) = f(c).$$

So f is continuous at c.

(2) Let $S_n(x) = \sum_{k=1}^{n} f_k(x)$. Then for each $n \in \mathbb{N}$, S_n is continuous on E. Since $\{S_n\}$ converges uniformly to S on E, by (1), S is continuous on E. □

So the series of Examples 11.1.5 and 11.2.14 does not converge uniformly on $\mathbb{R}$, because its sum is not continuous at 0. In the same way, x^n does not converge uniformly on $[0, 1]$.

The converse of Corollary 11.2.16 is false. The limit can be continuous even when the convergence is not uniform, as Example 11.2.4 shows. Dini's Theorem gives a case where it is true. On a compact set, a decreasing sequence of continuous functions with a continuous limit converges uniformly.

Theorem 11.2.17 (Dini). *Suppose $K \subset \mathbb{R}$ is compact and $\{f_n\}$ is a sequence of continuous functions $f_n : K \to \mathbb{R}$ such that $\{f_n\}$ converges pointwise on K to a continuous function f, and $f_n(x) \geq f_{n+1}(x)$ for all $x \in K$ and for all $n \in \mathbb{N}$. Then $\{f_n\}$ converges uniformly to f on K.*

PROOF. For each $n \in \mathbb{N}$, let $g_n = f_n - f$. Each g_n is continuous on K. Also $g_n(x) \geq g_{n+1}(x) \geq 0$ and $g_n(x) \to 0$ for every $x \in K$. It is enough to show $g_n \to 0$ uniformly on K.

Let $\varepsilon > 0$. For each $n \in \mathbb{N}$, let

$$K_n = \{x \in K : g_n(x) \geq \varepsilon\},$$

the set of points where g_n is still not within ε of 0. The goal is to show $K_N = \emptyset$ for some N.

Each K_n is closed. To see this, let c be a limit point of K_n. By Theorem 4.1.23, there is a sequence $\{x_k\}$ in K_n with $x_k \to c$. Then $c \in K$, since K is closed (Theorem 4.2.9). Since g_n is continuous and $g_n(x_k) \geq \varepsilon$ for all k, $g_n(c) = \lim_{k\to\infty} g_n(x_k) \geq \varepsilon$. So $c \in K_n$. By Theorem 4.1.24, K_n is closed. As a closed subset of K, it is compact, by Lemma 4.2.10.

The sets are nested. If $x \in K_{n+1}$, then $g_n(x) \geq g_{n+1}(x) \geq \varepsilon$, so $x \in K_n$. So $K_{n+1} \subset K_n$.

The sets have empty intersection. For each $x \in K$, $g_n(x) \to 0$, so $g_n(x) < \varepsilon$ for some n, and then $x \notin K_n$.

If every K_n were nonempty, Cantor's Intersection Theorem (Theorem 4.2.12) would make $\bigcap_{n=1}^{\infty} K_n$ nonempty. So $K_N = \emptyset$ for some $N \in \mathbb{N}$. Since the sets are nested, $K_n = \emptyset$ for all $n \geq N$. This means $0 \leq g_n(x) < \varepsilon$, that is, $|f_n(x) - f(x)| < \varepsilon$, for all $x \in K$ and all $n \geq N$. So $\{f_n\}$ converges uniformly to f on K. □

All three hypotheses matter. On $[0, 1)$, which is not compact, $f_n(x) = x^n$ decreases to the continuous function 0, but not uniformly. On $[0, 1]$, the same sequence decreases to a function that is not continuous. See Exercise 11.2.8 for the monotonicity hypothesis. The theorem also holds for increasing sequences. Apply it to $\{-f_n\}$.

Theorem 11.2.18. *Suppose $\{f_n\}$ is a sequence of functions $f_n : [a, b] \to \mathbb{R}$ with $f_n \in \mathcal{R}[a, b]$ for all $n \in \mathbb{N}$. Also, suppose the sequence $\{f_n\}$ converges uniformly to f on $[a, b]$. Then $f \in \mathcal{R}[a, b]$ and*

$$\int_a^b f = \lim_{n\to\infty} \int_a^b f_n;$$

i.e.,

$$\lim_{n\to\infty} \int_a^b f_n = \int_a^b \lim_{n\to\infty} f_n.$$

PROOF. For each $n \in \mathbb{N}$, let $\varepsilon_n = \sup_{x\in[a,b]} |f_n(x) - f(x)|$. By Theorem 11.2.7, $\varepsilon_n \to 0$. In particular, ε_n is finite for all large n. For such n, $|f(x)| \leq |f_n(x)| + \varepsilon_n$ for all x, so f is bounded, since f_n is.

With f bounded, the next step shows that f is integrable. Let $\varepsilon > 0$. Choose n with $\varepsilon_n < \frac{\varepsilon}{3(b-a)}$. Since $f_n \in \mathcal{R}[a, b]$, Riemann's Criterion (Theorem 9.2.1) gives a

partition $\mathcal{P} = \{x_0, \ldots, x_k\}$ of $[a,b]$ with $\mathcal{U}(\mathcal{P}, f_n) - \mathcal{L}(\mathcal{P}, f_n) < \frac{\varepsilon}{3}$. On each $[x_{i-1}, x_i]$, $f_n(x) - \varepsilon_n \le f(x) \le f_n(x) + \varepsilon_n$. So the supremum of f there is at most the supremum of f_n plus ε_n, and the infimum of f there is at least the infimum of f_n minus ε_n. Summing over i,

$$\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) \le \mathcal{U}(\mathcal{P}, f_n) - \mathcal{L}(\mathcal{P}, f_n) + 2\varepsilon_n(b-a) < \frac{\varepsilon}{3} + \frac{2\varepsilon}{3} = \varepsilon.$$

So $f \in \mathcal{R}[a,b]$ by Riemann's Criterion.

It remains to show that the integrals converge. For all large n, by Theorems 9.2.11 and 9.2.12 and Exercise 9.2.3,

$$\left| \int_a^b f - \int_a^b f_n \right| = \left| \int_a^b (f - f_n) \right| \le \int_a^b |f - f_n| \le \varepsilon_n (b-a).$$

Since $\varepsilon_n \to 0$, $\lim_{n\to\infty} \int_a^b f_n = \int_a^b f$. □

Corollary 11.2.19. *If $f_k \in \mathcal{R}[a,b]$ for all $k \in \mathbb{N}$ and if $S(x) = \sum f_k(x)$, $x \in [a,b]$, where the series converges uniformly on $[a,b]$, then*

$$\int_a^b S = \sum \int_a^b f_k.$$

Proof. Let $S_n = \sum_{k=1}^n f_k$. By Theorem 9.2.11, $S_n \in \mathcal{R}[a,b]$ and $\int_a^b S_n = \sum_{k=1}^n \int_a^b f_k$. These are the partial sums of $\sum \int_a^b f_k$. By Theorem 11.2.18, they converge to $\int_a^b S$. □

Example 11.2.20. Notice even though for $f_n(x) = x^n$, $\{f_n\}$ converges pointwise and not uniformly to

$$f(x) = \begin{cases} 0, & x \in [0,1), \\ 1, & x = 1, \end{cases}$$

it is still true that

$$\lim_{n\to\infty} \int_0^1 f_n = \lim_{n\to\infty} \frac{1}{n+1} = 0 = \int_0^1 f.$$

So uniform convergence is sufficient for the conclusion of Theorem 11.2.18, but not necessary. The next theorem gives a different sufficient condition.

Theorem 11.2.21 (Bounded Convergence Theorem)**.** *Let $\{f_n\}$ be a sequence of functions $f_n : [a,b] \to \mathbb{R}$ converging pointwise to f on $[a,b]$ with $f_n \in \mathcal{R}[a,b]$ for all $n \in \mathbb{N}$ and $f \in \mathcal{R}[a,b]$. Suppose there exists a positive constant M such that $|f_n(x)| \le M$ for all $x \in [a,b]$ and for all $n \in \mathbb{N}$. Then*

$$\int_a^b f = \lim_{n\to\infty} \int_a^b f_n.$$

This theorem is due to Cesare Arzelà. Its proof using only Riemann integrals is long. It follows quickly from the Bounded Convergence Theorem for the Lebesgue integral, Theorem 16.3.6, since Riemann integrable functions are Lebesgue integrable with the same integral. See Corollary 16.3.8. The hypothesis $f \in \mathcal{R}[a,b]$ cannot be dropped, as Example 11.1.6 shows. Without the bound M, the conclusion can fail, as Example 11.1.7 shows.

Theorem 11.2.22. *Suppose $\{f_n\}$ is a sequence of differentiable functions on $[a,b]$. If $\{f_n'\}$ converges uniformly on $[a,b]$ and $\{f_n(x_0)\}$ converges for some $x_0 \in [a,b]$, then $\{f_n\}$ converges uniformly to a function f on $[a,b]$ with*

$$f'(x) = \lim_{n\to\infty} f_n'(x)$$

for all $x \in [a,b]$.

PROOF. The proof starts with an estimate that controls $f_n - f_m$ on all of $[a,b]$. Let $\varepsilon > 0$. Since $\{f_n(x_0)\}$ converges and $\{f_n'\}$ converges uniformly, there is an $N \in \mathbb{N}$ such that for all $m, n \geq N$,

$$|f_n(x_0) - f_m(x_0)| < \frac{\varepsilon}{2} \quad \text{and} \quad |f_n'(t) - f_m'(t)| < \frac{\varepsilon}{2(b-a)} \text{ for all } t \in [a,b].$$

Fix $m, n \geq N$, and let $h = f_n - f_m$. Then h is differentiable on $[a,b]$ and $|h'(t)| < \frac{\varepsilon}{2(b-a)}$ for all t. Let $x, y \in [a,b]$ with $x \neq y$. By the Mean Value Theorem, $h(x) - h(y) = h'(t)(x-y)$ for some t between x and y. So

$$|h(x) - h(y)| \leq \frac{\varepsilon}{2(b-a)}|x-y| \quad \text{for all } x, y \in [a,b]. \tag{11.2.1}$$

This also holds when $x = y$.

The estimate first shows that $\{f_n\}$ converges uniformly. Take $y = x_0$ in (11.2.1). Since $|x - x_0| \leq b - a$, for all $x \in [a,b]$,

$$|f_n(x) - f_m(x)| = |h(x)| \leq |h(x) - h(x_0)| + |h(x_0)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

This holds for all $m, n \geq N$. So by Theorem 11.2.5, $\{f_n\}$ converges uniformly on $[a,b]$ to a function f.

It remains to compute the derivative of f. Fix $c \in [a,b]$ and let $E = [a,b] \setminus \{c\}$. For $x \in E$, define

$$q_n(x) = \frac{f_n(x) - f_n(c)}{x - c} \quad \text{and} \quad q(x) = \frac{f(x) - f(c)}{x - c}.$$

Then $q_n(x) \to q(x)$ for every $x \in E$, and $\lim_{x\to c} q_n(x) = f_n'(c)$ for each n. Taking $y = c$ in (11.2.1), for all $m, n \geq N$ and all $x \in E$,

$$|q_n(x) - q_m(x)| = \frac{|h(x) - h(c)|}{|x - c|} \leq \frac{\varepsilon}{2(b-a)}.$$

Since $\varepsilon > 0$ was arbitrary, $\{q_n\}$ converges uniformly on E, by Theorem 11.2.5. Its limit is q. The point c is a limit point of E. So by Theorem 11.2.15,

$$f'(c) = \lim_{x\to c} q(x) = \lim_{n\to\infty} \left(\lim_{x\to c} q_n(x) \right) = \lim_{n\to\infty} f_n'(c). \qquad \square$$

Uniform convergence of $\{f_n\}$ alone is not enough. In Example 11.1.8, $\left|\frac{\sin nx}{n}\right| \leq \frac{1}{n}$, so $f_n \to 0$ uniformly on $\mathbb{R}$. But $f_n'(0) = 1$ for all n.

Exercises

11.2.1. Prove Theorem 11.2.7. [Hint, p. 356]

11.2.2. Show the series $\sum_{k=1}^{\infty} \left(\frac{x}{2}\right)^k$ does not converge uniformly on $(-2, 2)$. [Hint, p. 356]

11.2.3. For $n \in \mathbb{N}$, let $f_n(x) = x^n(1 + x^n)$, $x \in [0, 1]$. Prove the sequence $\{f_n\}$ does not converge uniformly on $[0, 1]$. [Hint, p. 357]

11.2.4. Prove the series $\sum_{k=0}^{\infty} x(1 - x)^k$ does not converge uniformly for $0 \leq x \leq 1$. [Hint, p. 357]

11.2.5. If $\sum_{k=0}^{\infty} a_k$ converges absolutely, prove that $\sum_{k=0}^{\infty} a_k x^k$ converges uniformly on $[-1, 1]$. [Hint, p. 357]

11.2.6. If $f \in \mathcal{R}[0, 1]$, use the Bounded Convergence Theorem to prove that

$$\lim_{n\to\infty} \int_0^1 x^n f(x)\, dx = 0.$$

[Hint, p. 357]

11.2.7. Suppose $\{f_n\}$ is a sequence of continuous functions on $[a, b]$ that converges uniformly on $[a, b]$. For each $x \in [a, b]$, set $g(x) = \sup_{n\in\mathbb{N}}\{f_n(x)\}$.
(a) Prove that g is continuous on $[a, b]$.
(b) Show by example that this conclusion may be false if the sequence $\{f_n\}$ converges only pointwise on $[a, b]$.

[Hint, p. 357]

11.2.8. Find continuous functions $f_n : [0, 1] \to \mathbb{R}$ that converge pointwise to 0, with $0 \leq f_n(x) \leq 1$, but not uniformly. (So Dini's Theorem needs the monotonicity hypothesis.) [Hint, p. 357]

11.2.9. Give a second proof of Dini's Theorem, by contradiction. If the convergence is not uniform, find an $\varepsilon > 0$ and points $x_n \in K$ with $g_n(x_n) \geq \varepsilon$ for all n. Use the Bolzano–Weierstrass Theorem to get a subsequence $x_{n_j} \to c \in K$. Then show $g_m(c) \geq \varepsilon$ for every m.

11.2.10. Prove Theorem 11.2.22 under the stronger hypothesis that each f_n' is continuous. [Hint, p. 357]

11.2.11. Suppose each $f_n : E \to \mathbb{R}$ is bounded and $f_n \to f$ uniformly on E. Prove that f is bounded, and that there is one $M > 0$ with $|f_n(x)| \leq M$ for all $n \in \mathbb{N}$ and all $x \in E$. [Hint, p. 357]

11.3. The Weierstrass Function*

This section can be skipped. Nothing later in the book depends on it.

A continuous function can fail to be differentiable at a point, as $|x|$ does at 0. It can fail at infinitely many points, as a zigzag does at each corner. For much of the nineteenth century, many mathematicians believed that a continuous function must still be differentiable at most points. Weierstrass's example shows this is false. His function is continuous on all of $\mathbb{R}$ and differentiable at no point. It is the sum of a uniformly convergent series of cosine waves. Each wave oscillates much faster than the one before, and its height is half as large. See Figure 11.4.

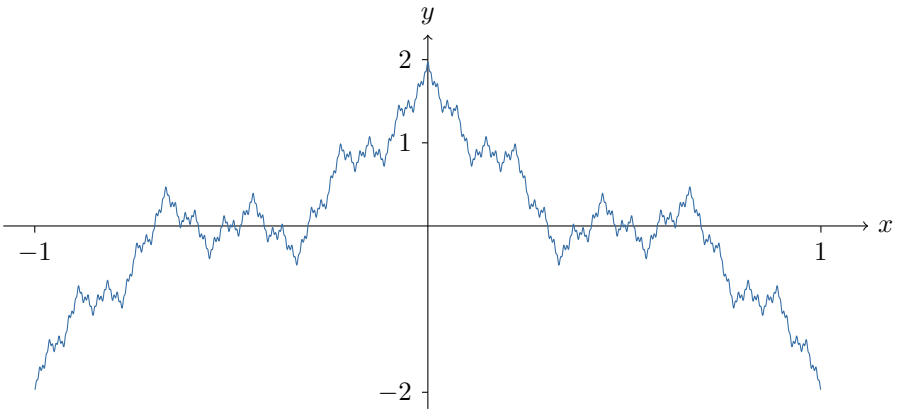


FIGURE 11.4. The partial sum $\sum_{k=0}^{5} \frac{\cos(3^k \pi x)}{2^k}$. This uses $a = 3$, which is easier to draw than the values $a > 3\pi + 2$ in Example 11.3.1. The full sum with $a = 3$ is also differentiable nowhere, by a theorem of Hardy.

Example 11.3.1. Consider the function

$$f(x) = \sum_{k=0}^{\infty} \frac{\cos(a^k \pi x)}{2^k},$$

where a is an odd positive integer satisfying $a > 3\pi + 2$. For example, $a = 13$ works.

Since $\left|\frac{\cos(a^k \pi x)}{2^k}\right| \leq \frac{1}{2^k}$ and $\sum \frac{1}{2^k}$ converges, by the Weierstrass M-Test, $\sum \frac{\cos(a^k \pi x)}{2^k}$ converges uniformly and absolutely, and hence by Corollary 11.2.16, f is continuous on $\mathbb{R}$.

Now fix $x \in \mathbb{R}$. The rest of the example shows f is not differentiable at x. For $n \in \mathbb{N}$, set

$$S_n(x) = \sum_{k=0}^{n-1} \frac{\cos(a^k \pi x)}{2^k},$$

and

$$R_n(x) = \sum_{k=n}^{\infty} \frac{\cos(a^k \pi x)}{2^k}.$$

Then for $h > 0$,

$$\frac{f(x+h) - f(x)}{h} = \frac{S_n(x+h) - S_n(x)}{h} + \frac{R_n(x+h) - R_n(x)}{h}.$$

The first step shows that the S_n part is not too large. By the Mean Value Theorem, for each k there exists a $\xi \in (0, h)$ with

$$\frac{\cos(a^k \pi(x+h)) - \cos(a^k \pi x)}{h} = -a^k \pi \sin(a^k \pi(x+\xi)).$$

Since

$$|-a^k \pi \sin(a^k \pi(x+\xi))| \leq a^k \pi,$$

$$\begin{aligned}\left|\frac{S_n(x+h)-S_n(x)}{h}\right| &\le \sum_{k=0}^{n-1}\frac{1}{2^k}\left|\frac{\cos(a^k\pi(x+h))-\cos(a^k\pi x)}{h}\right| \\ &\le \pi\sum_{k=0}^{n-1}\left(\frac{a}{2}\right)^k \\ &= \frac{\pi\left(1-\left(\frac{a}{2}\right)^n\right)}{1-\frac{a}{2}} \\ &< \frac{2\pi}{a-2}\left(\frac{a}{2}\right)^n.\end{aligned}$$

The second step chooses h so that the R_n part is large. Write $a^n x = k_n + \delta_n$, where $k_n \in \mathbb{Z}$ and $-\frac{1}{2} \le \delta_n < \frac{1}{2}$. Set

$$h_n = \frac{1-\delta_n}{a^n}.$$

Since $-\frac{1}{2} \le \delta_n < \frac{1}{2}$, $\frac{3}{2} \ge 1-\delta_n > \frac{1}{2}$. Therefore,

$$\frac{2}{3}a^n \le \frac{1}{h_n} < 2a^n.$$

In particular $h_n > 0$ and $h_n \to 0$ as $n \to \infty$.

With this choice of h, the third step shows that the R_n part is large. For $k \ge n$,

$$\begin{aligned}a^k\pi(x+h_n) &= a^{k-n}a^n\pi(x+h_n) \\ &= a^{k-n}\pi(a^n x + (1-\delta_n)) \\ &= a^{k-n}\pi(k_n+1).\end{aligned}$$

Since a^{k-n} is odd and k_n is an integer,

$$\cos(a^k\pi(x+h_n)) = \cos(a^{k-n}\pi(k_n+1)) = (-1)^{k_n+1}.$$

Also,

$$a^k\pi x = a^{k-n}a^n\pi x = a^{k-n}\pi(k_n+\delta_n).$$

Using the trigonometric identity

$$\cos(A+B) = \cos A\cos B - \sin A\sin B,$$

and the fact that $\sin(k_n a^{k-n}\pi) = 0$,

$$\begin{aligned}\cos(a^k\pi x) &= \cos(a^{k-n}\pi k_n + a^{k-n}\pi\delta_n) \\ &= \cos(a^{k-n}k_n\pi)\cos(a^{k-n}\delta_n\pi) \\ &= (-1)^{k_n}\cos(a^{k-n}\delta_n\pi).\end{aligned}$$

Therefore,

$$\begin{aligned}\cos(a^k\pi(x+h_n)) - \cos(a^k\pi x) &= (-1)^{k_n+1} - (-1)^{k_n}\cos(a^{k-n}\delta_n\pi) \\ &= (-1)^{k_n+1}(1+\cos(a^{k-n}\delta_n\pi)).\end{aligned}$$

So

$$\left|\frac{R_n(x+h_n)-R_n(x)}{h_n}\right| = \left|\sum_{k=n}^{\infty}\frac{1}{2^k}\cdot\frac{\cos(a^k\pi(x+h_n))-\cos(a^k\pi x)}{h_n}\right|$$

$$= \left| \sum_{k=n}^{\infty} \frac{(-1)^{k_n+1}}{h_n} \cdot \frac{1+\cos(a^{k-n}\delta_n\pi)}{2^k} \right|$$
$$= \frac{1}{h_n} \sum_{k=n}^{\infty} \frac{1+\cos(a^{k-n}\delta_n\pi)}{2^k}$$
$$\geq \frac{1}{h_n} \cdot \frac{1+\cos(\delta_n\pi)}{2^n}.$$

The last step keeps only the term $k = n$. The other terms are nonnegative. Since $-\frac{1}{2} \leq \delta_n < \frac{1}{2}$, $\cos(\delta_n\pi) \geq 0$. Therefore,

$$\left| \frac{R_n(x+h_n) - R_n(x)}{h_n} \right| \geq \frac{1}{h_n} \cdot \frac{1}{2^n} \geq \frac{2}{3} \left(\frac{a}{2}\right)^n.$$

The last step puts the two parts together. Therefore, by using the Reverse Triangle Inequality,

$$\left| \frac{f(x+h_n) - f(x)}{h_n} \right| \geq \left| \frac{R_n(x+h_n) - R_n(x)}{h_n} \right| - \left| \frac{S_n(x+h_n) - S_n(x)}{h_n} \right|$$
$$\geq \frac{2}{3} \left(\frac{a}{2}\right)^n - \frac{2\pi}{a-2} \left(\frac{a}{2}\right)^n$$
$$= \left(\frac{a}{2}\right)^n \left(\frac{2}{3} - \frac{2\pi}{a-2}\right).$$

Since $\frac{a}{2} > 1$ and $\frac{2}{3} - \frac{2\pi}{a-2} > 0$,

$$\left| \frac{f(x+h_n) - f(x)}{h_n} \right| \to \infty$$

as $n \to \infty$. If $f'(x)$ existed, then by Theorem 6.1.18 these difference quotients would converge to $f'(x)$, since $h_n \to 0$ and $h_n \neq 0$. A convergent sequence is bounded. So f is not differentiable at x. Since $x \in \mathbb{R}$ was arbitrary, f is differentiable at no point of $\mathbb{R}$.

11.4. The Weierstrass Approximation Theorem*

Section 11.3 built a continuous function with no derivative anywhere. This section proves that even such a function is a uniform limit of polynomials, the smoothest functions there are. The proof, due to Sergei Bernstein, writes down the polynomials explicitly. This section can be skipped. Nothing later depends on it.

Definition 11.4.1. Let $f : [0,1] \to \mathbb{R}$ and $n \in \mathbb{N}$. The *Bernstein polynomial* of f of degree n is

$$B_n f(x) = \sum_{k=0}^{n} f\left(\frac{k}{n}\right) \binom{n}{k} x^k (1-x)^{n-k}.$$

Write $p_{n,k}(x) = \binom{n}{k} x^k (1-x)^{n-k}$. For $x \in [0,1]$, these numbers are at least 0. So $B_n f(x)$ is a weighted average of the values $f\left(\frac{k}{n}\right)$. The next lemma shows the weights add to 1 and are concentrated where $\frac{k}{n}$ is close to x.

Lemma 11.4.2. *For $n \in \mathbb{N}$ and $x \in \mathbb{R}$,*

$$\sum_{k=0}^{n} p_{n,k}(x) = 1, \qquad \sum_{k=0}^{n} k p_{n,k}(x) = nx, \qquad \sum_{k=0}^{n} (k-nx)^2 p_{n,k}(x) = nx(1-x).$$

PROOF. The first sum is $(x + (1-x))^n = 1$, by the Binomial Theorem (Exercise 9.3.3). For $k \geq 1$, $k\binom{n}{k} = n\binom{n-1}{k-1}$. So

$$\sum_{k=0}^{n} k p_{n,k}(x) = nx \sum_{k=1}^{n} \binom{n-1}{k-1} x^{k-1}(1-x)^{n-k} = nx\big(x + (1-x)\big)^{n-1} = nx.$$

In the same way, $k(k-1)\binom{n}{k} = n(n-1)\binom{n-2}{k-2}$ for $k \geq 2$, so $\sum k(k-1)p_{n,k}(x) = n(n-1)x^2$. (When $n = 1$ both sides are 0.) Then

$$\sum_{k=0}^{n} (k-nx)^2 p_{n,k} = \sum k(k-1)p_{n,k} + \sum k p_{n,k} - 2nx \sum k p_{n,k} + n^2x^2 \sum p_{n,k},$$

which is $n(n-1)x^2 + nx - 2n^2x^2 + n^2x^2 = nx - nx^2$. □

Theorem 11.4.3 (Weierstrass Approximation Theorem)**.** *Let $f : [a,b] \to \mathbb{R}$ be continuous. For every $\varepsilon > 0$, there is a polynomial p with $|f(x) - p(x)| < \varepsilon$ for all $x \in [a,b]$. If $[a,b] = [0,1]$, then $B_n f \to f$ uniformly on $[0,1]$.*

PROOF. First take $[a,b] = [0,1]$. By Theorem 6.3.15, there is an M with $|f(x)| \leq M$ for all x. Let $\varepsilon > 0$. By Corollary 6.4.8, f is uniformly continuous, so there is a $\delta > 0$ with $|f(s) - f(t)| < \frac{\varepsilon}{2}$ whenever $|s - t| < \delta$. Let $x \in [0,1]$. By Lemma 11.4.2,

$$|B_n f(x) - f(x)| = \left| \sum_{k=0}^{n} \left(f\left(\tfrac{k}{n}\right) - f(x) \right) p_{n,k}(x) \right| \leq \sum_{k=0}^{n} \left| f\left(\tfrac{k}{n}\right) - f(x) \right| p_{n,k}(x).$$

Split the sum. When $\left|\frac{k}{n} - x\right| < \delta$, the term is at most $\frac{\varepsilon}{2} p_{n,k}(x)$, and these terms add to at most $\frac{\varepsilon}{2}$. When $\left|\frac{k}{n} - x\right| \geq \delta$, the term is at most $2M p_{n,k}(x)$, and $1 \leq \frac{(k-nx)^2}{n^2\delta^2}$. So these terms add to at most

$$\frac{2M}{n^2\delta^2} \sum_{k=0}^{n} (k-nx)^2 p_{n,k}(x) = \frac{2Mx(1-x)}{n\delta^2} \leq \frac{M}{2n\delta^2}.$$

If $n > \frac{M}{\varepsilon\delta^2}$, then $|B_n f(x) - f(x)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon$ for every $x \in [0,1]$. So $B_n f \to f$ uniformly.

For a general $[a,b]$, let $g(t) = f(a + (b-a)t)$ for $t \in [0,1]$. It is continuous, so there is a polynomial q with $|g(t) - q(t)| < \varepsilon$ on $[0,1]$. Let $p(x) = q\left(\frac{x-a}{b-a}\right)$, which is a polynomial. For $x \in [a,b]$, $t = \frac{x-a}{b-a} \in [0,1]$ and $|f(x) - p(x)| = |g(t) - q(t)| < \varepsilon$. □

Compare this with Taylor polynomials. The function f of Example 12.2.10 has derivatives of every order, but its Taylor polynomials at 0 are all 0. They do not approximate f at all. Bernstein polynomials need only continuity. They converge slowly, though. For $f(x) = x^2$, $B_n f(x) = x^2 + \frac{x(1-x)}{n}$ (Exercise 11.4.1).

Exercises

11.4.1. Find $B_n f$ for $f(x) = 1$, $f(x) = x$ and $f(x) = x^2$. [Solution, p. 358]

11.4.2. Let $f : [0,1] \to \mathbb{R}$ be continuous with $\int_0^1 f(x)x^n\,dx = 0$ for all $n \in \mathbb{N}_0$. Prove that $f(x) = 0$ for all $x \in [0,1]$. [Solution, p. 358]

11.4.3. Prove that there is no sequence of polynomials that converges uniformly to e^x on all of $\mathbb{R}$. [Hint, p. 358]

11.4.4. Show that the conclusion of Theorem 11.4.3 fails for $f(x) = \frac{1}{x}$ on $(0,1)$. [Hint, p. 358]

11.4.5. Let $f : [a,b] \to \mathbb{R}$ have a continuous derivative. Prove that there are polynomials p_n with $p_n \to f$ and $p_n' \to f'$ uniformly on $[a,b]$. [Hint, p. 358]

11.5. Picard's Theorem*

A differential equation $y' = f(t,y)$ with $y(t_0) = y_0$ asks for a function whose derivative at each t depends on t and on the value of the function. This section proves that, under a mild condition on f, there is exactly one solution. The proof builds the solution as a uniform limit, using the Weierstrass M-Test and Theorem 11.2.18. For the equation $y' = y$, the approximations are exactly the partial sums of the series for e^t. This section can be skipped. Nothing later depends on it.

Throughout, $a < b$, $t_0 \in [a,b]$ and $y_0 \in \mathbb{R}$. The function f is defined for $t \in [a,b]$ and $y \in \mathbb{R}$, and it satisfies two conditions.

(1) For each $y \in \mathbb{R}$, the function $t \mapsto f(t,y)$ is continuous on $[a,b]$.
(2) There is a constant L with $|f(t,y) - f(t,z)| \le L|y - z|$ for all $t \in [a,b]$ and $y, z \in \mathbb{R}$.

The second condition says f is Lipschitz in y, uniformly in t. When $t < t_0$, $\int_{t_0}^t$ means $-\int_t^{t_0}$.

Lemma 11.5.1. *(1) If $y : [a,b] \to \mathbb{R}$ is continuous, then $t \mapsto f(t, y(t))$ is continuous on $[a,b]$.*
(2) A continuous function $y : [a,b] \to \mathbb{R}$ is differentiable with $y'(t) = f(t,y(t))$ for all $t \in [a,b]$ and $y(t_0) = y_0$ if and only if

$$y(t) = y_0 + \int_{t_0}^t f(s, y(s))\,ds \quad \text{for all } t \in [a,b].$$

Proof. (1) Let $t \in [a,b]$. For $s \in [a,b]$,

$$\begin{aligned} |f(s,y(s)) - f(t,y(t))| &\le |f(s,y(s)) - f(s,y(t))| + |f(s,y(t)) - f(t,y(t))| \\ &\le L|y(s) - y(t)| + |f(s,y(t)) - f(t,y(t))|. \end{aligned}$$

As $s \to t$, the first term tends to 0 since y is continuous. The second tends to 0 by condition (1), applied with the fixed number $y(t)$.

(2) If y satisfies the integral equation, then $y(t_0) = y_0$, and $y'(t) = f(t,y(t))$ by (1) and Theorem 9.3.7. Conversely, if $y'(t) = f(t,y(t))$, then y' is continuous by (1), and Theorem 9.3.6 gives $y(t) - y(t_0) = \int_{t_0}^t y'(s)\,ds$. □

Theorem 11.5.2 (Picard's Theorem)**.** *There is exactly one differentiable function $y : [a, b] \to \mathbb{R}$ with $y'(t) = f(t, y(t))$ for all $t \in [a, b]$ and $y(t_0) = y_0$.*

PROOF. The proof builds approximate solutions one at a time. Let $\varphi_0(t) = y_0$ for all t, and for $n \in \mathbb{N}_0$ let

$$\varphi_{n+1}(t) = y_0 + \int_{t_0}^{t} f(s, \varphi_n(s))\, ds.$$

By Lemma 11.5.1 (1) and Theorem 9.3.7, each φ_n is continuous, so each step makes sense. These are the *Picard iterates*.

The next step bounds the distance between successive approximations. The function $t \mapsto f(t, y_0)$ is continuous on $[a, b]$, so it is bounded, say by M (Theorem 6.3.15). Induction on n shows that

$$|\varphi_{n+1}(t) - \varphi_n(t)| \le \frac{ML^n|t - t_0|^{n+1}}{(n+1)!} \quad \text{for all } t \in [a, b].$$

For $n = 0$, $|\varphi_1(t) - y_0| = \left|\int_{t_0}^{t} f(s, y_0)\, ds\right| \le M|t - t_0|$. If the estimate holds for n, then by condition (2),

$$\begin{aligned} |\varphi_{n+2}(t) - \varphi_{n+1}(t)| &\le \left|\int_{t_0}^{t} L|\varphi_{n+1}(s) - \varphi_n(s)|\, ds\right| \\ &\le \left|\int_{t_0}^{t} \frac{ML^{n+1}|s - t_0|^{n+1}}{(n+1)!}\, ds\right| = \frac{ML^{n+1}|t - t_0|^{n+2}}{(n+2)!}. \end{aligned}$$

These bounds make the approximations converge uniformly. Let $h = b - a$. Then $|\varphi_{n+1}(t) - \varphi_n(t)| \le \frac{ML^n h^{n+1}}{(n+1)!}$ for all t. These bounds form a convergent series, by the Ratio Test. By Theorem 11.2.10, $\varphi_n = y_0 + \sum_{k=0}^{n-1}(\varphi_{k+1} - \varphi_k)$ converges uniformly on $[a, b]$ to a function y, which is continuous by Corollary 11.2.16.

The next step shows that the limit y is a solution. By condition (2), $|f(s, \varphi_n(s)) - f(s, y(s))| \le L|\varphi_n(s) - y(s)|$, so $f(s, \varphi_n(s)) \to f(s, y(s))$ uniformly. By Theorem 11.2.18, letting $n \to \infty$ in the definition of φ_{n+1} gives $y(t) = y_0 + \int_{t_0}^{t} f(s, y(s))\, ds$. By Lemma 11.5.1 (2), y is a solution.

It remains to show that y is the only solution. Let z be another solution, and let $w(t) = |y(t) - z(t)|$. By Lemma 11.5.1, w is continuous, and by condition (2),

$$w(t) \le \left|\int_{t_0}^{t} Lw(s)\, ds\right|.$$

Let K be the maximum of w on $[a, b]$. Then $w(t) \le KL|t - t_0|$, and by induction $w(t) \le \frac{KL^n|t-t_0|^n}{n!} \le \frac{K(Lh)^n}{n!}$ for every n. This tends to 0 by Exercise 10.2.5. So $w = 0$ and $z = y$. □

Example 11.5.3. Take $f(t, y) = y$, $t_0 = 0$ and $y_0 = 1$ on any interval $[a, b]$ containing 0. Here $L = 1$. The Picard iterates are $\varphi_0(t) = 1$, $\varphi_1(t) = 1 + \int_0^t 1\, ds = 1 + t$, $\varphi_2(t) = 1 + \int_0^t (1 + s)\, ds = 1 + t + \frac{t^2}{2}$, and by induction

$$\varphi_n(t) = \sum_{k=0}^{n} \frac{t^k}{k!}.$$

These are the partial sums of the series for e^t in Section 12.5. So Picard's Theorem builds e^t a third way, as the unique solution of $y' = y$ with $y(0) = 1$.

The Lipschitz condition cannot simply be dropped. The equation $y' = 3y^{2/3}$ with $y(0) = 0$ has two solutions (Exercise 11.5.2). Without the Lipschitz condition, continuity of f in both variables still gives a solution near t_0, by a theorem of Giuseppe Peano, but the proof needs more than this book develops. The same method also works for systems of equations, such as $y' = v$, $v' = -y$. That system is the equation $y'' + y = 0$ of Sections 9.5 and 12.4, and Lemma 12.4.4 is its uniqueness statement.

Exercises

11.5.1. Find the Picard iterates for $y' = 2ty$, $y(0) = 1$. What is the solution? [Solution, p. 358]

11.5.2. Show that $y(t) = 0$ and $y(t) = t^3$ both solve $y' = 3y^{2/3}$, $y(0) = 0$. Which hypothesis of Theorem 11.5.2 fails? [Solution, p. 358]

11.5.3. Let $g : [a, b] \to [a, b]$ satisfy $|g(x) - g(y)| \le k|x - y|$ for all x, y, where $0 \le k < 1$. Let $x_1 \in [a, b]$ and $x_{n+1} = g(x_n)$. Prove that $\{x_n\}$ converges to a point c with $g(c) = c$, and that g has no other fixed point. [Hint, p. 358]

11.5.4. In the proof of Theorem 11.5.2, prove that $|y(t) - \varphi_n(t)| \le \frac{M(Lh)^n h}{n!} e^{Lh}$ for all $t \in [a, b]$. [Hint, p. 358]

11.5.5. Let $p, q : [a, b] \to \mathbb{R}$ be continuous. Prove that $y' = p(t)y + q(t)$, $y(t_0) = y_0$ has exactly one solution on $[a, b]$. [Hint, p. 358]

Notes

In his *Cours d'analyse* of 1821, Cauchy stated that a convergent series of continuous functions has a continuous sum [**15**]. In 1826 Abel pointed out that the series $\sin x - \frac{1}{2}\sin 2x + \frac{1}{3}\sin 3x - \cdots$ is a counterexample [**1**]. Its sum jumps at every odd multiple of π. Philipp Ludwig von Seidel [**66**] and George Gabriel Stokes [**70**] each found the missing hypothesis in 1847. Weierstrass used uniform convergence in an 1841 manuscript on power series, which was published only in 1894 [**77**]. His Berlin lectures made the idea standard. Ulisse Dini proved Theorem 11.2.17 in his 1878 book [**23**]. Cesare Arzelà proved Theorem 11.2.21 in 1885 [**3**].

Weierstrass presented the function of Example 11.3.1 to the Berlin Academy in 1872 [**78**]. Paul du Bois-Reymond published it first, in 1875 [**26**]. Weierstrass's function was $\sum b^k \cos(a^k \pi x)$ with $0 < b < 1$, a an odd integer and $ab > 1 + \frac{3}{2}\pi$. Example 11.3.1 is the case $b = \frac{1}{2}$. G. H. Hardy proved in 1916 that $ab \ge 1$ is enough [**36**]. Bernard Bolzano had found a continuous nowhere differentiable function around 1830, but it was not published until a century later. Weierstrass proved Theorem 11.4.3 in 1885. The proof with Bernstein polynomials is from 1912. Émile Picard introduced his iteration in 1890, and Ernst Lindelöf gave the version with a Lipschitz condition in 1894, so Theorem 11.5.2 is often called the Picard–Lindelöf Theorem. Peano's existence theorem is from 1886.

CHAPTER 12

Power Series Expansions

A power series is a series of functions whose terms are constant multiples of $(x-c)^k$. It is the infinite analogue of a polynomial. This chapter shows that power series behave almost as well as polynomials. Each one converges on an interval centered at c, and uniformly on smaller closed intervals. Inside that interval its sum is continuous, and it can be integrated and differentiated term by term. So the theorems of Chapter 11 pay off here.

Section 12.3 turns the question around. Given a function f, is it the sum of a power series? The only candidate is the Taylor series of f. Taylor's Theorem measures how far f is from the partial sums of that series. Sometimes the answer is yes, as for $\sin x$, e^x and $\ln(1+x)$. Sometimes it is no, even for a function with derivatives of every order. The starred Sections 12.4 and 12.5 use power series to define the trigonometric functions and the exponential function a second way.

12.1. Radius of Convergence

Definition 12.1.1. Let $\{a_k\}$ be a sequence of real numbers, and let $c \in \mathbb{R}$. A series of the form

$$\sum_{k=0}^{\infty} a_k(x-c)^k = a_0 + a_1(x-c) + a_2(x-c)^2 + \cdots,$$

is called a *power series* in (powers of) $(x-c)$. When $x = c$, it is assumed $(x-c)^0 = 1$. When $c = 0$, the series is called a power series in x. The numbers a_k are called the *coefficients* of the power series.

The radius of convergence is defined with a limit superior. The next lemma collects the facts about limits superior that this chapter needs. Here ∞ is allowed as a value, with $t \cdot \infty = \infty$ for $t > 0$.

Lemma 12.1.2. *Let $\{x_k\}$ and $\{z_k\}$ be sequences of nonnegative real numbers.*

(1) If $x_k \le z_k$ for all $k \ge K$, for some $K \in \mathbb{N}$, then $\limsup x_k \le \limsup z_k$.
(2) If $t > 0$, then $\limsup tx_k = t \limsup x_k$.
(3) If $y_k \to y$ with $0 < y < \infty$, then $\limsup x_k y_k = y \limsup x_k$.

Proof. (1) For $n \ge K$, every element of $\{x_k : k \ge n\}$ is at most the matching element of $\{z_k : k \ge n\}$. So $\sup\{x_k : k \ge n\} \le \sup\{z_k : k \ge n\}$. Let $n \to \infty$.

(2) For each n, $\sup\{tx_k : k \ge n\} = t \sup\{x_k : k \ge n\}$. Let $n \to \infty$.

(3) Let $0 < \varepsilon < y$. There is a $K \in \mathbb{N}$ with $y - \varepsilon < y_k < y + \varepsilon$ for all $k \ge K$. Then $(y-\varepsilon)x_k \le x_k y_k \le (y+\varepsilon)x_k$ for $k \ge K$. By (1) and (2),

$$(y-\varepsilon)\limsup x_k \le \limsup x_k y_k \le (y+\varepsilon)\limsup x_k.$$

If $\limsup x_k = \infty$, the left inequality gives $\limsup x_k y_k = \infty$. Otherwise let $\varepsilon \to 0$. □

Lemma 12.1.3. *If $b > 0$, then $\sqrt[k]{b} \to 1$. Also $\sqrt[k]{k} \to 1$.*

PROOF. Suppose $b \geq 1$, and let $h_k = \sqrt[k]{b} - 1 \geq 0$. By the Binomial Theorem (Exercise 9.3.3), $b = (1 + h_k)^k \geq 1 + kh_k$. So $0 \leq h_k \leq \frac{b-1}{k}$, and $h_k \to 0$ by the Squeeze Theorem. If $0 < b < 1$, then $\sqrt[k]{b} = 1/\sqrt[k]{1/b} \to 1$ by the first case and the limit laws.

For the second limit, let $h_k = \sqrt[k]{k} - 1 \geq 0$. For $k \geq 2$, the Binomial Theorem gives $k = (1 + h_k)^k \geq \binom{k}{2} h_k^2 = \frac{k(k-1)}{2} h_k^2$. So $0 \leq h_k \leq \sqrt{\frac{2}{k-1}}$, and $h_k \to 0$. □

Definition 12.1.4. Given a power series $\sum a_k(x-c)^k$, the *radius of convergence* R is defined as

$$\frac{1}{R} = \limsup \sqrt[k]{|a_k|}.$$

If $\limsup \sqrt[k]{|a_k|} = \infty$, set $R = 0$. If $\limsup \sqrt[k]{|a_k|} = 0$, set $R = \infty$.

If the ratios $\frac{|a_{k+1}|}{|a_k|}$ have a limit, that limit is also $\frac{1}{R}$, by the next theorem. This is usually the easier way to find R. But the ratios do not always have a limit. See Example 12.1.9.

Theorem 12.1.5. *Let $a_k > 0$ for all $k \in \mathbb{N}$. Then*

$$\liminf \frac{a_{k+1}}{a_k} \leq \liminf \sqrt[k]{a_k} \leq \limsup \sqrt[k]{a_k} \leq \limsup \frac{a_{k+1}}{a_k}.$$

In particular, if $\frac{a_{k+1}}{a_k} \to L$, then $\sqrt[k]{a_k} \to L$.

PROOF. The middle inequality is Theorem 3.6.10. For the right inequality, let $\beta = \limsup \frac{a_{k+1}}{a_k}$. If $\beta = \infty$, there is nothing to prove. Otherwise let $c > \beta$. By Theorem 3.6.11, there is a $K \in \mathbb{N}$ with $a_{k+1} \leq c a_k$ for all $k \geq K$. By induction, $a_k \leq a_K c^{k-K}$ for all $k \geq K$. So for $k \geq K$,

$$\sqrt[k]{a_k} \leq \sqrt[k]{a_K c^{-K}} \cdot c.$$

The right side tends to c, by Lemma 12.1.3. So $\limsup \sqrt[k]{a_k} \leq c$, by Lemma 12.1.2 (1) and Corollary 3.6.14. Since $c > \beta$ was arbitrary, $\limsup \sqrt[k]{a_k} \leq \beta$. The left inequality is proved the same way, with $c < \liminf \frac{a_{k+1}}{a_k}$ and the inequalities reversed. The last statement follows from Corollary 3.6.14. □

Corollary 12.1.6. *If $a_k \neq 0$ for all large k, and $\lim_{k\to\infty} \frac{|a_{k+1}|}{|a_k|}$ exists (as a real number or ∞), then this limit equals $\frac{1}{R}$.*

PROOF. Changing finitely many coefficients does not change $\limsup \sqrt[k]{|a_k|}$, by Lemma 12.1.2 (1). So assume $a_k \neq 0$ for all k. Apply Theorem 12.1.5 to $\{|a_k|\}$. (If the limit is ∞, the left inequality gives $\liminf \sqrt[k]{|a_k|} = \infty$.) □

Theorem 12.1.7. *Given a power series $\sum a_k(x-c)^k$ with radius of convergence R, $0 < R \leq \infty$, the series*

(1) converges absolutely for all x with $|x - c| < R$, and

(2) diverges for all x with $|x-c| > R$.
(3) Furthermore, if $0 < \rho < R$, then the series converges uniformly for all x with $|x-c| \leq \rho$.

PROOF. (1) By the root test (Theorem 10.3.4), $\sum a_k(x-c)^k$ converges absolutely if $\limsup \sqrt[k]{|a_k||x-c|^k} < 1$. By Lemma 12.1.2 (2), this lim sup is $|x-c| \cdot \frac{1}{R}$ when $x \neq c$, where $|x-c| \cdot \frac{1}{R} = 0$ if $R = \infty$. So the series converges if $|x-c| \cdot \frac{1}{R} < 1$, or if $|x-c| < R$.

(2) By the root test, $\sum a_k(x-c)^k$ diverges if $\limsup \sqrt[k]{|a_k||x-c|^k} > 1$. So the series diverges if $|x-c| \cdot \frac{1}{R} > 1$, or if $|x-c| > R$.

(3) Suppose $0 < \rho < R$. Choose $\beta \in (\rho, R)$. Since

$$\limsup \sqrt[k]{|a_k|} = \frac{1}{R} < \frac{1}{\beta},$$

there exists a $K \in \mathbb{N}$ such that $\sqrt[k]{|a_k|} < \frac{1}{\beta}$ for all $k \geq K$. Hence for $k \geq K$ and $|x-c| \leq \rho$,

$$|a_k(x-c)^k| \leq |a_k|\rho^k < \left(\frac{\rho}{\beta}\right)^k.$$

But $\frac{\rho}{\beta} < 1$, and thus $\sum \left(\frac{\rho}{\beta}\right)^k$ converges. Therefore, by the Weierstrass M-Test (Theorem 11.2.10), the series converges uniformly on $|x-c| \leq \rho$. To apply the M-Test, take $M_k = |a_k|\rho^k$ for all k. Then $\sum M_k$ converges by the Comparison Test, since $M_k < \left(\frac{\rho}{\beta}\right)^k$ for $k \geq K$. □

So a power series converges on an interval centered at c, called its *interval of convergence*. The theorem says nothing about the endpoints $c \pm R$. As the next example shows, anything can happen there.

Example 12.1.8. The series $\sum_{k=0}^{\infty} x^k$ has radius of convergence $R = 1$ and diverges when $x = -1$ or $x = 1$.

The series $\sum_{k=1}^{\infty} \frac{x^k}{k}$ has radius of convergence $R = 1$, converges when $x = -1$ but diverges when $x = 1$.

The series $\sum_{k=1}^{\infty} \frac{x^k}{k^2}$ has radius of convergence $R = 1$ and converges when $x = -1$ or $x = 1$.

The radii follow from Lemma 12.1.3, since $\sqrt[k]{k} \to 1$ and $\sqrt[k]{k^2} = \left(\sqrt[k]{k}\right)^2 \to 1$. At $x = \pm 1$, use the nth Term Test, the Alternating Series Test and Example 10.2.12.

Example 12.1.9. Consider the series

$$1 + 2x + 3^2x^2 + 2^3x^3 + 3^4x^4 + \cdots = \sum_{k=0}^{\infty} a_k x^k,$$

where

$$a_k = \begin{cases} 3^k, & k \in \mathbb{E}, \\ 2^k, & k \in \mathbb{O}. \end{cases}$$

So $\limsup \sqrt[k]{|a_k|} = 3$, implying $R = \frac{1}{3}$. The series diverges at $x = \frac{1}{3}$ and $x = -\frac{1}{3}$. There, the terms with k even are 1, so the terms do not tend to 0. The ratios $\frac{a_{k+1}}{a_k}$ have no limit. They tend to 0 along even k and to ∞ along odd k. So Corollary 12.1.6 does not apply here.

Example 12.1.10. The series $\sum_{k=0}^{\infty} k!x^k$ has radius of convergence $R = 0$ and therefore, the power series converges only for $x = 0$. By Corollary 12.1.6, $\frac{1}{R} = \lim_{k\to\infty} \frac{(k+1)!}{k!} = \lim_{k\to\infty}(k+1) = \infty$. For $x \neq 0$, the ratios of consecutive terms are $\frac{|(k+1)!x^{k+1}|}{|k!x^k|} = (k+1)|x| \to \infty$, so the series diverges by Theorem 10.3.4.

Proof Technique (Finding a Radius of Convergence). Start with the ratios $\frac{|a_{k+1}|}{|a_k|}$. If they have a limit, it is $\frac{1}{R}$, by Corollary 12.1.6. This works for coefficients built from factorials and powers, like $\frac{k^2}{3^k}$ or $\frac{k!}{k^k}$. If the ratios have no limit, use the definition, $\frac{1}{R} = \limsup \sqrt[k]{|a_k|}$. This works for coefficients given by powers, and for coefficients that follow different rules for even and odd k. If the series has gaps, like $\sum \frac{x^{3k}}{8^k}$, many coefficients are 0 and the ratios are undefined. Then substitute. With $u = x^3$, the series $\sum \frac{u^k}{8^k}$ converges for $|u| < 8$. So the original converges for $|x| < 2$ and diverges for $|x| > 2$, and $R = 2$. Finally, the endpoints $c \pm R$ always need separate work.

Definition 12.1.11. A function $f(x) = \sum_{k=0}^{\infty} a_k(x-c)^k$ on $|x-c| < R$ where R is the radius of convergence of $\sum a_k(x-c)^k$ is called a *real analytic function*.

Remark 12.1.12. The usual definition is local. A function $f : I \to \mathbb{R}$ on an open interval I is real analytic on I if for every $d \in I$, f is the sum of a power series centered at d on some interval $(d-r, d+r) \subset I$. Every function in Definition 12.1.11 is analytic in this sense, but the proof is not given here. It requires rewriting a power series centered at c as one centered at d.

Exercises

12.1.1. Find the radius of convergence of the power series $\sum_{k=0}^{\infty} \frac{1}{4^k}(x+1)^{2k}$. [Hint, p. 359]

12.1.2. Find the radius of convergence of the power series $\sum_{k=0}^{\infty} \frac{1}{8^k}x^{3k}$. [Hint, p. 359]

12.1.3. Find the radius of convergence and the interval of convergence of each power series. Check the endpoints.

(a) $\displaystyle\sum_{k=1}^{\infty} \frac{k}{2^k}x^k$

(b) $\displaystyle\sum_{k=0}^{\infty} \frac{x^k}{k!}$

(c) $\displaystyle\sum_{k=1}^{\infty} \frac{(x-3)^k}{k\,5^k}$

(d) $\displaystyle\sum_{k=1}^{\infty} \frac{k!}{k^k} x^k$

[Hint, p. 359]

12.1.4. Complete the proof of Theorem 12.1.5 by proving the left inequality.

[Hint, p. 359]

12.2. Integrals and Derivatives of Power Series

Inside its interval of convergence, a power series can be treated like a polynomial. It can be integrated and differentiated term by term. At an endpoint where the series converges, Abel's Theorem says the sum is still continuous from the inside.

Theorem 12.2.1 (Abel's Theorem)**.** *Suppose* $f(x) = \sum_{k=0}^{\infty} a_k x^k$ *has radius of convergence* $R = 1$*, and* $\sum_{k=0}^{\infty} a_k$ *converges. Then*

$$\lim_{x \to 1^-} f(x) = \sum_{k=0}^{\infty} a_k.$$

Proof. The plan is to show that $\sum a_k x^k$ converges uniformly on $[0, 1]$. Then its sum is continuous on $[0, 1]$, in particular at 1.

Let $\varepsilon > 0$. Since $\sum a_k$ converges, the Cauchy Criterion (Theorem 10.2.1) gives an $N \in \mathbb{N}$ such that

$$\left| \sum_{k=n}^{j} a_k \right| < \varepsilon \quad \text{for all } j \geq n > N.$$

Fix $m \geq n > N$ and $x \in [0, 1]$. For $j = n, \ldots, m$, let $B_j = \sum_{k=n}^{j} a_k$, and let $B_{n-1} = 0$. Then $|B_j| < \varepsilon$ and $a_k = B_k - B_{k-1}$ for $n \leq k \leq m$. Summing by parts, exactly as in the proof of Theorem 10.2.15,

$$\sum_{k=n}^{m} a_k x^k = \sum_{k=n}^{m} (B_k - B_{k-1}) x^k = \sum_{k=n}^{m-1} B_k \left(x^k - x^{k+1} \right) + B_m x^m.$$

Since $0 \leq x \leq 1$, each $x^k - x^{k+1}$ is nonnegative. So

$$\left| \sum_{k=n}^{m} a_k x^k \right| \leq \varepsilon \left(\sum_{k=n}^{m-1} \left(x^k - x^{k+1} \right) + x^m \right) = \varepsilon x^n \leq \varepsilon.$$

This holds for all $m \geq n > N$ and all $x \in [0, 1]$. By Corollary 11.2.6, $\sum a_k x^k$ converges uniformly on $[0, 1]$. Let $g(x) = \sum_{k=0}^{\infty} a_k x^k$ for $x \in [0, 1]$. Each term is continuous, so g is continuous on $[0, 1]$, by Corollary 11.2.16. Since $g = f$ on $[0, 1)$ and $g(1) = \sum_{k=0}^{\infty} a_k$,

$$\lim_{x \to 1^-} f(x) = \lim_{x \to 1^-} g(x) = g(1) = \sum_{k=0}^{\infty} a_k. \qquad \square$$

Example 12.2.2. Consider the series $\sum_{k=0}^{\infty} (-1)^k t^k$. This series has radius of convergence $R = 1$. The series converges to $f(t) = \frac{1}{1+t}$ for all t with $|t| < 1$.

Since the convergence is uniform on $|t| \le |x|$ when $|x| < 1$ (Theorem 12.1.7), by Corollary 11.2.19,

$$\begin{aligned}\ln(1+x) &= \int_0^x \frac{1}{1+t}\,dt \\ &= \sum_{k=0}^{\infty}(-1)^k \int_0^x t^k\,dt \\ &= \sum_{k=0}^{\infty}\frac{(-1)^k}{k+1}x^{k+1} \\ &= \sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k}x^k,\end{aligned}$$

for all x with $|x| < 1$. The first equality holds because both sides are 0 at $x = 0$ and have derivative $\frac{1}{1+x}$, by Theorem 9.4.4 and Theorem 9.3.7. When $x < 0$, $\int_0^x$ means $-\int_x^0$. The series $\sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k}x^k$ has radius of convergence $R = 1$ and also converges when $x = 1$. Thus by Abel's Theorem, and since ln is continuous at 2,

$$\ln 2 = \sum_{k=1}^{\infty}\frac{(-1)^{k+1}}{k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots.$$

So the sum s in Example 10.3.7 is $\ln 2$.

Theorem 12.2.3. *Suppose $\sum_{k=0}^{\infty} a_k(x-c)^k$ has radius of convergence $R > 0$ and*

$$f(x) = \sum_{k=0}^{\infty} a_k(x-c)^k, \quad |x-c| < R.$$

Then

(1) $\sum_{k=1}^{\infty} k a_k(x-c)^{k-1}$ has radius of convergence R, and
(2) $f'(x) = \sum_{k=1}^{\infty} k a_k(x-c)^{k-1}$ for all x, $|x-c| < R$.

PROOF. (1) Without loss of generality, take $c = 0$. Consider the differentiated series $\sum k a_k x^{k-1}$. For $x \neq 0$,

$$\lim_{k\to\infty} \sqrt[k]{|x|^{k-1}} = \lim_{k\to\infty} \frac{|x|}{\sqrt[k]{|x|}} = |x|.$$

Therefore, since $\sqrt[k]{k} \to 1$ (Lemma 12.1.3),

$$\lim_{k\to\infty} \sqrt[k]{k|x|^{k-1}} = |x|.$$

Also, for $x \neq 0$, by Lemma 12.1.2 (3),

$$\begin{aligned}\limsup \sqrt[k]{|a_k|k|x|^{k-1}} &= \limsup \left(\sqrt[k]{|a_k|} \cdot \sqrt[k]{k|x|^{k-1}}\right) \\ &= |x| \limsup \sqrt[k]{|a_k|} \\ &= |x| \cdot \frac{1}{R}.\end{aligned}$$

This is less than 1 when $|x| < R$ and greater than 1 when $|x| > R$. Thus by the Root Test (Theorem 10.3.4) the differentiated series converges for all $|x| < R$ and diverges for all $|x| > R$. Thus the radius of convergence of $\sum k a_k x^{k-1}$ is R.

(2) Choose $\rho \in (0, R)$. Then by Theorem 12.1.7, $\sum k a_k x^{k-1}$ converges uniformly for all $|x| \leq \rho$. Thus by Theorem 11.2.22,

$$f'(x) = \sum_{k=1}^{\infty} k a_k x^{k-1}.$$

In detail, apply Theorem 11.2.22 on $[-\rho, \rho]$ to the partial sums $S_n(x) = \sum_{k=0}^{n} a_k x^k$. Their derivatives $S_n'(x) = \sum_{k=1}^{n} k a_k x^{k-1}$ converge uniformly on $[-\rho, \rho]$, and $S_n(0) = a_0$ converges. So $f'(x) = \lim_{n\to\infty} S_n'(x)$ for $|x| < \rho$. Every x with $|x| < R$ satisfies $|x| < \rho$ for some $\rho < R$. □

Corollary 12.2.4. *Suppose $\sum_{k=0}^{\infty} a_k (x-c)^k$ has radius of convergence $R > 0$ and*

$$f(x) = \sum_{k=0}^{\infty} a_k (x-c)^k, \quad |x-c| < R.$$

Then f has derivatives of all orders in $|x-c| < R$, and for every $n \in \mathbb{N}$,

$$f^{(n)}(x) = \sum_{k=n}^{\infty} \frac{k!}{(k-n)!} a_k (x-c)^{k-n}.$$

In particular,

$$f^{(n)}(c) = n! a_n.$$

Proof. The formula for $f^{(n)}$ follows by induction on n. The case $n = 1$ is Theorem 12.2.3. If the formula holds for n, then $f^{(n)}$ is the sum of a power series with radius of convergence R, by part (1) of that theorem applied n times. Applying the theorem to it gives the formula for $n+1$, since $(k-n)\frac{k!}{(k-n)!} = \frac{k!}{(k-n-1)!}$. Setting $x = c$ leaves only the term $k = n$, which is $n! a_n$. □

Definition 12.2.5. A function $f : I \to \mathbb{R}$ defined on an open interval I is said to be *infinitely differentiable* on I if $f^{(n)}(x)$ exists for all $x \in I$ for all $n \in \mathbb{N}$. The set of infinitely differentiable functions on an open interval I is denoted by $C^\infty(I)$. Since differentiable functions are continuous, $C^\infty(I)$ is the intersection of the sets $C^{(n)}(I)$ of Definition 7.1.4.

Corollary 12.2.6. *If f is analytic, $f \in C^\infty(I)$, where $I = (c-R, c+R)$ as in Definition 12.1.11.*

Example 12.2.7. For $|x| < 1$,

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k.$$

Thus by Corollary 12.2.4,

$$\frac{d^n}{dx^n} \frac{1}{1-x} = \frac{n!}{(1-x)^{n+1}} = \sum_{k=0}^{\infty} \frac{(k+n)!}{k!} x^k.$$

For example, with $n = 1$, $\frac{1}{(1-x)^2} = \sum_{k=0}^{\infty}(k+1)x^k$ for $|x| < 1$.

Corollary 12.2.8. *Suppose $\sum a_k(x-c)^k$ and $\sum b_k(x-c)^k$ are two power series which converge for all $|x-c| < R$ for some $R > 0$. Then*

$$\sum_{k=0}^{\infty} a_k(x-c)^k = \sum_{k=0}^{\infty} b_k(x-c)^k, \quad |x-c| < R,$$

if and only if $a_k = b_k$ for all $k \in \mathbb{N}_0$.

PROOF. ($\Leftarrow$) Clearly, if $a_k = b_k$ for all $k \in \mathbb{N}_0$, then the two power series are equal.

($\Rightarrow$) Set $f(x) = \sum_{k=0}^{\infty} a_k(x-c)^k$ and $g(x) = \sum_{k=0}^{\infty} b_k(x-c)^k$. Since $f(x) = g(x)$ for all x with $|x-c| < R$, then $f^{(n)}(x) = g^{(n)}(x)$ for all $n \in \mathbb{N}_0$ and all x, $|x-c| < R$. In particular, $f^{(n)}(c) = g^{(n)}(c)$ for all $n \in \mathbb{N}_0$. Thus by Corollary 12.2.4, $a_k = b_k$ for all $k \in \mathbb{N}_0$. □

The next example needs one more fact about e^x. It grows faster than every power of x.

Lemma 12.2.9. *For all $u \geq 0$ and all $m \in \mathbb{N}_0$, $e^u \geq \frac{u^m}{m!}$.*

PROOF. Recall that e^u is increasing, $\frac{d}{du}e^u = e^u$ and $e^0 = 1$ (Theorem 9.4.6). For $m \in \mathbb{N}_0$ and $u \geq 0$, let $g_m(u) = e^u - \sum_{k=0}^{m} \frac{u^k}{k!}$. The proof shows $g_m(u) \geq 0$ for $u \geq 0$ by induction on m. For $m = 0$, $g_0(u) = e^u - 1 \geq 0$, since e^u is increasing and $e^0 = 1$. If $g_{m-1} \geq 0$ on $[0, \infty)$, then $g_m' = g_{m-1} \geq 0$ there, so g_m is increasing on $[0, \infty)$ by Theorem 7.2.7. Since $g_m(0) = 0$, $g_m(u) \geq 0$ for all $u \geq 0$. So $e^u \geq \sum_{k=0}^{m} \frac{u^k}{k!} \geq \frac{u^m}{m!}$. □

Example 12.2.10. $f \in C^\infty(I)$ does not imply f has a power series representation with positive radius of convergence. For example, let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{cases} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

See Figure 12.1. The proof rests on one limit, and every derivative of f at 0 comes down to it. Let P be a polynomial of degree d. Then

$$\lim_{x \to 0} P\left(\frac{1}{x}\right) e^{-1/x^2} = 0. \tag{12.2.1}$$

To see this, let C be the sum of the absolute values of the coefficients of P. For $0 < |x| \leq 1$, $\left|P\left(\frac{1}{x}\right)\right| \leq C|x|^{-d}$. By Lemma 12.2.9 with $u = \frac{1}{x^2}$ and $m = d+1$, $e^{-1/x^2} \leq (d+1)!\, x^{2d+2}$. So $\left|P\left(\frac{1}{x}\right) e^{-1/x^2}\right| \leq C(d+1)!\, |x|^{d+2}$, which tends to 0.

Since

$$\lim_{x \to 0} e^{-1/x^2} = 0,$$

f is continuous at 0. For $x \neq 0$,

$$f'(x) = \frac{2e^{-1/x^2}}{x^3}$$

and

$$\begin{aligned} f'(0) &= \lim_{h\to 0} \frac{f(h)-f(0)}{h} \\ &= \lim_{h\to 0} \frac{e^{-1/h^2}}{h} \\ &= 0, \end{aligned}$$

by (12.2.1) with $P(t) = t$. Thus

$$f'(x) = \begin{cases} \frac{2}{x^3} e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0. \end{cases}$$

Now assume for $n = k$,

$$f^{(k)}(x) = \begin{cases} P\left(\frac{1}{x}\right) e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

where P is a polynomial of degree $3k$. So for $x \neq 0$,

$$\begin{aligned} f^{(k+1)}(x) &= P'\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right) e^{-1/x^2} + P\left(\frac{1}{x}\right) \cdot \frac{2}{x^3} e^{-1/x^2} \\ &= Q\left(\frac{1}{x}\right) e^{-1/x^2}, \end{aligned}$$

where

$$Q\left(\frac{1}{x}\right) = P'\left(\frac{1}{x}\right) \cdot \left(-\frac{1}{x^2}\right) + P\left(\frac{1}{x}\right) \cdot \frac{2}{x^3}.$$

So Q is a polynomial of degree $3(k+1)$. Indeed, $Q(t) = -t^2 P'(t) + 2t^3 P(t)$. Also, for $x = 0$,

$$\begin{aligned} f^{(k+1)}(0) &= \lim_{h\to 0} \frac{f^{(k)}(h) - f^{(k)}(0)}{h} \\ &= \lim_{h\to 0} \frac{P\left(\frac{1}{h}\right) e^{-1/h^2}}{h} \\ &= 0, \end{aligned}$$

by (12.2.1) with the polynomial $tP(t)$ in place of P. Thus by induction, for all $n \in \mathbb{N}_0$,

$$f^{(n)}(x) = \begin{cases} P\left(\frac{1}{x}\right) e^{-1/x^2}, & x \neq 0, \\ 0, & x = 0, \end{cases}$$

where P is a polynomial of degree $3n$. (For $n = 0$, $P = 1$.)

If there exists an $R > 0$ such that

$$f(x) = \sum_{k=0}^{\infty} a_k x^k, \quad |x| < R,$$

then $n!\, a_n = f^{(n)}(0) = 0$ for all $n \in \mathbb{N}_0$, by Corollary 12.2.4. So every $a_n = 0$, and $f(x) = 0$ for $|x| < R$. But $f(x) > 0$ for $x \neq 0$. So f cannot be written as a power series with positive radius of convergence that converges to f on an interval $(-r, r)$.

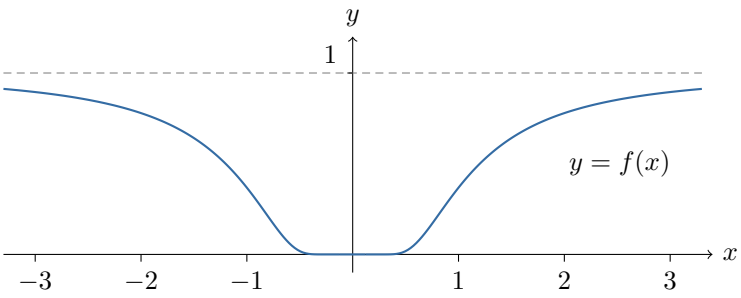


FIGURE 12.1. The function f of Example 12.2.10. It is extremely flat at 0. Every derivative of f at 0 is 0, but f is not the zero function.

Exercises

12.2.1. Suppose $f(x) = \sum_{k=0}^{\infty} a_k(x-c)^k$ has radius of convergence $R > 0$. For $|x-c| < R$, set $F(x) = \int_c^x f(t)\,dt$. Prove that $F(x) = \sum_{k=0}^{\infty} \frac{a_k}{k+1}(x-c)^{k+1}$, $|x-c| < R$.

(a) Use the fact that $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$ to show

$$\arctan x = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1}$$

for $x \in (-1, 1)$.

(b) Use Abel's theorem to show $\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \frac{\pi}{4}$. Then explain why the formula in (a) also holds at $x = \pm 1$.

[Hint, p. 359]

12.2.2. Show that $\lim_{x\to 1^-} \sum_{k=0}^{\infty}(-1)^k x^k$ exists, although $\sum_{k=0}^{\infty}(-1)^k$ diverges. So the converse of Abel's Theorem is false. [Hint, p. 360]

12.2.3. Find power series centered at 0, with their radii of convergence, for $\frac{x}{(1-x)^2}$ and for $\ln(1-x^2)$. [Hint, p. 360]

12.3. Taylor Series

By Corollary 12.2.4, if f is the sum of a power series centered at c, the coefficients must be $a_k = \frac{f^{(k)}(c)}{k!}$. So the Taylor series below is the only power series centered at c that could converge to f. Whether it does is a question about the remainder $f - T_n(f, c)$.

Definition 12.3.1. Let I be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$ and $n \in \mathbb{N}$. Suppose $f^{(n)}(x)$ exists for all $x \in I$. The polynomial

$$T_n(f, c)(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k$$

is the *Taylor polynomial* of order n of f at c. If $f \in C^\infty(I)$, the series

$$T(f,c)(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}(x-c)^k,$$

is called the *Taylor series* of f at c. If $c = 0$, the Taylor series of f is the *Maclaurin series* of f. The Taylor polynomial $T_1(f,c)$ is the linear approximation to f at c.

Remark 12.3.2. Differentiating term by term, the jth derivative of $\frac{(x-c)^k}{k!}$ at $x = c$ is 1 if $j = k$ and 0 if $j \neq k$. So $T_n(f,c)$ is the unique polynomial of degree at most n whose value and first n derivatives at c agree with those of f.

Example 12.3.3. Consider $\sqrt{25.7}$. Let $f(x) = \sqrt{x}$, $c = 25$. Then $f(25) = 5$, $f'(25) = \frac{1}{10}$ and $f''(25) = -\frac{1}{500}$. Then

$$\begin{aligned} T_2(f,25)(x) &= f(c) + f'(c)(x-c) + \frac{f''(c)}{2}(x-c)^2 \\ &= 5 + \frac{1}{10}(x-25) - \frac{1}{1000}(x-25)^2. \end{aligned}$$

So

$$\begin{aligned} f(25.7) &\approx 5 + \frac{1}{10}(25.7-25) - \frac{1}{1000}(25.7-25)^2 \\ &= 5 + \frac{7}{100} - \frac{49}{100000} \\ &= 5.06951. \end{aligned}$$

Example 12.3.4. For the function $f(x) = e^{-1/x^2}$ of Example 12.2.10, $T_n(f,0)(x) = 0$ for all $x \in \mathbb{R}$ and for all $n \in \mathbb{N}_0$. So T_n converges but not to f.

Definition 12.3.5. Define the function

$$R_n(x) = R_n(f,c)(x) = f(x) - T_n(f,c)(x)$$

as the *remainder* or error function between f and $T_n(f,c)$.

Theorem 12.3.6. *Suppose $f \in C^\infty(I)$ and $c \in I$. Then for $x \in I$,*

$$f(x) = \sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}(x-c)^k$$

if and only if

$$\lim_{n\to\infty} R_n(f,c)(x) = 0.$$

Proof. The nth partial sum of the Taylor series at x is $T_n(f,c)(x) = f(x) - R_n(f,c)(x)$. It converges to $f(x)$ if and only if $R_n(f,c)(x) \to 0$. □

Theorem 12.3.7 (Taylor's Theorem, Lagrange's Form of the Remainder)**.** *Let I be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$ and $n \in \mathbb{N}$. If $f^{(n+1)}(t)$ exists for every $t \in I$, then for any $x \in I$, there exists a ξ between x and c such that*

$$R_n(x) = R_n(f,c)(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}(x-c)^{n+1}.$$

Notice continuity of $f^{(n+1)}$ is not required.

PROOF. If $x = c$, both sides are 0 with $\xi = c$. So suppose $x \neq c$. By symmetry, it is enough to treat $x > c$. Let M be the number with

$$f(x) = T_n(f, c)(x) + M(x - c)^{n+1}.$$

The goal is to show $M = \frac{f^{(n+1)}(\xi)}{(n+1)!}$ for some $\xi \in (c, x)$. Define $g : I \to \mathbb{R}$ by

$$g(t) = f(t) - T_n(f, c)(t) - M(t - c)^{n+1}.$$

Three facts about g are needed. First, $g(x) = 0$, by the choice of M. Second, $g^{(j)}(c) = 0$ for $j = 0, 1, \ldots, n$. This holds because f and $T_n(f, c)$ have the same derivatives of order at most n at c (Remark 12.3.2), and every derivative of $(t-c)^{n+1}$ of order at most n is 0 at $t = c$. Third, since $T_n(f, c)$ is a polynomial of degree at most n, its $(n + 1)$st derivative is 0. So

$$g^{(n+1)}(t) = f^{(n+1)}(t) - (n + 1)!\, M \quad \text{for all } t \in I.$$

Now apply Rolle's Theorem $n + 1$ times. Let $x_0 = x$. The goal is to find points $c < x_{n+1} < x_n < \cdots < x_1 < x_0$ with $g^{(j)}(x_j) = 0$ for each j. Suppose x_j has been found for some $j \leq n$. Then $g^{(j)}(c) = 0 = g^{(j)}(x_j)$, and $g^{(j)}$ is differentiable on I because $j \leq n$. So Rolle's Theorem (Theorem 7.2.1) gives an $x_{j+1} \in (c, x_j)$ with $g^{(j+1)}(x_{j+1}) = 0$. This starts with $j = 0$, since $g(c) = 0 = g(x)$.

Let $\xi = x_{n+1}$. Then $\xi \in (c, x)$ and $0 = g^{(n+1)}(\xi) = f^{(n+1)}(\xi) - (n + 1)!\, M$. So $M = \frac{f^{(n+1)}(\xi)}{(n+1)!}$, and $R_n(x) = M(x - c)^{n+1}$ is the formula claimed. □

Proof Technique (Applying a Theorem Repeatedly). Rolle's Theorem turns two zeros of g into one zero of g'. The proof of Taylor's Theorem uses this $n + 1$ times. The auxiliary function g is built so that the process never stops early. It has a zero at x, and $g, g', \ldots, g^{(n)}$ all vanish at c. So each new zero x_j of $g^{(j)}$, together with the zero of $g^{(j)}$ at c, gives the next one. When a single application of a theorem gives too little, look for a way to set up the next application with what the last one produced. Compare the Auxiliary Functions box in Section 7.2.

Example 12.3.8. Let $f(x) = \sqrt{1 + x}$, $c = 0$. So $f(0) = 1$, $f'(0) = \frac{1}{2}$ and $f''(0) = -\frac{1}{4}$. Thus

$$T_2(f, 0)(x) = 1 + \frac{1}{2}x - \frac{1}{8}x^2.$$

Therefore,

$$\sqrt{1 + x} = 1 + \frac{1}{2}x - \frac{1}{8}x^2 + R_2(f, 0)(x).$$

For $x > 0$, the Lagrange Form of the Remainder gives a $\xi \in (0, x)$ such that

$$\begin{aligned} R_2(f, 0)(x) &= \frac{f'''(\xi)}{3!}x^3 \\ &= \frac{1}{16}(1 + \xi)^{-5/2}x^3. \end{aligned}$$

Since $(1+\xi)^{-5/2} < 1$ (because $\xi > 0$),

$$\left|\sqrt{1+x} - T_2(f,0)(x)\right| < \frac{1}{16}x^3.$$

Let $x = 0.4$. Then

$$T_2(f,0)(0.4) = 1 + \frac{1}{5} - \frac{1}{50} = 1.18,$$

and

$$\left|\sqrt{1.4} - 1.18\right| < \frac{1}{16}(0.4)^3 = \frac{1}{250} = 0.004.$$

So $\sqrt{1.4} \approx 1.18$, accurate to two decimal places.

Notice $0 < R_2(f,0)(x) < \frac{1}{16}x^3$ for $x > 0$. Thus

$$1 + \frac{1}{2}x - \frac{1}{8}x^2 < \sqrt{1+x} < 1 + \frac{1}{2}x - \frac{1}{8}x^2 + \frac{1}{16}x^3.$$

Theorem 12.3.9 (Integral Form of the Remainder)**.** *Let I be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$ and $n \in \mathbb{N}$. If $f^{(n+1)}(t)$ exists for all $t \in I$ and $f^{(n+1)} \in \mathcal{R}_{\text{loc}}(I)$, then*

$$R_n(x) = R_n(f,c)(x) = \frac{1}{n!}\int_c^x f^{(n+1)}(t)(x-t)^n\,dt, \quad x \in I.$$

Proof. Induct on n. Suppose $n = 1$. Then, by the Fundamental Theorem of Calculus

$$\begin{aligned} R_1(x) &= f(x) - f(c) - f'(c)(x-c) \\ &= \int_c^x f'(t)\,dt - f'(c)\int_c^x dt \\ &= \int_c^x [f'(t) - f'(c)]\,dt. \end{aligned}$$

Using integration by parts with $u = f'(t) - f'(c)$, $du = f''(t)\,dt$, $dv = dt$ and $v = t - x$,

$$\begin{aligned} R_1(x) &= [f'(t) - f'(c)](t-x)\Big|_c^x - \int_c^x f''(t)(t-x)\,dt \\ &= \int_c^x (x-t)f''(t)\,dt. \end{aligned}$$

Now assume the result holds for $n = k$. Suppose $f^{(k+2)}$ exists on I and $f^{(k+2)} \in \mathcal{R}_{\text{loc}}(I)$. Then $f^{(k+1)}$ is differentiable, hence continuous, hence in $\mathcal{R}_{\text{loc}}(I)$. So the result for $n = k$ applies. So

$$\begin{aligned} R_{k+1}(x) &= f(x) - T_{k+1}(f,c)(x) \\ &= f(x) - T_k(f,c)(x) - \frac{f^{(k+1)}(c)}{(k+1)!}(x-c)^{k+1} \\ &= R_k(x) - \frac{f^{(k+1)}(c)}{(k+1)!}(x-c)^{k+1} \\ &= \frac{1}{k!}\int_c^x (x-t)^k f^{(k+1)}(t)\,dt - \frac{1}{k!}f^{(k+1)}(c)\int_c^x (x-t)^k\,dt \end{aligned}$$

$$= \frac{1}{k!}\int_c^x [f^{(k+1)}(t) - f^{(k+1)}(c)](x-t)^k\,dt$$
$$= \frac{1}{(k+1)!}\int_c^x f^{(k+2)}(t)(x-t)^{k+1}\,dt.$$

The fourth line uses $\int_c^x (x-t)^k\,dt = \frac{(x-c)^{k+1}}{k+1}$. The last line is integration by parts with $u = f^{(k+1)}(t) - f^{(k+1)}(c)$ and $v = -\frac{(x-t)^{k+1}}{k+1}$. The boundary terms vanish, since $u = 0$ at $t = c$ and $v = 0$ at $t = x$. When $x < c$, read $\int_c^x$ as $-\int_x^c$. The same computations hold. Thus by induction, the result holds. □

Corollary 12.3.10 (Cauchy's Form of the Remainder)**.** *Let I be an open interval, let $f : I \to \mathbb{R}$, and let $c \in I$ and $n \in \mathbb{N}$. If $f^{(n+1)}(t)$ is continuous for all $t \in I$, then for each $x \in I$, there exists a ξ between c and x such that*

$$R_n(x) = R_n(f,c)(x) = \frac{f^{(n+1)}(\xi)}{n!}(x-c)(x-\xi)^n.$$

Proof. Since $f^{(n+1)}(t)(x-t)^n$ is continuous on the interval between c and x, by the Mean Value Theorem for Integrals (Theorem 9.6.1), there exists a ξ between c and x such that

$$\int_c^x f^{(n+1)}(t)(x-t)^n\,dt = (x-c)f^{(n+1)}(\xi)(x-\xi)^n.$$

The result then follows from the Integral Form of the Remainder (Theorem 12.3.9). □

Proof Technique (Bounding a Remainder). To show $R_n(x) \to 0$, bound the formula for $R_n(x)$ by something that does not depend on ξ, and then let $n \to \infty$. The point ξ is not known. It depends on both x and n. So the bound on $f^{(n+1)}(\xi)$ must hold for every ξ between c and x. For $\sin x$, $|f^{(n+1)}(\xi)| \le 1$ for every ξ. In Example 12.3.12, $\frac{1}{1+\xi}$ is bounded by $\frac{1}{1+x}$ because $\xi > x$. A bound that is finite for each n, but grows with n, is not enough. If one form of the remainder gives a bound that does not tend to 0, try another form.

Example 12.3.11. Let $f(x) = \sin x$, $c = 0$. Then

$$f^{(n)}(x) = \begin{cases} (-1)^k \cos x, & n = 2k+1, \\ (-1)^k \sin x, & n = 2k. \end{cases}$$

Thus $f^{(n)}(0) = 0$ for all $n \in \mathbb{E}$ and $f^{(n)}(0) = (-1)^k$, $n = 2k+1$, $k \in \mathbb{N}_0$. Therefore,

$$T(\sin x, 0) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!}x^{2k+1}.$$

Notice $R = \infty$ for this power series. Let $x \in \mathbb{R}$. Then by Lagrange's Form of the Remainder, there exists a ξ between 0 and x such that

$$R_n(x) = \frac{f^{(n+1)}(\xi)}{(n+1)!}x^{n+1}.$$

Since $|f^{(n+1)}(\xi)| \leq 1$,

$$|R_n(x)| \leq \frac{|x|^{n+1}}{(n+1)!} \to 0$$

by Exercise 10.2.5. Thus $R_n(x) \to 0$ for all $x \in \mathbb{R}$ and so

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1}, \quad x \in \mathbb{R}.$$

See Figure 12.2.

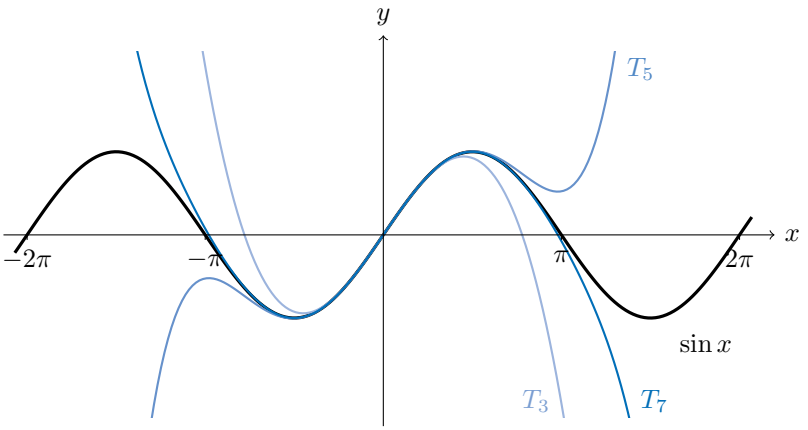


FIGURE 12.2. The function $\sin x$ (black) and its Taylor polynomials T_3, T_5 and T_7 at 0. Each one is accurate on a longer interval than the one before.

Example 12.3.12. Let $f(x) = \ln(x+1)$. Then $f(0) = \ln(1) = 0$, and so, by the Fundamental Theorem of Calculus,

$$f'(x) = \frac{1}{1+x}.$$

Thus for $n \in \mathbb{N}$,

$$f^{(n)}(x) = (-1)^{n+1} \frac{(n-1)!}{(1+x)^n}.$$

In particular, for $n \in \mathbb{N}$,

$$f^{(n)}(0) = (-1)^{n+1}(n-1)!.$$

So

$$\begin{aligned} T(\ln(1+x), 0) &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}(n-1)!}{n!} x^n \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n. \end{aligned}$$

Since $\sqrt[n]{\left|\frac{(-1)^{n+1}}{n}\right|} \to 1$, $R = 1$.

Suppose $x \in (0,1]$. Then Lagrange's Form of the Remainder gives a $\xi \in (0,x)$ with

$$R_n(x) = \frac{(-1)^n}{(n+1)(1+\xi)^{n+1}} x^{n+1}.$$

Since $(1+\xi) > 1$,

$$|R_n(x)| \leq \frac{1}{n+1} x^{n+1} \leq \frac{1}{n+1} \to 0.$$

So $R_n(x) \to 0$ for all $x \in [0,1]$.

Now suppose $x \in (-1,0)$. By Cauchy's Form of the Remainder, there exists a $\xi \in (x,0)$ such that

$$R_n(x) = \frac{(-1)^{n+2}}{(1+\xi)^{n+1}} (x-\xi)^n x.$$

Therefore

$$|R_n(x)| \leq \frac{|x|}{(1+\xi)} \left[\frac{\xi - x}{1+\xi}\right]^n.$$

Consider the function $\varphi(t)$ defined on $[x,0]$ by

$$\varphi(t) = \frac{t-x}{1+t}.$$

Thus

$$\varphi'(t) = \frac{1+x}{(1+t)^2} \geq 0$$

for $t \in [x,0]$. Therefore

$$\varphi(t) \leq \varphi(0) = -x = |x|,$$

for all $t \in [x,0]$. Thus since $1+\xi > 1+x > 0$,

$$|R_n(x)| \leq \frac{|x|}{1+\xi} |x|^n \leq \frac{|x|^{n+1}}{1+x}.$$

Since $|x| < 1$, $R_n(x) \to 0$. Therefore

$$\ln(1+x) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{n} x^n, \quad x \in (-1,1].$$

This agrees with Example 12.2.2, which found the same series by integrating term by term.

Example 12.3.13. If $f(x) = \frac{1}{1+x}$,

$$f(x) = \sum_{k=0}^{\infty} (-1)^k x^k, \quad x \in (-1,1).$$

Notice f has a singularity at $x = -1$, and the radius of convergence is $R = 1$, which is equal to the distance from $c = 0$ to the singularity $x = -1$.

For $g(x) = \frac{1}{1+x^2}$,

$$g(x) = \sum_{k=0}^{\infty} (-1)^k x^{2k}, \quad x \in (-1,1).$$

Here $R = 1$ but g has no real singularities. However, g has complex singularities at $\pm i$, and the distance in the complex plane from $c = 0$ to i is 1, the radius of

convergence of the power series. The explanation belongs to complex analysis, where power series are studied for complex x.

Exercises

12.3.1. Without expanding the right side, find constants a_0, a_1, a_2, a_3, a_4 such that

$$x^4 + 3x^2 - 2x + 5 = a_4(x-1)^4 + a_3(x-1)^3 + a_2(x-1)^2 + a_1(x-1) + a_0.$$

[Hint, p. 360]

12.3.2. Use a third order Taylor polynomial to estimate $\ln 1.2$. Use a remainder to show this is accurate to three decimal places. [Hint, p. 360]

12.3.3. Suppose $f(x) = \sum_{k=0}^{\infty} a_k x^k$, $|x| < R$, where $R > 0$. Prove that $f(x)$ is even if and only if $a_k = 0$ for all odd k. [Hint, p. 361]

12.3.4. Prove that the Taylor series of e^x centered at 0, $T(e^x, 0) = \sum_{k=0}^{\infty} \frac{x^k}{k!}$, converges to e^x for all $x \in \mathbb{R}$. [Hint, p. 361]

12.3.5. Find the Taylor series of $\cos x$ at 0, and prove that it converges to $\cos x$ for all $x \in \mathbb{R}$. [Hint, p. 361]

12.3.6. Let $\alpha \in \mathbb{R}$, and let $f(x) = (1+x)^\alpha$ for $x > -1$. Define $\binom{\alpha}{k} = \frac{\alpha(\alpha-1)\cdots(\alpha-k+1)}{k!}$, with $\binom{\alpha}{0} = 1$.

(a) Show that $T(f, 0)(x) = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k$, and that its radius of convergence is 1 unless $\alpha \in \mathbb{N}_0$.

(b) Use Cauchy's Form of the Remainder, as in Example 12.3.12, to prove that $(1+x)^\alpha = \sum_{k=0}^{\infty} \binom{\alpha}{k} x^k$ for $|x| < 1$. This is the *binomial series*.

[Hint, p. 361]

12.4. Trigonometric Functions from Power Series*

The functions $\sin x$ and $\cos x$ have been used throughout the book with their properties assumed. See Remark 3.1.1 and the introduction to Chapter 7. This section defines them and proves those properties. Section 9.5 did the same thing starting from an integral. Either section can be read without the other, and a reader willing to assume the familiar properties of sin and cos can skip both. The end of this section shows the two give the same functions.

Example 12.3.11 showed that if sin has its familiar derivatives, then it equals its Taylor series. This section turns that around. The series becomes the definition. Everything else follows from the derivatives, and from one fact about the differential equation $y'' + y = 0$.

Definition 12.4.1. For $x \in \mathbb{R}$, define

$$\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k+1)!} x^{2k+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots$$

and

$$\cos x = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!} x^{2k} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots.$$

These are the *sine* and *cosine* functions.

Remark 12.4.2. Here sin and cos are names of functions, just as f is, and $\sin x$ and $\cos x$ are their values at x, just as $f(x)$ is. By tradition the parentheses are usually left off, so $\sin x$ means $\sin(x)$. Writing sin alone, as in "$\sin' = \cos$," refers to the whole function. Parentheses are used when the input is more than a single symbol, as in $\sin(x+y)$. Also by tradition, $\sin^2 x$ means $(\sin x)^2$, not $\sin(\sin x)$. The same conventions hold for tan and arctan.

Both series converge for every x. Each term has absolute value at most $\frac{|x|^n}{n!}$ for some n, with each n used once. The series $\sum \frac{|x|^n}{n!}$ converges by the Ratio Test (Theorem 10.2.13), since $\frac{|x|^{n+1}/(n+1)!}{|x|^n/n!} = \frac{|x|}{n+1} \to 0$. So both series converge absolutely by the Comparison Test (Theorem 10.2.7). A power series that converges everywhere has radius of convergence ∞, by Theorem 12.1.7 (2).

Theorem 12.4.3. *The functions* sin *and* cos *are differentiable on* $\mathbb{R}$*, and*

$$\frac{d}{dx}\sin x = \cos x \quad \textit{and} \quad \frac{d}{dx}\cos x = -\sin x.$$

Also $\sin 0 = 0$*,* $\cos 0 = 1$*,* $\sin(-x) = -\sin x$ *and* $\cos(-x) = \cos x$ *for all* $x \in \mathbb{R}$*.*

Proof. By Theorem 12.2.3, the series can be differentiated term by term. For sin,

$$\frac{d}{dx}\sin x = \sum_{k=0}^{\infty} \frac{(-1)^k(2k+1)}{(2k+1)!}x^{2k} = \sum_{k=0}^{\infty} \frac{(-1)^k}{(2k)!}x^{2k} = \cos x.$$

For cos, the term with $k = 0$ is constant, so

$$\begin{aligned}\frac{d}{dx}\cos x &= \sum_{k=1}^{\infty} \frac{(-1)^k(2k)}{(2k)!}x^{2k-1} = \sum_{k=1}^{\infty} \frac{(-1)^k}{(2k-1)!}x^{2k-1} \\ &= -\sum_{j=0}^{\infty} \frac{(-1)^j}{(2j+1)!}x^{2j+1} = -\sin x,\end{aligned}$$

where $j = k-1$. Setting $x = 0$ in the series gives $\sin 0 = 0$ and $\cos 0 = 1$. The series for sin has only odd powers and the series for cos has only even powers. So $\sin(-x) = -\sin x$ and $\cos(-x) = \cos x$. □

So $\sin'' = -\sin$ and $\cos'' = -\cos$. Both functions solve the differential equation $y'' + y = 0$. The next lemma says a solution of this equation is determined by its value and its derivative at 0. It is the key to every identity in this section.

Lemma 12.4.4. *Let* $y : \mathbb{R} \to \mathbb{R}$ *be twice differentiable with* $y'' + y = 0$ *on* $\mathbb{R}$*. If* $y(0) = 0$ *and* $y'(0) = 0$*, then* $y(x) = 0$ *for all* $x \in \mathbb{R}$*.*

Proof. Let $E(x) = y(x)^2 + y'(x)^2$. Then

$$E'(x) = 2y(x)y'(x) + 2y'(x)y''(x) = 2y'(x)\big(y(x) + y''(x)\big) = 0.$$

By Theorem 7.2.7 (5), E is constant. Since $E(0) = 0$, $E(x) = 0$ for all x. A sum of two squares is 0 only when both are 0. So $y(x) = 0$. □

Theorem 12.4.5. *Let $y : \mathbb{R} \to \mathbb{R}$ be twice differentiable with $y'' + y = 0$ on $\mathbb{R}$. Then*

$$y(x) = y(0)\cos x + y'(0)\sin x$$

for all $x \in \mathbb{R}$.

PROOF. Let $z(x) = y(x) - y(0)\cos x - y'(0)\sin x$. By Theorem 12.4.3, $z'' = -z$, so $z'' + z = 0$. Also $z(0) = y(0) - y(0) = 0$ and $z'(0) = y'(0) - y'(0) = 0$. By Lemma 12.4.4, $z = 0$. □

Theorem 12.4.6. *For all $x, y \in \mathbb{R}$,*

(1) $\sin^2 x + \cos^2 x = 1$, *so* $|\sin x| \le 1$ *and* $|\cos x| \le 1$,
(2) $\sin(x + y) = \sin x \cos y + \cos x \sin y$, *and*
(3) $\cos(x + y) = \cos x \cos y - \sin x \sin y$.

PROOF. (1) Let $E(x) = \sin^2 x + \cos^2 x$. By Theorem 12.4.3, $E'(x) = 2\sin x \cos x - 2\cos x \sin x = 0$. So E is constant, and $E(x) = E(0) = 1$.

(2) Fix y, and let $f(x) = \sin(x + y)$. By the Chain Rule, $f'(x) = \cos(x + y)$ and $f''(x) = -\sin(x + y) = -f(x)$. Also $f(0) = \sin y$ and $f'(0) = \cos y$. By Theorem 12.4.5, $f(x) = \sin y \cos x + \cos y \sin x$.

(3) Fix y, and let $g(x) = \cos(x + y)$. Then $g'' = -g$, $g(0) = \cos y$ and $g'(0) = -\sin y$. By Theorem 12.4.5, $g(x) = \cos y \cos x - \sin y \sin x$. □

Corollary 12.4.7. *For all $x, y \in \mathbb{R}$,*

(1) $\sin 2x = 2\sin x \cos x$ *and* $\cos 2x = \cos^2 x - \sin^2 x = 1 - 2\sin^2 x$,
(2) $\sin x \sin y = \frac{1}{2}\big[\cos(x - y) - \cos(x + y)\big]$,
(3) $\cos x \cos y = \frac{1}{2}\big[\cos(x - y) + \cos(x + y)\big]$ *and* $\sin x \cos y = \frac{1}{2}\big[\sin(x + y) + \sin(x - y)\big]$, *and*
(4) $\sin x - \sin y = 2\cos\frac{x+y}{2}\sin\frac{x-y}{2}$.

PROOF. (1) Take $y = x$ in Theorem 12.4.6, and use $\cos^2 x = 1 - \sin^2 x$.

(2) and (3) By Theorems 12.4.3 and 12.4.6, $\cos(x - y) = \cos x \cos y + \sin x \sin y$ and $\sin(x - y) = \sin x \cos y - \cos x \sin y$. Add or subtract these and the formulas for $\cos(x + y)$ and $\sin(x + y)$.

(4) Let $u = \frac{x+y}{2}$ and $v = \frac{x-y}{2}$. Then $x = u + v$ and $y = u - v$. By (3), $\sin x - \sin y = \sin(u + v) - \sin(u - v) = 2\cos u \sin v$. □

These are the identities used in Examples 6.4.3 and 11.3.1 and Theorem 10.2.21.

Next comes π. The plan is to find the first positive zero of cos. To locate it, compare sin and cos with their Taylor polynomials.

Lemma 12.4.8. *For all $x \ge 0$,*

(1) $x - \frac{x^3}{6} \le \sin x \le x$, *and*
(2) $1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720} \le \cos x \le 1 - \frac{x^2}{2} + \frac{x^4}{24}$.

PROOF. Each inequality has the form $f(x) \ge 0$, where $f(0) = 0$ and f' is the function in the previous inequality. If $f' \ge 0$ on $[0, \infty)$ and $f(0) = 0$, then $f \ge 0$ on $[0, \infty)$ by Theorem 7.2.7. Start with $1 - \cos x \ge 0$, which holds by Theorem 12.4.6 (1). Then, in order,

$$f_1(x) = x - \sin x, \qquad f_1'(x) = 1 - \cos x \ge 0,$$

$$f_2(x) = \cos x - 1 + \tfrac{x^2}{2}, \qquad f_2'(x) = f_1(x) \geq 0,$$
$$f_3(x) = \sin x - x + \tfrac{x^3}{6}, \qquad f_3'(x) = f_2(x) \geq 0,$$
$$f_4(x) = 1 - \tfrac{x^2}{2} + \tfrac{x^4}{24} - \cos x, \qquad f_4'(x) = f_3(x) \geq 0,$$
$$f_5(x) = x - \tfrac{x^3}{6} + \tfrac{x^5}{120} - \sin x, \qquad f_5'(x) = f_4(x) \geq 0,$$
$$f_6(x) = \cos x - 1 + \tfrac{x^2}{2} - \tfrac{x^4}{24} + \tfrac{x^6}{720}, \qquad f_6'(x) = f_5(x) \geq 0.$$

Each f_j is 0 at 0. So $f_j \geq 0$ on $[0, \infty)$ for each j. Now $f_1, f_3 \geq 0$ give (1), and $f_4, f_6 \geq 0$ give (2). □

Theorem 12.4.9. *There is a number $x_0 \in \left(\frac{3}{2}, 2\right)$ with $\cos x_0 = 0$ and $\cos x > 0$ for $0 \leq x < x_0$.*

Proof. By Lemma 12.4.8 (2), $\cos 2 \leq 1 - 2 + \frac{16}{24} = -\frac{1}{3} < 0$. Let $p(x) = 1 - \frac{x^2}{2} + \frac{x^4}{24} - \frac{x^6}{720}$. For $0 < x \leq \frac{3}{2}$, $x^2 < 6$, so

$$p'(x) = -x + \frac{x^3}{6} - \frac{x^5}{120} = -x\left(1 - \frac{x^2}{6}\right) - \frac{x^5}{120} < 0.$$

So p is decreasing on $\left[0, \frac{3}{2}\right]$, and for $0 \leq x \leq \frac{3}{2}$,

$$\cos x \geq p(x) \geq p\left(\tfrac{3}{2}\right) = 1 - \frac{9}{8} + \frac{81}{384} - \frac{729}{46080} > 0.$$

(The value is about 0.07.) Let $Z = \{x \in [0, 2] : \cos x = 0\}$. Since $\cos \frac{3}{2} > 0 > \cos 2$, Z is not empty by Theorem 6.3.10. Let $x_0 = \inf Z$. Since $\cos > 0$ on $\left[0, \frac{3}{2}\right]$, $x_0 \geq \frac{3}{2}$. For each $n \in \mathbb{N}$ there is a $z_n \in Z$ with $x_0 \leq z_n < x_0 + \frac{1}{n}$. Then $z_n \to x_0$, and $\cos x_0 = \lim \cos z_n = 0$ by Theorem 6.3.9. So $x_0 \in Z$, and $x_0 \neq \frac{3}{2}$ and $x_0 \neq 2$. Finally, $\cos x \neq 0$ for $0 \leq x < x_0$. If $\cos x < 0$ for such an x, then cos would have a zero in $(0, x)$ by Theorem 6.3.10, which is smaller than x_0. So $\cos x > 0$ for $0 \leq x < x_0$. □

Definition 12.4.10. Define $\pi = 2x_0$, where x_0 is the number in Theorem 12.4.9. So π is twice the smallest positive zero of cos, and $3 < \pi < 4$.

Theorem 12.4.11. *(1) $\cos \frac{\pi}{2} = 0$, $\sin \frac{\pi}{2} = 1$, $\cos \pi = -1$ and $\sin \pi = 0$.*
(2) $\sin\left(x + \frac{\pi}{2}\right) = \cos x$ and $\cos\left(x + \frac{\pi}{2}\right) = -\sin x$ for all $x \in \mathbb{R}$.
(3) $\sin(x + \pi) = -\sin x$ and $\cos(x + \pi) = -\cos x$ for all $x \in \mathbb{R}$.
(4) $\sin(x + 2\pi) = \sin x$ and $\cos(x + 2\pi) = \cos x$ for all $x \in \mathbb{R}$.
(5) $\sin x > 0$ for $0 < x < \pi$.
(6) $\sin x = 0$ exactly when $x = k\pi$ for some $k \in \mathbb{Z}$, and $\cos x = 0$ exactly when $x = \frac{\pi}{2} + k\pi$ for some $k \in \mathbb{Z}$.

Proof. (1) By definition, $\cos \frac{\pi}{2} = 0$. On $\left[0, \frac{\pi}{2}\right)$, $\sin' = \cos > 0$. So sin is strictly increasing on $\left[0, \frac{\pi}{2}\right]$ by Theorem 7.2.7, and $\sin \frac{\pi}{2} > \sin 0 = 0$. Since $\sin^2 \frac{\pi}{2} = 1 - \cos^2 \frac{\pi}{2} = 1$, $\sin \frac{\pi}{2} = 1$. By Corollary 12.4.7 (1), $\cos \pi = 1 - 2\sin^2 \frac{\pi}{2} = -1$ and $\sin \pi = 2 \sin \frac{\pi}{2} \cos \frac{\pi}{2} = 0$.

(2) By Theorem 12.4.6 and (1), $\sin\left(x + \frac{\pi}{2}\right) = \sin x \cdot 0 + \cos x \cdot 1 = \cos x$ and $\cos\left(x + \frac{\pi}{2}\right) = \cos x \cdot 0 - \sin x \cdot 1 = -\sin x$.

(3) Apply (2) twice. Then $\sin(x+\pi) = \cos\left(x+\frac{\pi}{2}\right) = -\sin x$ and $\cos(x+\pi) = -\sin\left(x+\frac{\pi}{2}\right) = -\cos x$.

(4) Apply (3) twice.

(5) If $0 < x \le \frac{\pi}{2}$, then $\sin x > 0$ by the proof of (1). If $\frac{\pi}{2} < x < \pi$, then $0 < \pi - x < \frac{\pi}{2}$. By (3) and Theorem 12.4.3, $\sin x = -\sin(x-\pi) = \sin(\pi - x) > 0$.

(6) By (1) and (3), $\sin(k\pi) = \pm\sin 0 = 0$ for all $k \in \mathbb{Z}$. Conversely, suppose $\sin x = 0$. Let k be the integer with $k\pi \le x < (k+1)\pi$, and let $t = x - k\pi \in [0, \pi)$. By (3), $\sin t = \pm\sin x = 0$. By (5), $t = 0$. So $x = k\pi$. By (2), $\cos x = \sin\left(x+\frac{\pi}{2}\right)$. So $\cos x = 0$ exactly when $x + \frac{\pi}{2} = m\pi$ for some $m \in \mathbb{Z}$, that is, when $x = \frac{\pi}{2} + (m-1)\pi$. □

This π is the familiar one. The point $(\cos t, \sin t)$ lies on the unit circle, by Theorem 12.4.6 (1). As t goes from 0 to 2π, it goes around the circle once, starting and ending at $(1,0)$. See Exercise 12.4.7. Its velocity $(-\sin t, \cos t)$ has length 1. So the arc of the circle from $(1,0)$ to $(\cos t, \sin t)$ has length t, and the whole circle has length 2π. Arc length is not developed in this book, so this is only a remark. But it explains why the functions defined here are the sine and cosine of trigonometry, measured in radians.

Two limits from Chapter 6 now come for free. By Theorem 12.4.3,

$$\lim_{x\to 0}\frac{\sin x}{x} = \lim_{x\to 0}\frac{\sin x - \sin 0}{x - 0} = \cos 0 = 1 \quad \text{and} \quad \lim_{x\to 0}\frac{\cos x - 1}{x} = -\sin 0 = 0.$$

Compare Exercise 6.1.7, which uses a geometric inequality instead.

Definition 12.4.12. For $x \ne \frac{\pi}{2} + k\pi$, $k \in \mathbb{Z}$, define $\tan x = \frac{\sin x}{\cos x}$.

Theorem 12.4.13. *(1)* $\frac{d}{dx}\tan x = \frac{1}{\cos^2 x} = 1 + \tan^2 x$ *wherever* tan *is defined.*

(2) tan *is strictly increasing on* $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ *and maps it onto* $\mathbb{R}$.

(3) The inverse $\arctan : \mathbb{R} \to \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ *is differentiable, and* $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$.

(4) $\arctan x = \int_0^x \frac{dt}{1+t^2}$ *for all* $x \in \mathbb{R}$. *In particular,* $\pi = 4\int_0^1 \frac{dt}{1+t^2}$.

Proof. (1) By the Quotient Rule (Theorem 7.1.5) and Theorem 12.4.6 (1),

$$\frac{d}{dx}\tan x = \frac{\cos x \cos x - \sin x(-\sin x)}{\cos^2 x} = \frac{1}{\cos^2 x} = \frac{\cos^2 x + \sin^2 x}{\cos^2 x} = 1 + \tan^2 x.$$

(2) By (1), $\tan' > 0$ on $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$. So tan is strictly increasing there by Theorem 7.2.7. As $x \to \frac{\pi}{2}^-$, $\sin x \to 1$ and $\cos x \to 0$ with $\cos x > 0$. So $\tan x \to \infty$. Since $\tan(-x) = -\tan x$, $\tan x \to -\infty$ as $x \to -\frac{\pi}{2}^+$. Given $y \in \mathbb{R}$, choose $a < b$ in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ with $\tan a < y < \tan b$. By Theorem 6.3.10, $\tan c = y$ for some $c \in (a,b)$. So tan maps $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ onto $\mathbb{R}$.

(3) By Theorem 7.2.9, arctan is differentiable on $\mathbb{R}$. If $y = \tan x$ with $|x| < \frac{\pi}{2}$, then

$$\arctan'(y) = \frac{1}{\tan'(x)} = \frac{1}{1+\tan^2 x} = \frac{1}{1+y^2}.$$

(4) Let $F(x) = \int_0^x \frac{dt}{1+t^2}$. By Theorem 9.3.7, $F'(x) = \frac{1}{1+x^2} = \arctan'(x)$. So $F - \arctan$ is constant by Theorem 7.2.7 (5). Both are 0 at 0. So $F = \arctan$. Last, $\cos^2\frac{\pi}{4} - \sin^2\frac{\pi}{4} = \cos\frac{\pi}{2} = 0$ by Corollary 12.4.7 (1), and both $\sin\frac{\pi}{4}$ and $\cos\frac{\pi}{4}$ are positive. So $\sin\frac{\pi}{4} = \cos\frac{\pi}{4}$, $\tan\frac{\pi}{4} = 1$ and $\arctan 1 = \frac{\pi}{4}$. □

The inverse cosine was found in Example 7.2.10, and the inverse sine is Exercise 12.4.4.

The functions of this section are the ones built in Section 9.5. Write S and C for the sine and cosine of that section. By Theorem 9.5.10, $S'' + S = 0$, $S(0) = 0$ and $S'(0) = C(0) = 1$. So $S = \sin$ by Theorem 12.4.5. In the same way, $C = \cos$. In Section 9.5, π was defined as $2 \sup \arctan$, and Theorem 9.5.11 (4) shows that $\frac{\pi}{2}$ is the smallest positive zero of C. So the two definitions of π agree. By Theorem 12.4.13 (4), the two definitions of arctan agree too.

Exercises

12.4.1. Prove that $\sin^2 x = \frac{1-\cos 2x}{2}$ and $\cos^2 x = \frac{1+\cos 2x}{2}$ for all $x \in \mathbb{R}$. Then prove that $\sin 3x = 3\sin x - 4\sin^3 x$. [Hint, p. 361]

12.4.2. Prove that $\sin \frac{\pi}{4} = \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2}$, $\sin \frac{\pi}{6} = \frac{1}{2}$ and $\cos \frac{\pi}{6} = \frac{\sqrt{3}}{2}$. [Solution, p. 361]

12.4.3. Prove that $\tan(x+y) = \frac{\tan x + \tan y}{1 - \tan x \tan y}$ whenever $\tan x$, $\tan y$ and $\tan(x+y)$ are defined and $\tan x \tan y \neq 1$. [Hint, p. 362]

12.4.4. Prove that sin is strictly increasing on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ and maps it onto $[-1, 1]$. Let $\arcsin : [-1,1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ be the inverse. Prove that $\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$ for $-1 < x < 1$, and that $\arcsin x + \arccos x = \frac{\pi}{2}$ for $-1 \leq x \leq 1$, where arccos is as in Example 7.2.10. [Solution, p. 362]

12.4.5. Let $\omega > 0$, and let $y : \mathbb{R} \to \mathbb{R}$ be twice differentiable with $y'' + \omega^2 y = 0$ on $\mathbb{R}$. Prove that $y(x) = y(0)\cos \omega x + \frac{y'(0)}{\omega} \sin \omega x$ for all $x \in \mathbb{R}$. [Solution, p. 362]

12.4.6. Suppose $p \in \mathbb{R}$ and $\sin(x+p) = \sin x$ for all $x \in \mathbb{R}$. Prove that $p = 2k\pi$ for some $k \in \mathbb{Z}$. [Hint, p. 362]

12.4.7. Let $a, b \in \mathbb{R}$ with $a^2 + b^2 = 1$. Prove that there is exactly one $t \in [0, 2\pi)$ with $\cos t = a$ and $\sin t = b$. [Solution, p. 362]

12.4.8. Use Lemma 12.4.8 to prove that $3.1 < \pi < 3.2$. [Hint, p. 362]

12.4.9. Prove that $\arctan x = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1}$ for $|x| < 1$. Then use Theorem 12.2.1 to prove that

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \cdots .$$

[Hint, p. 363]

12.4.10. Prove that sin is not a polynomial. [Hint, p. 363]

12.5. The Exponential Function from Its Series*

Section 9.4 defined ln as an integral and e^x as its inverse. This section goes the other way. It defines e^x by its power series and gets ln as the inverse. It can be read without Section 9.4. At the end, the two approaches are shown to give the same functions. A reader willing to assume the familiar properties of e^x and $\ln x$ can skip both sections.

Definition 12.5.1. For $x \in \mathbb{R}$, define

$$E(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!} = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots .$$

The series converges for every x. For $x \neq 0$, the ratio of consecutive terms is $\frac{|x|}{k+1} \to 0$, so it converges absolutely by the Ratio Test (Theorem 10.2.13). So its radius of convergence is ∞, by Theorem 12.1.7 (2).

Theorem 12.5.2. *(1) E is differentiable on $\mathbb{R}$, $E' = E$ and $E(0) = 1$.*
(2) $E(x)E(-x) = 1$ for all $x \in \mathbb{R}$.
(3) $E(x+y) = E(x)E(y)$ for all $x, y \in \mathbb{R}$.
(4) $E(x) > 0$ for all x, and $E(x) \geq 1 + x$ for all $x \geq 0$.
(5) E is strictly increasing, $\lim_{x\to\infty} E(x) = \infty$ and $\lim_{x\to-\infty} E(x) = 0$.
(6) E maps $\mathbb{R}$ onto $(0, \infty)$.

Proof. (1) By Theorem 12.2.3,

$$E'(x) = \sum_{k=1}^{\infty} \frac{kx^{k-1}}{k!} = \sum_{k=1}^{\infty} \frac{x^{k-1}}{(k-1)!} = \sum_{j=0}^{\infty} \frac{x^j}{j!} = E(x).$$

Setting $x = 0$ leaves only the first term, so $E(0) = 1$.

(2) Let $g(x) = E(x)E(-x)$. By the Product Rule and the Chain Rule, $g'(x) = E(x)E(-x) - E(x)E(-x) = 0$. So g is constant by Theorem 7.2.7, and $g(x) = g(0) = 1$.

(3) Fix y, and let $h(x) = E(x+y)E(-x)$. Then $h'(x) = E(x+y)E(-x) - E(x+y)E(-x) = 0$. So $h(x) = h(0) = E(y)$. Multiply $E(x+y)E(-x) = E(y)$ by $E(x)$ and use (2).

(4) If $x \geq 0$, every term of the series is at least 0, so $E(x) \geq 1 + x \geq 1$. If $x < 0$, then $E(x) = \frac{1}{E(-x)} > 0$ by (2).

(5) Since $E' = E > 0$, E is strictly increasing by Theorem 7.2.7. Since $E(x) \geq 1 + x$ for $x \geq 0$, $E(x) \to \infty$ as $x \to \infty$. Then $E(x) = \frac{1}{E(-x)} \to 0$ as $x \to -\infty$.

(6) Let $y > 0$. By (5), there are $a < b$ with $E(a) < y < E(b)$. Since E is continuous, Theorem 6.3.10 gives a c with $E(c) = y$. □

Definition 12.5.3. Let $e = E(1)$. Let $L : (0, \infty) \to \mathbb{R}$ be the inverse of E.

Remark 12.5.4. Here E and L are names of functions, just as f is, and $E(x)$ and $L(x)$ are their values. The traditional names for these functions are exp and ln. By custom ln is written without parentheses, so $\ln x$ means $\ln(x)$, and ln alone, as in "ln is increasing," refers to the whole function. Parentheses are used when the input is more than a single symbol, as in $\ln(1 + x)$.

Theorem 12.5.5. *(1) L is differentiable, $L'(x) = \frac{1}{x}$ and $L(1) = 0$. So $L(x) = \int_1^x \frac{1}{t}\, dt$ for all $x > 0$.*
(2) $L(ab) = L(a) + L(b)$ for all $a, b > 0$.
(3) $E(r) = e^r$ for all $r \in \mathbb{Q}$.
(4) $e = \lim_{n\to\infty} \left(1 + \frac{1}{n}\right)^n$.

PROOF. (1) By Theorem 7.2.9, L is differentiable on $(0,\infty)$. If $x = E(y)$, then $L'(x) = \frac{1}{E'(y)} = \frac{1}{E(y)} = \frac{1}{x}$. Since $E(0) = 1$, $L(1) = 0$. The function $\int_1^x \frac{1}{t}\,dt$ also has derivative $\frac{1}{x}$ and is 0 at 1, by Theorem 9.3.7. So the two are equal by Theorem 7.2.7 (5).

(2) Let $a = E(s)$ and $b = E(t)$. Then $ab = E(s+t)$ by Theorem 12.5.2 (3). So $L(ab) = s + t = L(a) + L(b)$.

(3) By (3) of Theorem 12.5.2 and induction, $E(nx) = E(x)^n$ for $n \in \mathbb{N}$. So $E(n) = e^n$, and $E(-n) = \frac{1}{E(n)} = e^{-n}$. Let $r = \frac{m}{n}$ with $m \in \mathbb{Z}$ and $n \in \mathbb{N}$. Then $E(r)^n = E(m) = e^m$. Since $E(r) > 0$, it is the positive nth root of e^m, which is $e^{m/n}$ by Theorem 1.8.13.

(4) By (2) and induction, $L(b^n) = nL(b)$ for $b > 0$ and $n \in \mathbb{N}$. So $\left(1 + \frac{1}{n}\right)^n = E\left(nL\left(1 + \frac{1}{n}\right)\right)$. Since $L'(1) = 1$, Theorem 6.1.18 with $x_n = 1 + \frac{1}{n}$ gives

$$nL\left(1 + \frac{1}{n}\right) = \frac{L\left(1 + \frac{1}{n}\right) - L(1)}{1/n} \to 1.$$

Since E is continuous, $\left(1 + \frac{1}{n}\right)^n \to E(1) = e$ by Theorem 6.3.9. □

Part (1) says L is the function $\ln x = \int_1^x \frac{1}{t}\,dt$ of Section 9.4. Part (4) says e is the number e defined in Example 9.4.1. So E, the inverse of L, is the function exp of Definition 9.4.5, and $e^x = \exp(x) = E(x)$. The two approaches give the same functions. A reader of this section can go on to Definition 9.4.7 to define a^x and $\log_a$. The facts used there from Theorems 9.4.4 and 9.4.6 follow from Theorems 12.5.2 and 12.5.5 in the same way.

The proof of (3) in Theorem 12.5.2 is the same trick as Lemma 12.4.4. A function is pinned down by a differential equation and one value. See Exercise 9.4.4.

Remark 12.5.6. Compare the three series

$$E(x) = \sum_{k=0}^{\infty} \frac{x^k}{k!}, \qquad \cos x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!}, \qquad \sin x = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}.$$

The series for cos and sin are the even and odd parts of the series for E, with alternating signs. Without the signs, the even and odd parts give the hyperbolic functions $\cosh x$ and $\sinh x$ (Exercise 12.5.4). The signs come from powers of $i = \sqrt{-1}$, since $i^2 = -1$, $i^3 = -i$ and $i^4 = 1$. If complex numbers are allowed in the series, then

$$E(ix) = \cos x + i \sin x.$$

This is Euler's formula. Making it precise requires series of complex numbers, which belong to complex analysis. This book stays with real numbers.

Exercises

12.5.1. Let $x \geq 0$ and $n \in \mathbb{N}_0$. Prove that $E(x) \geq \sum_{k=0}^{n} \frac{x^k}{k!}$. Use this to prove that $\lim_{x\to\infty} \frac{x^n}{E(x)} = 0$. [Hint, p. 363]

12.5.2. Prove that $0 < e - \sum_{k=0}^{n} \frac{1}{k!} < \frac{1}{n!\,n}$ for all $n \in \mathbb{N}$. Use $n = 7$ to prove that $2.718 < e < 2.719$. [Solution, p. 363]

12.5.3. Prove that e is irrational. (Suppose $e = \frac{p}{q}$ with $p, q \in \mathbb{N}$. Multiply the inequality of Exercise 12.5.2 with $n = q$ by $q!$.) [Solution, p. 363]

12.5.4. Let $\cosh x = \frac{E(x)+E(-x)}{2}$ and $\sinh x = \frac{E(x)-E(-x)}{2}$. Prove that $\cosh x = \sum_{k=0}^{\infty} \frac{x^{2k}}{(2k)!}$ and $\sinh x = \sum_{k=0}^{\infty} \frac{x^{2k+1}}{(2k+1)!}$. [Hint, p. 363]

12.5.5. Let $x \geq 0$. Use the Binomial Theorem to prove that $\left(1 + \frac{x}{n}\right)^n \leq E(x)$ for all $n \in \mathbb{N}$. [Hint, p. 363]

Notes

Brook Taylor published the series that bears his name in 1715 [**72**]. Colin Maclaurin used the special case $c = 0$ in 1742 [**53**]. Joseph-Louis Lagrange gave his form of the remainder in 1797 [**41**]. Cauchy gave his form in 1823, along with the function of Example 12.2.10, which shows that a Taylor series can converge to the wrong function [**16**]. Cauchy stated the formula $\frac{1}{R} = \limsup \sqrt[k]{|a_k|}$ in his *Cours d'analyse* [**15**]. Jacques Hadamard rediscovered it and made it widely known in 1892 [**35**], and it is now called the Cauchy–Hadamard formula. Abel proved Theorem 12.2.1 in 1826, in the same paper that first proved the binomial series of Exercise 12.3.6 converges [**1**]. Nicholas Mercator published the series for $\ln(1 + x)$ in 1668 [**54**]. Gottfried Wilhelm Leibniz published the series $\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \cdots$ in 1682 [**45**]. Madhava of Sangamagrama had found it around 1400. Madhava also knew the series for $\sin x$ and $\cos x$. Isaac Newton found them again in 1669. Leonhard Euler made sin and cos functions of a real variable in his *Introductio in analysin infinitorum* of 1748. Defining them by their series, as in Section 12.4, makes them part of analysis instead of geometry.

Part III

Measure and Lebesgue Integrals

Part III builds Lebesgue measure and the Lebesgue integral. Chapter 13 sets up the general language of σ-algebras and measures. Chapter 14 constructs Lebesgue measure on $\mathbb{R}$, the main example. Chapters 15 and 16 define measurable functions and their integrals, and prove the convergence theorems that make the Lebesgue integral more useful than the Riemann integral. Chapter 17 connects integration and differentiation. Part III assumes Parts I and II. Sections marked with a star can be skipped. The starred Chapter 18 introduces the spaces L^p.

CHAPTER 13

σ-Algebras and Measures

The Riemann integral of Chapter 9 measures area by cutting $[a, b]$ into finitely many intervals. That works well for continuous functions and badly for limits. Lebesgue's idea was to cut up the range of f instead, and measure the sets $\{x : c < f(x) \le d\}$. Those sets need not be intervals. They can be complicated sets like the Cantor set or the rationals. So the first job is to assign a length, or *measure*, to sets far more general than intervals.

Which sets can be measured, and what rules should measure obey? This chapter answers both questions in general. A σ-algebra is a collection of sets that is closed under complements and countable unions. A measure is a function on a σ-algebra that adds up correctly over countably many disjoint pieces. Lebesgue measure on $\mathbb{R}$, built in Chapter 14, is the main example. The same language is the foundation of probability theory. A probability space is a measure space whose total measure is 1.

This chapter uses a few things from Parts I and II. They are sets and De Morgan's Laws (Section 1.1), countable sets (Chapter 2), open and closed sets and the structure of open sets (Section 4.1), and series of nonnegative terms (Theorem 10.2.6).

13.1. Algebras and σ-Algebras

Throughout Part III, X denotes a nonempty set. For $E \subset X$, the complement E^c means $X \setminus E$. The power set $\mathcal{P}(X)$ is the collection of all subsets of X (Definition 1.1.19).

Definition 13.1.1. Let X be a nonempty set. An *algebra* of sets on X is a nonempty collection $\mathcal{A}$ of subsets of X such that

(1) if $E_1, \dots, E_n \in \mathcal{A}$, then $\bigcup_{i=1}^n E_i \in \mathcal{A}$; and
(2) if $E \in \mathcal{A}$, then $E^c \in \mathcal{A}$.

An algebra $\mathcal{A}$ on a nonempty set X is a *σ-algebra* if for any sequence $\{E_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$, $\bigcup_{n\in\mathbb{N}} E_n \in \mathcal{A}$.

The σ stands for "sum," an old word for union. An algebra is closed under finite unions. A σ-algebra is closed under countable unions.

Proposition 13.1.2. *An algebra $\mathcal{A}$ on a nonempty set X has the property that if $E_1, \dots, E_n \in \mathcal{A}$, then $\bigcap_{i=1}^n E_i \in \mathcal{A}$. Similarly, a σ-algebra is closed under countable intersections.*

Proof. Using De Morgan's law, $\bigcap_{i=1}^n E_i = (\bigcup_{i=1}^n E_i^c)^c$. Since $\mathcal{A}$ is an algebra, $E_i^c \in \mathcal{A}$ for each $i = 1, \dots, n$, implying $\bigcup_{i=1}^n E_i^c \in \mathcal{A}$. Therefore, $(\bigcup_{i=1}^n E_i^c)^c \in \mathcal{A}$. So $\bigcap_{i=1}^n E_i \in \mathcal{A}$. The countable case is the same, using Theorem 1.1.14 for a countable family. □

Remark 13.1.3. If $\mathcal{A}$ is an algebra on a nonempty set X, then $\emptyset, X \in \mathcal{A}$. Indeed, since $\mathcal{A}$ is nonempty, there is an $E \in \mathcal{A}$, implying $E^c \in \mathcal{A}$. So $E \cap E^c = \emptyset \in \mathcal{A}$. Thus $\emptyset^c = X \in \mathcal{A}$. Also, if $E, F \in \mathcal{A}$, then $E \setminus F = E \cap F^c \in \mathcal{A}$.

Theorem 13.1.4. *An algebra $\mathcal{A}$ is a σ-algebra if $\{F_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$ with $F_i \cap F_j = \emptyset$ for all $i \neq j \in \mathbb{N}$ implies $\bigcup_{n\in\mathbb{N}} F_n \in \mathcal{A}$.*

PROOF. Suppose $\{E_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$. Set

$$F_k = E_k \setminus \left[\bigcup_{i=1}^{k-1} E_i\right] = E_k \cap \left[\bigcup_{i=1}^{k-1} E_i\right]^c.$$

Since $E_i \in \mathcal{A}$, $i = 1, \ldots, k-1$, $\bigcup_{i=1}^{k-1} E_i \in \mathcal{A}$ and since $\mathcal{A}$ is an algebra, $\left[\bigcup_{i=1}^{k-1} E_i\right]^c \in \mathcal{A}$. Thus $F_k \in \mathcal{A}$. Now the sets F_k are pairwise disjoint. So $\bigcup_{n\in\mathbb{N}} F_n \in \mathcal{A}$. But $\bigcup_{n\in\mathbb{N}} F_n = \bigcup_{n\in\mathbb{N}} E_n$. So $\bigcup_{n\in\mathbb{N}} E_n \in \mathcal{A}$ and $\mathcal{A}$ is a σ-algebra.

To see the two claims about the sets F_k, note that F_k is the set of points of E_k that are in none of the earlier sets $E_1, \ldots, E_{k-1}$. If $j < k$, then $F_k \cap E_j = \emptyset$, and $F_j \subset E_j$, so $F_j \cap F_k = \emptyset$. Each $F_n \subset E_n$, so $\bigcup F_n \subset \bigcup E_n$. If $x \in \bigcup E_n$, let k be the smallest index with $x \in E_k$. Then $x \in F_k$. □

Proof Technique (Making Sets Disjoint). Measures add up correctly over disjoint sets. So many proofs start by replacing a sequence $E_1, E_2, \ldots$ with disjoint sets that have the same union. Take $F_1 = E_1$ and $F_k = E_k \setminus (E_1 \cup \cdots \cup E_{k-1})$. Each new set keeps only the points not already covered. The union does not change, $F_k \subset E_k$, and the sets F_k are disjoint. If the sets increase, $E_1 \subset E_2 \subset \cdots$, this simplifies to $F_k = E_k \setminus E_{k-1}$. The proofs of Theorems 13.1.4 and 13.3.9 use this trick.

Example 13.1.5. If X is a nonempty set, $\mathcal{P}(X)$ and $\{\emptyset, X\}$ are σ-algebras.

PROOF. $\mathcal{P}(X)$ contains every subset of X. The complement of any subset of X is a subset of X and the union of countably many subsets of X is a subset of X, so $\mathcal{P}(X)$ is closed under complements and countable unions. Since $X^c = \emptyset$, $\emptyset^c = X$, and $X \cup \emptyset = X$, $\{\emptyset, X\}$ is a σ-algebra. □

Example 13.1.6. If X is uncountable, then

$$\mathcal{A} = \{E \subset X : E \text{ is countable or } E^c \text{ is countable}\}$$

is a σ-algebra, called the *σ-algebra of countable or co-countable sets*. Here "countable" means at most countable, so finite sets are included.

PROOF. Let $E \in \mathcal{A}$. If E is countable, then $(E^c)^c = E$ is countable and so $E^c \in \mathcal{A}$. If E^c is countable, then $E^c \in \mathcal{A}$ directly. Let $\{E_n\}_{n\in\mathbb{N}} \subset \mathcal{A}$. If E_n is countable for all $n \in \mathbb{N}$, then $\bigcup_{n\in\mathbb{N}} E_n$ is countable (Theorem 2.2.6) and so $\bigcup_{n\in\mathbb{N}} E_n \in \mathcal{A}$. If there exists an $n_0 \in \mathbb{N}$ with E_{n_0} uncountable, then $E_{n_0}^c$ is countable and

$$\left(\bigcup_{n\in\mathbb{N}} E_n\right)^c = \bigcap_{n\in\mathbb{N}} E_n^c \subset E_{n_0}^c.$$

A subset of a countable set is countable. Therefore, $\left(\bigcup_{n\in\mathbb{N}} E_n\right)^c$ is countable. So $\bigcup_{n\in\mathbb{N}} E_n \in \mathcal{A}$. □

Proof Technique (Showing a Collection Is a σ-Algebra). There are three things to check. The collection is nonempty, usually by showing X or $\emptyset$ is in it. It is closed under complements. It is closed under countable unions. Check each one directly from the definition of the collection, as in Example 13.1.6. For the union, it is often easier to check disjoint unions, or increasing unions, and then use Theorem 13.1.4 or Exercise 13.1.3. Closure under intersections and differences then comes for free, by Proposition 13.1.2 and Remark 13.1.3.

Lemma 13.1.7. *The intersection of a family of σ-algebras on a set X is a σ-algebra.*

Proof. See Exercise 13.1.1. □

Lemma 13.1.7 gives a way to build σ-algebras. Start with any collection $\mathcal{E}$ of subsets of X. There is at least one σ-algebra containing $\mathcal{E}$, namely $\mathcal{P}(X)$. The intersection of all of them is a σ-algebra that contains $\mathcal{E}$, and it is contained in every σ-algebra that contains $\mathcal{E}$. So it is the smallest one.

Definition 13.1.8. If $\mathcal{E} \subset \mathcal{P}(X)$, the smallest σ-algebra containing $\mathcal{E}$, denoted $\sigma(\mathcal{E})$, is called the *σ-algebra generated by $\mathcal{E}$*. That is, $\sigma(\mathcal{E})$ is the intersection of all σ-algebras on X that contain $\mathcal{E}$.

Example 13.1.9. Let $X = \{a, b, c\}$. For $\mathcal{E}_1 = \{\{a\}\}$, $\sigma(\mathcal{E}_1) = \{\emptyset, \{a\}, \{b, c\}, X\}$. For $\mathcal{E}_2 = \{\{a\}, \{b\}\}$, $\sigma(\mathcal{E}_2) = \mathcal{P}(X)$.

Proof. If $\{a\} \in \sigma(\mathcal{E}_1)$, then $\{a\}^c = \{b, c\} \in \sigma(\mathcal{E}_1)$. So $\{a\} \cup \{b, c\} = X \in \sigma(\mathcal{E}_1)$ and $X^c = \emptyset \in \sigma(\mathcal{E}_1)$. So $\sigma(\mathcal{E}_1)$ contains $\{\emptyset, \{a\}, \{b, c\}, X\}$. This collection is itself a σ-algebra, since it is closed under complements and unions. So $\sigma(\mathcal{E}_1) = \{\emptyset, \{a\}, \{b, c\}, X\}$.

If $\{a\}, \{b\} \in \sigma(\mathcal{E}_2)$, then $\{a\} \cup \{b\} = \{a, b\} \in \sigma(\mathcal{E}_2)$. Then $\{a, b\}^c = \{c\} \in \sigma(\mathcal{E}_2)$. Also, $\{a\}^c = \{b, c\} \in \sigma(\mathcal{E}_2)$ and $\{b\}^c = \{a, c\} \in \sigma(\mathcal{E}_2)$. Lastly, $\{a, b\} \cup \{c\} = X \in \sigma(\mathcal{E}_2)$, implying $X^c = \emptyset \in \sigma(\mathcal{E}_2)$. So every subset of X is in $\sigma(\mathcal{E}_2)$, and $\sigma(\mathcal{E}_2) = \mathcal{P}(X)$. □

Proposition 13.1.10. *If $\mathcal{E} \subset \mathcal{P}(X)$, $\sigma(\mathcal{E})$ is unique.*

Proof. Assume $\mathcal{M}_1$ and $\mathcal{M}_2$ are two smallest σ-algebras containing $\mathcal{E}$. Then by Lemma 13.1.7, $\mathcal{M}_1 \cap \mathcal{M}_2$ is a σ-algebra containing $\mathcal{E}$. But $\mathcal{M}_1 \cap \mathcal{M}_2 \subset \mathcal{M}_1$ and $\mathcal{M}_1 \cap \mathcal{M}_2 \subset \mathcal{M}_2$. Since $\mathcal{M}_1$ and $\mathcal{M}_2$ are both smallest σ-algebras containing $\mathcal{E}$, $\mathcal{M}_1 \cap \mathcal{M}_2 = \mathcal{M}_1$ and $\mathcal{M}_1 \cap \mathcal{M}_2 = \mathcal{M}_2$. Thus $\mathcal{M}_1 = \mathcal{M}_2$. □

Lemma 13.1.11. *If $\mathcal{E} \subset \sigma(\mathcal{F})$, then $\sigma(\mathcal{E}) \subset \sigma(\mathcal{F})$.*

Proof. $\sigma(\mathcal{F})$ is a σ-algebra containing $\mathcal{E}$. Therefore, it must contain the smallest σ-algebra containing $\mathcal{E}$, $\sigma(\mathcal{E})$. □

Proof Technique (The Good Sets Principle). How can a statement be proved for every set in $\sigma(\mathcal{E})$, when nobody can list those sets? Let $\mathcal{G}$ be the collection of "good" sets, the sets for which the statement is true. Show two things. First, $\mathcal{G}$ is a σ-algebra. Second, $\mathcal{E} \subset \mathcal{G}$. Then $\sigma(\mathcal{E}) \subset \mathcal{G}$, because $\sigma(\mathcal{E})$ is the smallest σ-algebra containing $\mathcal{E}$. So every set in $\sigma(\mathcal{E})$ is good. Lemma 13.1.11 is the special case where the good sets are the sets in $\sigma(\mathcal{F})$. This argument appears again and again in Part III. For example, it shows that every Borel set is Lebesgue measurable.

Definition 13.1.12. A family of sets $\mathcal{R} \subset \mathcal{P}(X)$ is called a *ring* if $E_1, \ldots, E_n \in \mathcal{R}$ implies $\bigcup_{i=1}^n E_i \in \mathcal{R}$, and if $E, F \in \mathcal{R}$, then $E \setminus F \in \mathcal{R}$. A ring that is closed under countable unions is called a *σ-ring*.

Proposition 13.1.13. *Let $\mathcal{R}$ be a ring on a set X.*

(1) $\mathcal{R}$ is closed under finite intersections. Similarly, if $\mathcal{R}$ is a σ-ring, $\mathcal{R}$ is closed under countable intersections.
(2) If $\mathcal{R}$ is a σ-ring, then $\mathcal{R}$ is a σ-algebra if and only if $X \in \mathcal{R}$.
(3) If $\mathcal{R}$ is a σ-ring, then $\{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$ is a σ-algebra.

PROOF. See Exercise 13.1.2. □

Exercises

13.1.1. Prove that the intersection of a family of σ-algebras on a set X is a σ-algebra. [Solution, p. 363]

13.1.2. Prove Proposition 13.1.13.
(a) Prove that a ring $\mathcal{R}$ is closed under finite intersections.
(b) Prove that if $\mathcal{R}$ is a σ-ring, then $\mathcal{R}$ is a σ-algebra if and only if $X \in \mathcal{R}$.
(c) Prove that if $\mathcal{R}$ is a σ-ring, then $\{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$ is a σ-algebra.

[Solution, p. 363]

13.1.3. Prove that an algebra $\mathcal{A}$ is a σ-algebra if and only if it is closed under countable increasing unions; i.e. if $\{E_i\}_{i=1}^\infty \subset \mathcal{A}$ and $E_1 \subset E_2 \subset \cdots$, then $\bigcup_{i=1}^\infty E_i \in \mathcal{A}$.

13.1.4. Let X be a finite set. Prove that every algebra on X is a σ-algebra. Find $\sigma(\mathcal{E})$ when $X = \{1, 2, 3, 4\}$ and $\mathcal{E} = \{\{1, 2\}, \{2, 3\}\}$. [Hint, p. 364]

13.1.5. Let X be uncountable, and let $\mathcal{E}$ be the collection of one-point subsets of X. Prove that $\sigma(\mathcal{E})$ is the σ-algebra of countable or co-countable sets.

[Hint, p. 364]

13.1.6. Give an example of two σ-algebras on $X = \{1, 2, 3\}$ whose union is not a σ-algebra. [Hint, p. 364]

13.2. Borel Sets

On $\mathbb{R}$ the most important σ-algebra is the one generated by the open sets. Every set that can be built from open sets by countably many unions, intersections and complements is in it. Chapter 14 shows that Lebesgue measure assigns a length to every such set.

Definition 13.2.1. If $X = \mathbb{R}$, the σ-algebra generated by the family of open sets in $\mathbb{R}$ is called the *Borel σ-algebra* and is denoted $\mathcal{B}_{\mathbb{R}}$. The members of $\mathcal{B}_{\mathbb{R}}$ are called *Borel sets*. $\mathcal{B}_{\mathbb{R}}$ thus contains open sets, closed sets, countable intersections of open sets, countable unions of closed sets, and so forth.

Definition 13.2.2. A countable intersection of open sets is called a G_δ *set*, a countable union of closed sets is called an F_σ *set*, a countable union of G_δ sets is called a $G_{\delta\sigma}$ set, a countable intersection of F_σ sets is called an $F_{\sigma\delta}$ set, and so forth.

Example 13.2.3. Every countable subset of $\mathbb{R}$ is an F_σ set, since it is a countable union of one-point sets, which are closed. In particular, $\mathbb{Q}$ is an F_σ set. Its complement, the set of irrational numbers, is a G_δ set, by De Morgan's Laws. A half-open interval $(a, b]$ is both, since $(a, b] = \bigcap_{n\in\mathbb{N}} \left(a, b + \frac{1}{n}\right)$ and $(a, b] = \bigcup_{n\in\mathbb{N}} \left[a + \frac{1}{n}, b\right]$. All of these sets are Borel sets.

Example 13.2.4. If $f : \mathbb{R} \to \mathbb{R}$, the set of points at which f is continuous is a G_δ.

Proof. See Exercise 13.2.1, or Theorem 6.6.2 in the starred Section 6.6. □

Theorem 13.2.5. *$\mathcal{B}_{\mathbb{R}}$ is generated by each of the following:*

(1) the open intervals $\mathcal{E}_1 = \{(a, b) : a < b\}$;
(2) the closed intervals $\mathcal{E}_2 = \{[a, b] : a < b\}$;
(3) the half open intervals $\mathcal{E}_3 = \{(a, b] : a < b\}$ or $\mathcal{E}_4 = \{[a, b) : a < b\}$;
(4) the open rays $\mathcal{E}_5 = \{(a, \infty) : a \in \mathbb{R}\}$ or $\mathcal{E}_6 = \{(-\infty, a) : a \in \mathbb{R}\}$; and
(5) the closed rays $\mathcal{E}_7 = \{[a, \infty) : a \in \mathbb{R}\}$ or $\mathcal{E}_8 = \{(-\infty, a] : a \in \mathbb{R}\}$.

Proof. The goal is to show $\mathcal{B}_{\mathbb{R}} = \sigma(\mathcal{E}_i)$ for $i = 1, \ldots, 8$. First, note that the elements of $\mathcal{E}_1$, $\mathcal{E}_2$, $\mathcal{E}_5$, $\mathcal{E}_6$, $\mathcal{E}_7$, and $\mathcal{E}_8$ are open or closed, so they are Borel sets. Since $(a, b] = \bigcap_{n\in\mathbb{N}} \left(a, b + \frac{1}{n}\right)$ and $[a, b) = \bigcap_{n\in\mathbb{N}} \left(a - \frac{1}{n}, b\right)$, all elements of $\mathcal{E}_3$ and $\mathcal{E}_4$ are G_δ sets. So $\mathcal{E}_i \subset \mathcal{B}_{\mathbb{R}}$ for $i = 1, \ldots, 8$, and so by Lemma 13.1.11, $\sigma(\mathcal{E}_i) \subset \mathcal{B}_{\mathbb{R}}$ for $i = 1, \ldots, 8$.

Now by Theorem 4.1.12, every open set is a countable union of open intervals, so by Lemma 13.1.11, $\mathcal{B}_{\mathbb{R}} \subset \sigma(\mathcal{E}_1)$. Some of these intervals may be unbounded. But $(a, \infty) = \bigcup_{n\in\mathbb{N}}(a, a + n)$, $(-\infty, b) = \bigcup_{n\in\mathbb{N}}(b - n, b)$ and $\mathbb{R} = \bigcup_{n\in\mathbb{N}}(-n, n)$, so every open interval is in $\sigma(\mathcal{E}_1)$.

Since $(a, b) = \bigcup_{n\in\mathbb{N}} \left[a + \frac{1}{n}, b - \frac{1}{n}\right]$, $\mathcal{E}_1 \subset \sigma(\mathcal{E}_2)$. So $\mathcal{B}_{\mathbb{R}} \subset \sigma(\mathcal{E}_2)$. (Here only the n with $a + \frac{1}{n} < b - \frac{1}{n}$ are used.)

Since $(a, b) = \bigcup_{n\in\mathbb{N}} \left(a, b - \frac{1}{n}\right]$ and $(a, b) = \bigcup_{n\in\mathbb{N}} \left[a + \frac{1}{n}, b\right)$, $\mathcal{E}_1 \subset \sigma(\mathcal{E}_3)$ and $\mathcal{E}_1 \subset \sigma(\mathcal{E}_4)$. So $\mathcal{B}_{\mathbb{R}} \subset \sigma(\mathcal{E}_3)$ and $\mathcal{B}_{\mathbb{R}} \subset \sigma(\mathcal{E}_4)$.

Now $(a, b] = (a, \infty) \cap (b, \infty)^c$ and $[a, b) = (-\infty, a)^c \cap (-\infty, b)$, and so $\mathcal{E}_3 \subset \sigma(\mathcal{E}_5)$ and $\mathcal{E}_4 \subset \sigma(\mathcal{E}_6)$. So $\mathcal{B}_{\mathbb{R}} \subset \sigma(\mathcal{E}_5)$ and $\mathcal{B}_{\mathbb{R}} \subset \sigma(\mathcal{E}_6)$.

Lastly, $[a, b) = [a, \infty) \cap [b, \infty)^c$ and $(a, b] = (-\infty, a]^c \cap (-\infty, b]$, and so $\mathcal{E}_4 \subset \sigma(\mathcal{E}_7)$ and $\mathcal{E}_3 \subset \sigma(\mathcal{E}_8)$. So $\mathcal{B}_{\mathbb{R}} \subset \sigma(\mathcal{E}_7)$ and $\mathcal{B}_{\mathbb{R}} \subset \sigma(\mathcal{E}_8)$. □

Exercises

13.2.1. Let $f : \mathbb{R} \to \mathbb{R}$. Prove that the set of points at which f is continuous is a G_δ. [Hint, p. 364]

13.2.2. Let $F \subset \mathbb{R}$ be closed. For $n \in \mathbb{N}$, let $U_n = \bigcup_{y \in F} \left(y - \frac{1}{n}, y + \frac{1}{n}\right)$. Prove that $F = \bigcap_{n \in \mathbb{N}} U_n$. Conclude that every closed set is a G_δ set and every open set is an F_σ set. [Hint, p. 364]

13.2.3. Prove that the Cantor set of Chapter 5 is a Borel set. [Hint, p. 365]

13.2.4. Prove that $\mathcal{B}_\mathbb{R}$ is generated by the compact subsets of $\mathbb{R}$. [Hint, p. 365]

13.3. Measures

A measure assigns a size to each set in a σ-algebra. It must give the empty set size 0, and it must add up over countably many disjoint sets. Sizes may be infinite. So a measure takes values in $[0, \infty]$, with the usual conventions. For $a \in [0, \infty]$, $a + \infty = \infty + a = \infty$, and $a \leq \infty$. A series $\sum a_n$ of terms in $[0, \infty]$ always has a sum in $[0, \infty]$. It is ∞ if some $a_n = \infty$ or if the partial sums are not bounded (Theorem 10.2.6), and otherwise it is the usual sum.

Definition 13.3.1. Let X be a set with σ-algebra $\mathcal{M}$. A *measure* on $\mathcal{M}$ (or on $(X, \mathcal{M})$ or on X if $\mathcal{M}$ is understood) is a function $\mu : \mathcal{M} \to [0, \infty]$ such that

(1) $\mu(\emptyset) = 0$; and
(2) if $\{E_n\}_{n \in \mathbb{N}}$ is a sequence of pairwise disjoint sets in $\mathcal{M}$, then $\mu\left(\bigcup_{n \in \mathbb{N}} E_n\right) = \sum_{n \in \mathbb{N}} \mu(E_n)$.

Property (2) is called *countable additivity*.

Definition 13.3.2. If μ is a measure on $(X, \mathcal{M})$, $(X, \mathcal{M}, \mu)$ is called a *measure space* and the sets in $\mathcal{M}$ are called *measurable sets*. If $\mu(X) = 1$, then μ is a *probability measure* and $(X, \mathcal{M}, \mu)$ is a *probability space*.

Remark 13.3.3. A measure is also *finitely additive*. If $E_1, \ldots, E_n \in \mathcal{M}$ are disjoint, set $E_k = \emptyset$ for $k > n$. Then countable additivity and $\mu(\emptyset) = 0$ give $\mu\left(\bigcup_{k=1}^n E_k\right) = \sum_{k=1}^n \mu(E_k)$.

Definition 13.3.4. Let $(X, \mathcal{M}, \mu)$ be a measure space. If $\mu(X) < \infty$, μ is called *finite*. If $X = \bigcup_{n \in \mathbb{N}} E_n$, where $E_n \in \mathcal{M}$ and $\mu(E_n) < \infty$ for all $n \in \mathbb{N}$, μ is called *σ-finite*. If $E \in \mathcal{M}$ with $E = \bigcup_{n \in \mathbb{N}} E_n$, where $E_n \in \mathcal{M}$ and $\mu(E_n) < \infty$ for all $n \in \mathbb{N}$, E is called σ-finite for μ. If for each $E \in \mathcal{M}$ with $\mu(E) = \infty$, there exists an $F \in \mathcal{M}$ with $F \subset E$ and $0 < \mu(F) < \infty$, μ is called *semifinite*.

Example 13.3.5. Let $X \neq \emptyset$, $\mathcal{M} = \mathcal{P}(X)$, and $f : X \to [0, \infty]$ be a function. Then for $E \in \mathcal{M}$, define $\mu(E) = \sum_{x \in E} f(x)$. (Note: $\sum_{x \in E} f(x) = \sup\{\sum_{x \in F} f(x) : F \subset E \text{ is finite}\}$.) Then μ is a measure. μ is semifinite if and only if $f(x) < \infty$ for all $x \in X$. μ is σ-finite if and only if μ is semifinite and $\{x : f(x) > 0\}$ is countable. See Exercise 13.3.1.

Example 13.3.6. In Example 13.3.5, if $f(x) = 1$ for all $x \in X$, μ is called *counting measure*. Then $\mu(E)$ is the number of elements of E if E is finite, and $\mu(E) = \infty$ if

E is infinite. If, for some $x_0 \in X$, $f(x_0) = 1$ and $f(x) = 0$ for all $x \neq x_0$, μ is called the *Dirac measure* at x_0. Then $\mu(E) = 1$ if $x_0 \in E$ and $\mu(E) = 0$ if $x_0 \notin E$.

Example 13.3.7. Flip a fair coin twice. The set of possible outcomes is

$$X = \{HH, HT, TH, TT\}.$$

Let $\mathcal{M} = \mathcal{P}(X)$ and let $P(E)$ be the number of elements of E divided by 4. This is Example 13.3.5 with $f(x) = \frac{1}{4}$ for every x. So $(X, \mathcal{M}, P)$ is a probability space. The event "at least one head" is the set $E = \{HH, HT, TH\}$, and $P(E) = \frac{3}{4}$.

Example 13.3.8. Let X be an uncountable set and let $\mathcal{M}$ be the σ-algebra of countable or co-countable sets. Then the function μ on $\mathcal{M}$ defined by $\mu(E) = 0$ if E is countable and $\mu(E) = 1$ if E is co-countable is a measure.

PROOF. Since $\emptyset$ is countable, $\mu(\emptyset) = 0$. No set is both countable and co-countable, since then $X = E \cup E^c$ would be countable. So μ is well defined.

Next, suppose $E, F \in \mathcal{M}$ where E and F are both co-countable. Then $(E \cap F)^c = E^c \cup F^c$ is countable, and so $(E \cap F)^c \neq X$. Thus $E \cap F \neq \emptyset$.

Now, let $\{E_n\}_{n\in\mathbb{N}}$ be a sequence of disjoint sets in $\mathcal{M}$. By the previous argument, at most one element of $\{E_n\}_{n\in\mathbb{N}}$ can be co-countable. If this is the case, call that element E_{n_0}. Then $\mu(E_{n_0}) = 1$ and $\mu(E_n) = 0$ for all $n \neq n_0$. By the proof of Example 13.1.6, $\bigcup_{n\in\mathbb{N}} E_n$ is co-countable and so $\mu\left(\bigcup_{n\in\mathbb{N}} E_n\right) = 1$. Thus $\mu\left(\bigcup_{n\in\mathbb{N}} E_n\right) = 1 = \sum_{n\in\mathbb{N}} \mu(E_n)$.

If E_n is countable for all $n \in \mathbb{N}$, then $\bigcup_{n\in\mathbb{N}} E_n$ is countable, and $\mu\left(\bigcup_{n\in\mathbb{N}} E_n\right) = 0 = \sum_{n\in\mathbb{N}} \mu(E_n)$. □

Theorem 13.3.9. *Let $(X, \mathcal{M}, \mu)$ be a measure space.*

(1) (Monotonicity) *If $E, F \in \mathcal{M}$ with $E \subset F$, then $\mu(E) \leq \mu(F)$.*
(2) (Subadditivity) *If $\{E_n\}_{n\in\mathbb{N}} \subset \mathcal{M}$, then $\mu\left(\bigcup_{n\in\mathbb{N}} E_n\right) \leq \sum_{n\in\mathbb{N}} \mu(E_n)$.*
(3) (Continuity from below) *If $\{E_n\}_{n\in\mathbb{N}} \subset \mathcal{M}$ and $E_1 \subset E_2 \subset \cdots$, then $\mu\left(\bigcup_{n\in\mathbb{N}} E_n\right) = \lim_{n\to\infty} \mu(E_n)$.*
(4) (Continuity from above) *If $\{E_n\}_{n\in\mathbb{N}} \subset \mathcal{M}$, $E_1 \supset E_2 \supset \cdots$, and $\mu(E_1) < \infty$, then $\mu\left(\bigcap_{n\in\mathbb{N}} E_n\right) = \lim_{n\to\infty} \mu(E_n)$.*

PROOF. (1) Write $F = E \cup (F \setminus E)$, a disjoint union of sets in $\mathcal{M}$. By finite additivity, $\mu(F) = \mu(E) + \mu(F \setminus E) \geq \mu(E)$.

(2) Make the sets disjoint. Let $F_1 = E_1$ and $F_k = E_k \setminus \bigcup_{i=1}^{k-1} E_i$ for $k \geq 2$. As in the proof of Theorem 13.1.4, the sets F_k are in $\mathcal{M}$, they are disjoint, $F_k \subset E_k$, and $\bigcup F_k = \bigcup E_k$. By countable additivity and (1),

$$\mu\left(\bigcup_{n\in\mathbb{N}} E_n\right) = \sum_{k\in\mathbb{N}} \mu(F_k) \leq \sum_{k\in\mathbb{N}} \mu(E_k).$$

(3) Let $E_0 = \emptyset$ and $F_k = E_k \setminus E_{k-1}$ for $k \in \mathbb{N}$. These sets are disjoint, $\bigcup_{k=1}^{n} F_k = E_n$ for each n, and $\bigcup_{k\in\mathbb{N}} F_k = \bigcup_{n\in\mathbb{N}} E_n$. So

$$\mu\left(\bigcup_{n\in\mathbb{N}} E_n\right) = \sum_{k=1}^{\infty} \mu(F_k) = \lim_{n\to\infty} \sum_{k=1}^{n} \mu(F_k) = \lim_{n\to\infty} \mu(E_n),$$

using finite additivity in the last step.

(4) Let $F_n = E_1 \setminus E_n$. Then $F_1 \subset F_2 \subset \cdots$ and $\bigcup_{n\in\mathbb{N}} F_n = E_1 \setminus \bigcap_{n\in\mathbb{N}} E_n$. By (3),

$$\mu\left(E_1 \setminus \bigcap_{n\in\mathbb{N}} E_n\right) = \lim_{n\to\infty} \mu(E_1 \setminus E_n).$$

If $A \subset E_1$ and $A \in \mathcal{M}$, then $\mu(E_1) = \mu(A) + \mu(E_1 \setminus A)$, and all three numbers are finite, since $\mu(E_1) < \infty$. So $\mu(E_1 \setminus A) = \mu(E_1) - \mu(A)$. Using this with $A = \bigcap E_n$ and with $A = E_n$,

$$\mu(E_1) - \mu\left(\bigcap_{n\in\mathbb{N}} E_n\right) = \mu(E_1) - \lim_{n\to\infty} \mu(E_n).$$

Subtracting $\mu(E_1)$ from both sides gives the result. □

In (3) and (4), the limits exist in $[0,\infty]$, since $\{\mu(E_n)\}$ is monotone by (1).

The hypothesis $\mu(E_1) < \infty$ in (4) cannot be dropped. Let μ be counting measure on $\mathbb{N}$, and let $E_n = \{n, n+1, n+2, \dots\}$. Then $\mu(E_n) = \infty$ for every n, but $\bigcap_{n\in\mathbb{N}} E_n = \emptyset$. It is enough that $\mu(E_n) < \infty$ for some n. Start the sequence there.

Chapter 14 asks whether there is a measure m on a σ-algebra containing every interval, with $m(I)$ equal to the length of I for every interval I. The answer is yes. But Section 14.4 shows that such a measure cannot be defined on every subset of $\mathbb{R}$, if it is also unchanged by translation. That is why measures are defined on σ-algebras and not on all of $\mathcal{P}(X)$.

Exercises

13.3.1. Let $X \neq \emptyset$, $\mathcal{M} = \mathcal{P}(X)$, and $f : X \to [0,\infty]$. For $E \in \mathcal{M}$, define $\mu(E) = \sum_{x\in E} f(x)$, as in Example 13.3.5.

(a) Prove that $(X, \mathcal{M}, \mu)$ is a measure space.

(b) Prove that μ is semifinite if and only if $f(x) < \infty$ for all $x \in X$.

(c) Prove that μ is σ-finite if and only if μ is semifinite and $\{x : f(x) > 0\}$ is countable.

[Solution, p. 365]

13.3.2. Let $\mathcal{M}$ be a σ-algebra on $\mathbb{R}$ and assume that the function $\mu : \mathcal{M} \to [0,\infty]$ has the following properties.

(i) If $E_1, E_2 \in \mathcal{M}$ with $E_1 \cap E_2 = \emptyset$, then $\mu(E_1 \cup E_2) = \mu(E_1) + \mu(E_2)$.

(ii) If $\{E_n\}_{n=1}^{\infty}$ is a sequence in $\mathcal{M}$ such that $E_{n+1} \subset E_n$ for all n and $\bigcap_{n=1}^{\infty} E_n = \emptyset$, then $\lim_{n\to\infty} \mu(E_n) = 0$.

Prove that $(\mathbb{R}, \mathcal{M}, \mu)$ is a measure space. [Solution, p. 365]

13.3.3. Let $(X, \mathcal{M}, \mu)$ be a measure space. Prove that if $E_1, E_2 \in \mathcal{M}$, then $\mu(E_1 \cup E_2) + \mu(E_1 \cap E_2) = \mu(E_1) + \mu(E_2)$. [Hint, p. 366]

13.3.4. Let $(X, \mathcal{M}, \mu)$ be a measure space and let $A \in \mathcal{M}$. Prove that $\mu_A(E) = \mu(E \cap A)$ defines a measure on $\mathcal{M}$. [Hint, p. 366]

13.3.5. Let $\mu_1, \mu_2, \dots$ be measures on $(X, \mathcal{M})$ and let $c_n \geq 0$. Prove that $\mu = \sum_{n=1}^{\infty} c_n \mu_n$ is a measure on $(X, \mathcal{M})$. [Hint, p. 366]

13.3.6. Let δ_k be the Dirac measure at k on $(\mathbb{R}, \mathcal{P}(\mathbb{R}))$. Find a formula for $\mu(E)$, where $\mu = \sum_{k=1}^{\infty} 2^{-k}\delta_k$, and show μ is a probability measure. [Hint, p. 366]

13.3.7. (Borel–Cantelli Lemma) Let $(X, \mathcal{M}, \mu)$ be a measure space and let $E_1, E_2, \ldots \in \mathcal{M}$ with $\sum_{k=1}^{\infty} \mu(E_k) < \infty$. Let E be the set of points that belong to infinitely many of the sets E_k. Prove that $E = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$, so $E \in \mathcal{M}$, and that $\mu(E) = 0$. [Solution, p. 366]

Notes

Émile Borel's 1898 book assigned lengths to the sets that can be built from intervals by countable unions and complements [7]. Henri Lebesgue's 1902 thesis turned this into a theory of measure and integration [42]. The notation G_δ and F_σ is due to Felix Hausdorff [37]. The G stands for *Gebiet*, a German word for an open set, and the F for *fermé*, French for closed. The δ and σ stand for the German words *Durchschnitt* (intersection) and *Summe* (union). Maurice Fréchet defined measures on abstract sets in 1915 [32]. Andrey Kolmogorov based probability theory on measure theory in 1933 [40]. The lemma in Exercise 13.3.7 is named for Émile Borel, who used it in probability in 1909, and Francesco Paolo Cantelli, who stated it in general in 1917.

CHAPTER 14

Lebesgue Measure

Chapter 13 described measures in general. This chapter builds the one that matters most for analysis. Lebesgue measure gives every interval its length, it does not change when a set is translated, and it is a measure on a σ-algebra that contains every Borel set.

The construction has two steps. First, Section 14.1 defines the outer measure $m^*(A)$ of every set $A \subset \mathbb{R}$, by covering A with countably many open intervals as efficiently as possible. This is the idea behind sets of measure zero in Definition 9.2.5. Outer measure is defined for every set, but it is not additive. Second, Section 14.2 uses a condition due to Carathéodory to pick out the *measurable* sets, the ones that split every other set additively. They form a σ-algebra, and on it m^* is a measure. Section 14.3 shows every measurable set is a Borel set up to a set of measure zero. Section 14.4, which is marked with a star, shows that some sets are not measurable.

This chapter uses a few results from earlier chapters. They are the Heine–Borel Theorem (Theorem 4.2.14), countable unions of countable sets (Theorem 2.2.6), sets of measure zero (Definition 9.2.5), and, from Chapter 13, Theorem 13.1.4, Theorem 13.2.5 and the Good Sets Principle.

14.1. Outer Measure

Remark 14.1.1. It seems like a properly defined measure m on $(\mathbb{R}, \mathcal{M})$ should have the following properties.

(1) $\mathcal{M} = \mathcal{P}(\mathbb{R})$.
(2) m is a measure.
(3) For an interval I, $m(I) = \ell(I)$, where $\ell(I)$ is the length of I.
(4) m is translation invariant; i.e. if $E + y = \{x + y : x \in E\}$, $m(E + y) = m(E)$.

However, in order for (2), (3), and (4) to be true, there will be certain sets $P \in \mathcal{P}(\mathbb{R})$ that will not be measurable. The set of measurable sets will end up containing $\mathcal{B}_\mathbb{R}$. (See Theorem 14.4.6. The measurable sets form a σ-algebra strictly larger than $\mathcal{B}_\mathbb{R}$, as Remark 14.3.2 explains.)

Definition 14.1.2. For an interval I with endpoints $a \leq b$, the *length* of I is $\ell(I) = b - a$. An unbounded interval has length ∞. Let $A \subset \mathbb{R}$ and consider all countable collections $\{I_n\}$ of open intervals with $A \subset \bigcup_n I_n$. Define the *outer measure* of A, $m^*(A)$, to be

$$m^*(A) = \inf_{A \subset \bigcup I_n} \sum \ell(I_n).$$

The infimum is over a nonempty set, since the intervals $(-n, n)$ cover every set. Its value may be ∞. A set E has measure zero in the sense of Definition 9.2.5 exactly

when $m^*(E) = 0$. Indeed, both say that for every $\varepsilon > 0$ there is a countable cover of E by open intervals with total length less than ε.

Lemma 14.1.3. *If $A \subset B$, $m^*(A) \le m^*(B)$.*

PROOF. See Exercise 14.1.1. □

Theorem 14.1.4. *The outer measure of an interval is its length.*

PROOF. The proof starts with closed bounded intervals, since every other interval can be squeezed between two of them. Let $a \le b$. For $\varepsilon > 0$, the single interval $(a-\varepsilon, b+\varepsilon)$ covers $[a,b]$. So $m^*([a,b]) \le b-a+2\varepsilon$ for every $\varepsilon > 0$, and $m^*([a,b]) \le b-a$.

For the reverse inequality, let $\{I_n\}$ be a countable collection of open intervals covering $[a,b]$. It is enough to show $\sum \ell(I_n) \ge b-a$. If some I_n is unbounded, the sum is ∞. So assume each I_n is bounded. By the Heine–Borel Theorem, finitely many of them cover $[a,b]$. Choose a chain from this finite list as follows. See Figure 14.1. Let (a_1,b_1) be one of them that contains a. If $b_1 > b$, stop. Otherwise $b_1 \in [a,b]$, and $b_1 \notin (a_1,b_1)$, so another interval (a_2,b_2) in the list contains b_1. Then $a_2 < b_1 < b_2$. If $b_2 > b$, stop. Otherwise continue in the same way. The right endpoints $b_1 < b_2 < \cdots$ increase, so no interval is chosen twice. Since the list is finite, the process stops, with an interval (a_k,b_k) such that $b_k > b$. Since $a_{i+1} < b_i$ for each i,

$$\sum_{i=1}^{k}(b_i - a_i) = (b_k - a_1) + \sum_{i=1}^{k-1}(b_i - a_{i+1}) > b_k - a_1 > b - a.$$

The chosen intervals are distinct members of $\{I_n\}$, so $\sum \ell(I_n) \ge \sum_{i=1}^{k}(b_i - a_i) > b-a$. Taking the infimum over all covers, $m^*([a,b]) \ge b-a$.

Next come the other bounded intervals. Let I be a bounded interval with endpoints $a < b$. Then $I \subset [a,b]$, so $m^*(I) \le b-a$ by Lemma 14.1.3. For $0 < \varepsilon < \frac{b-a}{2}$, the closed interval $[a+\varepsilon, b-\varepsilon]$ is contained in I, so $m^*(I) \ge b-a-2\varepsilon$. So $m^*(I) = b-a = \ell(I)$.

The last case is an unbounded interval. If I is unbounded, then for every $M > 0$ it contains a closed interval J with $\ell(J) = M$. So $m^*(I) \ge m^*(J) = M$ for every $M > 0$, and $m^*(I) = \infty = \ell(I)$. □

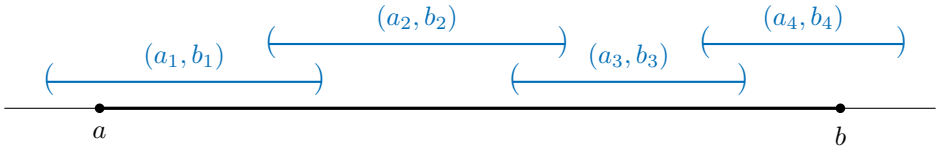


FIGURE 14.1. The chain of intervals in the proof of Theorem 14.1.4. Each interval contains the right endpoint of the one before. The last one reaches past b, so the lengths add up to more than $b-a$.

Theorem 14.1.5. *m^* is translation invariant. That is, $m^*(A+y) = m^*(A)$ for all $A \subset \mathbb{R}$ and $y \in \mathbb{R}$.*

PROOF. See Exercise 14.1.2. □

Theorem 14.1.6. *Let $\{A_n\}$ be a countable collection of sets of real numbers. Then*

$$m^*\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leq \sum_{n\in\mathbb{N}} m^*(A_n).$$

PROOF. If $\sum m^*(A_n) = \infty$, there is nothing to prove. So assume each $m^*(A_n)$ is finite. Let $\varepsilon > 0$. For each n, the definition of $m^*(A_n)$ as an infimum gives a countable collection $\{I_{n,i}\}_{i\in\mathbb{N}}$ of open intervals that covers A_n with

$$\sum_{i\in\mathbb{N}} \ell(I_{n,i}) < m^*(A_n) + \frac{\varepsilon}{2^n}.$$

The collection of all the intervals $I_{n,i}$ is countable, as a countable union of countable collections (Theorem 2.2.6). It covers $\bigcup_n A_n$. List it as one sequence $J_1, J_2, \ldots$. Any partial sum $\sum_{k=1}^{K} \ell(J_k)$ uses finitely many intervals $I_{n,i}$, all with $n \leq N$ for some N. Since the lengths are nonnegative,

$$\sum_{k=1}^{K} \ell(J_k) \leq \sum_{n=1}^{N}\sum_{i\in\mathbb{N}} \ell(I_{n,i}) < \sum_{n=1}^{N}\left(m^*(A_n) + \frac{\varepsilon}{2^n}\right) < \sum_{n\in\mathbb{N}} m^*(A_n) + \varepsilon.$$

So $\sum_k \ell(J_k) \leq \sum_n m^*(A_n) + \varepsilon$, and therefore $m^*\left(\bigcup_n A_n\right) \leq \sum_n m^*(A_n) + \varepsilon$. Since $\varepsilon > 0$ was arbitrary, the result follows. □

Proof Technique (Splitting ε Over Countably Many Steps). The proof of Theorem 14.1.6 makes countably many choices, one for each A_n, and each choice costs a small error. If every error were ε, the total error would be infinite. So the nth choice is allowed an error of only $\frac{\varepsilon}{2^n}$. Since $\sum_{n=1}^{\infty} \frac{1}{2^n} = 1$, the total error is at most ε. This is the countable version of Splitting ε from Chapter 3. Example 9.2.7 used the same idea. Any convergent series of positive terms with sum 1 would do, but $\frac{1}{2^n}$ is the usual choice.

Corollary 14.1.7. *If A is countable, $m^*(A) = 0$.*

PROOF. Enumerate the elements of A, say $A = \{x_1, x_2, \ldots\}$. Let $A_n = \{x_n\}$ for $n \in \mathbb{N}$. So

$$m^*(A) \leq m^*\left(\bigcup_{n\in\mathbb{N}} A_n\right) \leq \sum_{n\in\mathbb{N}} m^*(A_n) = 0.$$

Here $m^*(\{x_n\}) = m^*([x_n, x_n]) = 0$ by Theorem 14.1.4. A finite set is handled the same way. □

Example 14.1.8. The set $[0, 1]$ is not countable. This gives a second proof of Theorem 2.3.2.

PROOF. $m^*([0, 1]) = 1 \neq 0$. Thus $[0, 1]$ is not countable. □

Example 14.1.9. Let $A = \mathbb{Q} \cap [0, 1]$. Let $\{I_n\}$ be a finite collection of open intervals covering A. Then $\sum \ell(I_n) \geq 1$. However, $m^*(A) = 0$. So countable covers, not just finite ones, are essential in Definition 14.1.2.

PROOF. See Exercise 14.1.3. □

Theorem 14.1.10. *Given any set A, for all $\varepsilon > 0$, there exists an open set U such that $A \subset U$ and $m^*(U) \leq m^*(A) + \varepsilon$.*

PROOF. See Exercise 14.1.4. □

Exercises

14.1.1. Prove Lemma 14.1.3. That is, prove that if $A \subset B$, then $m^*(A) \leq m^*(B)$. [Solution, p. 366]

14.1.2. Prove that m^* is translation invariant. [Solution, p. 366]

14.1.3. Let $A = \mathbb{Q} \cap [0, 1]$. Let $\{I_n\}$ be a finite collection of open intervals covering A. Prove $\sum \ell(I_n) \geq 1$.

14.1.4. If $A \subset \mathbb{R}$, prove that for all $\varepsilon > 0$, there exists an open set U such that $A \subset U$ and $m^*(U) \leq m^*(A) + \varepsilon$. [Solution, p. 367]

14.1.5. Prove that if $m^*(A) = 0$, then $m^*(A \cup B) = m^*(B)$.

14.1.6. Prove that $m^*(A)$ does not change if, in Definition 14.1.2, open intervals are replaced by closed intervals. [Hint, p. 367]

14.1.7. For $c > 0$ and $A \subset \mathbb{R}$, let $cA = \{cx : x \in A\}$. Prove that $m^*(cA) = c\, m^*(A)$. [Hint, p. 367]

14.2. Lebesgue Measurable Sets

Outer measure is defined for every set, but it is not a measure on $\mathcal{P}(\mathbb{R})$. There are disjoint sets A and B with $m^*(A \cup B) < m^*(A) + m^*(B)$, although they cannot be described explicitly (see Section 14.4). The fix is to keep only the sets E that split every set A additively. These are the measurable sets.

Definition 14.2.1. A set $E \subset \mathbb{R}$ is said to be *measurable* if for each set $A \subset \mathbb{R}$,

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c).$$

If E is measurable, define the *Lebesgue measure* $m(E)$ to be the outer measure of E, $m(E) = m^*(E)$. The collection of measurable sets is denoted $\mathcal{M}$.

Remark 14.2.2. Since $A = (A \cap E) \cup (A \cap E^c)$, $m^*(A) \leq m^*(A \cap E) + m^*(A \cap E^c)$ by Theorem 14.1.6. So E is measurable if for each set $A \subset \mathbb{R}$,

$$m^*(A) \geq m^*(A \cap E) + m^*(A \cap E^c).$$

This is automatic when $m^*(A) = \infty$. So it only needs to be checked for sets A with $m^*(A) < \infty$.

Proof Technique (Proving a Set Is Measurable). To prove E is measurable, fix an arbitrary "test set" $A \subset \mathbb{R}$. By Remark 14.2.2, only one inequality is needed, $m^*(A) \geq m^*(A \cap E) + m^*(A \cap E^c)$, and only for $m^*(A) < \infty$. Then either use what is known about E to split A, as in the proofs of Lemma 14.2.5 and Theorem 14.2.8, or go back to a cover of A by intervals with total length

close to $m^*(A)$ and split the cover, as in the proof of Lemma 14.2.9. The test set A is arbitrary. It does not have to be measurable, and usually nothing is known about it.

Lemma 14.2.3. *If $m^*(E) = 0$, then E is measurable.*

PROOF. Let A be any set. Then $A \cap E \subset E$ and so $m^*(A \cap E) \leq m^*(E) = 0$. Also, $A \supset A \cap E^c$, and so

$$m^*(A) \geq m^*(A \cap E^c) = m^*(A \cap E^c) + m^*(A \cap E).$$

Thus E is measurable. □

Lemma 14.2.4. *If $F \subset E$ and E is measurable with $m(E) = 0$, F is measurable and $m(F) = 0$.*

PROOF. Let A be a set. Since E is measurable,

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c).$$

Since $A \cap E \subset E$, $m^*(A \cap E) = 0$. So $m^*(A \cap E^c) = m^*(A)$. Thus, since $A \cap F \subset A \cap E$ and $A \cap F^c \subset A$,

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c) \geq m^*(A \cap F) + m^*(A) \geq m^*(A \cap F) + m^*(A \cap F^c).$$

So F is measurable. Since $F \subset E$, $m(F) \leq m(E)$ and so $m(F) = 0$. (There is also a shorter proof. By Lemma 14.1.3, $m^*(F) \leq m^*(E) = 0$, so F is measurable by Lemma 14.2.3.) □

Lemma 14.2.5. *If E_1 and E_2 are measurable, $E_1 \cup E_2$ is measurable.*

PROOF. Let $A \subset \mathbb{R}$. Split A with E_1, and then split the part of A outside E_1 with E_2. Since E_1 is measurable,

$$m^*(A) = m^*(A \cap E_1) + m^*(A \cap E_1^c).$$

Since E_2 is measurable, using the test set $A \cap E_1^c$,

$$m^*(A \cap E_1^c) = m^*(A \cap E_1^c \cap E_2) + m^*(A \cap E_1^c \cap E_2^c).$$

By De Morgan's Laws, $E_1^c \cap E_2^c = (E_1 \cup E_2)^c$. Also $A \cap (E_1 \cup E_2) = (A \cap E_1) \cup (A \cap E_1^c \cap E_2)$, so by subadditivity,

$$m^*(A \cap (E_1 \cup E_2)) \leq m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2).$$

Putting these together,

$$\begin{aligned} m^*(A) &= m^*(A \cap E_1) + m^*(A \cap E_1^c \cap E_2) + m^*(A \cap (E_1 \cup E_2)^c) \\ &\geq m^*(A \cap (E_1 \cup E_2)) + m^*(A \cap (E_1 \cup E_2)^c). \end{aligned}$$

So $E_1 \cup E_2$ is measurable. □

Corollary 14.2.6. *The family $\mathcal{M}$ of measurable sets of real numbers forms an algebra.*

PROOF. Using Lemma 14.2.5 and an induction argument shows that if $E_1, \dots, E_n$ are measurable, then $\bigcup_{i=1}^n E_i$ is measurable. Also, if E is measurable, then for any set A,

$$m^*(A) = m^*(A \cap E) + m^*(A \cap E^c) = m^*(A \cap E^c) + m^*(A \cap (E^c)^c).$$

So E^c is measurable and $\mathcal{M}$ is an algebra. It is nonempty, since $\emptyset$ is measurable by Lemma 14.2.3. □

Lemma 14.2.7. *Let A be any set and let $E_1, \dots, E_n$ be a finite sequence of disjoint measurable sets. Then*

$$m^*\left(A \cap \left[\bigcup_{i=1}^n E_i\right]\right) = \sum_{i=1}^n m^*(A \cap E_i).$$

PROOF. Induct on n. For $n = 1$ there is nothing to prove. Suppose the statement holds for $n = k$, and let $B = A \cap \bigcup_{i=1}^{k+1} E_i$. Test B with the measurable set E_{k+1}. Since the sets E_i are disjoint, $B \cap E_{k+1} = A \cap E_{k+1}$ and $B \cap E_{k+1}^c = A \cap \bigcup_{i=1}^k E_i$. So

$$m^*(B) = m^*(A \cap E_{k+1}) + m^*\left(A \cap \bigcup_{i=1}^k E_i\right) = m^*(A \cap E_{k+1}) + \sum_{i=1}^k m^*(A \cap E_i),$$

using the case $n = k$. This is the case $n = k + 1$. □

Theorem 14.2.8. *The collection $\mathcal{M}$ of measurable subsets of $\mathbb{R}$ is a σ-algebra.*

PROOF. By Corollary 14.2.6, $\mathcal{M}$ is an algebra. By Theorem 13.1.4, it is enough to show that the union $E = \bigcup_{i=1}^\infty E_i$ of a sequence of pairwise disjoint measurable sets is measurable. Let $A \subset \mathbb{R}$, and let $F_n = \bigcup_{i=1}^n E_i$. Then $F_n \in \mathcal{M}$ and $E^c \subset F_n^c$. Testing A with F_n and using Lemma 14.2.7,

$$m^*(A) = m^*(A \cap F_n) + m^*(A \cap F_n^c) \geq \sum_{i=1}^n m^*(A \cap E_i) + m^*(A \cap E^c).$$

This holds for every $n \in \mathbb{N}$. Letting $n \to \infty$ and then using subadditivity on $A \cap E = \bigcup_i (A \cap E_i)$,

$$m^*(A) \geq \sum_{i=1}^\infty m^*(A \cap E_i) + m^*(A \cap E^c) \geq m^*(A \cap E) + m^*(A \cap E^c). \tag{14.2.1}$$

So $E \in \mathcal{M}$. □

Lemma 14.2.9. *The interval (a, ∞) is measurable.*

PROOF. Let $A \subset \mathbb{R}$ with $m^*(A) < \infty$. Let $A_1 = A \cap (a, \infty)$ and $A_2 = A \cap (-\infty, a]$. Let $\varepsilon > 0$, and choose a countable collection $\{I_n\}$ of open intervals covering A with $\sum \ell(I_n) < m^*(A) + \varepsilon$. Cut each interval at a. Let $I_n' = I_n \cap (a, \infty)$ and $I_n'' = I_n \cap (-\infty, a]$. Each is an interval or empty, and $\ell(I_n) = \ell(I_n') + \ell(I_n'')$. The sets I_n' cover A_1 and the sets I_n'' cover A_2. They need not be open, but by Lemma 14.1.3 and Theorems 14.1.4 and 14.1.6,

$$m^*(A_1) \leq \sum m^*(I_n') = \sum \ell(I_n') \quad \text{and} \quad m^*(A_2) \leq \sum \ell(I_n'').$$

Adding,

$$m^*(A \cap (a,\infty)) + m^*(A \cap (a,\infty)^c) \leq \sum \left(\ell(I'_n) + \ell(I''_n)\right) = \sum \ell(I_n) < m^*(A) + \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, (a,∞) is measurable, by Remark 14.2.2. □

Theorem 14.2.10. *Every Borel set is measurable.*

PROOF. By Theorem 13.2.5 (4), $\mathcal{B}_{\mathbb{R}}$ is generated by $\{(a,\infty) : a \in \mathbb{R}\}$. Since, by Lemma 14.2.9, $(a,\infty) \in \mathcal{M}$ and $\mathcal{M}$ is a σ-algebra, $\mathcal{B}_{\mathbb{R}} \subset \mathcal{M}$ by Lemma 13.1.11. □

In particular, open sets, closed sets, G_δ sets and F_σ sets are measurable, and so is every interval. By Theorem 14.1.4, $m(I) = \ell(I)$ for every interval I.

Theorem 14.2.11. *If E is a measurable set, then $E+y$ is measurable and $m(E+y) = m(E)$.*

PROOF. This is a direct result of Theorem 14.1.5. Let $A \subset \mathbb{R}$. Then $(A-y)\cap E = (A \cap (E+y)) - y$, and similarly for complements. Since E is measurable and m^* is translation invariant,

$$\begin{aligned} m^*(A) = m^*(A-y) &= m^*((A-y)\cap E) + m^*((A-y)\cap E^c) \\ &= m^*(A \cap (E+y)) + m^*(A \cap (E+y)^c). \end{aligned}$$

So $E+y$ is measurable, and $m(E+y) = m^*(E+y) = m^*(E) = m(E)$. □

Theorem 14.2.12. *Let $\mathcal{M}$ be the collection of measurable subsets of $\mathbb{R}$. $(\mathbb{R}, \mathcal{M}, m)$ is a measure space.*

PROOF. First, note that since $\emptyset$ is countable, by Corollary 14.1.7, $m(\emptyset) = 0$. Now, let $\{E_i\}_{i=1}^{\infty}$ be a disjoint sequence of measurable sets, and let $E = \bigcup_i E_i$. Taking $A = E$ in (14.2.1) gives $m(E) \geq \sum_{i=1}^{\infty} m(E_i)$. The reverse inequality is subadditivity. So

$$m\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} m(E_i).$$

So m is a measure and $(\mathbb{R}, \mathcal{M}, m)$ is a measure space. □

So Lebesgue measure has properties (2), (3) and (4) of Remark 14.1.1, on the σ-algebra $\mathcal{M}$. Every result of Section 13.3 applies to it. For example, m is continuous from below, and from above on sets of finite measure.

Exercises

14.2.1. Prove that every countable set is measurable with measure 0. Prove that $\mathbb{R} \setminus \mathbb{Q}$ is measurable, and find $m\big([0,1] \setminus \mathbb{Q}\big)$. [Hint, p. 367]

14.2.2. Let E be measurable and let $F \subset \mathbb{R}$ with $m^*(E \setminus F) = m^*(F \setminus E) = 0$. Prove that F is measurable and $m(F) = m(E)$. [Hint, p. 367]

14.2.3. Prove that if E_1 and E_2 are measurable, then $m(E_1 \cup E_2) + m(E_1 \cap E_2) = m(E_1) + m(E_2)$. (Compare Exercise 13.3.3.)

14.2.4. Let E be measurable and let $A \subset \mathbb{R}$. Prove that $m^*(A \cup E) + m^*(A \cap E) = m^*(A) + m(E)$. [Hint, p. 367]

14.2.5. Let E be measurable with $m(E) > 0$. Prove that E is uncountable. [Hint, p. 367]

14.3. Approximation by Open and Closed Sets

Measurable sets can be very complicated. The next theorem says they are never far from simple sets. Every measurable set lies between an F_σ set and a G_δ set that differ by a set of measure zero. When $m(E) < \infty$, E is even close to a finite union of intervals.

Theorem 14.3.1. *Let $E \subset \mathbb{R}$. The following are equivalent.*

(1) E is measurable.
(2) For all $\varepsilon > 0$, there exists an open set $U \supset E$ such that $m^*(U \setminus E) < \varepsilon$.
(3) For all $\varepsilon > 0$, there exists a closed set $F \subset E$ such that $m^*(E \setminus F) < \varepsilon$.
(4) There exists a G_δ set $G \supset E$ such that $m^*(G \setminus E) = 0$.
(5) There exists an F_σ set $F \subset E$ such that $m^*(E \setminus F) = 0$.

If $m^*(E) < \infty$, the above are equivalent to:

(6) for all $\varepsilon > 0$, there exists a finite union O of open intervals such that $m^*(O \triangle E) < \varepsilon$, where $O \triangle E = (O \setminus E) \cup (E \setminus O)$.

Proof. See Exercises 14.3.1 and 14.3.2. □

Remark 14.3.2. By (4), every measurable set is a Borel set with a set of measure zero removed. So $\mathcal{M}$ is the σ-algebra generated by $\mathcal{B}_\mathbb{R}$ together with the sets of measure zero. There are measurable sets that are not Borel sets. Every subset of the Cantor set is measurable with measure 0, by Example 9.2.8 and Lemma 14.2.3. A counting argument from set theory shows there are more subsets of the Cantor set than there are Borel sets, so some of them are not Borel sets. That argument is beyond this book. Exercise 17.6.3 gives a different argument, using the Cantor function.

Exercises

14.3.1. Prove that (1)–(5) in Theorem 14.3.1 are equivalent. [Hint, p. 367]

14.3.2. Prove that (6) is equivalent to (1)–(5) when $m^*(E) < \infty$. [Hint, p. 368]

14.3.3. Let E be measurable with $m(E) < \infty$. Prove that for every $\varepsilon > 0$ there is a compact set $K \subset E$ with $m(E \setminus K) < \varepsilon$. [Hint, p. 368]

14.3.4. Let E be measurable with $m(E) > 0$. Prove that for every $\alpha < 1$ there is an open interval I with $m(E \cap I) > \alpha\, \ell(I)$. [Hint, p. 368]

14.4. A Nonmeasurable Set*

This section can be skipped. Nothing later depends on it.

Is every subset of $\mathbb{R}$ measurable? This section shows the answer is no. The proof uses the Axiom of Choice. It works inside $[0, 1)$, where it is convenient to add numbers "around a circle," wrapping past 1 back to 0.

Definition 14.4.1. If x and y are real numbers in $[0,1)$, define the *sum modulo* 1 of x and y to be $x+y$ if $x+y<1$ and to be $x+y-1$ if $x+y\geq 1$. Denote the sum modulo 1 of x and y by $x \overset{\circ}{+} y$.

Remark 14.4.2. $\overset{\circ}{+}$ is commutative and associative. If for each $x\in[0,1)$, the angle $2\pi x$ is assigned to x, $\overset{\circ}{+}$ corresponds to the addition of angles.

Definition 14.4.3. If $E\subset[0,1)$ and $y\in[0,1)$, define the *translation modulo* 1 of E to be the set

$$E \overset{\circ}{+} y = \{z : z = x \overset{\circ}{+} y \text{ for some } x\in E\}.$$

Lemma 14.4.4. *Let $E\subset[0,1)$ be a measurable set. Then for each $y\in[0,1)$, the set $E \overset{\circ}{+} y$ is measurable and $m(E \overset{\circ}{+} y) = m(E)$.*

Proof. Split E at $1-y$. Let $E_1 = E\cap[0,1-y)$ and $E_2 = E\cap[1-y,1)$. Both are measurable, they are disjoint, and $m(E) = m(E_1)+m(E_2)$. On E_1, adding y modulo 1 is ordinary addition, so $E_1 \overset{\circ}{+} y = E_1 + y$. On E_2, it wraps around, so $E_2 \overset{\circ}{+} y = E_2 + (y-1)$. By Theorem 14.2.11, both sets are measurable, with $m(E_1 \overset{\circ}{+} y) = m(E_1)$ and $m(E_2 \overset{\circ}{+} y) = m(E_2)$. They are disjoint, since $E_1 \overset{\circ}{+} y \subset [y,1)$ and $E_2 \overset{\circ}{+} y \subset [0,y)$. Their union is $E \overset{\circ}{+} y$. So $E \overset{\circ}{+} y$ is measurable and

$$m(E \overset{\circ}{+} y) = m(E_1 \overset{\circ}{+} y) + m(E_2 \overset{\circ}{+} y) = m(E_1) + m(E_2) = m(E). \qquad \square$$

Axiom 14.4.5 (Axiom of Choice)**.** *Let $\{E_\alpha\}_{\alpha\in\Lambda}$ be a family of nonempty sets. There exists a set $\{x_\alpha\}_{\alpha\in\Lambda}$ such that $x_\alpha\in E_\alpha$ for all $\alpha\in\Lambda$.*

The proof below also uses equivalence classes. If $\sim$ is an equivalence relation on a set S (Definition 1.1.17), the *equivalence class* of $x\in S$ is $[x] = \{y\in S : y\sim x\}$. Two equivalence classes are either equal or disjoint, and every element of S is in its own class. So the classes split S into disjoint pieces.

Theorem 14.4.6. *There exists a set $P\subset\mathbb{R}$ such that P is nonmeasurable in the Lebesgue sense.*

Proof. For $x,y\in[0,1)$, write $x\sim y$ if $x-y\in\mathbb{Q}$. This is an equivalence relation on $[0,1)$. Its equivalence classes are disjoint, and their union is $[0,1)$. By the Axiom of Choice, there is a set $P\subset[0,1)$ that contains exactly one element of each equivalence class. So two different elements of P never differ by a rational number.

List the rational numbers in $[0,1)$ as $r_0, r_1, r_2, \ldots$, with $r_0 = 0$, and let $P_i = P \overset{\circ}{+} r_i$. So $P_0 = P$.

First, the sets P_i are disjoint. To see this, suppose $x\in P_i\cap P_j$. Then $x = p \overset{\circ}{+} r_i = q \overset{\circ}{+} r_j$ for some $p,q\in P$. So $p-q$ is $r_j - r_i$, or $r_j - r_i \pm 1$. In every case $p-q\in\mathbb{Q}$, so $p\sim q$, and therefore $p=q$. Then $p \overset{\circ}{+} r_i = p \overset{\circ}{+} r_j$, which forces $r_i = r_j$, so $i=j$.

Next, the sets P_i cover $[0,1)$. To see this, let $x\in[0,1)$. Then $x\sim p$ for some $p\in P$. Let $r = x-p$ if $x\geq p$, and $r = x-p+1$ if $x<p$. Then $r\in\mathbb{Q}\cap[0,1)$, so $r = r_i$ for some i, and $x = p \overset{\circ}{+} r_i \in P_i$.

Now suppose P were measurable. By Lemma 14.4.4, each P_i would be measurable with $m(P_i) = m(P)$. By countable additivity,

$$1 = m([0,1)) = \sum_{i=0}^{\infty} m(P_i) = \sum_{i=0}^{\infty} m(P).$$

The right side is 0 if $m(P) = 0$ and ∞ if $m(P) > 0$. Neither is 1. So P is not measurable. □

Corollary 14.4.7. *Define P as before. Let μ be a measure on a σ-algebra $\mathcal{N}$ of subsets of $\mathbb{R}$. Suppose $\mathcal{N}$ contains P and every interval, and that $E + y \in \mathcal{N}$ and $\mu(E + y) = \mu(E)$ for all $E \in \mathcal{N}$ and $y \in \mathbb{R}$. Then $\mu([0,1)) = 0$ or $\mu([0,1)) = \infty$.*

Proof. The proof of Lemma 14.4.4 uses only these properties of m. So it applies to μ, and $\mu(P_i) = \mu(P)$ for every i. As in the proof of Theorem 14.4.6, $\mu([0,1)) = \sum_{i=0}^{\infty} \mu(P)$, which is 0 or ∞. □

In particular, no measure on all of $\mathcal{P}(\mathbb{R})$ is translation invariant and gives $[0,1)$ measure 1. So properties (1)–(4) in Remark 14.1.1 cannot all hold. Lebesgue measure gives up (1).

Exercises

14.4.1. Prove that the set P in the proof of Theorem 14.4.6 has $m^*(P) > 0$.
[Hint, p. 368]

14.4.2. Use the sets P_i to show that m^* is not countably additive on $\mathcal{P}(\mathbb{R})$. That is, find disjoint sets $A_1, A_2, \ldots$ with $m^*\left(\bigcup A_i\right) < \sum m^*(A_i)$.
[Hint, p. 368]

14.4.3. Let $E \subset [0,1)$ be measurable with $m(E) > 0$. Prove that E contains a nonmeasurable set.
[Hint, p. 369]

Notes

Émile Borel showed in 1898 how to assign lengths to the sets now called Borel sets [**7**]. Henri Lebesgue's 1902 thesis defined outer measure by countable covers, as in Definition 14.1.2, and built the measure that bears his name [**42**]. Lebesgue called a set measurable when its outer and inner measures agree. The condition in Definition 14.2.1 is due to Constantin Carathéodory in 1914 [**14**]. It works for any outer measure, which is why it is now the standard definition. Giuseppe Vitali constructed the first nonmeasurable set in 1905 [**74**]. Ernst Zermelo had stated the Axiom of Choice the year before [**79**]. Robert Solovay showed in 1970 that the Axiom of Choice cannot be avoided completely here. Without it, it is consistent that every subset of $\mathbb{R}$ is Lebesgue measurable [**68**].

CHAPTER 15

Measurable Functions

The Lebesgue integral of f will be built by measuring the sets where f takes values in a given range. So f must be a function for which those sets are measurable. Such functions are called measurable. This chapter shows that the class of measurable functions is very large. It contains the continuous functions, it is closed under the usual arithmetic, and, unlike the Riemann integrable functions, it is closed under pointwise limits.

Section 15.1 defines measurable functions and the phrase "almost everywhere." Section 15.2 introduces simple functions and step functions, and shows that every measurable function on $[a, b]$ is close to a step function and to a continuous function. Section 15.3 shows that pointwise convergence is close to uniform convergence. Section 15.4, which is marked with a star, proves Lusin's Theorem, the sharpest form of "measurable functions are nearly continuous."

This chapter uses a few results from earlier chapters. They are Lebesgue measure and its properties (Chapter 14), the continuity of measure (Theorem 13.3.9), limits superior and inferior (Section 3.6), uniform convergence (Section 11.2), and Theorem 6.3.16.

15.1. Measurable Functions and "Almost Everywhere"

Measurable functions may take the values ∞ and $-\infty$. The *extended real numbers* are $[-\infty, \infty] = \mathbb{R} \cup \{-\infty, \infty\}$, with $-\infty < x < \infty$ for every $x \in \mathbb{R}$. An *extended real-valued function* is a function $f : D \to [-\infty, \infty]$. The set $\{x \in D : f(x) > \alpha\}$ is written $\{x : f(x) > \alpha\}$, or just $\{f > \alpha\}$, when the domain D is clear.

Theorem 15.1.1. *Let $D \subset \mathbb{R}$ be measurable and let $f : D \to [-\infty, \infty]$. The following are equivalent.*

(1) For each $\alpha \in \mathbb{R}$, the set $\{x : f(x) > \alpha\}$ is measurable.
(2) For each $\alpha \in \mathbb{R}$, the set $\{x : f(x) \geq \alpha\}$ is measurable.
(3) For each $\alpha \in \mathbb{R}$, the set $\{x : f(x) < \alpha\}$ is measurable.
(4) For each $\alpha \in \mathbb{R}$, the set $\{x : f(x) \leq \alpha\}$ is measurable.

These statements imply:

(5) for each $\alpha \in [-\infty, \infty]$, the set $\{x : f(x) = \alpha\}$ is measurable.

Proof. ((1)⇒(4)) Now $\{x : f(x) \leq \alpha\} = D \setminus \{x : f(x) > \alpha\}$, and so $\{x : f(x) \leq \alpha\}$ is measurable.

((4)⇒(1)) Now $\{x : f(x) > \alpha\} = D \setminus \{x : f(x) \leq \alpha\}$, and so $\{x : f(x) > \alpha\}$ is measurable.

((2)⇒(3)) Now $\{x : f(x) < \alpha\} = D \setminus \{x : f(x) \geq \alpha\}$, and so $\{x : f(x) < \alpha\}$ is measurable.

((3)⇒(2)) Now $\{x : f(x) \geq \alpha\} = D \setminus \{x : f(x) < \alpha\}$, and so $\{x : f(x) \geq \alpha\}$ is measurable.

((1)⇒(2)) Since $\{x : f(x) \geq \alpha\} = \bigcap_{n=1}^{\infty} \left\{x : f(x) > \alpha - \frac{1}{n}\right\}$, $\{x : f(x) \geq \alpha\}$ is measurable.

((2)⇒(1)) Since $\{x : f(x) > \alpha\} = \bigcup_{n=1}^{\infty} \left\{x : f(x) \geq \alpha + \frac{1}{n}\right\}$, $\{x : f(x) > \alpha\}$ is measurable. So (1)–(4) are equivalent.

((2),(4)⇒(5) if $\alpha \in \mathbb{R}$) Let $\alpha \in \mathbb{R}$. Then $\{x : f(x) = \alpha\} = \{x : f(x) \geq \alpha\} \cap \{x : f(x) \leq \alpha\}$. So $\{x : f(x) = \alpha\}$ is measurable.

((2)⇒(5) if $\alpha = \infty$) Let $\alpha = \infty$. Since $\{x : f(x) = \infty\} = \bigcap_{n=1}^{\infty} \{x : f(x) \geq n\}$, $\{x : f(x) = \infty\}$ is measurable.

((4)⇒(5) if $\alpha = -\infty$) Let $\alpha = -\infty$. Since $\{x : f(x) = -\infty\} = \bigcap_{n=1}^{\infty} \{x : f(x) \leq -n\}$, $\{x : f(x) = -\infty\}$ is measurable. □

Definition 15.1.2. An extended real-valued function is *measurable* if its domain is measurable and if it satisfies one of (1)–(4) in Theorem 15.1.1.

Example 15.1.3. Let $f : \mathbb{R} \to \mathbb{R}$ be continuous. For each α, the set $\{f > \alpha\} = f^{-1}[(\alpha, \infty)]$ is open, by Theorem 6.3.16. Open sets are measurable, so f is measurable. If $f : [a, b] \to \mathbb{R}$ is increasing, then each set $\{f > \alpha\}$ is an interval, so f is measurable. The same holds for decreasing functions.

Measurability is a weak version of continuity. A continuous function pulls open sets back to open sets. A measurable function pulls rays, and so all Borel sets, back to measurable sets (Exercise 15.1.3).

Theorem 15.1.4. *Let $c \in \mathbb{R}$ and let D be measurable and $f, g : D \to \mathbb{R}$ be measurable. Then the functions $f + c$, cf, $f + g$, $f - g$, $g - f$, f^2, and fg are also measurable.*

Proof. Now $\{x : f(x) + c < \alpha\} = \{x : f(x) < \alpha - c\}$. So $f + c$ is measurable since f is measurable.

If $c = 0$, $0f(x) = 0$ for all x, and so $\{x : cf(x) < \alpha\} = \emptyset$ or $\{x : cf(x) < \alpha\} = D$. Either way, $\{x : cf(x) < \alpha\}$ is measurable. If $c > 0$, $\{x : cf(x) < \alpha\} = \left\{x : f(x) < \frac{\alpha}{c}\right\}$ is measurable. If $c < 0$, $\{x : cf(x) < \alpha\} = \left\{x : f(x) > \frac{\alpha}{c}\right\}$ is measurable. So for all $c \in \mathbb{R}$, $\{x : cf(x) < \alpha\}$ is measurable. Thus cf is measurable.

If $f(x) + g(x) < \alpha$, then $f(x) < \alpha - g(x)$. So there exists an $r \in \mathbb{Q}$ such that $f(x) < r < \alpha - g(x)$ (Theorem 1.8.11). Hence

$$\{x : f(x) + g(x) < \alpha\} = \bigcup_{r \in \mathbb{Q}} \big(\{x : f(x) < r\} \cap \{x : g(x) < \alpha - r\}\big).$$

Since $\mathbb{Q}$ is countable, this set is measurable. So $f + g$ is measurable.

Since $-g = (-1)g$ is measurable, $f + (-g) = f - g$ is measurable. This implies $(-1)(f - g) = g - f$ is also measurable.

Lastly, $\{x : f^2(x) > \alpha\} = \{x : f(x) > \sqrt{\alpha}\} \cup \{x : f(x) < -\sqrt{\alpha}\}$ for $\alpha \geq 0$ and $\{x : f^2(x) > \alpha\} = D$ if $\alpha < 0$, and so f^2 is measurable. Therefore

$$fg = \frac{1}{2}\left[(f + g)^2 - f^2 - g^2\right]$$

is measurable. □

Theorem 15.1.5. *Let $\{f_n\}$ be a sequence of measurable functions with the same domain D. Then the functions* $\sup\{f_1, \ldots, f_n\}$, $\inf\{f_1, \ldots, f_n\}$, $\sup_{n\in\mathbb{N}} f_n$, $\inf_{n\in\mathbb{N}} f_n$, $\limsup f_n$, *and* $\liminf f_n$ *are all measurable.*

PROOF. For each $x \in D$, define $h(x) = \sup\{f_1(x), \ldots, f_n(x)\}$. Then $\{x : h(x) > \alpha\} = \bigcup_{i=1}^{n}\{x : f_i(x) > \alpha\}$. So h is measurable. Similarly, for each $x \in D$, define $g(x) = \sup_{n\in\mathbb{N}} f_n(x)$. Then $\{x : g(x) > \alpha\} = \bigcup_{n=1}^{\infty}\{x : f_n(x) > \alpha\}$. So g is measurable. A similar argument establishes the fact that $\inf\{f_1, \ldots, f_n\}$ and $\inf_{n\in\mathbb{N}} f_n$ are measurable. For example, $\{x : \inf_n f_n(x) < \alpha\} = \bigcup_{n=1}^{\infty}\{x : f_n(x) < \alpha\}$. Since $\limsup f_n = \inf_{n\in\mathbb{N}}\left(\sup_{k\geq n} f_k\right)$ and $\liminf f_n = \sup_{n\in\mathbb{N}}\left(\inf_{k\geq n} f_k\right)$, $\limsup f_n$ and $\liminf f_n$ are measurable. □

Corollary 15.1.6. *If $\{f_n\}$ is a sequence of measurable functions with domain D and $f_n(x) \to f(x)$ for every $x \in D$, then f is measurable.*

PROOF. Here $f = \limsup f_n$, which is measurable by Theorem 15.1.5. □

This is a big improvement on the Riemann integrable functions. In Example 11.1.6, each f_n is Riemann integrable but the pointwise limit is not. Each f_n there is measurable, so the limit is measurable.

Definition 15.1.7. A property is said to hold *almost everywhere* (a.e.) if the set of points where it fails to hold is a set of measure zero.

For example, the Dirichlet function of Example 6.1.9 is 0 almost everywhere, since it is nonzero only on $\mathbb{Q}$, which has measure zero. Two functions that are equal almost everywhere will have the same integral. So in Lebesgue theory, sets of measure zero can usually be ignored.

Theorem 15.1.8. *If f is a measurable function and $f = g$ a.e., then g is measurable. Here f and g have the same domain D.*

PROOF. Let E be the set $\{x : f(x) \neq g(x)\}$. By hypothesis, $m(E) = 0$. Now

$$\{x : g(x) > \alpha\} = \left[\{x : f(x) > \alpha\} \cup \{x \in E : g(x) > \alpha\}\right] \setminus \{x \in E : g(x) \leq \alpha\}.$$

Now the sets $\{x \in E : g(x) > \alpha\}$ and $\{x \in E : g(x) \leq \alpha\}$ are both subsets of a measure zero set E, so by Lemma 14.2.4, they are both measurable. Also, since f is measurable, $\{x : f(x) > \alpha\}$ is measurable. So $\{x : g(x) > \alpha\}$ is measurable, implying g is measurable. □

Definition 15.1.9. If A is any set, the *characteristic function* χ_A of the set A is defined by

$$\chi_A(x) = \begin{cases} 1, & x \in A, \\ 0, & x \notin A. \end{cases}$$

Here $\chi_A : \mathbb{R} \to \mathbb{R}$. In probability, χ_A is called the *indicator function* of A and is often written $\mathbf{1}_A$.

Lemma 15.1.10. *Let A and B be disjoint measurable sets. Then $\chi_{A\cup B} = \chi_A + \chi_B$ and $\chi_{A\cap B} = \chi_A \cdot \chi_B$. (The second identity holds for any sets A and B.)*

PROOF. See Exercise 15.1.1. □

Lemma 15.1.11. *χ_A is measurable if and only if A is measurable.*

PROOF. The set $\{x : \chi_A(x) > \alpha\}$ is $\mathbb{R}$ if $\alpha < 0$, it is A if $0 \le \alpha < 1$, and it is $\emptyset$ if $\alpha \ge 1$. The sets $\mathbb{R}$ and $\emptyset$ are measurable. So every set $\{x : \chi_A(x) > \alpha\}$ is measurable if and only if A is measurable. □

Exercises

15.1.1. Let A and B be disjoint measurable sets. Prove that $\chi_{A\cup B} = \chi_A + \chi_B$ and $\chi_{A\cap B} = \chi_A \cdot \chi_B$. [Solution, p. 369]

15.1.2. Let f be an extended real-valued function on $\mathbb{R}$ such that $\{x : f(x) > \alpha\}$ is measurable for each $\alpha \in \mathbb{Q}$. Prove f is measurable.

15.1.3. Let $\mathcal{M}$ be the σ-algebra of Lebesgue measurable sets on $\mathbb{R}$, let E be measurable, and let $f : E \to \mathbb{R}$ be a measurable function. Prove that $\{A \subset \mathbb{R} : f^{-1}[A] \in \mathcal{M}\}$ is a σ-algebra containing all the open subsets of $\mathbb{R}$. Conclude that $f^{-1}[B]$ is measurable for every Borel set B. [Solution, p. 369]

15.1.4. Let $f : D \to \mathbb{R}$ be measurable and let $g : \mathbb{R} \to \mathbb{R}$ be continuous. Prove that $g \circ f$ is measurable. Conclude that $|f|$ and $|f|^p$ for $p > 0$ are measurable. [Hint, p. 370]

15.1.5. Let $f, g : D \to \mathbb{R}$ be measurable. Prove directly from the definition that $\max\{f, g\}$ is measurable. Then prove it again using $\max\{f, g\} = \frac{1}{2}(f + g + |f - g|)$. [Hint, p. 370]

15.1.6. Give an example of a measurable function $f : [0, 1] \to \mathbb{R}$ that is not equal almost everywhere to any continuous function. [Hint, p. 370]

15.2. Simple Functions and Step Functions

Simple functions are to the Lebesgue integral what step functions are to the Riemann integral. A step function is constant on each of finitely many intervals. A simple function is constant on each of finitely many measurable sets. The lemmas below show that a measurable function on $[a, b]$ can be approximated by simple functions, simple functions by step functions, and step functions by continuous functions. Theorem 15.2.7 puts these together.

Definition 15.2.1. A real-valued function φ on a measurable set D is *simple* if it is measurable and assumes only a finite number of values. If φ is simple, and $\{a_1, \ldots, a_n\}$ is the set of values φ takes, $\varphi = \sum_{i=1}^n a_i \chi_{A_i}$ where for $i = 1, \ldots, n$, $A_i = \{x : \varphi(x) = a_i\}$ is measurable. This representation is called the *canonical representation* of φ. Notice $A_i \cap A_j = \emptyset$ for $i \ne j$.

Definition 15.2.2. A real-valued function g is a *step function* if $g = \sum_{i=1}^n c_i \chi_{I_i}$, where $I_1, \ldots, I_n$ are pairwise disjoint intervals and $c_1, \ldots, c_n \in \mathbb{R}$.

Intervals are measurable, so every step function is a simple function. The converse is false. The function $\chi_{\mathbb{Q}}$ is simple, but it is not a step function, since $\mathbb{Q}$ is not a finite union of intervals.

Lemma 15.2.3. *Let f be a measurable function on $[a,b]$ that takes the values $\pm\infty$ only on a set of measure zero. Then for all $\varepsilon > 0$, there exists an $M > 0$ such that $m(\{x : |f(x)| > M\}) < \varepsilon$.*

PROOF. See Exercise 15.2.1. □

Lemma 15.2.4. *Let f be a measurable function on $[a,b]$. Then for all $\varepsilon > 0$ and for all $M > 0$, there exists a simple function φ such that $|f(x) - \varphi(x)| < \varepsilon$ except when $|f(x)| \geq M$. If $\alpha \leq f \leq \beta$, φ can be chosen so that $\alpha \leq \varphi \leq \beta$.*

PROOF. See Exercise 15.2.2. □

Lemma 15.2.5. *Let φ be a simple function on $[a,b]$. Then for all $\varepsilon > 0$, there exists a step function g defined on $[a,b]$ such that $m(\{x : g(x) \neq \varphi(x)\}) < \varepsilon$. If $\alpha \leq \varphi \leq \beta$, g can be chosen so that $\alpha \leq g \leq \beta$.*

PROOF. See Exercise 15.2.3. □

Lemma 15.2.6. *Let g be a step function on $[a,b]$. Then for all $\varepsilon > 0$, there exists a continuous function h on $[a,b]$ such that $m(\{x : g(x) \neq h(x)\}) < \varepsilon$. If $\alpha \leq g \leq \beta$, h can be chosen so that $\alpha \leq h \leq \beta$.*

PROOF. See Exercise 15.2.4. □

Theorem 15.2.7. *Let f be a measurable function defined on an interval $[a,b]$, and assume f takes the values $\pm\infty$ only on a set of measure zero. Then for all $\varepsilon > 0$, there exists a step function g and a continuous function h such that $m(\{x : |f(x) - g(x)| \geq \varepsilon\}) < \varepsilon$ and $m(\{x : |f(x) - h(x)| \geq \varepsilon\}) < \varepsilon$. If $\alpha \leq f \leq \beta$, g and h may be chosen so that $\alpha \leq g \leq \beta$ and $\alpha \leq h \leq \beta$.*

PROOF. See Exercise 15.2.5. □

Proof Technique (Approximating in Stages). Theorem 15.2.7 goes from a measurable function to a continuous one in three steps, measurable to simple to step to continuous. Each step allows a small error. Spend $\frac{\varepsilon}{3}$ on each, or $\frac{\varepsilon}{2}$ on each of the last two after the first is done. When a hard approximation is needed, look for a chain of easier ones. This is the same idea as the $\varepsilon/3$ argument in Chapter 11. Here the errors are measured by the size of the bad set instead of by a supremum.

Exercises

15.2.1. Prove Lemma 15.2.3. [Hint, p. 370]

15.2.2. Prove Lemma 15.2.4. [Hint, p. 370]

15.2.3. Prove Lemma 15.2.5. [Hint, p. 370]

15.2.4. Prove Lemma 15.2.6. [Hint, p. 370]

15.2.5. Prove Theorem 15.2.7. [Solution, p. 370]

15.3. Egoroff's Theorem

Remark 15.3.1. *Littlewood's Three Principles.*

(1) Every measurable set is nearly a finite union of intervals. (Theorem 14.3.1.)
(2) Every measurable function is nearly continuous. (Theorem 15.2.7 and Lusin's Theorem, Theorem 15.4.2.)
(3) Every convergent sequence of measurable functions is nearly uniformly convergent. (Weak form, Theorems 15.3.2 and 15.3.4. Stronger form, Egoroff's Theorem, Theorem 15.3.5.)

Theorem 15.3.2. *Let E be a measurable set of finite measure and $\{f_n\}$ a sequence of measurable real-valued functions defined on E. Let f be a real-valued function such that for all $x \in E$, $f_n(x) \to f(x)$. Then for all $\varepsilon > 0$ and for all $\delta > 0$, there exists a measurable set $A \subset E$ with $m(A) < \delta$ and an $N \in \mathbb{N}$ such that for all $x \in E \setminus A$ and all $n \geq N$, $|f_n(x) - f(x)| < \varepsilon$.*

Proof. By Corollary 15.1.6, f is measurable, so each $f_n - f$ is measurable. For each $n \in \mathbb{N}$, set $G_n = \{x \in E : |f_n(x) - f(x)| \geq \varepsilon\}$ and set

$$E_N = \bigcup_{n=N}^{\infty} G_n = \{x \in E : |f_n(x) - f(x)| \geq \varepsilon \text{ for some } n \geq N\}.$$

Now $E_{N+1} \subset E_N$, and for each $x \in E$, there exists an E_{N_0} such that $x \notin E_{N_0}$, since $f_n(x) \to f(x)$. Thus, $\bigcap E_N = \emptyset$ and so by Theorem 13.3.9 (4), $\lim_{N\to\infty} m(E_N) = 0$. This uses $m(E_1) \leq m(E) < \infty$. Now, let $\delta > 0$. So there exists an $N \in \mathbb{N}$ such that $m(E_N) < \delta$. Let $A = E_N$. Then $m(A) < \delta$ and

$$E \setminus A = \{x \in E : |f_n(x) - f(x)| < \varepsilon \text{ for all } n \geq N\}. \qquad \square$$

Definition 15.3.3. If there exists a subset $B \subset E$ with $m(B) = 0$ such that $f_n \to f$ pointwise on $E \setminus B$, then $f_n \to f$ *a.e.* on E.

Theorem 15.3.4. *Let E be a measurable set of finite measure, and $\{f_n\}$ a sequence of measurable real-valued functions that converge to a real-valued function f a.e. on E. Then for all $\varepsilon > 0$ and for all $\delta > 0$, there exists a measurable set $A \subset E$ with $m(A) < \delta$ and an $N \in \mathbb{N}$ such that for all $x \in E \setminus A$ and all $n \geq N$, $|f_n(x) - f(x)| < \varepsilon$.*

Proof. Let $B \subset E$ with $m(B) = 0$ be a set off of which $f_n \to f$. Apply Theorem 15.3.2 on $E \setminus B$. It gives a measurable $A' \subset E \setminus B$ with $m(A') < \delta$, and an N. Let $A = A' \cup B$. Then $m(A) < \delta$, and $E \setminus A = (E \setminus B) \setminus A'$. $\square$

Theorem 15.3.5 (Egoroff's Theorem)**.** *If $\{f_n\}$ is a sequence of measurable real-valued functions that converge to a real-valued function f a.e. on a measurable set E of finite measure, then for all $\eta > 0$, there exists a measurable subset $A \subset E$ with $m(A) < \eta$ such that $f_n \to f$ uniformly on $E \setminus A$.*

Proof. Let $\eta > 0$. For each $k \in \mathbb{N}$, apply Theorem 15.3.4 with $\varepsilon = \frac{1}{k}$ and $\delta = \frac{\eta}{2^k}$. It gives a measurable set $A_k \subset E$ with $m(A_k) < \frac{\eta}{2^k}$, and an $N_k \in \mathbb{N}$, such that

$$|f_n(x) - f(x)| < \frac{1}{k} \quad \text{for all } x \in E \setminus A_k \text{ and all } n \geq N_k.$$

Let $A = \bigcup_{k\in\mathbb{N}} A_k$. Then A is measurable, and $m(A) \le \sum_k m(A_k) < \eta$ by subadditivity. The set A does not depend on any ε. Now let $\varepsilon > 0$, and choose k with $\frac{1}{k} < \varepsilon$. If $x \in E \setminus A$, then $x \notin A_k$. So $|f_n(x) - f(x)| < \frac{1}{k} < \varepsilon$ for all $n \ge N_k$ and all $x \in E \setminus A$. So $f_n \to f$ uniformly on $E \setminus A$. □

The conclusion of Egoroff's Theorem is called *almost uniform convergence*. The hypothesis $m(E) < \infty$ cannot be dropped. Let $f_n = \chi_{[n,\infty)}$ on $\mathbb{R}$. Then $f_n \to 0$ everywhere. But if $m(A) < 1$, then $[n, \infty) \setminus A$ is nonempty for every n, so $\sup_{x\notin A} |f_n(x)| = 1$ for every n.

Exercises

15.3.1. Let $f_n(x) = x^n$ on $[0, 1]$. Find the pointwise limit f. Given $\eta > 0$, find a set A with $m(A) < \eta$ such that $f_n \to f$ uniformly on $[0, 1] \setminus A$. [Hint, p. 371]

15.3.2. Show that in Egoroff's Theorem the set A cannot always be taken to have measure zero. [Hint, p. 371]

15.4. Lusin's Theorem*

This section can be skipped. Nothing later depends on it.

Theorem 15.2.7 says a measurable function f on $[a, b]$ is close to a continuous function h, in the sense that $|f - h|$ is small except on a small set. Lusin's Theorem says more. There is a continuous function that *equals* f except on a small set. The proof needs one lemma, which extends a continuous function from a closed set to a whole interval.

Lemma 15.4.1. *Let $F \subset [a, b]$ be closed and nonempty, and let $g : F \to \mathbb{R}$ be continuous. Then there is a continuous function $\varphi : [a, b] \to \mathbb{R}$ with $\varphi = g$ on F.*

Proof. The set $U = [a, b] \setminus F$ is open relative to $[a, b]$. By Theorem 4.1.12, applied to $(a - 1, b + 1) \setminus F$, it is a union of countably many disjoint intervals, the *gaps* of F. Each gap is an open interval (c, d) with $c, d \in F$, or else an interval $[a, d)$ or $(c, b]$ that contains a or b and has one endpoint in F. Define $\varphi = g$ on F. On a gap (c, d) with both endpoints in F, let φ be linear, with $\varphi(c) = g(c)$ and $\varphi(d) = g(d)$. On a gap containing a or b, let φ be constant, equal to g at the endpoint in F.

Then φ is continuous at every point of a gap, and continuous from each side at the endpoints of each gap, since it is linear or constant there. It remains to check continuity at a point $x \in F$ from a side where x is a limit of other gaps. Let $\varepsilon > 0$. Since F is compact, g is uniformly continuous on F (Theorem 6.4.7). Choose $\delta > 0$ with $|g(s) - g(t)| < \varepsilon$ for $s, t \in F$ with $|s - t| < \delta$. Only finitely many gaps have length at least $\frac{\delta}{2}$, since the gaps are disjoint and lie in $[a, b]$. Each of these is either a gap with endpoint x, which is handled above, or is at a positive distance from x. So there is a $\delta' \in \left(0, \frac{\delta}{2}\right)$ such that every point $y \in [a, b]$ with $|y - x| < \delta'$ lies in F, or in a gap with endpoint x, or in a gap of length less than $\frac{\delta}{2}$. In the last case, the endpoint of that gap on the side of x is within δ' of x, so every endpoint of the gap that lies in F is within $\delta' + \frac{\delta}{2} < \delta$ of x. The value $\varphi(y)$ lies between the values of g at those endpoints. So

$$|\varphi(y) - g(x)| \le \max\{|g(c) - g(x)| : c \text{ an endpoint of the gap in } F\} < \varepsilon.$$

If $y \in F$, then $|\varphi(y) - g(x)| = |g(y) - g(x)| < \varepsilon$. Together with continuity from the side of any gap with endpoint x, this shows φ is continuous at x. □

Theorem 15.4.2 (Lusin's Theorem)**.** *Let f be a measurable real-valued function on an interval $[a, b]$. Then for all $\delta > 0$, there exists a continuous function φ on $[a, b]$ such that $m(\{x : f(x) \neq \varphi(x)\}) < \delta$.*

PROOF. For each $n \in \mathbb{N}$, Theorem 15.2.7 with $\varepsilon = \frac{1}{2^n}$ gives a continuous function h_n on $[a, b]$ with $m(E_n) < \frac{1}{2^n}$, where $E_n = \{x : |f(x) - h_n(x)| \geq \frac{1}{2^n}\}$. Choose N with $\frac{1}{2^{N-1}} < \frac{\delta}{2}$, and let $B = \bigcup_{n \geq N} E_n$. Then $m(B) \leq \sum_{n \geq N} \frac{1}{2^n} = \frac{1}{2^{N-1}} < \frac{\delta}{2}$. For $x \in [a, b] \setminus B$ and $n \geq N$, $|f(x) - h_n(x)| < \frac{1}{2^n}$. So $h_n \to f$ uniformly on $[a, b] \setminus B$.

By Theorem 14.3.1 (3), there is a closed set $F \subset [a, b] \setminus B$ with $m\big(([a, b] \setminus B) \setminus F\big) < \frac{\delta}{2}$. Then $m([a, b] \setminus F) < \delta$. (If F is empty, then $m([a, b]) < \delta$, and any continuous φ works.) On F, the continuous functions h_n converge uniformly to f. So f restricted to F is continuous, by Corollary 11.2.16. By Lemma 15.4.1, there is a continuous φ on $[a, b]$ that equals f on F. Then $\{x : f(x) \neq \varphi(x)\} \subset [a, b] \setminus F$, which has measure less than δ. □

Exercises

15.4.1. Show that $m(\{f \neq \varphi\}) < \delta$ in Lusin's Theorem cannot be replaced by $m(\{f \neq \varphi\}) = 0$. [Hint, p. 371]

15.4.2. Show that if $|f| \leq M$ on $[a, b]$, then the function φ in Lusin's Theorem can be chosen with $|\varphi| \leq M$. [Hint, p. 371]

Notes

Lebesgue introduced measurable functions in his 1902 thesis [**42**]. Dmitri Egoroff proved his theorem in 1911 [**27**]. Carlo Severini had found it independently a year earlier. Nikolai Lusin proved his theorem in 1912 [**52**], although Lebesgue and Vitali had stated forms of it earlier. John Edensor Littlewood stated his three principles in his 1944 lectures [**51**]. His advice was that if a problem would be easy when one of the principles held exactly, the word "nearly" is usually good enough.

CHAPTER 16

The Lebesgue Integral

The Riemann integral squeezes a function between step functions, which are constant on intervals. The Lebesgue integral squeezes it between simple functions, which are constant on measurable sets. This one change gives an integral that applies to far more functions. It also behaves well under limits, which the Riemann integral does not.

The integral is built in four stages. Section 16.1 rewrites the upper and lower Riemann integrals in terms of step functions. Section 16.2 integrates simple functions. Section 16.3 integrates bounded measurable functions on sets of finite measure. It shows that every Riemann integrable function is included, proves the Bounded Convergence Theorem, and finally proves Lebesgue's Theorem from Chapter 9. Sections 16.4 and 16.5 integrate nonnegative functions and then general functions, and prove the three big convergence theorems.

This chapter uses a few results from earlier chapters. They are upper and lower sums and Riemann's Criterion (Chapter 9), Lebesgue measure (Chapter 14), and measurable functions, simple functions and step functions (Chapter 15), including Theorem 15.3.2.

16.1. The Riemann Integral Revisited

Recall the upper and lower sums $\mathcal{U}(\mathcal{P}, f)$ and $\mathcal{L}(\mathcal{P}, f)$ and the upper and lower integrals $\overline{\int_a^b} f$ and $\underline{\int_a^b} f$ from Section 9.1. An upper sum is the area under a step function that lies above f. This section makes that precise. The upper integral turns out to be the infimum of the integrals of all step functions above f. The Lebesgue integral will be defined the same way, with step functions replaced by simple functions.

Step functions were defined in Definition 15.2.2. In this chapter, a *step function on* $[a, b]$ is a function $\psi = \sum_{i=1}^{n} c_i \chi_{I_i}$, restricted to $[a, b]$, where $I_1, \ldots, I_n$ are pairwise disjoint subintervals of $[a, b]$. A single point is an interval (Definition 1.8.5). So a step function may take any values at finitely many points.

Lemma 16.1.1. *Let $\psi = \sum_{i=1}^{n} c_i \chi_{I_i}$ be a step function on $[a, b]$, where $I_1, \ldots, I_n$ are pairwise disjoint subintervals of $[a, b]$. Then $\psi \in \mathcal{R}[a, b]$ and*

$$\int_a^b \psi = \sum_{i=1}^{n} c_i \, \ell(I_i).$$

Proof. By Theorem 9.2.11, it is enough to show that $\chi_I \in \mathcal{R}[a, b]$ with $\int_a^b \chi_I = \ell(I)$ for each subinterval I of $[a, b]$. Let I have endpoints $c \le d$. If $c = d$, then χ_I is zero except at one point. So $\int_a^b \chi_I = 0 = \ell(I)$ by Exercise 9.2.2.

Suppose $c < d$. On $[c,d]$, χ_I equals the constant 1 except possibly at c and d. So $\chi_I \in \mathcal{R}[c,d]$ and $\int_c^d \chi_I = d - c$, by Exercise 9.2.4. If $a < c$, then on $[a,c]$, χ_I equals 0 except possibly at c. So $\chi_I \in \mathcal{R}[a,c]$ and $\int_a^c \chi_I = 0$. Similarly, if $d < b$, then $\chi_I \in \mathcal{R}[d,b]$ and $\int_d^b \chi_I = 0$. By Theorem 9.2.13, applied once or twice, $\chi_I \in \mathcal{R}[a,b]$ and $\int_a^b \chi_I = d - c = \ell(I)$. □

Proposition 16.1.2. *Let f be a bounded real-valued function defined on the interval $[a,b]$. Then*

$$\overline{\int_a^b} f = \inf\left\{\int_a^b \psi : \psi \text{ is a step function on } [a,b] \text{ with } \psi \geq f\right\}$$

and

$$\underline{\int_a^b} f = \sup\left\{\int_a^b \varphi : \varphi \text{ is a step function on } [a,b] \text{ with } \varphi \leq f\right\}.$$

Proof. Let J denote the infimum on the right side of the first equation. First, let ψ be a step function on $[a,b]$ with $\psi \geq f$. By Lemma 16.1.1, $\psi \in \mathcal{R}[a,b]$. For each partition $\mathcal{P}$ of $[a,b]$, the supremum of ψ on each subinterval of $\mathcal{P}$ is at least the supremum of f there. So $\mathcal{U}(\mathcal{P},\psi) \geq \mathcal{U}(\mathcal{P},f)$. Taking the infimum over all $\mathcal{P}$,

$$\int_a^b \psi = \overline{\int_a^b} \psi \geq \overline{\int_a^b} f.$$

Since ψ was arbitrary, $J \geq \overline{\int_a^b} f$.

Next, let $\mathcal{P} = \{x_0, x_1, \ldots, x_n\}$ be a partition of $[a,b]$, and let $M_i = \sup\{f(x) : x_{i-1} \leq x \leq x_i\}$. Define $\psi_{\mathcal{P}}$ by

$$\psi_{\mathcal{P}}(x) = \begin{cases} M_i, & x \in (x_{i-1}, x_i),\ i = 1, \ldots, n, \\ f(x_i), & x = x_i,\ i = 0, \ldots, n. \end{cases}$$

Then $\psi_{\mathcal{P}}$ is a step function on $[a,b]$, since the open intervals (x_{i-1}, x_i) and the single points $\{x_i\}$ are pairwise disjoint intervals. Also $\psi_{\mathcal{P}} \geq f$. By Lemma 16.1.1,

$$\int_a^b \psi_{\mathcal{P}} = \sum_{i=1}^n M_i(x_i - x_{i-1}) + \sum_{i=0}^n f(x_i) \cdot 0 = \mathcal{U}(\mathcal{P},f).$$

So $J \leq \mathcal{U}(\mathcal{P},f)$ for every partition $\mathcal{P}$, and $J \leq \overline{\int_a^b} f$. Therefore $J = \overline{\int_a^b} f$.

The proof for the lower integral is the same, with M_i replaced by $m_i = \inf\{f(x) : x_{i-1} \leq x \leq x_i\}$ and the inequalities reversed. □

Look again at the Dirichlet function $f = \chi_{\mathbb{Q}}$ on $[0,1]$ from Example 9.1.9. Every nondegenerate interval contains rational and irrational numbers. So a step function $\psi \geq f$ is at least 1 except at finitely many points, and $\int_0^1 \psi \geq 1$. A step function $\varphi \leq f$ is at most 0 except at finitely many points, and $\int_0^1 \varphi \leq 0$. This agrees with $\overline{\int_0^1} f = 1$ and $\underline{\int_0^1} f = 0$. Step functions are too coarse to see that $\mathbb{Q}$ is a very thin set. Simple functions are not. The function $\chi_{\mathbb{Q}}$ is itself a simple function. In Example 16.2.6 its Lebesgue integral will be 0.

Exercises

16.1.1. Let $f : [a, b] \to \mathbb{R}$ be bounded. Prove that $f \in \mathcal{R}[a, b]$ if and only if for every $\varepsilon > 0$ there are step functions φ and ψ on $[a, b]$ with $\varphi \le f \le \psi$ and $\int_a^b (\psi - \varphi) < \varepsilon$. [Hint, p. 372]

16.1.2. Let $f(x) = x$ on $[0, 1]$. Show that $\int_0^1 \psi > \frac{1}{2}$ for every step function $\psi \ge f$ on $[0, 1]$. So the infimum in Proposition 16.1.2 need not be attained. [Hint, p. 372]

16.1.3. Let ψ be a step function on $[a, b]$. Prove that there is a partition $\{x_0, \dots, x_n\}$ of $[a, b]$ such that ψ is constant on each open interval (x_{i-1}, x_i). Conversely, prove that every function with this property is a step function on $[a, b]$. [Hint, p. 372]

16.2. The Integral of a Simple Function

Recall from Definition 15.2.1 that a simple function is a measurable function with finitely many values. Its canonical representation is $\varphi = \sum_{i=1}^n a_i \chi_{A_i}$, where $a_1, \dots, a_n$ are the distinct values of φ and $A_i = \{\varphi = a_i\}$. The integral should be the sum of each value times the measure of the set where it is taken. Some of the sets A_i may have infinite measure. So this section uses the convention $0 \cdot \infty = 0$. It only integrates simple functions that are zero outside a set of finite measure.

Definition 16.2.1. Let φ be a simple function with canonical representation $\varphi = \sum_{i=1}^n a_i \chi_{A_i}$. If φ vanishes outside a set of finite measure, define the *integral* of φ by

$$\int \varphi = \int \varphi \, dm = \sum_{i=1}^n a_i m(A_i).$$

If E is a measurable set, define

$$\int_E \varphi = \int \varphi \cdot \chi_E.$$

Here "φ vanishes outside a set of finite measure" means $m(\{\varphi \ne 0\}) < \infty$. Then $m(A_i) < \infty$ whenever $a_i \ne 0$. If $a_i = 0$, the term $a_i m(A_i)$ is 0, even if $m(A_i) = \infty$. If φ is defined on a measurable set D, then $E \subset D$, and $\varphi \cdot \chi_E$ is again a simple function that vanishes outside a set of finite measure.

Lemma 16.2.2. *Let $\varphi = \sum_{i=1}^n a_i \chi_{E_i}$ with $E_i \cap E_j = \emptyset$ for $i \ne j$. Suppose each set E_i is a measurable set of finite measure. Then*

$$\int \varphi = \sum_{i=1}^n a_i m(E_i).$$

Proof. Assume φ has canonical representation $\varphi = \sum_{k=1}^m \hat{a}_k \chi_{A_k}$. Suppose $\hat{a}_k \ne 0$. Since the E_i are disjoint, $x \in A_k$ exactly when x lies in some E_i with $a_i = \hat{a}_k$. So the set

$$A_k = \{x : \varphi(x) = \hat{a}_k\} = \bigcup_{a_i = \hat{a}_k} E_i.$$

Hence, by finite additivity,

$$\hat{a}_k m(A_k) = \sum_{a_i = \hat{a}_k} a_i m(E_i).$$

If $\hat{a}_k = 0$, then $\hat{a}_k m(A_k) = 0$. The terms $a_i m(E_i)$ with $a_i = 0$ are also 0. Each i with $a_i \neq 0$ has $a_i = \hat{a}_k$ for exactly one k. So

$$\int \varphi = \sum_{k=1}^{m} \hat{a}_k m(A_k) = \sum_{i=1}^{n} a_i m(E_i). \qquad \square$$

Theorem 16.2.3. *Let φ and ψ be simple functions which vanish outside a set of finite measure. Then*

$$\int (a\varphi + b\psi) = a \int \varphi + b \int \psi$$

for all $a, b \in \mathbb{R}$, and if $\varphi \geq \psi$ a.e., then

$$\int \varphi \geq \int \psi.$$

PROOF. Let $\{A_j\}$ and $\{B_i\}$ be the sets in the canonical representations of φ and ψ, respectively. The sets $A_j \cap B_i$ are measurable and pairwise disjoint. Let $E_1, \ldots, E_N$ be those sets $A_j \cap B_i$ that are nonempty and on which φ or ψ is not zero. Each E_k lies in $\{\varphi \neq 0\}$ or in $\{\psi \neq 0\}$, so $m(E_k) < \infty$. On E_k, φ and ψ are constant, say $\varphi = c_k$ and $\psi = d_k$. Off $\bigcup_k E_k$, $\varphi = \psi = 0$. So

$$\varphi = \sum_{k=1}^{N} c_k \chi_{E_k} \quad \text{and} \quad \psi = \sum_{k=1}^{N} d_k \chi_{E_k}.$$

So

$$a\varphi + b\psi = \sum_{k=1}^{N} (ac_k + bd_k) \chi_{E_k}.$$

Thus, by Lemma 16.2.2,

$$\begin{aligned} \int (a\varphi + b\psi) &= \sum_{k=1}^{N} (ac_k + bd_k) m(E_k) = a \sum_{k=1}^{N} c_k m(E_k) + b \sum_{k=1}^{N} d_k m(E_k) \\ &= a \int \varphi + b \int \psi. \end{aligned}$$

Lastly, suppose $\varphi \geq \psi$ a.e. Then $\varphi - \psi = \sum_{k=1}^{N} (c_k - d_k) \chi_{E_k}$. If $c_k - d_k < 0$, then $E_k \subset \{\varphi < \psi\}$, so $m(E_k) = 0$. So every term of $\sum_{k=1}^{N} (c_k - d_k) m(E_k)$ is at least 0. Thus

$$\int \varphi - \int \psi = \int (\varphi - \psi) = \sum_{k=1}^{N} (c_k - d_k) m(E_k) \geq 0. \qquad \square$$

Corollary 16.2.4. *If $\varphi = \sum_{i=1}^{n} a_i \chi_{E_i}$, where $E_1, \ldots, E_n$ are measurable sets of finite measure that need not be disjoint, then*

$$\int \varphi = \sum_{i=1}^{n} a_i m(E_i).$$

PROOF. Each χ_{E_i} is a simple function by Lemma 15.1.11, and $\int \chi_{E_i} = m(E_i)$ by Lemma 16.2.2. So this is a direct result of Theorem 16.2.3. □

Proof Technique (Putting Two Simple Functions on the Same Sets). To combine or compare two simple functions φ and ψ, write them over the same disjoint sets, as in the proof of Theorem 16.2.3. Intersect the sets of the two canonical representations. Then both functions are constant on each piece. Sums, products, maxima and inequalities can then be checked one piece at a time. This is the measurable version of a common refinement of two partitions (Definition 8.1.4).

Step functions on $[a, b]$ are simple functions, since intervals are measurable. Their two integrals agree.

Corollary 16.2.5. *Let ψ be a step function on $[a, b]$. Then ψ is a simple function and*

$$\int_{[a,b]} \psi = \int_a^b \psi.$$

PROOF. Write $\psi = \sum_{i=1}^n c_i \chi_{I_i}$, where $I_1, \ldots, I_n$ are pairwise disjoint subintervals of $[a, b]$. Then $\psi \chi_{[a,b]} = \psi$. Intervals are measurable and $m(I_i) = \ell(I_i)$, by Theorems 14.1.4 and 14.2.10. By Lemma 16.2.2 and Lemma 16.1.1,

$$\int_{[a,b]} \psi = \sum_{i=1}^n c_i \, \ell(I_i) = \int_a^b \psi. \qquad \square$$

Example 16.2.6. Let $f = \chi_{\mathbb{Q}}$ on $[0, 1]$, the Dirichlet function. It is a simple function with canonical representation $f = 1 \cdot \chi_{\mathbb{Q} \cap [0,1]} + 0 \cdot \chi_{[0,1] \setminus \mathbb{Q}}$. The set $\mathbb{Q} \cap [0, 1]$ is countable, so it has measure zero by Corollary 14.1.7. So

$$\int_{[0,1]} f = 1 \cdot m(\mathbb{Q} \cap [0, 1]) + 0 \cdot m([0, 1] \setminus \mathbb{Q}) = 0.$$

This function is not Riemann integrable (Example 9.1.9).

Exercises

16.2.1. Let $\varphi(x) = \lfloor x \rfloor + \chi_{\mathbb{Q}}(x)$ for $x \in [0, 3]$. Show that φ is a simple function, write its canonical representation, and compute $\int_{[0,3]} \varphi$. [Hint, p. 372]

16.2.2. Let φ be a simple function that vanishes outside a set of finite measure. Prove that $|\varphi|$ is such a function, and that $\left|\int \varphi\right| \leq \int |\varphi|$. [Hint, p. 373]

16.2.3. Let φ and ψ be simple functions. Prove that $\max\{\varphi, \psi\}$, $\min\{\varphi, \psi\}$ and $\varphi\psi$ are simple functions. [Hint, p. 373]

16.2.4. Let φ be a simple function that vanishes outside a set of finite measure, and let A and B be disjoint measurable sets. Prove that $\int_{A \cup B} \varphi = \int_A \varphi + \int_B \varphi$. [Hint, p. 373]

16.2.5. Let $\varphi \geq 0$ be a simple function that vanishes outside a set of finite measure. Prove that $\nu(E) = \int_E \varphi$ defines a measure ν on the Lebesgue measurable sets (Definition 13.3.1). [Hint, p. 373]

16.3. Bounded Functions on Sets of Finite Measure

Let f be a bounded function on a measurable set E with $m(E) < \infty$. Every simple function on E vanishes outside E, a set of finite measure, so it has an integral. Following Proposition 16.1.2, compare the infimum of $\int_E \psi$ over simple functions $\psi \geq f$ with the supremum of $\int_E \varphi$ over simple functions $\varphi \leq f$. These are the *upper and lower Lebesgue integrals* of f. For the Riemann integral, the upper and lower integrals can differ for many functions. For the Lebesgue integral, they agree exactly when f is measurable. In this section all simple functions are defined on E.

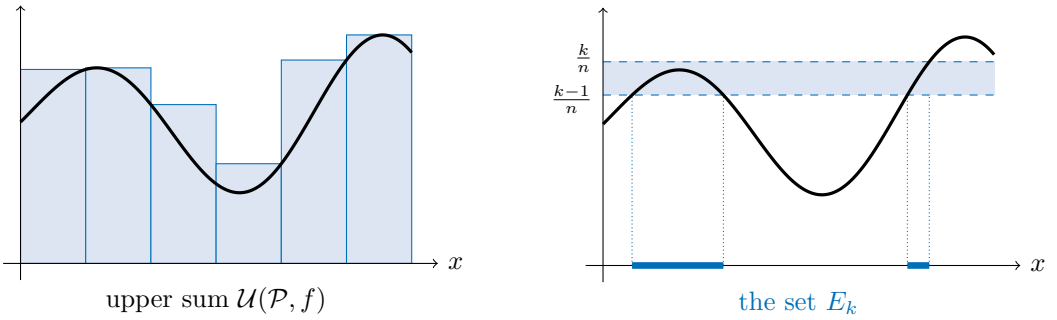


FIGURE 16.1. Riemann and Lebesgue cut up the area under a graph in different ways. On the left, the domain is cut into intervals, and f is replaced on each one by its supremum. On the right, the range is cut into slices. The set $E_k = \left\{\frac{k-1}{n} < f \leq \frac{k}{n}\right\}$, shown in bold, is a union of two intervals here. For a general measurable function it is only a measurable set. This is the set used in the proof of Theorem 16.3.1.

Theorem 16.3.1. *Let f be defined and bounded on a measurable set E with $m(E) < \infty$. Then f is measurable if and only if*

$$\inf_{f \leq \psi} \int_E \psi = \sup_{f \geq \varphi} \int_E \varphi,$$

where ψ and φ range over the simple functions on E with $\psi \geq f$ and $\varphi \leq f$.

PROOF. If φ and ψ are simple functions with $\varphi \leq f \leq \psi$, then $\int_E \varphi \leq \int_E \psi$ by Theorem 16.2.3. So the supremum is always at most the infimum. See Figure 16.1.

($\Rightarrow$) Suppose f is measurable, and let $|f(x)| \leq M$ for all $x \in E$. Let $n \in \mathbb{N}$. For each integer k, let

$$E_k = \left\{x \in E : \frac{k-1}{n} < f(x) \leq \frac{k}{n}\right\} = \left\{f > \frac{k-1}{n}\right\} \cap \left\{f \leq \frac{k}{n}\right\}.$$

Each E_k is measurable. The sets E_k are pairwise disjoint and their union is E. Since $|f| \le M$, E_k is empty unless $-Mn \le k < Mn+1$. So only finitely many E_k are nonempty. Define

$$\psi_n = \sum_k \frac{k}{n}\chi_{E_k} \quad \text{and} \quad \varphi_n = \sum_k \frac{k-1}{n}\chi_{E_k},$$

where both sums are over the finitely many k with $E_k \neq \emptyset$. These are simple functions, and $\varphi_n \le f \le \psi_n$. Also $\psi_n - \varphi_n = \frac{1}{n}$ on E. So

$$0 \le \inf_{f\le\psi} \int_E \psi - \sup_{f\ge\varphi} \int_E \varphi \le \int_E \psi_n - \int_E \varphi_n = \int_E (\psi_n - \varphi_n) = \frac{m(E)}{n}.$$

This holds for every $n \in \mathbb{N}$ and $m(E) < \infty$. So the infimum and the supremum are equal.

($\Leftarrow$) Suppose the infimum and the supremum are equal. Then for each $n \in \mathbb{N}$ there are simple functions φ_n and ψ_n with

$$\varphi_n \le f \le \psi_n \quad \text{and} \quad \int_E \psi_n - \int_E \varphi_n < \frac{1}{n}.$$

Let $\varphi^* = \sup_{n\in\mathbb{N}} \varphi_n$ and $\psi^* = \inf_{n\in\mathbb{N}} \psi_n$. These are measurable by Theorem 15.1.5, and $\varphi^* \le f \le \psi^*$. In particular they are real-valued, since f is bounded. The goal is to show $\varphi^* = \psi^*$ a.e.

For $j \in \mathbb{N}$, let $\Delta_j = \left\{\psi^* - \varphi^* > \frac{1}{j}\right\}$, a measurable set. Fix $n \in \mathbb{N}$. Since $\varphi_n \le \varphi^*$ and $\psi^* \le \psi_n$, the set Δ_j is contained in $G = \left\{\psi_n - \varphi_n > \frac{1}{j}\right\}$. Now $\frac{1}{j}\chi_G \le \psi_n - \varphi_n$ on E, and both sides are simple. By Theorem 16.2.3,

$$\frac{1}{j}\, m(\Delta_j) \le \frac{1}{j}\, m(G) \le \int_E (\psi_n - \varphi_n) < \frac{1}{n}.$$

So $m(\Delta_j) < \frac{j}{n}$ for every $n \in \mathbb{N}$, and $m(\Delta_j) = 0$. The set where $\varphi^* < \psi^*$ is $\bigcup_{j\in\mathbb{N}} \Delta_j$, which has measure zero. So $\psi^* = \varphi^*$ a.e. Since $\varphi^* \le f \le \psi^*$, $f = \psi^*$ a.e. By Theorem 15.1.8, f is measurable. □

Definition 16.3.2. If f is a bounded measurable function on a measurable set E with $m(E) < \infty$, the *Lebesgue integral* of f over E is

$$\int_E f = \inf_{\psi\ge f} \int_E \psi,$$

where ψ ranges over all simple functions $\psi \ge f$ on E. By Theorem 16.3.1, this also equals $\sup_{\varphi\le f} \int_E \varphi$. The integral is also written $\int_E f\, dm$ or $\int_E f(x)\, dm(x)$.

If f is itself simple, this agrees with Definition 16.2.1. Indeed, $\psi = f$ is one of the functions in the infimum, and every simple $\psi \ge f$ has $\int_E \psi \ge \int_E f$ by Theorem 16.2.3. On an interval, $\int_{[a,b]} f$ denotes the Lebesgue integral and $\int_a^b f$ the Riemann integral. The next theorem shows they agree whenever the Riemann integral exists.

Theorem 16.3.3. *Let f be a bounded function defined on $[a,b]$. If $f \in \mathcal{R}[a,b]$, then f is measurable and*

$$\int_a^b f = \int_{[a,b]} f.$$

PROOF. By Corollary 16.2.5, every step function on $[a,b]$ is a simple function, with the same Riemann and Lebesgue integrals. So by Proposition 16.1.2,

$$\underline{\int_a^b} f = \sup_{\substack{\varphi \le f \\ \varphi \text{ step}}} \int_{[a,b]} \varphi \le \sup_{\substack{\varphi \le f \\ \varphi \text{ simple}}} \int_{[a,b]} \varphi$$

$$\le \inf_{\substack{\psi \ge f \\ \psi \text{ simple}}} \int_{[a,b]} \psi \le \inf_{\substack{\psi \ge f \\ \psi \text{ step}}} \int_{[a,b]} \psi = \overline{\int_a^b} f.$$

The first and last inequalities hold because there are more simple functions than step functions. The middle one holds by the first paragraph of the proof of Theorem 16.3.1. Since $f \in \mathcal{R}[a,b]$, the two ends are equal to $\int_a^b f$. So all of these quantities are equal. By Theorem 16.3.1, f is measurable, and then the middle quantities equal $\int_{[a,b]} f$. □

The converse is false. The Dirichlet function is measurable but not Riemann integrable (Example 16.2.6). So the Lebesgue integral extends the Riemann integral. From now on, $\int_{[a,b]} f$ can be computed for a Riemann integrable f by any method from Chapter 9, including the Fundamental Theorem of Calculus.

To integrate over a measurable subset $A \subset E$, restrict f to A. The next lemma says this is the same as integrating $f\chi_A$ over E.

Lemma 16.3.4. *Let f be a bounded measurable function on a measurable set E with $m(E) < \infty$, and let $A \subset E$ be measurable. Then $\int_A f = \int_E f\chi_A$.*

PROOF. The function $f\chi_A$ is bounded and measurable by Lemma 15.1.11 and Theorem 15.1.4. Let ψ be a simple function on A with $\psi \ge f$ on A. Extend ψ to E by setting it equal to 0 on $E \setminus A$. The extension is simple, it is at least $f\chi_A$, and its integral over E is $\int_A \psi$. So $\int_E f\chi_A \le \int_A \psi$. Taking the infimum over ψ, $\int_E f\chi_A \le \int_A f$.

Conversely, let ψ be a simple function on E with $\psi \ge f\chi_A$. Then $\psi \ge f$ on A, and $\psi \ge 0$ on $E \setminus A$. So $\int_A \psi \le \int_A \psi + \int_{E\setminus A} \psi = \int_E \psi$, using $\chi_E = \chi_A + \chi_{E\setminus A}$ and Theorem 16.2.3. Thus $\int_A f \le \int_A \psi \le \int_E \psi$. Taking the infimum over ψ, $\int_A f \le \int_E f\chi_A$. □

Theorem 16.3.5. *If f and g are bounded measurable functions defined on a set E of finite measure, then each of the following hold.*

(1) $\int_E (af + bg) = a\int_E f + b\int_E g$ for all $a, b \in \mathbb{R}$.
(2) If $f = g$ a.e., then $\int_E f = \int_E g$.
(3) If $f \le g$ a.e., then $\int_E f \le \int_E g$.
(4) $\left|\int_E f\right| \le \int_E |f|$.
(5) If $\alpha \le f(x) \le \beta$ for all $x \in E$, then $\alpha m(E) \le \int_E f \le \beta m(E)$.
(6) If A and B are disjoint measurable subsets of E, then $\int_{A\cup B} f = \int_A f + \int_B f$.

PROOF. (1) If ψ is a simple function, so is $a\psi$, and conversely for $a \ne 0$. Hence, for $a > 0$, by Theorem 16.2.3,

$$\int_E af = \inf_{a\psi \ge af} \int_E a\psi = a \inf_{\psi \ge f} \int_E \psi = a\int_E f.$$

If $a < 0$, then $a\varphi \geq af$ exactly when $\varphi \leq f$. So, by Theorem 16.3.1,

$$\int_E af = \inf_{a\varphi \geq af} \int_E a\varphi = \inf_{\varphi \leq f} a \int_E \varphi = a \sup_{\varphi \leq f} \int_E \varphi = a \int_E f.$$

If $a = 0$, both sides are 0.

If ψ_1 is a simple function with $f \leq \psi_1$ and ψ_2 is a simple function with $g \leq \psi_2$, then $\psi_1 + \psi_2$ is a simple function with $f + g \leq \psi_1 + \psi_2$. Hence

$$\int_E (f+g) \leq \int_E (\psi_1 + \psi_2) = \int_E \psi_1 + \int_E \psi_2.$$

Since the infimum of the right side is $\int_E f + \int_E g$,

$$\int_E (f+g) \leq \int_E f + \int_E g.$$

Similarly, if φ_1 and φ_2 are simple functions with $\varphi_1 \leq f$ and $\varphi_2 \leq g$, $\varphi_1 + \varphi_2$ is a simple function with $\varphi_1 + \varphi_2 \leq f + g$. So

$$\int_E (f+g) \geq \int_E (\varphi_1 + \varphi_2) = \int_E \varphi_1 + \int_E \varphi_2.$$

Since the supremum of the right side is $\int_E f + \int_E g$,

$$\int_E (f+g) \geq \int_E f + \int_E g.$$

So

$$\int_E (f+g) = \int_E f + \int_E g.$$

Here $f + g$ is bounded and measurable by Theorem 15.1.4, and the second inequality uses Theorem 16.3.1 for $f+g$. Combining the two parts, $\int_E (af+bg) = \int_E af + \int_E bg = a\int_E f + b\int_E g$.

(2) Since $f - g = 0$ a.e., if $\psi \geq f - g$, then $\psi \geq 0$ a.e. So $\int_E \psi \geq 0$ by Theorem 16.2.3. Thus $\int_E (f-g) \geq 0$. Also, if $\varphi \leq f - g$, then $\varphi \leq 0$ a.e., so $\int_E \varphi \leq 0$. Thus $\int_E (f-g) \leq 0$ by Theorem 16.3.1. So $\int_E (f-g) = 0$, and $\int_E f = \int_E g$ by (1).

(3) Since $f - g \leq 0$ a.e., if $\varphi \leq f - g$ is a simple function, then $\varphi \leq 0$ a.e., and $\int_E \varphi \leq 0$. So $\int_E (f-g) \leq 0$ by Theorem 16.3.1. By (1), $\int_E f \leq \int_E g$.

(4) Since $-|f| \leq f \leq |f|$, (1) and (3) give $-\int_E |f| \leq \int_E f \leq \int_E |f|$. So $\left|\int_E f\right| \leq \int_E |f|$. Here $|f| = \max\{f, -f\}$ is measurable by Theorem 15.1.5.

(5) Since $\alpha \leq f(x) \leq \beta$ for all $x \in E$, (3) gives

$$\int_E \alpha \leq \int_E f \leq \int_E \beta.$$

But the constants α and β are simple functions, so $\int_E \alpha = \alpha m(E)$ and $\int_E \beta = \beta m(E)$. So

$$\alpha m(E) \leq \int_E f \leq \beta m(E).$$

(6) By Lemma 15.1.10, Lemma 16.3.4 and (1),

$$\int_{A\cup B} f = \int_E f \cdot (\chi_A + \chi_B) = \int_E f \cdot \chi_A + \int_E f \cdot \chi_B = \int_A f + \int_B f. \qquad \square$$

Theorem 16.3.6 (Bounded Convergence Theorem)**.** *Let $\{f_n\}$ be a sequence of measurable functions defined on a set E of finite measure, and suppose that there is a real number M such that $|f_n(x)| \le M$ for all $n \in \mathbb{N}$ and for all $x \in E$. If $\lim_{n\to\infty} f_n(x) = f(x)$ for each $x \in E$, then*

$$\int_E f = \lim_{n\to\infty} \int_E f_n.$$

Proof. By Corollary 15.1.6, f is measurable. Also $|f| \le M$, so $\int_E f$ is defined. If $M = 0$, then every f_n and f are 0, and there is nothing to prove. So assume $M > 0$. Let $\varepsilon > 0$. Choose $\eta > 0$ so small that $\eta\, m(E) < \frac{\varepsilon}{2}$, and let $\delta = \frac{\varepsilon}{4M}$. By Theorem 15.3.2, there are a measurable set $A \subset E$ with $m(A) < \delta$ and an $N \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \eta$ for all $x \in E \setminus A$ and all $n \ge N$.

Let $n \ge N$. Split E into the good set $E \setminus A$, where $|f_n - f|$ is small, and the bad set A, which is small. On $E \setminus A$, $0 \le |f_n - f| \le \eta$. On A, $0 \le |f_n - f| \le 2M$. By Theorem 16.3.5 (1), (4), (6) and (5),

$$\begin{aligned}\left|\int_E f_n - \int_E f\right| = \left|\int_E (f_n - f)\right| &\le \int_E |f_n - f| \\ &= \int_{E\setminus A} |f_n - f| + \int_A |f_n - f| \\ &\le \eta\, m(E \setminus A) + 2M\, m(A) < \frac{\varepsilon}{2} + 2M \cdot \frac{\varepsilon}{4M} = \varepsilon.\end{aligned}$$

So $\int_E f_n \to \int_E f$. □

Corollary 16.3.7. *Let $\{f_n\}$ and M be as in Theorem 16.3.6, and let f be a bounded measurable function on E. If $f_n \to f$ a.e. on E, then $\int_E f = \lim_{n\to\infty} \int_E f_n$.*

Proof. By Definition 15.3.3, there is a set $B \subset E$ with $m(B) = 0$ such that $f_n \to f$ at every point of $E \setminus B$. If g is bounded and measurable on E, say $|g| \le K$, then $\left|\int_B g\right| \le K\, m(B) = 0$ by Theorem 16.3.5 (4) and (5). So $\int_E g = \int_{E\setminus B} g$ by Theorem 16.3.5 (6). Apply this to each f_n and to f, and apply Theorem 16.3.6 on $E \setminus B$. □

> **Proof Technique (Ignoring Sets of Measure Zero).** Integrals do not see sets of measure zero. If $f = g$ a.e., then $\int_E f = \int_E g$. So a hypothesis that holds a.e. is usually as good as one that holds everywhere. The standard move has three steps. Find a set B of measure zero off which the hypothesis holds. Prove the result on $E \setminus B$. Then note that $\int_E h = \int_{E\setminus B} h$ for every function h in sight. The proof of Corollary 16.3.7 works this way. There is one caution. The functions must still be measurable on all of E, and bounded if the integral being used needs that, or the integrals over E are not defined.

Arzelà's Bounded Convergence Theorem for Riemann integrals, Theorem 11.2.21, was stated in Chapter 11 without proof. It now follows quickly.

Corollary 16.3.8 (Arzelà)**.** *Let $f_n, f \in \mathcal{R}[a,b]$ with $f_n(x) \to f(x)$ for each $x \in [a,b]$. Suppose there is an M with $|f_n(x)| \le M$ for all $x \in [a,b]$ and all $n \in \mathbb{N}$. Then $\int_a^b f_n \to \int_a^b f$.*

PROOF. By Theorem 16.3.3, each f_n and f is measurable, with the same Riemann and Lebesgue integrals. Apply Theorem 16.3.6 with $E = [a,b]$. □

The Lebesgue version is stronger. It does not need the hypothesis $f \in \mathcal{R}[a,b]$, since the limit of measurable functions is always measurable. In Example 11.1.6, the limit f is not Riemann integrable. But $f = 1$ a.e., so $\int_{[0,1]} f = 1 = \lim \int_0^1 f_n$.

Finally, Lebesgue's Theorem, Theorem 9.2.9, can be proved. The proof measures how badly f fails to be continuous at each point.

Definition 16.3.9. Let $f : [a,b] \to \mathbb{R}$ be bounded. For $x \in [a,b]$ and $\delta > 0$, let

$$\omega_f(x,\delta) = \sup\{|f(s) - f(t)| : s,t \in [a,b],\ |s-x| < \delta,\ |t-x| < \delta\}.$$

This decreases as δ decreases. The *oscillation* of f at x is $\omega_f(x) = \inf_{\delta>0} \omega_f(x,\delta)$.

Lemma 16.3.10. *Let $f : [a,b] \to \mathbb{R}$ be bounded and let $x \in [a,b]$. Then f is continuous at x if and only if $\omega_f(x) = 0$.*

PROOF. Suppose f is continuous at x, and let $\varepsilon > 0$. Choose $\delta > 0$ with $|f(s) - f(x)| < \frac{\varepsilon}{2}$ for all $s \in [a,b]$ with $|s-x| < \delta$. If s and t are such points, then $|f(s) - f(t)| \le |f(s) - f(x)| + |f(x) - f(t)| < \varepsilon$. So $\omega_f(x) \le \omega_f(x,\delta) \le \varepsilon$. Since ε was arbitrary, $\omega_f(x) = 0$.

Conversely, suppose $\omega_f(x) = 0$, and let $\varepsilon > 0$. Choose $\delta > 0$ with $\omega_f(x,\delta) < \varepsilon$. If $s \in [a,b]$ and $|s-x| < \delta$, take $t = x$ in the definition to get $|f(s) - f(x)| \le \omega_f(x,\delta) < \varepsilon$. So f is continuous at x. □

Theorem 16.3.11 (Lebesgue's Theorem)**.** *Let $f : [a,b] \to \mathbb{R}$ be bounded. Then $f \in \mathcal{R}[a,b]$ if and only if the set of points at which f has discontinuities has measure zero.*

PROOF. Let D be the set of points of $[a,b]$ at which f is discontinuous. For $k \in \mathbb{N}$, let $D_k = \left\{x \in [a,b] : \omega_f(x) \ge \frac{1}{k}\right\}$. By Lemma 16.3.10, $D = \bigcup_{k\in\mathbb{N}} D_k$. Recall from Section 14.1 that D has measure zero exactly when $m^*(D) = 0$.

($\Rightarrow$) Suppose $f \in \mathcal{R}[a,b]$. Fix $k \in \mathbb{N}$ and let $\varepsilon > 0$. By Riemann's Criterion (Theorem 9.2.1), there is a partition $\mathcal{P} = \{x_0, \ldots, x_n\}$ of $[a,b]$ with

$$\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) = \sum_{i=1}^{n} (M_i - m_i)\Delta x_i < \frac{\varepsilon}{k}.$$

Let $x \in D_k$ with $x \notin \mathcal{P}$. Then $x \in (x_{i-1}, x_i)$ for some i. For small $\delta > 0$, the interval $(x-\delta, x+\delta)$ lies inside (x_{i-1}, x_i), and so $\omega_f(x,\delta) \le M_i - m_i$. Thus $M_i - m_i \ge \omega_f(x) \ge \frac{1}{k}$. Let S be the set of indices i with $M_i - m_i \ge \frac{1}{k}$. Then

$$\frac{1}{k} \sum_{i\in S} \Delta x_i \le \sum_{i\in S} (M_i - m_i)\Delta x_i < \frac{\varepsilon}{k},$$

so $\sum_{i\in S}\Delta x_i < \varepsilon$. This shows $D_k \subset \mathcal{P} \cup \bigcup_{i\in S}(x_{i-1}, x_i)$. The finite set $\mathcal{P}$ has outer measure zero. By Theorems 14.1.4 and 14.1.6, $m^*(D_k) \le \sum_{i\in S}\Delta x_i < \varepsilon$. Since ε was arbitrary, $m^*(D_k) = 0$. By subadditivity again, $m^*(D) \le \sum_k m^*(D_k) = 0$.

($\Leftarrow$) Suppose $m^*(D) = 0$, and let $|f| \le K$ on $[a, b]$. For each $n \in \mathbb{N}$, let $\mathcal{P}_n$ be the partition of $[a, b]$ into 2^n subintervals of equal length. Let m_i and M_i be the infimum and supremum of f on the ith subinterval $[x_{i-1}, x_i]$ of $\mathcal{P}_n$. Define step functions φ_n and ψ_n on $[a, b]$ by $\varphi_n = m_i$ and $\psi_n = M_i$ on (x_{i-1}, x_i), and $\varphi_n = \psi_n = f$ at the points of $\mathcal{P}_n$. Then $\varphi_n \le f \le \psi_n$. As in the proof of Proposition 16.1.2, $\int_a^b \psi_n = \mathcal{U}(\mathcal{P}_n, f)$ and $\int_a^b \varphi_n = \mathcal{L}(\mathcal{P}_n, f)$. By Corollary 16.2.5 and Theorem 16.2.3,

$$\mathcal{U}(\mathcal{P}_n, f) - \mathcal{L}(\mathcal{P}_n, f) = \int_{[a,b]} (\psi_n - \varphi_n).$$

Let $Q = \bigcup_n \mathcal{P}_n$, a countable set. Let $x \in [a, b]$ with $x \notin D \cup Q$, and let $\varepsilon > 0$. Since f is continuous at x, there is a $\delta > 0$ with $|f(s) - f(x)| < \varepsilon$ for all $s \in [a, b]$ with $|s - x| < \delta$. Choose N with $\frac{b-a}{2^N} < \delta$. For $n \ge N$, x lies in an open subinterval (x_{i-1}, x_i) of $\mathcal{P}_n$ of length less than δ. So $[x_{i-1}, x_i] \subset (x - \delta, x + \delta)$, and $f(x) - \varepsilon \le m_i \le M_i \le f(x) + \varepsilon$. So $0 \le \psi_n(x) - \varphi_n(x) \le 2\varepsilon$ for all $n \ge N$. Thus $\psi_n - \varphi_n \to 0$ at every point outside $D \cup Q$. The set $D \cup Q$ has measure zero. Also $|\psi_n - \varphi_n| \le 2K$. By Corollary 16.3.7,

$$\mathcal{U}(\mathcal{P}_n, f) - \mathcal{L}(\mathcal{P}_n, f) = \int_{[a,b]} (\psi_n - \varphi_n) \to 0.$$

So for every $\varepsilon > 0$ there is a partition $\mathcal{P}_n$ with $\mathcal{U}(\mathcal{P}_n, f) - \mathcal{L}(\mathcal{P}_n, f) < \varepsilon$. By Riemann's Criterion, $f \in \mathcal{R}[a, b]$. □

In words, a bounded function on $[a, b]$ is Riemann integrable exactly when it is continuous almost everywhere. Compare this with the Lebesgue integral. A bounded function on $[a, b]$ has a Lebesgue integral exactly when it is measurable (Theorem 16.3.1). Measurability is a much weaker condition than continuity almost everywhere.

Exercises

16.3.1. Show that the hypothesis $m(E) < \infty$ cannot be dropped from Theorem 16.3.6. Let $f_n = \frac{1}{n}\chi_{[0,n]}$ on $\mathbb{R}$. Show that each f_n is a simple function that vanishes outside a set of finite measure, that $|f_n| \le 1$, and that $f_n \to 0$ uniformly on $\mathbb{R}$. But $\int f_n = 1$ for every n.

16.3.2. Show that the bound M cannot be dropped from Theorem 16.3.6. Let $f_n = n\chi_{(0,1/n)}$ on $[0, 1]$. Show that $f_n \to 0$ at every point of $[0, 1]$, but $\int_{[0,1]} f_n = 1$ for every n. [Solution, p. 373]

16.3.3. Use Corollary 16.3.7 to find

$$\lim_{n\to\infty} \int_{[0,1]} \frac{1}{1+x^n}\,dm \quad \text{and} \quad \lim_{n\to\infty} \int_{[0,1]} \frac{nx}{1+n^2x^2}\,dm.$$

Check the second answer by computing the integral with the Fundamental Theorem of Calculus. [Hint, p. 373]

16.3.4. Let f be a bounded measurable function on a set E of finite measure. Suppose $\int_A f = 0$ for every measurable set $A \subset E$. Prove that $f = 0$ a.e. on E. [Hint, p. 374]

16.3.5. Use Lebesgue's Theorem to show that Thomae's function (Example 6.3.2), restricted to $[0,1]$, is Riemann integrable. Then use Theorem 16.3.3 and Theorem 16.3.5 (2) to show that its integral is 0. [Hint, p. 374]

16.3.6. Let C be the Cantor set (Chapter 5). Show that χ_C is discontinuous exactly at the points of C. Conclude that $\chi_C \in \mathcal{R}[0,1]$, and find $\int_0^1 \chi_C$. [Hint, p. 374]

16.3.7. Use Lebesgue's Theorem to give new proofs of two earlier results. First, if $f, g \in \mathcal{R}[a,b]$, then $fg \in \mathcal{R}[a,b]$ (Theorem 9.2.11). Second, if $f \in \mathcal{R}[a,b]$ and h is continuous on $\mathbb{R}$, then $h \circ f \in \mathcal{R}[a,b]$ (Theorem 9.2.3). [Hint, p. 374]

16.3.8. Let $f \in \mathcal{R}[a,b]$ and let $g : [a,b] \to \mathbb{R}$ be bounded with $g = f$ a.e. Show by example that g need not be Riemann integrable. Show that g is measurable and $\int_{[a,b]} g = \int_a^b f$. [Hint, p. 374]

16.4. The Integral of a Nonnegative Function

Now drop both restrictions, for nonnegative functions. A nonnegative measurable function may be unbounded, it may take the value ∞, and it may be nonzero on a set of infinite measure. Its integral is defined as the supremum of the integrals of the bounded functions below it that vanish outside a set of finite measure. This supremum may be ∞. The conventions $c \cdot \infty = \infty$ for $c > 0$ and $0 \cdot \infty = 0$ are used throughout.

Definition 16.4.1. If f is a nonnegative measurable function defined on a measurable set E, define

$$\int_E f = \sup_{h \le f} \int_E h,$$

where h ranges over the bounded measurable functions on E with $m(\{x : h(x) \neq 0\}) < \infty$. For such an h, let $E_h = \{x \in E : h(x) \neq 0\}$, and define $\int_E h = \int_{E_h} h$. If E' is any measurable set with $E_h \subset E' \subset E$ and $m(E') < \infty$, then $\int_{E'} h = \int_{E_h} h + \int_{E' \setminus E_h} h = \int_{E_h} h$ by Theorem 16.3.5 (6) and (5), since $h = 0$ on $E' \setminus E_h$. So $\int_E h$ can be computed on any such E'. The value $\int_E f$ lies in $[0, \infty]$.

If f is bounded and $m(E) < \infty$, this agrees with Definition 16.3.2. Indeed, $h = f$ is allowed, and every allowed $h \le f$ has $\int_E h \le \int_E f$ by Theorem 16.3.5 (3).

Theorem 16.4.2. *If f and g are nonnegative measurable functions and E is a measurable set, then the following are true.*

(1) $\int_E cf = c \int_E f$ *for all* $c > 0$.
(2) $\int_E (f + g) = \int_E f + \int_E g$.
(3) If $f \le g$ *a.e., then* $\int_E f \le \int_E g$.

Proof. In this proof, h, k and ℓ always denote bounded measurable functions on E that vanish outside a set of finite measure.

(1) Well, an application of Theorem 16.3.5 (1) gives

$$\int_E cf = \sup_{ch \le cf} \int_E ch = c \sup_{h \le f} \int_E h = c \int_E f.$$

Here ch ranges over all allowed functions as h does, since $c \neq 0$. So (1) holds.

(2) The function $f + g$ is measurable. The proof of Theorem 15.1.4 works for functions with values in $[0, \infty]$. First, let $h \le f$ and $k \le g$. Then $h+k$ is bounded and measurable, it vanishes outside $E_h \cup E_k$, which has finite measure, and $h + k \le f + g$. Computing all three integrals on $E_h \cup E_k$, Theorem 16.3.5 (1) gives

$$\int_E h + \int_E k = \int_E (h + k) \le \int_E (f + g).$$

Take the supremum over h, and then over k. This gives $\int_E f + \int_E g \le \int_E (f + g)$.

For the reverse inequality, let $\ell \le f + g$. Split ℓ into a part below f and a part below g. Let $h = \min\{f, \ell\}$ and $k = \ell - h$. Since $\min\{0, \ell\} \le h \le \ell$, the function h is real-valued, and $|h| \le |\ell|$. So h and k are measurable by Theorems 15.1.4 and 15.1.5. At a point where $\ell \le f$, $h = \ell$ and $k = 0 \le g$. At a point where $\ell > f$, f is finite, $h = f$ and $k = \ell - f \le g$. So $h \le f$ and $k \le g$ everywhere. Also $0 \le k \le |\ell|$. Both h and k vanish where ℓ vanishes. Hence

$$\int_E \ell = \int_E h + \int_E k \le \int_E f + \int_E g.$$

Taking the supremum over ℓ gives $\int_E (f + g) \le \int_E f + \int_E g$.

(3) Let $h \le f$. The set $B = \{h > g\}$ is contained in $\{f > g\}$, so $m(B) = 0$. Let $\tilde{h} = h$ on $E \setminus B$ and $\tilde{h} = 0$ on B. Then $\tilde{h}$ is bounded and measurable, it vanishes outside E_h, and $\tilde{h} \le g$ everywhere, since $g \ge 0$. Also $\tilde{h} = h$ a.e. So, computing on E_h, Theorem 16.3.5 (2) gives the equality below, and Definition 16.4.1 gives the inequality.

$$\int_E h = \int_E \tilde{h} \le \int_E g.$$

Taking the supremum over h gives $\int_E f \le \int_E g$. □

Theorem 16.4.3. *Let f be a nonnegative measurable function defined on a measurable set E. Then $\int_E f = 0$ if and only if $f = 0$ a.e. on E.*

Proof. See Exercise 16.4.1. □

As in Lemma 16.3.4, the integral over a subset $A \subset E$ is the integral of $f\chi_A$ over E. It follows that sets of measure zero can be ignored.

Lemma 16.4.4. *Let f be a nonnegative measurable function on a measurable set E, and let $A \subset E$ be measurable.*

(1) $\int_A f = \int_E f\chi_A$.
(2) If $A \subset B \subset E$ with B measurable, then $\int_A f \le \int_B f$.
(3) If $m(A) = 0$, then $\int_A f = 0$ and $\int_E f = \int_{E \setminus A} f$.

Proof. (1) The proof is the same as for Lemma 16.3.4, with suprema in place of infima. An allowed $h \le f$ on A, extended by 0 to E, is an allowed function below

$f\chi_A$ with the same integral. Conversely, if $h \le f\chi_A$ on E, then $h \le 0$ on $E \setminus A$. So $\int_E h \le \int_A h$, and the restriction of h to A is allowed and below f.

(2) Since $f\chi_A \le f\chi_B$, this follows from (1) and Theorem 16.4.2 (3).

(3) Since $f\chi_A = 0$ a.e. and $f\chi_{E\setminus A} = f$ a.e., this follows from (1) and Theorem 16.4.2 (3). □

Lemma 16.4.5 (Fatou's Lemma)**.** *If $\{f_n\}$ is a sequence of nonnegative measurable functions and $f_n \to f$ a.e. on a set E, where $f \ge 0$ on E, then*

$$\int_E f \le \liminf \int_E f_n.$$

PROOF. The function f equals the measurable function $\liminf f_n$ a.e., so it is measurable by Theorems 15.1.5 and 15.1.8. Let $B \subset E$ be a set of measure zero off which $f_n \to f$. By Lemma 16.4.4 (3), removing B from E does not change $\int_E f$ or any $\int_E f_n$. So it is enough to treat the case where $f_n(x) \to f(x)$ for every $x \in E$.

Let h be a bounded measurable function with $h \le f$ that vanishes outside a set $E' \subset E$ of finite measure. Say $|h| \le K$. It is enough to show $\int_E h \le \liminf \int_E f_n$. Then the supremum over h gives the result.

For each n, let $h_n = \min\{h, f_n\}$. This is measurable, $-K \le h_n \le h \le K$, and $h_n = 0$ off E'. Let $x \in E'$. If $f(x) > h(x)$, then $f_n(x) > h(x)$ for large n, so $h_n(x) = h(x)$ for large n. If $f(x) = h(x)$, then $h_n(x) = \min\{h(x), f_n(x)\} \to h(x)$. Either way $h_n \to h$ on E'. By the Bounded Convergence Theorem on E',

$$\int_E h = \int_{E'} h = \lim_{n\to\infty} \int_{E'} h_n.$$

Each h_n is an allowed function with $h_n \le f_n$. So $\int_{E'} h_n = \int_E h_n \le \int_E f_n$ by Definition 16.4.1. Thus $\int_E h \le \liminf \int_E f_n$. □

Example 16.4.6. The $\le$ in $\int_E f \le \liminf \int_E f_n$ in the statement of Fatou's Lemma is important. Consider the sequence $\{f_n\}$ defined by $f_n(x) = 1$ if $n \le x < n+1$ with $f_n(x) = 0$ otherwise. It satisfies the hypotheses of Fatou's Lemma and $\int_{[0,\infty)} f < \liminf \int_{[0,\infty)} f_n$. Here $f = 0$ and every $\int_{[0,\infty)} f_n = 1$. See Figure 16.2 and Exercise 16.4.2.

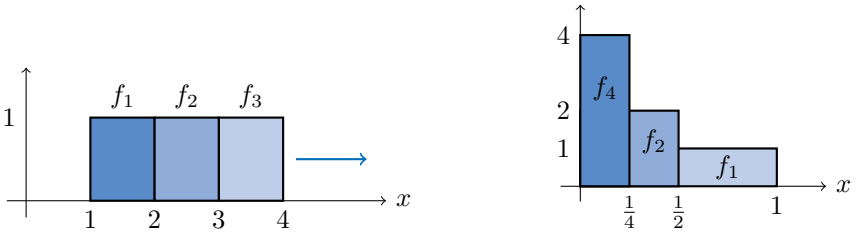


FIGURE 16.2. Two ways the limit of the integrals can miss the integral of the limit. On the left, $f_n = \chi_{[n,n+1)}$ slides off to infinity. On the right, $f_n = n\chi_{(0,1/n)}$ gets taller and thinner. In both, every f_n has integral 1, but $f_n \to 0$ at every point.

Theorem 16.4.7 (Monotone Convergence Theorem)**.** *Let $\{f_n\}$ be an increasing sequence of nonnegative measurable functions, and let $f_n \to f$ a.e. on a set E, where $f \geq 0$ on E. Then*

$$\int_E f = \lim_{n\to\infty} \int_E f_n.$$

PROOF. By Fatou's Lemma (Lemma 16.4.5),

$$\int_E f \leq \liminf \int_E f_n.$$

But for each $n \in \mathbb{N}$, $f_n \leq f$ a.e., since $\{f_n\}$ increases to f a.e., and so $\int_E f_n \leq \int_E f$ by Theorem 16.4.2 (3). Thus

$$\limsup \int_E f_n \leq \int_E f.$$

Hence

$$\int_E f = \lim_{n\to\infty} \int_E f_n. \qquad \square$$

Example 16.4.8. The Monotone Convergence Theorem does not always hold for decreasing sequences. For example, the sequence of functions $\{f_n\}$ defined by $f_n(x) = 0$ for $x < n$ and $f_n(x) = 1$ for $x \geq n$ is decreasing but the conclusion of the Monotone Convergence Theorem does not hold. Here $f_n \to 0$ everywhere, but $\int_{\mathbb{R}} f_n = \infty$ for every n. See Exercise 16.4.3.

Corollary 16.4.9. *Let $\{u_n\}$ be a sequence of nonnegative measurable functions, and let $f = \sum_{n=1}^{\infty} u_n$ on a set E. Then*

$$\int_E f = \sum_{n=1}^{\infty} \int_E u_n.$$

PROOF. Define $s_k = \sum_{n=1}^{k} u_n$. Now $\{s_k\}$ is an increasing sequence and converges to f. So by the Monotone Convergence Theorem and Theorem 16.4.2 (2),

$$\int_E f = \lim_{k\to\infty} \int_E s_k = \lim_{k\to\infty} \int_E \sum_{n=1}^{k} u_n = \lim_{k\to\infty} \sum_{n=1}^{k} \int_E u_n = \sum_{n=1}^{\infty} \int_E u_n. \qquad \square$$

Compare Corollary 16.4.9 with Corollary 11.2.19, its Riemann version. There the series had to converge uniformly. Here the terms only need to be nonnegative.

Theorem 16.4.10. *Let f be a nonnegative measurable function and $\{E_i\}$ a disjoint sequence of measurable sets. Let $E = \bigcup_{i\in\mathbb{N}} E_i$. Then*

$$\int_E f = \sum_{i=1}^{\infty} \int_{E_i} f.$$

PROOF. Let $u_i = f \cdot \chi_{E_i}$. Then $f \cdot \chi_E = \sum_{i=1}^{\infty} u_i$, since the sets E_i are disjoint. Thus by Corollary 16.4.9 and Lemma 16.4.4 (1),

$$\int_E f = \sum_{i=1}^{\infty} \int_E u_i = \sum_{i=1}^{\infty} \int_{E_i} f. \qquad \square$$

So if $f \geq 0$ is measurable, then $\nu(A) = \int_A f$ defines a measure ν on the measurable sets (Definition 13.3.1). Compare Exercise 16.2.5.

Definition 16.4.11. A nonnegative measurable function f is called *integrable* over the measurable set E if

$$\int_E f < \infty.$$

Lemma 16.4.12. *If a nonnegative measurable function f is integrable over E, then $f < \infty$ a.e. on E.*

PROOF. Let $A = \{f = \infty\}$, which is measurable by Theorem 15.1.1, and for $k \in \mathbb{N}$ let $A_k = A \cap [-k, k]$. For each $n \in \mathbb{N}$, the function $n\chi_{A_k}$ is bounded and measurable, vanishes outside A_k, and is at most f. So $n\, m(A_k) \leq \int_E f$ for every n. Since $\int_E f < \infty$, $m(A_k) = 0$. So $m(A) = 0$, since $A = \bigcup_k A_k$. □

Theorem 16.4.13. *Let f and g be two nonnegative measurable functions. If f is integrable over E and $g \leq f$ on E, then g is also integrable over E and*

$$\int_E (f - g) = \int_E f - \int_E g.$$

(Where $f = g = \infty$, set $f - g = 0$. By Lemma 16.4.12, this happens only on a set of measure zero.)

PROOF. With this convention, $f = (f - g) + g$ at every point of E. The function $f - g$ is nonnegative. It is measurable, since it is measurable on the set where $f < \infty$ and the rest of E has measure zero. By Theorem 16.4.2 (2), $\int_E f = \int_E (f - g) + \int_E g$. Since $\int_E f < \infty$, $\int_E (f - g) + \int_E g < \infty$, and thus $\int_E g < \infty$. So $\int_E (f - g) < \infty$ too, and subtracting $\int_E g$ from both sides gives the result. □

Theorem 16.4.14. *Let f be a nonnegative function which is integrable over a set E. Then for all $\varepsilon > 0$, there exists a $\delta > 0$ such that for all measurable sets $A \subset E$ with $m(A) < \delta$,*

$$\int_A f < \varepsilon.$$

PROOF. Let $\varepsilon > 0$. For $N \in \mathbb{N}$, let $f_N = \min\{f, N\}$. Each f_N is measurable and bounded by N, the sequence $\{f_N\}$ increases, and $f_N \to f$ at every point of E, including points where $f = \infty$. By the Monotone Convergence Theorem, $\int_E f_N \to \int_E f$. Since $\int_E f < \infty$, there is an N with $\int_E f - \int_E f_N < \frac{\varepsilon}{2}$. The function $f - f_N$ is defined everywhere, since f_N is finite. By Theorem 16.4.13, $\int_E (f - f_N) = \int_E f - \int_E f_N < \frac{\varepsilon}{2}$.

Let $\delta = \frac{\varepsilon}{2N}$, and let $A \subset E$ be measurable with $m(A) < \delta$. On A, $f = (f - f_N) + f_N$, and $f_N \leq N$. By Theorem 16.4.2 (2), Lemma 16.4.4 (2) and Theorem 16.3.5 (5),

$$\int_A f = \int_A (f - f_N) + \int_A f_N \leq \int_E (f - f_N) + N\, m(A) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon. \qquad \square$$

This property is called the *absolute continuity* of the integral. It returns in Chapter 17, where it shows that an integral $F(x) = \int_{[a,x]} f$ is absolutely continuous (Theorem 17.5.8).

For nonnegative functions, the Lebesgue integral also includes the improper Riemann integral of Section 9.7.

Proposition 16.4.15. *Let $a \in \mathbb{R}$ and $b \in \mathbb{R} \cup \{\infty\}$ with $a < b$. Let $f \in \mathcal{R}_{\text{loc}}[a,b)$ with $f \geq 0$. Then f is measurable on $[a,b)$, and*

$$\int_{[a,b)} f = \lim_{s \to b^-} \int_a^s f,$$

where both sides may be ∞. So f is integrable over $[a,b)$ if and only if the improper integral $\int_a^b f$ converges, and then the two integrals are equal.

PROOF. Since $f \geq 0$, $\int_a^s f$ increases with s. So the limit exists in $[0,\infty]$ and equals $\lim_{n\to\infty} \int_a^{s_n} f$ for any increasing sequence $s_n \to b$ with $a < s_n < b$. Fix such a sequence. By Theorem 16.3.3, f is measurable on each $[a,s_n]$, with $\int_{[a,s_n]} f = \int_a^{s_n} f$. So $\{f > \alpha\} = \bigcup_n \{x \in [a,s_n] : f(x) > \alpha\}$ is measurable, and f is measurable. The functions $f\chi_{[a,s_n]}$ are nonnegative and measurable, they increase with n, and they converge to f on $[a,b)$. By the Monotone Convergence Theorem and Lemma 16.4.4 (1),

$$\int_{[a,b)} f = \lim_{n\to\infty} \int_{[a,s_n]} f = \lim_{n\to\infty} \int_a^{s_n} f. \qquad \square$$

The same holds on $(a,b]$, with the same proof. For example, $\int_{[0,\infty)} e^{-x}\, dm(x) = \lim_{s\to\infty}(1 - e^{-s}) = 1$, and by the p-Test (Theorem 9.7.7), $\frac{1}{x^p}$ is integrable over $[1,\infty)$ exactly when $p > 1$.

Exercises

16.4.1. Let f be a nonnegative measurable function defined on E. Prove that $\int_E f = 0$ if and only if $f = 0$ a.e. on E. This is Theorem 16.4.3.
[Hint, p. 374]

16.4.2. Show that the sequence $\{f_n\}$ defined by $f_n(x) = 1$ if $n \leq x < n+1$ with $f_n(x) = 0$ otherwise satisfies the hypotheses of Fatou's Lemma and that $\int_{[0,\infty)} f < \liminf \int_{[0,\infty)} f_n$.
[Solution, p. 374]

16.4.3. Show that the Monotone Convergence Theorem does not always hold for a decreasing sequence of functions by considering the sequence of functions $\{f_n\}$ defined by $f_n(x) = 0$ for $x < n$ and $f_n(x) = 1$ for $x \geq n$.
[Solution, p. 375]

16.4.4. Let f be a nonnegative integrable function over $\mathbb{R}$. Use the Monotone Convergence Theorem to prove the function $F(x) = \int_{(-\infty,x]} f$ is continuous.
[Hint, p. 375]

16.4.5. (Chebyshev's Inequality) Let f be a nonnegative measurable function on E, and let $\alpha > 0$. Prove that $m(\{f \geq \alpha\}) \leq \frac{1}{\alpha} \int_E f$.
[Hint, p. 375]

16.4.6. Let f be a nonnegative measurable function on E. For $n \in \mathbb{N}$, let $\varphi_n = \frac{1}{2^n}\lfloor 2^n f \rfloor$ on $\{f < n\} \cap [-n,n]$, let $\varphi_n = n$ on $\{f \geq n\} \cap [-n,n]$, and let $\varphi_n = 0$ elsewhere. Prove that each φ_n is a simple function that vanishes outside a set of finite measure, that $\{\varphi_n\}$ increases, and that $\varphi_n \to f$ on E. Conclude that $\int_E f = \lim_{n\to\infty} \int_E \varphi_n$.
[Hint, p. 375]

16.4.7. Apply Corollary 16.4.9 to $u_n(x) = x^n(1-x)$ on $[0,1]$ to show that $\sum_{n=0}^{\infty} \frac{1}{(n+1)(n+2)} = 1$. [Hint, p. 375]

16.4.8. Let f be a measurable function on E with $f(x) > 0$ for all $x \in E$, and suppose $m(E) > 0$. Prove that $\int_E f > 0$. [Hint, p. 375]

16.4.9. For which $p > 0$ is $\frac{1}{x^p}$ integrable over $(0,1]$? For which $p > 0$ is it integrable over $(0,\infty)$? [Hint, p. 375]

16.5. The General Lebesgue Integral

A general measurable function is split into its positive and negative parts, and each part is integrated separately. This is the same idea as absolute convergence of series (Exercise 10.3.4). The integral exists when both parts have finite integrals.

Definition 16.5.1. Let f be an extended real-valued function. Define $f^+(x) = \max\{f(x), 0\}$ and define $f^-(x) = \max\{-f(x), 0\}$. These are the *positive part* and the *negative part* of f. Both are nonnegative.

Lemma 16.5.2. *If f is measurable, f^+ and f^- are measurable. Also, $f = f^+ - f^-$ and $|f| = f^+ + f^-$.*

Proof. See Exercise 16.5.1. □

Definition 16.5.3. A measurable function f is said to be *integrable* over E if f^+ and f^- are both integrable over E. In this case, define

$$\int_E f = \int_E f^+ - \int_E f^-.$$

The set of functions that are integrable over E is denoted $L^1(E)$. For an interval, write $L^1[a,b]$ for $L^1([a,b])$. A function in $L^1(E)$ is also called *Lebesgue integrable* over E.

By Lemma 16.5.2 and Theorem 16.4.2 (2), $\int_E |f| = \int_E f^+ + \int_E f^-$. So f is integrable if and only if f is measurable and $\int_E |f| < \infty$. If f is bounded and $m(E) < \infty$, this definition agrees with Definition 16.3.2, by Theorem 16.3.5 (1). If $f \geq 0$, then $f^- = 0$ and it agrees with Definition 16.4.1.

Theorem 16.5.4. *Let f and g be integrable over E. Then each of the following hold.*

(1) cf is integrable over E and $\int_E cf = c\int_E f$ for all $c \in \mathbb{R}$.
(2) $f+g$ is integrable over E and $\int_E (f+g) = \int_E f + \int_E g$.
(3) If $f \leq g$ a.e., then $\int_E f \leq \int_E g$.
(4) If A and B are disjoint measurable sets contained in E, then $\int_{A\cup B} f = \int_A f + \int_B f$.

Proof. (1) This part is an exercise. See Exercise 16.5.2.

(2) By Lemma 16.4.12, applied to $|f|$ and $|g|$, the functions f and g are finite a.e. Changing f and g on a set of measure zero does not change any of the integrals involved, by Theorem 16.4.2 (3). So assume f and g are real-valued. Notice that

if f_1 and f_2 are nonnegative integrable functions on E with $f = f_1 - f_2$, then $f^+ + f_2 = f^- + f_1$. Thus, by Theorem 16.4.2 (2), $\int_E f^+ + \int_E f_2 = \int_E f^- + \int_E f_1$. Therefore, since all four integrals are finite, $\int_E f = \int_E f^+ - \int_E f^- = \int_E f_1 - \int_E f_2$.

Now, if f and g are integrable, so are $f^+ + g^+$ and $f^- + g^-$, and since $f + g = f^+ + g^+ - (f^- + g^-)$,

$$\begin{aligned}\int_E (f+g) &= \int_E (f^+ + g^+) - \int_E (f^- + g^-) \\ &= \int_E f^+ + \int_E g^+ - \int_E f^- - \int_E g^- \\ &= \int_E f + \int_E g.\end{aligned}$$

Here $f + g$ is integrable, since it is measurable and $|f+g| \le |f| + |g|$, so $\int_E |f+g| \le \int_E |f| + \int_E |g| < \infty$.

(3) This follows from (1) and (2). The function $g - f$ is integrable by (1) and (2), and $(g-f)^- = 0$ a.e. So $\int_E (g-f)^- = 0$, and $\int_E g - \int_E f = \int_E (g-f) = \int_E (g-f)^+ \ge 0$.

(4) Now, applying Lemma 16.4.4 (1) to f^+ and f^-, and using Lemma 15.1.10 and (2),

$$\begin{aligned}\int_{A\cup B} f &= \int_E f \cdot \chi_{A\cup B} \\ &= \int_E f \cdot (\chi_A + \chi_B) \\ &= \int_E f \cdot \chi_A + \int_E f \cdot \chi_B \\ &= \int_A f + \int_B f.\end{aligned}$$

□

Remark 16.5.5. Note that $f + g$ is not defined at points where $f = \infty$ and $g = -\infty$. However, since f and g are integrable, the set of such points must have measure zero, by Lemma 16.4.12. So $\int_E (f+g)$ still exists.

Proof Technique (Building an Integral in Stages). The integral in this chapter was defined in four stages, for simple functions, bounded functions on sets of finite measure, nonnegative functions, and integrable functions. Each basic property was proved once at each stage, in Theorems 16.2.3, 16.3.5, 16.4.2 and 16.5.4. A supremum or infimum carries a property from simple functions to bounded ones, and from bounded functions to nonnegative ones. The decomposition $f = f^+ - f^-$ carries it to integrable functions. To prove a new fact about the Lebesgue integral, follow the same path. Prove it for simple functions, where it is usually a finite computation. Then pass it up the stages. Another route skips the middle stage. Choose simple functions φ_n that increase to $f \ge 0$ (Exercise 16.4.6) and apply the Monotone Convergence Theorem.

Theorem 16.5.6 (Lebesgue Dominated Convergence Theorem)**.** *Let g be integrable over E and let $\{f_n\}$ be a sequence of measurable functions such that $|f_n| \le g$ on E.*

If $f_n \to f$ a.e. on E, then

$$\int_E f = \lim_{n\to\infty} \int_E f_n.$$

PROOF. Each f_n is integrable, since $|f_n| \le g$ and Theorem 16.4.13 applies to f_n^+ and f_n^-. The function f is measurable, since it equals $\limsup f_n$ a.e. Also $|f| \le g$ a.e., so f is integrable. By Lemma 16.4.12, $g < \infty$ a.e. Removing a set of measure zero from E changes none of the integrals. So assume that g is finite and that $f_n \to f$ at every point of E.

Then $g + f_n$ and $g - f_n$ are nonnegative measurable functions, and they converge to $g + f$ and $g - f$. Apply Fatou's Lemma to $g + f_n$. Using Theorem 16.5.4 (2),

$$\int_E g + \int_E f = \int_E (g+f) \le \liminf \int_E (g+f_n) = \int_E g + \liminf \int_E f_n.$$

Since $\int_E g$ is finite, $\int_E f \le \liminf \int_E f_n$. Now apply Fatou's Lemma to $g - f_n$.

$$\int_E g - \int_E f = \int_E (g-f) \le \liminf \int_E (g-f_n) = \int_E g - \limsup \int_E f_n.$$

So $\limsup \int_E f_n \le \int_E f$. Together,

$$\limsup \int_E f_n \le \int_E f \le \liminf \int_E f_n,$$

so $\lim_{n\to\infty} \int_E f_n$ exists and equals $\int_E f$. □

The Bounded Convergence Theorem is the special case where $m(E) < \infty$ and g is the constant M. In both examples of Figure 16.2, no integrable g lies above every f_n. Any such g would be at least 1 on $[1, \infty)$ in the first example, and at least $\frac{1}{2x}$ on $(0, 1)$ in the second.

Theorem 16.5.7. *Let $\{g_n\}$ be a sequence of integrable functions with $g_n \to g$ a.e. on E, where g is an integrable function. Let $\{f_n\}$ be a sequence of measurable functions such that $|f_n| \le g_n$ and $f_n \to f$ a.e. on E. If*

$$\int_E g = \lim_{n\to\infty} \int_E g_n,$$

then

$$\int_E f = \lim_{n\to\infty} \int_E f_n.$$

PROOF. See Exercise 16.5.4 (a). □

Proof Technique (Choosing a Convergence Theorem). To compute $\lim \int_E f_n$, first find the pointwise limit f, a.e. Then look for a theorem that moves the limit inside the integral.

- If $f_n \ge 0$ and $\{f_n\}$ increases, use the Monotone Convergence Theorem. No integrability is needed, and the answer may be ∞.
- If $|f_n| \le g$ for one integrable g, use the Dominated Convergence Theorem. On a set of finite measure, a constant g works when the f_n are uniformly bounded. This is the Bounded Convergence Theorem.

- If only an inequality is needed, Fatou's Lemma may be enough.

If none of these applies, check whether the conclusion is even true. The examples in Figure 16.2 fail all three tests, and the conclusion fails for them. Sometimes a change of variables or an integration by parts produces a sequence that one of the theorems does handle.

Example 16.5.8. The Bounded Convergence Theorem, Monotone Convergence Theorem and the Lebesgue Dominated Convergence Theorem can be used to exactly evaluate limits of integrals of sequences of functions without evaluating the integral of the function itself. For example,

$$\lim_{n\to\infty} \int_{[1,\infty)} e^{-x} \sin^n x \, dm(x)$$

can be evaluated without evaluating the actual integrals. (Exercise 16.5.6 gives two more.)

Set $f_n(x) = e^{-x} \sin^n x$. Define $g(x) = e^{-x}$, which is integrable on $[1,\infty)$ by Proposition 16.4.15. Since $|\sin^n x| \leq 1$ for all $n \in \mathbb{N}$ and for all $x \in [1,\infty)$, $|f_n(x)| \leq g(x)$ for all $n \in \mathbb{N}$ and for all $x \in [1,\infty)$. So the Lebesgue Dominated Convergence Theorem can be applied. If $|\sin x| < 1$, then $\sin^n x \to 0$. If $\sin x = 1$, then $\sin^n x = 1$ for every n. If $\sin x = -1$, then $\sin^n x$ alternates between -1 and 1 and has no limit. The last two cases happen only at the countably many points $x = \frac{(2k-1)\pi}{2}$, $k \in \mathbb{N}$. So $f_n \to 0$ a.e. on $[1,\infty)$, and

$$\lim_{n\to\infty} \int_{[1,\infty)} e^{-x} \sin^n x \, dm(x) = \int_{[1,\infty)} 0 \, dm(x) = 0.$$

The improper Riemann integral and the Lebesgue integral differ for functions that change sign. In Example 9.7.11, the improper integral $\int_\pi^\infty \frac{\sin x}{x}\, dx$ converges, but $\int_\pi^\infty \frac{|\sin x|}{x}\, dx$ diverges. By Proposition 16.4.15, $\int_{[\pi,\infty)} \frac{|\sin x|}{x}\, dm(x) = \infty$. So $\frac{\sin x}{x}$ is not Lebesgue integrable over $[\pi,\infty)$. The improper integral converges only because positive and negative parts cancel, the same way a conditionally convergent series does. The Lebesgue integral requires absolute convergence. When $\int_a^b |f|$ converges as an improper integral, the two integrals agree (Exercise 16.5.8).

Exercises

16.5.1. Prove that if f is measurable, so are f^+ and f^- and that $f = f^+ - f^-$ and $|f| = f^+ + f^-$. [Hint, p. 375]

16.5.2. Let $c \in \mathbb{R}$. Prove that if f is integrable over E, then so is cf and $\int_E cf = c\int_E f$. [Solution, p. 375]

16.5.3. Let f be integrable over E. Prove that $\left|\int_E f\right| \leq \int_E |f|$. [Hint, p. 376]

16.5.4. (a) Let $\{g_n\}$ be a sequence of integrable functions with $g_n \to g$ a.e. on E, where g is integrable. Let $\{f_n\}$ be a sequence of measurable functions such that $|f_n| \leq g_n$ and $f_n \to f$ a.e. on E. Prove that if $\int_E g = \lim_{n\to\infty} \int_E g_n$, then $\int_E f = \lim_{n\to\infty} \int_E f_n$. This is Theorem 16.5.7.

(b) Under the same hypotheses, prove that $\int_E |f_n - f| \to 0$.

[Hint, p. 376]

16.5.5. Let $\{f_n\}$ be a sequence of measurable functions on a set E such that $\{f_n\}$ converges to an integrable function f a.e. on E. Prove $\int_E |f - f_n| \to 0$ if and only if $\int_E |f_n| \to \int_E |f|$. [Hint, p. 376]

16.5.6. Evaluate each of the following.

(a) $\lim_{n\to\infty} \int_{[0,1]} \frac{nx^{n-1}}{x+1}\, dm$

(b) $\lim_{n\to\infty} \int_{[1,\infty)} \frac{\sin^3(x/n)}{x^3}\, dm$

[Hint, p. 376]

16.5.7. Evaluate $\lim_{n\to\infty} \int_{[0,\infty)} e^{-x} \cos(x/n)\, dm(x)$ explicitly and state the names of any theorems used to do so. [Hint, p. 376]

16.5.8. Let $f \in \mathcal{R}_{\text{loc}}[a, b)$, and suppose the improper integral $\int_a^b |f|$ converges. Prove that $f \in L^1[a, b)$ and $\int_{[a,b)} f = \int_a^b f$. [Hint, p. 376]

16.5.9. Let $f \in L^1[a, b]$, and let $F(x) = \int_{[a,x]} f$ for $x \in [a, b]$. Use the Dominated Convergence Theorem to prove that F is continuous. Then use Theorem 16.4.14 to prove that F is uniformly continuous.

Notes

Bernhard Riemann defined his integral in 1854, in the paper on trigonometric series that qualified him to lecture at Göttingen. It was published in 1868 [**59**]. Gaston Darboux introduced upper and lower sums in 1875 [**19**]. Henri Lebesgue defined his integral in his 1902 thesis [**42**], where he also proved the Bounded Convergence Theorem. Lebesgue's Theorem on Riemann integrability appears in his 1904 lectures [**43**]. Cesare Arzelà had proved the bounded convergence theorem for Riemann integrals in 1885 [**3**], with a much harder proof. Pierre Fatou proved his lemma in his 1906 thesis on trigonometric series and power series [**30**]. The Monotone Convergence Theorem is due to Lebesgue, and in the general form given here to Beppo Levi in 1906 [**46**]. Lebesgue proved the Dominated Convergence Theorem in 1908 [**44**].

CHAPTER 17

Differentiation and Integration

The Fundamental Theorem of Calculus in Section 9.3 has two halves. If f is continuous, then $\int_a^x f$ has derivative f. If F' is Riemann integrable, then $\int_a^b F' = F(b) - F(a)$. This chapter proves versions of both halves for the Lebesgue integral. The first half holds for every integrable f, almost everywhere. The second half needs the right class of functions F. It is not enough for F to be continuous and differentiable almost everywhere. The right class is the absolutely continuous functions.

Section 17.1, which is marked with a star, proves the Vitali Covering Lemma, a tool for choosing disjoint intervals. Section 17.2 uses it to show that monotone functions are differentiable almost everywhere. Section 17.3 returns to functions of bounded variation, which are differences of monotone functions. Section 17.4 differentiates integrals. Section 17.5 introduces absolute continuity and proves the Fundamental Theorem of Calculus for the Lebesgue integral. Section 17.6, also starred, builds the Cantor function. It is continuous and increasing, and its derivative is 0 almost everywhere, but it is not constant. Section 17.7, also starred, builds Volterra's function. It is differentiable everywhere with a bounded derivative that is not Riemann integrable.

This chapter uses a few results from earlier chapters. They are limits superior and inferior (Section 3.6), the Fundamental Theorem of Calculus and the Mean Value Theorem for Integrals (Chapter 9), the definitions in Section 8.1, outer measure (Chapter 14), and the integral and the convergence theorems (Chapter 16). Section 17.6 uses the Cantor set (Chapter 5). Section 17.7 uses Lebesgue's Theorem (Theorem 9.2.9).

17.1. The Vitali Covering Lemma*

This section can be skipped. Later sections use the statement of the Vitali Covering Lemma, Lemma 17.1.2, but not its proof.

To show that a set is small, cover it by intervals with small total length. Sometimes the opposite is needed. Many intervals around the points of a set E are given, and the task is to pick finitely many *disjoint* ones that cover almost all of E. This is possible when each point of E has arbitrarily short intervals around it.

Definition 17.1.1. Let $\mathcal{I}$ be a collection of intervals. Then $\mathcal{I}$ covers a set E *in the sense of Vitali* if for all $\varepsilon > 0$ and for all $x \in E$, there exists an $I \in \mathcal{I}$ such that $x \in I$ and $\ell(I) < \varepsilon$. The intervals may be open, half open, or closed, but must not be degenerate intervals consisting of only one point. Such a collection is called a *Vitali covering* of E.

Lemma 17.1.2 (Vitali Covering Lemma)**.** *Let E be a set with $m^*(E) < \infty$ and $\mathcal{I}$ a collection of intervals that cover E in the sense of Vitali. Then for all $\varepsilon > 0$, there exists a finite disjoint collection $\{I_1, \dots, I_N\}$ of intervals in $\mathcal{I}$ such that*

$$m^*\left(E \setminus \bigcup_{n=1}^{N} I_n\right) < \varepsilon.$$

PROOF. The proof has three steps. The first step makes two reductions. Replace each interval of $\mathcal{I}$ by its closure. This changes no lengths. It changes a union of finitely many intervals by finitely many points, which does not change outer measure. If the closures are disjoint, so are the original intervals. So assume every interval in $\mathcal{I}$ is closed. Next, by Theorem 14.1.10, there is an open set $U \supset E$ with $m(U) < \infty$. Discard the intervals of $\mathcal{I}$ that are not contained in U. What is left still covers E in the sense of Vitali. Indeed, each $x \in E$ has a neighborhood $(x - r, x + r) \subset U$, and every interval of $\mathcal{I}$ that contains x and is shorter than r lies in it.

The second step chooses disjoint intervals one at a time, taking a long one at each stage. Choose any $I_1 \in \mathcal{I}$. Suppose $I_1, \dots, I_n$ have been chosen and are pairwise disjoint. If $E \subset I_1 \cup \cdots \cup I_n$, stop. The lemma holds with $N = n$. Otherwise, let

$$k_n = \sup\{\ell(I) : I \in \mathcal{I} \text{ and } I \text{ is disjoint from } I_1, \dots, I_n\}.$$

This set of lengths is not empty. A point $x \in E$ outside the closed set $I_1 \cup \cdots \cup I_n$ has a neighborhood that misses this set, and some interval of $\mathcal{I}$ that contains x lies in that neighborhood. Also $k_n \le m(U) < \infty$. Choose $I_{n+1} \in \mathcal{I}$, disjoint from $I_1, \dots, I_n$, with $\ell(I_{n+1}) > \frac{1}{2}k_n$.

If the process never stops, it produces pairwise disjoint intervals $I_1, I_2, \dots$ in U. By countable additivity, $\sum_n \ell(I_n) \le m(U) < \infty$. In particular, $\ell(I_n) \to 0$. Let $\varepsilon > 0$. Choose N with $\sum_{n=N+1}^{\infty} \ell(I_n) < \frac{\varepsilon}{5}$.

The third step shows that the part of E left uncovered is small. Let $R = E \setminus \bigcup_{n=1}^{N} I_n$. For each n, let J_n be the closed interval with the same midpoint as I_n and five times its length. The claim is that $R \subset \bigcup_{n=N+1}^{\infty} J_n$. Then

$$m^*(R) \le \sum_{n=N+1}^{\infty} \ell(J_n) = 5 \sum_{n=N+1}^{\infty} \ell(I_n) < \varepsilon.$$

Let $x \in R$. As in the second step, there is an $I \in \mathcal{I}$ with $x \in I$ that is disjoint from $I_1, \dots, I_N$. The interval I must meet some I_n. Otherwise I would be disjoint from $I_1, \dots, I_n$ for every n, and so $\ell(I) \le k_n < 2\ell(I_{n+1})$ for every n. This would force $\ell(I) = 0$, but I is not degenerate. Let n_0 be the smallest index with $I \cap I_{n_0} \neq \emptyset$. Then $n_0 > N$. Also I is disjoint from $I_1, \dots, I_{n_0-1}$, so $\ell(I) \le k_{n_0-1} < 2\ell(I_{n_0})$. Let z be the midpoint of I_{n_0}. Since $x \in I$ and I meets I_{n_0},

$$|x - z| \le \ell(I) + \frac{1}{2}\ell(I_{n_0}) < \frac{5}{2}\ell(I_{n_0}).$$

So $x \in J_{n_0}$. See Figure 17.1. □

Proof Technique (Greedy Choices). The proof of Lemma 17.1.2 picks intervals one at a time, each time taking one that is nearly as long as possible

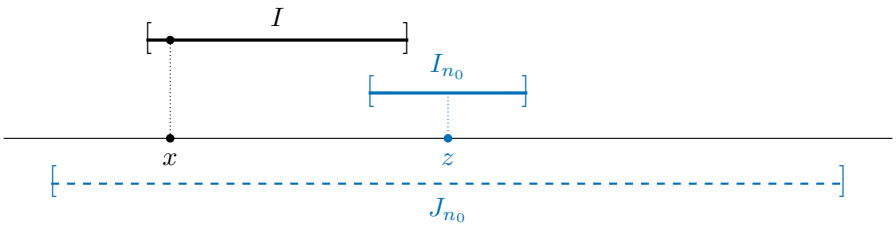


FIGURE 17.1. The key step in the proof of the Vitali Covering Lemma. The interval I contains x, meets I_{n_0}, and is less than twice as long as I_{n_0}. So x lies in the interval J_{n_0} that has the same midpoint z as I_{n_0} and five times its length.

among those still allowed. This is a *greedy* construction. The longest allowed interval may not exist, since a supremum need not be attained. So the proof settles for one longer than half the supremum. The payoff comes at the end. Any interval that was passed over is not much longer than one that was chosen, so a fixed enlargement of the chosen intervals covers everything that was missed.

Exercises

17.1.1. Let $E = [0,1]$ and $\mathcal{I} = \left\{\left[0, \frac{2}{3}\right], \left[\frac{1}{3}, 1\right]\right\}$. Show that $\mathcal{I}$ covers E, but every disjoint subcollection of $\mathcal{I}$ misses a part of E of measure at least $\frac{1}{3}$. Which hypothesis of Lemma 17.1.2 fails? [Solution, p. 376]

17.1.2. Under the hypotheses of Lemma 17.1.2, prove that there is a countable disjoint collection $\{I_n\}$ of intervals in $\mathcal{I}$ with $m^*\left(E \setminus \bigcup_n I_n\right) = 0$. [Hint, p. 376]

17.1.3. Let $E \subset \mathbb{R}$. Show that the closed intervals with rational endpoints that meet E form a Vitali covering of E. [Hint, p. 377]

17.2. Differentiating Monotone Functions

A continuous function can fail to be differentiable at every point, as the Weierstrass function in Section 11.3 shows. A monotone function cannot be that bad. This section proves that a monotone function is differentiable almost everywhere. To handle points where the derivative might not exist, the proof uses four one-sided substitutes that always exist.

Definition 17.2.1. Let f be a real-valued function defined on an interval containing x. Define the *(Dini) derivatives* of f as follows:

$$D^+f(x) = \limsup_{h \to 0^+} \frac{f(x+h) - f(x)}{h}, \qquad D^-f(x) = \limsup_{h \to 0^+} \frac{f(x) - f(x-h)}{h},$$

$$D_+f(x) = \liminf_{h \to 0^+} \frac{f(x+h) - f(x)}{h}, \qquad D_-f(x) = \liminf_{h \to 0^+} \frac{f(x) - f(x-h)}{h}.$$

By Definition 7.1.1, f is differentiable at an interior point x exactly when $D^+f(x) = D_+f(x) = D^-f(x) = D_-f(x) \neq \pm\infty$, and then $f'(x)$ is the common value of the derivatives at x. If $D^+f(x) = D_+f(x)$, f has a right-hand derivative, denoted $f'(x+)$. If $D^-f(x) = D_-f(x)$, f has a left-hand derivative, denoted $f'(x-)$.

Here $\limsup_{h\to 0^+} g(h)$ means $\inf_{\delta>0}\sup\{g(h) : 0 < h < \delta\}$, and $\liminf_{h\to 0^+} g(h)$ means $\sup_{\delta>0}\inf\{g(h) : 0 < h < \delta\}$. As δ decreases, $\sup\{g(h) : 0 < h < \delta\}$ decreases. So, as with sequences in Section 3.6, these always exist as values in $[-\infty, \infty]$. Also $D_+f(x) \leq D^+f(x)$ and $D_-f(x) \leq D^-f(x)$. The right derivatives D^+f and D_+f look only to the right of x, and the left derivatives D^-f and D_-f only to the left. For example, if $f(x) = |x|$, then $D^+f(0) = D_+f(0) = 1$ and $D^-f(0) = D_-f(0) = -1$. If f is increasing, every difference quotient is at least 0, so all four Dini derivatives are at least 0.

Theorem 17.2.2. *If f is continuous on $[a, b]$ and one of its derivatives is everywhere nonnegative on (a, b), then f is nondecreasing on $[a, b]$. That is, $f(x) \leq f(y)$ for $x \leq y$.*

Proof. See Exercise 17.2.1, which treats D_+f. The other three cases are similar. □

Theorem 17.2.3 (Lebesgue's Differentiation Theorem)**.** *Let f be an increasing real-valued function on the interval $[a, b]$. Then f is differentiable almost everywhere, the derivative f' is measurable and*

$$\int_{[a,b]} f' \leq f(b) - f(a).$$

(Here f' is defined only almost everywhere. Its values on a set of measure zero do not affect measurability or the integral.)

Proof. The proof has three steps. The first step shows that $D^+f \leq D_-f$ a.e. Let E be the set of $x \in (a, b)$ with $D^+f(x) > D_-f(x)$. For rational numbers $u > v$, let

$$E_{u,v} = \{x \in (a, b) : D^+f(x) > u > v > D_-f(x)\}.$$

Between two extended real numbers there is a rational number. So E is the union of the countably many sets $E_{u,v}$, and it is enough to show $m^*(E_{u,v}) = 0$. Fix $u > v$ and let $s = m^*(E_{u,v}) \leq b - a$. Since f is increasing, $D_-f \geq 0$. So $v > 0$. Let $\varepsilon > 0$, and choose an open set $U \supset E_{u,v}$ with $m(U) < s + \varepsilon$ (Theorem 14.1.10). Replace U by $U \cap (a, b)$ if needed, so that $U \subset (a, b)$.

Let $x \in E_{u,v}$. Since $D_-f(x) < v$, there are arbitrarily small $h > 0$ with $f(x) - f(x - h) < vh$. If h is small, $[x - h, x] \subset U$. These intervals form a Vitali covering of $E_{u,v}$. By the Vitali Covering Lemma, there are disjoint intervals $I_n = [x_n - h_n, x_n]$, $n = 1, \ldots, N$, of this kind with $m^*\big(E_{u,v} \setminus \bigcup_n I_n\big) < \varepsilon$. Let A be the set of points of $E_{u,v}$ in the interiors $(x_n - h_n, x_n)$. Removing the finitely many endpoints does not change outer measure, so by subadditivity $m^*(A) > s - \varepsilon$. The intervals I_n are disjoint and lie in U, so $\sum_n h_n \leq m(U)$. Thus

$$\sum_{n=1}^{N} \big[f(x_n) - f(x_n - h_n)\big] < v \sum_{n=1}^{N} h_n \leq v\, m(U) < v(s + \varepsilon). \tag{17.2.1}$$

Now let $y \in A$. Since $D^+f(y) > u$, there are arbitrarily small $k > 0$ with $f(y+k) - f(y) > uk$. If k is small, $[y, y+k]$ lies inside the interval $(x_n - h_n, x_n)$ that contains y. These intervals form a Vitali covering of A. By the Vitali Covering Lemma again, there are disjoint intervals $[y_j, y_j + k_j]$, $j = 1, \dots, M$, of this kind that cover all of A except a set of outer measure less than ε. So $\sum_j k_j > m^*(A) - \varepsilon > s - 2\varepsilon$, and

$$\sum_{j=1}^{M} \big[f(y_j + k_j) - f(y_j)\big] > u \sum_{j=1}^{M} k_j > u(s - 2\varepsilon). \tag{17.2.2}$$

Each interval $[y_j, y_j + k_j]$ lies inside some I_n. Fix n, and list the intervals $[y_j, y_j + k_j]$ inside I_n from left to right. They are disjoint and f is increasing, so the sum of the increases of f over them is at most the increase $f(x_n) - f(x_n - h_n)$ over all of I_n. Summing over n, and using (17.2.1) and (17.2.2),

$$u(s - 2\varepsilon) < \sum_{j=1}^{M} \big[f(y_j + k_j) - f(y_j)\big] \le \sum_{n=1}^{N} \big[f(x_n) - f(x_n - h_n)\big] < v(s + \varepsilon).$$

This holds for every $\varepsilon > 0$, so $us \le vs$. Since $u > v$ and $s \ge 0$, $s = 0$.

The second step shows that all four Dini derivatives agree a.e. The function $\tilde{f}(x) = -f(-x)$ is increasing on $[-b, -a]$. Directly from the definitions, $D^+\tilde{f}(x) = D^-f(-x)$ and $D_-\tilde{f}(x) = D_+f(-x)$. So the first step, applied to $\tilde{f}$, gives $D^-f \le D_+f$ a.e. Together with the first step,

$$D^+f \le D_-f \le D^-f \le D_+f \le D^+f \quad \text{a.e.}$$

So the four Dini derivatives are equal a.e. Call the common value $g(x)$. It lies in $[0, \infty]$.

The third step shows that f' is measurable and that $\int_{[a,b]} f' \le f(b) - f(a)$. Extend f to $[a, b+1]$ by $f(x) = f(b)$ for $x > b$. For $n \in \mathbb{N}$ and $x \in [a, b]$, let

$$g_n(x) = n\left[f\left(x + \frac{1}{n}\right) - f(x)\right].$$

Each g_n is measurable, since increasing functions are measurable (Example 15.1.3). Also $g_n \ge 0$, and $g_n \to g$ a.e. So g is measurable, and by Fatou's Lemma,

$$\int_{[a,b]} g \le \liminf \int_{[a,b]} g_n.$$

Monotone functions are Riemann integrable (Theorem 9.2.2), so the integrals of g_n are Riemann integrals by Theorem 16.3.3. The upper and lower sums of $x \mapsto f\left(x + \frac{1}{n}\right)$ for a partition $\mathcal{P}$ of $[a, b]$ are the upper and lower sums of f for the shifted partition $\mathcal{P} + \frac{1}{n}$. So $\int_a^b f\left(x + \frac{1}{n}\right) dx = \int_{a+1/n}^{b+1/n} f$. By Theorem 9.2.13,

$$\int_{[a,b]} g_n = n\int_b^{b+1/n} f - n\int_a^{a+1/n} f = f(b) - n\int_a^{a+1/n} f \le f(b) - f(a),$$

since $f = f(b)$ on $\left[b, b + \frac{1}{n}\right]$ and $f \ge f(a)$ on $\left[a, a + \frac{1}{n}\right]$. Thus $\int_{[a,b]} g \le f(b) - f(a) < \infty$. By Lemma 16.4.12, $g < \infty$ a.e. So at almost every x, the four Dini derivatives are equal and finite. There f is differentiable and $f'(x) = g(x)$. □

The inequality can be strict. Let $f = \chi_{[1/2,1]}$ on $[0,1]$. Then $f' = 0$ except at $\frac{1}{2}$, so $\int_{[0,1]} f' = 0 < 1 = f(1) - f(0)$. The jump at $\frac{1}{2}$ counts in $f(1) - f(0)$ but not in the integral. Section 17.6 gives a *continuous* increasing function with the same behavior.

Proof Technique (Splitting a Bad Set Using Rationals). To show that an inequality $A(x) \le B(x)$ holds almost everywhere, look at the bad set $\{A > B\}$. Whenever $A(x) > B(x)$, there are rational numbers $u > v$ with $A(x) > u > v > B(x)$. So the bad set is a countable union of the sets $\{A > u > v > B\}$, and it is enough to show each of these has measure zero. Fixing u and v turns a comparison between two varying quantities into two comparisons with fixed numbers, which are much easier to use. The first step of the proof of Theorem 17.2.3 works this way. The same trick, with one rational number, appeared in the proof of Theorem 15.1.4.

Exercises

17.2.1. If f is continuous on $[a,b]$ and D_+f is everywhere nonnegative on (a,b), prove f is nondecreasing on $[a,b]$. Then prove Theorem 17.2.2 when $D^-f \ge 0$, and explain why this proves Theorem 17.2.2 for all four Dini derivatives. [Hint, p. 377]

17.2.2. Find the four Dini derivatives at 0 of $f(x) = x \sin \frac{1}{x}$, $f(0) = 0$, and of the Dirichlet function $\chi_{\mathbb{Q}}$. [Hint, p. 377]

17.2.3. Prove that $D^+(-f)(x) = -D_+f(x)$ and $D^-(-f)(x) = -D_-f(x)$. [Hint, p. 377]

17.2.4. Let f be decreasing on $[a,b]$. Prove that f is differentiable a.e. and $\int_{[a,b]} f' \ge f(b) - f(a)$. [Hint, p. 377]

17.2.5. Let $\{r_n\}$ be an enumeration of the rational numbers in $(0,1)$, and let $f(x) = \sum_{n:\, r_n \le x} 2^{-n}$ for $x \in [0,1]$. Prove that f is increasing and discontinuous at every rational number in $(0,1)$. Conclude that f is differentiable almost everywhere. [Hint, p. 377]

17.3. Functions of Bounded Variation Revisited

Functions of bounded variation were studied in Chapter 8, which was optional. Here are the definitions again. For a partition $\mathcal{P} = \{x_0, \ldots, x_k\}$ of $[a,b]$, let $V(\mathcal{P}) = \sum_{i=1}^{k} |f(x_i) - f(x_{i-1})|$. The total variation of f on $[a,b]$ is $V_f[a,b] = \sup_{\mathcal{P}} V(\mathcal{P})$, and f is of bounded variation, written $f \in BV[a,b]$, if $V_f[a,b] < \infty$ (Definition 8.1.5). The Jordan Decomposition Theorem (Theorem 8.2.5) says $f \in BV[a,b]$ exactly when f is a difference of two increasing functions. This section gives a second proof, using the variation up and the variation down separately. Then Theorem 17.2.3 applies to functions of bounded variation.

Definition 17.3.1. Let f be a real-valued function on $[a,b]$, and let $\mathcal{P} = \{x_0, x_1, \ldots, x_k\}$ be a partition of $[a,b]$. Define

$$p(\mathcal{P}) = \sum_{i=1}^{k} [f(x_i) - f(x_{i-1})]^+, \qquad n(\mathcal{P}) = \sum_{i=1}^{k} [f(x_i) - f(x_{i-1})]^-,$$

and

$$V(\mathcal{P}) = p(\mathcal{P}) + n(\mathcal{P}) = \sum_{i=1}^{k} |f(x_i) - f(x_{i-1})|,$$

where r^+ is r if $r \geq 0$ and 0 if $r \leq 0$ and set $r^- = |r| - r^+$. Note $f(b) - f(a) = p(\mathcal{P}) - n(\mathcal{P})$.

Set

$$P_f[a,b] = \sup_{\mathcal{P}} p(\mathcal{P}), \qquad N_f[a,b] = \sup_{\mathcal{P}} n(\mathcal{P}), \qquad \text{and} \qquad V_f[a,b] = \sup_{\mathcal{P}} V(\mathcal{P}),$$

where the supremum is taken over all possible partitions of $[a,b]$.

Notice $P_f[a,b] \leq V_f[a,b] \leq P_f[a,b] + N_f[a,b]$. The quantities $P_f[a,b]$, $N_f[a,b]$ and $V_f[a,b]$ are the *positive*, *negative*, and *total variations* of f over $[a,b]$. If $V_f[a,b] < \infty$, f is of bounded variation over $[a,b]$, written $f \in BV[a,b]$.

Lemma 17.3.2. *If $f \in BV[a,b]$, then*

$$V_f[a,b] = P_f[a,b] + N_f[a,b] \quad \textit{and} \quad f(b) - f(a) = P_f[a,b] - N_f[a,b].$$

Proof. For any partition $\mathcal{P}$ of $[a,b]$,

$$p(\mathcal{P}) = n(\mathcal{P}) + f(b) - f(a).$$

Both $p(\mathcal{P})$ and $n(\mathcal{P})$ are at most $V(\mathcal{P}) \leq V_f[a,b] < \infty$. Taking the supremum over $\mathcal{P}$ on both sides gives

$$P_f[a,b] = N_f[a,b] + f(b) - f(a).$$

Also,

$$V(\mathcal{P}) = p(\mathcal{P}) + n(\mathcal{P}) = 2p(\mathcal{P}) - \big(f(b) - f(a)\big).$$

So

$$V_f[a,b] = 2P_f[a,b] - \big(f(b) - f(a)\big) = P_f[a,b] + N_f[a,b]. \qquad \square$$

Theorem 17.3.3 (Jordan Decomposition Theorem)**.** *A function f is of bounded variation on $[a,b]$ if and only if it is the difference of two monotone real-valued functions on $[a,b]$.*

Proof. ($\Rightarrow$) Let f be of bounded variation on $[a,b]$ and set $g(x) = P_f[a,x]$ and $h(x) = N_f[a,x]$, with $g(a) = h(a) = 0$. Then g and h are monotone increasing functions (Exercise 17.3.1) which are real-valued, since $0 \leq P_f[a,x] \leq V_f[a,x] \leq V_f[a,b] < \infty$ and $0 \leq N_f[a,x] \leq V_f[a,x] \leq V_f[a,b] < \infty$, by Theorem 8.2.1. But by Lemma 17.3.2 on $[a,x]$, $f(x) = g(x) - h(x) + f(a)$. Since $h - f(a)$ is a monotone function, f is the difference of two monotone functions.

($\Leftarrow$) If $f = g - h$ on $[a,b]$ with g and h increasing, then for a partition $a = x_0 < x_1 < \cdots < x_n = b$ of $[a,b]$,

$$\begin{aligned}\sum_{i=1}^{n} |f(x_i) - f(x_{i-1})| &\le \sum_{i=1}^{n} [g(x_i) - g(x_{i-1})] + \sum_{i=1}^{n} [h(x_i) - h(x_{i-1})] \\ &= g(b) - g(a) + h(b) - h(a).\end{aligned}$$

Hence $V_f[a,b] \le g(b) - g(a) + h(b) - h(a) < \infty$. □

This is a second proof of Theorem 8.2.5. The decomposition there was $f = V_f - (V_f - f)$, where $V_f(x) = V_f[a,x]$. Exercise 17.3.3 shows how the two decompositions are related.

Corollary 17.3.4. *If f is of bounded variation on $[a,b]$, then $f'(x)$ exists a.e. on $[a,b]$, and f' is integrable over $[a,b]$.*

PROOF. This is an immediate consequence of Theorem 17.3.3 and Theorem 17.2.3. Write $f = g - h$ with g and h increasing. Then $f' = g' - h'$ wherever g and h are both differentiable, which is almost everywhere. Also $|f'| \le g' + h'$ a.e., and $\int_{[a,b]} (g' + h') \le g(b) - g(a) + h(b) - h(a) < \infty$. □

Theorem 17.3.5. *Let $\{f_n\}$ be a sequence of functions on $[a,b]$ with $f_n \to f$ on $[a,b]$. Then*

$$V_f[a,b] \le \liminf V_{f_n}[a,b].$$

PROOF. See Exercise 17.3.2. □

Exercises

17.3.1. Let $f : [a,b] \to \mathbb{R}$ and $a \le x < y \le b$. Prove that $P_f[a,x] \le P_f[a,y]$ and $N_f[a,x] \le N_f[a,y]$. [Hint, p. 378]

17.3.2. Let $\{f_n\}$ be a sequence of functions on $[a,b]$ with $f_n \to f$ on $[a,b]$. Prove $V_f[a,b] \le \liminf V_{f_n}[a,b]$. Then show that the inequality can be strict. [Hint, p. 378]

17.3.3. Let $f \in BV[a,b]$. Prove that $P_f[a,x] = \frac{1}{2}\big(V_f[a,x] + f(x) - f(a)\big)$ and $N_f[a,x] = \frac{1}{2}\big(V_f[a,x] - f(x) + f(a)\big)$. [Hint, p. 378]

17.3.4. Find $P_f[0,2\pi]$, $N_f[0,2\pi]$ and $V_f[0,2\pi]$ for $f(x) = \sin x$. [Hint, p. 378]

17.3.5. Let f be of bounded variation on $[a,b]$. Prove $V_f[a,b] \ge \int_{[a,b]} |f'|$. [Hint, p. 378]

17.4. Differentiating an Integral

The first half of the Fundamental Theorem of Calculus says that the derivative of an integral is the integrand. For Riemann integrals this needs continuity at the point (Theorem 9.3.7). For Lebesgue integrals it holds almost everywhere, for every integrable function.

Definition 17.4.1. If f is an integrable function on $[a,b]$, the function F defined on $[a,b]$ by

$$F(x) = \int_{[a,x]} f$$

is called the *indefinite integral* of f. More generally, a function of the form $F(x) = F(a) + \int_{[a,x]} f$, where $F(a)$ is a constant, is called an indefinite integral of f.

Lemma 17.4.2. *If f is integrable on $[a,b]$, then the indefinite integral F of f is a continuous function of bounded variation.*

PROOF. Let $\varepsilon > 0$. By Theorem 16.4.14, applied to $|f|$, there is a $\delta > 0$ such that $\int_A |f| < \varepsilon$ for every measurable set $A \subset [a,b]$ with $m(A) < \delta$. Let $x, y \in [a,b]$ with $x < y < x + \delta$. By Theorem 16.5.4 (4), $\int_{[a,y]} f = \int_{[a,x]} f + \int_{(x,y]} f$. So, by Exercise 16.5.3,

$$|F(y) - F(x)| = \left| \int_{(x,y]} f \right| \leq \int_{(x,y]} |f| < \varepsilon.$$

So F is continuous, and in fact uniformly continuous.

Now let $a = x_0 < x_1 < \cdots < x_n = b$ be a partition of $[a,b]$. Then

$$\begin{aligned} \sum_{i=1}^{n} |F(x_i) - F(x_{i-1})| &= \sum_{i=1}^{n} \left| \int_{(x_{i-1},x_i]} f \right| \\ &\leq \sum_{i=1}^{n} \int_{(x_{i-1},x_i]} |f| \\ &\leq \int_{[a,b]} |f| < \infty. \end{aligned}$$

Thus $V_F[a,b] \leq \int_{[a,b]} |f| < \infty$. □

Lemma 17.4.3. *If f is integrable on $[a,b]$ and*

$$\int_{[a,x]} f = 0$$

for all $x \in [a,b]$, then $f(x) = 0$ a.e. on $[a,b]$.

PROOF. The first goal is to show that $\int_A f = 0$ for every measurable set $A \subset [a,b]$. The proof starts with intervals and builds up. Integrals over sets of measure zero are 0, by Lemma 16.4.4 (3) applied to f^+ and f^-.

The first case is an interval. If $a \leq c < d \leq b$, then $\int_{(c,d)} f = \int_{[a,d]} f - \int_{[a,c]} f - \int_{\{d\}} f = 0$.

Open sets come next, since they are unions of intervals. Let O be open. Then $O \cap (a,b)$ is open, so by Theorem 4.1.12 it is the union of countably many disjoint open intervals $(c_k, d_k) \subset (a,b)$. Applying Theorem 16.4.10 to f^+ and to f^-, and subtracting,

$$\int_{O \cap (a,b)} f = \sum_k \int_{(c_k,d_k)} f = 0.$$

Every measurable set is nearly a G_δ set, so measurable sets come last. Let $A \subset [a,b]$ be measurable. By Theorem 14.3.1 (4), there is a G_δ set $G \supset A$ with $m(G \setminus A) = 0$. Write $G = \bigcap_k O_k$ with each O_k open. Replace O_k by $O_1 \cap \cdots \cap O_k$, so that $O_1 \supset O_2 \supset \cdots$. Then $f\chi_{O_k \cap (a,b)} \to f\chi_{G \cap (a,b)}$ at every point, and $|f\chi_{O_k \cap (a,b)}| \leq |f|$. By the Dominated Convergence Theorem,

$$\int_{G \cap (a,b)} f = \lim_{k \to \infty} \int_{O_k \cap (a,b)} f = 0.$$

The sets A and $G \cap (a,b)$ differ by a subset of $(G \setminus A) \cup \{a,b\}$, which has measure zero. So $\int_A f = 0$.

It remains to show that $f = 0$ a.e. Let $A = \{x \in [a,b] : f(x) > 0\}$. Then $\int_{[a,b]} f^+ = \int_A f = 0$, so $f^+ = 0$ a.e. by Theorem 16.4.3. Similarly, taking $A = \{f < 0\}$ gives $f^- = 0$ a.e. So $f = 0$ a.e. □

Proof Technique (From Intervals to All Measurable Sets). The proof of Lemma 17.4.3 proves a statement about every measurable set by climbing a ladder. Intervals come first, because the hypothesis is about intervals. Open sets are countable disjoint unions of intervals. G_δ sets are decreasing limits of open sets. Measurable sets are G_δ sets minus sets of measure zero. Each rung needs one tool, here countable additivity, the Dominated Convergence Theorem, and Theorem 14.3.1. The Good Sets Principle of Chapter 13 is another way to make the same climb.

Lemma 17.4.4. *If f is bounded and measurable on $[a,b]$ and*

$$F(x) = \int_{[a,x]} f + F(a),$$

then $F'(x) = f(x)$ a.e. on $[a,b]$.

Proof. By Lemma 17.4.2, F is continuous and of bounded variation. So by Corollary 17.3.4, $F'(x)$ exists for almost every $x \in [a,b]$. Let $|f| \leq K$. Extend f by 0 to $(b, b+1]$, so that the formula for F makes sense on $[a, b+1]$ and $F = F(b)$ on $[b, b+1]$. For $n \in \mathbb{N}$ and $x \in [a,b]$, let $h = \frac{1}{n}$ and

$$f_n(x) = \frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_{(x,x+h]} f.$$

Then $|f_n| \leq K$, and $f_n \to F'$ a.e. on $[a,b]$. The function $g = \limsup f_n$ is measurable, $|g| \leq K$, and $g = F'$ a.e. Let $c \in [a,b]$. By the Bounded Convergence Theorem (Corollary 16.3.7),

$$\int_{[a,c]} F' = \int_{[a,c]} g = \lim_{n \to \infty} \int_{[a,c]} f_n = \lim_{h \to 0^+} \frac{1}{h} \int_a^c \big(F(x+h) - F(x)\big)\, dx.$$

The last integral is a Riemann integral, since F is continuous. By u-Substitution with $\varphi(t) = t + h$ (Theorem 9.6.3) and Theorem 9.2.13,

$$\int_a^c \big(F(x+h) - F(x)\big)\, dx = \int_{a+h}^{c+h} F - \int_a^c F = \int_c^{c+h} F - \int_a^{a+h} F.$$

By the Mean Value Theorem for Integrals (Theorem 9.6.1), $\frac{1}{h}\int_c^{c+h} F = F(\xi_h)$ for some $\xi_h \in [c, c+h]$. Since F is continuous, this tends to $F(c)$. In the same way, $\frac{1}{h}\int_a^{a+h} F \to F(a)$. So

$$\int_{[a,c]} F' = F(c) - F(a) = \int_{[a,c]} f.$$

Hence $\int_{[a,c]}(F' - f) = 0$ for all $c \in [a,b]$. By Lemma 17.4.3, $F' = f$ a.e. on $[a,b]$. □

Theorem 17.4.5. *Let f be an integrable function on $[a,b]$ and suppose that*

$$F(x) = F(a) + \int_{[a,x]} f.$$

Then $F'(x) = f(x)$ a.e. on $[a,b]$.

Proof. First suppose $f \geq 0$. For $n \in \mathbb{N}$, let $f_n = \min\{f, n\}$. Split the integral into a bounded part and a remainder,

$$F(x) = F(a) + \int_{[a,x]} f_n + G_n(x), \quad \text{where } G_n(x) = \int_{[a,x]} (f - f_n).$$

Since $f - f_n \geq 0$, G_n is increasing. By Theorem 17.2.3, G_n' exists and is at least 0 almost everywhere. By Lemma 17.4.4, the derivative of $\int_{[a,x]} f_n$ is $f_n(x)$ almost everywhere. So $F' \geq f_n$ a.e. Discarding countably many sets of measure zero, one for each n, gives $F'(x) \geq f_n(x)$ for every n, for almost every x. Letting $n \to \infty$, $F' \geq f$ a.e. So

$$\int_{[a,b]} F' \geq \int_{[a,b]} f = F(b) - F(a).$$

On the other hand, F is increasing, since $f \geq 0$. So $\int_{[a,b]} F' \leq F(b) - F(a)$ by Theorem 17.2.3. Thus $\int_{[a,b]}(F' - f) = 0$. Since $F' - f \geq 0$ a.e., Theorem 16.4.3 gives $F' = f$ a.e.

For general f, write $f = f^+ - f^-$. Then $F(x) = F(a) + \int_{[a,x]} f^+ - \int_{[a,x]} f^-$. By the first case, the two integrals have derivatives f^+ and f^- almost everywhere. So $F' = f^+ - f^- = f$ a.e. □

Compare Theorem 9.3.7, which needs f to be continuous at x and then gives $F'(x) = f(x)$ at that point. Here f is only integrable, and the conclusion holds almost everywhere. It can fail at particular points. If $f = \chi_{[0,1/2]}$ on $[0,1]$, then $F(x) = \min\left\{x, \frac{1}{2}\right\}$, which is not differentiable at $\frac{1}{2}$.

Exercises

17.4.1. Let f be integrable on $[a,b]$. Prove that $\lim_{h\to 0^+} \frac{1}{h}\int_{[x,x+h]} f = f(x)$ for almost every $x \in [a,b)$. [Hint, p. 378]

17.4.2. (Lebesgue's Density Theorem, one-sided.) Let $E \subset [a,b]$ be measurable. Prove that $\lim_{h\to 0^+} \frac{m(E\cap[x,x+h])}{h}$ is 1 for almost every $x \in E$ and 0 for almost every $x \in [a,b) \setminus E$. [Hint, p. 378]

17.4.3. Let f be measurable on $[a,b]$ with $|f| \leq K$. Prove that its indefinite integral satisfies $|F(x) - F(y)| \leq K|x - y|$. [Hint, p. 378]

17.4.4. Let f be integrable on $[a,b]$. Suppose $\int_{[c,d]} f = 0$ whenever $c < d$ are rational numbers in $[a,b]$. Prove that $f = 0$ a.e. [Hint, p. 379]

17.4.5. Let $f = \chi_{\mathbb{Q}}$ on $[0,1]$. Find its indefinite integral F. Show that $F'(x) \neq f(x)$ for every rational x. [Hint, p. 379]

17.5. Absolute Continuity

Which functions F satisfy $F(x) = F(a) + \int_{[a,x]} F'$? Theorem 17.4.5 says this happens exactly when F is an indefinite integral of some integrable function. Being continuous and differentiable almost everywhere is not enough, as the Cantor function in Section 17.6 shows. The right condition strengthens uniform continuity. Uniform continuity controls the change of f over one short interval. Absolute continuity controls the total change over many short intervals.

Definition 17.5.1. A real-valued function f defined on $[a,b]$ is said to be *absolutely continuous* on $[a,b]$ if for all $\varepsilon > 0$, there exists a $\delta > 0$ such that

$$\sum_{i=1}^{n} |f(x_i') - f(x_i)| < \varepsilon$$

for every finite collection $\{(x_i, x_i')\}$ of nonoverlapping subintervals of $[a,b]$ with $\sum_{i=1}^{n} |x_i' - x_i| < \delta$. If f is absolutely continuous on $[a,b]$, write $f \in AC[a,b]$ or $f \in AC$.

Here *nonoverlapping* means the open intervals (x_i, x_i') are pairwise disjoint. Taking a single interval shows that an absolutely continuous function is uniformly continuous. Every Lipschitz function is absolutely continuous (Exercise 17.5.5). By Theorem 16.4.14, every indefinite integral is absolutely continuous (Exercise 17.5.2).

Lemma 17.5.2. *If f is absolutely continuous on $[a,b]$, then f is of bounded variation on $[a,b]$.*

Proof. Let $\delta > 0$ be as in Definition 17.5.1 for $\varepsilon = 1$. Choose $K \in \mathbb{N}$ with $\frac{b-a}{K} < \delta$, and let $c_j = a + j\frac{b-a}{K}$ for $j = 0, \ldots, K$. Let $\mathcal{P}$ be a partition of $[a,b]$, and let $\mathcal{P}^* = \mathcal{P} \cup \{c_0, \ldots, c_K\}$. By Lemma 8.1.7, $V(\mathcal{P}) \leq V(\mathcal{P}^*)$. For each j, the subintervals of $\mathcal{P}^*$ that lie in $[c_{j-1}, c_j]$ are nonoverlapping with total length less than δ. So their terms in $V(\mathcal{P}^*)$ add up to less than 1. There are K values of j, so $V(\mathcal{P}) \leq V(\mathcal{P}^*) < K$. Thus $V_f[a,b] \leq K$. □

Corollary 17.5.3. *If f is absolutely continuous on $[a,b]$, then f has a derivative a.e. on $[a,b]$, and f' is integrable over $[a,b]$.*

Proof. This is a direct consequence of Lemma 17.5.2 and Corollary 17.3.4. □

Example 17.5.4. The function

$$f(x) = \begin{cases} x \sin(1/x), & x \in (0,1], \\ 0, & x = 0, \end{cases}$$

is continuous but not absolutely continuous since it is not of bounded variation. See Exercise 8.1.4 and Lemma 17.5.2.

Lemma 17.5.5. *If f is absolutely continuous on $[a,b]$ and $f'(x) = 0$ a.e., then f is a constant function.*

PROOF. Let $c \in (a,b]$. The goal is to show $f(c) = f(a)$. Let $\varepsilon > 0$ and $\eta > 0$, and let $\delta > 0$ be as in Definition 17.5.1 for this ε. Let E be the set of $x \in (a,c)$ with $f'(x) = 0$. Then $m(E) = c - a$.

Let $x \in E$. Since $f'(x) = 0$, there are arbitrarily small $h > 0$ with $[x, x+h] \subset (a,c)$ and $|f(x+h) - f(x)| < \eta h$. These intervals form a Vitali covering of E. By the Vitali Covering Lemma, there are disjoint intervals $[x_1, y_1], \ldots, [x_n, y_n]$ of this kind with $m^*\big(E \setminus \bigcup_k [x_k, y_k]\big) < \delta$. Number them so that

$$y_0 = a < x_1 < y_1 < x_2 < y_2 < \cdots < x_n < y_n < c = x_{n+1}.$$

Since $m(E) = c - a$, the chosen intervals have total length $\sum_k (y_k - x_k) > c - a - \delta$. So the gaps $[y_k, x_{k+1}]$, $k = 0, \ldots, n$, have total length less than δ. By absolute continuity,

$$\sum_{k=0}^{n} |f(x_{k+1}) - f(y_k)| < \varepsilon.$$

On the chosen intervals,

$$\sum_{k=1}^{n} |f(y_k) - f(x_k)| < \eta \sum_{k=1}^{n} (y_k - x_k) \le \eta(c-a).$$

The difference $f(c) - f(a)$ is the sum of the changes of f over the gaps and over the chosen intervals. So

$$|f(c) - f(a)| \le \sum_{k=0}^{n} |f(x_{k+1}) - f(y_k)| + \sum_{k=1}^{n} |f(y_k) - f(x_k)| < \varepsilon + \eta(c-a).$$

Since ε and η were arbitrary, $f(c) = f(a)$. □

Definition 17.5.6. A monotone function f on $[a,b]$ is called *singular* if $f' = 0$ a.e. on $[a,b]$.

Theorem 17.5.7. *A monotone increasing function is the sum of an absolutely continuous function and a singular function.*

PROOF. See Exercise 17.5.1. □

Theorem 17.5.8. *A function F is an indefinite integral if and only if it is absolutely continuous.*

PROOF. ($\Rightarrow$) This direction is an exercise. See Exercise 17.5.2.

($\Leftarrow$) Suppose that F is absolutely continuous on $[a,b]$. Then F is of bounded variation and so $F(x) = F_1(x) - F_2(x)$, where F_1 and F_2 are monotone increasing functions. Hence $F'(x)$ exists a.e. and $|F'(x)| \le F_1'(x) + F_2'(x)$ a.e. Thus

$$\int_{[a,b]} |F'(x)|\, dm(x) \le F_1(b) + F_2(b) - F_1(a) - F_2(a) < \infty.$$

So $F'(x)$ is integrable.

Let

$$G(x) = \int_{[a,x]} F'.$$

Then G is an indefinite integral and so $G \in AC[a,b]$. Thus $f = F - G \in AC[a,b]$ by Exercise 17.5.3. Now $f'(x) = F'(x) - G'(x) = 0$ a.e., by Theorem 17.4.5, and so f is constant by Lemma 17.5.5. Since $G(a) = 0$, the constant is $F(a)$. Thus

$$F(x) = \int_{[a,x]} F' + F(a). \qquad \square$$

Corollary 17.5.9 (Fundamental Theorem of Calculus for Lebesgue Integrals)**.** *Every absolutely continuous function is the indefinite integral of its derivative. That is, if $F \in AC[a,b]$, then F' is integrable and $F(x) = F(a) + \int_{[a,x]} F'$ for all $x \in [a,b]$.*

Proof. This is a direct consequence of Theorem 17.5.8 and Theorem 17.4.5. $\square$

Together, Theorem 17.4.5 and Corollary 17.5.9 are the two halves of the Fundamental Theorem of Calculus for the Lebesgue integral. Compare Theorem 9.3.6. There F is differentiable everywhere and F' is assumed Riemann integrable. Here F need only be absolutely continuous, and F' is automatically integrable. For example, if F is differentiable everywhere on $[a,b]$ with $|F'| \le M$, then F is Lipschitz (Exercise 7.2.7), so it is absolutely continuous, and $\int_{[a,b]} F' = F(b) - F(a)$. This holds even if F' is not Riemann integrable.

Theorem 17.5.7 splits an increasing function into the part that is recovered by integrating its derivative and a singular part that the derivative cannot see. Jumps are singular, as in the example $\chi_{[1/2,1]}$ after Theorem 17.2.3. Section 17.6 gives a continuous singular function that is not constant.

Proof Technique (Showing a Function Is Absolutely Continuous). The definition is rarely used directly. Instead, try these.

- If $|f(x) - f(y)| \le M|x - y|$, then f is absolutely continuous with $\delta = \frac{\varepsilon}{M}$.
- If f is an indefinite integral, it is absolutely continuous by Theorem 17.5.8.
- Sums, differences and products of absolutely continuous functions are absolutely continuous (Exercise 17.5.3).

To show that f is not absolutely continuous, show that it is not of bounded variation, as in Example 17.5.4, or show that $f(b) - f(a) \ne \int_{[a,b]} f'$, as for the Cantor function.

Exercises

17.5.1. Prove a monotone increasing function is the sum of an absolutely continuous function and a singular function. [Hint, p. 379]

17.5.2. Prove if a function F is an indefinite integral, then it is absolutely continuous. [Hint, p. 379]

17.5.3. Let $f, g \in AC[a,b]$ and $c \in \mathbb{R}$. Prove that $f + g$, cf and fg are in $AC[a,b]$. [Hint, p. 379]

17.5.4. Let f be absolutely continuous on $[a,b]$. Prove $V_f[a,b] = \int_{[a,b]} |f'|$.
[Hint, p. 380]

17.5.5. A function f satisfies a Lipschitz condition on an interval $[a,b]$ if there is a constant M such that $|f(x) - f(y)| \le M|x-y|$ for all $x, y \in [a,b]$.
(a) Prove that a function f satisfying a Lipschitz condition is absolutely continuous.
(b) Prove that an absolutely continuous function f satisfies a Lipschitz condition if and only if $|f'|$ is bounded a.e. That is, there is an M with $|f'| \le M$ a.e.
(c) Let f be continuous on $[a,b]$. Prove that f satisfies a Lipschitz condition if one of its Dini derivatives is bounded on (a,b). Show by example that continuity is needed.
[Hint, p. 380]

17.5.6. Show that $f(x) = \sqrt{x}$ is absolutely continuous on $[0,1]$ but does not satisfy a Lipschitz condition there.
[Hint, p. 380]

17.5.7. Let $f \in AC[a,b]$ and let $Z \subset [a,b]$ with $m(Z) = 0$. Prove that $m^*(f[Z]) = 0$.
[Hint, p. 380]

17.5.8. (Integration by Parts) Let $f, g \in AC[a,b]$. Prove that $\int_{[a,b]} fg' = f(b)g(b) - f(a)g(a) - \int_{[a,b]} f'g$.
[Hint, p. 380]

17.6. The Cantor Function*

This section can be skipped. Nothing later depends on it.

This section builds a function ϕ on $[0,1]$ that is continuous and increasing, with $\phi(0) = 0$ and $\phi(1) = 1$, and whose derivative is 0 at every point outside the Cantor set C. Since $m(C) = 0$, ϕ is singular. So $\int_{[0,1]} \phi' = 0 < 1 = \phi(1) - \phi(0)$, and ϕ is not absolutely continuous. Continuity and differentiability almost everywhere are not enough for the Fundamental Theorem of Calculus.

The idea is to spread one unit of increase evenly over C. At stage n, spread it evenly over C_n. Recall from Definition 5.1.1 and Lemma 5.1.2 that C_n is the union of 2^n disjoint closed intervals $J_{n,1}, \ldots, J_{n,2^n}$, numbered from left to right, each of length 3^{-n}. So $m(C_n) = \left(\frac{2}{3}\right)^n$. Each $J_{n,i}$ contains exactly 2^{m-n} of the intervals of C_m for $m \ge n$, since each interval splits in two at each stage.

Lemma 17.6.1. *For $n \in \mathbb{N}_0$ and $x \in [0,1]$, let*

$$\phi_n(x) = \left(\frac{3}{2}\right)^n \int_{[0,x]} \chi_{C_n}.$$

Then the following hold.

(1) ϕ_n is continuous and increasing on $[0,1]$, with $\phi_n(0) = 0$ and $\phi_n(1) = 1$.
(2) If $J_{n,i} = [s,t]$, then $\phi_n(s) = \frac{i-1}{2^n}$ and $\phi_n(t) = \frac{i}{2^n}$.
(3) If $m \ge n$, then $\phi_m = \phi_n$ at the endpoints of each $J_{n,i}$ and on $[0,1] \setminus C_n$.
(4) If $m \ge n$, then $|\phi_m(x) - \phi_n(x)| \le \frac{1}{2^n}$ for all $x \in [0,1]$.

Proof. (1) By Lemma 17.4.2, ϕ_n is continuous. It is increasing since $\chi_{C_n} \ge 0$. Also $\phi_n(0) = 0$ and $\phi_n(1) = \left(\frac{3}{2}\right)^n m(C_n) = 1$.

(2) The set $C_n \cap [0,s]$ is $J_{n,1} \cup \cdots \cup J_{n,i-1}$, together with the point s. So $\phi_n(s) = \left(\frac{3}{2}\right)^n (i-1)3^{-n} = \frac{i-1}{2^n}$. The same argument with $J_{n,i}$ included gives $\phi_n(t) = \frac{i}{2^n}$.

(3) Let $m \geq n$. The set $C_m \cap [0,s]$ consists of the $(i-1)2^{m-n}$ intervals of C_m inside $J_{n,1}, \ldots, J_{n,i-1}$, together with the point s. So $\phi_m(s) = \left(\frac{3}{2}\right)^m (i-1)2^{m-n}3^{-m} = \frac{i-1}{2^n} = \phi_n(s)$. The same holds at t. Each interval of $[0,1] \setminus C_n$ lies between two consecutive intervals $J_{n,i}$ and $J_{n,i+1}$. Since $C_m \subset C_n$, both ϕ_m and ϕ_n are constant there, and they agree at the right endpoint of $J_{n,i}$. So they agree on the whole gap.

(4) Off C_n the two functions are equal, by (3). On $J_{n,i}$, both are increasing and, by (2) and (3), take values between $\frac{i-1}{2^n}$ and $\frac{i}{2^n}$. So they differ by at most $\frac{1}{2^n}$. □

Definition 17.6.2. By Lemma 17.6.1 (4) and the Cauchy Criterion (Theorem 11.2.5), the sequence $\{\phi_n\}$ converges uniformly on $[0,1]$. Its limit $\phi = \lim_{n\to\infty} \phi_n$ is the *Cantor function*.

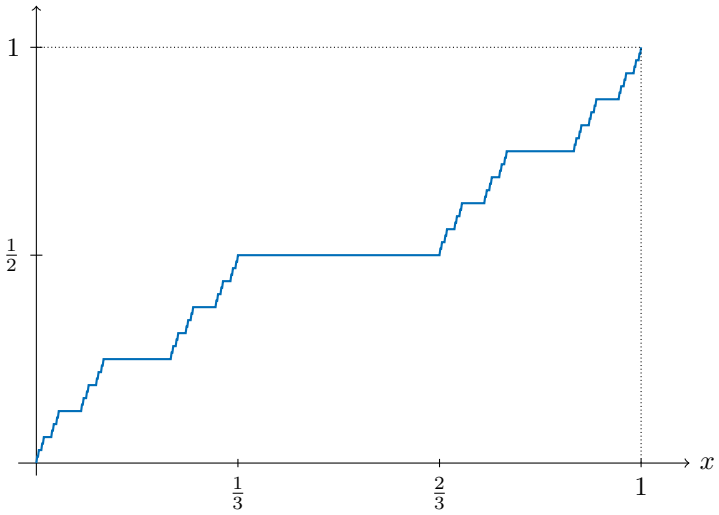


FIGURE 17.2. The Cantor function. The graph shown is ϕ_6, which equals ϕ off C_6 and is within $\frac{1}{64}$ of ϕ everywhere. The function is constant on every interval removed in building the Cantor set, yet it climbs from 0 to 1.

Theorem 17.6.3. *The Cantor function ϕ is continuous and increasing on $[0,1]$, with $\phi(0) = 0$ and $\phi(1) = 1$. It is constant on each interval of $[0,1] \setminus C$, and $\phi'(x) = 0$ for every $x \in (0,1) \setminus C$. So ϕ is singular, $\int_{[0,1]} \phi' = 0 < \phi(1) - \phi(0)$, and ϕ is not absolutely continuous.*

PROOF. The limit is continuous by Corollary 11.2.16, and increasing since each ϕ_n is. Also $\phi(0) = 0$ and $\phi(1) = 1$. Let $x \in (0,1) \setminus C$. Then $x \notin C_n$ for some n. The set C_n is closed, so x lies in an open interval O that misses C_n. By Lemma 17.6.1 (3), $\phi_m = \phi_n$ on O for all $m \geq n$, so $\phi = \phi_n$ on O. Since ϕ_n is constant on O, so is ϕ, and $\phi'(x) = 0$. The Cantor set has measure zero (Example 9.2.8), so $\phi' = 0$ a.e. If ϕ

were absolutely continuous, Corollary 17.5.9 would give $\phi(1) - \phi(0) = \int_{[0,1]} \phi' = 0$, which is false. □

The failure of absolute continuity can also be seen directly. The set C_n is a union of 2^n nonoverlapping intervals of total length $\left(\frac{2}{3}\right)^n$, which tends to 0. But ϕ increases by $\frac{1}{2^n}$ on each of them, by Lemma 17.6.1 (2) and (3), for a total of 1. So no δ works for $\varepsilon = 1$ in Definition 17.5.1.

Exercises

17.6.1. Prove that $\phi(1-x) = 1 - \phi(x)$ and $\phi\left(\frac{x}{3}\right) = \frac{1}{2}\phi(x)$ for $x \in [0,1]$. Conclude that $\int_{[0,1]} \phi = \frac{1}{2}$. [Hint, p. 380]

17.6.2. Let $x = \sum_{k=1}^{\infty} \frac{d_k}{3^k}$ with each $d_k \in \{0,2\}$. Prove that $x \in C$ and $\phi(x) = \sum_{k=1}^{\infty} \frac{d_k/2}{2^k}$. Conclude that ϕ maps C onto $[0,1]$. This gives another proof that C is uncountable (Theorem 5.2.8). [Hint, p. 381]

17.6.3. Let $\psi(x) = x + \phi(x)$ on $[0,1]$.

(a) Prove that ψ is continuous and strictly increasing, and maps $[0,1]$ onto $[0,2]$. Conclude that its inverse $\psi^{-1} : [0,2] \to [0,1]$ is continuous.

(b) Prove that ψ maps each interval of $[0,1] \setminus C$ onto an interval of the same length. Conclude that $m(\psi[C]) = 1$.

(c) Use Exercise 14.4.3, applied to $\psi[C] \cap [0,1)$ or to a translate of $\psi[C] \cap [1,2]$, to find a nonmeasurable set $A \subset \psi[C]$. Let $B = \psi^{-1}[A]$. Prove that B is measurable but is not a Borel set.

This gives an example for Remark 14.3.2. [Hint, p. 381]

17.7. Volterra's Function*

The book has two famous strange functions so far. The Weierstrass function (Section 11.3) is continuous but differentiable nowhere. The Cantor function (Section 17.6) is continuous and increasing with derivative 0 almost everywhere, but it is not constant. This section builds a third. Volterra's function is differentiable at every point, and its derivative is bounded, but its derivative is not Riemann integrable. So the Fundamental Theorem of Calculus for the Riemann integral (Theorem 9.3.6) cannot even be stated for it. The Lebesgue version (Corollary 17.5.9) still holds. This section can be skipped. Nothing later depends on it.

The construction has three ingredients. The first is a closed set with positive measure that contains no intervals. The second is the function $x^2 \sin \frac{1}{x}$, whose derivative oscillates near 0. The third is a way to glue copies of that function into the gaps of the set.

Example 17.7.1 (A Fat Cantor Set)**.** Start with $[0,1]$. At step 1, remove the open middle interval of length $\frac{1}{4}$. At step n, there are 2^{n-1} closed intervals left, all of the same length. Remove from the middle of each one an open interval of length 4^{-n}. Let K be the set of points that are never removed.

Before step n, the total length removed is $\sum_{j=1}^{n-1} 2^{j-1} 4^{-j} = \frac{1}{2} - 2^{-n}$. So each remaining interval has length $\ell_n = \frac{1/2 + 2^{-n}}{2^{n-1}} = 2^{-n} + 2^{1-2n}$, which is more than 4^{-n}.

So each step can be carried out. The set K is closed, since its complement is the union of the removed open intervals with $(-\infty, 0)$ and $(1, \infty)$. It contains no open interval, since every point of K lies in a remaining interval of length ℓ_n for every n, and $\ell_n \to 0$. The removed intervals are disjoint, and Lebesgue measure is countably additive (Theorem 14.2.12). So

$$m(K) = 1 - \sum_{n=1}^{\infty} 2^{n-1} 4^{-n} = 1 - \frac{1}{2} = \frac{1}{2}.$$

So K is a closed set with no intervals in it, like the Cantor set, but it has positive measure.

Next, let $g(x) = x^2 \sin \frac{1}{x}$ for $x \neq 0$ and $g(0) = 0$. By Exercise 17.7.1, g is differentiable on $\mathbb{R}$ with

$$g'(x) = 2x \sin \frac{1}{x} - \cos \frac{1}{x} \quad (x \neq 0), \qquad g'(0) = 0,$$

and $|g'(x)| \leq 3$ for $|x| \leq 1$. Also $|g(x)| \leq x^2$. At the points $x = \frac{1}{2k\pi}$, $g'(x) = -1$. These points come arbitrarily close to 0.

Definition 17.7.2. Let (α, β) be one of the open intervals removed in Example 17.7.1, and let $\ell = \beta - \alpha$. The function g' is continuous on $\left(0, \frac{\ell}{2}\right]$ and has zeros arbitrarily close to 0. Let c be the largest zero of g' in $\left(0, \frac{\ell}{2}\right]$. Define V on $[\alpha, \beta]$ by

$$V(x) = \begin{cases} g(x - \alpha), & \alpha \leq x \leq \alpha + c, \\ g(c), & \alpha + c \leq x \leq \beta - c, \\ g(\beta - x), & \beta - c \leq x \leq \beta. \end{cases}$$

Volterra's function $F : [0, 1] \to \mathbb{R}$ is $F(x) = 0$ for $x \in K$, and $F(x) = V(x)$ on each removed interval, with the V built for that interval.

The largest zero c exists. The zeros of g' in $\left[\delta, \frac{\ell}{2}\right]$ form a closed set for each small $\delta > 0$, so the supremum of the zeros is one of them. On each removed interval, V is a copy of g starting at each end, joined by a flat piece. The flat piece starts at a zero of g', so the pieces fit together smoothly.

Theorem 17.7.3. *Volterra's function F has the following properties.*

(1) F is differentiable at every point of $[0, 1]$, and $|F'(x)| \leq 3$ for all x.
(2) $F'(x) = 0$ for $x \in K$, and F' is not continuous at any point of K.
(3) F' is not Riemann integrable on $[0, 1]$.
(4) F' is Lebesgue integrable, and $F(x) = \int_{[0,x]} F'$ for every $x \in [0, 1]$.

Proof. Parts (1) and (2) are proved together. Let (α, β) be a removed interval. On $(\alpha, \alpha + c)$, $V'(x) = g'(x - \alpha)$. On $(\alpha + c, \beta - c)$, $V'(x) = 0$. On $(\beta - c, \beta)$, $V'(x) = -g'(\beta - x)$. At $\alpha + c$ and $\beta - c$, both one-sided derivatives are 0, since $g'(c) = 0$. So V is differentiable on (α, β) with $|V'| \leq 3$, since $c \leq \frac{1}{2}$.

Let $x \in K$. If $y \in K$, then $F(y) - F(x) = 0$. If y is in a removed interval (α, β), let d be the distance from y to the nearer of α and β. On the outer pieces, $|V(y)| \leq d^2$, since $|g(t)| \leq t^2$. On the flat piece, $d \geq c$, so $|V(y)| = |g(c)| \leq c^2 \leq d^2$.

Since $x \notin (\alpha, \beta)$, $|y - x| \geq d$. So in every case $|F(y) - F(x)| \leq |y - x|^2$. Thus $\left|\frac{F(y)-F(x)}{y-x}\right| \leq |y - x| \to 0$, and $F'(x) = 0$.

Now let $x \in K$ and $\delta > 0$. Choose n with $\ell_n < \delta$. The remaining interval of step n that contains x lies in $(x-\delta, x+\delta)$, and at step n an interval (α, β) is removed from its middle. For every large k, $\alpha + \frac{1}{2k\pi}$ lies in $(\alpha, \alpha + c)$, and there $F' = g'\left(\frac{1}{2k\pi}\right) = -1$. So every neighborhood of x contains points where $F' = -1$, while $F'(x) = 0$. So F' is not continuous at x.

(3) By (1), F' is bounded. By (2), F' is discontinuous at every point of K, and $m(K) = \frac{1}{2} > 0$. By Theorem 9.2.9, F' is not Riemann integrable.

(4) For $x, y \in [0, 1]$, the Mean Value Theorem and (1) give $|F(x) - F(y)| \leq 3|x - y|$. So F is absolutely continuous. Given $\varepsilon > 0$, take $\delta = \frac{\varepsilon}{3}$ in Definition 17.5.1. By Corollary 17.5.9, F' is Lebesgue integrable and $F(x) = F(0) + \int_{[0,x]} F'$. Since $0 \in K$, $F(0) = 0$. □

So F has a derivative everywhere, the derivative is bounded, and F is the integral of its derivative. But the Riemann integral cannot see this, because F' is discontinuous on a set of positive measure. This is one of the problems that led Lebesgue to his integral. In his 1904 lectures, he cited Volterra's example as a reason the Riemann integral is not enough.

Exercises

17.7.1. Let $g(x) = x^2 \sin \frac{1}{x}$ for $x \neq 0$ and $g(0) = 0$. Prove that g is differentiable on $\mathbb{R}$ with $g'(0) = 0$, that $|g'(x)| \leq 3$ for $|x| \leq 1$, and that $\lim_{x\to 0} g'(x)$ does not exist. [Solution, p. 381]

17.7.2. Let $0 < a < 1$. Change Example 17.7.1 so that the intervals removed at step n have length $a3^{-n}$. Prove that each step can be carried out, and that the resulting set is closed, contains no open interval, and has measure $1 - a$. [Solution, p. 381]

17.7.3. Let K be the set of Example 17.7.1. Prove that χ_K is discontinuous at every point of K and continuous everywhere else. Conclude that χ_K is Lebesgue integrable but not Riemann integrable on $[0, 1]$. [Hint, p. 382]

17.7.4. Prove that $\int_{[0,1]} F' = 0$ for Volterra's function F. Then find a removed interval (α, β) and explain why $\int_\alpha^\beta F' = 0$ as a Riemann integral, even though F' is not Riemann integrable on $[0, 1]$. [Hint, p. 382]

Notes

Ulisse Dini introduced the four derivatives that bear his name in 1878 [**23**]. Camille Jordan introduced functions of bounded variation in 1881 [**39**]. In his 1904 lectures [**43**], Henri Lebesgue proved that continuous monotone functions are differentiable almost everywhere, and that the derivative of an indefinite integral equals the integrand almost everywhere. Giuseppe Vitali introduced absolute continuity in 1905, and proved that the absolutely continuous functions are exactly the indefinite integrals [**75**]. He proved his covering lemma in 1908 [**76**]. Georg Cantor described the Cantor function in 1884 [**12**]. The same year, Ludwig Scheeffer used it as a

counterexample to a claimed extension of the Fundamental Theorem of Calculus [**64**]. Vito Volterra published the function of Section 17.7 in 1881, while he was still a student. Henry John Stephen Smith had described closed sets with no intervals and positive measure in 1875, and they are sometimes called Smith–Volterra–Cantor sets.

CHAPTER 18

L^p Spaces*

Chapter 16 defined $L^1(E)$, the functions whose absolute value has a finite integral. This chapter replaces $|f|$ by $|f|^p$. The resulting spaces $L^p(E)$ are where much of modern analysis takes place, including Fourier analysis, probability and the study of differential equations. Section 18.1 defines them and proves the inequalities of Hölder and Minkowski. Section 18.2 proves that L^p is complete, the analogue of the completeness of $\mathbb{R}$. Section 18.3 shows that step functions and continuous functions are dense in $L^p[a,b]$.

The whole chapter is marked with a star. It can be skipped. Nothing else in the book depends on it. It uses Chapters 15 and 16, and the power functions x^p for real p (see Section 9.4).

Throughout, E is a measurable subset of $\mathbb{R}$. Two functions that are equal almost everywhere on E have the same integrals over E. So in this chapter they are treated as the same function. This is what makes $\|f\|_p = 0$ mean $f = 0$.

18.1. The Spaces L^p

Definition 18.1.1. Let $1 \le p < \infty$. A measurable function f on E is in $L^p(E)$ if $\int_E |f|^p < \infty$. For such f, the L^p *norm* of f is

$$\|f\|_p = \left(\int_E |f|^p\right)^{1/p}.$$

A measurable function f is in $L^\infty(E)$ if there is an M with $|f| \le M$ almost everywhere on E. Then $\|f\|_\infty$ is the infimum of all such M. Write $L^p[a,b]$ for $L^p([a,b])$.

For $p = 1$ this is the space $L^1(E)$ of Definition 16.5.3. The infimum in the definition of $\|f\|_\infty$ is attained, so $|f| \le \|f\|_\infty$ almost everywhere (Exercise 18.1.4). The number q with $\frac{1}{p} + \frac{1}{q} = 1$ is called the *conjugate exponent* of p. For $p = 1$, take $q = \infty$, and for $p = 2$, $q = 2$.

Lemma 18.1.2 (Young's Inequality)**.** *Let $1 < p < \infty$, let $\frac{1}{p} + \frac{1}{q} = 1$, and let $a, b \ge 0$. Then $ab \le \frac{a^p}{p} + \frac{b^q}{q}$.*

Proof. If $b = 0$, there is nothing to prove. Fix $b > 0$, and let $h(a) = \frac{a^p}{p} + \frac{b^q}{q} - ab$ for $a \ge 0$. Then $h'(a) = a^{p-1} - b$, which is negative for $a < a_0 = b^{1/(p-1)}$ and positive for $a > a_0$. So h has its smallest value at a_0, by Theorem 7.2.7. Since $\frac{1}{p-1} = \frac{q}{p}$, $a_0^p = b^q$ and $a_0 b = b^{q/p+1} = b^q$, because $\frac{q}{p} + 1 = q\left(\frac{1}{p} + \frac{1}{q}\right) = q$. So $h(a_0) = \frac{b^q}{p} + \frac{b^q}{q} - b^q = 0$, and $h \ge 0$. □

Theorem 18.1.3 (Hölder's Inequality)**.** *Let $1 \le p \le \infty$ and let q be its conjugate exponent. If $f \in L^p(E)$ and $g \in L^q(E)$, then $fg \in L^1(E)$ and*

$$\int_E |fg| \le \|f\|_p \|g\|_q.$$

PROOF. If $p = 1$, then $|fg| \le |f|\|g\|_\infty$ almost everywhere, and integrating gives the result. The case $p = \infty$ is the same with f and g exchanged. Let $1 < p < \infty$. If $\|f\|_p = 0$, then $f = 0$ almost everywhere, and both sides are 0. The same holds if $\|g\|_q = 0$. Otherwise, let $F = \frac{|f|}{\|f\|_p}$ and $G = \frac{|g|}{\|g\|_q}$. Then $\int_E F^p = \int_E G^q = 1$. By Lemma 18.1.2, $FG \le \frac{F^p}{p} + \frac{G^q}{q}$. Integrating, $\int_E FG \le \frac{1}{p} + \frac{1}{q} = 1$. Multiply by $\|f\|_p \|g\|_q$. □

Theorem 18.1.4 (Minkowski's Inequality)**.** *Let $1 \le p \le \infty$. If $f, g \in L^p(E)$, then $f + g \in L^p(E)$ and $\|f + g\|_p \le \|f\|_p + \|g\|_p$.*

PROOF. The cases $p = 1$ and $p = \infty$ follow from $|f+g| \le |f|+|g|$ (Exercise 18.1.4). Let $1 < p < \infty$. Since $|f + g| \le 2\max\{|f|, |g|\}$, $|f + g|^p \le 2^p(|f|^p + |g|^p)$. So $f + g \in L^p(E)$. If $\|f + g\|_p = 0$ there is nothing to prove. Let q be the conjugate exponent, so $(p-1)q = p$. Then $|f+g|^{p-1} \in L^q(E)$, with $\||f+g|^{p-1}\|_q = \|f+g\|_p^{p/q}$. By Theorem 18.1.3,

$$\int_E |f+g|^p \le \int_E |f||f+g|^{p-1} + \int_E |g||f+g|^{p-1} \le \big(\|f\|_p + \|g\|_p\big)\|f+g\|_p^{p/q}.$$

Divide by $\|f + g\|_p^{p/q}$. Since $p - \frac{p}{q} = 1$, this gives $\|f + g\|_p \le \|f\|_p + \|g\|_p$. □

So $L^p(E)$ is closed under addition, and clearly under multiplication by constants. It is a vector space, and $\|\cdot\|_p$ behaves like a length. It satisfies $\|cf\|_p = |c|\|f\|_p$ and the triangle inequality, and $\|f\|_p = 0$ only when $f = 0$ almost everywhere. The distance between f and g is $\|f - g\|_p$. For $p = 2$, Hölder's Inequality is the *Cauchy–Schwarz Inequality* (Exercise 18.1.3).

On a set of finite measure, larger p means a smaller space (Exercise 18.1.1). On $(0, \infty)$, no space contains another (Exercise 18.1.2).

Exercises

18.1.1. Let $m(E) < \infty$ and $1 \le p < r < \infty$. Prove that $L^r(E) \subset L^p(E)$ and that $\|f\|_p \le m(E)^{1/p-1/r}\|f\|_r$. (Apply Theorem 18.1.3 to $|f|^p \cdot 1$ with exponent $\frac{r}{p}$.) Prove also that $L^\infty(E) \subset L^p(E)$. [Solution, p. 382]

18.1.2. On $E = (0, \infty)$, find a function in $L^1(E)$ that is not in $L^2(E)$, and a function in $L^2(E)$ that is not in $L^1(E)$. [Solution, p. 382]

18.1.3. Prove that $\left(\int_E |fg|\right)^2 \le \int_E f^2 \int_E g^2$ for $f, g \in L^2(E)$. [Hint, p. 382]

18.1.4. Let $f, g \in L^\infty(E)$. Prove that $|f| \le \|f\|_\infty$ almost everywhere, and that $\|f + g\|_\infty \le \|f\|_\infty + \|g\|_\infty$. [Hint, p. 382]

18.2. Completeness

Definition 18.2.1. Let $1 \le p \le \infty$. A sequence $\{f_n\}$ in $L^p(E)$ *converges in* L^p to $f \in L^p(E)$ if $\|f_n - f\|_p \to 0$. It is a *Cauchy sequence in* L^p if for every $\varepsilon > 0$ there is an N with $\|f_n - f_m\|_p < \varepsilon$ for all $m, n \ge N$.

For $p = \infty$, convergence in L^∞ is uniform convergence off a set of measure zero. For $p < \infty$, convergence in L^p is a different kind of convergence from those in Chapter 15. The sequence $f_n = n\chi_{(0,1/n)}$ converges to 0 at every point, but $\|f_n\|_1 = 1$ for all n, so it does not converge to 0 in L^1. (See Figure 16.2.) In the other direction, a sequence can converge in L^p without converging at any point (Exercise 18.2.1). The next theorem shows that some subsequence always converges almost everywhere.

Theorem 18.2.2 (Riesz–Fischer Theorem)**.** *Let $1 \le p < \infty$. Every Cauchy sequence in $L^p(E)$ converges in L^p to a function in $L^p(E)$. Also, some subsequence converges to it almost everywhere.*

Proof. Let $\{f_n\}$ be Cauchy in $L^p(E)$. Choose $n_1 < n_2 < \cdots$ with $\|f_n - f_m\|_p < 2^{-k}$ for all $m, n \ge n_k$. Then $\|f_{n_{k+1}} - f_{n_k}\|_p < 2^{-k}$. Let

$$g_K = |f_{n_1}| + \sum_{k=1}^{K} |f_{n_{k+1}} - f_{n_k}| \quad \text{and} \quad g = \lim_{K\to\infty} g_K.$$

By Theorem 18.1.4, $\|g_K\|_p \le \|f_{n_1}\|_p + 1$ for every K. The functions g_K^p increase to g^p, so by Theorem 16.4.7, $\int_E g^p \le (\|f_{n_1}\|_p + 1)^p < \infty$. In particular, $g < \infty$ almost everywhere.

Where $g(x) < \infty$, the series $f_{n_1}(x) + \sum_k \big(f_{n_{k+1}}(x) - f_{n_k}(x)\big)$ converges absolutely. Its partial sums are $f_{n_{K+1}}(x)$. So $f_{n_k}(x)$ converges to a number $f(x)$ almost everywhere. Set $f = 0$ elsewhere. Then f is measurable, and $|f| \le g$, so $f \in L^p(E)$. Also $|f - f_{n_k}|^p \le (2g)^p$, which is integrable, and $|f - f_{n_k}|^p \to 0$ almost everywhere. By Theorem 16.5.6, $\|f - f_{n_k}\|_p \to 0$.

Finally, let $\varepsilon > 0$. Choose N with $\|f_n - f_m\|_p < \varepsilon$ for $m, n \ge N$, and choose k with $n_k \ge N$ and $\|f - f_{n_k}\|_p < \varepsilon$. For $n \ge N$, $\|f - f_n\|_p \le \|f - f_{n_k}\|_p + \|f_{n_k} - f_n\|_p < 2\varepsilon$. So $f_n \to f$ in L^p. □

In the language of later courses, $L^p(E)$ is a *Banach space*, a complete normed vector space. The theorem is also true for $p = \infty$ (Exercise 18.2.2). For the Riemann integral, nothing like it holds. A sequence of Riemann integrable functions can be Cauchy for $\int |f_n - f_m|$ without converging to any Riemann integrable function. This is one of the main reasons the Lebesgue integral replaced the Riemann integral.

Exercises

18.2.1. For $n \in \mathbb{N}$, write $n = 2^k + j$ with $k \in \mathbb{N}_0$ and $0 \le j < 2^k$, and let $f_n = \chi_{[j2^{-k},(j+1)2^{-k}]}$ on $[0,1]$. Prove that $f_n \to 0$ in $L^p[0,1]$ for every $p < \infty$, but that $\{f_n(x)\}$ converges for no $x \in [0,1]$. Find a subsequence that converges to 0 almost everywhere. [Solution, p. 382]

18.2.2. Prove that every Cauchy sequence in $L^\infty(E)$ converges in L^∞. [Hint, p. 382]

18.2.3. A sequence $\{f_n\}$ of measurable functions on E *converges in measure* to f if $m(\{x \in E : |f_n(x) - f(x)| \geq \varepsilon\}) \to 0$ for every $\varepsilon > 0$. Use Exercise 16.4.5 to prove that convergence in L^p implies convergence in measure. Then show that the sequence of Exercise 18.2.1 converges in measure but not almost everywhere. [Hint, p. 382]

18.3. Approximation in L^p

Theorem 18.3.1. *Let $1 \leq p < \infty$ and $f \in L^p[a,b]$. For every $\varepsilon > 0$, there is a step function g on $[a,b]$ with $\|f - g\|_p < \varepsilon$.*

PROOF. The proof has two steps. The first step cuts f off at height N, which makes it bounded. For $N \in \mathbb{N}$, let $f_N = f$ where $|f| \leq N$, $f_N = N$ where $f > N$, and $f_N = -N$ where $f < -N$. By Exercise 16.4.5, $m(\{|f|^p \geq N^p\}) \leq N^{-p} \int |f|^p \to 0$, so $|f| < \infty$ almost everywhere. So $f_N \to f$ almost everywhere. Also $|f - f_N|^p \leq |f|^p$, which is integrable. By Theorem 16.5.6, $\|f - f_N\|_p \to 0$. Choose N with $\|f - f_N\|_p < \frac{\varepsilon}{2}$.

The second step approximates the bounded function f_N by a step function. Let $\eta > 0$. By Theorem 15.2.7, there is a step function g with $-N \leq g \leq N$ and $m(A) < \eta$, where $A = \{x : |f_N(x) - g(x)| \geq \eta\}$. Off A, $|f_N - g| < \eta$. On A, $|f_N - g| \leq 2N$. So

$$\int_{[a,b]} |f_N - g|^p \leq \eta^p (b-a) + (2N)^p \eta.$$

Choose η so small that this is less than $\left(\frac{\varepsilon}{2}\right)^p$. Then $\|f_N - g\|_p < \frac{\varepsilon}{2}$, and $\|f - g\|_p < \varepsilon$ by Theorem 18.1.4. □

The same proof, with the continuous function h of Theorem 15.2.7 in place of g, shows that continuous functions are dense too (Exercise 18.3.1). So every function in $L^p[a,b]$, however wild, is close in the L^p norm to a continuous one. Together with Theorem 18.2.2, this says $L^p[a,b]$ is exactly what is obtained by filling in the gaps in the continuous functions under the norm $\|\cdot\|_p$, much as $\mathbb{R}$ fills in the gaps in $\mathbb{Q}$ in Appendix B.

Exercises

18.3.1. Let $1 \leq p < \infty$ and $f \in L^p[a,b]$. Prove that for every $\varepsilon > 0$ there is a continuous function h on $[a,b]$ with $\|f - h\|_p < \varepsilon$. [Solution, p. 382]

18.3.2. (Riemann–Lebesgue Lemma) Let $f \in L^1[a,b]$. Prove that

$$\lim_{n\to\infty} \int_{[a,b]} f(x) \sin nx \, dx = 0.$$

(First take f to be a step function.) [Hint, p. 383]

Notes

Frigyes Riesz introduced the spaces L^p in 1910. Riesz and Ernst Fischer each proved the completeness of L^2 in 1907, which is why Theorem 18.2.2 carries both names. Leonard James Rogers proved the inequality now named for Otto Hölder in 1888, and Hölder found it again in 1889. Hermann Minkowski proved his inequality for sums in 1896. Stefan Banach made complete normed spaces the subject of a general theory in the 1920s. The Riemann–Lebesgue Lemma of Exercise 18.3.2 goes back to Riemann's 1854 paper on trigonometric series, and Lebesgue proved it for his integral.

APPENDIX A

Construction of the Real Numbers by Cuts

This appendix proves Theorem 1.8.2. It builds an ordered field with the least upper bound property out of $\mathbb{Q}$. The construction is due to Richard Dedekind.

The idea comes from Section 1.4. The rational numbers p with $p \leq 0$ or $p^2 < 2$ form a left piece of $\mathbb{Q}$. The positive rational numbers with $p^2 > 2$ form a right piece. No rational number sits between the two pieces. Dedekind's idea was to let the left piece itself be the missing number. So a real number will be a set of rational numbers, namely all the rational numbers to its left. A rational number q gives such a set too, the set of all rational numbers less than q.

Only $\mathbb{Q}$ is assumed here, with its usual operations and order. So $\mathbb{Q}$ is an ordered field (Example 1.6.7), and the results of Sections 1.5 to 1.7 hold in $\mathbb{Q}$. Nothing from Section 1.8 or later is used, since those results assume that $\mathbb{R}$ exists. In this appendix the letters p, q, r, s, t and u always denote rational numbers, and Greek letters denote cuts.

A.1. Cuts

Definition A.1.1. A set $\alpha \subset \mathbb{Q}$ is a *cut* if it has the following three properties.

(1) $\alpha \neq \emptyset$ and $\alpha \neq \mathbb{Q}$.
(2) If $p \in \alpha$ and $q < p$, then $q \in \alpha$.
(3) α has no largest element. That is, if $p \in \alpha$, then $p < r$ for some $r \in \alpha$.

In this appendix, $\mathbb{R}$ denotes the set of all cuts.

Remark A.1.2. Property (2) has two consequences that are used constantly. Let α be a cut. If $p \in \alpha$ and $u \notin \alpha$, then $p < u$. Indeed, $u = p$ is impossible, and $u < p$ would put u in α. Also, if $u \notin \alpha$ and $u < s$, then $s \notin \alpha$. Indeed, $s \in \alpha$ would put u in α. So everything below a member of α is in α, and everything above a nonmember is outside α.

Example A.1.3. Let $q \in \mathbb{Q}$. Then $q^* = \{p \in \mathbb{Q} : p < q\}$ is a cut, called a *rational cut*. To see this, note that $q - 1 \in q^*$ and $q \notin q^*$. Property (2) is the transitivity of the order. For (3), if $p < q$, then $p < \frac{p+q}{2} < q$ by Exercise 1.4.3. Notice that q is the smallest element of $\mathbb{Q} \setminus q^*$.

Example A.1.4. Let $\alpha = \{p \in \mathbb{Q} : p \leq 0 \text{ or } p^2 < 2\}$. Then α is a cut, and $\alpha \neq q^*$ for every $q \in \mathbb{Q}$.

First, $0 \in \alpha$ and $2 \notin \alpha$. For (2), let $p \in \alpha$ and $q < p$. If $q \leq 0$, then $q \in \alpha$. If $q > 0$, then $0 < q < p$. So $p^2 < 2$, and $q^2 < p^2 < 2$ by Exercise 1.6.3. So $q \in \alpha$. For (3), let $p \in \alpha$. If $p \leq 0$, then $1 \in \alpha$ and $1 > p$. If $p > 0$, then p is in the set A of Theorem 1.4.5, which has no largest element. So α is a cut.

The set $\mathbb{Q} \setminus \alpha$ consists of the $p > 0$ with $p^2 \geq 2$. By Theorem 1.4.4, $p^2 \neq 2$. So $\mathbb{Q} \setminus \alpha$ is the set B of Theorem 1.4.5, which has no smallest element. By Example A.1.3, $\mathbb{Q} \setminus q^*$ has a smallest element. So $\alpha \neq q^*$. This cut will play the role of $\sqrt{2}$. Compare Figure 1.1.

Lemma A.1.5. *Let $p, r \in \mathbb{Q}$ with $r > 0$. Then there exists an $n \in \mathbb{N}$ with $nr > p$.*

PROOF. If $p \leq 0$, take $n = 1$. Suppose $p > 0$. As in the proof of Lemma 1.4.3, write $p = \frac{a}{b}$ and $r = \frac{c}{d}$ with $a, b, c, d \in \mathbb{N}$. Let $n = ad + 1$. Since $b \geq 1$ and $c \geq 1$,

$$nr = \frac{(ad+1)c}{d} \geq \frac{ad+1}{d} > a \geq \frac{a}{b} = p. \qquad \square$$

Lemma A.1.6. *Let α be a cut and let $r \in \mathbb{Q}$ with $r > 0$. Then there exists a $p \in \alpha$ with $p + r \notin \alpha$.*

PROOF. Choose $s \in \alpha$ and $u \notin \alpha$. By Lemma A.1.5, $nr > u - s$ for some $n \in \mathbb{N}$. Then $s + nr > u$, so $s + nr \notin \alpha$ by Remark A.1.2. So the set $S = \{n \in \mathbb{N} : s + nr \notin \alpha\}$ is not empty. By the Well-Ordering Principle, it has a smallest element n_0. If $n_0 = 1$, let $p = s$. If $n_0 > 1$, let $p = s + (n_0 - 1)r$. Then $p \in \alpha$, since $n_0 - 1 \notin S$. In both cases $p \in \alpha$ and $p + r = s + n_0 r \notin \alpha$. $\square$

In words, steps of size r that start inside α must eventually leave it, and Lemma A.1.6 picks the last step inside. So the inside and the outside of a cut come within r of each other, for every $r > 0$. This lemma is the key to the negative of a cut in Appendix A.3 and to the reciprocal in Appendix A.4.

Exercises

A.1.1. Decide which of the following sets are cuts. Justify each answer.
(a) $\{p \in \mathbb{Q} : p^2 < 2\}$
(b) $\{p \in \mathbb{Q} : p \leq 1\}$
(c) $\{p \in \mathbb{Q} : p < 0 \text{ or } p^2 < 3\}$
(d) $\{p \in \mathbb{Q} : p < \frac{1}{n} \text{ for every } n \in \mathbb{N}\}$

[Hint, p. 383]

A.1.2. Let α be a cut. Prove that $\alpha = q^*$ for some $q \in \mathbb{Q}$ if and only if $\mathbb{Q} \setminus \alpha$ has a smallest element. [Solution, p. 383]

A.2. Order and the Least Upper Bound Property

Cuts are ordered by inclusion. A smaller real number has fewer rational numbers to its left.

Definition A.2.1. For cuts α and β, define $\alpha < \beta$ to mean $\alpha \subsetneq \beta$.

So $\alpha \leq \beta$ means $\alpha \subset \beta$. For rational cuts, $p^* < q^*$ exactly when $p < q$, by Theorem A.5.1.

Theorem A.2.2. *The relation $<$ is an order on $\mathbb{R}$.*

PROOF. An order must be transitive and must satisfy trichotomy. To check transitivity, let $\alpha < \beta$ and $\beta < \gamma$. Then $\alpha \subset \beta \subset \gamma$, so $\alpha \subset \gamma$. If $\alpha = \gamma$, then $\beta \subset \gamma = \alpha \subset \beta$, so $\alpha = \beta$, which is false. So $\alpha < \gamma$.

Next comes trichotomy. At most one of $\alpha < \beta$, $\alpha = \beta$ and $\beta < \alpha$ holds. Indeed, $\alpha < \beta$ and $\beta < \alpha$ would give $\alpha \subset \beta \subset \alpha$, so $\alpha = \beta$, which is false. Now suppose $\alpha \neq \beta$ and $\alpha \not< \beta$. Then $\alpha \not\subset \beta$. So there is a $p \in \alpha$ with $p \notin \beta$. Let $q \in \beta$. Since $p \notin \beta$, $q < p$ by Remark A.1.2. Since $p \in \alpha$, $q \in \alpha$. So $\beta \subset \alpha$. Since $\beta \neq \alpha$, $\beta < \alpha$. □

Theorem A.2.3. *The ordered set $\mathbb{R}$ has the least upper bound property. If $A \subset \mathbb{R}$ is nonempty and bounded above, then $\sup A$ is the union of the cuts in A.*

PROOF. Let β be an upper bound of A, and let $\gamma = \bigcup_{\alpha \in A} \alpha$.

The proof shows that γ is a cut, that it is an upper bound of A, and that no smaller cut is an upper bound of A. First, γ is a cut. For property (1), choose $\alpha_0 \in A$. Then $\emptyset \neq \alpha_0 \subset \gamma$. Each $\alpha \in A$ has $\alpha \leq \beta$, that is, $\alpha \subset \beta$. So $\gamma \subset \beta \neq \mathbb{Q}$, and $\gamma \neq \mathbb{Q}$. For (2), let $p \in \gamma$ and $q < p$. Then $p \in \alpha$ for some $\alpha \in A$. So $q \in \alpha \subset \gamma$. For (3), let $p \in \gamma$, say $p \in \alpha$ with $\alpha \in A$. Then $p < r$ for some $r \in \alpha$, and $r \in \gamma$.

Next, γ is an upper bound of A. Indeed, each $\alpha \in A$ satisfies $\alpha \subset \gamma$, so $\alpha \leq \gamma$.

Finally, no smaller cut is an upper bound of A. To see this, let $\delta < \gamma$. Then there is a $p \in \gamma$ with $p \notin \delta$. Say $p \in \alpha$ with $\alpha \in A$. Then $\alpha \not\subset \delta$, so $\alpha \not\leq \delta$. By Theorem A.2.2, $\delta < \alpha$. So δ is not an upper bound of A. Thus $\gamma = \sup A$. □

Notice how little work this took. The hole in $\mathbb{Q}$ at $\sqrt{2}$ was a gap between a left piece and a right piece. A cut is a left piece, so the gap is filled by definition. The real work in this appendix is the arithmetic.

Exercises

A.2.1. Let α and β be cuts with $\alpha < \beta$. Prove that there is a $q \in \mathbb{Q}$ with $\alpha < q^* < \beta$. [Solution, p. 383]

A.2.2. Let α be a cut. Prove that $\alpha = \sup\{q^* : q \in \alpha\}$. [Hint, p. 383]

A.3. Addition

Definition A.3.1. For cuts α and β, define

$$\alpha + \beta = \{r + s : r \in \alpha,\ s \in \beta\}.$$

Let $0^* = \{p \in \mathbb{Q} : p < 0\}$, the rational cut of 0.

Lemma A.3.2. *If α and β are cuts, then $\alpha + \beta$ is a cut.*

PROOF. Clearly $\alpha + \beta \neq \emptyset$. Choose $u \notin \alpha$ and $t \notin \beta$. If $r \in \alpha$ and $s \in \beta$, then $r < u$ and $s < t$, so $r + s < u + t$. So $u + t \notin \alpha + \beta$, and $\alpha + \beta \neq \mathbb{Q}$. For (2), let $p = r + s$ with $r \in \alpha$ and $s \in \beta$, and let $q < p$. Then $q - s < r$, so $q - s \in \alpha$, and $q = (q - s) + s \in \alpha + \beta$. For (3), choose $r' \in \alpha$ with $r' > r$. Then $r' + s \in \alpha + \beta$ and $r' + s > p$. □

Definition A.3.3. For a cut α, let $-\alpha$ be the set of all $p \in \mathbb{Q}$ such that $-p - r \notin \alpha$ for some rational $r > 0$.

In words, $p \in -\alpha$ when $-p$ lies outside α with some room to spare. For a rational cut, $-(q^*) = (-q)^*$. Indeed, $p \in -(q^*)$ means $-p - r \geq q$ for some $r > 0$, which happens exactly when $-p > q$, that is, $p < -q$. The room to spare matters. Without it, $-(q^*)$ would contain $-q$ as a largest element.

Theorem A.3.4. *Let α, β and γ be cuts.*

(1) $\alpha + \beta = \beta + \alpha$.
(2) $(\alpha + \beta) + \gamma = \alpha + (\beta + \gamma)$.
(3) $\alpha + 0^* = \alpha$.
(4) $-\alpha$ *is a cut, and* $\alpha + (-\alpha) = 0^*$.
(5) If $\beta \leq \gamma$, *then* $\alpha + \beta \leq \alpha + \gamma$.

Proof. Parts (1) and (2) are Exercise A.3.1. Part (5) holds since $\beta \subset \gamma$ gives $\alpha + \beta \subset \alpha + \gamma$.

(3) Let $r \in \alpha$ and $s \in 0^*$. Then $r + s < r$, so $r + s \in \alpha$. So $\alpha + 0^* \subset \alpha$. Conversely, let $p \in \alpha$. Choose $r \in \alpha$ with $r > p$. Then $p - r \in 0^*$, and $p = r + (p - r) \in \alpha + 0^*$. So $\alpha \subset \alpha + 0^*$.

(4) First, $-\alpha$ is a cut. Choose $u \notin \alpha$, and let $p = -u - 1$. Then $-p - 1 = u \notin \alpha$, so $p \in -\alpha$. Next, let $q \in \alpha$. For every $r > 0$, $-(q) - r = q - r < q$, so $q - r \in \alpha$. So $-q \notin -\alpha$, and $-\alpha \neq \mathbb{Q}$. For (2), let $p \in -\alpha$, with $-p - r \notin \alpha$, and let $q < p$. Then $-q - r > -p - r$, so $-q - r \notin \alpha$ by Remark A.1.2. So $q \in -\alpha$. For (3), let $p \in -\alpha$, with $-p - r \notin \alpha$. Let $t = p + \frac{r}{2}$. Then $-t - \frac{r}{2} = -p - r \notin \alpha$, so $t \in -\alpha$, and $t > p$.

Next, $\alpha + (-\alpha) \subset 0^*$. Let $r \in \alpha$ and $s \in -\alpha$, with $-s - t \notin \alpha$ for some $t > 0$. Then $r < -s - t$, so $r + s < -t < 0$.

Finally, $0^* \subset \alpha + (-\alpha)$. Let $v \in 0^*$, and let $w = -\frac{v}{2} > 0$. By Lemma A.1.6, there is a $p \in \alpha$ with $p + w \notin \alpha$. Let $q = -p - 2w$. Then $-q - w = p + w \notin \alpha$, so $q \in -\alpha$. And $p + q = -2w = v$. So $v \in \alpha + (-\alpha)$. □

Proof Technique (Proving Two Cuts Are Equal). Cuts are sets, so two cuts are equal when each is contained in the other. Usually one inclusion is easy, and it follows from the order of $\mathbb{Q}$, as in the first half of the proof of Theorem A.3.4 (3). The other inclusion needs room to move. Given p in one cut, it must be written as a sum or a product of elements of the other cuts. Property (3) of Definition A.1.1, or Lemma A.1.6, gives the room.

Corollary A.3.5. *Axioms (1)–(5) of Definition 1.6.1 hold for cuts, with* 0^* *as the zero. So does property (1) of Definition 1.6.6. Therefore Theorem 1.6.5 (1)–(4) and Theorem 1.6.8 (1) hold for cuts, since their proofs use only these facts. In particular,* $-\alpha$ *is the only cut with* $\alpha + (-\alpha) = 0^*$, $-(-\alpha) = \alpha$, *and* $\alpha > 0^*$ *if and only if* $-\alpha < 0^*$.

Proof. Axiom (1) is Lemma A.3.2. Axioms (2)–(5) are Theorem A.3.4 (1)–(4), using (1) to write $0^* + \alpha = \alpha$. Property (1) of Definition 1.6.6 is Theorem A.3.4 (5), together with (1). □

Exercises

A.3.1. Prove Theorem A.3.4 (1) and (2). [Solution, p. 383]

A.3.2. Let α be the cut of Example A.1.4. Prove that $-\alpha = \{p \in \mathbb{Q} : p < 0 \text{ and } p^2 > 2\}$. [Hint, p. 383]

A.4. Multiplication

Multiplying left pieces is harder than adding them, because multiplying by a negative number reverses the order. So multiplication is defined first for positive cuts. Then it is extended to all cuts by the usual rules of signs.

Lemma A.4.1. *Let α be a cut with $\alpha > 0^*$. Then $0 \in \alpha$, α contains a positive rational number, and every $u \notin \alpha$ is positive.*

PROOF. Since $0^* \subsetneqq \alpha$, there is a $p \in \alpha$ with $p \notin 0^*$, that is, $p \geq 0$. So $0 \in \alpha$. By property (3) of Definition A.1.1, some $r \in \alpha$ has $r > 0$. If $u \notin \alpha$, then $u > 0$ by Remark A.1.2, since $0 \in \alpha$. □

Definition A.4.2. For a cut $\alpha > 0^*$, let $\alpha^+ = \{p \in \alpha : p > 0\}$. For cuts $\alpha, \beta > 0^*$, define $\alpha\beta$ to be the set of all $p \in \mathbb{Q}$ such that

$$p \leq rs \quad \text{for some } r \in \alpha^+ \text{ and } s \in \beta^+.$$

Also define α^{-1} to be the set of all $p \in \mathbb{Q}$ such that $p \leq 0$, or $p > 0$ and $pu < 1$ for some $u \notin \alpha$. Let $1^* = \{p \in \mathbb{Q} : p < 1\}$, the rational cut of 1.

The definition of α^{-1} follows the definition of $-\alpha$. A positive p is in α^{-1} when $\frac{1}{p}$ lies outside α with room to spare, namely when $\frac{1}{p} > u$ for some $u \notin \alpha$. For example, if $q > 0$, then $(q^*)^{-1} = \left(\frac{1}{q}\right)^*$.

Lemma A.4.3. *Let $\alpha, \beta > 0^*$. Then $\alpha\beta$ and α^{-1} are cuts, $\alpha\beta > 0^*$, and $\alpha^{-1} > 0^*$.*

PROOF. First, $\alpha\beta$ is shown to be a cut. By Lemma A.4.1, α^+ and β^+ are not empty, so $\alpha\beta \neq \emptyset$. Choose $u \notin \alpha$ and $t \notin \beta$. Both are positive. If $r \in \alpha^+$ and $s \in \beta^+$, then $0 < r < u$ and $0 < s < t$, so $rs < ut$ by Theorem 1.6.8 (3). So $ut \notin \alpha\beta$, and $\alpha\beta \neq \mathbb{Q}$. Property (2) holds, since $q < p \leq rs$ gives $q \leq rs$. For (3), let $p \leq rs$ with $r \in \alpha^+$ and $s \in \beta^+$. Choose $r' \in \alpha$ with $r' > r$. Then $r's \in \alpha\beta$ and $r's > rs \geq p$. Finally, $0 \leq rs$, so $0 \in \alpha\beta \setminus 0^*$. Every negative number is less than rs, so $0^* \subset \alpha\beta$. Thus $\alpha\beta > 0^*$.

Next, α^{-1} is shown to be a cut. Clearly $0 \in \alpha^{-1}$. Let $t \in \alpha^+$. If $u \notin \alpha$, then $u > t$, so $\frac{1}{t} \cdot u > 1$. So $\frac{1}{t} \notin \alpha^{-1}$, and $\alpha^{-1} \neq \mathbb{Q}$. For (2), let $p \in \alpha^{-1}$ and $q < p$. If $q \leq 0$, then $q \in \alpha^{-1}$. If $q > 0$, then $p > 0$ and $pu < 1$ for some $u \notin \alpha$. Since $u > 0$, $qu < pu < 1$, so $q \in \alpha^{-1}$. For (3), let $p \in \alpha^{-1}$. If $p \leq 0$, choose $u \notin \alpha$. Then $\frac{1}{2u} > 0 \geq p$ and $\frac{1}{2u} \cdot u < 1$, so $\frac{1}{2u} \in \alpha^{-1}$. If $p > 0$, then $p < \frac{1}{u}$ for some $u \notin \alpha$. Let $p' = \frac{1}{2}\left(p + \frac{1}{u}\right)$. Then $p < p' < \frac{1}{u}$, so $p'u < 1$ and $p' \in \alpha^{-1}$. Thus α^{-1} is a cut. It contains 0, so $\alpha^{-1} > 0^*$. □

Lemma A.4.4. *Let $\alpha > 0^*$ and let $t \in \mathbb{Q}$ with $t > 1$. Then there exists a $p \in \alpha^+$ with $pt \notin \alpha$.*

PROOF. Choose $p_0 \in \alpha^+$, and let $r = p_0(t-1) > 0$. By Lemma A.1.6, there is a $p_1 \in \alpha$ with $p_1 + r \notin \alpha$. Let $p = \max\{p_0, p_1\}$. Then $p \in \alpha^+$. Since $p \geq p_1$, $p + r \notin \alpha$. Since $p \geq p_0$,

$$pt = p + p(t-1) \geq p + p_0(t-1) = p + r.$$

So $pt \notin \alpha$ by Remark A.1.2. □

Theorem A.4.5. *Let $\alpha, \beta, \gamma > 0^*$.*

(1) $\alpha\beta = \beta\alpha$.
(2) $(\alpha\beta)\gamma = \alpha(\beta\gamma)$.
(3) $1^*\alpha = \alpha$.
(4) $\alpha\alpha^{-1} = 1^*$.
(5) $\alpha(\beta + \gamma) = \alpha\beta + \alpha\gamma$.

PROOF. (1) The definition of $\alpha\beta$ is symmetric in α and β, since $rs = sr$.

(2) Let $p \in (\alpha\beta)\gamma$. Then $p \leq wt$ with $w \in (\alpha\beta)^+$ and $t \in \gamma^+$, and $w \leq rs$ with $r \in \alpha^+$ and $s \in \beta^+$. So $p \leq rst$. Conversely, if $p \leq rst$ with $r \in \alpha^+$, $s \in \beta^+$ and $t \in \gamma^+$, then $rs \in (\alpha\beta)^+$, so $p \in (\alpha\beta)\gamma$. So $(\alpha\beta)\gamma$ is the set of p with $p \leq rst$ for some $r \in \alpha^+$, $s \in \beta^+$ and $t \in \gamma^+$. The same argument gives the same description of $\alpha(\beta\gamma)$.

(3) Let $p \in 1^*\alpha$. Then $p \leq rs$ with $0 < r < 1$ and $s \in \alpha^+$. So $p < s$, and $p \in \alpha$. Conversely, let $p \in \alpha$. Choose $s_0 \in \alpha^+$. Then $\max\{p, s_0\} \in \alpha$, so there is an $s \in \alpha$ with $s > \max\{p, s_0\}$. In particular $s > 0$. If $p \leq 0$, then $p \leq \frac{1}{2}s$ with $\frac{1}{2} \in (1^*)^+$. If $p > 0$, then $r = \frac{p}{s}$ satisfies $0 < r < 1$ and $p = rs$. Either way $p \in 1^*\alpha$.

(4) Let $p \in \alpha\alpha^{-1}$. Then $p \leq rs$ with $r \in \alpha^+$, $s > 0$ and $su < 1$ for some $u \notin \alpha$. Since $r < u$, $rs < us < 1$. So $p < 1$, and $p \in 1^*$. Conversely, let $q \in 1^*$. If $q \leq 0$, then $q \in \alpha\alpha^{-1}$, since $\alpha\alpha^{-1} > 0^*$. Let $0 < q < 1$, and let $t = \frac{1}{2}\big(1 + \frac{1}{q}\big)$. Then $1 < t < \frac{1}{q}$. By Lemma A.4.4, there is a $p \in \alpha^+$ with $pt \notin \alpha$. Let $s = \frac{q}{p} > 0$. Then $s \cdot pt = qt < 1$ and $pt \notin \alpha$, so $s \in \alpha^{-1}$. Since $q = ps$, $q \in \alpha\alpha^{-1}$.

(5) First, $\beta + \gamma > 0^*$, since $\beta + \gamma \geq 0^* + \gamma = \gamma > 0^*$ by Theorem A.3.4. Next, every $w \in \beta + \gamma$ satisfies $w \leq s + t$ for some $s \in \beta^+$ and $t \in \gamma^+$. Indeed, $w = s_1 + t_1$ with $s_1 \in \beta$ and $t_1 \in \gamma$. Choose $s_0 \in \beta^+$ and $t_0 \in \gamma^+$, and let $s = \max\{s_0, s_1\}$ and $t = \max\{t_0, t_1\}$.

Let $p \in \alpha(\beta + \gamma)$. Then $p \leq rw$ with $r \in \alpha^+$ and $w \in (\beta + \gamma)^+$. Choose $s \in \beta^+$ and $t \in \gamma^+$ with $w \leq s + t$. Then $p \leq rs + rt$. Since $rs \in \alpha\beta$ and $rt \in \alpha\gamma$, $rs + rt \in \alpha\beta + \alpha\gamma$, and so $p \in \alpha\beta + \alpha\gamma$.

Conversely, let $p \in \alpha\beta + \alpha\gamma$. Then $p = x + y$ with $x \leq rs$ and $y \leq r't$, where $r, r' \in \alpha^+$, $s \in \beta^+$ and $t \in \gamma^+$. Let $R = \max\{r, r'\} \in \alpha^+$. Then

$$p \leq rs + r't \leq Rs + Rt = R(s + t).$$

Since $s + t \in (\beta + \gamma)^+$, $p \in \alpha(\beta + \gamma)$. □

Definition A.4.6. For cuts α and β that are not both positive, define

$$\alpha\beta = \begin{cases} (-\alpha)(-\beta), & \alpha < 0^* \text{ and } \beta < 0^*, \\ -\big((-\alpha)\beta\big), & \alpha < 0^* < \beta, \\ -\big(\alpha(-\beta)\big), & \beta < 0^* < \alpha, \\ 0^*, & \alpha = 0^* \text{ or } \beta = 0^*. \end{cases}$$

For $\alpha < 0^*$, define $\alpha^{-1} = -\big((-\alpha)^{-1}\big)$.

If $\alpha < 0^*$, then $-\alpha > 0^*$ by Corollary A.3.5. So every product on the right side is a product of positive cuts, from Definition A.4.2. These are the familiar rules of signs.

Theorem A.4.7. *With the addition of Definition A.3.1, the multiplication of Definitions A.4.2 and A.4.6, and the order of Definition A.2.1,* $\mathbb{R}$ *is an ordered field. Its zero is* 0^* *and its one is* 1^*.

Proof. Axioms (1)–(5) of Definition 1.6.1 hold by Corollary A.3.5. Every product of cuts is a cut, by Lemma A.4.3, Theorem A.3.4 (4) and Definition A.4.6. So (6) holds. Each case of Definition A.4.6 is symmetric in α and β, by Theorem A.4.5 (1). So (7) holds.

For (9), note $1^* \neq 0^*$, since $0 \in 1^*$ and $0 \notin 0^*$. If $\alpha > 0^*$, then $1^*\alpha = \alpha$ by Theorem A.4.5 (3). If $\alpha < 0^*$, then $1^*\alpha = -\big(1^*(-\alpha)\big) = -(-\alpha) = \alpha$. If $\alpha = 0^*$, then $1^*\alpha = 0^* = \alpha$.

For (10), let $\alpha \neq 0^*$. If $\alpha > 0^*$, then $\alpha\alpha^{-1} = 1^*$ by Theorem A.4.5 (4). If $\alpha < 0^*$, then $(-\alpha)^{-1} > 0^*$, so $\alpha^{-1} < 0^*$. So $\alpha\alpha^{-1} = (-\alpha)\big(-\alpha^{-1}\big) = (-\alpha)(-\alpha)^{-1} = 1^*$.

Axioms (8) and (11) follow from Theorem A.4.5 (2) and (5), by splitting into cases by signs. This is Exercise A.4.1.

Property (1) of Definition 1.6.6 holds by Corollary A.3.5. For (2), let $0^* \leq \alpha$ and $0^* \leq \beta$. If $\alpha = 0^*$ or $\beta = 0^*$, then $\alpha\beta = 0^*$. Otherwise $\alpha\beta > 0^*$ by Lemma A.4.3. □

Exercises

A.4.1. Complete the proof of Theorem A.4.7 by proving axioms (8) and (11) of Definition 1.6.1 for cuts of all signs. First show that $(-\alpha)\beta = -(\alpha\beta) = \alpha(-\beta)$ for all cuts α and β. [Solution, p. 383]

A.4.2. Let α be the cut of Example A.1.4. Prove that $\alpha\alpha = 2^*$. [Hint, p. 384]

A.5. The Rational Numbers as a Subfield

It remains to find $\mathbb{Q}$ inside $\mathbb{R}$. Each rational number q has its rational cut q^*. The next theorem shows that the rational cuts add, multiply and compare exactly as the rational numbers do.

Theorem A.5.1. *Let* $p, q \in \mathbb{Q}$.

(1) $p < q$ *if and only if* $p^* < q^*$.
(2) $(p+q)^* = p^* + q^*$.
(3) $(-q)^* = -(q^*)$.
(4) $(pq)^* = p^*q^*$.

Proof. (1) Let $p < q$. If $r \in p^*$, then $r < p < q$, so $r \in q^*$. Also $p \in q^*$ and $p \notin p^*$. So $p^* < q^*$. Conversely, let $p^* < q^*$. Then $p \neq q$. If $q < p$, then $q^* < p^*$ by the first half, which contradicts trichotomy. So $p < q$.

(2) If $r < p$ and $s < q$, then $r + s < p + q$. So $p^* + q^* \subset (p+q)^*$. Conversely, let $t < p + q$, and let $h = \frac{1}{2}(p + q - t) > 0$. Then $p - h \in p^*$, $q - h \in q^*$, and $(p - h) + (q - h) = t$. So $t \in p^* + q^*$.

(3) By (2), $q^* + (-q)^* = 0^*$. By Corollary A.3.5, $(-q)^* = -(q^*)$.

(4) First let $p, q > 0$. Then $p^*, q^* > 0^*$ by (1), and $(p^*)^+ = \{r : 0 < r < p\}$. If $r \in (p^*)^+$ and $s \in (q^*)^+$, then $rs < pq$. So every element of p^*q^* is less than pq, and $p^*q^* \subset (pq)^*$. Conversely, let $t < pq$. If $t \le 0$, then $t \le \frac{p}{2} \cdot \frac{q}{2}$, so $t \in p^*q^*$. If $t > 0$, let $k = \frac{t}{pq}$ and $c = \frac{1}{2}(1 + k)$, so that $0 < k < c < 1$. Let $r = cp$ and $s = \frac{t}{r} = \frac{kq}{c}$. Then $0 < r < p$, $0 < s < q$ and $rs = t$. So $t \in p^*q^*$.

If $p = 0$ or $q = 0$, both sides of (4) are 0^*. Suppose $p < 0 < q$. Then $p^* < 0^* < q^*$ by (1), and $-(p^*) = (-p)^*$ by (3). By Definition A.4.6, the positive case and (3),

$$p^*q^* = -\big((-p)^*q^*\big) = -\big(((-p)q)^*\big) = -\big((-(pq))^*\big) = (pq)^*.$$

The cases $p > 0 > q$ and $p, q < 0$ are Exercise A.5.2. □

Theorem A.5.2. *Let $\mathbb{Q}^* = \{q^* : q \in \mathbb{Q}\}$. The map $q \mapsto q^*$ is one-to-one from $\mathbb{Q}$ onto $\mathbb{Q}^*$, and $\mathbb{Q}^*$ is a subfield of $\mathbb{R}$.*

Proof. If $p \ne q$, then $p < q$ or $q < p$, so $p^* \ne q^*$ by Theorem A.5.1 (1). By Theorem A.5.1, $\mathbb{Q}^*$ contains 0^* and 1^*, and it contains the sum, the product and the negative of any of its elements. If $q \ne 0$, then $q^*\left(\frac{1}{q}\right)^* = 1^*$ by Theorem A.5.1 (4), so $(q^*)^{-1} = \left(\frac{1}{q}\right)^* \in \mathbb{Q}^*$ by Theorem 1.6.5 (7). The axioms of an ordered field hold in $\mathbb{Q}^*$, since they hold in $\mathbb{R}$. So $\mathbb{Q}^*$, with the operations and order of $\mathbb{R}$, is an ordered field contained in $\mathbb{R}$. □

So $\mathbb{Q}^*$ is a perfect copy of $\mathbb{Q}$ inside $\mathbb{R}$. Every sum, product and inequality of rational numbers matches the same statement about their cuts. From now on, identify each $q \in \mathbb{Q}$ with its cut q^*. Then $\mathbb{Q} \subset \mathbb{R}$, and $\mathbb{Q}$, with its usual operations and order, is a subfield of $\mathbb{R}$.

Strictly speaking, q and q^* are different objects. The number q is a fraction, and q^* is a set of fractions. But nothing in this book depends on what a real number is made of. It depends only on how real numbers add, multiply and compare, and in those ways q and q^* are the same.

Proof of Theorem 1.8.2. By Theorem A.4.7, $\mathbb{R}$ is an ordered field. By Theorem A.2.3, it has the least upper bound property. By Theorem A.5.2, after each $q \in \mathbb{Q}$ is identified with q^*, $\mathbb{Q}$ is a subfield of $\mathbb{R}$. □

Exercises

A.5.1. Let α be a cut and $q \in \mathbb{Q}$. Prove that $q \in \alpha$ if and only if $q^* < \alpha$. So, after identifying each rational number with its cut, every real number α is the set of rational numbers less than α. [Solution, p. 384]

A.5.2. Prove Theorem A.5.1 (4) when $p > 0 > q$ and when $p, q < 0$. [Hint, p. 384]

Notes

Richard Dedekind found the idea of a cut in 1858, while teaching calculus in Zurich. He did not want to rely on geometric intuition for the completeness of the line. He published the construction in 1872 in *Stetigkeit und irrationale Zahlen* [**20**].

Dedekind's cuts were pairs of sets, a left piece and a right piece. The version here keeps only the left piece and does not allow it a largest element, as in Rudin [**62**].

APPENDIX B

Construction of the Real Numbers by Cauchy Sequences

This appendix gives a second proof of Theorem 1.8.2, due to Georg Cantor. It builds $\mathbb{R}$ out of Cauchy sequences of rational numbers, as in Section 3.5.

In $\mathbb{Q}$, a Cauchy sequence need not converge. A sequence of rational numbers can close in on $\sqrt{2}$, with its terms getting closer and closer together, and still have no rational limit (Exercise B.1.1). Cantor's idea was to let such a sequence stand for the number it ought to converge to. Two Cauchy sequences whose difference tends to 0 should stand for the same number. So a real number will be a whole class of Cauchy sequences, all of them closing in on the same point.

As in Appendix A, only $\mathbb{Q}$ is assumed, with its usual operations and order. The results of Sections 1.5 to 1.7 hold in $\mathbb{Q}$, and nothing from Section 1.8 or later is used. In particular, every ε in this appendix is a positive rational number. Appendix A is not needed, except for Lemma A.1.5, the Archimedean Principle for $\mathbb{Q}$. In this appendix, x, y and z denote sequences $\{x_n\}$, $\{y_n\}$ and $\{z_n\}$ of rational numbers, and p, q, r and c denote rational numbers.

B.1. Cauchy Sequences of Rational Numbers

Definition B.1.1. A sequence $x = \{x_n\}$ of rational numbers is a *Cauchy sequence in* $\mathbb{Q}$ if for every rational $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $|x_m - x_n| < \varepsilon$ for all $m, n \geq N$. It is a *null sequence* if for every rational $\varepsilon > 0$, there exists an $N \in \mathbb{N}$ such that $|x_n| < \varepsilon$ for all $n \geq N$. Let $\mathcal{C}$ denote the set of all Cauchy sequences in $\mathbb{Q}$.

Sums, differences and products of sequences are taken term by term. So $x + y$ is the sequence $\{x_n + y_n\}$, and xy is $\{x_n y_n\}$. A rational number q also stands for the constant sequence $q, q, q, \ldots$. Every constant sequence is Cauchy. By Lemma A.1.5, the sequence $\left\{\frac{1}{n}\right\}$ is a null sequence. Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N\varepsilon > 1$. Then $\left|\frac{1}{n}\right| \leq \frac{1}{N} < \varepsilon$ for $n \geq N$.

Lemma B.1.2. *Let $x, y \in \mathcal{C}$, and let z and w be null sequences.*

(1) There is a rational $M > 0$ with $|x_n| \leq M$ for all $n \in \mathbb{N}$.
(2) The sequences $x + y$, $x - y$ and xy are in $\mathcal{C}$.
(3) The sequences z, $z + w$ and xz are null sequences, and $z \in \mathcal{C}$.

Proof. (1) Choose N with $|x_m - x_n| < 1$ for all $m, n \geq N$. Then $|x_n| < 1 + |x_N|$ for $n \geq N$. Let $M = \max\{|x_1|, \ldots, |x_{N-1}|, 1 + |x_N|\}$, as in the proof of Theorem 3.5.2.

(2) Let $\varepsilon > 0$. Choose M as in (1) for both x and y. Choose N with $|x_m - x_n| < \frac{\varepsilon}{2M}$ and $|y_m - y_n| < \frac{\varepsilon}{2M}$ for all $m, n \geq N$. Then for $m, n \geq N$,

$$|x_m y_m - x_n y_n| = |x_m(y_m - y_n) + y_n(x_m - x_n)| \leq M|y_m - y_n| + M|x_m - x_n| < \varepsilon.$$

So $xy \in \mathcal{C}$. The proofs for $x+y$ and $x-y$ are the same, with $\frac{\varepsilon}{2}$ in place of $\frac{\varepsilon}{2M}$ and $|(x_m \pm y_m) - (x_n \pm y_n)| \le |x_m - x_n| + |y_m - y_n|$.

(3) Let $\varepsilon > 0$. Choose N with $|z_n| < \frac{\varepsilon}{2}$ and $|w_n| < \frac{\varepsilon}{2}$ for $n \ge N$. Then $|z_n + w_n| < \varepsilon$ and $|z_m - z_n| \le |z_m| + |z_n| < \varepsilon$ for $m, n \ge N$. So $z+w$ is null and $z \in \mathcal{C}$. With M as in (1), choose N with $|z_n| < \frac{\varepsilon}{M}$ for $n \ge N$. Then $|x_n z_n| \le M|z_n| < \varepsilon$. So xz is null. □

Definition B.1.3. For $x, y \in \mathcal{C}$, write $x \sim y$ if $x - y$ is a null sequence.

Lemma B.1.4. *The relation $\sim$ is an equivalence relation on $\mathcal{C}$.*

Proof. Since $x - x$ is the constant sequence 0, $x \sim x$. If $x \sim y$, then $y \sim x$, since $|y_n - x_n| = |x_n - y_n|$. If $x \sim y$ and $y \sim z$, then $x - z = (x - y) + (y - z)$ is null by Lemma B.1.2 (3). So $x \sim z$. □

Definition B.1.5. For $x \in \mathcal{C}$, let $[x] = \{y \in \mathcal{C} : y \sim x\}$ be its equivalence class. In this appendix, $\mathbb{R}$ denotes the set of all equivalence classes $[x]$ with $x \in \mathcal{C}$. For $q \in \mathbb{Q}$, $[q]$ denotes the class of the constant sequence $q, q, q, \ldots$.

So $[x] = [y]$ exactly when $x \sim y$, and $[x] = [0]$ exactly when x is a null sequence. For example, $\left[\left\{\frac{1}{n}\right\}\right] = [0]$ and $\left[\left\{1 + \frac{1}{n}\right\}\right] = [1]$.

Exercises

B.1.1. Define $x_1 = 2$ and $x_{n+1} = \frac{x_n}{2} + \frac{1}{x_n}$. Prove that each x_n is rational, that $\{x_n\}$ is a Cauchy sequence in $\mathbb{Q}$, and that no rational number L has $|x_n - L| \to 0$. [Solution, p. 384]

B.1.2. Let x be a sequence of rational numbers, and let $L \in \mathbb{Q}$. Suppose that for every rational $\varepsilon > 0$ there is an N with $|x_n - L| < \varepsilon$ for all $n \ge N$. Prove that $x \in \mathcal{C}$ and that $[x] = [L]$. [Hint, p. 385]

B.2. Addition and Multiplication

Definition B.2.1. For $[x], [y] \in \mathbb{R}$, define $[x] + [y] = [x + y]$ and $[x][y] = [xy]$.

Lemma B.2.2. *These operations are well defined. That is, if $x \sim x'$ and $y \sim y'$, then $x + y \sim x' + y'$ and $xy \sim x'y'$.*

Proof. By Lemma B.1.2 (2), all four sequences are in $\mathcal{C}$. Now $(x+y)-(x'+y') = (x - x') + (y - y')$ is a sum of two null sequences, so it is null. Also

$$xy - x'y' = x(y - y') + y'(x - x').$$

Each term is a Cauchy sequence times a null sequence, so it is null by Lemma B.1.2 (3). So their sum is null. □

> **Proof Technique (Checking That an Operation Is Well Defined).** A real number in this appendix is a class with many representatives. The sum $[x] + [y]$ is computed from one representative of each class. If another choice of representatives gave a different class, the sum would not make sense. So every definition made through representatives needs a check that the answer does not

depend on the choice. Lemma B.2.2 is that check for sums and products, and Lemma B.3.2 (1) is that check for the order. The same issue comes up whenever objects are defined as equivalence classes, as in the definition of $\mathbb{Q}$ itself, where $\frac{1}{2} = \frac{2}{4}$.

Lemma B.2.3. *Let $x \in \mathcal{C}$ be a sequence that is not null. Then there are a rational $c > 0$ and an $N \in \mathbb{N}$ such that either $x_n \geq c$ for all $n \geq N$, or $x_n \leq -c$ for all $n \geq N$.*

Proof. Since x is not null, there is a rational $\varepsilon > 0$ such that for every N, some $n \geq N$ has $|x_n| \geq \varepsilon$. Let $c = \frac{\varepsilon}{2}$. Choose N with $|x_m - x_n| < c$ for all $m, n \geq N$, and then choose $k \geq N$ with $|x_k| \geq 2c$. If $x_k \geq 2c$, then $x_n > x_k - c \geq c$ for all $n \geq N$. If $x_k \leq -2c$, then $x_n < x_k + c \leq -c$ for all $n \geq N$. □

Theorem B.2.4. *With the operations of Definition B.2.1, $\mathbb{R}$ is a field. Its zero is $[0]$ and its one is $[1]$.*

Proof. Axioms (1) and (6) of Definition 1.6.1 hold by Lemmas B.1.2 and B.2.2. Axioms (2), (3), (7), (8) and (11) hold because they hold term by term in $\mathbb{Q}$. For example,

$$[x]\big([y] + [z]\big) = [x(y+z)] = [xy + xz] = [x][y] + [x][z],$$

since $x_n(y_n + z_n) = x_n y_n + x_n z_n$ for every n. For (4) and (5), $[0] + [x] = [x]$ and $[x] + [-x] = [0]$, where $-x = \{-x_n\}$. For (9), $[1][x] = [x]$, and $[1] \neq [0]$, since the constant sequence 1 is not null.

For (10), let $[x] \neq [0]$. Then x is not null. By Lemma B.2.3, there are a rational $c > 0$ and an N with $|x_n| \geq c$ for all $n \geq N$. Define $y_n = \frac{1}{x_n}$ for $n \geq N$ and $y_n = 0$ for $n < N$. For $m, n \geq N$,

$$|y_m - y_n| = \frac{|x_n - x_m|}{|x_m||x_n|} \leq \frac{|x_m - x_n|}{c^2}.$$

Given $\varepsilon > 0$, choose $N' \geq N$ with $|x_m - x_n| < c^2\varepsilon$ for all $m, n \geq N'$. Then $|y_m - y_n| < \varepsilon$ for $m, n \geq N'$. So $y \in \mathcal{C}$. Also $x_n y_n = 1$ for all $n \geq N$, so $xy - 1$ is null, and $[x][y] = [1]$. □

Exercises

B.2.1. Give an example of an $x \in \mathcal{C}$ with $x_n \neq 0$ for every n such that $\left\{\frac{1}{x_n}\right\}$ is not in $\mathcal{C}$. Then prove that if $[x] \neq [0]$ and $x_n \neq 0$ for every n, then $\left\{\frac{1}{x_n}\right\} \in \mathcal{C}$ and $[x]^{-1} = \left[\left\{\frac{1}{x_n}\right\}\right]$. [Solution, p. 385]

B.3. Order

Definition B.3.1. A sequence $x \in \mathcal{C}$ is *positive* if there are a rational $c > 0$ and an $N \in \mathbb{N}$ with $x_n \geq c$ for all $n \geq N$. For $[x], [y] \in \mathbb{R}$, define $[x] < [y]$ to mean that $y - x$ is positive.

Lemma B.3.2. *Let $x, y \in \mathcal{C}$.*

(1) If x is positive and $y \sim x$, then y is positive.
(2) Exactly one of the following holds. The sequence x is positive, x is null, or $-x$ is positive.
(3) If x and y are positive, so are $x + y$ and xy.

PROOF. (1) Let $x_n \geq c$ for $n \geq N$. Choose $N' \geq N$ with $|y_n - x_n| < \frac{c}{2}$ for $n \geq N'$. Then $y_n > x_n - \frac{c}{2} \geq \frac{c}{2}$ for $n \geq N'$.

(2) If x is null, then $|x_n| < c$ for large n, for every $c > 0$. So neither x nor $-x$ is positive. If x and $-x$ were both positive, then $x_n \geq c$ and $-x_n \geq c'$ for large n, which is impossible. So at most one of the three holds. If x is not null, Lemma B.2.3 shows that x or $-x$ is positive.

(3) If $x_n \geq c$ for $n \geq N$ and $y_n \geq c'$ for $n \geq N'$, then for $n \geq \max\{N, N'\}$, $x_n + y_n \geq c + c'$ and $x_n y_n \geq cc'$. □

Theorem B.3.3. *With the order of Definition B.3.1, $\mathbb{R}$ is an ordered field. An element $[x]$ is positive in the sense of Definition 1.6.6 if and only if the sequence x is positive.*

PROOF. If $x \sim x'$ and $y \sim y'$, then $y - x \sim y' - x'$. So by Lemma B.3.2 (1), whether $y - x$ is positive does not depend on the choice of x and y in their classes. So $<$ is well defined.

Next, $<$ satisfies trichotomy. To see this, let $a = [x]$ and $b = [y]$. By Lemma B.3.2 (2), exactly one of the following holds. The sequence $y - x$ is positive, $y - x$ is null, or $x - y$ is positive. These say $a < b$, $a = b$ and $b < a$.

The relation is also transitive. If $[x] < [y]$ and $[y] < [z]$, then $z - x = (z - y) + (y - x)$ is positive by Lemma B.3.2 (3). So $[x] < [z]$.

It remains to check the two properties in Definition 1.6.6. Property (1) holds since $(y + z) - (x + z) = y - x$. So $[x] \leq [y]$ implies $[x] + [z] \leq [y] + [z]$.

For property (2), let $[0] \leq [x]$ and $[0] \leq [y]$. If either is $[0]$, then $[x][y] = [0]$. Otherwise x and y are positive, so xy is positive by Lemma B.3.2 (3), and $[0] < [x][y]$.

Finally, $[0] < [x]$ means $x - 0 = x$ is positive. □

Lemma B.3.4. *Let $x, y \in \mathcal{C}$. If there is an N with $x_n \leq y_n$ for all $n \geq N$, then $[x] \leq [y]$.*

PROOF. Suppose $[y] < [x]$. Then $x - y$ is positive, so there are $c > 0$ and N' with $x_n - y_n \geq c$ for $n \geq N'$. For $n \geq \max\{N, N'\}$, this contradicts $x_n \leq y_n$. So $[x] \leq [y]$. □

Theorem B.3.5. *For all $p, q \in \mathbb{Q}$, $[p + q] = [p] + [q]$ and $[pq] = [p][q]$, and $p < q$ if and only if $[p] < [q]$. So the map $q \mapsto [q]$ is one-to-one, and $\{[q] : q \in \mathbb{Q}\}$ is a subfield of $\mathbb{R}$.*

PROOF. The first two statements hold by Definition B.2.1, since the operations on constant sequences are the operations of $\mathbb{Q}$. The class $[q] - [p] = [q - p]$ is positive exactly when the constant sequence $q - p$ is at least some $c > 0$, that is, when $q - p > 0$. So $p < q$ if and only if $[p] < [q]$. In particular, $p \neq q$ implies $[p] \neq [q]$. Also $[q]^{-1} = \left[\frac{1}{q}\right]$ for $q \neq 0$. So $\{[q] : q \in \mathbb{Q}\}$ contains $[0]$ and $[1]$ and is closed under

sums, products, negatives and reciprocals. As in the proof of Theorem A.5.2, it is a subfield of $\mathbb{R}$. □

As in Appendix A.5, identify each $q \in \mathbb{Q}$ with $[q]$. Then $\mathbb{Q}$ is a subfield of $\mathbb{R}$. In particular, if $x \in \mathcal{C}$, then each term x_n is also a real number, namely $[x_n]$.

Exercises

B.3.1. Let $x \in \mathcal{C}$, and let $|x|$ denote the sequence $\{|x_n|\}$. Prove that $|x| \in \mathcal{C}$ and that $\big|[x]\big| = \big[|x|\big]$. [Solution, p. 385]

B.3.2. Find $x, y \in \mathcal{C}$ with $x_n < y_n$ for every n but $[x] = [y]$. So Lemma B.3.4 cannot be improved to strict inequalities. [Hint, p. 385]

B.4. Completeness

The field $\mathbb{R}$ is now an ordered field containing $\mathbb{Q}$. It remains to prove the least upper bound property. The proof uses bisection, as in the proof of the Nested Interval Theorem (Theorem 3.4.4). The endpoints are rational numbers, so the sequences of endpoints are Cauchy sequences in $\mathbb{Q}$, and they define a real number. Two lemmas come first. The first says $\mathbb{Q}$ is spread throughout $\mathbb{R}$. The second says a Cauchy sequence converges to its own class.

Lemma B.4.1. *(1) For every $a \in \mathbb{R}$, there is an $n \in \mathbb{N}$ with $a < n$.*
(2) For all $a, b \in \mathbb{R}$ with $a < b$, there is a $q \in \mathbb{Q}$ with $a < q < b$.

Proof. (1) Let $a = [x]$. By Lemma B.1.2 (1), $x_n \le M$ for all n, for some rational M. By Lemma B.3.4, $a \le M$. By Lemma A.1.5, $M < n$ for some $n \in \mathbb{N}$. So $a < n$.

(2) Let $a = [x]$ and $b = [y]$. Since $y - x$ is positive, there are $c > 0$ and N_1 with $y_n - x_n \ge c$ for all $n \ge N_1$. Choose $N \ge N_1$ with $|x_m - x_n| < \frac{c}{4}$ and $|y_m - y_n| < \frac{c}{4}$ for all $m, n \ge N$. Let $q = \frac{1}{2}(x_N + y_N)$. For $n \ge N$,

$$q - x_n = \frac{1}{2}(y_N - x_N) + (x_N - x_n) > \frac{c}{2} - \frac{c}{4} = \frac{c}{4},$$

and in the same way $y_n - q = \frac{1}{2}(y_N - x_N) + (y_n - y_N) > \frac{c}{4}$. So the sequences $q - x$ and $y - q$ are positive. That is, $a < q$ and $q < b$. □

Lemma B.4.2. *Let $x \in \mathcal{C}$ and $a = [x]$. For every rational $\varepsilon > 0$, there is an $N \in \mathbb{N}$ such that $|a - x_n| \le \varepsilon$ for all $n \ge N$.*

Proof. Choose N with $|x_m - x_n| < \varepsilon$ for all $m, n \ge N$. Fix $n \ge N$. The class of the sequence $\{x_m - x_n\}_{m=1}^{\infty}$ is $a - x_n$. For $m \ge N$, $-\varepsilon < x_m - x_n < \varepsilon$. By Lemma B.3.4, $-\varepsilon \le a - x_n \le \varepsilon$. So $|a - x_n| \le \varepsilon$ by Theorem 1.7.2 (4). □

Theorem B.4.3. *The ordered field $\mathbb{R}$ has the least upper bound property.*

Proof. Let $E \subset \mathbb{R}$ be nonempty and bounded above, say by B. Choose $e_0 \in E$. By Lemma B.4.1 (1), there are $j, k \in \mathbb{N}$ with $B < k$ and $-e_0 < j$. Let $a_0 = -j$ and $b_0 = k$. Then $a_0 < e_0 \le B < b_0$. So a_0 is not an upper bound of E, and b_0 is an upper bound of E.

The proof finds $\sup E$ by bisection. Suppose rational numbers $a_n < b_n$ have been chosen, where a_n is not an upper bound of E and b_n is. Let $m = \frac{1}{2}(a_n + b_n)$. If m is an upper bound of E, let $a_{n+1} = a_n$ and $b_{n+1} = m$. Otherwise let $a_{n+1} = m$ and $b_{n+1} = b_n$. By induction, for all $n \in \mathbb{N}_0$, a_n is not an upper bound of E, b_n is an upper bound of E,

$$a_n \le a_{n+1} < b_{n+1} \le b_n, \quad \text{and} \quad b_n - a_n = \frac{b_0 - a_0}{2^n}.$$

Next, the endpoints form Cauchy sequences in $\mathbb{Q}$. To see this, let $d = b_0 - a_0$. If $m \ge n \ge 1$, then $a_n \le a_m < b_m \le b_n$. So $|a_m - a_n|$ and $|b_m - b_n|$ are at most $\frac{d}{2^n} < \frac{d}{n}$, using $2^n > n$ (Exercise 1.2.9). Given $\varepsilon > 0$, choose $N \in \mathbb{N}$ with $N\varepsilon > d$ by Lemma A.1.5. Then $|a_m - a_n| < \varepsilon$ and $|b_m - b_n| < \varepsilon$ for all $m, n \ge N$. Also $b_n - a_n < \frac{d}{n}$, so $\{b_n - a_n\}$ is null. Thus the sequences $a = \{a_n\}$ and $b = \{b_n\}$ are in $\mathcal{C}$, and $a \sim b$. Let $s = [a] = [b]$.

The class s is the candidate for $\sup E$. First, s is an upper bound of E. To see this, let $e \in E$, and suppose $e > s$. By Lemma B.4.1 (2), there is a rational ε with $0 < \varepsilon < e - s$. By Lemma B.4.2, there is an n with $|s - b_n| \le \varepsilon$. Then $b_n \le s + \varepsilon < e$. This contradicts the fact that b_n is an upper bound of E. So $e \le s$.

Finally, no smaller number is an upper bound of E. To see this, let $t < s$. Choose a rational ε with $0 < \varepsilon < s - t$, and an n with $|s - a_n| \le \varepsilon$. Then $a_n \ge s - \varepsilon > t$. Since a_n is not an upper bound of E, some $e \in E$ has $e > a_n > t$. So t is not an upper bound of E. Therefore $s = \sup E$. □

Second proof of Theorem 1.8.2. By Theorem B.3.3, $\mathbb{R}$ is an ordered field. By Theorem B.4.3, it has the least upper bound property. By Theorem B.3.5, after each $q \in \mathbb{Q}$ is identified with $[q]$, $\mathbb{Q}$ is a subfield of $\mathbb{R}$. □

Exercises

B.4.1. Let $\{a_k\}$ be a Cauchy sequence in $\mathbb{R}$, in the sense of Definition 3.5.1. Prove directly that it converges, without the least upper bound property. Choose $q_k \in \mathbb{Q}$ with $|a_k - q_k| < \frac{1}{k}$, show that $q = \{q_k\}$ is in $\mathcal{C}$, and show that $a_k \to [q]$. [Solution, p. 385]

B.4.2. Prove that every $a \in \mathbb{R}$ is the limit of a strictly increasing sequence of rational numbers. [Hint, p. 386]

B.5. Uniqueness of the Real Numbers*

This section can be skipped. Nothing else depends on it.

Appendices A and B build two different objects. A real number is a set of rational numbers in one, and a class of sequences of rational numbers in the other. This section shows that the difference does not matter. Any two ordered fields with the least upper bound property are copies of each other. Every sum, product and inequality in one matches a sum, product and inequality in the other. So Theorem 1.8.2 describes $\mathbb{R}$ completely, and it makes sense to speak of the real numbers.

First, every ordered field contains a copy of $\mathbb{Q}$. For an ordered field $\mathbb{F}$ and $n \in \mathbb{N}$, let $n_{\mathbb{F}} = 1 + 1 + \cdots + 1$, with n terms. That is, $1_{\mathbb{F}} = 1$ and $(n+1)_{\mathbb{F}} = n_{\mathbb{F}} + 1$. Let $0_{\mathbb{F}} = 0$ and $(-n)_{\mathbb{F}} = -n_{\mathbb{F}}$. This defines $m_{\mathbb{F}}$ for every $m \in \mathbb{Z}$.

Lemma B.5.1. *Let $\mathbb{F}$ be an ordered field. For $m \in \mathbb{Z}$ and $n \in \mathbb{N}$, define $\iota_{\mathbb{F}}\left(\frac{m}{n}\right) = m_{\mathbb{F}}\, n_{\mathbb{F}}^{-1}$. Then $\iota_{\mathbb{F}} : \mathbb{Q} \to \mathbb{F}$ is well defined, and for all $p, q \in \mathbb{Q}$,*

$$\iota_{\mathbb{F}}(p+q) = \iota_{\mathbb{F}}(p) + \iota_{\mathbb{F}}(q), \quad \iota_{\mathbb{F}}(pq) = \iota_{\mathbb{F}}(p)\,\iota_{\mathbb{F}}(q), \quad \text{and} \quad p < q \iff \iota_{\mathbb{F}}(p) < \iota_{\mathbb{F}}(q).$$

Proof. For all $m, k \in \mathbb{Z}$, $(m+k)_{\mathbb{F}} = m_{\mathbb{F}} + k_{\mathbb{F}}$ and $(mk)_{\mathbb{F}} = m_{\mathbb{F}}\, k_{\mathbb{F}}$, and $m < k$ implies $m_{\mathbb{F}} < k_{\mathbb{F}}$. These are proved by induction, in Exercise B.5.1. In particular, $n_{\mathbb{F}} > 0$ for $n \in \mathbb{N}$, so $n_{\mathbb{F}}^{-1}$ exists.

If $\frac{m}{n} = \frac{j}{k}$ with $n, k \in \mathbb{N}$, then $mk = nj$. So $m_{\mathbb{F}}\, k_{\mathbb{F}} = n_{\mathbb{F}}\, j_{\mathbb{F}}$, and dividing by $n_{\mathbb{F}}\, k_{\mathbb{F}}$ gives $m_{\mathbb{F}}\, n_{\mathbb{F}}^{-1} = j_{\mathbb{F}}\, k_{\mathbb{F}}^{-1}$. So $\iota_{\mathbb{F}}$ is well defined. For sums,

$$\iota_{\mathbb{F}}\Big(\frac{m}{n} + \frac{j}{k}\Big) = \iota_{\mathbb{F}}\Big(\frac{mk + nj}{nk}\Big) = \frac{m_{\mathbb{F}}\, k_{\mathbb{F}} + n_{\mathbb{F}}\, j_{\mathbb{F}}}{n_{\mathbb{F}}\, k_{\mathbb{F}}} = \frac{m_{\mathbb{F}}}{n_{\mathbb{F}}} + \frac{j_{\mathbb{F}}}{k_{\mathbb{F}}}.$$

Products are similar. For the order, $\frac{m}{n} < \frac{j}{k}$ if and only if $mk < nj$, which holds if and only if $m_{\mathbb{F}}\, k_{\mathbb{F}} < n_{\mathbb{F}}\, j_{\mathbb{F}}$. Since $n_{\mathbb{F}}\, k_{\mathbb{F}} > 0$, this holds if and only if $\frac{m_{\mathbb{F}}}{n_{\mathbb{F}}} < \frac{j_{\mathbb{F}}}{k_{\mathbb{F}}}$, by Theorem 1.6.8 (3) and (6). □

If $\mathbb{F}$ also has the least upper bound property, then the proofs of the Archimedean Principle (Theorem 1.8.8), of Theorem 1.8.9, and of the density of the rational numbers (Theorem 1.8.11) work in $\mathbb{F}$ word for word. They use only the axioms of an ordered field, the least upper bound property and the Well-Ordering Principle, with each integer n read as $n_{\mathbb{F}}$. So for all $x, y \in \mathbb{F}$ with $x < y$, there is a $q \in \mathbb{Q}$ with $x < \iota_{\mathbb{F}}(q) < y$. It follows that two elements x and y of $\mathbb{F}$ are equal if and only if

$$\{q \in \mathbb{Q} : \iota_{\mathbb{F}}(q) < x\} = \{q \in \mathbb{Q} : \iota_{\mathbb{F}}(q) < y\}.$$

Indeed, if $x < y$, a rational number between them belongs to the second set and not the first. So an element of $\mathbb{F}$ is determined by the rational numbers below it.

Theorem B.5.2. *Let $\mathbb{F}$ and $\mathbb{G}$ be ordered fields with the least upper bound property. Then there is a one-to-one function φ from $\mathbb{F}$ onto $\mathbb{G}$ such that for all $x, y \in \mathbb{F}$,*

$$\varphi(x+y) = \varphi(x) + \varphi(y), \quad \varphi(xy) = \varphi(x)\varphi(y), \quad \text{and} \quad x < y \iff \varphi(x) < \varphi(y).$$

Also $\varphi(\iota_{\mathbb{F}}(q)) = \iota_{\mathbb{G}}(q)$ for every $q \in \mathbb{Q}$.

Proof. Write $\iota = \iota_{\mathbb{F}}$ and $\kappa = \iota_{\mathbb{G}}$. For $x \in \mathbb{F}$, let $C(x) = \{q \in \mathbb{Q} : \iota(q) < x\}$. By the Archimedean Principle in $\mathbb{F}$, there are $j, k \in \mathbb{N}$ with $-j_{\mathbb{F}} < x < k_{\mathbb{F}}$. So $-j \in C(x)$, and every $q \in C(x)$ has $q < k$, so $\kappa(q) < k_{\mathbb{G}}$. Define

$$\varphi(x) = \sup\{\kappa(q) : q \in C(x)\},$$

which exists in $\mathbb{G}$ by the least upper bound property.

The proof has six steps. The first step shows that for $q \in \mathbb{Q}$, $\kappa(q) < \varphi(x)$ if and only if $\iota(q) < x$. Suppose $\iota(q) < x$. By density in $\mathbb{F}$, there is an $r \in \mathbb{Q}$ with $\iota(q) < \iota(r) < x$. Then $r \in C(x)$ and $q < r$, so $\kappa(q) < \kappa(r) \le \varphi(x)$. Conversely, suppose $\kappa(q) < \varphi(x)$. Then $\kappa(q)$ is not an upper bound of $\{\kappa(r) : r \in C(x)\}$. So $\kappa(q) < \kappa(r)$ for some $r \in C(x)$. Then $q < r$, and $\iota(q) < \iota(r) < x$.

The second step shows that φ sends each rational number of $\mathbb{F}$ to the matching rational number of $\mathbb{G}$. For $p, q \in \mathbb{Q}$, the first step gives $\kappa(q) < \varphi(\iota(p))$ if and only if $\iota(q) < \iota(p)$, that is, $q < p$, that is, $\kappa(q) < \kappa(p)$. So $\varphi(\iota(p))$ and $\kappa(p)$ have the same rational numbers below them, and $\varphi(\iota(p)) = \kappa(p)$. In particular $\varphi(0) = 0$.

The third step shows that φ preserves order. Let $x < y$. By density in $\mathbb{F}$, there is a $q \in \mathbb{Q}$ with $x < \iota(q) < y$. By the first step, $\kappa(q) < \varphi(y)$, and $\kappa(q) \not< \varphi(x)$. So $\varphi(x) \leq \kappa(q) < \varphi(y)$. Conversely, if $\varphi(x) < \varphi(y)$, then $x < y$, since $x = y$ and $y < x$ are ruled out. In particular, φ is one-to-one.

The fourth step shows that φ is onto. Define $\psi : \mathbb{G} \to \mathbb{F}$ in the same way, with the roles of $\mathbb{F}$ and $\mathbb{G}$ exchanged. By the first step for ψ and then for φ, for every $q \in \mathbb{Q}$,

$$\iota(q) < \psi(\varphi(x)) \iff \kappa(q) < \varphi(x) \iff \iota(q) < x.$$

So $\psi(\varphi(x)) = x$ for every $x \in \mathbb{F}$. In the same way, $\varphi(\psi(g)) = g$ for every $g \in \mathbb{G}$. So φ maps $\mathbb{F}$ onto $\mathbb{G}$.

The fifth step shows that φ preserves sums. Let $x, y \in \mathbb{F}$ and $q \in \mathbb{Q}$. Suppose $\kappa(q) < \varphi(x) + \varphi(y)$. Then $\kappa(q) - \varphi(y) < \varphi(x)$. By density in $\mathbb{G}$, there is an $r \in \mathbb{Q}$ with $\kappa(q) - \varphi(y) < \kappa(r) < \varphi(x)$. Then $\kappa(q - r) < \varphi(y)$ and $\kappa(r) < \varphi(x)$. By the first step, $\iota(q - r) < y$ and $\iota(r) < x$. Adding, $\iota(q) < x + y$, and so $\kappa(q) < \varphi(x + y)$ by the first step. The same argument in $\mathbb{F}$ shows that $\kappa(q) < \varphi(x + y)$ implies $\kappa(q) < \varphi(x) + \varphi(y)$. So $\varphi(x + y) = \varphi(x) + \varphi(y)$. In particular, $\varphi(-x) = -\varphi(x)$, since $\varphi(x) + \varphi(-x) = \varphi(0) = 0$.

The sixth step shows that φ preserves products. First let $x, y > 0$. Then $\varphi(x), \varphi(y) > 0$ by the second and third steps. Let $q \in \mathbb{Q}$. If $q \leq 0$, then $\kappa(q) \leq 0 < \varphi(x)\varphi(y)$, and $\iota(q) \leq 0 < xy$, so $\kappa(q) < \varphi(xy)$. Now let $q > 0$, and suppose $\kappa(q) < \varphi(x)\varphi(y)$. Then $0 < \kappa(q)\varphi(y)^{-1} < \varphi(x)$. By density in $\mathbb{G}$, there is an $r \in \mathbb{Q}$ with $\kappa(q)\varphi(y)^{-1} < \kappa(r) < \varphi(x)$. Then $r > 0$, $\kappa(r) < \varphi(x)$ and $\kappa\left(\frac{q}{r}\right) < \varphi(y)$. By the first step, $0 < \iota(r) < x$ and $0 < \iota\left(\frac{q}{r}\right) < y$. Multiplying, $\iota(q) < xy$, so $\kappa(q) < \varphi(xy)$. The same argument in $\mathbb{F}$ gives the converse. So $\varphi(xy) = \varphi(x)\varphi(y)$ when $x, y > 0$. The other cases follow from this one, since $\varphi(0) = 0$ and $\varphi(-x) = -\varphi(x)$. For example, if $x < 0 < y$, then $\varphi(xy) = -\varphi((-x)y) = -\varphi(-x)\varphi(y) = \varphi(x)\varphi(y)$. □

Applied to the ordered fields of Appendices A and B, Theorem B.5.2 matches each cut with a class of Cauchy sequences. The cut of rational numbers below $\sqrt{2}$ is matched with the class of every sequence of rational numbers that converges to $\sqrt{2}$. Since the two constructions give copies of the same ordered field, the rest of the book never needs to know which one was used.

Exercises

B.5.1. Let $\mathbb{F}$ be an ordered field. Prove that for all $m, k \in \mathbb{Z}$, $(m + k)_{\mathbb{F}} = m_{\mathbb{F}} + k_{\mathbb{F}}$, $(mk)_{\mathbb{F}} = m_{\mathbb{F}}\, k_{\mathbb{F}}$, and $m < k$ implies $m_{\mathbb{F}} < k_{\mathbb{F}}$. [Solution, p. 386]

B.5.2. Let $\varphi : \mathbb{R} \to \mathbb{R}$ be one-to-one and onto, with $\varphi(x + y) = \varphi(x) + \varphi(y)$ and $\varphi(xy) = \varphi(x)\varphi(y)$ for all $x, y \in \mathbb{R}$. Prove that $\varphi(x) = x$ for every $x \in \mathbb{R}$. First show that $\varphi(q) = q$ for $q \in \mathbb{Q}$. Then use Theorem 1.8.13 to show that φ preserves order. [Hint, p. 386]

Notes

Charles Méray gave a construction of the real numbers from Cauchy sequences of rational numbers in 1869. Cantor gave his in 1872, in a paper on trigonometric series [**8**], the same year as Dedekind's cuts. Eduard Heine published Cantor's construction, with credit to Cantor, in the same year [**38**]. The same idea, applied to an arbitrary metric space instead of $\mathbb{Q}$, gives the completion of a metric space. Theorem B.5.2 shows that the least upper bound property, together with the axioms of an ordered field, describes $\mathbb{R}$ completely. Edward V. Huntington gave one of the first such complete sets of axioms in 1903.

APPENDIX C

Hints and Solutions

This appendix gives hints and solutions for the exercises marked [Hint] or [Solution] in the text. They appear in the same order as the exercises. Most solutions come from homework and tests in MAT 315 and MAT 815. Try each exercise before reading its hint, and use the hint before reading a solution. There is almost always more than one correct proof, so a proof that differs from the one here may still be right. Each heading links back to its exercise.

Chapter 1. The Real Number System

Exercise 1.1.1. $A \cup B = \{1,2,3,4,6,8\}$, $A \cap C = \{3\}$, $B \setminus C = \{2,4,8\}$, and $(A \cup B)^c = X \setminus (A \cup B) = \{5,7\}$. Since $A^c = \{4,5,6,7,8\}$, $A^c \cap C = \{6\}$.

Exercise 1.1.3. Let $x \in (A \cap B)^c$. Then $x \in X$ and $x \notin A \cap B$. So $x \notin A$ or $x \notin B$. If $x \notin A$, then $x \in A^c$. If $x \notin B$, then $x \in B^c$. Either way, $x \in A^c \cup B^c$. So $(A \cap B)^c \subset A^c \cup B^c$.

Now let $x \in A^c \cup B^c$. If $x \in A^c$, then $x \notin A$, so $x \notin A \cap B$. If $x \in B^c$, then $x \notin B$, so again $x \notin A \cap B$. Either way, $x \in (A \cap B)^c$. So $A^c \cup B^c \subset (A \cap B)^c$. By Definition 1.1.3, $(A \cap B)^c = A^c \cup B^c$.

Exercise 1.1.5. Let $x \in A$. Since $A \subset B$, $x \in B$. Since $B \subset C$, $x \in C$. So every element of A is an element of C, and $A \subset C$.

Exercise 1.1.7. Let $A = \{1,2\}$, $B = \{1\}$ and $C = \{2\}$. Then $A \cup B = \{1,2\} = A \cup C$, but $B \neq C$.

Exercise 1.2.2. Let m be even and n be odd. Then $m = 2j$ and $n = 2k+1$ for some $j, k \in \mathbb{Z}$. So $m + n = 2j + 2k + 1 = 2(j+k) + 1$. Since $j + k \in \mathbb{Z}$, $m + n$ is odd.

Exercise 1.2.3. ($\Rightarrow$) Suppose n is odd. Then $n = 2k+1$ for some $k \in \mathbb{Z}$. So $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$. Since $2k^2 + 2k \in \mathbb{Z}$, n^2 is odd.

($\Leftarrow$) Use the contrapositive. Suppose n is not odd. By Corollary 1.2.13, n is even. By Theorem 1.2.5 (3), $n^2 = n \cdot n$ is even. By Corollary 1.2.13 again, n^2 is not odd. So if n^2 is odd, then n is odd.

Exercise 1.2.4. *Hint.* Write $n = 3q + r$ with $r \in \{0,1,2\}$, and compute n^2 in each case.

Exercise 1.2.6. For $n = 1$, the left side is 1 and the right side is $\frac{1\cdot 2\cdot 3}{6} = 1$. Now suppose the formula holds for some $k \in \mathbb{N}$. Then

$$\sum_{i=1}^{k+1} i^2 = \frac{k(k+1)(2k+1)}{6} + (k+1)^2$$
$$= \frac{(k+1)\big(2k^2+k+6k+6\big)}{6} = \frac{(k+1)(k+2)(2k+3)}{6}.$$

This is the formula with $n = k+1$, since $k+2 = (k+1)+1$ and $2k+3 = 2(k+1)+1$. By the Principle of Mathematical Induction, the formula holds for all $n \in \mathbb{N}$.

Exercise 1.2.9. For $n = 1$, $2^1 = 2 > 1$. Now suppose $2^k > k$ for some $k \in \mathbb{N}$. Then $2^{k+1} = 2 \cdot 2^k > 2k = k + k \geq k + 1$, since $k \geq 1$. So $2^{k+1} > k + 1$. By the Principle of Mathematical Induction, $2^n > n$ for all $n \in \mathbb{N}$.

Exercise 1.2.10. *Hint.* For the inductive step, use $(k+1)! = (k+1)\,k!$ and $k+1 \geq 2$.

Exercise 1.3.1. Since p divides $n^2 = n \cdot n$, Lemma 1.3.3 with $a = b = n$ shows that p divides n.

Exercise 1.3.2. *Hint.* Write $n = dm$ with $2 \leq d \leq m$. Then $d^2 \leq n$, and any prime factor of d is a prime factor of n. For 97, only 2, 3, 5 and 7 need to be checked.

Exercise 1.4.1. *Hint.* Use Exercise 1.2.4.

Exercise 1.4.2. *Hint.* Follow the proof of Theorem 1.4.4. If $6n^2 = m^2$, show that m and then n must be even.

Exercise 1.4.3. Define $c = \frac{p+q}{2}$. Then $c \in \mathbb{Q}$ and $p < c < q$. Indeed, $p + q \in \mathbb{Q}$, and the quotient of a rational number by 2 is rational. Since $p < q$, $2p = p + p < p + q$, so $p < c$. Similarly $p + q < q + q = 2q$, so $c < q$.

Exercise 1.4.4. Assume by way of contradiction that $q \in \mathbb{Q}$ and $q^2 = p$. Since $(-q)^2 = q^2$ and $0^2 \neq p$, assume $q > 0$. Write $q = \frac{m}{n}$ with $m, n \in \mathbb{N}$. Then $m^2 = pn^2$, and $m \geq 2$ since $m^2 \geq p \geq 2$. Count how many times p appears in prime factorizations. In m^2, each prime appears twice as often as in m, so p appears an even number of times. In pn^2, p appears one more time than in n^2, so an odd number of times. This contradicts the uniqueness in Theorem 1.3.4.

Exercise 1.5.2. Let $\alpha = \inf E$ and $\beta = \sup E$. Since $E \neq \emptyset$, there is an $x \in E$. Since α is a lower bound of E, $\alpha \leq x$. Since β is an upper bound of E, $x \leq \beta$. If $\alpha = x$ or $x = \beta$, then $\alpha \leq \beta$ follows at once. Otherwise $\alpha < x$ and $x < \beta$, so $\alpha < \beta$ by transitivity. Either way, $\inf E \leq \sup E$.

Exercise 1.5.3. *Hint.* To show 1 is the least upper bound, suppose $q < 1$ is an upper bound and find an n with $1 - \frac{1}{n} > q$.

Exercise 1.6.1. (2) Now $x + y = x = x + 0$, so by Theorem 1.6.5 (1), $y = 0$.

(3) Now $x + y = 0 = x + (-x)$, so by Theorem 1.6.5 (1), $y = -x$.

(4) Now $-x + x = 0$, so by Theorem 1.6.5 (3), applied to $-x$ in place of x and x in place of y, $x = -(-x)$.

Exercise 1.6.2. (2) Suppose $x < y$. Adding $(-x) + (-y)$ to both sides gives $-y < -x$, by the axiom that $x < y$ implies $x + z < y + z$. Conversely, if $-y < -x$, adding $x + y$ to both sides gives $x < y$.

(3) Since $z > y$, $z + (-y) > y + (-y) = 0$. Thus $x(z + (-y)) > 0$, implying $xz = x(z + (-y)) + xy > 0 + xy = xy$.

(4) Now $-(x(z + (-y))) = (-x)(z + (-y)) > 0$, and so $x(z + (-y)) < 0$. Thus $xz = x(z + (-y)) + xy < 0 + xy = xy$.

Exercise 1.6.6. *Hint.* In the inductive step, multiply $(1+x)^k \geq 1+kx$ by $1+x \geq 0$.

Exercise 1.6.7. Let $E \subset \mathbb{F}$, where $E \neq \emptyset$ and E is bounded below. Since $\mathbb{F}$ is a field, for each $x \in E$, $-x \in \mathbb{F}$. Since $E \neq \emptyset$, there is an $x \in E$. So $-x \in -E$ and $-E$ is not empty. Since E is bounded below, there exists an $\alpha \in \mathbb{F}$ such that for all $x \in E$, $\alpha \leq x$. Again, since $\mathbb{F}$ is a field, $-\alpha \in \mathbb{F}$. Since for all $x \in E$, $\alpha \leq x$, for all $-x \in -E$, $-x \leq -\alpha$. So $-\alpha$ is an upper bound of $-E$. This proves (a). Since $\mathbb{F}$ has the least upper bound property, $\sup(-E) \in \mathbb{F}$, say $\beta = \sup(-E)$.

For (b), first note that $-\beta$ is a lower bound of E. Indeed, if $x \in E$, then $-x \in -E$, so $-x \leq \beta$ and $x \geq -\beta$. Suppose $y > -\beta$. Then $-y < \beta$ and so $-y$ is not an upper bound of $-E$, implying there exists a $z \in -E$ with $-y < z$. So $y > -z$ and $-z \in E$. Thus y is not a lower bound of E. So $-\beta = \inf E$.

Exercise 1.7.2. Write $x - y = (x - z) + (z - y)$. By the Triangle Inequality (Theorem 1.7.3 (2)),

$$|x - y| = |(x - z) + (z - y)| \leq |x - z| + |z - y|.$$

Exercise 1.8.1. *Hint.* If $x \neq 0$, try $\varepsilon = \frac{|x|}{2}$.

Solution. Suppose $x \neq 0$. Then $|x| > 0$ by Theorem 1.7.2 (1). Let $\varepsilon = \frac{|x|}{2} > 0$. By hypothesis, $|x| \leq \frac{|x|}{2}$. Subtracting $\frac{|x|}{2}$ from both sides gives $\frac{|x|}{2} \leq 0$, so $|x| \leq 0$. This contradicts $|x| > 0$. So $x = 0$.

Exercise 1.8.3. *Hint.* Show -1 is a lower bound. Then, for $\varepsilon > 0$, find an odd n with $-\frac{n}{n+1} < -1 + \varepsilon$.

Solution. First, -1 is a lower bound of E. If n is even, then $(-1)^n \frac{n}{n+1} = \frac{n}{n+1} \geq 0 > -1$. If n is odd, then $(-1)^n \frac{n}{n+1} = -\frac{n}{n+1} > -1$, since $\frac{n}{n+1} < 1$.

Now let $\gamma > -1$, and set $\varepsilon = \gamma + 1 > 0$. By the Archimedean Principle (Theorem 1.8.8), there is an $N \in \mathbb{N}$ with $N > \frac{1}{\varepsilon}$. Let $n = 2N - 1$, which is odd. Then

$$(-1)^n \frac{n}{n+1} = -\frac{2N-1}{2N} = -1 + \frac{1}{2N} < -1 + \frac{1}{N} < -1 + \varepsilon = \gamma.$$

So γ is not a lower bound of E. By Definition 1.5.4, $\inf E = -1$.

Exercise 1.8.4. *Hint.* Argue by contradiction. If $r + x = q \in \mathbb{Q}$, then $x = q - r$.

Exercise 1.8.5. *Hint.* (a) If $p < q$ are rational, try $p + \frac{q-p}{\sqrt{2}}$.

Exercise 1.8.6. *Hint.* Let $s = \sup A$. Rule out $s^2 < 2$ and $s^2 > 2$, using the ideas in the proof of Theorem 1.4.5.

Exercise 1.8.7. First, let $u_0 \in U$. Then for all $l \in L$, $l < u_0$. So u_0 is an upper bound for L, and since L is nonempty, L has a supremum in $\mathbb{R}$. Similarly, let $l_0 \in L$.

Then for all $u \in U$, $l_0 < u$. So l_0 is a lower bound for U, and since U is nonempty, U has an infimum in $\mathbb{R}$ by Exercise 1.6.7.

Now suppose $\sup L \notin L$. So, since $\mathbb{R} = L \cup U$, $\sup L \in U$. Note for $u \in U$, $u > l$ for all $l \in L$, so each $u \in U$ is an upper bound of L. So $\sup L \leq u$ for all $u \in U$. So $\sup L$ is a lower bound of U. Let k be a lower bound of U. Since $\sup L \in U$, $k \leq \sup L$. Thus for any lower bound k of U, $k \leq \sup L$. So $\sup L = \inf U$. Thus $\inf U \in U$. So if $\sup L \notin L$, then $\inf U \in U$, which is what was to be proved.

Chapter 2. Cardinality

Exercise 2.1.1. Suppose $f(n) = f(k)$. Now $f(n) > 0$ or $f(n) \leq 0$. First, suppose $f(n) > 0$. Then $f(k) > 0$, so n and k are both even, and $f(n) = \frac{n}{2} = \frac{k}{2} = f(k)$. So $k = n$. Now suppose $f(n) \leq 0$. Then $f(k) \leq 0$, so n and k are both odd, and so $f(n) = -\frac{n-1}{2} = -\frac{k-1}{2} = f(k)$. So $k = n$. Thus f is an injection.

Now let $n \in \mathbb{Z}$. If $n > 0$, $2n \in \mathbb{N}$ and $f(2n) = \frac{2n}{2} = n$. If $n \leq 0$, $-2n + 1 \in \mathbb{N}$ and $f(-2n + 1) = -\frac{-2n+1-1}{2} = n$. So f is a surjection. Therefore, f is a bijection and $\mathbb{N} \sim \mathbb{Z}$. So $\mathbb{Z}$ is countable.

Exercise 2.1.2. Define $g : \mathbb{Z} \to \mathbb{O}$ by $g(k) = 2k - 1$. Each $2k - 1 = 2(k - 1) + 1$ is odd, so g maps into $\mathbb{O}$. If $g(k) = g(j)$, then $2k - 1 = 2j - 1$, so $k = j$. So g is one-to-one. If $m \in \mathbb{O}$, then $m = 2j + 1$ for some $j \in \mathbb{Z}$, and $g(j + 1) = m$. So g is onto. Thus $\mathbb{Z} \sim \mathbb{O}$. By Exercise 2.1.1, $\mathbb{N} \sim \mathbb{Z}$. By Theorem 2.1.8 (3), $\mathbb{N} \sim \mathbb{O}$, so $\mathbb{O}$ is countable.

The same idea shows that the odd *positive* integers are countable. The function $n \mapsto \frac{n+1}{2}$ is a bijection from them onto $\mathbb{N}$.

Exercise 2.1.5. *Hint.* The function cannot be one-to-one.

Exercise 2.2.1. *Hint.* Use induction on n and Theorem 2.2.1, since $\mathbb{N}^{n+1} \sim \mathbb{N}^n \times \mathbb{N}$.

Solution. Use induction on n. For $n = 1$, $\mathbb{N}^1 = \mathbb{N}$, and the identity map is a bijection. So $\mathbb{N}^1$ is countable.

Now suppose $\mathbb{N}^n$ is countable, and let $g : \mathbb{N}^n \to \mathbb{N}$ be a bijection. Define $h : \mathbb{N}^{n+1} \to \mathbb{N} \times \mathbb{N}$ by

$$h(x_1, \ldots, x_n, x_{n+1}) = \big(g(x_1, \ldots, x_n), x_{n+1}\big).$$

If $h(x) = h(y)$, then $g(x_1, \ldots, x_n) = g(y_1, \ldots, y_n)$ and $x_{n+1} = y_{n+1}$. Since g is one-to-one, $x_i = y_i$ for $i = 1, \ldots, n$. So $x = y$ and h is one-to-one. If $(a, b) \in \mathbb{N} \times \mathbb{N}$, let $(x_1, \ldots, x_n) = g^{-1}(a)$ and $x_{n+1} = b$. Then $h(x_1, \ldots, x_{n+1}) = (a, b)$. So h is onto. Thus $\mathbb{N}^{n+1} \sim \mathbb{N} \times \mathbb{N}$. By Theorem 2.2.1, $\mathbb{N} \times \mathbb{N} \sim \mathbb{N}$. By Theorem 2.1.8 (3), $\mathbb{N}^{n+1} \sim \mathbb{N}$. By induction, $\mathbb{N}^n$ is countable for every $n \in \mathbb{N}$.

Exercise 2.3.2. *Hint.* If $\mathbb{R} \setminus \mathbb{Q}$ were countable, then $\mathbb{R} = \mathbb{Q} \cup (\mathbb{R} \setminus \mathbb{Q})$ would be countable by Theorem 2.2.6.

Solution. Suppose $\mathbb{R} \setminus \mathbb{Q}$ is countable. Let $E_1 = \mathbb{Q}$ and $E_n = \mathbb{R} \setminus \mathbb{Q}$ for $n \geq 2$. Each E_n is countable, so $\mathbb{R} = \bigcup_{n=1}^{\infty} E_n$ is countable by Theorem 2.2.6. Then $[0, 1]$ is an infinite subset of a countable set. By Theorem 2.2.4, $[0, 1]$ is countable. This contradicts Theorem 2.3.2. So $\mathbb{R} \setminus \mathbb{Q}$ is uncountable.

Exercise 2.4.1. *Hint.* Use the identity function, the inclusion $a \mapsto a$, and a composition of one-to-one functions.

Exercise 2.4.2. *Hint.* Each of the four sets lies between (a, b) and $[a, b]$. Use $x \mapsto a + (b - a)x$ and Example 2.4.3.

Exercise 2.4.3. Since $|x| < 1 + |x|$, $|f(x)| < 1$, so f maps $\mathbb{R}$ into $(-1, 1)$. Also $f(x)$ has the same sign as x, and $|f(x)| = \frac{|x|}{1+|x|}$.

To see that f is one-to-one, let $f(x) = f(y)$. Then x and y have the same sign, and $\frac{|x|}{1+|x|} = \frac{|y|}{1+|y|}$. Cross-multiplying gives $|x| + |x||y| = |y| + |x||y|$, so $|x| = |y|$. So $x = y$.

To see that f is onto, let $y \in (-1, 1)$, and let $x = \frac{y}{1-|y|}$. Then x has the sign of y and $|x| = \frac{|y|}{1-|y|}$, so $1 + |x| = \frac{1}{1-|y|}$ and $f(x) = \frac{y/(1-|y|)}{1/(1-|y|)} = y$.

So $\mathbb{R} \sim (-1, 1)$. The function $g(x) = \frac{x+1}{2}$ is a bijection from $(-1, 1)$ to $(0, 1)$. By Example 2.4.3, $(0, 1) \sim [0, 1]$, and by Theorem 2.4.4, $[0, 1] \sim 2^{\mathbb{N}}$. By Theorem 2.1.8, $\mathbb{R} \sim [0, 1] \sim 2^{\mathbb{N}}$.

Exercise 2.4.4. Let $\Phi(E) = \chi_E$. If $E \neq F$, there is a k in one set and not the other, and then $\chi_E(k) \neq \chi_F(k)$. So Φ is one-to-one. Let $s \in 2^{\mathbb{N}}$, and let $E = \{k \in \mathbb{N} : s(k) = 1\}$. Then $\chi_E = s$. So Φ is onto, and $\mathcal{P}(\mathbb{N}) \sim 2^{\mathbb{N}}$. By Exercise 2.4.3 and Theorem 2.1.8, $\mathcal{P}(\mathbb{N}) \sim \mathbb{R}$.

Exercise 2.4.5. *Hint.* If $A_m \sim A_n$ with $m < n$, show $\mathcal{P}(A_m) \lesssim A_m$. Then use Theorems 2.4.2 and 2.4.5.

Exercise 2.4.6. *Hint.* To invert the map, read off the odd-numbered and even-numbered terms of u. Then use Exercise 2.4.3.

Solution. Call the map Ψ. If $\Psi(s, t) = \Psi(s', t') = u$, then $s(k) = u(2k-1) = s'(k)$ and $t(k) = u(2k) = t'(k)$ for all k. So $(s, t) = (s', t')$. Given $u \in 2^{\mathbb{N}}$, let $s(k) = u(2k - 1)$ and $t(k) = u(2k)$. Then $\Psi(s, t) = u$. So Ψ is a bijection.

By Exercise 2.4.3, there is a bijection $\phi : \mathbb{R} \to 2^{\mathbb{N}}$. Then $(x, y) \mapsto (\phi(x), \phi(y))$ is a bijection from $\mathbb{R} \times \mathbb{R}$ to $2^{\mathbb{N}} \times 2^{\mathbb{N}}$. So $\mathbb{R} \times \mathbb{R} \sim 2^{\mathbb{N}} \times 2^{\mathbb{N}} \sim 2^{\mathbb{N}} \sim \mathbb{R}$ by Theorem 2.1.8.

Exercise 2.4.7. *Hint.* Constant functions show $\mathbb{R} \lesssim \mathcal{F}$. The functions that take only the values 0 and 1 show $\mathcal{P}(\mathbb{R}) \lesssim \mathcal{F}$. Then use Theorems 2.4.2 and 2.4.5.

Chapter 3. Sequences of Real Numbers

Exercise 3.1.1. (a) Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{10}{9\varepsilon}$. Then, for all $n \geq N$,

$$\left|x_n - \frac{5}{3}\right| = \left|\frac{5n}{3n+2} - \frac{5}{3}\right| = \left|\frac{15n - 15n - 10}{9n+6}\right| = \frac{10}{9n+6} \leq \frac{10}{9n} < \frac{10}{9\left(\frac{10}{9\varepsilon}\right)} = \varepsilon.$$

So $x_n \to \frac{5}{3}$.

(b) Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \left(\frac{5}{2\varepsilon}\right)^{2/3}$. Then, for all $n \geq N$,

$$|x_n - 6| = \left|\frac{12n^2 + 5\sqrt{n}}{2n^2} - 6\right| = \left|\frac{12n^2 + 5\sqrt{n} - 12n^2}{2n^2}\right| = \frac{5\sqrt{n}}{2n^2}$$
$$= \frac{5}{2n^{3/2}} < \frac{5}{2\left(\left(\frac{5}{2\varepsilon}\right)^{2/3}\right)^{3/2}} = \varepsilon.$$

So $x_n \to 6$.

(c) Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{1}{\varepsilon}$. Then, for all $n \geq N$,

$$|x_n - 1| = \left|1 - \frac{(-1)^n}{n} - 1\right| = \frac{1}{n} < \frac{1}{\frac{1}{\varepsilon}} = \varepsilon.$$

So $x_n \to 1$.

(d) Let $\varepsilon > 0$. Choose $N \in \mathbb{N}$ such that $N > \max\left\{75, \frac{16}{9\varepsilon}, \sqrt{\frac{67}{\varepsilon}}\right\}$. Then, for all $n \geq N$,

$$\left|x_n - \frac{8}{3}\right| = \left|\frac{(1+4n)^2}{6n^2+n-75} - \frac{8}{3}\right| = \left|\frac{48n^2+24n+3-48n^2-8n+600}{18n^2+3n-225}\right|$$
$$= \left|\frac{16n+603}{18n^2+3n-225}\right|.$$

Since for all $n \geq N$, $n > 75$, $3n - 225 > 0$. So $|18n^2+3n-225| = 18n^2+3n-225 > 18n^2$. So for all $n \geq N$,

$$\left|x_n - \frac{8}{3}\right| \leq \frac{16n+603}{18n^2} = \frac{8}{9n} + \frac{67}{2n^2} < \frac{8}{9\left(\frac{16}{9\varepsilon}\right)} + \frac{67}{2\left(\sqrt{\frac{67}{\varepsilon}}\right)^2} = \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Thus $x_n \to \frac{8}{3}$.

Exercise 3.2.2. Let $\varepsilon > 0$. Since $\{y_n\}$ is bounded, there exists an $M > 0$ such that for all $n \in \mathbb{N}$, $|y_n| \leq M$. Since $x_n \to 0$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - 0| < \frac{\varepsilon}{M}$. Thus, for all $n \geq N$,

$$|x_n y_n - 0| = |y_n||x_n - 0| \leq M|x_n - 0| < M\frac{\varepsilon}{M} = \varepsilon.$$

So $x_n y_n \to 0$.

Notice that if $\{y_n\}$ is unbounded, the conclusion may not be true. For example, consider $x_n = \frac{1}{n}$ and $y_n = n$. Then $x_n \to 0$, but $x_n y_n = 1$ for all $n \in \mathbb{N}$, and so $x_n y_n \to 1 \neq 0$.

Exercise 3.3.1. First, note that $x_1 = 2 > \frac{7}{6}$. Now, assume for some $k \in \mathbb{N}$, $x_k > \frac{7}{6}$. Then

$$x_{k+1} = \frac{4x_k + 7}{10} > \frac{\frac{14}{3} + 7}{10} = \frac{7}{6}.$$

So by induction, for all $n \in \mathbb{N}$, $x_n > \frac{7}{6}$. So $\{x_n\}$ is bounded below.

Next, notice that $x_1 = 2 > \frac{3}{2} = x_2$. Now assume for some $k \geq 2$ that $x_{k-1} > x_k$. Then

$$x_{k+1} = \frac{4x_k + 7}{10} < \frac{4x_{k-1} + 7}{10} = x_k.$$

Therefore, by induction, $x_{n-1} > x_n$ for all $n \geq 2$. Since $\{x_n\}$ is bounded below by $\frac{7}{6}$ and decreasing, it is also bounded above by x_1. So $\{x_n\}$ is bounded and monotone. Therefore, $\{x_n\}$ converges by Theorem 3.2.13, say $x_n \to a$.

Now, since $x_n \to a$, $x_{n+1} \to a$ and so $a = \frac{4a+7}{10}$. Therefore, $a = \frac{7}{6}$.

Exercise 3.4.1. *Hint.* Choose n_1 with $x_{n_1} > 1$. Given n_k, use that $\{x_n : n > n_k\}$ is not bounded above either.

Solution. First, for every $N \in \mathbb{N}$, the set $\{x_n : n > N\}$ is not bounded above. Otherwise there is an M with $x_n \leq M$ for all $n > N$. Then every term is at most $\max\{M, x_1, \ldots, x_N\}$, and $\{x_n\}$ is bounded above. That is a contradiction.

Now choose the indices one at a time. Since $\{x_n\}$ is not bounded above, 1 is not an upper bound, so there is an n_1 with $x_{n_1} > 1$. Suppose $n_1 < n_2 < \cdots < n_k$ have been chosen with $x_{n_j} > j$ for $j = 1, \ldots, k$. Since $\{x_n : n > n_k\}$ is not bounded above, there is an $n_{k+1} > n_k$ with $x_{n_{k+1}} > k + 1$. This defines a strictly increasing sequence $\{n_k\}$ with $x_{n_k} > k$ for all $k \in \mathbb{N}$. By Definition 3.4.1, $\{x_{n_k}\}$ is the required subsequence.

Exercise 3.5.1. *Hint.* For $m > n$, bound $x_m - x_n$ using $k! \geq 2^{k-1}$ (Exercise 1.2.10) and Exercise 1.2.7.

Exercise 3.5.2. Let $n \in \mathbb{N}$. Then

$$x_{2n} - x_n = \sum_{k=n+1}^{2n} \frac{1}{k} \geq \sum_{k=n+1}^{2n} \frac{1}{2n} = n \cdot \frac{1}{2n} = \frac{1}{2},$$

since each of the n terms is at least $\frac{1}{2n}$.

Suppose $\{x_n\}$ is Cauchy. With $\varepsilon = \frac{1}{2}$, there is an $N \in \mathbb{N}$ such that $|x_m - x_n| < \frac{1}{2}$ for all $m, n \geq N$. Take $n = N$ and $m = 2N$. Then $x_{2N} - x_N < \frac{1}{2}$, which contradicts the inequality above. So $\{x_n\}$ is not Cauchy. By the Cauchy Criterion (Theorem 3.5.2), $\{x_n\}$ diverges.

Exercise 3.6.1. Let $a = \limsup x_n$. If $a = \infty$, then ∞ is a cluster point by Theorem 3.6.12 (1), and it is the largest one. If $a = -\infty$, then $x_n \to -\infty$ by Theorem 3.6.12 (2), so every subsequence diverges to $-\infty$, and $-\infty$ is the only cluster point. So suppose $a \in \mathbb{R}$.

First, a is a cluster point. Choose $n_1 < n_2 < \cdots$ as follows. For each $k \in \mathbb{N}$, by Theorem 3.6.11 with $\varepsilon = \frac{1}{k}$, there exists an $N_k \in \mathbb{N}$ such that $x_n < a + \frac{1}{k}$ for all $n \geq N_k$. By the same theorem, there exists an $n_k \geq \max\{N_k, n_{k-1} + 1\}$ with $x_{n_k} > a - \frac{1}{k}$ (take $n_0 = 0$). Then $|x_{n_k} - a| < \frac{1}{k}$ for every k. So $x_{n_k} \to a$, and a is a cluster point of $\{x_n\}$.

Now, by way of contradiction, assume b is a cluster point of $\{x_n\}$ with $b > a$. Define $\varepsilon = \frac{b-a}{2} > 0$. Since b is a cluster point of $\{x_n\}$, there exists a subsequence $x_{n_j} \to b$. So there exists a $J \in \mathbb{N}$ such that for all $j \geq J$, $|x_{n_j} - b| < \frac{b-a}{2}$. This implies that for all $j \geq J$, $x_{n_j} > b - \frac{b-a}{2} = \frac{a+b}{2}$. But, by Theorem 3.6.11, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $x_n < a + \frac{b-a}{2} = \frac{a+b}{2}$. Choose $j_0 \geq J$ with $n_{j_0} \geq N$. Then

$$\frac{a+b}{2} < x_{n_{j_0}} < \frac{a+b}{2},$$

a contradiction. So a is the largest cluster point of $\{x_n\}$. A similar argument shows $\liminf x_n$ is the smallest cluster point of $\{x_n\}$. Alternatively, apply the result to $\{-x_n\}$ and use Theorem 3.6.10 (1).

Chapter 4. Topological Properties of the Real Line

Exercise 4.1.1. ($\Rightarrow$) Let $x \in U^\circ$. Then, by definition, $x \in U$ and so $U^\circ \subset U$. Now, let $x \in U$. Since U is open, there exists an $\varepsilon > 0$ such that $(x-\varepsilon, x+\varepsilon) \subset U$. So $x \in U^\circ$. So $U \subset U^\circ$ and consequently, $U = U^\circ$.

($\Leftarrow$) Let $x \in U$. Since $U = U^\circ$, $x \in U^\circ$. So there exists an $\varepsilon > 0$ such that $(x-\varepsilon, x+\varepsilon) \subset U$. So U is open.

Exercise 4.1.2. Let $a, b \in \mathbb{R}$ with $a < b$. First, consider the interval $(-\infty, b)$. Let $x \in (-\infty, b)$ and define $\varepsilon = \frac{b-x}{2} > 0$. Then $(x-\varepsilon, x+\varepsilon) \subset (-\infty, b)$ and $(-\infty, b)$ is open.

Next, consider the interval (a, ∞). Let $x \in (a, \infty)$ and define $\varepsilon = \frac{x-a}{2} > 0$. Then $(x-\varepsilon, x+\varepsilon) \subset (a, \infty)$ and (a, ∞) is open.

Last, consider the interval (a, b). Let $x \in (a, b)$ and define $\varepsilon = \min\left\{\frac{x-a}{2}, \frac{b-x}{2}\right\}$. Then $(x-\varepsilon, x+\varepsilon) \subset (a, b)$ and (a, b) is open. The interval $(-\infty, \infty) = \mathbb{R}$ is open, since every interval $(x-1, x+1)$ lies in $\mathbb{R}$.

Exercise 4.1.6. *Hint.* Every open interval contains rational and irrational numbers, by Theorem 1.8.11 and Exercise 1.8.5.

Exercise 4.1.7. ($\Rightarrow$) Let $x \in \overline{E}$. If $x \in E$, the constant sequence $x_n = x$ works. Otherwise x is a limit point of E. So for all $n \in \mathbb{N}$, there exists an $x_n \in E$ such that $|x_n - x| < 1/n$. Let $\varepsilon > 0$. Then by the Archimedean Principle, there exists an $N \in \mathbb{N}$ such that $N > 1/\varepsilon$. So for all $n \geq N$, $|x_n - x| < 1/n \leq 1/N < \varepsilon$. So $\{x_n : n \in \mathbb{N}\} \subset E$ and $x_n \to x$.

($\Leftarrow$) Assume $\{x_n\}$ is a sequence in E with $x_n \to x$. If $x \in E$, then $x \in \overline{E}$. Suppose $x \notin E$. Then $x_n \neq x$ for all n. Let $\delta > 0$. Since $x_n \to x$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - x| < \delta$. In particular, $x_N \in E$, $x_N \neq x$, and $|x_N - x| < \delta$. So x is a limit point of E, and $x \in \overline{E}$.

Exercise 4.1.9. *Hint.* If only finitely many points of E other than a lie in the interval, choose a smaller ε that misses all of them.

Exercise 4.1.10. *Hint.* Let A be an interval. Suppose U and V are disjoint open sets with $A \cap U \neq \emptyset$, $A \cap V \neq \emptyset$ and $(A \cap U) \cup (A \cap V) = A$. Choose $a \in A \cap U$ and $b \in A \cap V$. Assume $a < b$. Let $c = \sup\{t \in [a, b] : [a, t] \subset U\}$. Show that $c \in A$, $c \notin U$ and $c \notin V$.

Solution. Let A be an interval, and suppose A is not connected. Then there are disjoint open sets U and V with $A \cap U \neq \emptyset$, $A \cap V \neq \emptyset$ and $(A \cap U) \cup (A \cap V) = A$. Choose $a \in A \cap U$ and $b \in A \cap V$. Since $U \cap V = \emptyset$, $a \neq b$. By switching the names of U and V if needed, assume $a < b$. Since A is an interval, $[a, b] \subset A$.

Let $S = \{t \in [a, b] : [a, t] \subset U\}$. Then $a \in S$ and S is bounded above by b. Let $c = \sup S$. Then $a \leq c \leq b$, so $c \in A$. So $c \in U$ or $c \in V$.

Suppose $c \in U$. Then $c \neq b$, since $b \in V$. Since U is open, there is an $\varepsilon > 0$ with $(c-\varepsilon, c+\varepsilon) \subset U$, and ε can be chosen with $\varepsilon < b - c$. If $c = a$, then $[a, c + \frac{\varepsilon}{2}] \subset U$. If $c > a$, there is a $t \in S$ with $c - \varepsilon < t \leq c$, so $[a, t] \subset U$, and then $[a, c + \frac{\varepsilon}{2}] \subset [a, t] \cup (c-\varepsilon, c+\varepsilon) \subset U$. Either way $c + \frac{\varepsilon}{2} \in S$, which contradicts $c = \sup S$.

Suppose $c \in V$. Then $c \neq a$, since $a \in U$. Since V is open, there is an $\varepsilon > 0$ with $(c-\varepsilon, c+\varepsilon) \subset V$, and ε can be chosen with $\varepsilon < c - a$. Since $c = \sup S$, there

is a $t \in S$ with $c - \varepsilon < t \leq c$. Then $t \in [a, t] \subset U$ and $t \in (c - \varepsilon, c + \varepsilon) \subset V$. This contradicts $U \cap V = \emptyset$.

Both cases are impossible. So A is connected.

Exercise 4.2.2. Let $S = \{x_n : n \in \mathbb{N}\} \cup \{a\}$, and let $\mathcal{U}$ be an open cover of S. So there exists a $U_a \in \mathcal{U}$ such that $a \in U_a$. Since U_a is open, there exists a $\delta > 0$ such that $(a - \delta, a + \delta) \subset U_a$. Since $x_n \to a$, there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $|x_n - a| < \delta$. So for all $n \geq N$, $x_n \in (a - \delta, a + \delta) \subset U_a$. Now for $i = 1, \ldots, N - 1$, there is a $U_i \in \mathcal{U}$ such that $x_i \in U_i$. So $\{U_1, \ldots, U_{N-1}, U_a\}$ is a finite subcover of $\mathcal{U}$ that covers S. So S is compact.

Exercise 4.2.3. *Hint.* Use the Bolzano–Weierstrass Theorem (Theorem 3.4.5) and Theorem 4.2.15.

Solution. Let $\{x_n\}$ be a sequence in K. By Theorem 4.2.9, K is closed and bounded. Since K is bounded, so is $\{x_n\}$. By the Bolzano–Weierstrass Theorem, there is a subsequence with $x_{n_k} \to a$ for some $a \in \mathbb{R}$. Each x_{n_k} is in K, so $a \in \overline{K}$ by Exercise 4.1.7. Since K is closed, $\overline{K} = K$ by Theorem 4.1.26 (2). So $a \in K$.

Exercise 4.2.5. *Hint.* Take a finite subcover for each set and combine them.

Exercise 4.2.6. *Hint.* The intersection is closed and lies inside one compact set. Use Lemma 4.2.10.

Exercise 4.2.7. Let F be one of the sets F_α such that F is bounded. So F is closed and bounded. Therefore, by the Heine–Borel Theorem, F is compact. Now, assume by way of contradiction that $\bigcap_{\alpha \in \Lambda} F_\alpha = \emptyset$. So $\left(\bigcap_{\alpha \in \Lambda} F_\alpha\right)^c = \bigcup_{\alpha \in \Lambda} F_\alpha^c = \mathbb{R}$. Thus, $F \subset \bigcup_{\alpha \in \Lambda} F_\alpha^c$. But since each F_α is closed, each F_α^c is open. So $\{F_\alpha^c\}_{\alpha \in \Lambda}$ is an open cover of F. Since F is compact, there exist $\alpha_1, \ldots, \alpha_n$ such that $F \subset \bigcup_{i=1}^n F_{\alpha_i}^c$. So $F \cap [\bigcap_{i=1}^n F_{\alpha_i}] = \emptyset$, a contradiction to the finite intersection property. So $\bigcap_{\alpha \in \Lambda} F_\alpha \neq \emptyset$.

Let $\mathcal{U} = \{[n, \infty)\}_{n \in \mathbb{N}}$. Then each element of $\mathcal{U}$ is closed and unbounded, and every finite subcollection has nonempty intersection. But $\bigcap_{n \in \mathbb{N}} [n, \infty) = \emptyset$.

Chapter 5. The Cantor Set

Exercise 5.1.1. By Lemma 5.1.2 (1), C_n is the union of the 2^n disjoint intervals $J_{n,1}, \ldots, J_{n,2^n}$, and each has length $\frac{1}{3^n}$. So the sum of their lengths is $2^n \cdot \frac{1}{3^n} = \left(\frac{2}{3}\right)^n$.

The intervals removed in building $C_1, \ldots, C_n$ fill up the rest of $[0, 1]$, apart from endpoints. So their lengths add up to $1 - \left(\frac{2}{3}\right)^n$. As $n \to \infty$, $\left(\frac{2}{3}\right)^n \to 0$ by Example 3.1.7. So the lengths of all the removed intervals add up to 1. This is Theorem 5.1.5.

Exercise 5.1.2. *Hint.* For each n there are only finitely many endpoints. Use Theorem 2.2.6.

Exercise 5.2.3. *Hint.* Find a ternary expansion of $\frac{3}{4}$ that uses only the digits 0 and 2, and use Theorem 5.2.7.

Solution. Take the digits $n_k = 2$ for odd k and $n_k = 0$ for even k. The odd k are $k = 2j-1$ for $j \in \mathbb{N}$, so

$$\sum_{k=1}^{\infty} \frac{n_k}{3^k} = \sum_{j=1}^{\infty} \frac{2}{3^{2j-1}} = 6 \sum_{j=1}^{\infty} \left(\frac{1}{9}\right)^j = 6 \cdot \frac{1/9}{1-1/9} = \frac{3}{4},$$

using $\sum_{i=0}^{\infty} r^i = \frac{1}{1-r}$ for $|r| < 1$, as in the proof of Theorem 5.1.5. So $.2020\ldots$ is a ternary expansion of $\frac{3}{4}$ that uses only the digits 0 and 2. By Theorem 5.2.7, $\frac{3}{4} \in C$.

Exercise 5.2.4. Four points in the Cantor set include $1/3$ and $2/3$, which are both endpoints, and $1/4$ and $111/364$, which both have ternary expansions consisting of 0's and 2's. Indeed, $\frac{1}{4} = .020202\ldots$ and $\frac{111}{364} = .022020\,022020\ldots$ (base 3), so both are in C by Theorem 5.2.7. Every endpoint of an interval $J_{n,i}$ has the form $\frac{k}{3^n}$ by Lemma 5.1.2. In lowest terms, $\frac{1}{4}$ and $\frac{111}{364}$ have denominators 4 and 364, which are not powers of 3. So they are not endpoints.

Exercise 5.2.5. *Hint.* An interval inside C lies inside C_n for every n, so its length is at most 3^{-n}. For $C' = C$, use Theorem 5.1.4 and the fact that C is closed.

Chapter 6. Functions and Continuity

Exercise 6.1.1. *Hint.* (d) First show $|\sqrt{x} - \sqrt{c}| = \dfrac{|x-c|}{\sqrt{x}+\sqrt{c}}$.

Solution. (a) Let $\varepsilon > 0$. Define $\delta = \varepsilon$. Then for all x with $0 < |x-c| < \delta$, $|k-k| = 0 < \varepsilon$. So $\lim_{x\to c} k = k$.

(b) Let $\varepsilon > 0$. Define $\delta = \varepsilon$. Then for all x with $0 < |x-c| < \delta$, $|x-c| < \delta = \varepsilon$. So $\lim_{x\to c} x = c$.

(c) Let $\varepsilon > 0$. Define $\delta = \varepsilon$. Then, by the reverse triangle inequality, for all x with $0 < |x-c| < \delta$, $\big||x| - |c|\big| \le |x-c| < \delta = \varepsilon$. So $\lim_{x\to c} |x| = |c|$.

(d) Let $\varepsilon > 0$. Define $\delta = \sqrt{c}\,\varepsilon$. Then for all $x \ge 0$ with $0 < |x-c| < \delta$,

$$|\sqrt{x} - \sqrt{c}| = |\sqrt{x} - \sqrt{c}|\frac{\sqrt{x}+\sqrt{c}}{\sqrt{x}+\sqrt{c}} = \frac{|x-c|}{\sqrt{x}+\sqrt{c}} \le \frac{|x-c|}{\sqrt{c}} < \frac{\delta}{\sqrt{c}} = \varepsilon.$$

So $\lim_{x\to c} \sqrt{x} = \sqrt{c}$.

(e) Let $\varepsilon > 0$. Define $\delta = \frac{\varepsilon}{6}$. Then for all x with $0 < |x-3| < \delta$,

$$|(6x+7) - 25| = |6x - 18| = 6|x-3| < 6\delta = \varepsilon.$$

So $\lim_{x\to 3}(6x+7) = 25$.

(f) Let $\varepsilon > 0$. Define $\delta = \min\left\{1, \frac{\varepsilon}{10}\right\}$. Then for all x with $0 < |x-2| < \delta$,

$$|(x^2+5x) - 14| = |x+7||x-2|.$$

Since $|x-2| < \delta \le 1$, $-1 < x-2 < 1$, and it follows that $8 < x+7 < 10$. So $|x+7| < 10$. Therefore, for all x with $0 < |x-2| < \delta$,

$$|(x^2+5x) - 14| = |x+7||x-2| < 10|x-2| < 10\delta \le \varepsilon.$$

So $\lim_{x\to 2}(x^2+5x) = 14$.

(g) Let $\varepsilon > 0$. Define $\delta = \min\left\{\frac{1}{2}, \frac{23\varepsilon}{54}\right\}$. Then for all x with $0 < |x-4| < \delta$,

$$\left|\frac{1}{x^2+5x-24} - \frac{1}{12}\right| = \left|\frac{x^2+5x-36}{12(x^2+5x-24)}\right| = \frac{|x+9||x-4|}{12|x+8||x-3|}.$$

Since $|x-4| < \delta \le \frac{1}{2}$, it follows that $\frac{25}{2} < x+9 < \frac{27}{2}$, $\frac{23}{2} < x+8 < \frac{25}{2}$, and $\frac{1}{2} < x-3 < \frac{3}{2}$. So $|x+9| < \frac{27}{2}$, $\frac{1}{|x+8|} < \frac{2}{23}$, and $\frac{1}{|x-3|} < 2$. Therefore, for all x with $0 < |x-4| < \delta$,

$$\left|\frac{1}{x^2+5x-24} - \frac{1}{12}\right| < \frac{54|x-4|}{23 \cdot 12} < \frac{54|x-4|}{23} < \frac{54\delta}{23} \le \varepsilon.$$

So $\lim_{x\to 4} \frac{1}{x^2+5x-24} = \frac{1}{12}$.

(h) Let $\varepsilon > 0$. Define $\delta = \min\left\{\frac{1}{2}, \frac{\varepsilon}{2}\right\}$. Then for all x with $0 < |x-3| < \delta$,

$$\left|\frac{1}{x-2} - 1\right| = \frac{|x-3|}{|x-2|}.$$

Since $|x-3| < \delta \le \frac{1}{2}$, it follows that $\frac{1}{2} < x-2 < \frac{3}{2}$, and so $\frac{1}{|x-2|} < 2$. Therefore, for all x with $0 < |x-3| < \delta$,

$$\left|\frac{1}{x-2} - 1\right| = \frac{|x-3|}{|x-2|} < 2|x-3| < 2\delta \le \varepsilon.$$

So $\lim_{x\to 3} \frac{1}{x-2} = 1$.

Exercise 6.1.3. Let $L > 0$. Set $\varepsilon = L/2 > 0$. Then there exists a $\delta > 0$ such that for all $x \in E$ with $0 < |x-c| < \delta$, $|f(x) - L| < L/2$. So for all such x, $f(x) - L > -L/2$, or $f(x) > L/2 > 0$. If $L < 0$, apply this to $-f$, which has limit $-L > 0$ by Theorem 6.1.13.

Exercise 6.1.5. (1) Let $\varepsilon > 0$. There exist $\delta_1, \delta_2 > 0$ with $|f(x) - L_1| < \frac{\varepsilon}{2}$ for $x \in E$ with $0 < |x-c| < \delta_1$, and $|g(x) - L_2| < \frac{\varepsilon}{2}$ for $x \in E$ with $0 < |x-c| < \delta_2$. Let $\delta = \min\{\delta_1, \delta_2\}$. For $x \in E$ with $0 < |x-c| < \delta$, the Triangle Inequality gives $|f(x) + g(x) - (L_1 + L_2)| \le |f(x) - L_1| + |g(x) - L_2| < \varepsilon$.

(2) Since $\lim_{x\to c} f(x)$ exists, by Lemma 6.1.12 there exists an $M > 0$ and a $\delta_1 > 0$ such that for all x with $0 < |x-c| < \delta_1$, $|f(x)| \le M$. Let $\varepsilon > 0$. Since $\lim_{x\to c} f(x) = L_1$, there exists a $\delta_2 > 0$ such that for all x with $0 < |x-c| < \delta_2$, $|f(x) - L_1| < \frac{\varepsilon}{2(|L_2|+1)}$. Since $\lim_{x\to c} g(x) = L_2$, there exists a $\delta_3 > 0$ such that for all x with $0 < |x-c| < \delta_3$, $|g(x) - L_2| < \frac{\varepsilon}{2M}$. Set $\delta = \min\{\delta_1, \delta_2, \delta_3\}$. Then for all x with $0 < |x-c| < \delta$,

$$\begin{aligned} |f(x)g(x) - L_1L_2| &= |f(x)g(x) - f(x)L_2 + f(x)L_2 - L_1L_2| \\ &\le |f(x)||g(x) - L_2| + |L_2||f(x) - L_1| \\ &< M\left(\frac{\varepsilon}{2M}\right) + |L_2|\left(\frac{\varepsilon}{2(|L_2|+1)}\right) < \varepsilon. \end{aligned}$$

So $\lim_{x\to c} f(x)g(x) = L_1L_2$. Throughout, x ranges over E.

Exercise 6.1.7. *Hint.* For $0 < |x| < \frac{\pi}{2}$, $\cos x < \frac{\sin x}{x} < 1$.

Exercise 6.1.8. ($\Rightarrow$) Let $\varepsilon > 0$. Then there exists a $\delta > 0$ such that for all x with $0 < |x-c| < \delta$, $|f(x) - L| < \varepsilon$. So for all $x \in (c-\delta, c)$ and for all $x \in (c, c+\delta)$, $|f(x) - L| < \varepsilon$. So $\lim_{x\to c^-} f(x) = \lim_{x\to c^+} f(x) = L$.

($\Leftarrow$) Let $\varepsilon > 0$. Then there exists a $\delta_1 > 0$ such that for all x with $0 < x - c < \delta_1$, $|f(x) - L| < \varepsilon$. Also, there exists a $\delta_2 > 0$ such that for all x with $-\delta_2 < x - c < 0$,

$|f(x)-L|<\varepsilon$. Set $\delta=\min\{\delta_1,\delta_2\}$. Then for all x with $0<|x-c|<\delta$, $|f(x)-L|<\varepsilon$. So $\lim_{x\to c} f(x)=L$.

Exercise 6.1.9. *Hint.* For $c\neq 0$, use a sequence of rational numbers and a sequence of irrational numbers that converge to c, and Theorem 6.1.18.

Exercise 6.1.10. *Hint.* Look at $x_n=\frac{1}{2n\pi}$ and $y_n=\frac{1}{(2n+1)\pi}$.

Exercise 6.2.1. Let $\varepsilon>0$. Define $d=\frac{1}{\varepsilon}$. Then for all x with $x>d$,

$$\left|\frac{1}{x}-0\right|=\frac{1}{x}<\frac{1}{d}=\frac{1}{\frac{1}{\varepsilon}}=\varepsilon.$$

So $\lim_{x\to\infty}\frac{1}{x}=0$.

Exercise 6.2.3. *Hint.* For $x>0$, $\left|\frac{2x+1}{x+3}-2\right|=\frac{5}{x+3}<\frac{5}{x}$.

Solution. For $x>0$,

$$\left|\frac{2x+1}{x+3}-2\right|=\left|\frac{2x+1-2x-6}{x+3}\right|=\frac{5}{x+3}<\frac{5}{x}.$$

Let $\varepsilon>0$ and set $d=\frac{5}{\varepsilon}$. If $x>d$, then $\left|\frac{2x+1}{x+3}-2\right|<\frac{5}{x}<\frac{5}{d}=\varepsilon$. By Definition 6.2.5, $\lim_{x\to\infty}\frac{2x+1}{x+3}=2$.

Exercise 6.3.1. *Hint.* $|x\sin\frac{1}{x}|\leq|x|$.

Solution. By Definition 6.3.1, it is enough to show $\lim_{x\to 0} f(x)=f(0)=0$. Let $\varepsilon>0$ and set $\delta=\varepsilon$. If $0<|x|<\delta$, then

$$|f(x)-0|=|x|\left|\sin\frac{1}{x}\right|\leq|x|<\delta=\varepsilon,$$

since $|\sin t|\leq 1$ for all t. So $\lim_{x\to 0} f(x)=0=f(0)$, and f is continuous at 0.

Exercise 6.3.4. *Hint.* Apply the Intermediate Value Theorem to $f-g$.

Exercise 6.3.5. *Hint.* Choose rational numbers $q_n\to x$, and use Theorem 6.3.9.

Exercise 6.3.6. ($\Rightarrow$) Let O be an open set. For all $x\in f^{-1}[O]$, $f(x)\in O$, and so for all $x\in f^{-1}[O]$, there exists an $\varepsilon_x>0$ such that $(f(x)-\varepsilon_x,f(x)+\varepsilon_x)\subset O$. Since for all $x\in f^{-1}[O]\subset E$, f is continuous at x, there exists a $\delta_x>0$ such that for all $y\in E$ with $|x-y|<\delta_x$, $|f(x)-f(y)|<\varepsilon_x$. Set $U=\bigcup_{x\in f^{-1}[O]}(x-\delta_x,x+\delta_x)$, an open set. If $y\in E\cap U$, then $y\in E$ and there exists an $x\in f^{-1}[O]$ such that $y\in(x-\delta_x,x+\delta_x)$. So $f(y)\in(f(x)-\varepsilon_x,f(x)+\varepsilon_x)\subset O$. So $y\in f^{-1}[O]$ and $E\cap U\subset f^{-1}[O]$. Now let $x\in f^{-1}[O]$. Then $x\in U$. Also, since $f^{-1}[O]\subset E$, $x\in E$. So $x\in E\cap U$. So $f^{-1}[O]\subset E\cap U$ and $f^{-1}[O]=E\cap U$.

($\Leftarrow$) Let $\varepsilon>0$. Let $x\in E$. By the assumption, there exists an open set U such that $f^{-1}[(f(x)-\varepsilon,f(x)+\varepsilon)]=E\cap U$. So $x\in U$. Since U is open, there exists a $\delta>0$ such that $(x-\delta,x+\delta)\subset U$. Thus, for all $y\in E$ with $|x-y|<\delta$, $y\in E\cap U$. So $f(y)\in(f(x)-\varepsilon,f(x)+\varepsilon)$. Thus, for all $y\in E$ with $|x-y|<\delta$, $|f(x)-f(y)|<\varepsilon$. So f is continuous at x. Since $x\in E$ was arbitrary, f is continuous on all of E.

Exercise 6.4.1. Let $\varepsilon > 0$. Define $\delta = a^2\varepsilon$. Then for all $x, y \in [a, \infty)$ with $|x - y| < \delta$,

$$|f(x) - f(y)| = \left|\frac{1}{x} - \frac{1}{y}\right| = \frac{|x-y|}{xy} \leq \frac{|x-y|}{a^2} < \frac{\delta}{a^2} = \frac{a^2\varepsilon}{a^2} = \varepsilon.$$

So f is uniformly continuous on $[a, \infty)$.

Exercise 6.4.2. Assume, by way of contradiction, that f is uniformly continuous on $\mathbb{R}$. Let $\varepsilon = 1$. Then there exists a $\delta > 0$ such that for all $x, y \in \mathbb{R}$ with $|x - y| < \delta$, $|f(x) - f(y)| = |x^2 - y^2| < 1$. Set $x = \frac{1}{\delta} - \frac{\delta}{4}$ and $y = \frac{1}{\delta} + \frac{\delta}{4}$. Then $|x - y| = \frac{\delta}{2} < \delta$ but $|x^2 - y^2| = |x + y||x - y| = \frac{2}{\delta} \cdot \frac{\delta}{2} = 1$, a contradiction. So f is not uniformly continuous on $\mathbb{R}$.

Exercise 6.4.4. *Hint.* Show $|\sqrt{x} - \sqrt{y}| \leq \sqrt{|x - y|}$.

Exercise 6.4.5. *Hint.* Given ε, take δ from uniform continuity. Then use the Cauchy condition with δ in place of ε.

Exercise 6.4.6. *Hint.* Try $x_n = \frac{1}{2n\pi + \pi/2}$ and $y_n = \frac{1}{2n\pi - \pi/2}$.

Exercise 6.5.1. Suppose $f^{-1}(a) = f^{-1}(b)$. Then, since f is a function, $f(f^{-1}(a)) = f(f^{-1}(b))$. So $a = b$ and f^{-1} is one-to-one.

Exercise 6.5.2. Suppose $a, b \in E$ with $a \neq b$. Without loss of generality, $a < b$. If f is strictly increasing, then $f(a) < f(b)$. If f is strictly decreasing, then $f(a) > f(b)$. Either way, $f(a) \neq f(b)$. So f is one-to-one.

Exercise 6.6.1. *Hint.* For the Dirichlet function, every interval contains points where f is 0 and points where it is 1. For the step function, only intervals around 0 cause trouble.

Exercise 6.6.2. Let $q_1, q_2, q_3, \ldots$ be an enumeration of $\mathbb{Q}$. Each set $\mathbb{R} \setminus \{q_k\}$ is open, and

$$\mathbb{R} \setminus \mathbb{Q} = \bigcap_{k=1}^{\infty} (\mathbb{R} \setminus \{q_k\}).$$

So $\mathbb{R} \setminus \mathbb{Q}$ is a G_δ set.

If $x = \frac{m}{n}$ in lowest terms, then $x + 1 = \frac{m+n}{n}$ in lowest terms. So $f(x + 1) = f(x)$ for every x. Near a point $c \in (k, k + 1)$ with $k \in \mathbb{Z}$, $f(x) = g(x - k)$, where g is Thomae's function on $(0, 1)$. By Example 6.3.2, f is continuous at c when c is irrational and not continuous at c when c is rational. At an integer k, $f(k) = 1$, but every interval around k contains irrational numbers, where $f = 0$. So f is not continuous at k. Thus f is continuous exactly at the irrational numbers, which agrees with Theorem 6.6.2.

Exercise 6.6.3. Suppose $\mathbb{R}$ is countable, say $\mathbb{R} = \{x_1, x_2, x_3, \ldots\}$. Let $V_k = \mathbb{R} \setminus \{x_k\}$. Each V_k is open. It is dense, since every open interval contains points other than x_k. By Theorem 6.6.3, $\bigcap_{k=1}^{\infty} V_k$ is not empty. But a point in it differs from every x_k, so it is not in $\mathbb{R}$. This is a contradiction.

Exercise 6.6.4. *Hint.* If $G = \bigcap U_n$ is dense, then each U_n is dense. Apply Theorem 6.6.3 to all the U_n and V_n together. For the last part, add the sets $\mathbb{R} \setminus \{g_k\}$, as in the proof of Theorem 6.6.4.

Exercise 6.6.5. *Hint.* If $\mathbb{R} = \bigcup A_n$, apply Theorem 6.6.3 to $U_n = \mathbb{R} \setminus \overline{A_n}$. For the Cantor set, an open interval longer than 3^{-n} cannot lie inside C_n.

Chapter 7. The Derivative

Exercise 7.1.2. *Hint.* Use induction and the product rule.

Solution. For $n = 0$, $x^0 = 1$ is constant, so its derivative is 0. That is the formula with $n = 0$. For $n = 1$, the difference quotient of $f(x) = x$ at c is $\frac{x-c}{x-c} = 1$ for $x \neq c$, so $f'(c) = 1 = 1 \cdot c^0$.

Now suppose $\frac{d}{dx}x^k = kx^{k-1}$ for some $k \in \mathbb{N}$. Since $x^{k+1} = x \cdot x^k$, the product rule (Theorem 7.1.5 (2)) gives

$$\frac{d}{dx}x^{k+1} = 1 \cdot x^k + x \cdot kx^{k-1} = (k+1)x^k.$$

By induction, $\frac{d}{dx}x^n = nx^{n-1}$ for all $n \in \mathbb{N}$, and so for all $n \in \mathbb{N}_0$.

Exercise 7.1.3. *Hint.* For $n < 0$, write $x^n = \frac{1}{x^{-n}}$ and use the quotient rule.

Exercise 7.1.4. The difference quotient at 0 is $q(x) = \frac{|x|-0}{x-0} = \frac{|x|}{x}$. It equals 1 for $x > 0$ and -1 for $x < 0$. Suppose $\lim_{x\to 0} q(x) = L$. With $\varepsilon = 1$, there is a $\delta > 0$ such that $|q(x) - L| < 1$ whenever $0 < |x| < \delta$. Taking $x = \frac{\delta}{2}$ and $x = -\frac{\delta}{2}$ gives $|1 - L| < 1$ and $|-1 - L| < 1$. Then, by the Triangle Inequality,

$$2 = |1 - (-1)| \leq |1 - L| + |L - (-1)| < 1 + 1 = 2,$$

which is impossible. So the limit does not exist, and f is not differentiable at 0.

Exercise 7.1.5. *Hint.* Find $f'(0)$ from the definition. For $x \neq 0$, $f'(x) = 2x \sin\frac{1}{x} - \cos\frac{1}{x}$.

Exercise 7.2.1. *Hint.* Apply the Mean Value Theorem to e^t on the interval between 0 and x.

Solution. For $x = 0$, both sides equal 1. Since $\frac{d}{dt}e^t = e^t > 0$, e^t is strictly increasing by Theorem 7.2.7 (2).

Suppose $x > 0$. By the Mean Value Theorem (Theorem 7.2.4) on $[0, x]$, there is a $c \in (0, x)$ with $e^x - 1 = e^c x$. Since $c > 0$, $e^c > e^0 = 1$. So $e^x - 1 > x$.

Suppose $x < 0$. By the Mean Value Theorem on $[x, 0]$, there is a $c \in (x, 0)$ with $1 - e^x = e^c(0 - x)$. Since $c < 0$, $e^c < 1$, and $-x > 0$. So $1 - e^x < -x$, that is, $e^x > 1 + x$.

In every case, $e^x \geq 1 + x$.

Exercise 7.2.2. *Hint.* Show $f'(x) = 0$ for every x, since $\frac{|f(x+h)-f(x)|}{|h|} \leq M|h|^{\alpha-1}$. Then use Theorem 7.2.7.

Solution. Let $x \in (a, b)$. For $y \in (a, b)$ with $y \neq x$,

$$\left|\frac{f(y) - f(x)}{y - x}\right| \leq \frac{M|y - x|^\alpha}{|y - x|} = M|y - x|^{\alpha-1}.$$

Since $\alpha > 1$, $M|y - x|^{\alpha-1} \to 0$ as $y \to x$. By the Squeeze Theorem, $f'(x) = \lim_{y\to x} \frac{f(y)-f(x)}{y-x} = 0$. So $f'(x) = 0$ for every $x \in (a, b)$. By Theorem 7.2.7 (5), f is constant on (a, b).

Exercise 7.2.5. *Hint.* Use the Inverse Function Theorem (Theorem 7.2.9) with $f(x) = x^n$.

Exercise 7.2.7. Assume $M > 0$. If not, replace M by $M+1$. Let $x, y \in I$ with $x < y$. Since f is differentiable on I, it is continuous on $[x, y]$ by Theorem 7.1.3. By the Mean Value Theorem (Theorem 7.2.4), there is a $c \in (x, y)$ with $f(y) - f(x) = f'(c)(y - x)$. So

$$|f(y) - f(x)| = |f'(c)||y - x| \leq M|y - x|.$$

The same holds with the roles of x and y switched, and trivially when $x = y$. So f is Lipschitz continuous on I.

For the second part, $\frac{d}{dx} \sin x = \cos x$ and $|\cos x| \leq 1$ for all x. So $|\sin x - \sin y| \leq |x - y|$ for all $x, y \in \mathbb{R}$.

Exercise 7.2.8. *Hint.* Use Theorem 7.2.8.

Exercise 7.3.2. (a) Let $f(x) = \ln(1 + x)$ and $g(x) = x$ on $(0, 1)$. Then $f(x) \to 0$ and $g(x) \to 0$ as $x \to 0^+$, and $g'(x) = 1 \neq 0$. Since $\frac{f'(x)}{g'(x)} = \frac{1}{1+x} \to 1$, L'Hospital's Rule (Theorem 7.3.1) gives $\lim_{x\to 0^+} \frac{\ln(1+x)}{x} = 1$.

(b) First, $\lim_{x\to 0^+} \frac{\sin x}{2x} = \frac{1}{2}$. Indeed, $\sin x \to 0$ and $2x \to 0$, $\frac{d}{dx}(2x) = 2 \neq 0$, and $\frac{\cos x}{2} \to \frac{1}{2}$. Now let $f(x) = 1 - \cos x$ and $g(x) = x^2$ on $(0, \pi)$. Then $f(x) \to 0$, $g(x) \to 0$ and $g'(x) = 2x \neq 0$. Since $\frac{f'(x)}{g'(x)} = \frac{\sin x}{2x} \to \frac{1}{2}$, L'Hospital's Rule gives $\lim_{x\to 0^+} \frac{1-\cos x}{x^2} = \frac{1}{2}$.

(c) Applying the rule to $\frac{e^{-1/x}}{x}$ directly makes things worse. Instead, write $\frac{e^{-1/x}}{x} = \frac{1/x}{e^{1/x}}$. Let $f(x) = \frac{1}{x}$ and $g(x) = e^{1/x}$ on $(0, 1)$. Then $g(x) \to \infty$ as $x \to 0^+$, and $g'(x) = -\frac{e^{1/x}}{x^2} \neq 0$. Since

$$\frac{f'(x)}{g'(x)} = \frac{-1/x^2}{-e^{1/x}/x^2} = e^{-1/x} \to 0$$

as $x \to 0^+$, case (2) of L'Hospital's Rule gives $\lim_{x\to 0^+} \frac{e^{-1/x}}{x} = 0$.

Exercise 7.4.1. Let $a < x < b$ in I and set $t = \frac{b-x}{b-a}$. Then $0 < t < 1$, $1 - t = \frac{x-a}{b-a}$, and $ta + (1 - t)b = b - t(b - a) = x$. Multiplying by $b - a > 0$ and writing $b - a = (b - x) + (x - a)$ shows that

$$f(x) \leq tf(a) + (1 - t)f(b)$$

is the same as $(b - x)f(x) + (x - a)f(x) \leq (b - x)f(a) + (x - a)f(b)$, which is the same as $(b - x)\big(f(x) - f(a)\big) \leq (x - a)\big(f(b) - f(x)\big)$. Dividing by $(x - a)(b - x) > 0$, this is the same as

$$(*) \qquad \frac{f(x) - f(a)}{x - a} \leq \frac{f(b) - f(x)}{b - x}.$$

($\Rightarrow$) If f is convex, the first inequality holds, so $(*)$ holds.

($\Leftarrow$) Suppose $(*)$ holds whenever $a < x < b$. Let $a, b \in I$ and $0 \leq t \leq 1$. If $a = b$, $t = 0$ or $t = 1$, then $f(ta + (1 - t)b) = tf(a) + (1 - t)f(b)$. If $a > b$, switch the names of a and b and replace t by $1 - t$. So assume $a < b$ and $0 < t < 1$, and let $x = ta + (1 - t)b$. Then $a < x < b$ and $t = \frac{b-x}{b-a}$. By $(*)$ and the equivalences above, $f(x) \leq tf(a) + (1 - t)f(b)$. So f is convex.

The statement for concave functions follows in the same way with every inequality reversed, or by applying the first part to $-f$.

Exercise 7.4.2. *Hint.* Use Theorem 7.4.2 and the Mean Value Theorem.

Chapter 8. Functions of Bounded Variation

Exercise 8.1.1. *Hint.* Show that $|f'(x)| \leq \alpha + \beta$ for $0 < x \leq 1$ and that $|f(x)| \leq x$. Then f satisfies a Lipschitz condition, by the Mean Value Theorem. Use Theorem 8.1.12.

Exercise 8.1.2. *Hint.* Each term of a variation sum satisfies $|f(x_i) - f(x_{i-1})| \leq M(x_i - x_{i-1})$. These bounds add up to $M(b-a)$.

Solution. Choose $M > 0$ with $|f(x) - f(y)| \leq M|x - y|$ for all $x, y \in [a, b]$. Let $\mathcal{P} = \{x_0, \ldots, x_n\}$ be a partition of $[a, b]$. Then

$$V(\mathcal{P}) = \sum_{k=1}^{n} |f(x_k) - f(x_{k-1})| \leq \sum_{k=1}^{n} M(x_k - x_{k-1}) = M(b-a),$$

since the last sum telescopes. So $M(b-a)$ is an upper bound for the sums $V(\mathcal{P})$, and $V_f[a, b] \leq M(b-a) < \infty$. So f is of bounded variation on $[a, b]$.

Exercise 8.1.3. *Hint.* Use the partition $\{a, x, b\}$.

Solution. Let $x \in [a, b]$. If $x = a$, then $|f(x) - f(a)| = 0 \leq V_f[a, b]$. If $x = b$, then $|f(x) - f(a)| \leq V_f[a, b]$ by Remark 8.1.6. If $a < x < b$, then $\{a, x, b\}$ is a partition of $[a, b]$. So

$$|f(x) - f(a)| \leq |f(x) - f(a)| + |f(b) - f(x)| = V(\{a, x, b\}) \leq V_f[a, b].$$

In every case, $|f(x) - f(a)| \leq V_f[a, b]$. By the Triangle Inequality,

$$|f(x)| \leq |f(a)| + |f(x) - f(a)| \leq |f(a)| + V_f[a, b].$$

Since $V_f[a, b] < \infty$, this is one bound for every $x \in [a, b]$. So f is bounded.

Exercise 8.1.4. *Hint.* Use the points $\frac{2}{(2k+1)\pi}$. You may use that the sums $1 + \frac{1}{2} + \cdots + \frac{1}{n}$ are not bounded. To see this, group the terms $\frac{1}{3} + \frac{1}{4}$, then $\frac{1}{5} + \cdots + \frac{1}{8}$, and so on.

Exercise 8.1.5. *Hint.* Choose a partition whose points alternate between rational and irrational numbers. Each term of its variation sum is 1.

Exercise 8.2.1. *Hint.* f decreases on $[-1, 0]$ and increases on $[0, 1]$. On each piece the variation is the total change in f.

Solution. Let $f(x) = x^2$. Since $f \in C^{(1)}[-1, 1]$, f is of bounded variation on $[-1, 1]$ by Corollary 8.1.10. By definition, $V_f(-1) = 0$.

Let $-1 < x \leq 0$. Then f is decreasing on $[-1, x]$. By Theorem 8.1.8,

$$V_f(x) = V_f[-1, x] = |f(x) - f(-1)| = 1 - x^2.$$

Let $0 < x \leq 1$. By Theorem 8.2.1 on $[-1, x]$ with $c = 0$, and Theorem 8.1.8 on $[-1, 0]$ and $[0, x]$,

$$V_f(x) = V_f[-1, 0] + V_f[0, x] = 1 + x^2.$$

So

$$V_f(x) = \begin{cases} 1-x^2, & -1 \le x \le 0, \\ 1+x^2, & 0 < x \le 1, \end{cases} \qquad V_f(x) - f(x) = \begin{cases} 1-2x^2, & -1 \le x \le 0, \\ 1, & 0 < x \le 1. \end{cases}$$

Both functions are increasing on $[-1,1]$, by Theorems 8.2.2 and 8.2.4. This can also be checked directly. Then

$$x^2 = V_f(x) - \big(V_f(x) - f(x)\big)$$

writes f as the difference of two increasing functions.

Exercise 8.2.2. Let $h = f + g$ or $h = f - g$. Let $\mathcal{P} = \{x_0, \dots, x_n\}$ be a partition of $[a,b]$. By the Triangle Inequality, for each k,

$$|h(x_k) - h(x_{k-1})| \le |f(x_k) - f(x_{k-1})| + |g(x_k) - g(x_{k-1})|.$$

Adding over k, the sum $V(\mathcal{P})$ for h is at most the sum $V(\mathcal{P})$ for f plus the sum $V(\mathcal{P})$ for g. So it is at most $V_f[a,b] + V_g[a,b]$. This holds for every partition. So

$$V_h[a,b] \le V_f[a,b] + V_g[a,b] < \infty,$$

and h is of bounded variation on $[a,b]$.

Exercise 8.2.3. *Hint.* Use Exercise 8.1.3.

Exercise 8.2.4. *Hint.* Write $f = V_f - (V_f - f)$. Use Theorems 8.2.3 and 8.2.4.

Chapter 9. Integrals

Exercise 9.1.2. *Hint.* Each m_i and M_i lies in $[m, M]$, and $\sum \Delta x_i = b - a$.

Solution. Let $\mathcal{P} = \{x_0, \dots, x_n\}$ be a partition of $[a,b]$, and let $1 \le i \le n$. Since $m \le f(x)$ for all $x \in [x_{i-1}, x_i]$, m is a lower bound of $\{f(x) : x_{i-1} \le x \le x_i\}$. So $m \le m_i$. In the same way, $M_i \le M$. Also $m_i \le f(x_i) \le M_i$. So

$$m \le m_i \le M_i \le M.$$

Multiply by $\Delta x_i > 0$ and add over i. Since $\sum_{i=1}^n \Delta x_i = b - a$,

$$m(b-a) = \sum_{i=1}^n m\,\Delta x_i \le \sum_{i=1}^n m_i \Delta x_i \le \sum_{i=1}^n M_i \Delta x_i \le \sum_{i=1}^n M\,\Delta x_i = M(b-a).$$

The middle two sums are $\mathcal{L}(\mathcal{P}, f)$ and $\mathcal{U}(\mathcal{P}, f)$.

Exercise 9.1.3. *Hint.* Every subinterval contains rational and irrational numbers. So $m_i = 0$ and $M_i = x_i$. Compare $\mathcal{U}(\mathcal{P}, f)$ with the upper sums of $g(x) = x$.

Exercise 9.1.4. *Hint.* The function is constant on $[0, \frac{1}{3})$, $[\frac{1}{3}, \frac{2}{3})$ and $[\frac{2}{3}, 1)$. Use partitions with short subintervals around $\frac{1}{3}$, $\frac{2}{3}$ and 1.

Exercise 9.1.5. *Hint.* Use the regular partition into n subintervals and $\sum_{i=1}^n i = \frac{n(n+1)}{2}$.

Solution. Let $f(x) = 2x + 1$. For $n \in \mathbb{N}$, let $\mathcal{P}_n = \big\{0, \frac{1}{n}, \frac{2}{n}, \dots, 1\big\}$. Since f is increasing, its infimum and supremum on $\big[\frac{i-1}{n}, \frac{i}{n}\big]$ are the values at the endpoints. So

$$m_i = \frac{2(i-1)}{n} + 1 \quad \text{and} \quad M_i = \frac{2i}{n} + 1.$$

Since $\Delta x_i = \frac{1}{n}$ and $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$,

$$\mathcal{L}(\mathcal{P}_n, f) = \frac{2}{n^2}\sum_{i=1}^{n}(i-1) + 1 = \frac{n-1}{n} + 1 = 2 - \frac{1}{n}$$

and

$$\mathcal{U}(\mathcal{P}_n, f) = \frac{2}{n^2}\sum_{i=1}^{n} i + 1 = \frac{n+1}{n} + 1 = 2 + \frac{1}{n}.$$

By Definition 9.1.4 and Theorem 9.1.6, for every $n \in \mathbb{N}$,

$$2 - \frac{1}{n} = \mathcal{L}(\mathcal{P}_n, f) \le \underline{\int_0^1} f \le \overline{\int_0^1} f \le \mathcal{U}(\mathcal{P}_n, f) = 2 + \frac{1}{n}.$$

Letting $n \to \infty$ gives $\underline{\int_0^1} f = \overline{\int_0^1} f = 2$. So $\int_0^1 (2x+1)\,dx = 2$.

Exercise 9.1.6. *Hint.* For any partition, $x_{i-1}^2 \le \frac{1}{3}(x_{i-1}^2 + x_{i-1}x_i + x_i^2) \le x_i^2$. Also $\frac{1}{3}(x_{i-1}^2 + x_{i-1}x_i + x_i^2)\Delta x_i = \frac{1}{3}(x_i^3 - x_{i-1}^3)$. Sum over i.

Exercise 9.2.1. *Hint.* For $c \ge 0$, $\mathcal{U}(\mathcal{P}, cf) = c\mathcal{U}(\mathcal{P}, f)$ and $\mathcal{L}(\mathcal{P}, cf) = c\,\mathcal{L}(\mathcal{P}, f)$. For $c < 0$, multiplying by c swaps sup and inf. So $\mathcal{U}(\mathcal{P}, cf) = c\,\mathcal{L}(\mathcal{P}, f)$.

Solution. If $c = 0$, then cf is the zero function. Every upper and lower sum of cf is 0. So $cf \in \mathcal{R}[a,b]$ and $\int_a^b cf = 0 = c\int_a^b f$.

Suppose $c \neq 0$, and let $\mathcal{P} = \{x_0, \dots, x_n\}$ be a partition of $[a,b]$. Write m_i and M_i for the infimum and supremum of f on $[x_{i-1}, x_i]$.

Suppose $c > 0$. For $x \in [x_{i-1}, x_i]$, $cm_i \le cf(x) \le cM_i$. So the supremum of cf on $[x_{i-1}, x_i]$ is at most cM_i, and its infimum is at least cm_i. Adding over i,

$$c\,\mathcal{L}(\mathcal{P}, f) \le \mathcal{L}(\mathcal{P}, cf) \le \mathcal{U}(\mathcal{P}, cf) \le c\mathcal{U}(\mathcal{P}, f).$$

Suppose $c < 0$. Multiplying by c reverses inequalities, so $cM_i \le cf(x) \le cm_i$. In the same way,

$$c\mathcal{U}(\mathcal{P}, f) \le \mathcal{L}(\mathcal{P}, cf) \le \mathcal{U}(\mathcal{P}, cf) \le c\,\mathcal{L}(\mathcal{P}, f).$$

In both cases, call the outer numbers $A_{\mathcal{P}} \le B_{\mathcal{P}}$. Then $B_{\mathcal{P}} - A_{\mathcal{P}} = |c|\big(\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f)\big)$. Since $\mathcal{L}(\mathcal{P}, f) \le \int_a^b f \le \mathcal{U}(\mathcal{P}, f)$, also $A_{\mathcal{P}} \le c\int_a^b f \le B_{\mathcal{P}}$.

Let $\varepsilon > 0$. By Riemann's Criterion (Theorem 9.2.1), there is a partition $\mathcal{P}$ with $\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) < \frac{\varepsilon}{|c|}$. Then $\mathcal{U}(\mathcal{P}, cf) - \mathcal{L}(\mathcal{P}, cf) \le B_{\mathcal{P}} - A_{\mathcal{P}} < \varepsilon$. So $cf \in \mathcal{R}[a,b]$, by Riemann's Criterion. Both $\int_a^b cf$ and $c\int_a^b f$ lie in $[A_{\mathcal{P}}, B_{\mathcal{P}}]$, an interval of length less than ε. So they differ by less than ε. Since $\varepsilon > 0$ was arbitrary, $\int_a^b cf = c\int_a^b f$.

Exercise 9.2.2. *Hint.* Let $M = \max|f(c_i)|$. Choose a partition in which each c_i lies in at most two subintervals, each shorter than δ. Then $|\mathcal{U}(\mathcal{P}, f)|$ and $|\mathcal{L}(\mathcal{P}, f)|$ are at most $2nM\delta$.

Exercise 9.2.3. *Hint.* Show that $\mathcal{U}(\mathcal{P}, f) \le \mathcal{U}(\mathcal{P}, g)$ for every partition $\mathcal{P}$.

Solution. Let $\mathcal{P} = \{x_0, \dots, x_n\}$ be a partition of $[a,b]$. For $x \in [x_{i-1}, x_i]$, $f(x) \le g(x) \le \sup\{g(t) : x_{i-1} \le t \le x_i\}$. So the supremum of f on $[x_{i-1}, x_i]$ is at most

the supremum of g there. Multiplying by Δx_i and adding gives $\mathcal{U}(\mathcal{P}, f) \le \mathcal{U}(\mathcal{P}, g)$. Since $f \in \mathcal{R}[a, b]$,

$$\int_a^b f = \overline{\int_a^b} f \le \mathcal{U}(\mathcal{P}, f) \le \mathcal{U}(\mathcal{P}, g).$$

So $\int_a^b f$ is a lower bound of the upper sums of g. Therefore $\int_a^b f \le \overline{\int_a^b} g = \int_a^b g$.

Exercise 9.2.4. *Hint.* Apply Exercise 9.2.2 to $g - f$.

Exercise 9.2.5. *Hint.* If $f(c) > 0$, then $f > \frac{f(c)}{2}$ on an interval around c. Show that $\int_a^b f > 0$.

Exercise 9.3.1. *Hint.* By Theorem 9.3.7, $F'(x) = f(x)$ wherever f is continuous. At $x = 1$, find the one-sided derivatives of F.

Exercise 9.3.2. *Hint.* If $|f| \le M$, then $|F(x) - F(y)| \le M|x - y|$.

Solution. Since f is bounded, there is an $M > 0$ with $|f(t)| \le M$ for all $t \in [a, b]$. By Theorem 9.2.13, f is integrable on every closed interval in $[a, b]$, and

$$F(y) - F(x) = \int_x^y f(t)\,dt \quad \text{for } a \le x < y \le b.$$

(If $x = a$, this is the definition of $F(y)$, since $F(a) = 0$.) Since $-M \le f(t) \le M$ on $[x, y]$, Theorem 9.1.3 gives $-M(y - x) \le \mathcal{L}(\mathcal{P}, f) \le \int_x^y f \le \mathcal{U}(\mathcal{P}, f) \le M(y - x)$ for any partition $\mathcal{P}$ of $[x, y]$. So

$$|F(y) - F(x)| \le M|y - x| \quad \text{for all } x, y \in [a, b].$$

Let $\varepsilon > 0$ and $\delta = \frac{\varepsilon}{M}$. If $x, y \in [a, b]$ and $|x - y| < \delta$, then $|F(y) - F(x)| < M\delta = \varepsilon$. So F is continuous on $[a, b]$. In fact F is Lipschitz continuous, and so uniformly continuous.

Exercise 9.3.3. *Hint.* For the identity, use the common denominator $k!(n + 1 - k)!$. For the induction step, write $(a + b)^{n+1} = (a + b)(a + b)^n$, expand, and shift the index in one of the two sums.

Exercise 9.4.1. *Hint.* Use Theorem 9.4.10 to write each logarithm as a quotient of natural logarithms.

Exercise 9.4.2. *Hint.* On $[1, 2]$, $\frac{1}{t} \le 1$, and on $\left[\frac{3}{2}, 2\right]$, $\frac{1}{t} \le \frac{2}{3}$. For $\ln 3$, the function $\frac{1}{t}$ is decreasing, so its infimum on each subinterval is at the right endpoint. The lower sum is $\frac{1}{5} + \frac{1}{6} + \cdots + \frac{1}{12}$.

Exercise 9.4.3.

(a) For $x > 0$, $f(x) = e^{x \ln x}$. By the Chain Rule and the Product Rule,

$$f'(x) = e^{x \ln x}\left(\ln x + x \cdot \frac{1}{x}\right) = x^x(1 + \ln x).$$

(b) Since $x^x > 0$, $f'(x)$ has the sign of $1 + \ln x$. Since ln is strictly increasing and $\ln \frac{1}{e} = -1$, $f'(x) < 0$ for $0 < x < \frac{1}{e}$ and $f'(x) > 0$ for $x > \frac{1}{e}$. By Theorem 7.2.7, f is strictly decreasing on $\left(0, \frac{1}{e}\right]$ and strictly increasing on $\left[\frac{1}{e}, \infty\right)$. So the smallest value of f is $f\left(\frac{1}{e}\right) = e^{-1/e}$.

(c) Let $\{x_n\}$ be a sequence with $x_n > 0$ and $x_n \to 0$. By Theorem 9.4.13 (3) with $\alpha = 1$ and Theorem 6.2.8, $x_n \ln x_n \to 0$. Since exp is continuous at 0, $x_n^{x_n} = e^{x_n \ln x_n} \to e^0 = 1$ by Theorem 6.3.9. So $\lim_{x\to 0^+} x^x = 1$ by Theorem 6.2.8.

Exercise 9.4.4. Let $g(x) = f(x)e^{-x}$. By the Chain Rule, $\frac{d}{dx}e^{-x} = -e^{-x}$. So

$$g'(x) = f'(x)e^{-x} - f(x)e^{-x} = 0$$

for all $x \in \mathbb{R}$, since $f' = f$. By Theorem 7.2.7 (5), g is constant on $\mathbb{R}$. Since $g(0) = f(0) = 1$, $f(x)e^{-x} = 1$ for all x. Multiplying by e^x gives $f(x) = e^x$.

Exercise 9.4.5. *Hint.* First show $f(x) = f\left(\frac{x}{2}\right)^2 \geq 0$, and that f is never 0. Then show $f(r) = a^r$ for $r \in \mathbb{Q}$. Finish with the density of $\mathbb{Q}$ and the continuity of f and of $x \mapsto a^x$.

Exercise 9.4.6. *Hint.* (b) $\frac{d}{dx}\sinh x = \cosh x > 0$. Find the limits of $\sinh x$ as $x \to \pm\infty$ and use the Intermediate Value Theorem. (c) If $\sinh y = x$, then $u = e^y$ solves $u^2 - 2xu - 1 = 0$.

Exercise 9.4.7. For $x > -1$ with $x \neq 0$, $(1+x)^{1/x} = e^{\ln(1+x)/x}$ by Definition 9.4.7. Let $\{x_n\}$ be a sequence with $x_n > -1$, $x_n \neq 0$ and $x_n \to 0$. Since the derivative of ln at 1 is 1, Theorem 6.1.18 gives

$$\frac{\ln(1+x_n)}{x_n} = \frac{\ln(1+x_n) - \ln 1}{x_n} \to 1.$$

Since exp is continuous at 1, $(1+x_n)^{1/x_n} \to e^1 = e$ by Theorem 6.3.9. So $\lim_{x\to 0}(1+x)^{1/x} = e$ by Theorem 6.1.18.

For the second limit, let $\{x_n\}$ be a sequence with $x_n > 0$ and $x_n \to \infty$. Let $h_n = \frac{1}{x_n}$. Then $h_n > 0$ and $h_n \to 0$. By the first part, $\left(1 + \frac{1}{x_n}\right)^{x_n} = (1+h_n)^{1/h_n} \to e$. So $\lim_{x\to\infty}\left(1 + \frac{1}{x}\right)^x = e$ by Theorem 6.2.8.

Exercise 9.4.8. *Hint.* By Theorem 9.4.4 (4), $\ln(ta + (1-t)b) \geq t\ln a + (1-t)\ln b$. Write the right side as one logarithm and apply exp.

Exercise 9.5.1. Let $g(x) = \arctan x + \arctan\frac{1}{x}$ for $x > 0$. By Theorem 9.5.2 (1) and the Chain Rule,

$$g'(x) = \frac{1}{1+x^2} + \frac{1}{1+1/x^2}\cdot\left(-\frac{1}{x^2}\right) = \frac{1}{1+x^2} - \frac{1}{x^2+1} = 0.$$

So g is constant on $(0,\infty)$ by Theorem 7.2.7 (5). As $x \to \infty$, $\arctan x \to \frac{\pi}{2}$ by Theorem 9.5.4. Also $\frac{1}{x} \to 0$, so $\arctan\frac{1}{x} \to \arctan 0 = 0$, since arctan is continuous. So the constant is $\frac{\pi}{2}$. With $x = 1$, $2\arctan 1 = \frac{\pi}{2}$.

Exercise 9.5.2. *Hint.* Fix y. On the interval of x with $xy < 1$ that contains 0, compare the derivatives of both sides with respect to x. Both sides are equal at $x = 0$.

Exercise 9.5.3. *Hint.* Multiply each inequality by $1 + t^2$. Then integrate from 0 to 1, and use $\arctan 1 = \frac{\pi}{4}$.

Exercise 9.5.4. *Hint.* Let $f(x) = x - \sin x$. Then $f'(x) = 1 - \cos x \geq 0$, with equality only at isolated points. For the limit, $\frac{\sin x}{x}$ is a difference quotient of sin at 0.

Exercise 9.5.5. *Hint.* Use the Quotient Rule and Theorem 9.5.10. For the second part, divide $\sin^2 x + \cos^2 x = 1$ by $\cos^2 x$.

Exercise 9.5.6. For $-1 < x < 1$, the Product Rule, the Chain Rule and Theorem 9.5.12 (1) give

$$F'(x) = \frac{1}{2}\left(\sqrt{1-x^2} - \frac{x^2}{\sqrt{1-x^2}} + \frac{1}{\sqrt{1-x^2}}\right) = \frac{1}{2}\cdot\frac{(1-x^2) - x^2 + 1}{\sqrt{1-x^2}} = \sqrt{1-x^2}.$$

Let $G(x) = \int_0^x \sqrt{1-t^2}\,dt$ for $0 \leq x \leq 1$. By Theorem 9.3.7, $G'(x) = \sqrt{1-x^2} = F'(x)$ for $0 \leq x < 1$, and G is continuous on $[0,1]$. Since $F(0) = G(0) = 0$, Theorem 7.2.7 (5) gives $G = F$ on $[0,1)$. Both are continuous at 1. (F is, since arcsin is the inverse of a continuous strictly increasing function, by Theorem 6.5.7.) So

$$\int_0^1 \sqrt{1-t^2}\,dt = G(1) = F(1) = \frac{1}{2}\big(0 + \arcsin 1\big) = \frac{\pi}{4}.$$

The region under the graph of $\sqrt{1-t^2}$ on $[0,1]$ is a quarter of the unit disk.

Exercise 9.6.1. *Hint.* Let m and M be the minimum and maximum of f. Then $m\int_a^b g \leq \int_a^b fg \leq M\int_a^b g$. If $\int_a^b g > 0$, use the Intermediate Value Theorem. If $\int_a^b g = 0$, any c works.

Solution. By the Extreme Value Theorem (Theorem 6.3.15), there are $c_1, c_2 \in [a,b]$ with $m = f(c_1) \leq f(x) \leq f(c_2) = M$ for all $x \in [a,b]$. Since $g(x) \geq 0$,

$$mg(x) \leq f(x)g(x) \leq Mg(x) \quad \text{for all } x \in [a,b].$$

Since f is continuous, $f \in \mathcal{R}[a,b]$, and so $fg \in \mathcal{R}[a,b]$ by Theorem 9.2.11 (3). By Exercise 9.2.3 and Theorem 9.2.11 (2),

$$m\int_a^b g \leq \int_a^b fg \leq M\int_a^b g.$$

Also $\int_a^b g \geq 0$, by Exercise 9.2.3 with the zero function.

If $\int_a^b g = 0$, then $\int_a^b fg = 0$, and every $c \in [a,b]$ works.

Suppose $\int_a^b g > 0$, and let $k = \frac{\int_a^b fg}{\int_a^b g}$. Then $m \leq k \leq M$. If $k = m$, take $c = c_1$. If $k = M$, take $c = c_2$. Otherwise $f(c_1) < k < f(c_2)$. Apply the Intermediate Value Theorem (Theorem 6.3.10) to f on the closed interval with endpoints c_1 and c_2. It gives a c between c_1 and c_2, so in $[a,b]$, with $f(c) = k$. In every case, $\int_a^b fg = f(c)\int_a^b g$.

Exercise 9.6.2. *Hint.* Write $\int_{-a}^a f = \int_{-a}^0 f + \int_0^a f$. Show that $\int_{-a}^0 f(x)\,dx = \int_0^a f(-t)\,dt$. Reflect each partition of $[0,a]$ to a partition of $[-a,0]$. The upper and lower sums do not change.

Exercise 9.7.1. *Hint.* For $p \neq 1$, $\int_a^s x^{-p}\,dx = \frac{s^{1-p} - a^{1-p}}{1-p}$. For $p = 1$, it is $\ln s - \ln a$. Let $s \to \infty$. For the second part, integrate over $[t,a]$ and let $t \to 0^+$.

Solution. Two facts about powers are used. If $q > 0$, then $s^q \to \infty$ as $s \to \infty$, since s^q is increasing and $s^q \geq n$ for $s \geq n^{1/q}$. So $s^{-q} = \frac{1}{s^q} \to 0$ as $s \to \infty$, and $t^q = \frac{1}{(1/t)^q} \to 0$ as $t \to 0^+$. Also $\ln s \to \infty$ as $s \to \infty$, since ln is increasing and $\ln(2^n) = n \ln 2$ with $\ln 2 > 0$.

The function x^{-p} is continuous on $(0, \infty)$, so it is locally integrable there. An antiderivative is $\frac{x^{1-p}}{1-p}$ if $p \neq 1$, and $\ln x$ if $p = 1$. By the Fundamental Theorem of Calculus, for $0 < t < a < s$,

$$\int_a^s \frac{dx}{x^p} = \begin{cases} \dfrac{s^{1-p} - a^{1-p}}{1-p}, & p \neq 1, \\ \ln s - \ln a, & p = 1, \end{cases} \qquad \int_t^a \frac{dx}{x^p} = \begin{cases} \dfrac{a^{1-p} - t^{1-p}}{1-p}, & p \neq 1, \\ \ln a - \ln t, & p = 1. \end{cases}$$

Let $s \to \infty$. If $p > 1$, then $s^{1-p} \to 0$, and the first integral tends to $\frac{a^{1-p}}{p-1}$. If $p < 1$, then $s^{1-p} \to \infty$, and the first integral tends to ∞. If $p = 1$, $\ln s \to \infty$. So $\int_a^\infty \frac{dx}{x^p}$ converges if and only if $p > 1$.

Let $t \to 0^+$. If $p < 1$, then $t^{1-p} \to 0$, and the second integral tends to $\frac{a^{1-p}}{1-p}$. If $p > 1$, then $t^{1-p} = \frac{1}{t^{p-1}} \to \infty$, and the second integral tends to ∞. If $p = 1$, $-\ln t = \ln \frac{1}{t} \to \infty$. So $\int_0^a \frac{dx}{x^p}$ converges if and only if $p < 1$.

Exercise 9.7.2. *Hint.* By definition, $\lim_{s \to c^-} \int_a^s f$ and $\lim_{t \to c^+} \int_t^b f$ both exist. Take $s = c - \varepsilon$ and $t = c + \varepsilon$.

Solution. Since $\int_a^b f$ converges, the limits

$$L_1 = \lim_{s \to c^-} \int_a^s f \quad \text{and} \quad L_2 = \lim_{t \to c^+} \int_t^b f$$

exist, and $\int_a^b f = L_1 + L_2$. For $0 < \varepsilon < \min\{c - a, b - c\}$, let $G(\varepsilon) = \int_a^{c-\varepsilon} f + \int_{c+\varepsilon}^b f$. The goal is to show $G(\varepsilon) \to L_1 + L_2$ as $\varepsilon \to 0^+$.

Let $\eta > 0$. There is a $\delta_1 > 0$ such that $\left| \int_a^s f - L_1 \right| < \frac{\eta}{2}$ for all $s \in [a, c)$ with $c - s < \delta_1$. There is a $\delta_2 > 0$ such that $\left| \int_t^b f - L_2 \right| < \frac{\eta}{2}$ for all $t \in (c, b]$ with $t - c < \delta_2$. Let $\delta = \min\{\delta_1, \delta_2, c - a, b - c\}$. If $0 < \varepsilon < \delta$, take $s = c - \varepsilon$ and $t = c + \varepsilon$. Then

$$|G(\varepsilon) - (L_1 + L_2)| \leq \left| \int_a^{c-\varepsilon} f - L_1 \right| + \left| \int_{c+\varepsilon}^b f - L_2 \right| < \frac{\eta}{2} + \frac{\eta}{2} = \eta.$$

So P. V. $\int_a^b f = \lim_{\varepsilon \to 0^+} G(\varepsilon) = L_1 + L_2 = \int_a^b f$.

Exercise 9.7.3. *Hint.* Substitute $u = \ln x$ on $[2, s]$. Then use the p-Test.

Exercise 9.7.4. *Hint.* (a) Split at $t = 1$. On $(0, 1]$, $e^{-t} t^{x-1} \leq t^{x-1}$. On $[1, \infty)$, $e^{-t} t^{x+1} \to 0$, so $e^{-t} t^{x-1} \leq t^{-2}$ for large t. Use Theorems 9.7.7 and 9.7.8.

Exercise 9.7.5. *Hint.* If $c < c'$, then $\int_s^c f + \int_c^t f = \int_s^{c'} f + \int_{c'}^t f$ for $s < c$ and $t > c'$, by Theorem 9.2.13.

Exercise 9.8.1. *Hint.* Use Theorem 9.6.3 and $\cos\left(\frac{\pi}{2} - t\right) = \sin t$.

Exercise 9.8.2. Let $c_n = \frac{n!}{\sqrt{2\pi n}(n/e)^n}$. Then

$$\frac{\sqrt[n]{n!}}{n} = \sqrt[n]{c_n} \cdot \sqrt[n]{\sqrt{2\pi n}} \cdot \frac{1}{e}.$$

By Theorem 9.8.3, $c_n \to 1$, so $\frac{1}{2} \le c_n \le 2$ for large n. Then $\sqrt[n]{1/2} \le \sqrt[n]{c_n} \le \sqrt[n]{2}$, and both bounds tend to 1 by Lemma 12.1.3. So $\sqrt[n]{c_n} \to 1$. Also $\sqrt[n]{\sqrt{2\pi n}} = \sqrt{\sqrt[n]{2\pi} \cdot \sqrt[n]{n}} \to 1$ by Lemma 12.1.3. So $\frac{\sqrt[n]{n!}}{n} \to \frac{1}{e}$.

Exercise 9.8.3. *Hint.* In the proof of Theorem 9.8.3, a_n strictly decreases to $\sqrt{2\pi}$, and $b_n = a_n e^{-1/(4n)}$ strictly increases to it. For $n = 10$ the lower bound is about 3598696.

Exercise 9.8.4. Let $0 < a < b$. On $[\sqrt{a}, \sqrt{b}]$, the substitution $t = u^2$, $dt = 2u\,du$, gives

$$\int_a^b e^{-t} t^{-1/2}\,dt = \int_{\sqrt{a}}^{\sqrt{b}} e^{-u^2} \cdot \frac{1}{u} \cdot 2u\,du = 2\int_{\sqrt{a}}^{\sqrt{b}} e^{-u^2}\,du.$$

Let $a \to 0^+$ and $b \to \infty$. By Theorem 9.8.4, $\Gamma\left(\frac{1}{2}\right) = 2 \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\pi}$. By Exercise 9.7.4, $\Gamma(x+1) = x\Gamma(x)$. So $\Gamma\left(\frac{5}{2}\right) = \frac{3}{2} \cdot \frac{1}{2} \cdot \Gamma\left(\frac{1}{2}\right) = \frac{3\sqrt{\pi}}{4}$.

Exercise 9.8.5. The substitution $x = -u$ gives $\int_{-b}^{0} e^{-x^2}\,dx = \int_0^b e^{-u^2}\,du$. Letting $b \to \infty$ and adding $\int_0^\infty e^{-x^2}\,dx$ gives $\int_{-\infty}^{\infty} e^{-x^2}\,dx = 2 \cdot \frac{\sqrt{\pi}}{2} = \sqrt{\pi}$.

For the second integral, let $b > 0$ and integrate by parts with $u = x$ and $v = -\frac{1}{2}e^{-x^2}$, so $v' = xe^{-x^2}$. Then

$$\int_0^b x^2 e^{-x^2}\,dx = -\frac{b}{2}e^{-b^2} + \frac{1}{2}\int_0^b e^{-x^2}\,dx.$$

For $b \ge 1$, $0 < be^{-b^2} \le be^{-b}$, which tends to 0 by Theorem 9.4.13. So letting $b \to \infty$ gives $\int_0^\infty x^2 e^{-x^2}\,dx = \frac{1}{2} \cdot \frac{\sqrt{\pi}}{2} = \frac{\sqrt{\pi}}{4}$.

Exercise 9.8.6. *Hint.* Split $\sum \frac{1}{k^2}$ into the sum over even k and the sum over odd k. The even terms add up to $\frac{1}{4}\sum \frac{1}{k^2}$.

Exercise 9.8.7. *Hint.* (a) f is $\frac{1}{n!}$ times a polynomial with integer coefficients and only powers x^n through x^{2n}. Use $a - b(\pi - x) = bx$. (b) $f^{(2n+2)} = 0$. (c) On $(0, \pi)$, $0 < x(a - bx) < \pi a$.

Exercise 9.9.1. *Hint.* Write $\alpha = c_1 + (c_2 - c_1)I_c$, and argue as in Example 9.9.6.

Exercise 9.9.2. *Hint.* Use the Intermediate Value Theorem to find x_i with $\alpha(x_i) = \alpha(a) + \frac{i}{n}[\alpha(b) - \alpha(a)]$.

Solution. Let $n \in \mathbb{N}$. If $\alpha(b) = \alpha(a)$, then α is constant, since it is increasing. Then every partition with n subintervals works, since every $\Delta\alpha_i = 0$.

Suppose $\alpha(a) < \alpha(b)$. For $i = 0, 1, \ldots, n$, let

$$y_i = \alpha(a) + \frac{i}{n}[\alpha(b) - \alpha(a)].$$

Then $\alpha(a) = y_0 < y_1 < \cdots < y_n = \alpha(b)$. Let $x_0 = a$ and $x_n = b$. For $0 < i < n$, y_i is between $\alpha(a)$ and $\alpha(b)$. Since α is continuous, the Intermediate Value Theorem (Theorem 6.3.10) gives an $x_i \in (a, b)$ with $\alpha(x_i) = y_i$. So $\alpha(x_i) = y_i$ for every i.

The points are in order. If $x_i \le x_{i-1}$ for some i, then $\alpha(x_i) \le \alpha(x_{i-1})$, since α is increasing. That says $y_i \le y_{i-1}$, which is false. So $a = x_0 < x_1 < \cdots < x_n = b$, and $\mathcal{P} = \{x_0, \dots, x_n\}$ is a partition of $[a, b]$. For each i,

$$\Delta\alpha_i = y_i - y_{i-1} = \frac{1}{n}[\alpha(b) - \alpha(a)].$$

Exercise 9.9.3. *Hint.* Since α and β increase, the sup of α on $[x_{i-1}, x_i]$ is $\alpha(x_i)$ and the inf of β is $\beta(x_{i-1})$. Check that $\alpha(x_i)\beta(x_i) - \alpha(x_{i-1})\beta(x_{i-1}) = \alpha(x_i)\Delta\beta_i + \beta(x_{i-1})\Delta\alpha_i$, and sum over i.

Exercise 9.9.4. *Hint.* By Example 9.9.6, $\int_a^b f\, dI_{s_i} = f(s_i)$. Multiplying the integrator by $c \ge 0$ multiplies every Riemann–Stieltjes sum by c. Then use Theorem 9.9.10 (3).

Solution. The first step shows that scaling the integrator scales the integral. Let $\beta : [a, b] \to \mathbb{R}$ be increasing, let $f \in \mathcal{R}(\beta)$, and let $c \ge 0$. The claim is that $f \in \mathcal{R}(c\beta)$ and $\int_a^b f\, d(c\beta) = c\int_a^b f\, d\beta$. If $c = 0$, then $c\beta$ is constant. Every sum for $c\beta$ is 0, so the integral is 0. Suppose $c > 0$. For any partition $\mathcal{P}$, the increments of $c\beta$ are $c\Delta\beta_i$. So

$$\mathcal{U}(\mathcal{P}, f, c\beta) = c\,\mathcal{U}(\mathcal{P}, f, \beta) \quad \text{and} \quad \mathcal{L}(\mathcal{P}, f, c\beta) = c\,\mathcal{L}(\mathcal{P}, f, \beta).$$

Let $\varepsilon > 0$. Riemann's Criterion (Theorem 9.9.8) gives a partition $\mathcal{P}$ with $\mathcal{U}(\mathcal{P}, f, \beta) - \mathcal{L}(\mathcal{P}, f, \beta) < \frac{\varepsilon}{c}$. Then the difference for $c\beta$ is less than ε. So $f \in \mathcal{R}(c\beta)$. Also, for every $\mathcal{P}$, $\mathcal{L}(\mathcal{P}, f, \beta) \le \int_a^b f\, d\beta \le \mathcal{U}(\mathcal{P}, f, \beta)$. Multiplying by c, the number $c\int_a^b f\, d\beta$ lies between every lower sum and every upper sum for $c\beta$. So it lies between the lower and upper integrals, which both equal $\int_a^b f\, d(c\beta)$.

The second step handles a single jump. Since $a < s_i \le b$ and f is continuous, Example 9.9.6 gives $f \in \mathcal{R}(I_{s_i})$ with $\int_a^b f\, dI_{s_i} = f(s_i)$. Here $I_{s_i}(x) = I(x - s_i)$. By the first step, $f \in \mathcal{R}(c_i I_{s_i})$ with $\int_a^b f\, d(c_i I_{s_i}) = c_i f(s_i)$. Each $c_i I_{s_i}$ is increasing, since $c_i \ge 0$.

The third step adds the jumps together. Let $\alpha_k = \sum_{i=1}^k c_i I_{s_i}$. Induction on k shows that $f \in \mathcal{R}(\alpha_k)$ and $\int_a^b f\, d\alpha_k = \sum_{i=1}^k c_i f(s_i)$. The case $k = 1$ is the second step. If it holds for k, then $\alpha_{k+1} = \alpha_k + c_{k+1} I_{s_{k+1}}$ is a sum of two increasing functions. By Theorem 9.9.10 (3) and the second step,

$$\int_a^b f\, d\alpha_{k+1} = \int_a^b f\, d\alpha_k + \int_a^b f\, d(c_{k+1} I_{s_{k+1}}) = \sum_{i=1}^{k+1} c_i f(s_i).$$

Since $\alpha = \alpha_n$, this proves the theorem.

Exercise 9.9.5. *Hint.* On $[1, 5]$, $\lfloor x^2 \rfloor = 1 + \sum_{k=2}^{25} I(x - \sqrt{k})$. Adding a constant to the integrator does not change the integral. Use Theorem 9.9.15.

Exercise 9.9.6. *Hint.* Use Theorem 9.9.17.

Exercise 9.9.7. *Hint.* First show that $V_\alpha(x) = \int_a^x |\alpha'(t)|\, dt$, so that $V_\alpha \in C^{(1)}[a, b]$. Then apply Theorem 9.9.17 to V_α and to $V_\alpha - \alpha$.

Solution. By Corollary 8.1.10, $\alpha \in BV[a, b]$. So V_α is defined.

The first step shows that $V_\alpha(x) = \int_a^x |\alpha'(t)|\, dt$ for $x \in [a, b]$. Both sides are 0 at $x = a$. Fix $x \in (a, b]$. Since $|\alpha'|$ is continuous, it is integrable on $[a, x]$. Let

$\mathcal{P} = \{x_0, \ldots, x_n\}$ be a partition of $[a, x]$. By the Mean Value Theorem, $\alpha(x_i) - \alpha(x_{i-1}) = \alpha'(t_i)\Delta x_i$ for some $t_i \in (x_{i-1}, x_i)$. So

$$V(\mathcal{P}) = \sum_{i=1}^{n} |\alpha'(t_i)|\Delta x_i,$$

and $\mathcal{L}(\mathcal{P}, |\alpha'|) \le V(\mathcal{P}) \le \mathcal{U}(\mathcal{P}, |\alpha'|)$. The left inequality and $V(\mathcal{P}) \le V_\alpha[a, x]$ give $\int_a^x |\alpha'| = \underline{\int_a^x} |\alpha'| \le V_\alpha[a, x]$. For the other direction, let $\mathcal{Q}$ be any partition of $[a, x]$. By Lemmas 8.1.7 and 9.1.5,

$$V(\mathcal{P}) \le V(\mathcal{P} \cup \mathcal{Q}) \le \mathcal{U}(\mathcal{P} \cup \mathcal{Q}, |\alpha'|) \le \mathcal{U}(\mathcal{Q}, |\alpha'|).$$

So $V(\mathcal{P}) \le \overline{\int_a^x} |\alpha'| = \int_a^x |\alpha'|$ for every $\mathcal{P}$, and $V_\alpha[a, x] \le \int_a^x |\alpha'|$.

The second step writes α as a difference of two increasing functions with integrable derivatives. By the first step and Theorem 9.3.7, V_α is differentiable on $[a, b]$ with $V_\alpha' = |\alpha'|$, which is continuous and so in $\mathcal{R}[a, b]$. Let $\beta = V_\alpha - \alpha$. Then β is differentiable with $\beta' = |\alpha'| - \alpha' \in \mathcal{R}[a, b]$. Both V_α and β are increasing, by Theorems 8.2.2 and 8.2.4. By Theorem 9.9.17, $f \in \mathcal{R}(V_\alpha)$ and $f \in \mathcal{R}(\beta)$, with

$$\int_a^b f\, dV_\alpha = \int_a^b f|\alpha'| \quad \text{and} \quad \int_a^b f\, d\beta = \int_a^b f\big(|\alpha'| - \alpha'\big).$$

By Definition 9.9.20, $f \in \mathcal{R}(\alpha)$, and by Theorem 9.2.11,

$$\int_a^b f\, d\alpha = \int_a^b f|\alpha'| - \int_a^b f\big(|\alpha'| - \alpha'\big) = \int_a^b f\alpha'.$$

Chapter 10. Series of Real Numbers

Exercise 10.1.2.

(a) For $n \ge 2$, $\sum_{k=2}^{n} \left(\frac{2}{3}\right)^k = \sum_{k=1}^{n} \left(\frac{2}{3}\right)^k - \frac{2}{3}$. By Example 10.1.3, this tends to $\frac{2/3}{1-2/3} - \frac{2}{3} = 2 - \frac{2}{3} = \frac{4}{3}$.

(b) The partial sums are 5 times those of $\sum_{k=0}^{\infty} \left(-\frac{1}{3}\right)^k$, which tend to $\frac{1}{1+\frac{1}{3}} = \frac{3}{4}$ by Example 10.1.3. By Theorem 3.2.4 (2), the sum is $\frac{15}{4}$.

(c) The nth partial sum is $\sum_{k=1}^{n} \left(\frac{1}{3}\right)^k + \sum_{k=1}^{n} \left(\frac{1}{2}\right)^k$. By Example 10.1.3, these tend to $\frac{1/3}{2/3} = \frac{1}{2}$ and $\frac{1/2}{1/2} = 1$. By Theorem 3.2.4 (1), the sum is $\frac{3}{2}$.

Exercise 10.1.3. Let $r = \frac{1}{10^m}$, so $0 < r < 1$. The partial sums of $\sum_{k=1}^{\infty} \frac{D}{10^{mk}}$ are D times those of $\sum_{k=1}^{\infty} r^k$. By Example 10.1.3 and Theorem 3.2.4 (2),

$$\sum_{k=1}^{\infty} \frac{D}{10^{mk}} = D \cdot \frac{r}{1-r} = \frac{D}{10^m - 1}.$$

Since D and $10^m - 1$ are integers and $10^m - 1 \neq 0$, this is a rational number. With $m = 2$ and $D = 27$, $0.272727\ldots = \frac{27}{99} = \frac{3}{11}$.

Exercise 10.2.1. *Hint.* Write $a_k = S_k - S_{k-1}$, where S_k is the kth partial sum.

Solution. Let $s_n = \sum_{k=1}^{n} a_k$, and suppose $s_n \to s$. Let $\varepsilon > 0$. There is an $N \in \mathbb{N}$ with $|s_n - s| < \frac{\varepsilon}{2}$ for all $n \geq N$. Let $k \geq N+1$. Then $k-1 \geq N$, and $a_k = s_k - s_{k-1}$. So

$$|a_k| = |(s_k - s) + (s - s_{k-1})| \leq |s_k - s| + |s_{k-1} - s| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So $a_k \to 0$. The second statement is the contrapositive of the first.

Exercise 10.2.3. *Hint.* In (1), $\frac{L}{2} b_k < a_k < \frac{3L}{2} b_k$ for large k. In (2), $a_k < b_k$ for large k. Use Theorem 10.2.7.

Solution.

(1) Let $\varepsilon = \frac{L}{2}$. There is a $K \in \mathbb{N}$ such that $\left| \frac{a_k}{b_k} - L \right| < \frac{L}{2}$ for all $k \geq K$. Since $b_k > 0$, this gives

$$\frac{L}{2} b_k < a_k < \frac{3L}{2} b_k \quad \text{for all } k \geq K.$$

If $\sum b_k$ converges, then $\sum a_k$ converges by the Comparison Test (Theorem 10.2.7), with $M = \frac{3L}{2}$. If $\sum a_k$ converges, then $\sum b_k$ converges by the Comparison Test, since $b_k < \frac{2}{L} a_k$ for all $k \geq K$.

(2) With $\varepsilon = 1$, there is a $K \in \mathbb{N}$ with $\frac{a_k}{b_k} < 1$, that is, $0 < a_k < b_k$, for all $k \geq K$. If $\sum b_k$ converges, then $\sum a_k$ converges by the Comparison Test, with $M = 1$.

Exercise 10.2.4. *Hint.* Use the series $\sum \frac{1}{k}$ and $\sum \frac{1}{k^2}$. To find $\limsup \sqrt[k]{a_k}$, first show $\sqrt[k]{k} \to 1$. One way is to write $\sqrt[k]{k} = 1 + h_k$ with $h_k \geq 0$ and use the Binomial Theorem to get $k \geq \binom{k}{2} h_k^2$ for $k \geq 2$.

Solution. First, $\sqrt[k]{k} \to 1$. Since $k \geq 1$, $\sqrt[k]{k} \geq 1$. Write $\sqrt[k]{k} = 1 + h_k$ with $h_k \geq 0$. For $k \geq 2$, every term in the Binomial Theorem (Exercise 9.3.3) is nonnegative. So

$$k = (1 + h_k)^k \geq \binom{k}{2} h_k^2 = \frac{k(k-1)}{2} h_k^2.$$

So $0 \leq h_k^2 \leq \frac{2}{k-1}$. Let $\varepsilon > 0$. If $k > 1 + \frac{2}{\varepsilon^2}$, then $h_k^2 < \varepsilon^2$, so $h_k < \varepsilon$. So $h_k \to 0$, and $\sqrt[k]{k} \to 1$.

Now let $a_k = \frac{1}{k}$. Then $\sqrt[k]{a_k} = \frac{1}{\sqrt[k]{k}} \to 1$. Let $a_k = \frac{1}{k^2}$. Then $\sqrt[k]{a_k} = \left(\frac{1}{\sqrt[k]{k}} \right)^2 \to 1$. In both cases the sequence $\{\sqrt[k]{a_k}\}$ converges to 1, so $\alpha = \limsup \sqrt[k]{a_k} = 1$ by Corollary 3.6.14. But $\sum \frac{1}{k}$ diverges and $\sum \frac{1}{k^2}$ converges, by Example 10.2.12. So when $\alpha = 1$, the series may converge or diverge, and the Root Test gives no information.

Exercise 10.2.6. *Hint.* Let $b_k = \frac{1}{k}$ for even k and $b_k = \frac{1}{k^2}$ for odd k.

Exercise 10.2.7. *Hint.* If $\{b_k\}$ is decreasing, let $b = \lim_{k \to \infty} b_k$ and apply the Dirichlet Test to $\sum a_k (b_k - b)$.

Exercise 10.2.8. *Hint.* Since $a_k \to 0$, $a_k \leq 1$ for all large k, and then $a_k^2 \leq a_k$. For the example, try $a_k = \frac{1}{k^2}$.

Exercise 10.2.9. *Hint.* Use the identity $2 \sin \frac{t}{2} \cos kt = \sin\left((k + \frac{1}{2})t\right) - \sin\left((k - \frac{1}{2})t\right)$.

Solution. Since $\{b_k\}$ decreases to 0, $b_k \geq 0$ for all k. Let $t \neq 2p\pi$ for every $p \in \mathbb{Z}$. Then $\frac{t}{2}$ is not a multiple of π, so $\sin\frac{t}{2} \neq 0$. Let $A_n = \sum_{k=1}^{n} \cos kt$. By the identity $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$, with $A = \frac{t}{2}$ and $B = kt$,

$$2\sin\frac{t}{2}\cos kt = \sin\left(\left(k+\frac{1}{2}\right)t\right) - \sin\left(\left(k-\frac{1}{2}\right)t\right).$$

Adding over $k = 1, \ldots, n$, the sum telescopes. So

$$2\sin\frac{t}{2}\,A_n = \sin\left(\left(n+\frac{1}{2}\right)t\right) - \sin\frac{t}{2}, \quad \text{and} \quad |A_n| \leq \frac{2}{2\left|\sin\frac{t}{2}\right|} = \frac{1}{\left|\sin\frac{t}{2}\right|}.$$

So the partial sums A_n are bounded. By the Dirichlet Test (Theorem 10.2.16) with $a_k = \cos kt$, $\sum b_k \cos kt$ converges.

Exercise 10.2.10. *Hint.* (a) Use the Ratio Test. (c) Use the Integral Test. (d) Use the Root Test. (e) and (f) Use the Limit Comparison Test, with $\frac{1}{k}$ and with $\frac{1}{k^2}$. (g) Use the Alternating Series Test. (b) Use the Ratio Test and $\left(1+\frac{1}{k}\right)^k \to e$ from Example 9.4.1.

Exercise 10.2.11. *Hint.* Group the terms in blocks from 2^j to $2^{j+1}-1$. Show $s_n \leq t_m$ when $n < 2^{m+1}$, and $s_n \geq \frac{1}{2}t_m$ when $n \geq 2^m$, where $t_m = \sum_{j=0}^{m} 2^j a_{2^j}$.

Exercise 10.3.1. *Hint.* The partial sums of $\sum(b_k - b_{k+1})$ are $b_1 - b_{n+1}$. Use the Comparison Test.

Exercise 10.3.2. *Hint.* For large k, $|k^p a_k| \leq |A| + 1$. Compare with $\sum \frac{1}{k^p}$.

Solution. With $\varepsilon = 1$, there is a $K \in \mathbb{N}$ with $|k^p a_k - A| < 1$ for all $k \geq K$. Then $|k^p a_k| < |A| + 1$, so

$$0 \leq |a_k| \leq \frac{|A|+1}{k^p} \quad \text{for all } k \geq K.$$

Since $p > 1$, $\sum \frac{1}{k^p}$ converges (Example 10.2.12). By the Comparison Test (Theorem 10.2.7) with $M = |A| + 1$, $\sum |a_k|$ converges. So $\sum a_k$ converges absolutely.

Exercise 10.3.3. *Hint.* Since $a_k \to 0$, eventually $|a_k| \leq 1$ and $a_k^2 \leq |a_k|$. For the example, try $a_k = \frac{(-1)^k}{\sqrt{k}}$.

Exercise 10.3.4. *Hint.* Use $a_k^+ + a_k^- = |a_k|$, $a_k^+ - a_k^- = a_k$ and $0 \leq a_k^{\pm} \leq |a_k|$.

Solution. If $a_k \geq 0$, then $a_k^+ = a_k$ and $a_k^- = 0$. If $a_k < 0$, then $a_k^+ = 0$ and $a_k^- = -a_k$. In both cases,

$$a_k^+ + a_k^- = |a_k|, \quad a_k^+ - a_k^- = a_k, \quad \text{and} \quad 0 \leq a_k^{\pm} \leq |a_k|.$$

($\Rightarrow$) Suppose $\sum |a_k|$ converges. Since $0 \leq a_k^+ \leq |a_k|$, $\sum a_k^+$ converges by the Comparison Test (Theorem 10.2.7). In the same way, $\sum a_k^-$ converges.

($\Leftarrow$) Suppose $\sum a_k^+$ and $\sum a_k^-$ converge. By Theorem 10.2.4 (2), $\sum(a_k^+ + a_k^-) = \sum |a_k|$ converges. So $\sum a_k$ converges absolutely.

In this case, by Theorem 10.2.4 (1) with $c = -1$ and Theorem 10.2.4 (2),

$$\sum a_k = \sum(a_k^+ - a_k^-) = \sum a_k^+ - \sum a_k^-.$$

Exercise 10.3.5. *Hint.* Modify the construction in the proof of Theorem 10.3.9. Add positive terms until the total passes 1, then one negative term. Then add positive terms until the total passes 2, then one negative term, and so on.

Exercise 10.4.1. For $|x| < 1$, $\sum_{k=0}^{\infty} x^k$ converges absolutely to $\frac{1}{1-x}$. The Cauchy product of this series with itself has terms $c_n = \sum_{k=0}^{n} x^k x^{n-k} = (n+1)x^n$. By Theorem 10.4.2, $\sum_{n=0}^{\infty}(n+1)x^n = \frac{1}{1-x} \cdot \frac{1}{1-x} = \frac{1}{(1-x)^2}$.

Exercise 10.4.2. *Hint.* $|c_n| \le \sum_{k=0}^{n} |a_k||b_{n-k}|$, which is the nth term of the Cauchy product of $\sum |a_k|$ and $\sum |b_k|$.

Exercise 10.5.1. *Hint.* If p_i divides n, write $n = p_i j$ and $p_1 \cdots p_m = p_i c$. Then $p_i(j - c) = 1$.

Exercise 10.5.2. By the distributive law,

$$T_K = \left(\sum_{a=0}^{K} \frac{1}{2^a}\right)\left(\sum_{b=0}^{K} \frac{1}{3^b}\right).$$

By Example 10.1.3, the first factor tends to $\frac{1}{1-\frac{1}{2}} = 2$, and the second tends to $\frac{1}{1-\frac{1}{3}} = \frac{3}{2}$. By Theorem 3.2.4 (2), $T_K \to 2 \cdot \frac{3}{2} = 3$.

Exercise 10.5.3. *Hint.* By the proof of Theorem 10.5.1, $s_N \le \prod \frac{q}{q-1}$, the product over the primes $q \le N$. Take logarithms, and use $-\ln(1-x) \le 2x$ for $0 \le x \le \frac{1}{2}$.

Chapter 11. Sequences and Series of Functions

Exercise 11.2.1. *Hint.* If $|f_n(x) - f(x)| < \varepsilon$ for all $x \in E$, then $M_n \le \varepsilon$. If $M_n < \varepsilon$, then $|f_n(x) - f(x)| < \varepsilon$ for all $x \in E$.

Solution. ($\Rightarrow$) Suppose $f_n \to f$ uniformly on E. Let $\varepsilon > 0$. There is an $N \in \mathbb{N}$ such that $|f_n(x) - f(x)| < \frac{\varepsilon}{2}$ for all $n \ge N$ and all $x \in E$. For $n \ge N$, $\frac{\varepsilon}{2}$ is an upper bound of $\{|f_n(x) - f(x)| : x \in E\}$. So $0 \le M_n \le \frac{\varepsilon}{2} < \varepsilon$. So $M_n \to 0$.

($\Leftarrow$) Suppose $M_n \to 0$. Let $\varepsilon > 0$. There is an $N \in \mathbb{N}$ such that $M_n < \varepsilon$ for all $n \ge N$. Then for $n \ge N$ and $x \in E$,

$$|f_n(x) - f(x)| \le M_n < \varepsilon.$$

So $f_n \to f$ uniformly on E.

Exercise 11.2.2. *Hint.* By Corollary 11.2.6, the terms of a uniformly convergent series converge uniformly to 0. Find $\sup_{|x|<2} \left|\left(\frac{x}{2}\right)^k\right|$.

Solution. Suppose the series converges uniformly on $(-2, 2)$. By Corollary 11.2.6 with $\varepsilon = \frac{1}{2}$, there is an $N \in \mathbb{N}$ such that

$$\left|\sum_{k=n+1}^{m} \left(\frac{x}{2}\right)^k\right| < \frac{1}{2} \quad \text{for all } m > n \ge N \text{ and all } x \in (-2, 2).$$

Take $n = N$ and $m = N + 1$. Then $\left|\left(\frac{x}{2}\right)^{N+1}\right| < \frac{1}{2}$ for all $x \in (-2, 2)$. But let $x = 2 \cdot 2^{-1/(N+1)}$. Then $0 < x < 2$ and $\left(\frac{x}{2}\right)^{N+1} = \frac{1}{2}$. This is a contradiction. So the convergence is not uniform on $(-2, 2)$.

Exercise 11.2.3. *Hint.* Find the pointwise limit. It is not continuous at 1. Use Corollary 11.2.16.

Exercise 11.2.4. *Hint.* Sum the geometric series. The sum is 1 for $0 < x \leq 1$ and 0 at $x = 0$. Use Corollary 11.2.16.

Exercise 11.2.5. *Hint.* Use the Weierstrass M-Test (Theorem 11.2.10) with $M_k = |a_k|$.

Exercise 11.2.6. *Hint.* Let $|f| \leq M$. Then $|x^n f(x)| \leq M$. The limit is 0 on $[0, 1)$ and $f(1)$ at 1. Its integral is 0, by Exercise 9.2.2.

Solution. Since $f \in \mathcal{R}[0, 1]$, f is bounded. Choose $M > 0$ with $|f(x)| \leq M$ for all $x \in [0, 1]$. Let $g_n(x) = x^n f(x)$. Since x^n is continuous, $g_n \in \mathcal{R}[0, 1]$ by Theorem 9.2.11 (3). Also $|g_n(x)| \leq M$ for all $x \in [0, 1]$ and all $n \in \mathbb{N}$.

If $0 \leq x < 1$, then $x^n \to 0$ by Example 3.1.7, so $g_n(x) \to 0$. Also $g_n(1) = f(1)$ for all n. So $g_n \to g$ pointwise on $[0, 1]$, where $g(x) = 0$ for $x \in [0, 1)$ and $g(1) = f(1)$. By Exercise 9.2.2, $g \in \mathcal{R}[0, 1]$ and $\int_0^1 g = 0$. By the Bounded Convergence Theorem (Theorem 11.2.21),

$$\lim_{n\to\infty} \int_0^1 x^n f(x)\, dx = \int_0^1 g = 0.$$

Exercise 11.2.7. *Hint.* (a) Define a new sequence of functions $\{g_n\}$ by $g_n(x) = \max_{k\in\{1,\ldots,n\}} f_k(x)$ for each $x \in [a, b]$. Show $\{g_n\}$ is a sequence of continuous functions that converges uniformly to g. (b) Try $f_n(x) = \max\{0, 1 - n|x - \frac{1}{n}|\}$ on $[0, 1]$. These are tents of height 1 that move toward 0.

Exercise 11.2.8. *Hint.* Try the moving tents from the hint to Exercise 11.2.7 (b).

Solution. For $n \in \mathbb{N}$ and $x \in [0, 1]$, let

$$f_n(x) = \max\left\{0,\ 1 - n\left|x - \frac{1}{n}\right|\right\}.$$

The graph of f_n is a tent of height 1 over $\left[0, \frac{2}{n}\right]$, with its peak at $\frac{1}{n}$. Each f_n is continuous, since $x \mapsto 1 - n\left|x - \frac{1}{n}\right|$ is continuous and the maximum of two continuous functions is continuous. Also $0 \leq f_n(x) \leq 1$.

The sequence converges pointwise to 0. First, $f_n(0) = \max\{0, 1 - 1\} = 0$ for all n. Now let $0 < x \leq 1$. If $n > \frac{2}{x}$, then $x - \frac{1}{n} > \frac{1}{n}$, so $1 - n\left|x - \frac{1}{n}\right| < 0$ and $f_n(x) = 0$. So $f_n(x) \to 0$.

The convergence is not uniform. Since $f_n\left(\frac{1}{n}\right) = 1$, $\sup_{x\in[0,1]} |f_n(x) - 0| = 1$ for every n. By Theorem 11.2.7, $\{f_n\}$ does not converge uniformly to 0.

Here $[0, 1]$ is compact and the limit 0 is continuous. So Dini's Theorem fails only because the sequence is not decreasing. For example, $f_1(x) = x$, so $f_1\left(\frac{1}{n}\right) = \frac{1}{n} < 1 = f_n\left(\frac{1}{n}\right)$ for $n \geq 2$.

Exercise 11.2.10. *Hint.* Write $f_n(x) = f_n(x_0) + \int_{x_0}^x f_n'$ by the Fundamental Theorem of Calculus, and use Theorem 11.2.18.

Exercise 11.2.11. *Hint.* Choose N with $|f_n(x) - f(x)| < 1$ for all $n \geq N$ and $x \in E$. Then $|f|$ is at most 1 more than a bound for $|f_N|$. Treat $f_1, \ldots, f_{N-1}$ separately.

Exercise 11.4.1. By Lemma 11.4.2, $B_n 1 = \sum_k p_{n,k}(x) = 1$ and $B_n x = \sum_k \frac{k}{n} p_{n,k}(x) = \frac{nx}{n} = x$. In the proof of that lemma, $\sum_k k(k-1)p_{n,k}(x) = n(n-1)x^2$, so $\sum_k k^2 p_{n,k}(x) = n(n-1)x^2 + nx$. Thus

$$B_n x^2 = \sum_{k=0}^{n} \frac{k^2}{n^2} p_{n,k}(x) = \frac{n(n-1)x^2 + nx}{n^2} = x^2 + \frac{x(1-x)}{n}.$$

Exercise 11.4.2. By linearity, $\int_0^1 f(x)p(x)\,dx = 0$ for every polynomial p. Let M be a bound for $|f|$ on $[0,1]$ (Theorem 6.3.15), and let $\varepsilon > 0$. By Theorem 11.4.3, there is a polynomial p with $|f(x) - p(x)| < \varepsilon$ on $[0,1]$. Then

$$0 \le \int_0^1 f^2 = \int_0^1 f(f-p) + \int_0^1 fp = \int_0^1 f(f-p) \le M\varepsilon.$$

Since ε is arbitrary, $\int_0^1 f^2 = 0$. Suppose $f(c) \neq 0$ for some c. Then $f^2(c) > 0$, and since f^2 is continuous, $f^2(x) > \frac{f^2(c)}{2}$ on some interval $[\alpha, \beta] \subset [0,1]$ with $\alpha < \beta$. Then $\int_0^1 f^2 \ge \frac{f^2(c)}{2}(\beta - \alpha) > 0$. This is a contradiction. So $f = 0$.

Exercise 11.4.3. *Hint.* If $|p(x) - e^x| < 1$ for all x, then p is bounded on $(-\infty, 0]$. A polynomial that is not constant is unbounded there.

Exercise 11.4.4. *Hint.* A polynomial is bounded on $(0,1)$, but $\frac{1}{x}$ is not.

Exercise 11.4.5. *Hint.* Approximate f' by polynomials q_n, and let $p_n(x) = f(a) + \int_a^x q_n$.

Exercise 11.5.1. Here $f(t,y) = 2ty$. On any $[a,b]$ containing 0, it satisfies the hypotheses with $L = 2\max\{|a|,|b|\}$. The iterates are $\varphi_0 = 1$, $\varphi_1(t) = 1 + \int_0^t 2s\,ds = 1 + t^2$, $\varphi_2(t) = 1 + \int_0^t 2s(1+s^2)\,ds = 1 + t^2 + \frac{t^4}{2}$. By induction, $\varphi_n(t) = \sum_{k=0}^{n} \frac{t^{2k}}{k!}$. Indeed, if this holds for n, then

$$\varphi_{n+1}(t) = 1 + \int_0^t \sum_{k=0}^{n} \frac{2s^{2k+1}}{k!}\,ds = 1 + \sum_{k=0}^{n} \frac{t^{2k+2}}{(k+1)!} = \sum_{j=0}^{n+1} \frac{t^{2j}}{j!}.$$

These are the partial sums of the series for e^u with $u = t^2$. So the solution is $y(t) = e^{t^2}$.

Exercise 11.5.2. Clearly $y = 0$ works. For $y(t) = t^3$, $y(0) = 0$ and $y'(t) = 3t^2$. Also $y(t)^{2/3} = \left(\sqrt[3]{t^3}\right)^2 = t^2$. So $y' = 3y^{2/3}$. The Lipschitz condition fails. If $|3y^{2/3} - 3z^{2/3}| \le L|y - z|$ for all y, z, then taking $z = 0$ gives $3|y|^{2/3} \le L|y|$, so $3 \le L|y|^{1/3}$ for all $y \neq 0$. This is false for small y.

Exercise 11.5.3. *Hint.* Show $|x_{n+1} - x_n| \le k^{n-1}|x_2 - x_1|$, so $\{x_n\}$ is Cauchy. For uniqueness, $|c - d| = |g(c) - g(d)| \le k|c - d|$.

Exercise 11.5.4. *Hint.* Sum the estimate in the proof from $k = n$ to ∞, and use $(j + n + 1)! \ge n!\, j!$.

Exercise 11.5.5. *Hint.* Check the two conditions of Section 11.5 with $L = \max|p|$.

Chapter 12. Power Series Expansions

Exercise 12.1.1. *Hint.* Set $y = (x+1)^2$. The series becomes $\sum \left(\frac{y}{4}\right)^k$.

Solution. The radius of convergence is $R = 2$. Write the series as $\sum_{k=0}^{\infty} a_k (x+1)^k$. Then $a_{2j} = \frac{1}{4^j}$ and $a_k = 0$ for odd k. For $k = 2j$,

$$\sqrt[k]{|a_k|} = \left(4^{-j}\right)^{1/(2j)} = \frac{1}{2},$$

and for odd k, $\sqrt[k]{|a_k|} = 0$. So $\sup\left\{\sqrt[k]{|a_k|} : k \geq n\right\} = \frac{1}{2}$ for every n, and $\limsup \sqrt[k]{|a_k|} = \frac{1}{2}$. By Definition 12.1.4, $R = 2$.

This agrees with the substitution $y = (x+1)^2$. The series becomes the geometric series $\sum \left(\frac{y}{4}\right)^k$, which converges exactly when $(x+1)^2 < 4$, that is, $-3 < x < 1$. At $x = -3$ and $x = 1$ every term is 1, so the series diverges there.

Exercise 12.1.2. *Hint.* Set $y = x^3$. The series becomes $\sum \left(\frac{y}{8}\right)^k$.

Exercise 12.1.3. *Hint.* Use Corollary 12.1.6. In (d), the ratio of consecutive coefficients is $\left(\frac{k}{k+1}\right)^k \to \frac{1}{e}$. At $x = \pm e$, use $\left(1 + \frac{1}{k}\right)^k < e$, from Example 9.4.1, to show the terms do not tend to 0.

Exercise 12.1.4. *Hint.* Follow the proof of the right inequality. Take $0 < \beta < \liminf \frac{a_{k+1}}{a_k}$ and show that $a_k \geq c\beta^k$ for large k, for some $c > 0$.

Solution. Let $\gamma = \liminf \frac{a_{k+1}}{a_k}$. If $\gamma = 0$, there is nothing to prove, since every $\sqrt[k]{a_k} > 0$. Otherwise let $0 < b < \gamma$. By Definition 3.6.8, $\gamma = \sup_n \inf\left\{\frac{a_{k+1}}{a_k} : k \geq n\right\}$. Since $b < \gamma$, there is a $K \in \mathbb{N}$ with $\inf\left\{\frac{a_{k+1}}{a_k} : k \geq K\right\} > b$. So $a_{k+1} \geq b a_k$ for all $k \geq K$. By induction, $a_k \geq a_K b^{k-K}$ for all $k \geq K$. So for $k \geq K$,

$$\sqrt[k]{a_k} \geq z_k, \quad \text{where} \quad z_k = \sqrt[k]{a_K b^{-K}} \cdot b.$$

Since $a_K b^{-K} > 0$, $z_k \to b$ by Lemma 12.1.3. For $n \geq K$, each element of $\{\sqrt[k]{a_k} : k \geq n\}$ is at least the matching z_k. So $\inf\{\sqrt[k]{a_k} : k \geq n\} \geq \inf\{z_k : k \geq n\}$. Letting $n \to \infty$ gives $\liminf \sqrt[k]{a_k} \geq \liminf z_k = b$, by Corollary 3.6.14. Since this holds for every $b < \gamma$, $\liminf \sqrt[k]{a_k} \geq \gamma$. This also covers $\gamma = \infty$.

Exercise 12.2.1. *Hint.* Fix ρ with $|x - c| \leq \rho < R$. The series for f converges uniformly for $|t - c| \leq \rho$. Integrate it term by term, as in Example 12.2.2. (a) Expand $\frac{1}{1+t^2}$ as a geometric series.

Solution. Fix x with $|x - c| < R$, and choose ρ with $|x - c| \leq \rho < R$. By Theorem 12.1.7 (3), $\sum a_k (t-c)^k$ converges uniformly to $f(t)$ for $|t - c| \leq \rho$. In particular, it converges uniformly on the closed interval between c and x. Each term is continuous, so integrable. Suppose $x > c$. By Corollary 11.2.19 and the Fundamental Theorem of Calculus,

$$F(x) = \int_c^x f(t)\,dt = \sum_{k=0}^{\infty} \int_c^x a_k (t-c)^k\,dt = \sum_{k=0}^{\infty} \frac{a_k}{k+1} (x-c)^{k+1}.$$

If $x < c$, apply Corollary 11.2.19 on $[x, c]$ and multiply by -1, using Theorem 10.2.4 (1). Since $\int_x^c a_k(t-c)^k\,dt = -\frac{a_k}{k+1}(x-c)^{k+1}$, the same formula results. If $x = c$, both sides are 0.

(a) For $|t| < 1$, the geometric series with ratio $-t^2$ gives $\frac{1}{1+t^2} = \sum_{k=0}^{\infty}(-1)^k t^{2k}$. This is a power series in t whose coefficients are $(-1)^k$ for the power t^{2k} and 0 for odd powers. As in Exercise 12.1.1, $\limsup \sqrt[m]{|a_m|} = 1$, so its radius of convergence is 1. By the first part, for $|x| < 1$,

$$\int_0^x \frac{dt}{1+t^2} = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1}.$$

Since $\frac{d}{dx}\arctan x = \frac{1}{1+x^2}$ and $\arctan 0 = 0$, the Fundamental Theorem of Calculus gives $\int_0^x \frac{dt}{1+t^2} = \arctan x$. (For $x < 0$, $\int_0^x$ means $-\int_x^0$.) So $\arctan x = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1}$ for $x \in (-1, 1)$.

(b) Let $g(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} x^{2k+1} = \sum_{m=0}^{\infty} b_m x^m$, where $b_{2k+1} = \frac{(-1)^k}{2k+1}$ and $b_m = 0$ for even m. For odd m, $\sqrt[m]{|b_m|} = \frac{1}{\sqrt[m]{m}}$, which is at most 1 and tends to 1 by Lemma 12.1.3. So every tail of $\{\sqrt[m]{|b_m|}\}$ has supremum 1, and the radius of convergence is 1. The series $\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1}$ converges by the Alternating Series Test, since $\frac{1}{2k+1}$ decreases to 0. The partial sums of $\sum b_m$ are 0 followed by the partial sums of $\sum \frac{(-1)^k}{2k+1}$, each listed twice. So $\sum b_m$ converges to the same sum. By Abel's Theorem (Theorem 12.2.1) and (a),

$$\sum_{k=0}^{\infty} \frac{(-1)^k}{2k+1} = \lim_{x\to 1^-} g(x) = \lim_{x\to 1^-} \arctan x = \arctan 1 = \frac{\pi}{4},$$

since arctan is continuous. So the formula in (a) holds at $x = 1$. At $x = -1$, the series is $\sum_{k=0}^{\infty} \frac{(-1)^k(-1)^{2k+1}}{2k+1} = -\frac{\pi}{4} = \arctan(-1)$. So it holds at $x = -1$ too.

Exercise 12.2.2. *Hint.* For $|x| < 1$, the series is geometric.

Exercise 12.2.3. *Hint.* Differentiate the geometric series and multiply by x. For the logarithm, replace x by $-x^2$ in the series for $\ln(1+x)$ from Example 12.2.2.

Exercise 12.3.1. *Hint.* The right side is the Taylor polynomial of degree 4 at 1 of the left side $p(x)$. So $a_k = \frac{p^{(k)}(1)}{k!}$.

Exercise 12.3.2. *Hint.* Use the Taylor polynomial of $\ln(1+x)$ at 0 with $x = 0.2$. By Theorem 12.3.7, the error is at most $\frac{(0.2)^4}{4}$.

Solution. Let $f(x) = \ln(1+x)$ on $I = (-1, \infty)$, with $c = 0$. By Example 12.3.12, $f^{(n)}(x) = (-1)^{n+1}\frac{(n-1)!}{(1+x)^n}$. So $f(0) = 0$, $f'(0) = 1$, $f''(0) = -1$ and $f'''(0) = 2$, and

$$T_3(f, 0)(x) = x - \frac{x^2}{2} + \frac{x^3}{3}.$$

With $x = 0.2$,

$$\ln 1.2 \approx T_3(f, 0)(0.2) = 0.2 - 0.02 + \frac{0.008}{3} = \frac{137}{750} = 0.18266\ldots.$$

By Lagrange's Form of the Remainder (Theorem 12.3.7), there is a $\xi \in (0, 0.2)$ with

$$R_3(0.2) = \frac{f^{(4)}(\xi)}{4!}(0.2)^4 = \frac{-6}{24(1+\xi)^4}(0.2)^4 = -\frac{(0.2)^4}{4(1+\xi)^4}.$$

Since $(1+\xi)^4 > 1$,

$$|R_3(0.2)| < \frac{(0.2)^4}{4} = \frac{0.0016}{4} = 0.0004 < 0.0005.$$

So $\ln 1.2 \approx 0.18267$ with an error less than 0.0004, which is less than 0.0005. So the estimate is accurate to three decimal places. Since $R_3(0.2) < 0$, the estimate is a little too large. In fact $0.18226 < \ln 1.2 < 0.18267$.

Exercise 12.3.3. *Hint.* Compare the coefficients of $f(x)$ and $f(-x)$, using Corollary 12.2.8.

Exercise 12.3.4. *Hint.* By Theorem 12.3.7, $|R_n(x)| \le e^{|x|}\frac{|x|^{n+1}}{(n+1)!}$. Use Exercise 10.2.5.

Solution. Let $f(x) = e^x$. Since $\frac{d}{dx}e^x = e^x$ (Theorem 9.4.6), $f^{(n)}(x) = e^x$ for all n. So $f \in C^\infty(\mathbb{R})$ and $f^{(k)}(0) = e^0 = 1$ for all k. So $T(e^x, 0) = \sum_{k=0}^{\infty}\frac{x^k}{k!}$.

Let $x \in \mathbb{R}$. If $x = 0$, both sides equal 1. Suppose $x \neq 0$. By Lagrange's Form of the Remainder (Theorem 12.3.7) on $I = \mathbb{R}$, for each n there is a ξ between 0 and x with

$$R_n(x) = \frac{e^\xi}{(n+1)!}x^{n+1}.$$

Since $\xi \le |x|$ and E is increasing, $e^\xi \le e^{|x|}$. So

$$|R_n(x)| \le e^{|x|}\frac{|x|^{n+1}}{(n+1)!}.$$

By Exercise 10.2.5 with $p = |x|$, $\frac{|x|^{n+1}}{(n+1)!} \to 0$. So $R_n(x) \to 0$. By Theorem 12.3.6, $e^x = \sum_{k=0}^{\infty}\frac{x^k}{k!}$ for all $x \in \mathbb{R}$.

Exercise 12.3.5. *Hint.* Every derivative of $\cos x$ is $\pm\sin x$ or $\pm\cos x$, so it is bounded by 1. Then argue as in Exercise 12.3.4.

Exercise 12.3.6. *Hint.* (a) Compute $f^{(k)}(0)$. For the radius, use Corollary 12.1.6.

Exercise 12.4.1. *Hint.* Use Corollary 12.4.7 (1) and $\sin^2 x + \cos^2 x = 1$. For $\sin 3x$, write $3x = 2x + x$.

Exercise 12.4.2. The proof of Theorem 12.4.13 (4) showed $\sin\frac{\pi}{4} = \cos\frac{\pi}{4} > 0$. Since their squares add to 1, $2\sin^2\frac{\pi}{4} = 1$. So $\sin\frac{\pi}{4} = \cos\frac{\pi}{4} = \frac{1}{\sqrt{2}} = \frac{\sqrt{2}}{2}$.

Let $s = \sin\frac{\pi}{6}$. Then $s > 0$ by Theorem 12.4.11 (5). By Theorem 12.4.6 and Corollary 12.4.7 (1),

$$\begin{aligned}\sin 3x &= \sin 2x\cos x + \cos 2x \sin x \\ &= 2\sin x(1-\sin^2 x) + (1-2\sin^2 x)\sin x = 3\sin x - 4\sin^3 x.\end{aligned}$$

With $x = \frac{\pi}{6}$, this gives $1 = \sin\frac{\pi}{2} = 3s - 4s^3$. So

$$0 = 4s^3 - 3s + 1 = (s+1)(2s-1)^2.$$

Since $s > 0$, $s = \frac{1}{2}$. Also $\cos\frac{\pi}{6} > 0$ by Theorem 12.4.9, and $\cos^2\frac{\pi}{6} = 1 - \frac{1}{4} = \frac{3}{4}$. So $\cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$.

Exercise 12.4.3. *Hint.* Divide the numerator and denominator of $\frac{\sin(x+y)}{\cos(x+y)}$ by $\cos x \cos y$.

Exercise 12.4.4. By Theorem 12.4.9, $\cos x > 0$ for $0 \le x < \frac{\pi}{2}$, and cos is even. So $\cos x > 0$ for $|x| < \frac{\pi}{2}$. Let $-\frac{\pi}{2} \le a < b \le \frac{\pi}{2}$. By Theorem 7.2.4, $\sin b - \sin a = \cos\xi\,(b-a)$ for some $\xi \in (a,b)$. Since $|\xi| < \frac{\pi}{2}$, this is positive. So sin is strictly increasing on $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$. Since $\sin\left(\pm\frac{\pi}{2}\right) = \pm 1$, Theorem 6.3.10 shows that sin maps $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ onto $[-1,1]$.

Apply Theorem 7.2.9 to sin on $I = \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, where $\sin' = \cos \neq 0$ and $\sin[I] = (-1,1)$. If $y = \sin x$ with $|x| < \frac{\pi}{2}$, then $\cos x > 0$, so $\cos x = \sqrt{1-\sin^2 x} = \sqrt{1-y^2}$. Thus

$$\arcsin'(y) = \frac{1}{\cos x} = \frac{1}{\sqrt{1-y^2}}.$$

Let $y \in [-1,1]$ and $x = \arcsin y$. Then $\frac{\pi}{2} - x \in [0,\pi]$, and by Theorems 12.4.6 and 12.4.11,

$$\cos\left(\tfrac{\pi}{2} - x\right) = \cos\tfrac{\pi}{2}\cos x + \sin\tfrac{\pi}{2}\sin x = \sin x = y.$$

Since arccos is the inverse of cos on $[0,\pi]$, $\arccos y = \frac{\pi}{2} - x$. So $\arcsin y + \arccos y = \frac{\pi}{2}$.

Exercise 12.4.5. Let $u(t) = y\left(\frac{t}{\omega}\right)$. By the Chain Rule, $u'(t) = \frac{1}{\omega}y'\left(\frac{t}{\omega}\right)$ and

$$u''(t) = \frac{1}{\omega^2}y''\left(\frac{t}{\omega}\right) = -y\left(\frac{t}{\omega}\right) = -u(t).$$

So $u'' + u = 0$, with $u(0) = y(0)$ and $u'(0) = \frac{y'(0)}{\omega}$. By Theorem 12.4.5, $u(t) = y(0)\cos t + \frac{y'(0)}{\omega}\sin t$. Then $y(x) = u(\omega x) = y(0)\cos\omega x + \frac{y'(0)}{\omega}\sin\omega x$.

Exercise 12.4.6. *Hint.* Take $x = 0$, and then $x = \frac{\pi}{2}$. Use $\cos(k\pi) = (-1)^k$.

Exercise 12.4.7. There are two things to prove. Such a t exists, and it is unique. To find t, start with cos on $[0,\pi]$. On $(0,\pi)$, $\cos' = -\sin < 0$ by Theorem 12.4.11 (5). So cos is strictly decreasing on $[0,\pi]$, by Theorem 7.2.4 as in Exercise 12.4.4. Since $\cos 0 = 1$, $\cos\pi = -1$ and $|a| \le 1$, Theorem 6.3.10 gives an $s \in [0,\pi]$ with $\cos s = a$. Then $\sin s \ge 0$ and $\sin^2 s = 1 - a^2 = b^2$, so $\sin s = |b|$. If $b \ge 0$, let $t = s$. If $b < 0$, then $\sin s > 0$, so $0 < s < \pi$. Let $t = 2\pi - s \in (\pi, 2\pi)$. By Theorem 12.4.11 (4) and Theorem 12.4.3, $\cos t = \cos(-s) = a$ and $\sin t = \sin(-s) = -|b| = b$.

To show that t is unique, suppose $t_1, t_2 \in [0, 2\pi)$ both work. By Theorem 12.4.6 and Theorem 12.4.3,

$$\cos(t_1 - t_2) = \cos t_1 \cos t_2 + \sin t_1 \sin t_2 = a^2 + b^2 = 1$$

and $\sin(t_1 - t_2) = \sin t_1 \cos t_2 - \cos t_1 \sin t_2 = ba - ab = 0$. By Theorem 12.4.11 (6), $t_1 - t_2 = k\pi$ for some $k \in \mathbb{Z}$. By Theorem 12.4.11 (3), $\cos(k\pi) = (-1)^k$. So k is even. Since $|t_1 - t_2| < 2\pi$, $k = 0$ and $t_1 = t_2$.

Exercise 12.4.8. *Hint.* Show $\cos 1.6 < 0$ with the upper bound in Lemma 12.4.8 (2). For the lower bound, follow the proof of Theorem 12.4.9 with 1.55 in place of $\frac{3}{2}$.

Exercise 12.4.9. *Hint.* Both sides are 0 at $x = 0$ and have derivative $\frac{1}{1+x^2}$ for $|x| < 1$. At $x = 1$, the series converges by the Alternating Series Test.

Exercise 12.4.10. *Hint.* If p is a polynomial of degree n, then $p^{(m)} = 0$ for $m > n$. What is $\sin^{(4m)}$?

Exercise 12.5.1. *Hint.* For $x \geq 0$ every term of the series is at least 0. Then $E(x) \geq \frac{x^{n+1}}{(n+1)!}$.

Exercise 12.5.2. By Definition 12.5.3, $e = E(1) = \sum_{k=0}^{\infty} \frac{1}{k!}$. So $e - \sum_{k=0}^{n} \frac{1}{k!} = \sum_{k=n+1}^{\infty} \frac{1}{k!} > 0$. For $k \geq n+1$, $\frac{1}{k!} \leq \frac{1}{(n+1)!} \cdot \frac{1}{(n+1)^{k-n-1}}$, with strict inequality for $k \geq n+2$. So by the geometric series,

$$\sum_{k=n+1}^{\infty} \frac{1}{k!} < \frac{1}{(n+1)!} \sum_{j=0}^{\infty} \frac{1}{(n+1)^j} = \frac{1}{(n+1)!} \cdot \frac{n+1}{n} = \frac{1}{n!\,n}.$$

With $n = 7$, $\sum_{k=0}^{7} \frac{1}{k!} = \frac{685}{252} = 2.71825\ldots$ and $\frac{1}{7!\cdot 7} = \frac{1}{35280} < 0.00003$. So $2.718 < e < 2.71826 + 0.00003 < 2.719$.

Exercise 12.5.3. Suppose $e = \frac{p}{q}$ with $p, q \in \mathbb{N}$. Let

$$a = q! \left(e - \sum_{k=0}^{q} \frac{1}{k!} \right) = (q-1)!\, p - \sum_{k=0}^{q} \frac{q!}{k!}.$$

Each $\frac{q!}{k!}$ with $k \leq q$ is an integer, so $a \in \mathbb{Z}$. By Exercise 12.5.2 with $n = q$, $0 < a < \frac{q!}{q!\,q} = \frac{1}{q} \leq 1$. No integer lies strictly between 0 and 1. So e is irrational.

Exercise 12.5.4. *Hint.* Write out the series for $E(-x)$. Add or subtract it from the series for $E(x)$ term by term.

Exercise 12.5.5. *Hint.* By the Binomial Theorem, $\left(1 + \frac{x}{n}\right)^n = \sum_{k=0}^{n} \binom{n}{k} \frac{x^k}{n^k}$. Show $\binom{n}{k} \frac{1}{n^k} \leq \frac{1}{k!}$.

Chapter 13. σ-Algebras and Measures

Exercise 13.1.1. Let $\{\mathcal{M}_\alpha\}_{\alpha \in A}$ be a family of σ-algebras on a set X. Let $\mathcal{M} = \bigcap_{\alpha \in A} \mathcal{M}_\alpha$. First, $\mathcal{M}$ is nonempty. Each $\mathcal{M}_\alpha$ contains X by Remark 13.1.3, so $X \in \mathcal{M}$. Let $\{E_i\}_{i=1}^{\infty} \subset \mathcal{M}$. So for each $\alpha \in A$, $\{E_i\}_{i=1}^{\infty} \subset \mathcal{M}_\alpha$. Since for each $\alpha \in A$, $\mathcal{M}_\alpha$ is a σ-algebra, $\bigcup_{i=1}^{\infty} E_i \in \mathcal{M}_\alpha$ for all $\alpha \in A$. So $\bigcup_{i=1}^{\infty} E_i \in \mathcal{M}$.

Next, let $E \in \mathcal{M}$. So for all $\alpha \in A$, $E \in \mathcal{M}_\alpha$. Since for each $\alpha \in A$, $\mathcal{M}_\alpha$ is a σ-algebra, $E^c \in \mathcal{M}_\alpha$ for all $\alpha \in A$. So $E^c \in \mathcal{M}$. So $\mathcal{M}$ is a σ-algebra. Finite unions are covered too. Now $\emptyset = X^c \in \mathcal{M}$, and $E_1 \cup \cdots \cup E_n = \bigcup_{i=1}^{\infty} E_i$ with $E_i = \emptyset$ for $i > n$.

Exercise 13.1.2. (a) Let $E_1, E_2 \in \mathcal{R}$. Then $E_1 \cap E_2 = (E_1 \cup E_2) \setminus \big((E_1 \setminus E_2) \cup (E_2 \setminus E_1)\big) \in \mathcal{R}$. Indeed, a point of $E_1 \cup E_2$ lies in neither $E_1 \setminus E_2$ nor $E_2 \setminus E_1$ exactly when it lies in both sets. An inductive argument shows $\mathcal{R}$ is closed under finite intersections.

(b) ($\Rightarrow$) Let $E \in \mathcal{R}$. Such an E exists since a σ-algebra is nonempty. Then by the properties of a σ-ring, $E \setminus E = \emptyset \in \mathcal{R}$. Since $\mathcal{R}$ is a σ-algebra, $\emptyset^c = X \in \mathcal{R}$.

($\Leftarrow$) Since $\mathcal{R}$ is a σ-ring, it is closed under countable unions. Let $E \in \mathcal{R}$. Since $X \in \mathcal{R}$, $X \setminus E = E^c \in \mathcal{R}$, and so $\mathcal{R}$ is closed under complements. Also $\mathcal{R}$ is nonempty, since $X \in \mathcal{R}$. So $\mathcal{R}$ is a σ-algebra.

(c) Let $\mathcal{S} = \{E \subset X : E \cap F \in \mathcal{R} \text{ for all } F \in \mathcal{R}\}$. First, $X \in \mathcal{S}$, since $X \cap F = F \in \mathcal{R}$ for every $F \in \mathcal{R}$. So $\mathcal{S}$ is nonempty. Let $\{E_n\}_{n\in\mathbb{N}} \subset \mathcal{S}$. Let $F \in \mathcal{R}$. Since $E_n \cap F \in \mathcal{R}$ for all $n \in \mathbb{N}$,

$$\left[\bigcup_{n\in\mathbb{N}} E_n\right] \cap F = \bigcup_{n\in\mathbb{N}} (E_n \cap F) \in \mathcal{R}.$$

So $\bigcup_{n\in\mathbb{N}} E_n \in \mathcal{S}$.

Next, let $E \in \mathcal{S}$. Let $F \in \mathcal{R}$. Then $E^c \cap F = F \setminus (E \cap F) \in \mathcal{R}$. So $E^c \in \mathcal{S}$ and $\mathcal{S}$ is a σ-algebra.

Exercise 13.1.4. *Hint.* An algebra on a finite set has finitely many members, so a countable union of members is a finite union. For $\sigma(\mathcal{E})$, find the one-point sets using intersections and differences.

Exercise 13.1.5. *Hint.* The collection in Example 13.1.6 is a σ-algebra containing $\mathcal{E}$. Conversely, a countable set is a countable union of one-point sets.

Exercise 13.1.6. *Hint.* Try the σ-algebras generated by $\{1\}$ and by $\{2\}$.

Exercise 13.2.1. *Hint.* For $n \in \mathbb{N}$, let U_n be the set of points c for which there is a $\delta > 0$ with $|f(x) - f(y)| < \frac{1}{n}$ for all $x, y \in (c - \delta, c + \delta)$. Show U_n is open.

Solution. Let $S = \{x \in \mathbb{R} : f \text{ is continuous at } x\}$. Now for each $x \in S$ and $n \in \mathbb{N}$, there exists a $\delta_{x,n} > 0$ such that for all $y \in \mathbb{R}$ with $|x - y| < \delta_{x,n}$, $|f(x) - f(y)| < \frac{1}{n}$. Let $G_n = \bigcup_{x\in S}(x - \delta_{x,n}, x + \delta_{x,n})$. Let $G = \bigcap_{n\in\mathbb{N}} G_n$. Each G_n is a union of open intervals, so it is open. Thus G is a G_δ.

For $z \in S$, $z \in G_n$ for all $n \in \mathbb{N}$, and so $z \in G$. Thus $S \subset G$.

Now let $z \in G$. Let $\varepsilon > 0$. Choose an $N \in \mathbb{N}$ such that $N > \frac{2}{\varepsilon}$. Since $z \in G$, $z \in G_N$. So there exists an $x \in S$ such that $z \in (x - \delta_{x,N}, x + \delta_{x,N})$. In particular, $|f(z) - f(x)| < \frac{1}{N}$. Choose $\delta = \delta_{x,N} - |x - z| > 0$. So for all $y \in \mathbb{R}$ with $|z - y| < \delta$,

$$|x - y| \le |x - z| + |z - y| < |x - z| + \delta = \delta_{x,N}.$$

So for all $y \in \mathbb{R}$ with $|z - y| < \delta$,

$$|f(z) - f(y)| \le |f(z) - f(x)| + |f(x) - f(y)| < \frac{1}{N} + \frac{1}{N} = \frac{2}{N} < \varepsilon.$$

So f is continuous at z and thus $z \in S$. So $G \subset S$ and $G = S$.

Exercise 13.2.2. *Hint.* If $x \in U_n$ for every n, there are points $y_n \in F$ with $|x - y_n| < \frac{1}{n}$. Since F is closed, $x \in F$. For open sets, take complements.

Solution. Each U_n is a union of open intervals, so it is open by Theorem 4.1.11. If $y \in F$, then $y \in \left(y - \frac{1}{n}, y + \frac{1}{n}\right) \subset U_n$ for every n. So $F \subset \bigcap_n U_n$.

Conversely, let $x \in \bigcap_n U_n$. For each n there is a $y_n \in F$ with $|x - y_n| < \frac{1}{n}$. If $x = y_n$ for some n, then $x \in F$. Otherwise, let $\varepsilon > 0$ and choose n with $\frac{1}{n} < \varepsilon$. Then $y_n \in F$, $y_n \neq x$ and $|y_n - x| < \varepsilon$. So x is a limit point of F, and $x \in F$ by Theorem 4.1.24. So $\bigcap_n U_n \subset F$, and $F = \bigcap_n U_n$ is a G_δ set.

Now let U be open. Then U^c is closed, so $U^c = \bigcap_n V_n$ with each V_n open. By De Morgan's Laws (Theorem 1.1.14), $U = \bigcup_n V_n^c$. Each V_n^c is closed. So U is an F_σ set.

Exercise 13.2.3. *Hint.* The Cantor set is an intersection of closed sets.

Exercise 13.2.4. *Hint.* Compact sets are closed, so they are Borel sets. Conversely, every closed set F is the union of the compact sets $F \cap [-n, n]$.

Exercise 13.3.1. (a) First, $\mu(\emptyset) = \sum_{x\in\emptyset} f(x) = 0$. Let $\{E_i\}_{i=1}^{\infty} \subset \mathcal{M}$ be a disjoint sequence. Then

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{x\in\bigcup_{i=1}^{\infty} E_i} f(x) = \sum_{i=1}^{\infty}\sum_{x\in E_i} f(x) = \sum_{i=1}^{\infty} \mu(E_i).$$

So μ is a measure.

The middle equality needs a word, since each sum is a supremum over finite sets. Let $E = \bigcup_i E_i$. If $F \subset E$ is finite, then F meets only finitely many E_i, and the sets $F \cap E_i$ are disjoint. So

$$\sum_{x\in F} f(x) = \sum_{i=1}^{\infty}\sum_{x\in F\cap E_i} f(x) \leq \sum_{i=1}^{\infty} \mu(E_i).$$

Taking the supremum over F gives $\mu(E) \leq \sum_i \mu(E_i)$. Conversely, fix n and finite sets $F_i \subset E_i$ for $i = 1, \ldots, n$. Their union F is a finite subset of E. So $\sum_{i=1}^{n}\sum_{x\in F_i} f(x) = \sum_{x\in F} f(x) \leq \mu(E)$. Taking the supremum over each F_i in turn gives $\sum_{i=1}^{n} \mu(E_i) \leq \mu(E)$. Now let $n \to \infty$.

(b) ($\Rightarrow$) Suppose there exists an $x \in X$ such that $f(x) = \infty$. So $\{x\} \in \mathcal{P}(X) = \mathcal{M}$ and $\mu(\{x\}) = f(x) = \infty$. Now the only subsets of $\{x\}$ are $\{x\}$ and $\emptyset$. But $\mu(\{x\}) = \infty$ and $\mu(\emptyset) = 0$. So μ is not semifinite. So by contrapositive, if μ is semifinite, $f(x) < \infty$ for all $x \in X$.

($\Leftarrow$) Suppose $f(x) < \infty$ for all $x \in X$. Let $E \in \mathcal{M}$ with $\mu(E) = \sum_{x\in E} f(x) = \infty$. So there exists an $x \in E$ such that $f(x) > 0$. By the assumption, $f(x) < \infty$. Now $\{x\} \in \mathcal{P}(X) = \mathcal{M}$, $\{x\} \subset E$, and $\mu(\{x\}) = f(x)$. So $0 < \mu(\{x\}) < \infty$. Thus μ is semifinite.

(c) ($\Rightarrow$) Suppose μ is σ-finite. So there exists a sequence $\{E_n\}_{n\in\mathbb{N}} \subset \mathcal{M}$ with $\mu(E_n) < \infty$ for all $n \in \mathbb{N}$ such that $X = \bigcup_{n=1}^{\infty} E_n$. First, let $x \in X$. Then there exists an E_N with $x \in E_N$. So $f(x) \leq \mu(E_N) < \infty$. So $f(x) < \infty$ for all $x \in X$ and μ is semifinite by (b). Now, since for each $n \in \mathbb{N}$, $\mu(E_n) = \sum_{x\in E_n} f(x) < \infty$, the set $\{x \in E_n : f(x) > 0\}$ must be countable for all $n \in \mathbb{N}$. Indeed, let $k \in \mathbb{N}$. If F is a finite subset of $\{x \in E_n : f(x) > \frac{1}{k}\}$ with p elements, then $\frac{p}{k} \leq \sum_{x\in F} f(x) \leq \mu(E_n)$. So $\{x \in E_n : f(x) > \frac{1}{k}\}$ has at most $k\mu(E_n)$ elements. Then $\{x \in E_n : f(x) > 0\} = \bigcup_{k\in\mathbb{N}}\{x \in E_n : f(x) > \frac{1}{k}\}$ is a countable union of finite sets. So $\{x : f(x) > 0\} = \bigcup_{n\in\mathbb{N}}\{x \in E_n : f(x) > 0\}$ is countable by Theorem 2.2.6.

($\Leftarrow$) Suppose μ is semifinite and $\{x : f(x) > 0\}$ is countable. Enumerate the elements of $\{x : f(x) > 0\}$, $\{x_n\}_{n\in\mathbb{N}}$. Let $G = \{x : f(x) = 0\}$. Then $\mu(G) = 0$. Also, $\mu(\{x_n\}) < \infty$ for all $n \in \mathbb{N}$ by (b). Now $X = G \cup \left[\bigcup_{n\in\mathbb{N}}\{x_n\}\right]$. So μ is σ-finite. If $\{x : f(x) > 0\}$ is finite, list its elements and stop. The same argument works.

Exercise 13.3.2. First, the constant sequence $E_n = \emptyset$ satisfies the hypotheses of (ii). So $\mu(\emptyset) = \lim_{n\to\infty} \mu(E_n) = 0$. Note that from (i), by induction, it follows that for $\{E_i\}_{i=1}^{k}$ with $E_i \cap E_j = \emptyset$ for all $i \neq j$, $\mu\left(\bigcup_{i=1}^{k} E_i\right) = \sum_{i=1}^{k} \mu(E_i)$.

Now, let $\{E_i\}_{i=1}^{\infty}$ be a disjoint sequence in $\mathcal{M}$. First, if $\mu(E_k) = \infty$ for any $k \in \mathbb{N}$, $\infty = \mu(E_k) \leq \mu\left(\bigcup_{i=1}^{\infty} E_i\right)$ and $\sum_{i=1}^{\infty} \mu(E_i) = \infty$. The inequality holds since, by (i), $\mu\left(\bigcup_i E_i\right) = \mu(E_k) + \mu\left(\bigcup_{i \neq k} E_i\right) \geq \mu(E_k)$. So $\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \sum_{i=1}^{\infty} \mu(E_i)$. So assume $\mu(E_i) < \infty$ for all $i \in \mathbb{N}$. Let $\varepsilon > 0$. For all $k \in \mathbb{N}$, define $F_k = \bigcup_{i=k}^{\infty} E_i$. Then $\{F_n\}_{n=1}^{\infty}$ is a sequence in $\mathcal{M}$ with $F_{n+1} \subset F_n$ for all n and $\bigcap_{n=1}^{\infty} F_n = \emptyset$. So $\lim_{n\to\infty} \mu(F_n) = 0$. Thus there exists an $N \in \mathbb{N}$ such that for all $n \geq N$, $\mu(F_n) < \varepsilon$. So for all $n \geq N$, the finite sequence $\{E_i\}_{i=1}^{n-1} \cup \{F_n\} \subset \mathcal{M}$ is disjoint and $\bigcup_{i=1}^{\infty} E_i = \bigcup_{i=1}^{n-1} E_i \cup F_n$. So for all $n \geq N$,

$$\mu\left(\bigcup_{i=1}^{\infty} E_i\right) = \mu\left(\bigcup_{i=1}^{n-1} E_i \cup F_n\right) = \sum_{i=1}^{n-1} \mu(E_i) + \mu(F_n) < \sum_{i=1}^{n-1} \mu(E_i) + \varepsilon.$$

Thus, for all $n \geq N$, $\left|\mu\left(\bigcup_{i=1}^{\infty} E_i\right) - \sum_{i=1}^{n-1} \mu(E_i)\right| < \varepsilon$. So $\lim_{n\to\infty} \sum_{i=1}^{n-1} \mu(E_i) = \sum_{i=1}^{\infty} \mu(E_i) = \mu\left(\bigcup_{i=1}^{\infty} E_i\right)$. So $(\mathbb{R}, \mathcal{M}, \mu)$ is a measure space.

Exercise 13.3.3. *Hint.* Write $E_1 \cup E_2$, E_1 and E_2 as disjoint unions of the sets $E_1 \setminus E_2$, $E_1 \cap E_2$ and $E_2 \setminus E_1$. Then add.

Exercise 13.3.4. *Hint.* If the sets E_n are disjoint, so are the sets $E_n \cap A$.

Exercise 13.3.5. *Hint.* Countable additivity needs $\sum_n c_n \sum_k \mu_n(E_k) = \sum_k \sum_n c_n \mu_n(E_k)$. Show that both sides equal the supremum of the finite sums $\sum_{n \leq N} \sum_{k \leq K} c_n \mu_n(E_k)$.

Exercise 13.3.6. *Hint.* $\mu(E) = \sum_{k \in E \cap \mathbb{N}} 2^{-k}$. Use Exercise 13.3.5.

Exercise 13.3.7. A point x lies in infinitely many E_k exactly when, for every n, there is a $k \geq n$ with $x \in E_k$. That is, $x \in \bigcup_{k=n}^{\infty} E_k$ for every n. So $E = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} E_k$. A σ-algebra is closed under countable unions and countable intersections, so $E \in \mathcal{M}$. For each n, $E \subset \bigcup_{k=n}^{\infty} E_k$. By Theorem 13.3.9 (1) and (2),

$$\mu(E) \leq \sum_{k=n}^{\infty} \mu(E_k).$$

This is the tail of a convergent series, so it tends to 0 as $n \to \infty$. So $\mu(E) = 0$.

Chapter 14. Lebesgue Measure

Exercise 14.1.1. Since $A \subset B$, $\{\{I_n\} : B \subset \bigcup I_n\} \subset \{\{I_k\} : A \subset \bigcup I_k\}$. So

$$\left\{\sum \ell(I_n) : B \subset \bigcup I_n\right\} \subset \left\{\sum \ell(I_k) : A \subset \bigcup I_k\right\},$$

implying

$$\inf_{A \subset \bigcup I_k} \sum \ell(I_k) \leq \inf_{B \subset \bigcup I_n} \sum \ell(I_n).$$

Thus $m^*(A) \leq m^*(B)$.

Exercise 14.1.2. Let $E \subset \mathbb{R}$, $y \in \mathbb{R}$. Notice that $E \subset \bigcup I_n$, where each I_n is an open interval, if and only if $E + y \subset \bigcup J_n$, where $J_n = I_n + y$. Each J_n is an open interval and $\ell(I_n) = \ell(J_n)$, so $\sum \ell(I_n) = \sum \ell(J_n)$. Also notice that if $E + y \subset \bigcup J_n$, then there exists some cover $\{I_n\}$ of E with $I_n = J_n - y$ such

that $E + y \subset \bigcup(I_n + y)$. Again, since $\ell(I_n) = \ell(J_n)$, $\sum \ell(I_n) = \sum \ell(J_n)$. So $\{\sum \ell(I_n) : E \subset \bigcup I_n\} = \{\sum \ell(J_n) : E + y \subset \bigcup J_n\}$ and

$$\inf_{E \subset \bigcup I_n} \sum \ell(I_n) = \inf_{E+y \subset \bigcup J_n} \sum \ell(J_n).$$

So $m^*(E) = m^*(E + y)$.

Exercise 14.1.4. Let $\varepsilon > 0$. If $m^*(A) = \infty$, take $U = \mathbb{R}$. So assume $m^*(A) < \infty$. Since $m^*(A) = \inf_{A \subset \bigcup I_n} \sum \ell(I_n)$, there exists a sequence $\{I_n\}$ of open intervals such that $A \subset \bigcup I_n$ and $m^*(A) + \varepsilon \geq \sum \ell(I_n)$. Let $U = \bigcup I_n$. Then U is open, $A \subset U$ and

$$m^*(U) = m^*\left(\bigcup I_n\right) \leq \sum m^*(I_n) = \sum \ell(I_n) \leq m^*(A) + \varepsilon.$$

The first inequality is Theorem 14.1.6, and $m^*(I_n) = \ell(I_n)$ by Theorem 14.1.4.

Exercise 14.1.6. *Hint.* Each open interval lies in its closure, which has the same length. Each closed interval J_n lies in an open interval of length $\ell(J_n) + \frac{\varepsilon}{2^n}$.

Exercise 14.1.7. *Hint.* $\{I_n\}$ covers A if and only if $\{cI_n\}$ covers cA, and $\ell(cI_n) = c\,\ell(I_n)$. Compare Exercise 14.1.2.

Exercise 14.2.1. *Hint.* Use Corollary 14.1.7 and Lemma 14.2.3.

Solution. Let A be countable. By Corollary 14.1.7, $m^*(A) = 0$. By Lemma 14.2.3, A is measurable, and $m(A) = m^*(A) = 0$.

In particular, $\mathbb{Q}$ is measurable. Since $\mathcal{M}$ is a σ-algebra (Theorem 14.2.8), $\mathbb{R} \setminus \mathbb{Q} = \mathbb{Q}^c$ is measurable. The interval $[0, 1]$ is measurable by Theorem 14.2.10. So $[0, 1] \cap \mathbb{Q}$ and $[0, 1] \setminus \mathbb{Q} = [0, 1] \cap \mathbb{Q}^c$ are measurable and disjoint, and their union is $[0, 1]$. By additivity,

$$1 = m([0, 1]) = m([0, 1] \cap \mathbb{Q}) + m([0, 1] \setminus \mathbb{Q}) = 0 + m([0, 1] \setminus \mathbb{Q}).$$

So $m\big([0, 1] \setminus \mathbb{Q}\big) = 1$.

Exercise 14.2.2. *Hint.* Write $F = \big(E \setminus (E \setminus F)\big) \cup (F \setminus E)$.

Exercise 14.2.4. *Hint.* Apply the definition of measurability of E to the sets A and $A \cup E$.

Solution. Test the set $A \cup E$ with the measurable set E. Since $(A \cup E) \cap E = E$ and $(A \cup E) \cap E^c = A \cap E^c$,

$$m^*(A \cup E) = m^*(E) + m^*(A \cap E^c).$$

Add $m^*(A \cap E)$ to both sides. Then test A with E, which gives $m^*(A \cap E) + m^*(A \cap E^c) = m^*(A)$. So

$$m^*(A \cup E) + m^*(A \cap E) = m(E) + m^*(A \cap E) + m^*(A \cap E^c) = m(E) + m^*(A).$$

Only addition is used, so this holds even when some of the values are ∞.

Exercise 14.2.5. *Hint.* Countable sets have measure zero, by Corollary 14.1.7.

Exercise 14.3.1. *Hint.* First, show that for $m^*(E) < \infty$, (1)⇒(2). Then show for arbitrary sets E, (1)⇒(2)⇒(4)⇒(1). Finally, show (1)⇒(3)⇒(5)⇒(1).

Solution. Let $\varepsilon > 0$.

((1)⇒(2)) First assume $m^*(E) < \infty$. Then there exists an open set O with $E \subset O$ and $m(O) \leq m(E) + \frac{\varepsilon}{2}$, by Exercise 14.1.4. Since E is measurable, $m(O \setminus E) =$

$m(O \cap E^c) = m(O) - m(O \cap E) = m(O) - m(E) \leq \frac{\varepsilon}{2} < \varepsilon$. The subtraction is allowed because $m(E) < \infty$. Now assume $m^*(E) = \infty$. Let $E_n = E \cap [-n, n]$. So $m(E_n) \leq 2n < \infty$. So there exists an open set O_n such that $E_n \subset O_n$ and $m(O_n \setminus E_n) < \frac{\varepsilon}{2^n}$. Let $O = \bigcup_{n\in\mathbb{N}} O_n$. Then $E \subset O$ and

$$m^*(O \setminus E) = m(O \setminus E) = m\left(\left(\bigcup O_n\right) \setminus E\right) \leq m\left(\bigcup (O_n \setminus E_n)\right)$$
$$\leq \sum m(O_n \setminus E_n) < \sum_{n\in\mathbb{N}} \frac{\varepsilon}{2^n} = \varepsilon.$$

The first inequality holds since $E_n \subset E$, so $O_n \setminus E \subset O_n \setminus E_n$.

((2)⇒(4)) For each $n \in \mathbb{N}$, there exists an open set O_n containing E such that $m^*(O_n \setminus E) < \frac{1}{n}$. Let $G = \bigcap_{n\in\mathbb{N}} O_n$. So G is a G_δ containing E, and for all n, $m^*(G \setminus E) \leq m^*(O_n \setminus E) < \frac{1}{n}$. So $m^*(G \setminus E) = 0$.

((4)⇒(1)) Since G is a G_δ, G is measurable by Theorem 14.2.10. Now $m^*(G \cap E^c) = m^*(G \setminus E) = 0$ and so $G \cap E^c$ is measurable by Lemma 14.2.3. But $E = G \setminus (G \cap E^c)$ and so E is measurable.

((1)⇒(3)) Since E is measurable, E^c is measurable. So since (1)⇒(2), there exists an open set O containing E^c such that $m^*(O \setminus E^c) < \varepsilon$. Let $F = O^c$. So F is closed, $F \subset E$, and $E \setminus F = E \cap F^c = E \cap O = O \setminus E^c$. Thus $m^*(E \setminus F) = m(O \setminus E^c) < \varepsilon$.

((3)⇒(5)) For each $n \in \mathbb{N}$, there exists a closed set $F_n \subset E$ such that $m^*(E \setminus F_n) < \frac{1}{n}$. Let $F = \bigcup F_n$. So F is an F_σ contained in E, and for all n, $m^*(E \setminus F) \leq m^*(E \setminus F_n) < \frac{1}{n}$. So $m^*(E \setminus F) = 0$.

((5)⇒(1)) Since F is an F_σ, F is measurable. Now $m^*(E \setminus F) = 0$, so $E \setminus F$ is measurable by Lemma 14.2.3. But $E = F \cup (E \setminus F)$, and so E is measurable.

Exercise 14.3.2. *Hint.* For (2)⇒(6), write the open set U as a countable union of disjoint open intervals, by Theorem 4.1.12, and keep finitely many of them.

Exercise 14.3.3. *Hint.* By Theorem 14.3.1 (3), there is a closed set $F \subset E$ with $m(E \setminus F) < \frac{\varepsilon}{2}$. Let $K = F \cap [-n, n]$ for large n, and use Theorem 13.3.9 (4).

Solution. Let $\varepsilon > 0$. By Theorem 14.3.1 (3), there is a closed set $F \subset E$ with $m^*(E \setminus F) < \frac{\varepsilon}{2}$. The sets F and $E \setminus F$ are measurable, by Theorem 14.2.10. For $n \in \mathbb{N}$, let $K_n = F \cap [-n, n]$. It is closed by Theorem 4.1.17, and it lies in the compact set $[-n, n]$ (Theorem 4.2.14). So K_n is compact, by Lemma 4.2.10.

The sets $F \setminus K_n = F \setminus [-n, n]$ decrease, and their intersection is empty, since every x lies in some $[-n, n]$. Also $m(F \setminus K_1) \leq m(E) < \infty$. By continuity from above (Theorem 13.3.9 (4)), $m(F \setminus K_n) \to 0$. Choose n with $m(F \setminus K_n) < \frac{\varepsilon}{2}$. Since $K_n \subset F \subset E$, the set $E \setminus K_n$ is the disjoint union of $E \setminus F$ and $F \setminus K_n$. So

$$m(E \setminus K_n) = m(E \setminus F) + m(F \setminus K_n) < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

So $K = K_n$ works.

Exercise 14.3.4. *Hint.* Use (2) and Theorem 4.1.12.

Exercise 14.4.1. *Hint.* Write P_i as a union of translates of $P \cap [0, 1 - r_i)$ and $P \cap [1 - r_i, 1)$. So $m^*(P_i) \leq 2m^*(P)$. If $m^*(P) = 0$, subadditivity gives $m^*([0, 1)) = 0$.

Exercise 14.4.2. *Hint.* In the same way, $m^*(P) \leq 2m^*(P_i)$. So $\sum m^*(P_i) = \infty$, by Exercise 14.4.1. But $\bigcup P_i = [0, 1)$.

Exercise 14.4.3. *Hint.* Consider the sets $E \cap P_i$.

Solution. Use P, the rational numbers $r_0, r_1, \ldots$ and the sets $P_i = P \mathbin{\mathring{+}} r_i$ from the proof of Theorem 14.4.6. The sets P_i are disjoint and their union is $[0,1)$. So E is the disjoint union of the sets $E \cap P_i$. The claim is that one of them is not measurable.

Suppose every $E \cap P_i$ is measurable. By countable additivity, $m(E) = \sum_i m(E \cap P_i)$. Since $m(E) > 0$, some $F = E \cap P_k$ has $m(F) > 0$.

For each j, $F \mathbin{\mathring{+}} r_j$ is measurable with $m(F \mathbin{\mathring{+}} r_j) = m(F)$, by Lemma 14.4.4. Since $\mathring{+}$ is associative (Remark 14.4.2),

$$F \mathbin{\mathring{+}} r_j \subset P_k \mathbin{\mathring{+}} r_j = P \mathbin{\mathring{+}} (r_k \mathbin{\mathring{+}} r_j).$$

The number $r_k \mathbin{\mathring{+}} r_j$ is a rational number in $[0,1)$, so it is $r_{\sigma(j)}$ for some index $\sigma(j)$. These indices are distinct. If $r_k \mathbin{\mathring{+}} r_j = r_k \mathbin{\mathring{+}} r_{j'}$, then $r_j - r_{j'}$ is -1, 0 or 1. Since $r_j, r_{j'} \in [0,1)$, it is 0, so $j = j'$. So the sets $F \mathbin{\mathring{+}} r_j$ lie in the disjoint sets $P_{\sigma(j)}$, and they are pairwise disjoint. They all lie in $[0,1)$. By countable additivity and monotonicity,

$$\sum_{j=0}^{\infty} m(F) = m\left(\bigcup_{j=0}^{\infty} F \mathbin{\mathring{+}} r_j\right) \leq m([0,1)) = 1.$$

But the left side is ∞, since $m(F) > 0$. This contradiction shows that some $E \cap P_i$ is not measurable. It is a subset of E.

Chapter 15. Measurable Functions

Exercise 15.1.1. A little more is true. For any measurable sets A and B, $\chi_{A\cup B} = \chi_A + \chi_B - \chi_{A\cap B}$ and $\chi_{A\cap B} = \chi_A \cdot \chi_B$. Check the four cases for a point x.

Let $x \notin A$, $x \notin B$. Then $\chi_{A\cup B}(x) = 0 = 0 + 0 - 0 = \chi_A(x) + \chi_B(x) - \chi_{A\cap B}(x)$ and $\chi_{A\cap B}(x) = 0 = 0 \cdot 0 = \chi_A(x) \cdot \chi_B(x)$.

Let $x \in A$, $x \notin B$. Then $\chi_{A\cup B}(x) = 1 = 1 + 0 - 0 = \chi_A(x) + \chi_B(x) - \chi_{A\cap B}(x)$ and $\chi_{A\cap B}(x) = 0 = 1 \cdot 0 = \chi_A(x) \cdot \chi_B(x)$.

Let $x \notin A$, $x \in B$. Then $\chi_{A\cup B}(x) = 1 = 0 + 1 - 0 = \chi_A(x) + \chi_B(x) - \chi_{A\cap B}(x)$ and $\chi_{A\cap B}(x) = 0 = 0 \cdot 1 = \chi_A(x) \cdot \chi_B(x)$.

Let $x \in A$, $x \in B$. Then $\chi_{A\cup B}(x) = 1 = 1 + 1 - 1 = \chi_A(x) + \chi_B(x) - \chi_{A\cap B}(x)$ and $\chi_{A\cap B}(x) = 1 = 1 \cdot 1 = \chi_A(x) \cdot \chi_B(x)$.

If A and B are disjoint, then $A \cap B = \emptyset$, so $\chi_{A\cap B} = 0$ and $\chi_{A\cup B} = \chi_A + \chi_B$.

Exercise 15.1.3. Define $\mathcal{S} = \{A \subset \mathbb{R} : f^{-1}[A] \in \mathcal{M}\}$. First, $\mathbb{R} \in \mathcal{S}$, since $f^{-1}[\mathbb{R}] = E \in \mathcal{M}$. Let $A \in \mathcal{S}$. Then $f^{-1}[A] \in \mathcal{M}$. Since $\mathcal{M}$ is a σ-algebra, $E \setminus f^{-1}[A] = E \cap (f^{-1}[A])^c \in \mathcal{M}$ as well. Let $x \in f^{-1}[A^c]$. Then $f(x) \in A^c$. So $f(x) \notin A$ and $x \in E \setminus f^{-1}[A]$. Thus $f^{-1}[A^c] \subset E \setminus f^{-1}[A]$. Now let $x \in E \setminus f^{-1}[A]$. So $f(x) \notin A$, and $f(x) \in A^c$. Thus $x \in f^{-1}[A^c]$ and $E \setminus f^{-1}[A] \subset f^{-1}[A^c]$. So $f^{-1}[A^c] = E \setminus f^{-1}[A] \in \mathcal{M}$ and $A^c \in \mathcal{S}$.

Now let $\{A_n\} \subset \mathcal{S}$. So for each $n \in \mathbb{N}$, $f^{-1}[A_n] \in \mathcal{M}$, and since $\mathcal{M}$ is a σ-algebra, $\bigcup_{n\in\mathbb{N}} f^{-1}[A_n] \in \mathcal{M}$. Let $x \in f^{-1}\left[\bigcup_{n\in\mathbb{N}} A_n\right]$. Then $f(x) \in \bigcup_{n\in\mathbb{N}} A_n$. So there exists an $n_0 \in \mathbb{N}$ such that $f(x) \in A_{n_0}$. So $x \in f^{-1}[A_{n_0}] \subset \bigcup_{n\in\mathbb{N}} f^{-1}[A_n]$ and

$f^{-1}\left[\bigcup_{n\in\mathbb{N}} A_n\right] \subset \bigcup_{n\in\mathbb{N}} f^{-1}[A_n]$. Now let $x \in \bigcup_{n\in\mathbb{N}} f^{-1}[A_n]$. So there exists an $n_0 \in \mathbb{N}$ such that $x \in f^{-1}[A_{n_0}]$, implying $f(x) \in A_{n_0} \subset \bigcup_{n\in\mathbb{N}} A_n$. So $x \in f^{-1}\left[\bigcup_{n\in\mathbb{N}} A_n\right]$, and $\bigcup_{n\in\mathbb{N}} f^{-1}[A_n] \subset f^{-1}\left[\bigcup_{n\in\mathbb{N}} A_n\right]$. Thus

$$f^{-1}\left[\bigcup_{n\in\mathbb{N}} A_n\right] = \bigcup_{n\in\mathbb{N}} f^{-1}[A_n] \in \mathcal{M}.$$

So $\bigcup_{n\in\mathbb{N}} A_n \in \mathcal{S}$ and $\mathcal{S}$ is a σ-algebra.

It remains to show that $\mathcal{S}$ contains the open sets. Let $\alpha, \beta \in \mathbb{R}$. Since f is measurable, $f^{-1}[(\alpha,\infty)] = \{x : f(x) > \alpha\}$ and $f^{-1}[(-\infty,\beta)] = \{x : f(x) < \beta\}$ are measurable, by Theorem 15.1.1. So (α,∞) and $(-\infty,\beta)$ are in $\mathcal{S}$. Since $\mathcal{S}$ is a σ-algebra, $(\alpha,\beta) = (\alpha,\infty) \cap (-\infty,\beta) \in \mathcal{S}$ by Proposition 13.1.2. Also $\mathbb{R} \in \mathcal{S}$. So every open interval, bounded or not, is in $\mathcal{S}$. By Theorem 4.1.12, every nonempty open set is a countable union of open intervals, so it is in $\mathcal{S}$. Also $\emptyset = \mathbb{R}^c \in \mathcal{S}$.

Finally, $\mathcal{B}_\mathbb{R}$ is the smallest σ-algebra containing the open sets, and $\mathcal{S}$ is a σ-algebra containing the open sets. So $\mathcal{B}_\mathbb{R} \subset \mathcal{S}$. That is, $f^{-1}[B] \in \mathcal{M}$ for every Borel set B.

Exercise 15.1.4. *Hint.* $\{g \circ f > \alpha\} = f^{-1}\left[g^{-1}[(\alpha,\infty)]\right]$, and $g^{-1}[(\alpha,\infty)]$ is open. Use Exercise 15.1.3.

Exercise 15.1.5. *Hint.* $\{\max\{f,g\} > \alpha\} = \{f > \alpha\} \cup \{g > \alpha\}$. For the second proof, use Theorem 15.1.4 and Exercise 15.1.4.

Exercise 15.1.6. *Hint.* Try $\chi_{[0,\frac{1}{2}]}$. A set of measure zero contains no interval. So a continuous function equal to it a.e. agrees with it on a dense set.

Exercise 15.2.1. *Hint.* The sets $\{|f| > n\}$ decrease, and their intersection is $\{|f| = \infty\}$.

Exercise 15.2.2. *Hint.* Cut $[-M,M]$ into intervals of length less than ε, and let φ be constant on the set where f lies in each one.

Solution. Let $\varepsilon > 0$, $M > 0$. Choose $N \in \mathbb{N}$ such that $N > \frac{M}{\varepsilon}$. Define $A_i = \left\{x : \frac{(i+1)M}{N} > f(x) \geq \frac{iM}{N}\right\}$ for $-N \leq i \leq N-1$. Each A_i is measurable since f is. Note $\frac{(i+1)M - iM}{N} = \frac{M}{N} < \varepsilon$. Define $\varphi(x) = \sum_{i=-N}^{N-1} \frac{iM}{N}\chi_{A_i}$. So φ is a simple function. If $x \in [a,b]$ and $|f(x)| < M$, then there exists an A_i such that $x \in A_i$. So $|f(x) - \varphi(x)| < \left|\frac{(i+1)M}{N} - \frac{iM}{N}\right| < \varepsilon$. Indeed, $0 \leq f(x) - \frac{iM}{N} < \frac{M}{N}$.

Now suppose $\alpha \leq f \leq \beta$. Let $\psi = \min\{\max\{\varphi,\alpha\},\beta\}$. That is, $\psi(x) = \alpha$ where $\varphi(x) < \alpha$, $\psi(x) = \beta$ where $\varphi(x) > \beta$, and $\psi(x) = \varphi(x)$ otherwise. Then ψ takes finitely many values, each on a measurable set. So ψ is simple, and $\alpha \leq \psi \leq \beta$. Moving $\varphi(x)$ toward the interval $[\alpha,\beta]$, which contains $f(x)$, cannot increase its distance from $f(x)$. So $|f(x) - \psi(x)| \leq |f(x) - \varphi(x)| < \varepsilon$ whenever $|f(x)| < M$.

Exercise 15.2.3. *Hint.* Use the fact that if A is a measurable set with $m(A) < \infty$, there exists a finite union of open intervals U such that $m(A \triangle U) < \varepsilon$. This is (6) in Theorem 14.3.1.

Exercise 15.2.4. *Hint.* Near each jump of g, replace g by a short line segment.

Exercise 15.2.5. Let $\varepsilon > 0$. Since f takes on the values of $\pm\infty$ only on a set of measure zero, by Lemma 15.2.3 there exists an $M > 0$ such that $m(\{x : |f(x)| >$

$M\}) < \frac{\varepsilon}{3}$. Define $A = \{x : |f(x)| \le M\}$. Now there exists a simple function φ such that $|f(x)-\varphi(x)| < \varepsilon$ for all $x \in A$ by Lemma 15.2.4, applied with $M+1$ in place of M. Since φ is simple, there exists a step function g such that $m(\{x : g(x) \neq \varphi(x)\}) < \frac{\varepsilon}{3}$ by Lemma 15.2.5. But if $x \in A$ with $g(x) = \varphi(x)$, then $|f(x) - g(x)| < \varepsilon$. Thus

$$\{x : |f(x) - g(x)| \ge \varepsilon\} \subset \{x : g(x) \neq \varphi(x)\} \cup \{x : |f(x)| > M\},$$

implying

$$\begin{aligned} m(\{x : |f(x) - g(x)| \ge \varepsilon\}) &\le m(\{x : g(x) \neq \varphi(x)\}) + m(\{x : |f(x)| > M\}) \\ &< \frac{\varepsilon}{3} + \frac{\varepsilon}{3} < \varepsilon. \end{aligned}$$

Also, there exists a continuous function h such that $m(\{x : g(x) \neq h(x)\}) < \frac{\varepsilon}{3}$ by Lemma 15.2.6. But if $x \in A$ with $h(x) = g(x) = \varphi(x)$, then $|f(x) - h(x)| < \varepsilon$. So

$$\{x : |f(x) - h(x)| \ge \varepsilon\} \subset \{x : h(x) \neq g(x)\} \cup \{x : g(x) \neq \varphi(x)\} \cup \{x : |f(x)| > M\},$$

implying $m(\{x : |f(x) - h(x)| \ge \varepsilon\}) < \frac{\varepsilon}{3} + \frac{\varepsilon}{3} + \frac{\varepsilon}{3} = \varepsilon$.

If $\alpha \le f \le \beta$, φ can be chosen so that $\alpha \le \varphi \le \beta$. Since $\alpha \le \varphi \le \beta$, g can be chosen so that $\alpha \le g \le \beta$. Since $\alpha \le g \le \beta$, h can be chosen so that $\alpha \le h \le \beta$.

Exercise 15.3.1. *Hint.* Remove a short interval $(1-\delta, 1]$. On $[0, 1-\delta]$, $x^n \le (1-\delta)^n$.

Solution. For $0 \le x < 1$, $x^n \to 0$ by Example 3.1.7, and $1^n = 1$. So the pointwise limit is $f(x) = 0$ for $x \in [0, 1)$ and $f(1) = 1$.

Let $\eta > 0$, let $\delta = \min\left\{\frac{\eta}{2}, \frac{1}{2}\right\}$, and let $A = (1 - \delta, 1]$. Then $m(A) = \delta < \eta$, and $[0, 1] \setminus A = [0, 1 - \delta]$. For $x \in [0, 1 - \delta]$, $f(x) = 0$ and

$$|f_n(x) - f(x)| = x^n \le (1 - \delta)^n.$$

So $\sup_{x\in[0,1-\delta]} |f_n(x) - f(x)| \le (1-\delta)^n \to 0$, since $0 < 1-\delta < 1$. By Theorem 11.2.7, $f_n \to f$ uniformly on $[0, 1] \setminus A$.

Exercise 15.3.2. *Hint.* Use $f_n(x) = x^n$ on $[0, 1]$ again.

Exercise 15.4.1. *Hint.* Try $f = \chi_{[0,1/2]}$ on $[0, 1]$.

Solution. Let $f = \chi_{[0,1/2]}$ on $[0, 1]$. It is measurable by Lemma 15.1.11. Suppose φ is continuous on $[0, 1]$ and $Z = \{x : f(x) \neq \varphi(x)\}$ has $m(Z) = 0$. Then Z contains no interval of positive length, since such an interval has positive measure and m^* is monotone.

For $n \ge 2$, the interval $\left(\frac{1}{2} - \frac{1}{n}, \frac{1}{2}\right)$ is not contained in Z. So it contains a point s_n with $\varphi(s_n) = f(s_n) = 1$. In the same way, $\left(\frac{1}{2}, \frac{1}{2} + \frac{1}{n}\right)$ contains a point t_n with $\varphi(t_n) = f(t_n) = 0$. Both $s_n \to \frac{1}{2}$ and $t_n \to \frac{1}{2}$. Since φ is continuous, Theorem 6.3.9 gives

$$\varphi\left(\tfrac{1}{2}\right) = \lim_{n\to\infty} \varphi(s_n) = 1 \quad \text{and} \quad \varphi\left(\tfrac{1}{2}\right) = \lim_{n\to\infty} \varphi(t_n) = 0.$$

This is impossible. So no continuous φ equals f almost everywhere, and $m(\{f \neq \varphi\}) < \delta$ cannot be improved to $m(\{f \neq \varphi\}) = 0$.

Exercise 15.4.2. *Hint.* Replace φ by $\max\{-M, \min\{\varphi, M\}\}$. It is still continuous, and it equals f wherever φ does.

Chapter 16. The Lebesgue Integral

Exercise 16.1.1. *Hint.* Given a partition $\mathcal{P}$, the step functions equal to m_i and to M_i on (x_{i-1}, x_i) have integrals $\mathcal{L}(\mathcal{P}, f)$ and $\mathcal{U}(\mathcal{P}, f)$. For the converse, use Proposition 16.1.2.

Solution. ($\Rightarrow$) Suppose $f \in \mathcal{R}[a,b]$, and let $\varepsilon > 0$. By Riemann's Criterion (Theorem 9.2.1), there is a partition $\mathcal{P} = \{x_0, \dots, x_n\}$ with $\mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) < \varepsilon$. As in the proof of Proposition 16.1.2, let $\psi = M_i$ and $\varphi = m_i$ on (x_{i-1}, x_i), and let $\psi = \varphi = f$ at the points x_i. These are step functions on $[a,b]$ with $\varphi \le f \le \psi$. By Lemma 16.1.1, $\int_a^b \psi = \mathcal{U}(\mathcal{P}, f)$ and $\int_a^b \varphi = \mathcal{L}(\mathcal{P}, f)$. So, by Theorem 9.2.11,

$$\int_a^b (\psi - \varphi) = \mathcal{U}(\mathcal{P}, f) - \mathcal{L}(\mathcal{P}, f) < \varepsilon.$$

($\Leftarrow$) Let $\varepsilon > 0$, and choose step functions $\varphi \le f \le \psi$ with $\int_a^b (\psi - \varphi) < \varepsilon$. By Proposition 16.1.2, $\overline{\int_a^b} f \le \int_a^b \psi$ and $\underline{\int_a^b} f \ge \int_a^b \varphi$. With Theorem 9.1.6,

$$0 \le \overline{\int_a^b} f - \underline{\int_a^b} f \le \int_a^b \psi - \int_a^b \varphi = \int_a^b (\psi - \varphi) < \varepsilon.$$

Since $\varepsilon > 0$ was arbitrary, $\overline{\int_a^b} f = \underline{\int_a^b} f$, and $f \in \mathcal{R}[a,b]$.

Exercise 16.1.2. *Hint.* If $\psi \ge f$ is constant on (x_{i-1}, x_i), the constant is at least x_i. So $\int_0^1 \psi \ge \sum x_i \Delta x_i$. Compare with $\sum \frac{x_{i-1}+x_i}{2} \Delta x_i = \frac{1}{2}$.

Exercise 16.1.3. *Hint.* Use a, b and the endpoints of the intervals I_i in Definition 15.2.2.

Exercise 16.2.1. *Hint.* Since $m(\mathbb{Q}) = 0$, $\varphi = \lfloor x \rfloor$ except on a set of measure zero.

Solution. On $[0,1)$, φ is 1 at rational points and 0 at irrational points. On $[1,2)$ it is 2 or 1, and on $[2,3)$ it is 3 or 2. Also $\varphi(3) = 3 + 1 = 4$. So φ takes the values $0, 1, 2, 3, 4$, and

$$\begin{aligned}
A_0 &= \{\varphi = 0\} = [0,1) \setminus \mathbb{Q}, \\
A_1 &= \{\varphi = 1\} = \big([0,1) \cap \mathbb{Q}\big) \cup \big([1,2) \setminus \mathbb{Q}\big), \\
A_2 &= \{\varphi = 2\} = \big([1,2) \cap \mathbb{Q}\big) \cup \big([2,3) \setminus \mathbb{Q}\big), \\
A_3 &= \{\varphi = 3\} = [2,3) \cap \mathbb{Q}, \\
A_4 &= \{\varphi = 4\} = \{3\}.
\end{aligned}$$

Intervals and $\mathbb{Q}$ are measurable (Theorem 14.2.10 and Exercise 14.2.1), so each A_i is measurable. So φ is measurable, with finitely many values, and it is simple. Its canonical representation is

$$\varphi = 0 \cdot \chi_{A_0} + 1 \cdot \chi_{A_1} + 2 \cdot \chi_{A_2} + 3 \cdot \chi_{A_3} + 4 \cdot \chi_{A_4}.$$

A countable set has measure 0. So for an interval I, $m(I \cap \mathbb{Q}) = 0$ and $m(I \setminus \mathbb{Q}) = \ell(I)$. So $m(A_0) = m(A_1) = m(A_2) = 1$ and $m(A_3) = m(A_4) = 0$. The function vanishes outside $[0,3]$, so by Definition 16.2.1,

$$\int_{[0,3]} \varphi = 0 \cdot 1 + 1 \cdot 1 + 2 \cdot 1 + 3 \cdot 0 + 4 \cdot 0 = 3.$$

This agrees with $\int_0^3 \lfloor x \rfloor \, dx = 0 + 1 + 2 = 3$, since $\varphi = \lfloor x \rfloor$ except on a set of measure zero.

Exercise 16.2.2. *Hint.* If $\varphi = \sum a_i \chi_{A_i}$ is the canonical representation, then $|\varphi| = \sum |a_i| \chi_{A_i}$.

Solution. Let $\varphi = \sum_{i=1}^n a_i \chi_{A_i}$ be the canonical representation. On A_i, $|\varphi| = |a_i|$. So $|\varphi|$ takes finitely many values. For each α, the set $\{|\varphi| > \alpha\}$ is the union of the sets A_i with $|a_i| > \alpha$, which is measurable. So $|\varphi|$ is simple. Also $\{|\varphi| \neq 0\} = \{\varphi \neq 0\}$, which has finite measure.

Drop the terms with $a_i = 0$. The remaining sets A_i lie in $\{\varphi \neq 0\}$, so they have finite measure, and they are disjoint. Then $\varphi = \sum_{a_i \neq 0} a_i \chi_{A_i}$ and $|\varphi| = \sum_{a_i \neq 0} |a_i| \chi_{A_i}$. By Lemma 16.2.2 and the Triangle Inequality,

$$\left| \int \varphi \right| = \left| \sum_{a_i \neq 0} a_i m(A_i) \right| \leq \sum_{a_i \neq 0} |a_i| \, m(A_i) = \int |\varphi|.$$

Exercise 16.2.3. *Hint.* The sets $\{\varphi = a\} \cap \{\psi = b\}$ are measurable and there are finitely many. Each of the three functions is constant on each one.

Exercise 16.2.4. *Hint.* $\varphi \chi_{A \cup B} = \varphi \chi_A + \varphi \chi_B$. Use Theorem 16.2.3.

Exercise 16.2.5. *Hint.* If $\varphi = \sum a_i \chi_{A_i}$, then $\nu(E) = \sum a_i m(E \cap A_i)$. Use Exercises 13.3.4 and 13.3.5.

Exercise 16.3.2. Each $f_n = n \chi_{(0,1/n)}$ is a simple function that vanishes outside $\left(0, \frac{1}{n}\right)$. So

$$\int_{[0,1]} f_n = n \, m\left(\left(0, \tfrac{1}{n}\right)\right) = n \cdot \frac{1}{n} = 1.$$

The sequence converges to 0 at every point. First, $f_n(0) = 0$ for all n. If $0 < x \leq 1$ and $n > \frac{1}{x}$, then $x > \frac{1}{n}$, so $f_n(x) = 0$. So $f_n \to 0$ on $[0,1]$, and $\int_{[0,1]} 0 = 0 \neq 1 = \lim_{n\to\infty} \int_{[0,1]} f_n$.

The set $[0,1]$ has finite measure, and each f_n is measurable. What fails is the bound. Since $f_n\left(\frac{1}{2n}\right) = n$, no single M satisfies $|f_n| \leq M$ for all n.

Exercise 16.3.3. *Hint.* The first integrand tends to 1 on $[0,1)$. For the second, $2nx \leq 1 + n^2x^2$, so the integrands are bounded by $\frac{1}{2}$.

Solution. Both sequences consist of continuous, so measurable, functions on $[0,1]$.

Let $g_n(x) = \frac{1}{1+x^n}$. Then $0 < g_n \leq 1$. For $0 \leq x < 1$, $x^n \to 0$, so $g_n(x) \to 1$. At $x = 1$, $g_n(1) = \frac{1}{2}$. So $g_n \to 1$ except at one point, that is, a.e. By Corollary 16.3.7 with $M = 1$,

$$\lim_{n\to\infty} \int_{[0,1]} \frac{dm}{1+x^n} = \int_{[0,1]} 1 \, dm = 1.$$

Let $h_n(x) = \frac{nx}{1+n^2x^2}$. Since $(1 - nx)^2 \geq 0$, $2nx \leq 1 + n^2x^2$, so $0 \leq h_n \leq \frac{1}{2}$ on $[0,1]$. Also $h_n(0) = 0$, and for $x > 0$, $0 \leq h_n(x) \leq \frac{nx}{n^2x^2} = \frac{1}{nx} \to 0$. So $h_n \to 0$ on $[0,1]$. By Corollary 16.3.7 with $M = \frac{1}{2}$,

$$\lim_{n\to\infty} \int_{[0,1]} \frac{nx}{1+n^2x^2} \, dm = 0.$$

To check this, note that $\frac{1}{2n}\ln(1+n^2x^2)$ has derivative $h_n(x)$. By Theorem 16.3.3 and the Fundamental Theorem of Calculus,

$$\int_{[0,1]} h_n = \int_0^1 h_n = \frac{\ln(1+n^2)}{2n}.$$

By Lemma 12.2.9 with $u = \ln n$ and $m = 2$, $n \geq \frac{(\ln n)^2}{2}$, so $\ln n \leq \sqrt{2n}$. Since $1 + n^2 \leq 2n^2$,

$$0 \leq \frac{\ln(1+n^2)}{2n} \leq \frac{\ln 2 + 2\ln n}{2n} \leq \frac{\ln 2}{2n} + \frac{\sqrt{2}}{\sqrt{n}} \to 0.$$

Exercise 16.3.4. *Hint.* Consider $A = \left\{f > \frac{1}{n}\right\}$.

Solution. For $n \in \mathbb{N}$, let $A_n = \left\{x \in E : f(x) > \frac{1}{n}\right\}$, which is measurable. Since $f > \frac{1}{n}$ on A_n, Theorem 16.3.5 (5) on A_n gives

$$0 = \int_{A_n} f \geq \frac{1}{n}\, m(A_n).$$

So $m(A_n) = 0$. The set $\{f > 0\}$ is $\bigcup_n A_n$, so it has measure zero by subadditivity. In the same way, with $B_n = \left\{f < -\frac{1}{n}\right\}$, $0 = \int_{B_n} f \leq -\frac{1}{n}\, m(B_n)$, so $m(B_n) = 0$ and $m(\{f < 0\}) = 0$. So $\{f \neq 0\}$ has measure zero, that is, $f = 0$ a.e. on E.

Exercise 16.3.5. *Hint.* By Example 6.3.2, its discontinuities in $(0, 1)$ are the rational numbers.

Exercise 16.3.6. *Hint.* The complement of C is open, and C contains no interval. Also $m(C) = 0$.

Exercise 16.3.7. *Hint.* If f and g are continuous at x, so are fg and $h \circ f$.

Exercise 16.3.8. *Hint.* Try $f = 0$ and $g = \chi_{\mathbb{Q}}$ on $[0, 1]$. For the second part, use Theorem 15.1.8.

Exercise 16.4.1. *Hint.* Consider the sets $\{f > \frac{1}{n}\}$.

Solution. ($\Rightarrow$) Suppose that $f > 0$ on some set $S \subset E$ with $m(S) > 0$. Then there exists a set $A \subset S$ with $m(A) > 0$ and an $\varepsilon > 0$ such that $f \geq \varepsilon$ on A. Indeed, take $S = \{x \in E : f(x) > 0\}$, which is measurable. Then $S = \bigcup_{k\in\mathbb{N}}\{x \in S : f(x) > \frac{1}{k}\}$. If each of these sets had measure zero, then $m(S) = 0$ by subadditivity. So some $A = \{x \in S : f(x) > \frac{1}{k}\}$ has $m(A) > 0$. Take $\varepsilon = \frac{1}{k}$. So

$$\int_E f \geq \int_S f \geq \int_A f \geq \varepsilon m(A) > 0.$$

The inequalities use Lemma 16.4.4 (2) and Theorem 16.4.2 (3), with $\int_A \varepsilon = \varepsilon m(A)$. So by contrapositive, if $\int_E f = 0$, $f = 0$ a.e.

($\Leftarrow$) Suppose $f = 0$ a.e. on E. Then $\int_E f = \int_E 0 = 0$ by Theorem 16.4.2 (3), since $f \leq 0$ a.e. and $0 \leq f$ a.e.

Exercise 16.4.2. First, notice $\{f_n\}$ is a sequence of nonnegative measurable functions. Note that for each $x \in \mathbb{R}$, there exists an $N \in \mathbb{N}$ with $N > x$ such that for all $n \geq N$, $f_n(x) = 0$. Thus $f_n(x) \to f \equiv 0$. So the hypotheses of Fatou's Lemma are satisfied. Now $\int_{[0,\infty)} f = 0$. But $\int_{[0,\infty)} f_n = \int_{[n,n+1)} 1 = 1$ and so $\liminf \int_{[0,\infty)} f_n = 1$. Thus $\int_{[0,\infty)} f < \liminf \int_{[0,\infty)} f_n$.

Exercise 16.4.3. Each f_n is nonnegative and measurable, and the sequence decreases since $[n+1,\infty) \subset [n,\infty)$. For each $x \in \mathbb{R}$, there exists an $N \in \mathbb{N}$ with $N > x$ such that for all $n \geq N$, $f_n(x) = 0$. Thus $f_n(x) \to f \equiv 0$. So $\int_{\mathbb{R}} f = 0$. However, $\int_{\mathbb{R}} f_n = \int_{[n,\infty)} 1 = \infty$. So $0 = \int_{\mathbb{R}} f \neq \lim \int_{\mathbb{R}} f_n = \infty$ and the Monotone Convergence Theorem does not hold.

Exercise 16.4.4. *Hint.* For continuity from the right, consider $\int_{(x_n,\infty)} f$ for x_n decreasing to x.

Exercise 16.4.5. *Hint.* $\alpha\chi_{\{f\geq\alpha\}} \leq f$ on E.

Exercise 16.4.6. *Hint.* For $y \geq 0$, $\frac{\lfloor 2^{n+1}y \rfloor}{2^{n+1}} \geq \frac{\lfloor 2^n y \rfloor}{2^n}$. On $\{f < n\} \cap [-n,n]$, $0 \leq f - \varphi_n < 2^{-n}$. Finish with the Monotone Convergence Theorem.

Exercise 16.4.7. *Hint.* For $0 \leq x < 1$, $\sum_{n=0}^{\infty} u_n(x) = 1$. Also $\int_{[0,1]} u_n = \frac{1}{n+1} - \frac{1}{n+2}$.

Exercise 16.4.8. *Hint.* Use Theorem 16.4.3.

Exercise 16.4.9. *Hint.* Use Proposition 16.4.15, the remark after it, and the p-Test.

Exercise 16.5.1. *Hint.* For $\alpha \geq 0$, $\{f^+ > \alpha\} = \{f > \alpha\}$. For $\alpha < 0$, it is the whole domain.

Solution. Let f be measurable with domain D. If $\alpha \geq 0$, then $\{f^+ > \alpha\} = \{f > \alpha\}$. Indeed, if $f(x) > \alpha \geq 0$, then $f^+(x) = f(x) > \alpha$. If $f^+(x) > \alpha \geq 0$, then $f^+(x) > 0$, so $f^+(x) = f(x)$. If $\alpha < 0$, then $\{f^+ > \alpha\} = D$, since $f^+ \geq 0$. In both cases the set is measurable. So f^+ is measurable.

The function $-f$ is measurable, since $\{-f > \alpha\} = \{f < -\alpha\}$ is measurable by Theorem 15.1.1. Since $f^- = \max\{-f, 0\} = (-f)^+$, the first part shows f^- is measurable.

If $f(x) \geq 0$, then $f^+(x) = f(x)$ and $f^-(x) = 0$. If $f(x) < 0$, then $f^+(x) = 0$ and $f^-(x) = -f(x)$. In both cases,

$$f^+(x) - f^-(x) = f(x) \quad \text{and} \quad f^+(x) + f^-(x) = |f(x)|.$$

These also hold when $f(x) = \pm\infty$, since one of $f^+(x)$ and $f^-(x)$ is 0.

Exercise 16.5.2. First, notice that if $c \geq 0$, then $cf = (cf)^+ - (cf)^- = cf^+ - cf^-$. That is, $(cf)^+ = cf^+$ and $(cf)^- = cf^-$. If $c < 0$, then $cf = (cf)^+ - (cf)^- = -cf^- + cf^+$, with $(cf)^+ = -cf^-$ and $(cf)^- = -cf^+$. If $c = 0$, then $cf = 0$ and both sides are 0. So by Theorem 16.4.2 (1), for $c > 0$,

$$\begin{aligned}\int_E cf &= \int_E (cf)^+ - \int_E (cf)^- = \int_E cf^+ - \int_E cf^- \\ &= c\int_E f^+ - c\int_E f^- = c\left(\int_E f^+ - \int_E f^-\right) = c\int_E f.\end{aligned}$$

For $c < 0$, using (1) with $-c > 0$,

$$\begin{aligned}\int_E cf &= \int_E (cf)^+ - \int_E (cf)^- = \int_E -cf^- - \int_E -cf^+ \\ &= -c\int_E f^- - (-c)\int_E f^+ = c\left(-\int_E f^- + \int_E f^+\right) = c\int_E f.\end{aligned}$$

In each case $\int_E (cf)^+$ and $\int_E (cf)^-$ are finite multiples of $\int_E f^\pm$, so they are finite and cf is integrable.

Exercise 16.5.3. *Hint.* $\left|\int_E f^+ - \int_E f^-\right| \le \int_E f^+ + \int_E f^-$.

Exercise 16.5.4. *Hint.* (a) Apply Fatou's Lemma to $g_n + f_n$ and $g_n - f_n$. (b) Apply part (a) to $|f_n - f| \le g_n + g$.

Exercise 16.5.5. *Hint.* For ($\Rightarrow$), $\big||f_n| - |f|\big| \le |f_n - f|$. For ($\Leftarrow$), apply Exercise 16.5.4 (a) with $|f_n - f| \le |f_n| + |f|$.

Exercise 16.5.6. *Hint.* (a) Integrate by parts first.

Solution. (a) The integrand is continuous on $[0,1]$, so by Theorem 16.3.3 the Lebesgue integral equals the Riemann integral. Integrate by parts (Theorem 9.6.2) with $\frac{1}{x+1}$ and x^n.

$$\int_0^1 \frac{nx^{n-1}}{x+1}\,dx = \left[\frac{x^n}{x+1}\right]_0^1 + \int_0^1 \frac{x^n}{(x+1)^2}\,dx = \frac{1}{2} + \int_{[0,1]} \frac{x^n}{(x+1)^2}\,dm.$$

For $0 \le x < 1$, $\frac{x^n}{(x+1)^2} \to 0$, so the integrands tend to 0 a.e. on $[0,1]$. Also $0 \le \frac{x^n}{(x+1)^2} \le 1$, and 1 is integrable over $[0,1]$. By the Dominated Convergence Theorem (Theorem 16.5.6), the last integral tends to 0. So the limit is $\frac{1}{2}$.

(b) Set $f_n(x) = \frac{\sin^3(x/n)}{x^3}$. Then

$$\lim_{n\to\infty} f_n(x) = \lim_{n\to\infty} \frac{\sin^3(x/n)}{x^3} = \frac{\sin^3(0)}{x^3} = 0$$

since sine is continuous. Define $g(x) = \frac{1}{x^3}$, which is integrable on $[1,\infty)$ by Proposition 16.4.15 and the p-Test. Since $|\sin^3(x/n)| \le 1$ for all $n \in \mathbb{N}$ and for all $x \in [1,\infty)$, $|f_n(x)| \le g(x)$ for all $n \in \mathbb{N}$ and for all $x \in [1,\infty)$. So Lebesgue's Dominated Convergence Theorem can be applied. Thus

$$\lim_{n\to\infty} \int_{[1,\infty)} \frac{\sin^3(x/n)}{x^3}\,dm(x) = \int_{[1,\infty)} \lim_{n\to\infty} \frac{\sin^3(x/n)}{x^3}\,dm(x) = \int_{[1,\infty)} 0\,dm = 0.$$

Exercise 16.5.7. *Hint.* Dominate the integrands by e^{-x}, which is integrable over $[0,\infty)$ by the remark after Proposition 16.4.15.

Exercise 16.5.8. *Hint.* Use Proposition 16.4.15 for $|f|$, and the Dominated Convergence Theorem.

Chapter 17. Differentiation and Integration

Exercise 17.1.1. The union of the two intervals is $[0,1]$, so $\mathcal{I}$ covers E. The two intervals overlap on $\left[\frac{1}{3},\frac{2}{3}\right]$. So a disjoint subcollection has at most one of them. If it is $\left[0,\frac{2}{3}\right]$, it misses $\left(\frac{2}{3},1\right]$. If it is $\left[\frac{1}{3},1\right]$, it misses $\left[0,\frac{1}{3}\right)$. If it is empty, it misses all of $[0,1]$. In every case the part of E that is missed has measure at least $\frac{1}{3}$.

Here $m^*(E) = 1 < \infty$. The hypothesis that fails is that $\mathcal{I}$ covers E in the sense of Vitali. For example, the only interval of $\mathcal{I}$ that contains 0 has length $\frac{2}{3}$. So for $\varepsilon \le \frac{2}{3}$, no interval of $\mathcal{I}$ contains 0 and has length less than ε.

Exercise 17.1.2. *Hint.* Look at the third step of the proof of Lemma 17.1.2.

Exercise 17.1.3. *Hint.* Given $x \in E$ and $\varepsilon > 0$, use the density of $\mathbb{Q}$ to find rational numbers $p < x < q$ with $q - p < \varepsilon$.

Exercise 17.2.1. *Hint.* Suppose $f(c) > f(d)$ with $c < d$. Choose $\varepsilon > 0$ so that $g(x) = f(x) + \varepsilon x$ still has $g(c) > g(d)$, and choose r with $g(d) < r < g(c)$. Look at the largest $x \in [c, d]$ with $g(x) \geq r$.

Solution. Two facts are used. First, for any c, the difference quotients of $g(x) = f(x) + cx$ are those of f plus c. So each Dini derivative of g is the matching Dini derivative of f plus c. Second, it is enough to show $f(c) \leq f(d)$ whenever $a < c < d < b$. Indeed, if $a \leq x < y \leq b$, choose $c_k \to x$ and $d_k \to y$ with $a < c_k < d_k < b$. Then $f(c_k) \leq f(d_k)$, and $f(x) \leq f(y)$ by continuity.

The first case is $D_+ f \geq 0$. Suppose $a < c < d < b$ and $f(c) > f(d)$. Choose $\varepsilon > 0$ with $\varepsilon(d - c) < f(c) - f(d)$, and let $g(x) = f(x) + \varepsilon x$. Then $g(c) > g(d)$. Choose r with $g(d) < r < g(c)$. Let $x_0 = \sup\{x \in [c, d] : g(x) \geq r\}$. Since g is continuous, $g(x_0) \geq r$. (Take points of this set that tend to x_0.) So $x_0 \neq d$, and $c \leq x_0 < d$. For $x_0 < x \leq d$, $g(x) < r \leq g(x_0)$. So every difference quotient $\frac{g(x_0+h)-g(x_0)}{h}$ with $0 < h \leq d - x_0$ is negative. So $D^+ g(x_0) \leq 0$, and then $D_+ g(x_0) \leq 0$. But $D_+ g(x_0) = D_+ f(x_0) + \varepsilon \geq \varepsilon > 0$, since $x_0 \in (a, b)$. This contradiction shows $f(c) \leq f(d)$.

The second case is $D^- f \geq 0$. Suppose $a < c < d < b$ and $f(c) > f(d)$. Define ε, g and r as before. Let $x_1 = \inf\{x \in [c, d] : g(x) \leq r\}$. By continuity, $g(x_1) \leq r$. So $x_1 \neq c$, and $c < x_1 \leq d$. For $c \leq x < x_1$, $g(x) > r \geq g(x_1)$. So every difference quotient $\frac{g(x_1)-g(x_1-h)}{h}$ with $0 < h \leq x_1 - c$ is negative, and $D^- g(x_1) \leq 0$. But $D^- g(x_1) = D^- f(x_1) + \varepsilon \geq \varepsilon > 0$. This contradiction shows $f(c) \leq f(d)$.

These two cases cover all four Dini derivatives. The first argument showed $D^+ g(x_0) \leq 0$. So it also proves the theorem when $D^+ f \geq 0$. The second proves it when $D^- f \geq 0$. Since $D_+ f \leq D^+ f$ and $D_- f \leq D^- f$, the hypothesis $D_+ f \geq 0$ implies $D^+ f \geq 0$, and $D_- f \geq 0$ implies $D^- f \geq 0$. So the cases $D^+ f \geq 0$ and $D^- f \geq 0$ cover all four.

Exercise 17.2.2. *Hint.* For $x \sin \frac{1}{x}$, the difference quotient at 0 is $\sin \frac{1}{h}$. For $\chi_{\mathbb{Q}}$, treat rational and irrational h separately.

Exercise 17.2.3. *Hint.* For any function a_h of h, $\limsup(-a_h) = -\liminf a_h$.

Exercise 17.2.4. *Hint.* Apply Theorem 17.2.3 to $-f$.

Solution. The function $-f$ is increasing on $[a, b]$. By Theorem 17.2.3, $-f$ is differentiable a.e., $(-f)'$ is measurable, and

$$\int_{[a,b]} (-f)' \leq -f(b) - \big(-f(a)\big) = f(a) - f(b).$$

Wherever $-f$ is differentiable, so is f, with $f' = -(-f)'$. So f is differentiable a.e. Since $(-f)' \geq 0$ and its integral is finite, $(-f)'$ is integrable. By Theorem 16.5.4 (1), f' is integrable and

$$\int_{[a,b]} f' = -\int_{[a,b]} (-f)' \geq -\big(f(a) - f(b)\big) = f(b) - f(a).$$

Exercise 17.2.5. *Hint.* If $x < y$, then $f(y) - f(x) = \sum_{x < r_n \leq y} 2^{-n} \geq 0$. If $x < r_m$, then $f(r_m) - f(x) \geq 2^{-m}$.

Exercise 17.3.1. *Hint.* Add the point y to a partition of $[a, x]$.

Solution. If $x = a$, then $P_f[a, a] = N_f[a, a] = 0$ by convention, as in the proof of Theorem 17.3.3, and there is nothing to prove. Suppose $a < x < y$. Let $\mathcal{P} = \{x_0, \dots, x_k\}$ be a partition of $[a, x]$. Then $\mathcal{P}' = \mathcal{P} \cup \{y\}$ is a partition of $[a, y]$, with one more subinterval $[x, y]$. So

$$p(\mathcal{P}') = p(\mathcal{P}) + [f(y) - f(x)]^+ \geq p(\mathcal{P}).$$

So $p(\mathcal{P}) \leq p(\mathcal{P}') \leq P_f[a, y]$. Taking the supremum over all partitions $\mathcal{P}$ of $[a, x]$ gives $P_f[a, x] \leq P_f[a, y]$. The same argument with $n(\mathcal{P})$ gives $N_f[a, x] \leq N_f[a, y]$. (These hold in $[0, \infty]$, even if f is not of bounded variation.)

Exercise 17.3.2. *Hint.* Try $f_n(x) = \frac{1}{n} \sin(n^2 x)$ on $[0, \pi]$.

Exercise 17.3.3. *Hint.* Apply Lemma 17.3.2 on $[a, x]$, and solve the two equations for $P_f[a, x]$ and $N_f[a, x]$.

Exercise 17.3.4. *Hint.* $\sin x$ increases on $[0, \frac{\pi}{2}]$, decreases on $[\frac{\pi}{2}, \frac{3\pi}{2}]$, and increases on $[\frac{3\pi}{2}, 2\pi]$. Find V_f with Theorem 8.2.1, then use Exercise 17.3.3.

Exercise 17.3.5. *Hint.* Let $V_f(x) = V_f[a, x]$. Show that $|f'| \leq V_f'$ a.e.

Solution. Let $V_f(x) = V_f[a, x]$, with $V_f(a) = 0$. By Theorem 8.2.2, V_f is increasing. By Theorem 17.2.3, V_f is differentiable a.e., and

$$\int_{[a,b]} V_f' \leq V_f(b) - V_f(a) = V_f[a, b].$$

By Corollary 17.3.4, f is differentiable a.e., and f' is integrable.

Let $a \leq x < y \leq b$. By Remark 8.1.6 and Theorem 8.2.1,

$$|f(y) - f(x)| \leq V_f[x, y] = V_f(y) - V_f(x).$$

Let $x \in [a, b)$ be a point where both f and V_f are differentiable. For $h > 0$ small, divide by h with $y = x + h$. Then $\left|\frac{f(x+h)-f(x)}{h}\right| \leq \frac{V_f(x+h)-V_f(x)}{h}$. Letting $h \to 0^+$ gives $|f'(x)| \leq V_f'(x)$. So $|f'| \leq V_f'$ a.e. By Theorem 16.4.2 (3),

$$\int_{[a,b]} |f'| \leq \int_{[a,b]} V_f' \leq V_f[a, b].$$

Exercise 17.4.1. *Hint.* $\frac{1}{h} \int_{[x,x+h]} f = \frac{F(x+h)-F(x)}{h}$, where F is the indefinite integral of f. Use Theorem 17.4.5.

Exercise 17.4.2. *Hint.* Apply Exercise 17.4.1 to $f = \chi_E$.

Exercise 17.4.3. *Hint.* $|F(y) - F(x)| = \left|\int_{[x,y]} f\right| \leq \int_{[x,y]} |f|$.

Solution. Since f is measurable and $|f| \leq K$ on $[a, b]$, f is integrable, with $\int_{[a,b]} |f| \leq K(b - a)$. Let $F(x) = F(a) + \int_{[a,x]} f$. Let $a \leq x < y \leq b$. Since $[a, y]$ is the disjoint union of $[a, x]$ and $(x, y]$, Theorem 16.5.4 (4) gives $F(y) - F(x) = \int_{(x,y]} f$. By Exercise 16.5.3 and Theorem 16.3.5 (5),

$$|F(y) - F(x)| = \left|\int_{(x,y]} f\right| \leq \int_{(x,y]} |f| \leq K\, m\big((x, y]\big) = K(y - x).$$

By symmetry, and trivially when $x = y$, $|F(x) - F(y)| \leq K|x - y|$ for all $x, y \in [a, b]$.

Exercise 17.4.4. *Hint.* The indefinite integral F is continuous, and $F(c) = F(d)$ for rational $c < d$. So F is constant. Then use Lemma 17.4.3.

Solution. Let $F(x) = \int_{[a,x]} f$. By Lemma 17.4.2, F is continuous on $[a,b]$. Let $c < d$ be rational numbers in $[a,b]$. As in the proof of Exercise 17.4.3, $F(d) - F(c) = \int_{(c,d]} f$. The sets $[c,d]$ and $(c,d]$ differ by one point, which has measure zero, so $\int_{(c,d]} f = \int_{[c,d]} f = 0$. So F takes the same value at all rational points of $[a,b]$.

Fix a rational $q_0 \in [a,b]$. Every $x \in [a,b]$ is the limit of a sequence of rational numbers in $[a,b]$, by the density of $\mathbb{Q}$ (Theorem 1.8.11). Since F is continuous, $F(x) = F(q_0)$. So F is constant on $[a,b]$. Since $F(a) = 0$, $\int_{[a,x]} f = 0$ for every $x \in [a,b]$. By Lemma 17.4.3, $f = 0$ a.e. on $[a,b]$.

Exercise 17.4.5. *Hint.* $\chi_{\mathbb{Q}} = 0$ a.e., so $F = 0$.

Exercise 17.5.1. *Hint.* Let $g(x) = \int_{[a,x]} f'$. Use Theorem 17.2.3 on $[x,y]$ to show $f - g$ is increasing.

Solution. Let f be increasing on $[a,b]$. By Theorem 17.2.3, f' exists a.e., $f' \geq 0$, and $\int_{[a,b]} f' \leq f(b) - f(a) < \infty$. So f' is integrable. Let

$$g(x) = \int_{[a,x]} f' \quad \text{and} \quad h = f - g.$$

Then g is an indefinite integral, so $g \in AC[a,b]$ by Theorem 17.5.8 (see Exercise 17.5.2).

The function h is increasing. Let $a \leq x < y \leq b$. Then $g(y) - g(x) = \int_{(x,y]} f' = \int_{[x,y]} f'$, since one point has measure zero. The function f is increasing on $[x,y]$, so Theorem 17.2.3 on $[x,y]$ gives $\int_{[x,y]} f' \leq f(y) - f(x)$. So

$$h(y) - h(x) = \big(f(y) - f(x)\big) - \int_{[x,y]} f' \geq 0.$$

By Theorem 17.4.5, $g' = f'$ a.e. So $h' = f' - g' = 0$ a.e., and h is singular. Thus $f = g + h$ is the sum of an absolutely continuous function and a singular function.

Exercise 17.5.2. *Hint.* Use Theorem 16.4.14.

Solution. Let $F(x) = F(a) + \int_{[a,x]} f$, where f is integrable on $[a,b]$. Let $\varepsilon > 0$. By Theorem 16.4.14, applied to $|f|$, there is a $\delta > 0$ such that $\int_A |f| < \varepsilon$ for every measurable $A \subset [a,b]$ with $m(A) < \delta$.

Let $\{(x_i, x_i')\}_{i=1}^n$ be nonoverlapping subintervals of $[a,b]$ with $x_i < x_i'$ and $\sum_{i=1}^n (x_i' - x_i) < \delta$. As in Exercise 17.4.3, $F(x_i') - F(x_i) = \int_{(x_i,x_i']} f$. The half-open intervals $(x_i, x_i']$ are pairwise disjoint, since the open intervals are. Let A be their union. Then $m(A) = \sum_i (x_i' - x_i) < \delta$. By Exercise 16.5.3 and Theorem 16.4.10,

$$\sum_{i=1}^n |F(x_i') - F(x_i)| \leq \sum_{i=1}^n \int_{(x_i,x_i']} |f| = \int_A |f| < \varepsilon.$$

So F is absolutely continuous on $[a,b]$.

Exercise 17.5.3. *Hint.* Continuous functions on $[a,b]$ are bounded. For fg, use $|f(y)g(y) - f(x)g(x)| \leq |f(y)||g(y) - g(x)| + |g(x)||f(y) - f(x)|$.

Solution. Let $\{(x_i, x_i')\}$ denote a finite collection of nonoverlapping subintervals of $[a, b]$, and let $\varepsilon > 0$.

First, $f + g \in AC[a, b]$. Choose δ_1 for f and δ_2 for g in Definition 17.5.1, both with $\frac{\varepsilon}{2}$, and let $\delta = \min\{\delta_1, \delta_2\}$. If $\sum |x_i' - x_i| < \delta$, then by the Triangle Inequality

$$\sum |(f+g)(x_i') - (f+g)(x_i)| \le \sum |f(x_i') - f(x_i)| + \sum |g(x_i') - g(x_i)| < \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon.$$

Next, $cf \in AC[a, b]$. If $c = 0$, then $cf = 0$ is constant. Otherwise choose δ for f with $\frac{\varepsilon}{|c|}$. Then $\sum |cf(x_i') - cf(x_i)| = |c| \sum |f(x_i') - f(x_i)| < \varepsilon$.

Finally, $fg \in AC[a, b]$. An absolutely continuous function is continuous, so f and g are bounded on $[a, b]$ (Theorem 6.3.14). Choose $K > 0$ with $|f| \le K$ and $|g| \le K$. For $x, y \in [a, b]$,

$$\begin{aligned} |f(y)g(y) - f(x)g(x)| &\le |f(y)||g(y) - g(x)| + |g(x)||f(y) - f(x)| \\ &\le K\big(|g(y) - g(x)| + |f(y) - f(x)|\big). \end{aligned}$$

Choose δ for both f and g with $\frac{\varepsilon}{2K}$. If $\sum |x_i' - x_i| < \delta$, then

$$\begin{aligned} \sum |fg(x_i') - fg(x_i)| &\le K\left(\sum |g(x_i') - g(x_i)| + \sum |f(x_i') - f(x_i)|\right) \\ &< K\left(\frac{\varepsilon}{2K} + \frac{\varepsilon}{2K}\right) = \varepsilon. \end{aligned}$$

So $f + g$, cf and fg are in $AC[a, b]$.

Exercise 17.5.4. *Hint.* One inequality is Exercise 17.3.5. For the other, use Corollary 17.5.9 on each subinterval of a partition.

Exercise 17.5.5. *Hint.* (a) Take $\delta = \frac{\varepsilon}{M}$. (b) For ($\Leftarrow$), use Corollary 17.5.9 to write $f(y) - f(x) = \int_{[x,y]} f'$. (c) Apply Theorem 17.2.2 to $Mx + f(x)$ and $Mx - f(x)$.

Exercise 17.5.6. *Hint.* $\sqrt{x} = \int_{[0,x]} \frac{1}{2\sqrt{t}}$. Use Proposition 16.4.15 and Theorem 17.5.8. For the Lipschitz condition, look at $\frac{\sqrt{h}}{h}$ as $h \to 0^+$.

Exercise 17.5.7. *Hint.* Cover Z by open intervals of small total length, and use that f has a maximum and a minimum on each closed subinterval.

Exercise 17.5.8. *Hint.* By Exercise 17.5.3, $fg \in AC[a, b]$. Use $(fg)' = f'g + fg'$ a.e., and apply Corollary 17.5.9 to fg.

Exercise 17.6.1. *Hint.* Prove versions of both identities for the functions ϕ_n, using the symmetry of C_n, and let $n \to \infty$. For the integral, ϕ is continuous. Use Theorem 16.3.3 and the substitution $x \mapsto 1 - x$.

Solution. By Lemma 5.1.2 (2), C_n is the union of the intervals $[s, s + 3^{-n}]$, where $s = \sum_{k=1}^{n} \frac{d_k}{3^k}$ with each $d_k \in \{0, 2\}$. Also $\phi_n(x) = \left(\frac{3}{2}\right)^n m\big(C_n \cap [0, x]\big)$, by Definition 16.2.1.

First comes the symmetry $\phi(1 - x) = 1 - \phi(x)$. The reflection $t \mapsto 1 - t$ maps $[s, s + 3^{-n}]$ onto $[1 - s - 3^{-n}, 1 - s]$. Since $\sum_{k=1}^{n} \frac{2}{3^k} = 1 - 3^{-n}$,

$$1 - s - 3^{-n} = \sum_{k=1}^{n} \frac{2 - d_k}{3^k},$$

and each $2-d_k \in \{0,2\}$. So the reflection maps C_n onto itself. It maps $C_n \cap [0, 1-x]$ onto $C_n \cap [x,1]$. Both are finite unions of intervals, and the reflection does not change lengths. So

$$\phi_n(1-x) = \left(\tfrac{3}{2}\right)^n m\big(C_n \cap [x,1]\big) = \left(\tfrac{3}{2}\right)^n \Big(m(C_n) - m\big(C_n \cap [0,x]\big)\Big) = 1 - \phi_n(x),$$

using $m(C_n) = \left(\frac{2}{3}\right)^n$ and $m(\{x\}) = 0$. Letting $n \to \infty$ gives $\phi(1-x) = 1 - \phi(x)$.

Next comes the scaling $\phi\left(\frac{x}{3}\right) = \frac{1}{2}\phi(x)$. The map $t \mapsto \frac{t}{3}$ sends $[s, s+3^{-n}]$ to $\left[\frac{s}{3}, \frac{s}{3} + 3^{-(n+1)}\right]$, and $\frac{s}{3} = \sum_{k=1}^{n+1} \frac{d'_k}{3^k}$ with $d'_1 = 0$ and $d'_{k+1} = d_k$. These are exactly the intervals of C_{n+1} whose first digit is 0, which are the intervals of C_{n+1} inside $\left[0, \frac{1}{3}\right]$. So $\frac{1}{3}C_n = C_{n+1} \cap \left[0, \frac{1}{3}\right]$. For $x \in [0,1]$, this gives $C_{n+1} \cap \left[0, \frac{x}{3}\right] = \frac{1}{3}\big(C_n \cap [0,x]\big)$. Scaling by $\frac{1}{3}$ multiplies the lengths of intervals by $\frac{1}{3}$. So

$$\phi_{n+1}\left(\tfrac{x}{3}\right) = \left(\tfrac{3}{2}\right)^{n+1} \cdot \tfrac{1}{3}\, m\big(C_n \cap [0,x]\big) = \tfrac{1}{2}\phi_n(x).$$

Letting $n \to \infty$ gives $\phi\left(\frac{x}{3}\right) = \frac{1}{2}\phi(x)$.

Finally, the symmetry gives the integral. Since ϕ is continuous, $\int_{[0,1]} \phi = \int_0^1 \phi$, by Theorem 16.3.3. The substitution $x = 1 - t$ (Theorem 9.6.3) gives $\int_0^1 \phi(1-t)\,dt = \int_0^1 \phi(x)\,dx$. So, by the symmetry,

$$\int_0^1 \phi = \int_0^1 \phi(1-t)\,dt = \int_0^1 \big(1 - \phi(t)\big)\,dt = 1 - \int_0^1 \phi.$$

So $\int_{[0,1]} \phi = \frac{1}{2}$.

Exercise 17.6.2. *Hint.* The first n digits of x determine the interval $J_{n,i}$ of C_n that contains x. Use Lemma 17.6.1 (2).

Exercise 17.6.3. *Hint.* (c) If B were a Borel set, then $A = (\psi^{-1})^{-1}[B]$ would be a Borel set, by an argument like Exercise 15.1.3.

Exercise 17.7.1. For $x \neq 0$, the Product Rule and the Chain Rule give

$$g'(x) = 2x \sin\frac{1}{x} + x^2 \cos\frac{1}{x} \cdot \left(-\frac{1}{x^2}\right) = 2x \sin\frac{1}{x} - \cos\frac{1}{x}.$$

At 0, $\left|\frac{g(h)-g(0)}{h}\right| = \left|h \sin\frac{1}{h}\right| \leq |h| \to 0$, so $g'(0) = 0$. For $0 < |x| \leq 1$, $|g'(x)| \leq 2|x| + 1 \leq 3$.

Let $x_k = \frac{1}{2k\pi}$ and $y_k = \frac{1}{(2k+1)\pi}$. Both tend to 0. Since $\sin\frac{1}{x_k} = \sin\frac{1}{y_k} = 0$, $g'(x_k) = -\cos(2k\pi) = -1$ and $g'(y_k) = -\cos((2k+1)\pi) = 1$. Two sequences tending to 0 give different limits of g', so $\lim_{x\to 0} g'(x)$ does not exist, by Theorem 6.1.18.

Exercise 17.7.2. Before step n, the total length removed is $\sum_{j=1}^{n-1} 2^{j-1} a 3^{-j} = a\left(1 - \left(\frac{2}{3}\right)^{n-1}\right)$, by the geometric sum. The remaining length $1 - a + a\left(\frac{2}{3}\right)^{n-1}$ is split equally among 2^{n-1} intervals, each of length

$$\ell_n = \frac{1-a}{2^{n-1}} + \frac{a}{3^{n-1}} > \frac{a}{3^n}.$$

So each step can be carried out. As in Example 17.7.1, the set K_a that remains is closed, and it contains no open interval since $\ell_n \to 0$. The removed intervals are disjoint, with total length $\sum_{n=1}^{\infty} 2^{n-1} a 3^{-n} = a$. So $m(K_a) = 1 - a$.

Exercise 17.7.3. *Hint.* Every neighborhood of a point of K contains points of a removed interval, since K contains no open interval. For integrability, use Theorem 9.2.9 and $m(K) = \frac{1}{2}$.

Exercise 17.7.4. *Hint.* Use Theorem 17.7.3 (4) with $x = 1$. On $[\alpha, \beta]$, F' is bounded and continuous except at α and β.

Chapter 18. L^p Spaces

Exercise 18.1.1. Let $f \in L^r(E)$, and let $s = \frac{r}{p} > 1$, with conjugate exponent s'. Then $|f|^p \in L^s(E)$, since $(|f|^p)^s = |f|^r$, and $1 \in L^{s'}(E)$, since $m(E) < \infty$. By Theorem 18.1.3,

$$\int_E |f|^p \cdot 1 \le \left(\int_E |f|^r \right)^{p/r} m(E)^{1/s'} = \|f\|_r^p \, m(E)^{1-p/r}.$$

So $f \in L^p(E)$, and taking pth roots gives $\|f\|_p \le m(E)^{1/p-1/r} \|f\|_r$. If $f \in L^\infty(E)$, then $|f|^p \le \|f\|_\infty^p$ almost everywhere by Exercise 18.1.4, so $\int_E |f|^p \le \|f\|_\infty^p m(E) < \infty$.

Exercise 18.1.2. Let $f(x) = x^{-1/2}$ for $0 < x < 1$ and $f(x) = 0$ for $x \ge 1$. For $0 < \delta < 1$, $\int_\delta^1 x^{-1/2}\, dx = 2 - 2\sqrt{\delta}$. By Theorem 16.4.7 applied to $f\chi_{[1/n,1)}$, $\int_E f = 2$, so $f \in L^1(E)$. But $f^2 = \frac{1}{x}$ on $(0,1)$, and $\int_\delta^1 \frac{dx}{x} = -\ln \delta \to \infty$. So $f \notin L^2(E)$.

Let $g(x) = 0$ for $0 < x < 1$ and $g(x) = \frac{1}{x}$ for $x \ge 1$. Then $\int_1^b \frac{dx}{x^2} = 1 - \frac{1}{b} \to 1$, so $g \in L^2(E)$ by Theorem 16.4.7. But $\int_1^b \frac{dx}{x} = \ln b \to \infty$, so $g \notin L^1(E)$.

Exercise 18.1.3. *Hint.* Take $p = q = 2$ in Theorem 18.1.3.

Exercise 18.1.4. *Hint.* For each n, $|f| \le \|f\|_\infty + \frac{1}{n}$ off a set of measure zero. Take the union of these sets.

Exercise 18.2.1. For $2^k \le n < 2^{k+1}$, f_n is 1 on an interval of length 2^{-k} and 0 elsewhere. So $\|f_n\|_p^p = 2^{-k}$. As $n \to \infty$, $k \to \infty$, so $\|f_n\|_p \to 0$.

Fix $x \in [0,1]$ and $k \ge 1$. The intervals $[j2^{-k}, (j+1)2^{-k}]$ with $0 \le j < 2^k$ cover $[0,1]$, so $f_n(x) = 1$ for at least one n with $2^k \le n < 2^{k+1}$. Since there are at least two such intervals and x lies in at most two of them, if $k \ge 2$ then $f_n(x) = 0$ for another such n. So $f_n(x) = 1$ and $f_n(x) = 0$ each happen for infinitely many n, and $\{f_n(x)\}$ does not converge.

The subsequence $f_{2^k} = \chi_{[0,2^{-k}]}$ converges to 0 at every $x \in (0,1]$, so almost everywhere.

Exercise 18.2.2. *Hint.* Off one set of measure zero, $|f_n - f_m| \le \|f_n - f_m\|_\infty$ for all m, n. There the sequence is uniformly Cauchy.

Exercise 18.2.3. *Hint.* $\{|f_n - f| \ge \varepsilon\} = \{|f_n - f|^p \ge \varepsilon^p\}$.

Exercise 18.3.1. Repeat the proof of Theorem 18.3.1. The first step is unchanged. In the second step, Theorem 15.2.7 also gives a continuous h with $-N \le h \le N$

and $m(\{x : |f_N(x) - h(x)| \geq \eta\}) < \eta$. The same estimate gives $\int_{[a,b]} |f_N - h|^p \leq \eta^p(b-a) + (2N)^p\eta < \left(\frac{\varepsilon}{2}\right)^p$ for small η. Then $\|f - h\|_p \leq \|f - f_N\|_p + \|f_N - h\|_p < \varepsilon$.

Exercise 18.3.2. *Hint.* For $f = \chi_{[c,d]}$, the integral is $\frac{\cos nc - \cos nd}{n}$. Then use Theorem 18.3.1 with $p = 1$.

Appendix A. Construction of the Real Numbers by Cuts

Exercise A.1.1. *Hint.* (a) Check property (2). (b) Check property (3). (c) It is a cut. For property (3), use $r = \frac{3p+3}{p+3}$, as in Theorem 1.4.5. (d) First find the set, using Lemma A.1.5.

Exercise A.1.2. ($\Rightarrow$) If $\alpha = q^*$, then q is the smallest element of $\mathbb{Q} \setminus \alpha$, by Example A.1.3.

($\Leftarrow$) Let q be the smallest element of $\mathbb{Q} \setminus \alpha$. If $p \in \alpha$, then $p < q$ by Remark A.1.2, so $p \in q^*$. If $p < q$, then $p \notin \mathbb{Q} \setminus \alpha$, since q is the smallest element of that set. So $p \in \alpha$. Thus $\alpha = q^*$.

Exercise A.2.1. Since $\alpha \subsetneqq \beta$, there is a $p \in \beta$ with $p \notin \alpha$. Since β has no largest element, there is a $q \in \beta$ with $q > p$.

If $r \in \alpha$, then $r < p$ by Remark A.1.2, so $r < q$ and $r \in q^*$. So $\alpha \subset q^*$. Also $p \in q^*$ and $p \notin \alpha$. So $\alpha < q^*$.

If $r < q$, then $r \in \beta$, since $q \in \beta$. So $q^* \subset \beta$. Also $q \in \beta$ and $q \notin q^*$. So $q^* < \beta$.

Exercise A.2.2. *Hint.* By Theorem A.2.3, the supremum is the union of the cuts q^* with $q \in \alpha$.

Exercise A.3.1. (1) Since $r + s = s + r$ in $\mathbb{Q}$,

$$\alpha + \beta = \{r + s : r \in \alpha,\ s \in \beta\} = \{s + r : s \in \beta,\ r \in \alpha\} = \beta + \alpha.$$

(2) An element of $(\alpha + \beta) + \gamma$ is a sum $w + t$ with $w \in \alpha + \beta$ and $t \in \gamma$, and $w = r + s$ with $r \in \alpha$ and $s \in \beta$. So $(\alpha + \beta) + \gamma$ is the set of all numbers $(r + s) + t$ with $r \in \alpha$, $s \in \beta$ and $t \in \gamma$. In the same way, $\alpha + (\beta + \gamma)$ is the set of all numbers $r + (s + t)$. Since $(r + s) + t = r + (s + t)$ in $\mathbb{Q}$, the two cuts are equal.

Exercise A.3.2. *Hint.* By Example A.1.4, $\mathbb{Q} \setminus \alpha$ is the set B of Theorem 1.4.5. Use that B has no smallest element.

Exercise A.4.1. Only the additive facts of Corollary A.3.5 are used to rearrange sums, such as $-(\beta + \gamma) = (-\beta) + (-\gamma)$ and $-(-\gamma) = \gamma$.

The proof starts with the rule of signs. First, $(-\alpha)\beta = -(\alpha\beta)$ for all cuts α and β. If $\alpha = 0^*$ or $\beta = 0^*$, both sides are 0^*. Otherwise there are four cases, by Definition A.4.6.

- If $\alpha > 0^*$ and $\beta > 0^*$, then $-\alpha < 0^* < \beta$, so $(-\alpha)\beta = -(\alpha\beta)$ by definition.
- If $\alpha < 0^* < \beta$, then $\alpha\beta = -\big((-\alpha)\beta\big)$ by definition, so $(-\alpha)\beta = -(\alpha\beta)$.
- If $\beta < 0^* < \alpha$, then $-\alpha < 0^*$ and $\beta < 0^*$. So $(-\alpha)\beta = \alpha(-\beta)$ by definition, and $\alpha(-\beta) = -(\alpha\beta)$ since $\alpha\beta = -\big(\alpha(-\beta)\big)$.
- If $\alpha < 0^*$ and $\beta < 0^*$, then $\beta < 0^* < -\alpha$. So $(-\alpha)\beta = -\big((-\alpha)(-\beta)\big) = -(\alpha\beta)$.

Since multiplication is commutative, also $\alpha(-\beta) = -(\alpha\beta)$.

Next comes associativity. If one of α, β, γ is 0^*, both sides of $(\alpha\beta)\gamma = \alpha(\beta\gamma)$ are 0^*. If the identity holds for α, β and γ, it also holds with α replaced by $-\alpha$, since by the rule of signs

$$\big((-\alpha)\beta\big)\gamma = -\big((\alpha\beta)\gamma\big) = -\big(\alpha(\beta\gamma)\big) = (-\alpha)(\beta\gamma).$$

The same computation works for β and for γ. The identity holds when all three cuts are positive, by Theorem A.4.5 (2). Every nonzero cut is a positive cut or the negative of one. So the identity holds for all cuts.

Last comes distributivity. If $\alpha(\beta+\gamma) = \alpha\beta + \alpha\gamma$, then

$$(-\alpha)(\beta+\gamma) = -\big(\alpha\beta + \alpha\gamma\big) = (-\alpha)\beta + (-\alpha)\gamma.$$

So it is enough to treat $\alpha \geq 0^*$, and $\alpha = 0^*$ is clear. Let $\alpha > 0^*$. If $\beta, \gamma \geq 0^*$, the identity is Theorem A.4.5 (5), or is clear when β or γ is 0^*. If $\beta, \gamma \leq 0^*$, then $-\beta, -\gamma \geq 0^*$, and

$$\alpha(\beta+\gamma) = -\big(\alpha((-\beta)+(-\gamma))\big) = -\big(\alpha(-\beta) + \alpha(-\gamma)\big) = \alpha\beta + \alpha\gamma.$$

Finally, suppose one of β, γ is positive and the other is negative. Since addition is commutative, it is enough to take $\beta > 0^* > \gamma$. If $\beta+\gamma \geq 0^*$, then $\beta = (\beta+\gamma)+(-\gamma)$ is a sum of two cuts that are at least 0^*. So $\alpha\beta = \alpha(\beta+\gamma)+\alpha(-\gamma) = \alpha(\beta+\gamma) - \alpha\gamma$, which gives $\alpha(\beta+\gamma) = \alpha\beta + \alpha\gamma$. If $\beta+\gamma < 0^*$, then $-\gamma = \big(-(\beta+\gamma)\big) + \beta$ is a sum of two positive cuts. So

$$-(\alpha\gamma) = \alpha(-\gamma) = -\big(\alpha(\beta+\gamma)\big) + \alpha\beta,$$

which again gives $\alpha(\beta+\gamma) = \alpha\beta + \alpha\gamma$.

Exercise A.4.2. *Hint.* If $r, s > 0$ and $r^2, s^2 < 2$, then $(rs)^2 < 4$. For the other inclusion, let $0 < p < 2$. Apply Lemma A.4.4 with a rational $t > 1$ such that $t^2 \leq \frac{2}{p}$.

Exercise A.5.1. ($\Rightarrow$) Let $q \in \alpha$. If $r < q$, then $r \in \alpha$. So $q^* \subset \alpha$. Also $q \in \alpha$ and $q \notin q^*$. So $q^* < \alpha$.

($\Leftarrow$) Let $q^* < \alpha$. Then there is a $p \in \alpha$ with $p \notin q^*$, that is, $p \geq q$. If $q = p$, then $q \in \alpha$. If $q < p$, then $q \in \alpha$ by property (2) of Definition A.1.1.

Exercise A.5.2. *Hint.* Follow the case $p < 0 < q$ in the proof of Theorem A.5.1, using Definition A.4.6 and Theorem A.5.1 (3).

Appendix B. Construction of the Real Numbers by Cauchy Sequences

Exercise B.1.1. The proof starts with some basic facts. By induction, each x_n is a positive rational number. Let $d_n = x_n^2 - 2$. Then $d_1 = 2$, and

$$d_{n+1} = \frac{x_n^2}{4} + 1 + \frac{1}{x_n^2} - 2 = \left(\frac{x_n}{2} - \frac{1}{x_n}\right)^2 = \frac{d_n^2}{4x_n^2}.$$

So $d_n \geq 0$ for all n, and $d_n \neq 0$ by Theorem 1.4.4. So $x_n^2 > 2$ and $x_n > 1$. Also $x_n - x_{n+1} = \frac{x_n}{2} - \frac{1}{x_n} = \frac{d_n}{2x_n} > 0$. So $\{x_n\}$ decreases, and $1 < x_n \leq 2$.

The next step bounds d_n. Since $x_n^2 > 2$, $d_{n+1} < \frac{d_n^2}{8}$. So if $d_n \le 2$, then $d_{n+1} < \frac{d_n}{4}$. Since $d_1 = 2$, induction gives $d_n \le \frac{2}{4^{n-1}}$ for all n. Thus

$$0 < x_n - x_{n+1} = \frac{d_n}{2x_n} < \frac{d_n}{2} \le \frac{1}{4^{n-1}}.$$

This bound shows that $\{x_n\}$ is a Cauchy sequence. Let $m > n$. By Exercise 1.2.7,

$$0 < x_n - x_m = \sum_{k=n}^{m-1} (x_k - x_{k+1}) < \frac{1}{4^{n-1}} \sum_{j=0}^{m-n-1} \frac{1}{4^j} < \frac{1}{4^{n-1}} \cdot \frac{4}{3} < \frac{2}{n},$$

since $4^{n-1} \ge n$ by induction. Let $\varepsilon > 0$ be rational. By Lemma A.1.5, choose N with $N\varepsilon > 2$. Then $|x_m - x_n| < \frac{2}{N} < \varepsilon$ for all $m, n \ge N$. So $\{x_n\}$ is a Cauchy sequence in $\mathbb{Q}$.

It remains to show that $\{x_n\}$ has no rational limit. Suppose $L \in \mathbb{Q}$ and $|x_n - L| \to 0$. Since $0 < x_n \le 2$,

$$|x_n^2 - L^2| = |x_n - L|\,|x_n + L| \le (2 + |L|)\,|x_n - L|.$$

So $|x_n^2 - L^2| \to 0$. Also $|x_n^2 - 2| = d_n \le \frac{2}{4^{n-1}} \le \frac{2}{n} \to 0$. Since $|L^2 - 2| \le |L^2 - x_n^2| + |x_n^2 - 2|$ for every n, $|L^2 - 2|$ is less than every positive rational number. So $L^2 = 2$, which contradicts Theorem 1.4.4.

Exercise B.1.2. *Hint.* Write x as the sum of the null sequence $x - L$ and the constant sequence L.

Exercise B.2.1. An example shows that $\left\{\frac{1}{x_n}\right\}$ need not be a Cauchy sequence. Let $x_n = \frac{1}{n}$. Then x is a null sequence, so $x \in \mathcal{C}$, and $x_n \ne 0$. But $\frac{1}{x_n} = n$, and $|m - n| \ge 1$ whenever $m \ne n$. So $\{n\}$ is not a Cauchy sequence.

The second part of the exercise is the positive result. Suppose $[x] \ne [0]$ and $x_n \ne 0$ for every n. Then x is not null. By Lemma B.2.3, there are a rational $c > 0$ and an N with $|x_n| \ge c$ for all $n \ge N$. Let $y_n = \frac{1}{x_n}$. As in the proof of Theorem B.2.4, $|y_m - y_n| \le \frac{|x_m - x_n|}{c^2}$ for $m, n \ge N$, so $y \in \mathcal{C}$. Also $x_n y_n = 1$ for every n, so $[x][y] = [1]$. By Theorem 1.6.5 (7), $[y] = [x]^{-1}$.

Exercise B.3.1. By Theorem 1.7.3 (3), $\big||x_m| - |x_n|\big| \le |x_m - x_n|$. So $|x| \in \mathcal{C}$. By Lemma B.3.2 (2), there are three cases.

If x is positive, then $[x] > [0]$, so $\big|[x]\big| = [x]$. Also $x_n \ge c > 0$ for large n, so $|x_n| = x_n$ for large n. Then $|x| - x$ is eventually 0, so $|x| \sim x$ and $\big[|x|\big] = [x]$.

If x is null, then $[x] = [0]$, and $|x|$ is null too. So both sides are $[0]$.

If $-x$ is positive, then $[x] < [0]$, so $\big|[x]\big| = -[x] = [-x]$. Also $x_n \le -c < 0$ for large n, so $|x_n| = -x_n$ for large n. So $\big[|x|\big] = [-x]$.

Exercise B.3.2. *Hint.* Try a null sequence of positive numbers.

Exercise B.4.1. By Lemma B.4.1 (2), for each $k \in \mathbb{N}$ there is a $q_k \in \mathbb{Q}$ with $a_k - \frac{1}{k} < q_k < a_k$. So $|a_k - q_k| < \frac{1}{k}$.

First, the sequence q is in $\mathcal{C}$. To see this, let $\varepsilon > 0$ be rational. Choose K with $K\varepsilon > 3$ and with $|a_m - a_n| < \frac{\varepsilon}{3}$ for all $m, n \ge K$. For $m, n \ge K$,

$$|q_m - q_n| \le |q_m - a_m| + |a_m - a_n| + |a_n - q_n| < \frac{1}{m} + \frac{\varepsilon}{3} + \frac{1}{n} \le \varepsilon.$$

This is an inequality between rational numbers that holds in $\mathbb{R}$, so it holds in $\mathbb{Q}$, by Theorem B.3.5. So $q \in \mathcal{C}$. Let $s = [q]$.

Next, $\{a_k\}$ converges to s. To see this, let $\varepsilon > 0$ be a real number. By Lemma B.4.1 (2), there is a rational ε' with $0 < \varepsilon' < \varepsilon$. By Lemma B.4.2, there is an N with $|s - q_k| \leq \frac{\varepsilon'}{2}$ for all $k \geq N$. Choose $N' \geq N$ with $N'\varepsilon' > 2$. For $k \geq N'$,

$$|a_k - s| \leq |a_k - q_k| + |q_k - s| < \frac{1}{k} + \frac{\varepsilon'}{2} < \varepsilon' < \varepsilon.$$

So $a_k \to s$.

Exercise B.4.2. *Hint.* Choose q_1 and then each q_{k+1} in $\mathbb{Q}$ between $\max\{q_k, a - \frac{1}{k+1}\}$ and a, by Lemma B.4.1 (2).

Exercise B.5.1. First, $(m+1)_{\mathbb{F}} = m_{\mathbb{F}} + 1$ for every $m \in \mathbb{Z}$. For $m \geq 0$ this is the definition. For $m = -1$, $0_{\mathbb{F}} = 0 = -1_{\mathbb{F}} + 1$. For $m = -n$ with $n \geq 2$, the definition gives $n_{\mathbb{F}} = (n-1)_{\mathbb{F}} + 1$, so $(m+1)_{\mathbb{F}} = -(n-1)_{\mathbb{F}} = -n_{\mathbb{F}} + 1 = m_{\mathbb{F}} + 1$.

Next, sums are preserved. Fix $m \in \mathbb{Z}$. By induction on $k \in \mathbb{N}$, $(m+k)_{\mathbb{F}} = m_{\mathbb{F}} + k_{\mathbb{F}}$. The case $k = 1$ was just shown. If it holds for k, then

$$(m+k+1)_{\mathbb{F}} = (m+k)_{\mathbb{F}} + 1 = m_{\mathbb{F}} + k_{\mathbb{F}} + 1 = m_{\mathbb{F}} + (k+1)_{\mathbb{F}}.$$

The case $k = 0$ is clear. If $k = -n$ with $n \in \mathbb{N}$, then $(m-n)_{\mathbb{F}} + n_{\mathbb{F}} = m_{\mathbb{F}}$ by the case just proved, applied to $m - n$. So $(m-n)_{\mathbb{F}} = m_{\mathbb{F}} - n_{\mathbb{F}} = m_{\mathbb{F}} + (-n)_{\mathbb{F}}$. In particular, $(-j)_{\mathbb{F}} = -j_{\mathbb{F}}$ for every $j \in \mathbb{Z}$.

Products are preserved too. Fix $m \in \mathbb{Z}$. By induction on $k \in \mathbb{N}$, $(mk)_{\mathbb{F}} = m_{\mathbb{F}}\, k_{\mathbb{F}}$. For $k = 1$ this is clear. If it holds for k, then

$$(m(k+1))_{\mathbb{F}} = (mk+m)_{\mathbb{F}} = (mk)_{\mathbb{F}} + m_{\mathbb{F}} = m_{\mathbb{F}}\, k_{\mathbb{F}} + m_{\mathbb{F}} = m_{\mathbb{F}}\, (k+1)_{\mathbb{F}}.$$

For $k = 0$ both sides are 0. If $k = -n$ with $n \in \mathbb{N}$, then $(m(-n))_{\mathbb{F}} = -(mn)_{\mathbb{F}} = -(m_{\mathbb{F}}\, n_{\mathbb{F}}) = m_{\mathbb{F}}\, (-n)_{\mathbb{F}}$, by Theorem 1.6.5 (11).

Finally, order is preserved. By induction, $n_{\mathbb{F}} > 0$ for every $n \in \mathbb{N}$, since $1 > 0$ (Theorem 1.6.8 (5)) and a sum of two positive elements is positive. If $m < k$ in $\mathbb{Z}$, then $k - m \in \mathbb{N}$. So $k_{\mathbb{F}} - m_{\mathbb{F}} = (k-m)_{\mathbb{F}} > 0$, and $m_{\mathbb{F}} < k_{\mathbb{F}}$.

Exercise B.5.2. *Hint.* Show $\varphi(1) = 1$, and then $\varphi(q) = q$ for $q \in \mathbb{Q}$. Every $x \geq 0$ is a square, so $\varphi(x) \geq 0$. Then compare the rational numbers below x and below $\varphi(x)$, as in Appendix B.5.

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About the Author

Jeffrey T. Neugebauer is a Professor in the Department of Mathematics and Statistics at Eastern Kentucky University, where he has taught since 2011. He earned his Ph.D. in mathematics from Baylor University and his B.S. and M.S. from the University of Dayton. His research is on boundary value problems for ordinary and fractional differential equations, and he has published more than 55 research papers. This book grew out of his lecture notes for the undergraduate and graduate analysis courses at Eastern Kentucky University. Those notes have also been used at The Citadel.

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